Mathematics

Class 11 (English Medium)

Solved Exercises — Sindh Textbook Board

12 chapters — 67 exercises — 1502 questions

Unit 1: Complex Numbers

Exercise 1.1

Question: Evaluate: (i) i^10 . (-i^12)

Answer: Step 1: Recall the cycle of powers of i: i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, and the pattern repeats every 4

Step 2: Write i^10 = i^8 x i^2 = (i^4)^2 x i^2

Step 3: Substitute i^4 = 1 and i^2 = -1, so i^10 = 1 x (-1) = -1

Step 4: Write i^12 = (i^4)^3 = 1^3 = 1, so -i^12 = -1

Step 5: Multiply the two values, (-1) x (-1) = 1

Final answer: 1

Question: Evaluate: (ii) (-i)^16 . (-i)^6

Answer: Step 1: Recall (-i)^2 = -1 and (-i)^4 = 1, and the powers repeat in a cycle of 4

Step 2: Write (-i)^16 = ((-i)^4)^4 = 1^4 = 1

Step 3: Write (-i)^6 = (-i)^4 x (-i)^2

Step 4: Substitute (-i)^4 = 1 and (-i)^2 = -1, so (-i)^6 = 1 x (-1) = -1

Step 5: Multiply the two values, 1 x (-1) = -1

Final answer: -1

Question: Evaluate: (iii) (i)^11 . (-i)^14

Answer: Step 1: Recall i^4 = 1 and (-i)^4 = 1, so any power can be reduced by removing multiples of 4 from the exponent

Step 2: Write i^11 = i^8 x i^3 = (i^4)^2 x i^3 = 1 x (-i) = -i

Step 3: Write (-i)^14 = (-i)^12 x (-i)^2 = ((-i)^4)^3 x (-1)

Step 4: Substitute (-i)^4 = 1, so (-i)^14 = 1 x (-1) = -1

Step 5: Multiply the two values, (-i) x (-1) = i

Final answer: i

Question: (i) For what value of n, i^n is equal to 1, i, -1 or -i, where 1 <= n <= 4

Answer: Step 1: Compute each power in turn using i = sqrt(-1) and i^2 = -1

Step 2: For n = 1, i^1 = i

Step 3: For n = 2, i^2 = -1

Step 4: For n = 3, i^3 = i^2 x i = -1 x i = -i

Step 5: For n = 4, i^4 = i^2 x i^2 = (-1) x (-1) = 1

Step 6: So within 1 <= n <= 4, i^n equals i when n = 1, equals -1 when n = 2, equals -i when n = 3, and equals 1 when n = 4

Final answer: i^n = i for n = 1, i^n = -1 for n = 2, i^n = -i for n = 3, i^n = 1 for n = 4

Question: (ii) Show that i^n + i^(n+1) + i^(n+2) + i^(n+3) = 0, for all n in Z

Answer: Step 1: Take i^n common from all four terms, the left side becomes i^n (1 + i + i^2 + i^3)

Step 2: Substitute i^2 = -1 in the bracket

Step 3: Substitute i^3 = i^2 x i = -i in the bracket

Step 4: The bracket becomes 1 + i - 1 - i

Step 5: Add the real parts, 1 - 1 = 0, and add the imaginary parts, i - i = 0, so the bracket is 0

Step 6: Hence the whole expression is i^n x 0 = 0 for every integer n

Final answer: i^n + i^(n+1) + i^(n+2) + i^(n+3) = 0, hence proved

Question: Simplify the following: (i) (-6, 3) . (4, -2)

Answer: Step 1: For ordered pairs, multiplication is (a, b)(c, d) = (ac - bd, ad + bc)

Step 2: Here a = -6, b = 3, c = 4 and d = -2

Step 3: Compute the first component, ac - bd = (-6)(4) - (3)(-2) = -24 + 6 = -18

Step 4: Compute the second component, ad + bc = (-6)(-2) + (3)(4) = 12 + 12 = 24

Step 5: Combine the two components into one ordered pair

Final answer: (-18, 24)

Question: Simplify the following: (ii) (8, -4) / (-2, 2)

Answer: Step 1: Write the pairs in a + ib form, the division is (8 - 4i)/(-2 + 2i)

Step 2: Multiply numerator and denominator by the conjugate of the denominator, (-2 - 2i)

Step 3: Expand the numerator, (8 - 4i)(-2 - 2i) = -16 - 16i + 8i + 8i^2 =

-16 - 8i - 8 = -24 - 8i, using i^2 = -1

Step 4: Expand the denominator, (-2 + 2i)(-2 - 2i) = 4 + 4 = 8, since (c + di)(c - di) = c^2 + d^2

Step 5: Divide, (-24 - 8i)/8 = -3 - i

Step 6: Write back as an ordered pair

Final answer: (-3, -1), that is -3 - i

Question: Simplify the following: (iii) (2, 3) / (4, 5)

Answer: Step 1: Write the pairs in a + ib form, the division is (2 + 3i)/(4 + 5i)

Step 2: Multiply numerator and denominator by the conjugate of the denominator, (4 - 5i)

Step 3: Expand the numerator, (2 + 3i)(4 - 5i) = 8 - 10i + 12i - 15i^2 = 8 + 2i + 15 = 23 + 2i, using i^2 = -1

Step 4: Expand the denominator, (4 + 5i)(4 - 5i) = 16 + 25 = 41

Step 5: Divide, (23 + 2i)/41 = 23/41 + (2/41)i

Step 6: Write back as an ordered pair

Final answer: (23/41, 2/41)

Question: Simplify the following: (iv) (6 + 5i) - (4 + 3i)

Answer: Step 1: Open the bracket, changing the sign of every term of the second number, (6 + 5i) - (4 + 3i) = 6 + 5i - 4 - 3i

Step 2: Group the real parts together, 6 - 4 = 2

Step 3: Group the imaginary parts together, 5i - 3i = 2i

Step 4: Combine the two groups

Final answer: 2 + 2i

Question: Simplify the following: (v) (5 - 6i) + (3 + 4i) - (5i - 7)

Answer: Step 1: Open all brackets, changing signs where there is a minus before a bracket, giving 5 - 6i + 3 + 4i - 5i + 7

Step 2: Group the real parts together, 5 + 3 + 7 = 15

Step 3: Group the imaginary parts together, -6i + 4i - 5i = -7i

Step 4: Combine the two groups

Final answer: 15 - 7i

Question: Simplify the following: (vi) (4 + 5i)(6 + 7i)

Answer: Step 1: Multiply the two brackets term by term, (4 + 5i)(6 + 7i) = 4 x 6 + 4 x 7i + 5i x 6 + 5i x 7i

Step 2: Compute each product, 24 + 28i + 30i + 35i^2

Step 3: Substitute i^2 = -1, so 35i^2 = -35

Step 4: Group the real parts, 24 - 35 = -11

Step 5: Group the imaginary parts, 28i + 30i = 58i

Step 6: Combine the two groups

Final answer: -11 + 58i

Question: Simplify the following: (vii) (7 + 4i) / (8 + 5i)

Answer: Step 1: Multiply numerator and denominator by the conjugate of the denominator, (8 - 5i)

Step 2: Expand the numerator, (7 + 4i)(8 - 5i) = 56 - 35i + 32i - 20i^2 = 56 - 3i + 20 = 76 - 3i, using i^2 = -1

Step 3: Expand the denominator, (8 + 5i)(8 - 5i) = 64 + 25 = 89

Step 4: Divide both parts by 89

Final answer: 76/89 - (3/89)i

Question: Simplify the following: (viii) (5, -6) + (4, 8) - (3, -2)

Answer: Step 1: For ordered pairs, addition and subtraction are done component by component

Step 2: First add, (5, -6) + (4, 8) = (5 + 4, -6 + 8) = (9, 2)

Step 3: Now subtract, (9, 2) - (3, -2) = (9 - 3, 2 - (-2))

Step 4: Compute the first component, 9 - 3 = 6

Step 5: Compute the second component, 2 + 2 = 4

Final answer: (6, 4)

Question: Simplify: (i) (2 - i)^4

Answer: Step 1: First find the square, (2 - i)^2 = 4 - 4i + i^2 = 4 - 4i - 1 = 3 - 4i, using i^2 = -1

Step 2: Now (2 - i)^4 = ((2 - i)^2)^2 = (3 - 4i)^2

Step 3: Expand the square, (3 - 4i)^2 = 9 - 24i + 16i^2

Step 4: Substitute i^2 = -1, so 16i^2 = -16

Step 5: Group the real parts, 9 - 16 = -7

Step 6: The imaginary part stays -24i

Final answer: -7 - 24i

Question: Simplify: (ii) (-1 - i sqrt(3))^2

Answer: Step 1: Use the identity (a + b)^2 = a^2 + 2ab + b^2 with a = -1 and b = -i sqrt(3)

Step 2: Compute a^2 = (-1)^2 = 1

Step 3: Compute 2ab = 2(-1)(-i sqrt(3)) = 2i sqrt(3)

Step 4: Compute b^2 = (-i sqrt(3))^2 = i^2 x 3 = -3

Step 5: Add the three parts, 1 + 2i sqrt(3) - 3

Step 6: Group the real parts, 1 - 3 = -2

Final answer: -2 + 2 sqrt(3) i

Question: Simplify: (iii) (-1 + i sqrt(3))^2

Answer: Step 1: Use the identity (a + b)^2 = a^2 + 2ab + b^2 with a = -1 and b = i sqrt(3)

Step 2: Compute a^2 = (-1)^2 = 1

Step 3: Compute 2ab = 2(-1)(i sqrt(3)) = -2i sqrt(3)

Step 4: Compute b^2 = (i sqrt(3))^2 = i^2 x 3 = -3

Step 5: Add the three parts, 1 - 2i sqrt(3) - 3

Step 6: Group the real parts, 1 - 3 = -2

Final answer: -2 - 2 sqrt(3) i

Question: Simplify: (iv) (1 + i)^3

Answer: Step 1: Write the cube as (1 + i)^2 x (1 + i)

Step 2: Find the square, (1 + i)^2 = 1 + 2i + i^2 = 1 + 2i - 1 = 2i

Step 3: Multiply by (1 + i), giving 2i x (1 + i) = 2i + 2i^2

Step 4: Substitute i^2 = -1, so 2i^2 = -2

Step 5: Combine, -2 + 2i

Final answer: -2 + 2i

Question: Simplify: (v) (sqrt(2) + i)/(sqrt(2) - i)

Answer: Step 1: Multiply numerator and denominator by the conjugate of the denominator, (sqrt(2) + i)

Step 2: Expand the numerator, (sqrt(2) + i)^2 = 2 + 2 sqrt(2) i + i^2 = 2 + 2 sqrt(2) i - 1 = 1 + 2 sqrt(2) i

Step 3: Expand the denominator, (sqrt(2) - i)(sqrt(2) + i) = 2 + 1 = 3

Step 4: Divide both parts by 3

Final answer: 1/3 + (2 sqrt(2)/3) i

Question: Simplify: (vi) ((1 + i)/(1 - i)) . ((2 - i)/(1 - i))

Answer: Step 1: Simplify the first fraction, (1 + i)/(1 - i) = (1 + i)^2/((1 - i)(1 + i)) = (1 + 2i + i^2)/2 = 2i/2 = i

Step 2: Simplify the second fraction, (2 - i)/(1 - i) = (2 - i)(1 + i)/((1 - i)(1 + i))

Step 3: Expand the numerator, (2 - i)(1 + i) = 2 + 2i - i - i^2 = 2 + i + 1 = 3 + i, and the denominator is 2, so the fraction is (3 + i)/2

Step 4: Multiply the two simplified fractions, i x (3 + i)/2 = (3i + i^2)/2

Step 5: Substitute i^2 = -1, giving (3i - 1)/2

Final answer: -1/2 + (3/2) i

Question: Simplify: (vii) (2 + i)^2/(3 - 4i)

Answer: Step 1: Find the square in the numerator, (2 + i)^2 = 4 + 4i + i^2 = 3 + 4i

Step 2: The fraction becomes (3 + 4i)/(3 - 4i)

Step 3: Multiply numerator and denominator by the conjugate of the denominator, (3 + 4i)

Step 4: Expand the new numerator, (3 + 4i)^2 = 9 + 24i + 16i^2 = 9 + 24i - 16 = -7 + 24i

Step 5: Expand the new denominator, (3 - 4i)(3 + 4i) = 9 + 16 = 25

Step 6: Divide both parts by 25

Final answer: -7/25 + (24/25) i

Question: Simplify: (viii) 1/(2 - i)^2

Answer: Step 1: Find the square in the denominator, (2 - i)^2 = 4 - 4i + i^2 = 3 - 4i

Step 2: The fraction becomes 1/(3 - 4i)

Step 3: Multiply numerator and denominator by the conjugate of the denominator, (3 + 4i)

Step 4: The numerator becomes 3 + 4i

Step 5: The denominator becomes (3 - 4i)(3 + 4i) = 9 + 16 = 25

Step 6: Divide both parts by 25

Final answer: 3/25 + (4/25) i

Question: Show that z = 1 + i and z = 1 - i satisfy the equation z^2 - 2z + 2 = 0

Answer: Step 1: First take z = 1 + i and compute z^2 = (1 + i)^2 = 1 + 2i + i^2 = 2i

Step 2: Compute -2z = -2(1 + i) = -2 - 2i

Step 3: Substitute into the left side, z^2 - 2z + 2 = 2i + (-2 - 2i) + 2

Step 4: Group the real parts, -2 + 2 = 0, and group the imaginary parts, 2i - 2i = 0, so the total is 0 and z = 1 + i satisfies the equation

Step 5: Now take z = 1 - i and compute z^2 = (1 - i)^2 = 1 - 2i + i^2 = -2i

Step 6: Compute -2z = -2(1 - i) = -2 + 2i

Step 7: Substitute, z^2 - 2z + 2 = -2i + (-2 + 2i) + 2 = 0 by grouping real and imaginary parts as before

Final answer: Both z = 1 + i and z = 1 - i satisfy z^2 - 2z + 2 = 0, hence shown

Question: Find the conjugate and absolute value of: (i) 4 + 5i

Answer: Step 1: The conjugate of a + ib is a - ib, so the sign of the imaginary part is reversed

Step 2: Hence the conjugate of 4 + 5i is 4 - 5i

Step 3: The absolute value (modulus) is |z| = sqrt(a^2 + b^2)

Step 4: Substitute a = 4 and b = 5, |z| = sqrt(16 + 25) = sqrt(41)

Final answer: Conjugate = 4 - 5i, absolute value = sqrt(41)

Question: Find the conjugate and absolute value of: (ii) -1 + 7i

Answer: Step 1: The conjugate of a + ib is a - ib, so the sign of the imaginary part is reversed

Step 2: Hence the conjugate of -1 + 7i is -1 - 7i

Step 3: The absolute value is |z| = sqrt(a^2 + b^2)

Step 4: Substitute a = -1 and b = 7, |z| = sqrt(1 + 49) = sqrt(50) = 5 sqrt(2)

Final answer: Conjugate = -1 - 7i, absolute value = 5 sqrt(2)

Question: Find the conjugate and absolute value of: (iii) sqrt(3) i

Answer: Step 1: Write the number in a + ib form, 0 + sqrt(3) i, so a = 0

and b = sqrt(3)

Step 2: The conjugate reverses the sign of the imaginary part, giving -sqrt(3) i

Step 3: The absolute value is |z| = sqrt(a^2 + b^2) = sqrt(0 + 3)

Step 4: Simplify, sqrt(3)

Final answer: Conjugate = -sqrt(3) i, absolute value = sqrt(3)

Question: Find the conjugate and absolute value of: (iv) sqrt(7) - 3i

Answer: Step 1: Here a = sqrt(7) and b = -3

Step 2: The conjugate reverses the sign of the imaginary part, giving sqrt(7) + 3i

Step 3: The absolute value is |z| = sqrt(a^2 + b^2) = sqrt(7 + 9)

Step 4: Simplify, sqrt(16) = 4

Final answer: Conjugate = sqrt(7) + 3i, absolute value = 4

Question: Find the conjugate and absolute value of: (v) -3 - 4i

Answer: Step 1: Here a = -3 and b = -4

Step 2: The conjugate reverses the sign of the imaginary part, giving -3 + 4i

Step 3: The absolute value is |z| = sqrt(a^2 + b^2) = sqrt(9 + 16)

Step 4: Simplify, sqrt(25) = 5

Final answer: Conjugate = -3 + 4i, absolute value = 5

Question: Find the conjugate and absolute value of: (vi) (5 - 4i)^2

Answer: Step 1: First simplify the number, (5 - 4i)^2 = 25 - 40i + 16i^2

Step 2: Substitute i^2 = -1, giving 25 - 40i - 16 = 9 - 40i

Step 3: The conjugate reverses the sign of the imaginary part, giving 9 + 40i

Step 4: The absolute value is |z| = sqrt(9^2 + 40^2) = sqrt(81 + 1600) = sqrt(1681)

Step 5: Simplify, sqrt(1681) = 41

Final answer: Conjugate = 9 + 40i, absolute value = 41

Question: Find the conjugate and absolute value of: (vii) 2/3 - sqrt(-9)/sqrt(16)

Answer: Step 1: Simplify the roots, sqrt(-9) = 3i and sqrt(16) = 4

Step 2: So the number is 2/3 - (3i)/4 = 2/3 - (3/4)i, with a = 2/3 and b = -3/4

Step 3: The conjugate reverses the sign of the imaginary part, giving 2/3 + (3/4)i

Step 4: The absolute value is |z| = sqrt((2/3)^2 + (3/4)^2) = sqrt(4/9 + 9/16)

Step 5: Add the fractions with denominator 144, sqrt((64 + 81)/144) = sqrt(145)/12

Final answer: Conjugate = 2/3 + (3/4)i, absolute value = sqrt(145)/12

Question: Find the conjugate and absolute value of: (viii) ((1 + i)(1 + 2i))/(3 + i)

Answer: Step 1: Expand the numerator, (1 + i)(1 + 2i) = 1 + 2i + i + 2i^2 = 1 + 3i - 2 = -1 + 3i

Step 2: The fraction becomes (-1 + 3i)/(3 + i)

Step 3: Multiply numerator and denominator by the conjugate (3 - i),

numerator: (-1 + 3i)(3 - i) = -3 + i + 9i - 3i^2 = -3 + 10i + 3 = 10i

Step 4: The denominator is (3 + i)(3 - i) = 9 + 1 = 10

Step 5: Divide, 10i/10 = i, so the number is simply i

Step 6: The conjugate of i is -i, and the absolute value is |i| = sqrt(0^2 + 1^2) = 1

Final answer: Conjugate = -i, absolute value = 1

Question: Find real and imaginary parts of: (i) 2i(3 - 5i)

Answer: Step 1: Multiply out, 2i(3 - 5i) = 6i - 10i^2

Step 2: Substitute i^2 = -1, so -10i^2 = 10

Step 3: The result is 10 + 6i

Step 4: The part without i is the real part and the coefficient of i is the imaginary part

Final answer: Real part = 10, imaginary part = 6

Question: Find real and imaginary parts of: (ii) (sqrt(5) + i)/(sqrt(5) - i)

Answer: Step 1: Multiply numerator and denominator by the conjugate of the denominator, (sqrt(5) + i)

Step 2: Expand the numerator, (sqrt(5) + i)^2 = 5 + 2 sqrt(5) i + i^2 = 5 + 2 sqrt(5) i - 1 = 4 + 2 sqrt(5) i

Step 3: Expand the denominator, (sqrt(5) - i)(sqrt(5) + i) = 5 + 1 = 6

Step 4: Divide, (4 + 2 sqrt(5) i)/6 = 2/3 + (sqrt(5)/3) i

Step 5: Read off the two parts from the a + ib form

Final answer: Real part = 2/3, imaginary part = sqrt(5)/3

Question: If z = x + iy where Re(z) = 0 and |z| = 2, then find z

Answer: Step 1: Re(z) = 0 means the real part x = 0, so z = 0 + iy = iy, a purely imaginary number

Step 2: The condition |z| = 2 gives sqrt(x^2 + y^2) = 2

Step 3: Substitute x = 0, giving sqrt(y^2) = 2, that is |y| = 2

Step 4: So y = 2 or y = -2

Step 5: Both signs are possible because the modulus only fixes the distance from the origin

Final answer: z = 2i or z = -2i

Question: Solve the complex equation: (i) (x, y)(2, 3) = (-4, 7)

Answer: Step 1: Multiply the ordered pairs by (a, b)(c, d) = (ac - bd, ad + bc), giving (2x - 3y, 3x + 2y) = (-4, 7)

Step 2: Equate corresponding components, 2x - 3y = -4 and 3x + 2y = 7

Step 3: Multiply the first equation by 2 and the second by 3, giving 4x - 6y = -8 and 9x + 6y = 21

Step 4: Add the two equations to eliminate y, 13x = 13, so x = 1

Step 5: Substitute x = 1 in 2x - 3y = -4, giving 2 - 3y = -4, so -3y = -6 and y = 2

Step 6: Check: (1, 2)(2, 3) = (2 - 6, 3 + 4) = (-4, 7), which is correct

Final answer: x = 1, y = 2

Question: Solve the complex equation: (ii) (x + 3i)^2 = 2yi, where x and y are real

Answer: Step 1: Expand the square, (x + 3i)^2 = x^2 + 6xi + 9i^2 = x^2 + 6xi - 9

Step 2: Write both sides in a + ib form: left side is (x^2 - 9) + 6xi and right side is 0 + 2yi

Step 3: Equate the real parts, x^2 - 9 = 0, so x^2 = 9 and x = 3 or x = -3

Step 4: Equate the imaginary parts, 6x = 2y, so y = 3x

Step 5: If x = 3 then y = 9, and if x = -3 then y = -9

Final answer: (x, y) = (3, 9) or (x, y) = (-3, -9)

Question: Solve the complex equation: (iii) (x + 2yi)^2 = xi, where x and y are real

Answer: Step 1: Expand the square, (x + 2yi)^2 = x^2 + 4xyi + 4y^2 i^2 = x^2 + 4xyi - 4y^2

Step 2: Write both sides in a + ib form: left side is (x^2 - 4y^2) + 4xyi and right side is 0 + xi

Step 3: Equate the real parts, x^2 - 4y^2 = 0, so x^2 = 4y^2

Step 4: Equate the imaginary parts, 4xy = x, so x(4y - 1) = 0, giving x = 0 or y = 1/4

Step 5: If x = 0, then from x^2 = 4y^2 we get y = 0

Step 6: If y = 1/4, then x^2 = 4(1/16) = 1/4, so x = 1/2 or x = -1/2

Final answer: (x, y) = (0, 0), (1/2, 1/4) or (-1/2, 1/4)

Question: Represent the complex number (i) (2, -3) on the complex plane

Answer: Step 1: The pair (2, -3) means the complex number 2 - 3i with real part 2 and imaginary part -3

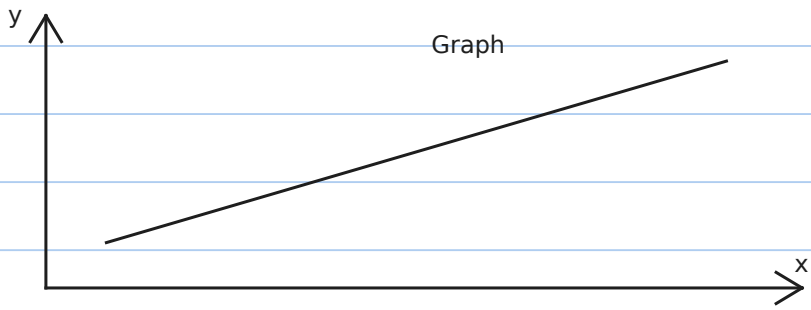
Step 2: On the complex plane the real part is measured along the horizontal (real) axis and the imaginary part along the vertical (imaginary) axis

Step 3: Move 2 units to the right of the origin along the real axis

Step 4: Move 3 units downwards because the imaginary part is negative

Step 5: Mark the point P(2, -3) in the fourth quadrant; the line from the origin to P represents the number

Final answer: The point (2, -3) in the fourth quadrant of the Argand diagram represents 2 - 3i



Question: Represent the complex number (ii) (3, 4) on the complex plane

Answer: Step 1: The pair (3, 4) means the complex number 3 + 4i with real part 3 and imaginary part 4

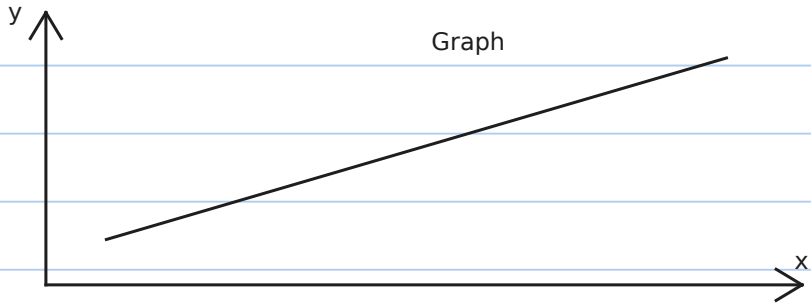
Step 2: On the complex plane the real part is measured along the horizontal (real) axis and the imaginary part along the vertical (imaginary) axis

Step 3: Move 3 units to the right of the origin along the real axis

Step 4: Move 4 units upwards because the imaginary part is positive

Step 5: Mark the point P(3, 4) in the first quadrant; the line from the origin to P represents the number, and its length is 5

Final answer: The point (3, 4) in the first quadrant of the Argand diagram represents 3 + 4i



Question: Represent the complex number (iii) (-5, 7) on the complex plane

Answer: Step 1: The pair (-5, 7) means the complex number -5 + 7i with real part -5 and imaginary part 7

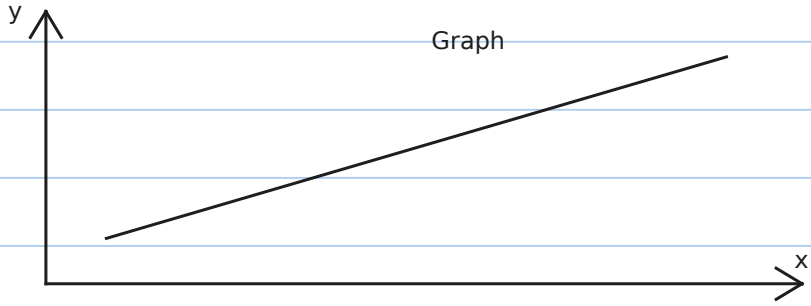
Step 2: On the complex plane the real part is measured along the horizontal (real) axis and the imaginary part along the vertical (imaginary) axis

Step 3: Move 5 units to the left of the origin because the real part is negative

Step 4: Move 7 units upwards because the imaginary part is positive

Step 5: Mark the point P(-5, 7) in the second quadrant; the line from the origin to P represents the number

Final answer: The point (-5, 7) in the second quadrant of the Argand diagram represents -5 + 7i



Question: Represent the complex number (iv) (-6, -2) on the complex plane

Answer: Step 1: The pair (-6, -2) means the complex number -6 - 2i with real part -6 and imaginary part -2

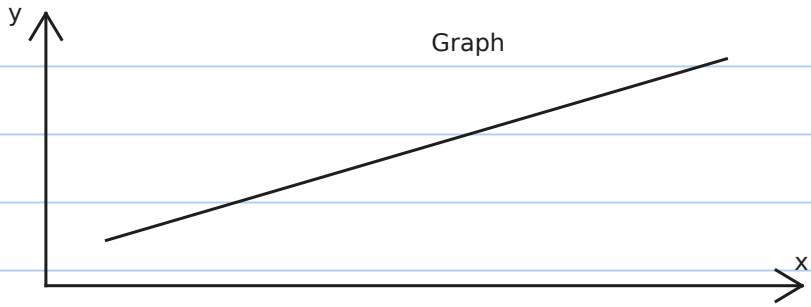
Step 2: On the complex plane the real part is measured along the horizontal (real) axis and the imaginary part along the vertical (imaginary) axis

Step 3: Move 6 units to the left of the origin because the real part is negative

Step 4: Move 2 units downwards because the imaginary part is negative

Step 5: Mark the point P(-6, -2) in the third quadrant; the line from the origin to P represents the number

Final answer: The point (-6, -2) in the third quadrant of the Argand diagram represents -6 - 2i



Question: Represent the complex number (v) (0, 6) on the complex plane

Answer: Step 1: The pair (0, 6) means the complex number 0 + 6i = 6i with real part 0 and imaginary part 6

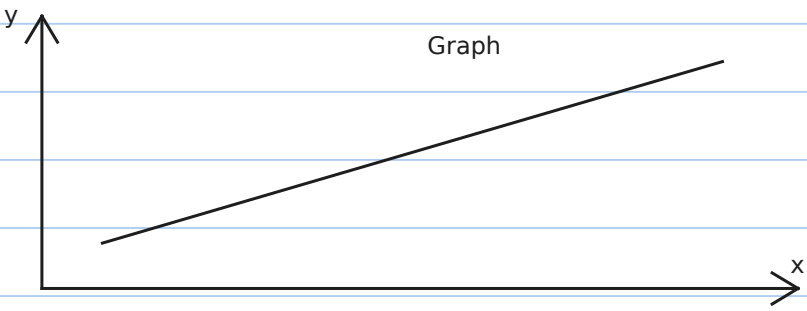
Step 2: Because the real part is zero, the point lies on the vertical (imaginary) axis itself

Step 3: Stay at 0 on the real axis

Step 4: Move 6 units upwards along the imaginary axis

Step 5: Mark the point P(0, 6) on the positive imaginary axis; it represents a purely imaginary number

Final answer: The point (0, 6) on the imaginary axis of the Argand diagram represents 6i



Question: Represent the complex number (vi) (-5, 0) on the complex plane

Answer: Step 1: The pair (-5, 0) means the complex number -5 + 0i = -5 with real part -5 and imaginary part 0

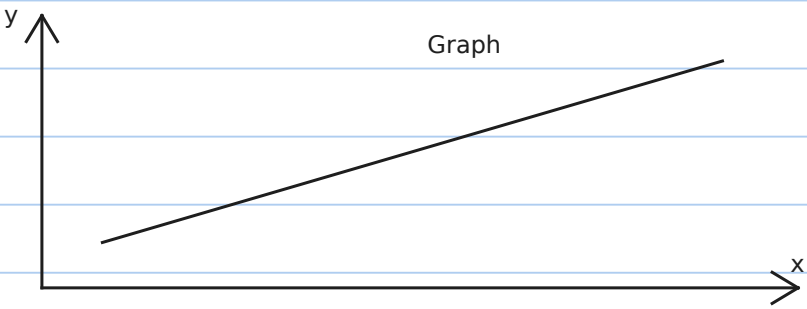
Step 2: Because the imaginary part is zero, the point lies on the horizontal (real) axis itself

Step 3: Move 5 units to the left of the origin along the real axis

Step 4: Stay at 0 on the imaginary axis

Step 5: Mark the point P(-5, 0) on the negative real axis; it represents a purely real number

Final answer: The point (-5, 0) on the real axis of the Argand diagram represents -5



Question: Find the length of the vector representing the complex number:

(i) -5 + 2i

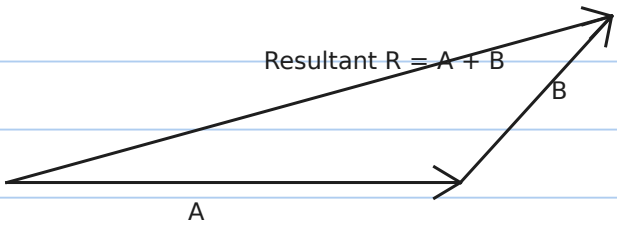
Answer: Step 1: The length of the vector representing z = a + ib is its modulus |z| = sqrt(a^2 + b^2)

Step 2: Here a = -5 and b = 2

Step 3: Substitute, length = sqrt((-5)^2 + 2^2) = sqrt(25 + 4)

Step 4: Add inside the root, sqrt(29)

Final answer: sqrt(29) units



Question: Find the length of the vector representing the complex number:

(ii) 7/3 + (8/3)i

Answer: Step 1: The length of the vector representing z = a + ib is its modulus |z| = sqrt(a^2 + b^2)

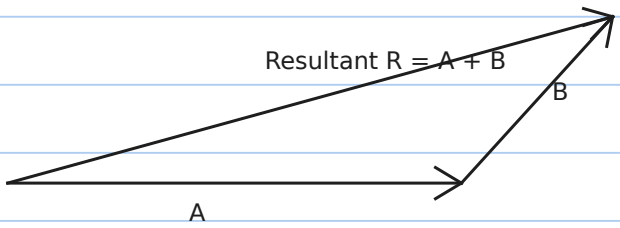
Step 2: Here a = 7/3 and b = 8/3

Step 3: Substitute, length = sqrt((7/3)^2 + (8/3)^2) = sqrt(49/9 + 64/9)

Step 4: Add the fractions, sqrt(113/9)

Step 5: Take the root of the denominator separately, sqrt(113)/3

Final answer: sqrt(113)/3 units



Question: Find the length of the vector representing the complex number: (iii) (1 + i)/sqrt(2)

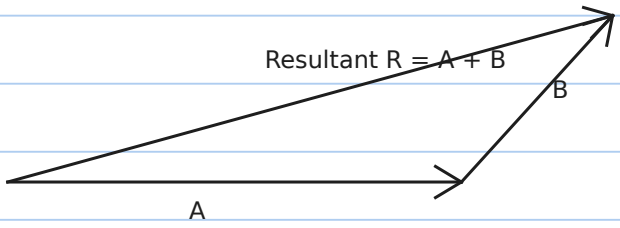
Answer: Step 1: Write the number as 1/sqrt(2) + (1/sqrt(2))i, so a = 1/sqrt(2) and b = 1/sqrt(2)

Step 2: The length is the modulus |z| = sqrt(a^2 + b^2)

Step 3: Substitute, length = sqrt(1/2 + 1/2) = sqrt(1)

Step 4: Simplify, sqrt(1) = 1

Final answer: 1 unit



Question: Find the length of the vector representing the complex number: (iv) (1 + sqrt(3) i)/2

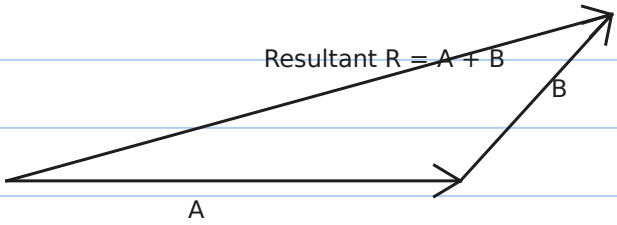
Answer: Step 1: Write the number as 1/2 + (sqrt(3)/2)i, so a = 1/2 and b = sqrt(3)/2

Step 2: The length is the modulus |z| = sqrt(a^2 + b^2)

Step 3: Substitute, length = sqrt(1/4 + 3/4) = sqrt(4/4) = sqrt(1)

Step 4: Simplify, sqrt(1) = 1

Final answer: 1 unit



Exercise 1.2

Question: Let z1 = 3 - 4i, z2 = 4 + 5i and z3 = -5 + 6i. Verify: (i) Addition of complex numbers is commutative

Answer: Step 1: Commutative property for addition means z1 + z2 = z2 + z1, so we compute both sides separately

Step 2: Left side, z1 + z2 = (3 - 4i) + (4 + 5i)

Step 3: Add the real parts, 3 + 4 = 7, and add the imaginary parts, -4i + 5i = i, so the left side is 7 + i

Step 4: Right side, z2 + z1 = (4 + 5i) + (3 - 4i)

Step 5: Add the real parts, 4 + 3 = 7, and add the imaginary parts, 5i - 4i = i, so the right side is 7 + i

Step 6: Both sides are equal, so addition is commutative for these numbers

Final answer: z1 + z2 = z2 + z1 = 7 + i, hence verified

Question: Let z1 = 3 - 4i, z2 = 4 + 5i and z3 = -5 + 6i. Verify: (ii) Multiplication of complex numbers is commutative

Answer: Step 1: Commutative property for multiplication means z1 z2 = z2 z1, so we compute both sides separately

Step 2: Left side, z1 z2 = (3 - 4i)(4 + 5i) = 12 + 15i - 16i - 20i^2

Step 3: Substitute i^2 = -1, so -20i^2 = 20, and group the parts, giving 12 + 20 + (15 - 16)i = 32 - i

Step 4: Right side, z2 z1 = (4 + 5i)(3 - 4i) = 12 - 16i + 15i - 20i^2

Step 5: Substitute i^2 = -1 and group the parts, giving 12 + 20 + (-16 + 15)i = 32 - i

Step 6: Both sides are equal, so multiplication is commutative for these numbers

Final answer: z1 z2 = z2 z1 = 32 - i, hence verified

Question: Let z1 = 3 - 4i, z2 = 4 + 5i and z3 = -5 + 6i. Verify: (iii) Addition of complex numbers is associative

Answer: Step 1: Associative property for addition means (z1 + z2) + z3 = z1 + (z2 + z3), so we compute both sides separately

Step 2: First z1 + z2 = (3 - 4i) + (4 + 5i) = 7 + i

Step 3: Left side, (z1 + z2) + z3 = (7 + i) + (-5 + 6i) = 2 + 7i

Step 4: First z2 + z3 = (4 + 5i) + (-5 + 6i) = -1 + 11i

Step 5: Right side, z1 + (z2 + z3) = (3 - 4i) + (-1 + 11i) = 2 + 7i

Step 6: Both sides are equal, so addition is associative for these numbers

Final answer: (z1 + z2) + z3 = z1 + (z2 + z3) = 2 + 7i, hence verified

Question: Let z1 = 3 - 4i, z2 = 4 + 5i and z3 = -5 + 6i. Verify: (iv) Multiplication of complex numbers is associative

Answer: Step 1: Associative property for multiplication means (z1 z2) z3 = z1 (z2 z3), so we compute both sides separately

Step 2: First z1 z2 = (3 - 4i)(4 + 5i) = 32 - i, as found by expanding and using i^2 = -1

Step 3: Left side, (z1 z2) z3 = (32 - i)(-5 + 6i) = -160 + 192i + 5i - 6i^2 = -160 + 197i + 6 = -154 + 197i

Step 4: First z2 z3 = (4 + 5i)(-5 + 6i) = -20 + 24i - 25i + 30i^2 = -20 - i - 30 = -50 - i

Step 5: Right side, z1 (z2 z3) = (3 - 4i)(-50 - i) = -150 - 3i + 200i + 4i^2 = -150 + 197i - 4 = -154 + 197i

Step 6: Both sides are equal, so multiplication is associative for these numbers

Final answer: (z1 z2) z3 = z1 (z2 z3) = -154 + 197i, hence verified

Question: Let z1 = 3 - 4i, z2 = 4 + 5i and z3 = -5 + 6i. Verify: (v) Multiplication of complex numbers is distributive over addition

Answer: Step 1: Distributive property means z1 (z2 + z3) = z1 z2 + z1 z3, so we compute both sides separately

Step 2: First z2 + z3 = (4 + 5i) + (-5 + 6i) = -1 + 11i

Step 3: Left side, z1 (z2 + z3) = (3 - 4i)(-1 + 11i) = -3 + 33i + 4i - 44i^2 = -3 + 37i + 44 = 41 + 37i

Step 4: Compute z1 z2 = 32 - i, as found by expanding and using i^2 = -1

Step 5: Compute z1 z3 = (3 - 4i)(-5 + 6i) = -15 + 18i + 20i - 24i^2 = -15

+ 38i + 24 = 9 + 38i

Step 6: Right side, z1 z2 + z1 z3 = (32 - i) + (9 + 38i) = 41 + 37i

Step 7: Both sides are equal, so multiplication is distributive over addition

Final answer: z1 (z2 + z3) = z1 z2 + z1 z3 = 41 + 37i, hence verified

Question: If z, z1 and z2 are complex numbers, prove that: (a) z + z-bar is real

Answer: Step 1: Let z = a + ib where a and b are real, then its conjugate is z-bar = a - ib

Step 2: Add them, z + z-bar = (a + ib) + (a - ib)

Step 3: Group the real parts, a + a = 2a

Step 4: Group the imaginary parts, ib - ib = 0

Step 5: The sum is 2a, which has no imaginary part

Step 6: A number with zero imaginary part is a real number

Final answer: z + z-bar = 2a, which is real, hence proved

Question: If z, z1 and z2 are complex numbers, prove that: (b) z - z-bar is imaginary

Answer: Step 1: Let z = a + ib where a and b are real, then its conjugate is z-bar = a - ib

Step 2: Subtract, z - z-bar = (a + ib) - (a - ib) = a + ib - a + ib

Step 3: Group the real parts, a - a = 0

Step 4: Group the imaginary parts, ib + ib = 2ib

Step 5: The result is 2ib, which has no real part

Step 6: A number with zero real part is a (purely) imaginary number

Final answer: z - z-bar = 2ib, which is imaginary, hence proved

Question: If z, z1 and z2 are complex numbers, prove that: (c) |z1 . z2| = |z1| |z2|

Answer: Step 1: Let z1 = a + ib and z2 = c + id where a, b, c, d are real

Step 2: Multiply, z1 z2 = (a + ib)(c + id) = (ac - bd) + (ad + bc)i

Step 3: Square the modulus of the product, |z1 z2|^2 = (ac - bd)^2 + (ad + bc)^2

Step 4: Expand, = a^2 c^2 - 2abcd + b^2 d^2 + a^2 d^2 + 2abcd + b^2 c^2, and the middle terms -2abcd and +2abcd cancel

Step 5: Group the remaining terms, = a^2(c^2 + d^2) + b^2(c^2 + d^2) = (a^2 + b^2)(c^2 + d^2)

Step 6: But |z1|^2 = a^2 + b^2 and |z2|^2 = c^2 + d^2, so |z1 z2|^2 = |z1|^2 |z2|^2

Step 7: Take the positive square root of both sides

Final answer: |z1 z2| = |z1| |z2|, hence proved

Question: If z, z1 and z2 are complex numbers, prove that: (d) |z1/z2| = |z1|/|z2|, z2 not equal to (0, 0)

Answer: Step 1: Let w = z1/z2, so that z1 = w z2

Step 2: Take the modulus of both sides, |z1| = |w z2|

Step 3: Use the product property of moduli, |z1| = |w| |z2|

Step 4: Divide both sides by |z2|, which is not zero because z2 is not (0, 0), giving |w| = |z1|/|z2|

Step 5: Replace w by z1/z2

Final answer: |z1/z2| = |z1|/|z2|, hence proved

Question: Find the additive inverse of: (i) 3 + 5i

Answer: Step 1: The additive inverse of z is the number which added to z gives the additive identity (0, 0)

Step 2: For z = a + ib the additive inverse is -a - ib

Step 3: Here a = 3 and b = 5, so the inverse is -3 - 5i

Step 4: Check: (3 + 5i) + (-3 - 5i) = 0 + 0i = (0, 0)

Final answer: -3 - 5i

Question: Find the additive inverse of: (ii) 6 - 5i

Answer: Step 1: The additive inverse of z is the number which added to z gives the additive identity (0, 0)

Step 2: For z = a + ib the additive inverse is -a - ib, so every sign is reversed

Step 3: Here a = 6 and b = -5, so the inverse is -6 + 5i

Step 4: Check: (6 - 5i) + (-6 + 5i) = 0 + 0i = (0, 0)

Final answer: -6 + 5i

Question: Find the additive inverse of: (iii) (11, 0)

Answer: Step 1: The additive inverse of (a, b) is (-a, -b)

Step 2: Here a = 11 and b = 0

Step 3: Reverse both signs, giving (-11, 0)

Step 4: Check: (11, 0) + (-11, 0) = (0, 0)

Final answer: (-11, 0)

Question: Find the additive inverse of: (iv) (2/3, 6)

Answer: Step 1: The additive inverse of (a, b) is (-a, -b)

Step 2: Here a = 2/3 and b = 6

Step 3: Reverse both signs, giving (-2/3, -6)

Step 4: Check: (2/3, 6) + (-2/3, -6) = (0, 0)

Final answer: (-2/3, -6)

Question: Find the additive inverse of: (v) (8/9, -4/5)

Answer: Step 1: The additive inverse of (a, b) is (-a, -b)

Step 2: Here a = 8/9 and b = -4/5

Step 3: Reverse both signs, giving (-8/9, 4/5)

Step 4: Check: (8/9, -4/5) + (-8/9, 4/5) = (0, 0)

Final answer: (-8/9, 4/5)

Question: Find the additive inverse of: (vi) (0, 3/8)

Answer: Step 1: The additive inverse of (a, b) is (-a, -b)

Step 2: Here a = 0 and b = 3/8

Step 3: Reverse both signs, giving (0, -3/8)

Step 4: Check: (0, 3/8) + (0, -3/8) = (0, 0)

Final answer: (0, -3/8)

Question: Find the multiplicative inverse of: (i) 3 + 5i

Answer: Step 1: The multiplicative inverse of z = a + ib is (a/(a^2 + b^2), -b/(a^2 + b^2))

Step 2: Here a = 3 and b = 5, so a^2 + b^2 = 9 + 25 = 34

Step 3: The first component is 3/34

Step 4: The second component is -5/34

Step 5: Check: (3 + 5i)(3/34 - (5/34)i) = (9 + 25)/34 = 1

Final answer: (3/34, -5/34), that is 3/34 - (5/34)i

Question: Find the multiplicative inverse of: (ii) 8i

Answer: Step 1: Write 8i as 0 + 8i, so a = 0 and b = 8

Step 2: Compute a^2 + b^2 = 0 + 64 = 64

Step 3: The inverse is (a/(a^2 + b^2), -b/(a^2 + b^2)) = (0/64, -8/64)

Step 4: Simplify -8/64 = -1/8

Step 5: Check: 8i x (-(1/8)i) = -i^2 = 1

Final answer: (0, -1/8), that is -(1/8)i

Question: Find the multiplicative inverse of: (iii) (10, 4)

Answer: Step 1: The multiplicative inverse of (a, b) is (a/(a^2 + b^2), -b/(a^2 + b^2))

Step 2: Here a = 10 and b = 4, so a^2 + b^2 = 100 + 16 = 116

Step 3: The first component is 10/116 = 5/58

Step 4: The second component is -4/116 = -1/29

Final answer: (5/58, -1/29)

Question: Find the multiplicative inverse of: (iv) (12, -7)

Answer: Step 1: The multiplicative inverse of (a, b) is (a/(a^2 + b^2), -b/(a^2 + b^2))

Step 2: Here a = 12 and b = -7, so a^2 + b^2 = 144 + 49 = 193

Step 3: The first component is 12/193

Step 4: The second component is -(-7)/193 = 7/193

Final answer: (12/193, 7/193)

Question: Find the multiplicative inverse of: (v) (-8, 0)

Answer: Step 1: The multiplicative inverse of (a, b) is (a/(a^2 + b^2), -b/(a^2 + b^2))

Step 2: Here a = -8 and b = 0, so a^2 + b^2 = 64 + 0 = 64

Step 3: The first component is -8/64 = -1/8

Step 4: The second component is 0/64 = 0

Final answer: (-1/8, 0)

Question: Find the multiplicative inverse of: (vi) (0, -3/4)

Answer: Step 1: The multiplicative inverse of (a, b) is (a/(a^2 + b^2), -b/(a^2 + b^2))

Step 2: Here a = 0 and b = -3/4, so a^2 + b^2 = 0 + 9/16 = 9/16

Step 3: The first component is 0/(9/16) = 0

Step 4: The second component is (3/4)/(9/16) = (3/4)(16/9) = 4/3

Step 5: Check: (0, -3/4) means -(3/4)i, and -(3/4)i x (4/3)i = -i^2 = 1

Final answer: (0, 4/3)

Question: Find the additive and multiplicative inverses by definition of: (i) (2, 3)

Answer: Step 1: For the additive inverse, let it be (c, d); by definition (2, 3) + (c, d) = (0, 0)

Step 2: This gives (2 + c, 3 + d) = (0, 0), so c = -2 and d = -3, and the additive inverse is (-2, -3)

Step 3: For the multiplicative inverse, let it be (c, d); by definition (2, 3)(c, d) = (1, 0)

Step 4: Multiply the pairs, (2c - 3d, 3c + 2d) = (1, 0), so 2c - 3d = 1 and 3c + 2d = 0

Step 5: Multiply the first equation by 2 and the second by 3, giving 4c - 6d = 2 and 9c + 6d = 0

Step 6: Add to eliminate d, 13c = 2, so c = 2/13

Step 7: Substitute in 3c + 2d = 0, giving 6/13 + 2d = 0, so d = -3/13

Final answer: Additive inverse = (-2, -3), multiplicative inverse = (2/13, -3/13)

Question: Find the additive and multiplicative inverses by definition of: (ii) (-4, 5)

Answer: Step 1: For the additive inverse, let it be (c, d); by definition (-4, 5) + (c, d) = (0, 0)

Step 2: This gives (-4 + c, 5 + d) = (0, 0), so c = 4 and d = -5, and the additive inverse is (4, -5)

Step 3: For the multiplicative inverse, let it be (c, d); by definition (-4, 5)(c, d) = (1, 0)

Step 4: Multiply the pairs, (-4c - 5d, 5c - 4d) = (1, 0), so -4c - 5d = 1 and 5c - 4d = 0

Step 5: From the second equation, c = 4d/5

Step 6: Substitute in the first equation, -4(4d/5) - 5d = 1, giving (-16d - 25d)/5 = 1, so -41d = 5 and d = -5/41

Step 7: Then c = 4(-5/41)/5 = -4/41

Final answer: Additive inverse = (4, -5), multiplicative inverse = (-4/41, -5/41)

Question: Verify the properties with z = 4 - 3i, z1 = 3 - 2i and z2 = 2 + 3i: (i) |z| = |-z| = |z-bar| = |-z-bar|

Answer: Step 1: Compute |z| = |4 - 3i| = sqrt(16 + 9) = sqrt(25) = 5

Step 2: Compute -z = -4 + 3i, so |-z| = sqrt(16 + 9) = 5

Step 3: Compute z-bar = 4 + 3i, so |z-bar| = sqrt(16 + 9) = 5

Step 4: Compute -z-bar = -4 - 3i, so |-z-bar| = sqrt(16 + 9) = 5

Step 5: All four values come out the same because squaring removes every sign change

Final answer: |z| = |-z| = |z-bar| = |-z-bar| = 5, hence verified

Question: Verify the properties with z = 4 - 3i, z1 = 3 - 2i and z2 = 2 + 3i: (ii) (z-bar)-bar = z

Answer: Step 1: Start with z = 4 - 3i

Step 2: Take its conjugate by reversing the sign of the imaginary part, z-bar = 4 + 3i

Step 3: Take the conjugate once more, (z-bar)-bar = 4 - 3i

Step 4: The result is exactly the original number z

Step 5: This shows that conjugating twice brings back the same number

Final answer: (z-bar)-bar = 4 - 3i = z, hence verified

Question: Verify the properties with z = 4 - 3i, z1 = 3 - 2i and z2 = 2 + 3i: (iii) z z-bar = |z|^2

Answer: Step 1: The conjugate is z-bar = 4 + 3i

Step 2: Multiply, z z-bar = (4 - 3i)(4 + 3i) = 16 + 12i - 12i - 9i^2

Step 3: Substitute i^2 = -1, so -9i^2 = 9, and the imaginary terms cancel, giving 16 + 9 = 25

Step 4: Compute the right side, |z|^2 = (sqrt(4^2 + 3^2))^2 = (sqrt(25))^2 = 25

Step 5: Left side and right side are both 25

Final answer: z z-bar = |z|^2 = 25, hence verified

Question: Verify the properties with z = 4 - 3i, z1 = 3 - 2i and z2 = 2 + 3i: (iv) (z1 + z2)-bar = z1-bar + z2-bar

Answer: Step 1: Add first, z1 + z2 = (3 - 2i) + (2 + 3i) = 5 + i

Step 2: Take the conjugate of the sum, (z1 + z2)-bar = 5 - i, which is the left side

Step 3: Take the conjugates separately, z1-bar = 3 + 2i and z2-bar = 2 - 3i

Step 4: Add them, z1-bar + z2-bar = (3 + 2i) + (2 - 3i) = 5 - i, which is the right side

Step 5: Both sides are equal

Final answer: (z1 + z2)-bar = z1-bar + z2-bar = 5 - i, hence verified

Question: Verify the properties with z = 4 - 3i, z1 = 3 - 2i and z2 = 2 + 3i: (v) (z1 z2)-bar = z1-bar . z2-bar

Answer: Step 1: Multiply first, z1 z2 = (3 - 2i)(2 + 3i) = 6 + 9i - 4i - 6i^2 = 6 + 5i + 6 = 12 + 5i

Step 2: Take the conjugate of the product, (z1 z2)-bar = 12 - 5i, which is the left side

Step 3: Take the conjugates separately, z1-bar = 3 + 2i and z2-bar = 2 - 3i

Step 4: Multiply them, z1-bar . z2-bar = (3 + 2i)(2 - 3i) = 6 - 9i + 4i - 6i^2 = 6 - 5i + 6 = 12 - 5i, which is the right side

Step 5: Both sides are equal

Final answer: (z1 z2)-bar = z1-bar . z2-bar = 12 - 5i, hence verified

Question: Verify the properties with z = 4 - 3i, z1 = 3 - 2i and z2 = 2 + 3i: (vi) (z1/z2)-bar = z1-bar/z2-bar

Answer: Step 1: Divide first, z1/z2 = (3 - 2i)/(2 + 3i) = (3 - 2i)(2 - 3i)/((2 + 3i)(2 - 3i))

Step 2: Expand the numerator, (3 - 2i)(2 - 3i) = 6 - 9i - 4i + 6i^2 = -13i, and the denominator is 4 + 9 = 13, so z1/z2 = -13i/13 = -i

Step 3: Take the conjugate, (z1/z2)-bar = i, which is the left side

Step 4: Now z1-bar/z2-bar = (3 + 2i)/(2 - 3i) = (3 + 2i)(2 + 3i)/((2 - 3i)(2 + 3i))

Step 5: Expand the numerator, (3 + 2i)(2 + 3i) = 6 + 9i + 4i + 6i^2 = 13i, and the denominator is 13, so the right side is 13i/13 = i

Step 6: Both sides are equal

Final answer: (z1/z2)-bar = z1-bar/z2-bar = i, hence verified

Question: If z1 = 3 - 2i and z2 = 2 - 3i, express in the form a + ib and also verify the Triangle inequality: (i) (z1 z2-bar)/(z1-bar)

Answer: Step 1: Write the conjugates, z2-bar = 2 + 3i and z1-bar = 3 + 2i

Step 2: Multiply the numerator, z1 z2-bar = (3 - 2i)(2 + 3i) = 6 + 9i - 4i - 6i^2 = 12 + 5i

Step 3: Divide by z1-bar, (12 + 5i)/(3 + 2i) = (12 + 5i)(3 - 2i)/((3 + 2i)(3 - 2i))

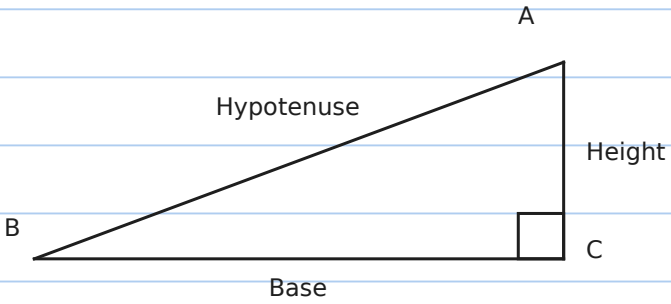
Step 4: Expand the numerator, (12 + 5i)(3 - 2i) = 36 - 24i + 15i - 10i^2 = 46 - 9i, and the denominator is 9 + 4 = 13

Step 5: So the value in a + ib form is 46/13 - (9/13)i

Step 6: For the Triangle inequality, |z1| = sqrt(9 + 4) = sqrt(13) and |z2| = sqrt(4 + 9) = sqrt(13), while z1 + z2 = 5 - 5i with |z1 + z2| = sqrt(50) = 5 sqrt(2)

Step 7: Check the bounds: ||z1| - |z2|| = 0 <= 5 sqrt(2) <= |z1| + |z2| = 2 sqrt(13), which is true since 5 sqrt(2) is about 7.07 and 2 sqrt(13) is about 7.21

Final answer: (z1 z2-bar)/(z1-bar) = 46/13 - (9/13)i, and |z1| - |z2| <= |z1 + z2| <= |z1| + |z2| holds, so the Triangle inequality is verified



Question: If z1 = 3 - 2i and z2 = 2 - 3i, express in the form a + ib and also verify the Triangle inequality: (ii) (z1-bar z2)/(z2-bar)

Answer: Step 1: Write the conjugates, z1-bar = 3 + 2i and z2-bar = 2 + 3i

Step 2: Multiply the numerator, z1-bar z2 = (3 + 2i)(2 - 3i) = 6 - 9i + 4i - 6i^2 = 12 - 5i

Step 3: Divide by z2-bar, (12 - 5i)/(2 + 3i) = (12 - 5i)(2 - 3i)/((2 + 3i)(2 - 3i))

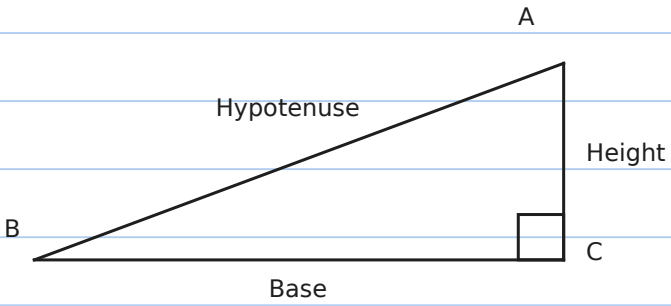
Step 4: Expand the numerator, (12 - 5i)(2 - 3i) = 24 - 36i - 10i + 15i^2 = 24 - 46i - 15 = 9 - 46i, and the denominator is 4 + 9 = 13

Step 5: So the value in a + ib form is 9/13 - (46/13)i

Step 6: For the Triangle inequality, |z1| = sqrt(13) and |z2| = sqrt(13), while |z1 + z2| = |5 - 5i| = 5 sqrt(2)

Step 7: Check the bounds: ||z1| - |z2|| = 0 <= 5 sqrt(2) <= |z1| + |z2| = 2 sqrt(13), which is true

Final answer: (z1-bar z2)/(z2-bar) = 9/13 - (46/13)i, and the Triangle inequality |z1| - |z2| <= |z1 + z2| <= |z1| + |z2| is verified



Question: If z1 = 3 - 2i and z2 = 2 - 3i, express in the form a + ib and also verify the Triangle inequality: (iii) ((z1 z2)-bar)/(z1)

Answer: Step 1: Multiply first, z1 z2 = (3 - 2i)(2 - 3i) = 6 - 9i - 4i + 6i^2 = -13i

Step 2: Take the conjugate of the product, (z1 z2)-bar = 13i

Step 3: Divide by z1, 13i/(3 - 2i) = 13i(3 + 2i)/((3 - 2i)(3 + 2i))

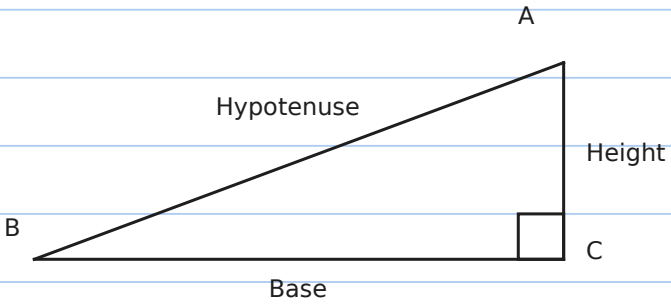
Step 4: Expand the numerator, 13i(3 + 2i) = 39i + 26i^2 = -26 + 39i, and the denominator is 9 + 4 = 13

Step 5: Divide, (-26 + 39i)/13 = -2 + 3i, which is in a + ib form

Step 6: For the Triangle inequality, |z1| = sqrt(13), |z2| = sqrt(13) and |z1 + z2| = 5 sqrt(2), and also |z1 - z2| = |1 + i| = sqrt(2) with ||z1| - |z2|| = 0 <= sqrt(2)

Step 7: Both bounds hold: 0 <= |z1 + z2| <= 2 sqrt(13) and 0 <= |z1 - z2|

Final answer: ((z1 z2)-bar)/(z1) = -2 + 3i, and the Triangle inequality is verified



Question: Find real and imaginary parts of: (i) (sqrt(2) + i)^(-2)

Answer: Step 1: The power -2 means the reciprocal of the square, so first find (sqrt(2) + i)^2 = 2 + 2 sqrt(2) i + i^2 = 1 + 2 sqrt(2) i

Step 2: Now take the reciprocal, (sqrt(2) + i)^(-2) = 1/(1 + 2 sqrt(2) i)

Step 3: Multiply numerator and denominator by the conjugate (1 - 2 sqrt(2) i)

Step 4: The numerator becomes 1 - 2 sqrt(2) i

Step 5: The denominator becomes 1 + 8 = 9, since (1)^2 + (2 sqrt(2))^2 = 1 + 8

Step 6: Divide, giving 1/9 - (2 sqrt(2)/9) i

Final answer: Real part = 1/9, imaginary part = -2 sqrt(2)/9

Question: Find real and imaginary parts of: (ii) (1 - sqrt(5) i)^(-1)

Answer: Step 1: The power -1 means the reciprocal, (1 - sqrt(5) i)^(-1) = 1/(1 - sqrt(5) i)

Step 2: Multiply numerator and denominator by the conjugate (1 + sqrt(5) i)

Step 3: The numerator becomes 1 + sqrt(5) i

Step 4: The denominator becomes 1 + 5 = 6

Step 5: Divide, giving 1/6 + (sqrt(5)/6) i

Final answer: Real part = 1/6, imaginary part = sqrt(5)/6

Question: Find real and imaginary parts of: (iii) (sqrt(3) - i)^(-2)

Answer: Step 1: First find the square, (sqrt(3) - i)^2 = 3 - 2 sqrt(3) i + i^2 = 2 - 2 sqrt(3) i

Step 2: Now take the reciprocal, (sqrt(3) - i)^(-2) = 1/(2 - 2 sqrt(3) i)

Step 3: Multiply numerator and denominator by the conjugate (2 + 2 sqrt(3) i)

Step 4: The numerator becomes 2 + 2 sqrt(3) i

Step 5: The denominator becomes 4 + 12 = 16

Step 6: Divide and simplify, (2 + 2 sqrt(3) i)/16 = 1/8 + (sqrt(3)/8) i

Final answer: Real part = 1/8, imaginary part = sqrt(3)/8

Question: Find real and imaginary parts of: (iv) (2i - sqrt(3))^(-1)

Answer: Step 1: Write the number as -sqrt(3) + 2i, so the reciprocal is 1/(-sqrt(3) + 2i)

Step 2: Multiply numerator and denominator by the conjugate (-sqrt(3) - 2i)

Step 3: The numerator becomes -sqrt(3) - 2i

Step 4: The denominator becomes 3 + 4 = 7

Step 5: Divide, giving -(sqrt(3)/7) - (2/7) i

Final answer: Real part = -sqrt(3)/7, imaginary part = -2/7

Question: Find real and imaginary parts of: (v) (1/(4i - 5))^(-1)

Answer: Step 1: The power -1 applied to a fraction gives back the original denominator, because the reciprocal of a reciprocal is the number itself

Step 2: So (1/(4i - 5))^(-1) = 4i - 5

Step 3: Write in a + ib form, -5 + 4i

Step 4: Read off the real part, -5

Step 5: Read off the imaginary part, 4

Final answer: Real part = -5, imaginary part = 4

Question: Find real and imaginary parts of: (vi) ((3 + 4i)/(5i - 4))^(-2)

Answer: Step 1: First simplify the inner fraction, (3 + 4i)/(-4 + 5i) = (3 + 4i)(-4 - 5i)/((-4 + 5i)(-4 - 5i))

Step 2: Expand the numerator, (3 + 4i)(-4 - 5i) = -12 - 15i - 16i - 20i^2 = 8 - 31i, and the denominator is 16 + 25 = 41, so the fraction is (8 - 31i)/41

Step 3: The required power -2 is the reciprocal of the square, so instead work with the reciprocal fraction (5i - 4)/(3 + 4i) and then square it

Step 4: Compute (5i - 4)/(3 + 4i) = (-4 + 5i)(3 - 4i)/((3 + 4i)(3 - 4i)) = (-12 + 16i + 15i - 20i^2)/25 = (8 + 31i)/25

Step 5: Square it, ((8 + 31i)/25)^2 = (64 + 496i + 961i^2)/625 = (64 - 961 + 496i)/625

Step 6: Simplify the real part, 64 - 961 = -897, giving -897/625 + (496/625) i

Final answer: Real part = -897/625, imaginary part = 496/625

Question: Find real and imaginary parts of: (vii) ((3i - 2)/(2 - 3i))^(-2)

Answer: Step 1: Notice that 3i - 2 = -(2 - 3i), because reversing all signs of

2 - 3i gives -2 + 3i = 3i - 2

Step 2: So the fraction is (3i - 2)/(2 - 3i) = -1

Step 3: Now apply the power, (-1)^(-2) = 1/((-1)^2) = 1/1 = 1

Step 4: Write 1 in a + ib form, 1 + 0i

Step 5: Read off the two parts

Final answer: Real part = 1, imaginary part = 0

Question: If x = 1 - 5i, show that x^2 - 2x + 26 = 0

Answer: Step 1: Compute x^2 = (1 - 5i)^2 = 1 - 10i + 25i^2

Step 2: Substitute i^2 = -1, giving x^2 = 1 - 10i - 25 = -24 - 10i

Step 3: Compute -2x = -2(1 - 5i) = -2 + 10i

Step 4: Substitute into the expression, x^2 - 2x + 26 = (-24 - 10i) + (-2 + 10i) + 26

Step 5: Group the real parts, -24 - 2 + 26 = 0

Step 6: Group the imaginary parts, -10i + 10i = 0

Step 7: The total is 0 + 0i = 0

Final answer: x^2 - 2x + 26 = 0 for x = 1 - 5i, hence shown

Question: If x = 2 + sqrt(3) i, find the value of x^4 - 4x^2 + 8x + 55

Answer: Step 1: First find x^2 = (2 + sqrt(3) i)^2 = 4 + 4 sqrt(3) i + 3i^2 = 4 + 4 sqrt(3) i - 3 = 1 + 4 sqrt(3) i

Step 2: Find x^4 by squaring x^2, x^4 = (1 + 4 sqrt(3) i)^2 = 1 + 8 sqrt(3) i + 48i^2 = 1 + 8 sqrt(3) i - 48 = -47 + 8 sqrt(3) i

Step 3: Compute -4x^2 = -4(1 + 4 sqrt(3) i) = -4 - 16 sqrt(3) i

Step 4: Compute 8x = 8(2 + sqrt(3) i) = 16 + 8 sqrt(3) i

Step 5: Add all the parts together with the constant 55

Step 6: Group the real parts, -47 - 4 + 16 + 55 = 20

Step 7: Group the imaginary parts, 8 sqrt(3) - 16 sqrt(3) + 8 sqrt(3) = 0

Final answer: x^4 - 4x^2 + 8x + 55 = 20

Exercise 1.3

Question: Solve the simultaneous linear equations with complex coefficients for w and z: (i) 3z + (2 + i)w = 11 - i, (2 - i)z - w = -1 + i

Answer: Step 1: Write the equations as (1) 3z + (2 + i)w = 11 - i and (2) (2 - i)z - w = -1 + i

Step 2: To eliminate w, multiply equation (2) by (2 + i): (2 - i)(2 + i)z - (2 + i)w = (-1 + i)(2 + i)

Step 3: Simplify the left coefficient, (2 - i)(2 + i) = 4 + 1 = 5

Step 4: Simplify the right side, (-1 + i)(2 + i) = -2 - i + 2i + i^2 = -3 + i, so the new equation is 5z - (2 + i)w = -3 + i

Step 5: Add this to equation (1) to cancel the w terms, 3z + 5z = (11 - i) + (-3 + i), giving 8z = 8, so z = 1

Step 6: Substitute z = 1 in equation (2), (2 - i) - w = -1 + i

Step 7: Solve for w, w = (2 - i) - (-1 + i) = 2 - i + 1 - i = 3 - 2i

Step 8: Check in equation (1): 3(1) + (2 + i)(3 - 2i) = 3 + (6 - 4i + 3i - 2i^2) = 3 + 8 - i = 11 - i, which is correct

Final answer: z = 1, w = 3 - 2i

Question: Solve the simultaneous linear equations with complex coefficients for w and z: (ii) 2z + (3 + i)w = 9 - i, (3 - i)z - iw = -1 + i

Answer: Step 1: Write the equations as (1) 2z + (3 + i)w = 9 - i and (2) (3 - i)z - iw = -1 + i

Step 2: From equation (2), iw = (3 - i)z + 1 - i, so w = -i[(3 - i)z + 1 - i] after dividing by i (since 1/i = -i)

Step 3: Expand, -i(3 - i) = -3i + i^2 = -1 - 3i and -i(1 - i) = -i + i^2 = -1 - i, so w = (-1 - 3i)z + (-1 - i)

Step 4: Substitute this w in equation (1), 2z + (3 + i)[(-1 - 3i)z + (-1 - i)] = 9 - i

Step 5: Expand the products, (3 + i)(-1 - 3i) = -3 - 9i - i - 3i^2 = -10i and (3 + i)(-1 - i) = -3 - 3i - i - i^2 = -2 - 4i

Step 6: Collect the z terms, 2z - 10iz - 2 - 4i = 9 - i, so z(2 - 10i) = 11 + 3i

Step 7: Divide, z = (11 + 3i)/(2 - 10i) = (11 + 3i)(2 + 10i)/((2 - 10i)(2 + 10i)) = (22 + 110i + 6i + 30i^2)/104 = (-8 + 116i)/104

Step 8: Simplify, z = (-2 + 29i)/26 = -1/13 + (29/26)i

Step 9: Find w from w = (-1 - 3i)z + (-1 - i): first (-1 - 3i)(-2 + 29i)/26 = (2 - 29i + 6i - 87i^2)/26 = (89 - 23i)/26

Step 10: Subtract (1 + i) = 26(1 + i)/26, giving w = (89 - 23i - 26 - 26i)/26 = (63 - 49i)/26

Final answer: z = -1/13 + (29/26)i, w = 63/26 - (49/26)i

Question: Write the polynomial as the product of linear factors: (i) z^2 + 81b^2

Answer: Step 1: Write the second term as a square, $81b^2 = (9b)^2$

Step 2: Use $i^2 = -1$ to write the sum of squares as a difference of squares, $z^2 + (9b)^2 = z^2 - (9bi)^2$, because $(9bi)^2 = 81b^2 i^2 = -81b^2$

Step 3: Apply the difference of two squares, $x^2 - y^2 = (x - y)(x + y)$, with $x = z$ and $y = 9bi$

Step 4: The factors are $(z - 9bi)$ and $(z + 9bi)$

Step 5: Check by multiplying back: $(z - 9bi)(z + 9bi) = z^2 - (9bi)^2 = z^2 + 81b^2$

Final answer: $z^2 + 81b^2 = (z - 9bi)(z + 9bi)$

Question: Write the polynomial as the product of linear factors: (ii) $z^3 + 3z^2 + z - 5$

Answer: Step 1: Try integer roots among the factors of 5, namely 1, -1, 5, -5, using the factor theorem

Step 2: Put $z = 1$: $1 + 3 + 1 - 5 = 0$, so $(z - 1)$ is a factor

Step 3: Divide by $(z - 1)$ by synthetic division with coefficients 1, 3, 1, -5: bring down 1, multiply by 1 and add successively to get the quotient coefficients 1, 4, 5 with remainder 0

Step 4: So the polynomial is $(z - 1)(z^2 + 4z + 5)$

Step 5: Solve $z^2 + 4z + 5 = 0$ by the quadratic formula, $z = [-4 +- sqrt(16 - 20)]/2 = [-4 +- sqrt(-4)]/2 = [-4 +- 2i]/2 = -2 +- i$

Step 6: So the quadratic gives the factors $(z + 2 - i)$ and $(z + 2 + i)$

Step 7: Collect all three linear factors

Final answer: $z^3 + 3z^2 + z - 5 = (z - 1)(z + 2 - i)(z + 2 + i)$

Question: Write the polynomial as the product of linear factors: (iii) 4z^2 + 9b^2

Answer: Step 1: Write both terms as squares, 4z^2 = (2z)^2 and 9b^2 = (3b)^2

Step 2: Use i^2 = -1 to write the sum of squares as a difference of squares, (2z)^2 + (3b)^2 = (2z)^2 - (3bi)^2, because (3bi)^2 = 9b^2 i^2 = -9b^2

Step 3: Apply the difference of two squares, x^2 - y^2 = (x - y)(x + y), with x = 2z and y = 3bi

Step 4: The factors are (2z - 3bi) and (2z + 3bi)

Step 5: Check by multiplying back: (2z - 3bi)(2z + 3bi) = 4z^2 - (3bi)^2 = 4z^2 + 9b^2

Final answer: 4z^2 + 9b^2 = (2z - 3bi)(2z + 3bi)

Question: Write the polynomial as the product of linear factors: (iv) z^3 + 3z^2 + 4z + 2

Answer: Step 1: Try integer roots among the factors of 2, namely 1, -1, 2, -2, using the factor theorem

Step 2: Put z = -1: -1 + 3 - 4 + 2 = 0, so (z + 1) is a factor

Step 3: Divide by (z + 1) by synthetic division with coefficients 1, 3, 4, 2 and root -1: bring down 1, multiply by -1 and add successively to get the quotient coefficients 1, 2, 2 with remainder 0

Step 4: So the polynomial is (z + 1)(z^2 + 2z + 2)

Step 5: Solve z^2 + 2z + 2 = 0 by the quadratic formula, z = [-2 +- sqrt(4 - 8)]/2 = [-2 +- sqrt(-4)]/2 = [-2 +- 2i]/2 = -1 +- i

Step 6: So the quadratic gives the factors (z + 1 - i) and (z + 1 + i)

Step 7: Collect all three linear factors

Final answer: z^3 + 3z^2 + 4z + 2 = (z + 1)(z + 1 - i)(z + 1 + i)

Question: Write the polynomial as the product of linear factors: (v) z^3 - 7z^2 + 19z - 13

Answer: Step 1: Try integer roots among the factors of 13, namely 1, -1, 13, -13, using the factor theorem

Step 2: Put z = 1: 1 - 7 + 19 - 13 = 0, so (z - 1) is a factor

Step 3: Divide by (z - 1) by synthetic division with coefficients 1, -7, 19, -13: bring down 1, multiply by 1 and add successively to get the quotient coefficients 1, -6, 13 with remainder 0

Step 4: So the polynomial is (z - 1)(z^2 - 6z + 13)

Step 5: Solve z^2 - 6z + 13 = 0 by the quadratic formula, z = [6 +- sqrt(36 - 52)]/2 = [6 +- sqrt(-16)]/2 = [6 +- 4i]/2 = 3 +- 2i

Step 6: So the quadratic gives the factors (z - 3 - 2i) and (z - 3 + 2i)

Step 7: Collect all three linear factors

Final answer: z^3 - 7z^2 + 19z - 13 = (z - 1)(z - 3 - 2i)(z - 3 + 2i)

Question: Write the polynomial as the product of linear factors: (vi) z^3 + 3z^2 + 19z + 17

Answer: Step 1: Try integer roots among the factors of 17, namely 1, -1, 17, -17, using the factor theorem

Step 2: Put z = -1: -1 + 3 - 19 + 17 = 0, so (z + 1) is a factor

Step 3: Divide by (z + 1) by synthetic division with coefficients 1, 3, 19, 17 and root -1: bring down 1, multiply by -1 and add successively to get the

quotient coefficients 1, 2, 17 with remainder 0

Step 4: So the polynomial is (z + 1)(z^2 + 2z + 17)

Step 5: Solve z^2 + 2z + 17 = 0 by the quadratic formula, z = [-2 +- sqrt(4 - 68)]/2 = [-2 +- sqrt(-64)]/2 = [-2 +- 8i]/2 = -1 +- 4i

Step 6: So the quadratic gives the factors (z + 1 - 4i) and (z + 1 + 4i)

Step 7: Collect all three linear factors

Final answer: z^3 + 3z^2 + 19z + 17 = (z + 1)(z + 1 - 4i)(z + 1 + 4i)

Question: Write the polynomial as the product of linear factors: (vii) z^3 - 3z^2 + 4z - 2

Answer: Step 1: Try integer roots among the factors of 2, namely 1, -1, 2, -2, using the factor theorem

Step 2: Put z = 1: 1 - 3 + 4 - 2 = 0, so (z - 1) is a factor

Step 3: Divide by (z - 1) by synthetic division with coefficients 1, -3, 4, -2: bring down 1, multiply by 1 and add successively to get the quotient coefficients 1, -2, 2 with remainder 0

Step 4: So the polynomial is (z - 1)(z^2 - 2z + 2)

Step 5: Solve z^2 - 2z + 2 = 0 by the quadratic formula, z = [2 +- sqrt(4 - 8)]/2 = [2 +- sqrt(-4)]/2 = [2 +- 2i]/2 = 1 +- i

Step 6: So the quadratic gives the factors (z - 1 - i) and (z - 1 + i)

Step 7: Collect all three linear factors

Final answer: z^3 - 3z^2 + 4z - 2 = (z - 1)(z - 1 - i)(z - 1 + i)

Question: Solve the quadratic equation by completing the squares, where z is a complex number: (i) z^2 - 4z + 5 = 0

Answer: Step 1: Move the constant to the right side, z^2 - 4z = -5

Step 2: Half of the coefficient of z is -2, and its square is 4; add 4 to both sides, z^2 - 4z + 4 = -5 + 4

Step 3: Factor the left side as a perfect square, (z - 2)^2 = -1

Step 4: Take the square root of both sides, z - 2 = +- sqrt(-1) = +- i

Step 5: Add 2 to both sides, z = 2 +- i

Final answer: z = 2 + i or z = 2 - i

Question: Solve the quadratic equation by completing the squares, where z is a complex number: (ii) z^2 + 12z + 52 = 0

Answer: Step 1: Move the constant to the right side, z^2 + 12z = -52

Step 2: Half of the coefficient of z is 6, and its square is 36; add 36 to both sides, z^2 + 12z + 36 = -52 + 36

Step 3: Factor the left side as a perfect square, (z + 6)^2 = -16

Step 4: Take the square root of both sides, z + 6 = +- sqrt(-16) = +- 4i

Step 5: Subtract 6 from both sides, z = -6 +- 4i

Final answer: z = -6 + 4i or z = -6 - 4i

Question: Solve the quadratic equation by completing the squares, where z is a complex number: (iii) 34z^2 - 6z = -1

Answer: Step 1: Bring all terms to standard form, 34z^2 - 6z + 1 = 0

Step 2: Divide every term by 34 to make the coefficient of z^2 equal to 1, z^2 - (3/17)z + 1/34 = 0

Step 3: Move the constant to the right side, z^2 - (3/17)z = -1/34

Step 4: Half of the coefficient of z is 3/34, and its square is 9/1156; add

9/1156 to both sides

Step 5: The left side becomes the perfect square (z - 3/34)^2, and the right side becomes -1/34 + 9/1156 = (-34 + 9)/1156 = -25/1156

Step 6: Take the square root of both sides, z - 3/34 = +- (5/34)i, since sqrt(25/1156) = 5/34

Step 7: Add 3/34 to both sides, z = (3 +- 5i)/34

Final answer: z = (3 + 5i)/34 or z = (3 - 5i)/34

Question: Solve the quadratic equation by completing the squares, where z is a complex number: (iv) z^2 - 6z + 34 = 0

Answer: Step 1: Move the constant to the right side, z^2 - 6z = -34

Step 2: Half of the coefficient of z is -3, and its square is 9; add 9 to both sides, z^2 - 6z + 9 = -34 + 9

Step 3: Factor the left side as a perfect square, (z - 3)^2 = -25

Step 4: Take the square root of both sides, z - 3 = +- sqrt(-25) = +- 5i

Step 5: Add 3 to both sides, z = 3 +- 5i

Final answer: z = 3 + 5i or z = 3 - 5i

Question: Solve the quadratic equation by completing the squares, where z is a complex number: (v) z^2 - 6z = -13

Answer: Step 1: Half of the coefficient of z is -3, and its square is 9

Step 2: Add 9 to both sides, z^2 - 6z + 9 = -13 + 9

Step 3: Factor the left side as a perfect square, (z - 3)^2 = -4

Step 4: Take the square root of both sides, z - 3 = +- sqrt(-4) = +- 2i

Step 5: Add 3 to both sides, z = 3 +- 2i

Final answer: z = 3 + 2i or z = 3 - 2i

Question: Solve the quadratic equation by completing the squares, where z is a complex number: (vi) z^2 + 64 = 0

Answer: Step 1: Move the constant to the right side, z^2 = -64

Step 2: There is no z term, so the square is already complete and no term needs to be added

Step 3: Take the square root of both sides, z = +- sqrt(-64)

Step 4: Write sqrt(-64) = sqrt(64) x sqrt(-1) = 8i

Step 5: Hence z = +- 8i

Final answer: z = 8i or z = -8i

Review Exercise 1

Question: Select the correct option: 1(i) For any complex number z, |z| is equal to (a) |z-bar| (b) |-z| (c) |-z-bar| (d) All of these

Answer: Step 1: Let z = a + ib, then |z| = sqrt(a^2 + b^2)

Step 2: z-bar = a - ib has |z-bar| = sqrt(a^2 + (-b)^2) = sqrt(a^2 + b^2), the same value

Step 3: -z = -a - ib has |-z| = sqrt((-a)^2 + (-b)^2) = sqrt(a^2 + b^2), again the same

Step 4: -z-bar = -a + ib has |-z-bar| = sqrt((-a)^2 + b^2) = sqrt(a^2 + b^2), again the same

Step 5: All three options equal |z| because squaring removes every sign

Final answer: option (d) All of these

Question: Select the correct option: 1(ii) If z1 and z2 are any two complex numbers, then (a) |z1 + z2| < |z1| + |z2| (b) |z1 + z2| <= |z1| + |z2| (c) |z1 - z2| < |z1| - |z2| (d) |z1 + z2| >= |z1| + |z2|

Answer: Step 1: This is the Triangle inequality for complex numbers

Step 2: In the Argand diagram z1 and z2 are two sides of a triangle and z1 + z2 is the third side

Step 3: The third side can never be longer than the sum of the other two sides, so |z1 + z2| <= |z1| + |z2|

Step 4: Equality holds when the two vectors point in the same direction, which is why the sign must be <= and not just <

Step 5: Options (a) and (d) miss the equality case or reverse the inequality, and (c) is not a valid form

Final answer: option (b) |z1 + z2| <= |z1| + |z2|

Question: Select the correct option: 1(iii) If z = 3i - 4, then z + z-bar = ... (a) 8 (b) -3i (c) -8 (d) 3i - 8

Answer: Step 1: Write z in a + ib form, z = -4 + 3i

Step 2: Its conjugate is z-bar = -4 - 3i

Step 3: Add them, z + z-bar = (-4 + 3i) + (-4 - 3i)

Step 4: Group the real parts, -4 - 4 = -8

Step 5: Group the imaginary parts, 3i - 3i = 0

Step 6: The sum is -8, which is twice the real part of z

Final answer: option (c) -8

Question: Select the correct option: 1(iv) If a > 0 and b < 0 then (a) ab > 0 (b) ab < 0 (c) ab = 0 (d) All of these

Answer: Step 1: a is a positive number and b is a negative number

Step 2: The product of a positive number and a negative number is always negative

Step 3: So ab < 0

Step 4: It cannot be positive, and it cannot be zero because neither a nor b is zero

Final answer: option (b) ab < 0

Question: Select the correct option: 1(v) If n is an even integer, then (i)^n is equal to: (a) i (b) -i (c) 1 or -1 (d) i or -i

Answer: Step 1: An even integer can be written as n = 2m for some integer m

Step 2: Then i^n = i^(2m) = (i^2)^m = (-1)^m

Step 3: If m is even, (-1)^m = 1, and if m is odd, (-1)^m = -1

Step 4: For example i^2 = -1 and i^4 = 1, both occur for even n

Step 5: So i^n can only be 1 or -1, never i or -i

Final answer: option (c) 1 or -1

Question: Select the correct option: 1(vi) If z is any real number x, then its conjugate is: (a) x (b) -x (c) xi (d) -xi

Answer: Step 1: A real number x can be written as the complex number x + 0i

Step 2: The conjugate reverses the sign of the imaginary part

Step 3: The imaginary part is 0, and -0 = 0, so the conjugate is x - 0i

Step 4: Hence a real number is its own conjugate, called self-conjugate

Final answer: option (a) x

Question: Select the correct option: 1(vii) (-1)^(-21/2) is equal to: (a) i (b) -i (c) 1 (d) -1

Answer: Step 1: First evaluate the positive power, (-1)^(21/2) = ((-1)^(1/2))^21

Step 2: Now (-1)^(1/2) = sqrt(-1) = i

Step 3: So we need i^21 = i^20 x i = (i^4)^5 x i = 1 x i = i

Step 4: The negative exponent means take the reciprocal, (-1)^(-21/2) = 1/i

Step 5: Multiply numerator and denominator by i, 1/i = i/i^2 = i/(-1) = -i

Final answer: option (b) -i

Question: Select the correct option: 1(viii) If z = (0, 1) then z^2 is: (a) 1 (b) -1 (c) i (d) -i

Answer: Step 1: The pair (0, 1) means z = 0 + 1i = i

Step 2: Square it, z^2 = i^2

Step 3: By the definition of the imaginary unit, i^2 = -1

Final answer: option (b) -1

Question: Select the correct option: 1(ix) The product of two conjugate complex numbers is: (a) A real number (b) An imaginary number (c) Always zero (d) A complex number

Answer: Step 1: Let the number be z = a + ib, so its conjugate is z-bar = a - ib

Step 2: Multiply, z z-bar = (a + ib)(a - ib) = a^2 - i^2 b^2

Step 3: Substitute i^2 = -1, giving a^2 + b^2

Step 4: The result a^2 + b^2 has no imaginary part and is never negative, in fact it equals |z|^2

Step 5: So the product is always a real number

Final answer: option (a) A real number

Question: Select the correct option: 1(x) Real part of (2 + i)/i is equal to: (a) 1 (b) 2 (c) -1 (d) 1/2

Answer: Step 1: Divide term by term, (2 + i)/i = 2/i + i/i = 2/i + 1

Step 2: Simplify 2/i by multiplying numerator and denominator by i, 2/i = 2i/i^2 = 2i/(-1) = -2i

Step 3: So (2 + i)/i = 1 - 2i

Step 4: The real part is the part without i, which is 1

Final answer: option (a) 1

Question: Select the correct option: 1(xi) The plane on which complex numbers are represented is called: (a) Coordinate plane (b) Complex plane (c) Cartesian plane (d) Real plane

Answer: Step 1: A complex number a + ib is represented by the point (a, b) in a plane

Step 2: In this plane the horizontal axis is the real axis and the vertical axis is the imaginary axis

Step 3: This diagram is called the Argand diagram

Step 4: The plane itself is therefore called the complex plane, also called the z-plane

Final answer: option (b) Complex plane

Question: Select the correct option: 1(xii) The modulus of a complex number z = x + iy is the distance of P(x, y) from: (a) x-axis (b) y-axis (c) Origin (d) (x, y)

Answer: Step 1: The point P(x, y) represents z in the complex plane

Step 2: The modulus |z| = sqrt(x^2 + y^2) is exactly the distance formula for the distance from P to the point (0, 0)

Step 3: The point (0, 0) is the origin

Step 4: So |z| is the length of the line segment joining P to the origin

Final answer: option (c) Origin

Question: Select the correct option: 1(xiii) If z1 = 2 + 3i, z2 = 1 - i then |z1 z2| = ... (a) sqrt(13) (b) sqrt(26) (c) sqrt(15) (d) 26

Answer: Step 1: Use the property |z1 z2| = |z1| |z2|, so there is no need to multiply first

Step 2: Compute |z1| = sqrt(4 + 9) = sqrt(13)

Step 3: Compute |z2| = sqrt(1 + 1) = sqrt(2)

Step 4: Multiply, |z1 z2| = sqrt(13) x sqrt(2) = sqrt(26)

Step 5: As a check, z1 z2 = (2 + 3i)(1 - i) = 5 + i and |5 + i| = sqrt(25 + 1) = sqrt(26)

Final answer: option (b) sqrt(26)

Question: Select the correct option: 1(xiv) If |x + 5i| = 3, then x is equal to (a) +-4 (b) +-4i (c) +-22i (d) None of these

Answer: Step 1: Here x is a real number and |x + 5i| = sqrt(x^2 + 25)

Step 2: Set this equal to 3, sqrt(x^2 + 25) = 3

Step 3: Square both sides, x^2 + 25 = 9

Step 4: Solve, x^2 = 9 - 25 = -16

Step 5: Take the square root, x = +- sqrt(-16) = +- 4i, so no real value of x satisfies the equation, and the values are imaginary

Final answer: option (b) +-4i

Question: Select the correct option: 1(xv) Multiplicative inverse of (3, 4) is: (a) (3/25, 4/25) (b) (3/25, -4/25) (c) (-3/25, -4/25) (d) (1, 0)

Answer: Step 1: The multiplicative inverse of (a, b) is (a/(a^2 + b^2), -b/(a^2 + b^2))

Step 2: Here a = 3 and b = 4, so a^2 + b^2 = 9 + 16 = 25

Step 3: The first component is 3/25

Step 4: The second component is -4/25, note the minus sign on the second component only

Final answer: option (b) (3/25, -4/25)

Question: Select the correct option: 1(xvi) If z = -3i + 4, then z-bar = ... (a) -3i - 4 (b) -3i + 4 (c) 3i + 4 (d) 3i - 4

Answer: Step 1: Write z in a + ib form, z = 4 - 3i

Step 2: The conjugate reverses the sign of the imaginary part only

Step 3: So z-bar = 4 + 3i

Step 4: Writing the imaginary term first, this is 3i + 4

Final answer: option (c) 3i + 4

Question: Select the correct option: 1(xvii) Real and imaginary parts of i(3 - 2i) are respectively: (a) 2 and -3 (b) 2 and 3 (c) 3 and -2 (d) -3 and -2

Answer: Step 1: Multiply out, i(3 - 2i) = 3i - 2i^2

Step 2: Substitute i^2 = -1, so -2i^2 = 2

Step 3: The result is 2 + 3i

Step 4: The real part is 2 and the imaginary part is 3

Final answer: option (b) 2 and 3

Question: Select the correct option: 1(xviii) If z = a + ib, then |z| = ... (a) sqrt(a - b) (b) sqrt(a^2 - b^2) (c) sqrt(a^2 + b^2) (d) sqrt(a - b)

Answer: Step 1: The modulus is the distance of the point (a, b) from the origin

Step 2: By the distance formula this distance is sqrt((a - 0)^2 + (b - 0)^2)

Step 3: Simplify, |z| = sqrt(a^2 + b^2)

Step 4: Both terms are added because both come from squares, so options with a minus are wrong

Final answer: option (c) sqrt(a^2 + b^2)

Question: Select the correct option: 1(xix) ((1 + i)/(1 - i))^12 = ... (a) -1 (b) 1 (c) i (d) -i

Answer: Step 1: Simplify the inner fraction, (1 + i)/(1 - i) = (1 + i)^2/((1 - i)(1 + i)) = (1 + 2i + i^2)/2 = 2i/2 = i

Step 2: So the expression becomes i^12

Step 3: Write i^12 = (i^4)^3

Step 4: Substitute i^4 = 1, giving 1^3 = 1

Final answer: option (b) 1

Question: Select the correct option: 1(xx) The value of i^(-7) is: (a) 1 (b) i (c) -1 (d) -i

Answer: Step 1: The negative exponent means take the reciprocal, i^(-7) = 1/i^7

Step 2: Reduce the power, i^7 = i^4 x i^3 = 1 x (-i) = -i

Step 3: So i^(-7) = 1/(-i)

Step 4: Multiply numerator and denominator by i, 1/(-i) = i/(-i^2) = i/1 = i, since i^2 = -1

Final answer: option (b) i

Question: Simplify: 2(i) (5 - 6i) + (5i) + (7 + 6i)

Answer: Step 1: Remove the brackets, 5 - 6i + 5i + 7 + 6i

Step 2: Group the real parts, 5 + 7 = 12

Step 3: Group the imaginary parts, -6i + 5i + 6i = 5i

Step 4: Combine the two groups

Final answer: 12 + 5i

Question: Simplify: 2(ii) (4 - i) + (-9 + 6i)

Answer: Step 1: Remove the brackets, 4 - i - 9 + 6i

Step 2: Group the real parts, 4 - 9 = -5

Step 3: Group the imaginary parts, -i + 6i = 5i

Step 4: Combine the two groups

Final answer: -5 + 5i

Question: Simplify: 2(iii) (9 + 11i) - (3 + 5i)

Answer: Step 1: Open the second bracket with changed signs, 9 + 11i - 3 - 5i

Step 2: Group the real parts, 9 - 3 = 6

Step 3: Group the imaginary parts, 11i - 5i = 6i

Step 4: Combine the two groups

Final answer: 6 + 6i

Question: Simplify: 2(iv) (-2 - 15i) - (-12 + 13i)

Answer: Step 1: Open the second bracket with changed signs, -2 - 15i + 12 - 13i

Step 2: Group the real parts, -2 + 12 = 10

Step 3: Group the imaginary parts, -15i - 13i = -28i

Step 4: Combine the two groups

Final answer: 10 - 28i

Question: Simplify: 2(v) (-3 + 2i)(3 - 8i)

Answer: Step 1: Multiply term by term, (-3)(3) + (-3)(-8i) + (2i)(3) + (2i)(-8i)

Step 2: Compute each product, -9 + 24i + 6i - 16i^2

Step 3: Substitute i^2 = -1, so -16i^2 = 16

Step 4: Group the real parts, -9 + 16 = 7

Step 5: Group the imaginary parts, 24i + 6i = 30i

Final answer: 7 + 30i

Question: Simplify: 2(vi) (3 - i)(4 + 3i)(5 - 2i)

Answer: Step 1: Multiply the first two brackets, (3 - i)(4 + 3i) = 12 + 9i - 4i - 3i^2 = 12 + 5i + 3 = 15 + 5i

Step 2: Multiply the result by the third bracket, (15 + 5i)(5 - 2i)

Step 3: Expand, 75 - 30i + 25i - 10i^2

Step 4: Substitute i^2 = -1, so -10i^2 = 10

Step 5: Group the real parts, 75 + 10 = 85, and the imaginary parts, -30i + 25i = -5i

Final answer: 85 - 5i

Question: Simplify: 2(vii) (4 - i)/(6 - 3i)

Answer: Step 1: Multiply numerator and denominator by the conjugate of the denominator, (6 + 3i)

Step 2: Expand the numerator, (4 - i)(6 + 3i) = 24 + 12i - 6i - 3i^2 = 24 + 6i + 3 = 27 + 6i

Step 3: Expand the denominator, (6 - 3i)(6 + 3i) = 36 + 9 = 45

Step 4: Divide both parts by 45, (27 + 6i)/45

Step 5: Simplify each fraction, 27/45 = 3/5 and 6/45 = 2/15

Final answer: 3/5 + (2/15)i, that is (9 + 2i)/15

Question: Simplify: 2(viii) (8 + 6i)/(6 - 2i)

Answer: Step 1: Multiply numerator and denominator by the conjugate of the denominator, (6 + 2i)

Step 2: Expand the numerator, (8 + 6i)(6 + 2i) = 48 + 16i + 36i + 12i^2 = 48 + 52i - 12 = 36 + 52i

Step 3: Expand the denominator, (6 - 2i)(6 + 2i) = 36 + 4 = 40

Step 4: Divide both parts by 40, (36 + 52i)/40

Step 5: Simplify each fraction, 36/40 = 9/10 and 52/40 = 13/10

Final answer: 9/10 + (13/10)i

Question: 3(a) Find each of the following: (i) z-bar for z = 3 - 15i

Answer: Step 1: The conjugate of a + ib is a - ib, so only the sign of the imaginary part changes

Step 2: Here a = 3 and b = -15

Step 3: Reverse the sign of b, giving b = 15 in the conjugate

Step 4: Hence z-bar = 3 + 15i

Final answer: z-bar = 3 + 15i

Question: 3(a) Find each of the following: (ii) (z1 - z2)-bar for z1 = 5 + i and z2 = -8 + 3i

Answer: Step 1: Subtract first, z1 - z2 = (5 + i) - (-8 + 3i) = 5 + i + 8 - 3i

Step 2: Group the parts, z1 - z2 = 13 - 2i

Step 3: Take the conjugate by reversing the sign of the imaginary part

Step 4: Hence (z1 - z2)-bar = 13 + 2i

Step 5: As a check, z1-bar - z2-bar = (5 - i) - (-8 - 3i) = 13 + 2i, the same value

Final answer: (z1 - z2)-bar = 13 + 2i

Question: 3(a) Find each of the following: (iii) (z1 z2)-bar for z1 = -10 + 5i and z2 = 5 - 10i

Answer: Step 1: Multiply first, z1 z2 = (-10 + 5i)(5 - 10i) = -50 + 100i + 25i - 50i^2

Step 2: Substitute i^2 = -1, so -50i^2 = 50

Step 3: Group the parts, real part -50 + 50 = 0 and imaginary part 100i + 25i = 125i, so z1 z2 = 125i

Step 4: Take the conjugate by reversing the sign of the imaginary part

Final answer: (z1 z2)-bar = -125i

Question: 3(a) Find each of the following: (iv) z/z-bar for z = 15 - 3i

Answer: Step 1: The conjugate is z-bar = 15 + 3i, so the fraction is (15 - 3i)/(15 + 3i)

Step 2: Multiply numerator and denominator by the conjugate of the denominator, (15 - 3i)

Step 3: Expand the numerator, (15 - 3i)^2 = 225 - 90i + 9i^2 = 225 - 90i - 9 = 216 - 90i

Step 4: Expand the denominator, (15 + 3i)(15 - 3i) = 225 + 9 = 234

Step 5: Divide both parts by 234 and simplify by dividing by 18, 216/234 = 12/13 and 90/234 = 5/13

Final answer: z/z-bar = 12/13 - (5/13)i

Question: 3(a) Find each of the following: (v) z1-bar - z2-bar for z1 = 5 + 4i and z2 = -8 + 5i

Answer: Step 1: Take the conjugates separately, z1-bar = 5 - 4i and z2-bar = -8 - 5i

Step 2: Subtract, z1-bar - z2-bar = (5 - 4i) - (-8 - 5i) = 5 - 4i + 8 + 5i

Step 3: Group the real parts, 5 + 8 = 13

Step 4: Group the imaginary parts, -4i + 5i = i

Final answer: z1-bar - z2-bar = 13 + i

Question: 3(b) Find the absolute value of the complex number: (i) -3 - 6i

Answer: Step 1: The absolute value is |z| = sqrt(a^2 + b^2)

Step 2: Here a = -3 and b = -6

Step 3: Substitute, |z| = sqrt(9 + 36) = sqrt(45)

Step 4: Simplify the root, sqrt(45) = sqrt(9 x 5) = 3 sqrt(5)

Final answer: 3 sqrt(5)

Question: 3(b) Find the absolute value of the complex number: (ii) -6 + 3i

Answer: Step 1: The absolute value is |z| = sqrt(a^2 + b^2)

Step 2: Here a = -6 and b = 3

Step 3: Substitute, |z| = sqrt(36 + 9) = sqrt(45)

Step 4: Simplify the root, sqrt(45) = 3 sqrt(5)

Final answer: 3 sqrt(5)

Question: 3(b) Find the absolute value of the complex number: (iii) (6 + 3i)/(10 + 2i)

Answer: Step 1: Use the property that the modulus of a quotient is the quotient of the moduli, |z1/z2| = |z1|/|z2|

Step 2: Compute the modulus of the numerator, |6 + 3i| = sqrt(36 + 9) = sqrt(45) = 3 sqrt(5)

Step 3: Compute the modulus of the denominator, |10 + 2i| = sqrt(100 + 4) = sqrt(104) = 2 sqrt(26)

Step 4: Divide, (3 sqrt(5))/(2 sqrt(26))

Step 5: Rationalise by multiplying numerator and denominator by sqrt(26), giving 3 sqrt(130)/52

Final answer: 3 sqrt(130)/52

Question: 3(b) Find the absolute value of the complex number: (iv) (1 + i)/(1 - i)

Answer: Step 1: Simplify the number first, (1 + i)/(1 - i) = (1 + i)^2/((1 - i)(1 + i)) = (1 + 2i + i^2)/2 = 2i/2 = i

Step 2: Now find the modulus of i = 0 + 1i

Step 3: Apply |z| = sqrt(a^2 + b^2) with a = 0 and b = 1

Step 4: |i| = sqrt(0 + 1) = 1

Step 5: As a check, |1 + i|/|1 - i| = sqrt(2)/sqrt(2) = 1, the same value

Final answer: 1

Question: Find the values of x and y if: (i) 5x + 3iy = -x + 2iy, where x and y are real

Answer: Step 1: Two complex numbers are equal only when their real parts are equal and their imaginary parts are equal

Step 2: Write both sides in a + ib form: the left side has real part 5x and imaginary part 3y, the right side has real part -x and imaginary part 2y

Step 3: Equate the real parts, 5x = -x

Step 4: Solve, 6x = 0, so x = 0

Step 5: Equate the imaginary parts, 3y = 2y

Step 6: Solve, 3y - 2y = 0, so y = 0

Step 7: Check: both sides become 0, which agrees

Final answer: x = 0, y = 0

Question: Find the values of x and y if: (ii) x^2 - 7x + 9iy = iy^2 + 20i - 12, where x and y are real

Answer: Step 1: Collect the right side in a + ib form: the real part is -12 and the imaginary part is y^2 + 20, because iy^2 + 20i = (y^2 + 20)i

Step 2: The left side has real part x^2 - 7x and imaginary part 9y

Step 3: Equate the real parts, x^2 - 7x = -12

Step 4: Bring all terms to one side, x^2 - 7x + 12 = 0

Step 5: Factorise, (x - 3)(x - 4) = 0, so x = 3 or x = 4

Step 6: Equate the imaginary parts, 9y = y^2 + 20

Step 7: Bring all terms to one side, y^2 - 9y + 20 = 0

Step 8: Factorise, (y - 4)(y - 5) = 0, so y = 4 or y = 5

Step 9: Each value of x can pair with each value of y, giving four combinations

Final answer: (x, y) = (3, 4), (3, 5), (4, 4) or (4, 5)

Question: Find real and imaginary parts of the following: (i) z = i - 42i - 3

Answer: Step 1: Group the like terms in the given expression

Step 2: Combine the imaginary terms, i - 42i = -41i

Step 3: The constant term -3 is the real part

Step 4: So z = -3 - 41i in a + ib form

Step 5: Read off the two parts

Final answer: Real part = -3, imaginary part = -41

Question: Find real and imaginary parts of the following: (ii) (3z + i)/(z + 4) for z = 3 + 2i

Answer: Step 1: Substitute z = 3 + 2i in the numerator, 3z + i = 3(3 + 2i) + i = 9 + 6i + i = 9 + 7i

Step 2: Substitute in the denominator, z + 4 = (3 + 2i) + 4 = 7 + 2i

Step 3: The fraction is (9 + 7i)/(7 + 2i)

Step 4: Multiply numerator and denominator by the conjugate (7 - 2i): numerator (9 + 7i)(7 - 2i) = 63 - 18i + 49i - 14i^2 = 63 + 31i + 14 = 77 + 31i

Step 5: The denominator is (7 + 2i)(7 - 2i) = 49 + 4 = 53

Step 6: So the value is 77/53 + (31/53)i

Final answer: Real part = 77/53, imaginary part = 31/53

Question: Find the additive and multiplicative inverse of: (i) 3 - 7i

Answer: Step 1: The additive inverse reverses both signs, giving -3 + 7i

Step 2: Check: (3 - 7i) + (-3 + 7i) = 0, as required

Step 3: For the multiplicative inverse use (a/(a^2 + b^2), -b/(a^2 + b^2)) with a = 3 and b = -7

Step 4: Compute a^2 + b^2 = 9 + 49 = 58

Step 5: The inverse is 3/58 - (-7)/58 i = 3/58 + (7/58)i

Step 6: Check: (3 - 7i)(3 + 7i)/58 = (9 + 49)/58 = 1, as required

Final answer: Additive inverse = -3 + 7i, multiplicative inverse = 3/58 + (7/58)i

Question: Find the additive and multiplicative inverse of: (ii) -2 + i

Answer: Step 1: The additive inverse reverses both signs, giving 2 - i

Step 2: Check: (-2 + i) + (2 - i) = 0, as required

Step 3: For the multiplicative inverse use (a/(a^2 + b^2), -b/(a^2 + b^2)) with a = -2 and b = 1

Step 4: Compute a^2 + b^2 = 4 + 1 = 5

Step 5: The inverse is -2/5 - (1/5)i

Step 6: Check: (-2 + i)(-2 - i)/5 = (4 + 1)/5 = 1, as required

Final answer: Additive inverse = 2 - i, multiplicative inverse = -2/5 - (1/5)i

Question: Find the additive and multiplicative inverse of: (iii) -2 - 5i

Answer: Step 1: The additive inverse reverses both signs, giving 2 + 5i

Step 2: Check: (-2 - 5i) + (2 + 5i) = 0, as required

Step 3: For the multiplicative inverse use (a/(a^2 + b^2), -b/(a^2 + b^2)) with a = -2 and b = -5

Step 4: Compute a^2 + b^2 = 4 + 25 = 29

Step 5: The inverse is -2/29 - (-5)/29 i = -2/29 + (5/29)i

Step 6: Check: (-2 - 5i)(-2 + 5i)/29 = (4 + 25)/29 = 1, as required

Final answer: Additive inverse = 2 + 5i, multiplicative inverse = -2/29 + (5/29)i

Question: Find the additive and multiplicative inverse of: (iv) 1/2 + (3/2)i

Answer: Step 1: The additive inverse reverses both signs, giving -1/2 - (3/2)i

Step 2: Check: the number and its additive inverse add to 0, as required

Step 3: For the multiplicative inverse use (a/(a^2 + b^2), -b/(a^2 + b^2)) with a = 1/2 and b = 3/2

Step 4: Compute a^2 + b^2 = 1/4 + 9/4 = 10/4 = 5/2

Step 5: The first component is (1/2)/(5/2) = 1/5 and the second component is -(3/2)/(5/2) = -3/5

Step 6: So the inverse is 1/5 - (3/5)i, that is (1 - 3i)/5

Final answer: Additive inverse = -1/2 - (3/2)i, multiplicative inverse = 1/5 - (3/5)i

Question: Find the additive and multiplicative inverse of: (v) -4 + sqrt(7) i

Answer: Step 1: The additive inverse reverses both signs, giving 4 - sqrt(7) i

Step 2: Check: the number and its additive inverse add to 0, as required

Step 3: For the multiplicative inverse use (a/(a^2 + b^2), -b/(a^2 + b^2)) with a = -4 and b = sqrt(7)

Step 4: Compute a^2 + b^2 = 16 + 7 = 23

Step 5: The inverse is -4/23 - (sqrt(7)/23) i

Step 6: Check: (-4 + sqrt(7) i)(-4 - sqrt(7) i)/23 = (16 + 7)/23 = 1, as required

Final answer: Additive inverse = 4 - sqrt(7) i, multiplicative inverse = -4/23 - (sqrt(7)/23) i

Question: Find the additive and multiplicative inverse of: (vi) 3 + 4i

Answer: Step 1: The additive inverse reverses both signs, giving -3 - 4i

Step 2: Check: (3 + 4i) + (-3 - 4i) = 0, as required

Step 3: For the multiplicative inverse use (a/(a^2 + b^2), -b/(a^2 + b^2)) with a = 3 and b = 4

Step 4: Compute a^2 + b^2 = 9 + 16 = 25

Step 5: The inverse is 3/25 - (4/25)i

Step 6: Check: (3 + 4i)(3 - 4i)/25 = (9 + 16)/25 = 1, as required

Final answer: Additive inverse = -3 - 4i, multiplicative inverse = 3/25 -

(4/25)i

Question: Solve the following equation: z^2 - (2 - 3i)z - 5 + i = 0

Answer: Step 1: The equation is a quadratic with a = 1, b = -(2 - 3i) and c = -5 + i, so we use the quadratic formula z = [-b +- sqrt(b^2 - 4ac)]/(2a)

Step 2: Compute b^2 = (2 - 3i)^2 = 4 - 12i + 9i^2 = 4 - 12i - 9 = -5 - 12i

Step 3: Compute -4ac = -4(1)(-5 + i) = 20 - 4i

Step 4: The discriminant is b^2 - 4ac = (-5 - 12i) + (20 - 4i) = 15 - 16i

Step 5: Find sqrt(15 - 16i): let it be p + qi with real p and q, then p^2 - q^2 = 15 and 2pq = -16, so pq = -8

Step 6: Also p^2 + q^2 = sqrt(15^2 + 16^2) = sqrt(481), so adding and subtracting gives p^2 = (sqrt(481) + 15)/2 and q^2 = (sqrt(481) - 15)/2

Step 7: Since pq is negative, p and q have opposite signs; take p = sqrt((sqrt(481) + 15)/2) (about 3.30) and q = -sqrt((sqrt(481) - 15)/2) (about -2.43), so sqrt(15 - 16i) = p + qi

Step 8: Substitute in the formula, z = [(2 - 3i) +- (p + qi)]/2

Step 9: The two roots are approximately (3.15 - 2.43i) and (-1.15 - 0.57i), and substituting either root back in the equation gives 0, which checks the work

Final answer: z = ((2 - 3i) + sqrt(15 - 16i))/2 or z = ((2 - 3i) - sqrt(15 - 16i))/2, with sqrt(15 - 16i) = sqrt((sqrt(481) + 15)/2) - i sqrt((sqrt(481) - 15)/2)

Question: Verify the following: (i) (2 + 3i)^2 = -5 + 12i

Answer: Step 1: Expand the left side using (a + b)^2 = a^2 + 2ab + b^2, (2 + 3i)^2 = 4 + 12i + 9i^2

Step 2: Substitute i^2 = -1, so 9i^2 = -9

Step 3: Group the real parts, 4 - 9 = -5

Step 4: The imaginary part is 12i

Step 5: The left side becomes -5 + 12i, which is exactly the right side

Final answer: (2 + 3i)^2 = -5 + 12i, hence verified

Question: Verify the following: (ii) 2(5 - 2i)^2 = 42 - 40i

Answer: Step 1: First expand the square, (5 - 2i)^2 = 25 - 20i + 4i^2

Step 2: Substitute i^2 = -1, so 4i^2 = -4, giving 25 - 20i - 4 = 21 - 20i

Step 3: Multiply by 2, 2(21 - 20i) = 42 - 40i

Step 4: The result is exactly the right side

Final answer: 2(5 - 2i)^2 = 42 - 40i, hence verified

Question: Verify the following: (iii) z^3 + i = (z - i)(z^2 + iz - 1)

Answer: Step 1: Expand the right side term by term: z x z^2 = z^3, z x iz = iz^2, z x (-1) = -z

Step 2: Continue: (-i) x z^2 = -iz^2, (-i) x iz = -i^2 z = z (since i^2 = -1), and (-i) x (-1) = i

Step 3: Add all six products, z^3 + iz^2 - z - iz^2 + z + i

Step 4: Cancel the pairs, iz^2 - iz^2 = 0 and -z + z = 0

Step 5: Only z^3 + i remains, which is exactly the left side

Final answer: (z - i)(z^2 + iz - 1) = z^3 + i, hence verified

Question: Verify the following: (iv) z^4 - 1 = (z + i)(z - i)(z + 1)(z - 1)

Answer: Step 1: Multiply the first pair, (z + i)(z - i) = z^2 - i^2 = z^2 + 1, since i^2 = -1

Step 2: Multiply the second pair, (z + 1)(z - 1) = z^2 - 1, by the difference of two squares

Step 3: Multiply the two results, (z^2 + 1)(z^2 - 1)

Step 4: Apply the difference of two squares again, (z^2)^2 - 1^2 = z^4 - 1

Step 5: The result is exactly the left side

Final answer: (z + i)(z - i)(z + 1)(z - 1) = z^4 - 1, hence verified

Question: Solve the quadratic equation by completing the squares: (i) z^2 - 5z + 7 = 0

Answer: Step 1: Move the constant to the right side, z^2 - 5z = -7

Step 2: Half of the coefficient of z is 5/2, and its square is 25/4; add 25/4 to both sides, z^2 - 5z + 25/4 = -7 + 25/4

Step 3: Factor the left side as a perfect square, (z - 5/2)^2 = -3/4, since -28/4 + 25/4 = -3/4

Step 4: Take the square root of both sides, z - 5/2 = +- sqrt(-3/4) = +- (sqrt(3)/2) i

Step 5: Add 5/2 to both sides, z = 5/2 +- (sqrt(3)/2) i

Final answer: z = (5 + sqrt(3) i)/2 or z = (5 - sqrt(3) i)/2

Question: Solve the quadratic equation by completing the squares: (ii) 2z^2 - 6z + 9 = 0

Answer: Step 1: Divide every term by 2 to make the coefficient of z^2 equal to 1, z^2 - 3z + 9/2 = 0

Step 2: Move the constant to the right side, z^2 - 3z = -9/2

Step 3: Half of the coefficient of z is 3/2, and its square is 9/4; add 9/4 to

both sides, z^2 - 3z + 9/4 = -9/2 + 9/4

Step 4: Factor the left side as a perfect square, (z - 3/2)^2 = -9/4, since -18/4 + 9/4 = -9/4

Step 5: Take the square root of both sides, z - 3/2 = +- (3/2) i

Step 6: Add 3/2 to both sides, z = 3/2 +- (3/2) i

Final answer: z = (3 + 3i)/2 or z = (3 - 3i)/2

Question: Solve the quadratic equation by completing the squares: (iii) 4z^2 + 25 = 0

Answer: Step 1: Divide every term by 4, z^2 + 25/4 = 0

Step 2: Move the constant to the right side, z^2 = -25/4

Step 3: There is no z term, so the square is already complete

Step 4: Take the square root of both sides, z = +- sqrt(-25/4)

Step 5: Simplify, sqrt(25/4) = 5/2 and sqrt(-1) = i, so z = +- (5/2) i

Final answer: z = (5/2)i or z = -(5/2)i

Question: Solve the quadratic equation by completing the squares: (iv) z^2 - 10z + 41 = 0

Answer: Step 1: Move the constant to the right side, z^2 - 10z = -41

Step 2: Half of the coefficient of z is -5, and its square is 25; add 25 to both sides, z^2 - 10z + 25 = -41 + 25

Step 3: Factor the left side as a perfect square, (z - 5)^2 = -16

Step 4: Take the square root of both sides, z - 5 = +- 4i

Step 5: Add 5 to both sides, z = 5 +- 4i

Final answer: z = 5 + 4i or z = 5 - 4i

Question: Solve the quadratic equation by completing the squares: (v) $4z^2 + 4z + 5 = 0$

Answer: Step 1: Divide every term by 4 to make the coefficient of z^2 equal to 1, $z^2 + z + 5/4 = 0$

Step 2: Move the constant to the right side, $z^2 + z = -5/4$

Step 3: Half of the coefficient of z is 1/2, and its square is 1/4; add 1/4 to both sides, $z^2 + z + 1/4 = -5/4 + 1/4$

Step 4: Factor the left side as a perfect square, $(z + 1/2)^2 = -1$

Step 5: Take the square root of both sides, $z + 1/2 = +- i$

Step 6: Subtract 1/2 from both sides, $z = -1/2 +- i$

Final answer: $z = (-1 + 2i)/2$ or $z = (-1 - 2i)/2$

Unit 2: Matrices and Determinants

Exercise 2.1

Question: Q1(i) Specify the type of the matrix [sqrt(5), 0, 0; 0, -1/3, 0; 0, 0, 2/5].

Answer: Step 1: Count the rows and columns: the matrix has 3 rows and 3 columns, so it is a square matrix of order 3

Step 2: Look at the off-diagonal elements: every element with i not equal to j is 0, for example a12 = 0, a23 = 0 and a31 = 0

Step 3: Look at the main diagonal elements: sqrt(5), -1/3 and 2/5 are non-zero but they are not all equal, so it is not a scalar matrix and not an identity matrix

Step 4: A square matrix whose off-diagonal elements are all zero is called a diagonal matrix, and it is also both upper triangular and lower triangular

Final answer: It is a square diagonal matrix of order 3, written as diag(sqrt(5), -1/3, 2/5).

Question: Q1(ii) Specify the type of the matrix [sqrt(3); 4/7].

Answer: Step 1: Count the rows and columns: the matrix has 2 rows and only 1 column, so its order is 2 x 1

Step 2: A matrix having only one column is called a column matrix, no matter how many rows it has

Step 3: It is not a square matrix because the number of rows is not equal

to the number of columns

Final answer: It is a column matrix of order 2 x 1.

Question: Q1(iii) Specify the type of the matrix [i/2, 0, i/3].

Answer: Step 1: Count the rows and columns: the matrix has only 1 row and 3 columns, so its order is 1 x 3

Step 2: A matrix having only one row is called a row matrix, no matter how many columns it has

Step 3: The entries i/2 and i/3 are complex numbers, which is allowed in a row matrix

Final answer: It is a row matrix of order 1 x 3.

Question: Q1(iv) Specify the type of the matrix [0, i; i, 0].

Answer: Step 1: The matrix has 2 rows and 2 columns, so it is a square matrix of order 2

Step 2: Find its transpose: interchanging rows and columns gives A^t = [0, i; i, 0] = A, so the matrix equals its own transpose

Step 3: A square matrix with A^t = A is called a symmetric matrix

Step 4: Also check the Hermitian condition: the conjugate transpose is (Abar)^t = [0, -i; -i, 0] = -A, so it also satisfies the skew-Hermitian condition

Final answer: It is a square symmetric matrix of order 2 (it is also skew-Hermitian).

Question: Q1(v) Specify the type of the matrix [1, 0, 0, 0; 0, 1, 0, 0; 0, 0, 1, 0; 0, 0, 0, 1].

Answer: Step 1: The matrix has 4 rows and 4 columns, so it is a square

matrix of order 4

Step 2: Every off-diagonal element is 0, so it is a diagonal matrix

Step 3: Every main diagonal element is equal to 1, which is the special condition for a unit matrix

Final answer: It is the identity (unit) matrix I4 of order 4; it is also a scalar matrix and a diagonal matrix.

Question: Q1(vi) Specify the type of the matrix [2, 0; 0, -2].

Answer: Step 1: The matrix has 2 rows and 2 columns, so it is a square matrix of order 2

Step 2: The off-diagonal elements a12 = 0 and a21 = 0 are both zero, so it is a diagonal matrix

Step 3: The diagonal elements 2 and -2 are not equal, so it is not a scalar matrix and not an identity matrix

Final answer: It is a diagonal matrix of order 2, written as diag(2, -2).

Question: Q1(vii) Specify the type of the matrix [sqrt(7), 0, 0; 0, sqrt(7), 0; 0, 0, sqrt(7)].

Answer: Step 1: The matrix has 3 rows and 3 columns, so it is a square matrix of order 3

Step 2: All off-diagonal elements are 0, so it is a diagonal matrix

Step 3: All main diagonal elements are equal to the same scalar sqrt(7), which is the extra condition for a scalar matrix

Final answer: It is a scalar matrix of order 3 with scalar sqrt(7) (every scalar matrix is also a diagonal matrix).

Question: Q1(viii) Specify the type of the matrix [0, i, 2i; -i, 0, -4i; -2i, 4i,

0].

Answer: Step 1: The matrix is square of order 3 and all its main diagonal elements are 0

Step 2: Take the conjugate of every element: Abar = [0, -i, -2i; i, 0, 4i; 2i, -4i, 0]

Step 3: Transpose the conjugate: (Abar)^t = [0, i, 2i; -i, 0, -4i; -2i, 4i, 0], which is exactly the original matrix A

Step 4: Since (Abar)^t = A, the matrix satisfies the Hermitian condition; note that its diagonal elements are 0, which is a real number, as a Hermitian matrix requires

Step 5: Also A^t = [0, -i, -2i; i, 0, 4i; 2i, -4i, 0] = -A, so in the ordinary sense it is also skew-symmetric

Final answer: It is a Hermitian matrix of order 3 (and also a skew-symmetric matrix).

Question: Q1(ix) Specify the type of the matrix [2, -i, 5i; i, 3, 7i; -5i, -7i, 4].

Answer: Step 1: The matrix is square of order 3 and its main diagonal elements 2, 3, 4 are all real numbers

Step 2: Take the conjugate of every element: Abar = [2, i, -5i; -i, 3, -7i; 5i, 7i, 4]

Step 3: Transpose the conjugate: (Abar)^t = [2, -i, 5i; i, 3, 7i; -5i, -7i, 4], which is exactly A

Step 4: Check one symmetric pair by hand: a13 = 5i and a31 = -5i are conjugates of each other, and a23 = 7i and a32 = -7i are also conjugates, as the Hermitian condition needs

Final answer: It is a Hermitian matrix of order 3.

Question: Q2 A newspaper agent records the papers sold on each day of one week: Daily Dawn: Mon 80, Tue 90, Wed 100, Thu 95, Fri 85, Sat 75, Sun 70; Daily Jang: Mon 100, Tue 110, Wed 90, Thu 95, Fri 105, Sat 85, Sun 80. Write this information in matrix form and write its order.

Answer: Step 1: Decide the arrangement: put each newspaper in one row and each day of the week in one column, so the two rows are Daily Dawn and Daily Jang and the seven columns are Mon to Sun

Step 2: Fill the first row with the Daily Dawn sales in day order: 80, 90, 100, 95, 85, 75, 70

Step 3: Fill the second row with the Daily Jang sales in day order: 100, 110, 90, 95, 105, 85, 80

Step 4: Count rows and columns of the completed array: 2 rows and 7 columns

Final answer: The information in matrix form is [80, 90, 100, 95, 85, 75, 70; 100, 110, 90, 95, 105, 85, 80], and its order is 2 x 7.

Question: Q3(i) Find the unknowns from the equality [a, -4i; 8i, 6i] = [7i, b; c, d].

Answer: Step 1: Two matrices are equal only when they have the same order and every pair of corresponding elements is equal; both matrices here are 2 x 2

Step 2: Compare the (1,1) elements: a = 7i

Step 3: Compare the (1,2) elements: b = -4i

Step 4: Compare the (2,1) elements: c = 8i

Step 5: Compare the (2,2) elements: d = 6i

Final answer: a = 7i, b = -4i, c = 8i, d = 6i.

Question: Q3(ii) Find the unknowns from the equality [2, -3, 5; a, 9, 0; b, c, -1] = [d, e, g; -2, f, 0; -4, 7, -1].

Answer: Step 1: Both matrices are 3 x 3, so equate corresponding elements one position at a time

Step 2: From the first row: d = 2, e = -3 and g = 5

Step 3: From the (2,1) element: a = -2

Step 4: From the (2,2) element: f = 9 (the (2,3) elements 0 = 0 agree automatically)

Step 5: From the third row: b = -4 and c = 7 (the (3,3) elements -1 = -1 agree automatically)

Final answer: a = -2, b = -4, c = 7, d = 2, e = -3, f = 9, g = 5.

Question: Q3(iii) Find the unknowns from the equality [x + y, 0, z; 9, 2x + y, 6; a, 1, 3] = [3, 0, 3x; 9, 4, 6; y, 1, 3].

Answer: Step 1: Equate corresponding elements: the (1,1) elements give x + y = 3 ...(A)

Step 2: The (1,3) elements give z = 3x ...(B)

Step 3: The (2,2) elements give 2x + y = 4 ...(C), and the (3,1) elements give a = y ...(D); all other positions agree automatically

Step 4: Subtract (A) from (C): (2x + y) - (x + y) = 4 - 3, so x = 1

Step 5: Put x = 1 in (A): 1 + y = 3, so y = 2

Step 6: Put x = 1 in (B): z = 3(1) = 3, and from (D) a = y = 2

Final answer: x = 1, y = 2, z = 3, a = 2.

Question: Q4(i) Find the transpose of the matrix [-4, 3, 6].

Answer: Step 1: The given matrix is a row matrix of order 1 x 3 with the

single row -4, 3, 6

Step 2: For the transpose, write this row as a column, keeping the order of the elements unchanged

Step 3: The transpose therefore has order 3 x 1

Final answer: The transpose is [-4; 3; 6].

Question: Q4(ii) Find the transpose of the matrix [2i, 5i, -3i; 0, -6i, 2i].

Answer: Step 1: The given matrix has order 2 x 3, so its transpose will have order 3 x 2

Step 2: Write the first row 2i, 5i, -3i as the first column of the transpose

Step 3: Write the second row 0, -6i, 2i as the second column of the transpose

Final answer: The transpose is [2i, 0; 5i, -6i; -3i, 2i].

Question: Q4(iii) Find the transpose of the matrix [8; 3; -4].

Answer: Step 1: The given matrix is a column matrix of order 3 x 1 with the single column 8, 3, -4

Step 2: For the transpose, write this column as a row, keeping the order of the elements unchanged

Step 3: The transpose therefore has order 1 x 3

Final answer: The transpose is [8, 3, -4].

Question: Q4(iv) Find the transpose of the matrix [-2i, i; 5i, 7i; 2i, -5i].

Answer: Step 1: The given matrix has order 3 x 2, so its transpose will have order 2 x 3

Step 2: Write the first row -2i, i as the first column of the transpose

Step 3: Write the second row 5i, 7i as the second column of the transpose

Step 4: Write the third row 2i, -5i as the third column of the transpose

Final answer: The transpose is [-2i, 5i, 2i; i, 7i, -5i].

Question: Q4(v) Find the transpose of the matrix [-8, 7; 3, 0].

Answer: Step 1: The given matrix is square of order 2, so its transpose is also of order 2

Step 2: Interchange the rows and columns: the element a12 = 7 and the element a21 = 3 swap their positions, while the diagonal elements -8 and 0 stay in place

Final answer: The transpose is [-8, 3; 7, 0].

Question: Q4(vi) Find the transpose of the matrix [-7, -10, 8; 4, 5, 9; -1, 2, 3].

Answer: Step 1: The given matrix is square of order 3, so its transpose is also of order 3

Step 2: Write the first row -7, -10, 8 as the first column of the transpose

Step 3: Write the second row 4, 5, 9 as the second column of the transpose

Step 4: Write the third row -1, 2, 3 as the third column of the transpose

Final answer: The transpose is [-7, 4, -1; -10, 5, 2; 8, 9, 3].

Question: Q5(i) Write A = [aij] of order (2,3) in tabular (expanded) form.

Answer: Step 1: The order (2,3) means 2 rows and 3 columns, so the elements are aij with i = 1, 2 and j = 1, 2, 3

Step 2: The first row contains the elements with i = 1: a11, a12, a13

Step 3: The second row contains the elements with i = 2: a21, a22, a23

Final answer: A = [a11, a12, a13; a21, a22, a23].

Question: Q5(ii) Write X = [xij] of order (3,4) in tabular (expanded) form.

Answer: Step 1: The order (3,4) means 3 rows and 4 columns, so the elements are xij with i = 1, 2, 3 and j = 1, 2, 3, 4

Step 2: The first row is x11, x12, x13, x14 and the second row is x21, x22, x23, x24

Step 3: The third row is x31, x32, x33, x34

Final answer: X = [x11, x12, x13, x14; x21, x22, x23, x24; x31, x32, x33, x34].

Question: Q5(iii) Write B = [bik] of order (4,4) in tabular (expanded) form.

Answer: Step 1: The order (4,4) means 4 rows and 4 columns, so the elements are bik with i = 1, 2, 3, 4 and k = 1, 2, 3, 4, where i is the row number and k is the column number

Step 2: The first two rows are b11, b12, b13, b14 and b21, b22, b23, b24

Step 3: The last two rows are b31, b32, b33, b34 and b41, b42, b43, b44

Final answer: B = [b11, b12, b13, b14; b21, b22, b23, b24; b31, b32, b33, b34; b41, b42, b43, b44].

Question: Q6(i) State whether the matrix [0, -5, -6; 5, 0, 7; 6, -7, 0] is symmetric or skew-symmetric.

Answer: Step 1: Find the transpose by interchanging rows and columns: A^t = [0, 5, 6; -5, 0, -7; -6, 7, 0]

Step 2: Compare with -A: -A = [0, 5, 6; -5, 0, -7; -6, 7, 0]

Step 3: A^t and -A are identical, so A^t = -A; also note the main diagonal elements are all 0, which must happen in a skew-symmetric matrix

Final answer: The matrix is skew-symmetric; it is not symmetric.

Question: Q6(ii) State whether the matrix [7, 5, 8; 5, -1, 6; 8, 6, -1] is symmetric or skew-symmetric.

Answer: Step 1: Find the transpose: A^t = [7, 5, 8; 5, -1, 6; 8, 6, -1]

Step 2: The transpose is identical to the original matrix, so A^t = A

Step 3: Check the symmetric pairs directly: a12 = a21 = 5, a13 = a31 = 8 and a23 = a32 = 6

Final answer: The matrix is symmetric; it is not skew-symmetric.

Question: Q6(iii) State whether the matrix [0, 1, 3; -1, 0, 5; -3, -5, 0] is symmetric or skew-symmetric.

Answer: Step 1: Find the transpose: A^t = [0, -1, -3; 1, 0, -5; 3, 5, 0]

Step 2: Compare with -A: -A = [0, -1, -3; 1, 0, -5; 3, 5, 0]

Step 3: A^t = -A, and every symmetric pair is a negative pair, for example a12 = 1 and a21 = -1, while the diagonal is all zero

Final answer: The matrix is skew-symmetric; it is not symmetric.

Question: Q6(iv) State whether the matrix [1, 2, 5; 2, 5, -7; 5, -7, 3] is symmetric or skew-symmetric.

Answer: Step 1: Find the transpose: A^t = [1, 2, 5; 2, 5, -7; 5, -7, 3]

Step 2: The transpose equals the original matrix, so A^t = A

Step 3: Check the pairs: a12 = a21 = 2, a13 = a31 = 5 and a23 = a32 = -7

Final answer: The matrix is symmetric; it is not skew-symmetric.

Question: Q6(v) State whether the matrix [0, -5, 4; 5, 0, -1; -4, 1, 0] is symmetric or skew-symmetric.

Answer: Step 1: Find the transpose: A^t = [0, 5, -4; -5, 0, 1; 4, -1, 0]

Step 2: Compare with -A: -A = [0, 5, -4; -5, 0, 1; 4, -1, 0]

Step 3: A^t = -A, the diagonal elements are all 0, and each symmetric pair sums to zero, for example a13 = 4 and a31 = -4

Final answer: The matrix is skew-symmetric; it is not symmetric.

Question: Q7(i) Find the index of the nilpotent matrix [0, 1, 0; 0, 0, 1; 0, 0, 0].

Answer: Step 1: Let A = [0, 1, 0; 0, 0, 1; 0, 0, 0]; the index is the smallest positive power p with A^p = 0, so compute the powers one by one

Step 2: Compute A^2 = A x A entry by entry: the (1,3) entry is (0)(0) + (1)(1) + (0)(0) = 1, and every other entry works out to 0, so A^2 = [0, 0, 1; 0, 0, 0; 0, 0, 0]

Step 3: A^2 is not the zero matrix, so the index is more than 2

Step 4: Compute A^3 = A^2 x A: the only non-zero entry of A^2 is in position (1,3), and row 3 of A is all zero, so every entry of A^3 is 0, giving A^3 = [0, 0, 0; 0, 0, 0; 0, 0, 0] = 0

Final answer: A is nilpotent of index 3.

Question: Q7(ii) Find the index of the nilpotent matrix [1, -3, -4; -1, 3, 4; 1, -3, -4].

Answer: Step 1: Let A = [1, -3, -4; -1, 3, 4; 1, -3, -4]; note A itself is not zero, so the index is at least 2; also notice row 3 equals row 1 and row 2 is the negative of row 1

Step 2: Compute A^2: the (1,1) entry is (1)(1) + (-3)(-1) + (-4)(1) = 1 + 3 - 4 = 0

Step 3: The (1,2) entry is (1)(-3) + (-3)(3) + (-4)(-3) = -3 - 9 + 12 = 0, and the (1,3) entry is (1)(-4) + (-3)(4) + (-4)(-4) = -4 - 12 + 16 = 0

Step 4: The same calculation in rows 2 and 3 gives 0 in every position, because each row of A is a multiple of row 1, so A^2 = [0, 0, 0; 0, 0, 0; 0, 0, 0] = 0

Final answer: A is nilpotent of index 2.

Question: Q7(iii) Find the index of the nilpotent matrix [1, 1, 3; 5, 2, 6; -2, -1, -3].

Answer: Step 1: Let A = [1, 1, 3; 5, 2, 6; -2, -1, -3]; compute its powers in order

Step 2: Compute A^2 row by row: row 1 gives (1+5-6), (1+2-3), (3+6-9) = 0, 0, 0; row 2 gives (5+10-12), (5+4-6), (15+12-18) = 3, 3, 9; row 3 gives (-2-5+6), (-2-2+3), (-6-6+9) = -1, -1, -3, so A^2 = [0, 0, 0; 3, 3, 9; -1, -1, -3]

Step 3: A^2 is not zero, so the index is more than 2

Step 4: Compute A^3 = A^2 x A: row 2 of the product is 3 x (row 1 of A) + 3 x (row 2 of A) + 9 x (row 3 of A) = (3+15-18), (3+6-9), (9+18-27) = 0, 0, 0, and row 3 is -1 times the same combination, also 0, while row 1 is already 0, so A^3 = 0

Final answer: A is nilpotent of index 3.

Question: Q8(i) Find the period of the periodic matrix [0, 1; -1, 0].

Answer: Step 1: Let A = [0, 1; -1, 0]; the period is the smallest positive integer k with A^(k+1) = A, so compute successive powers

Step 2: A^2 = [0, 1; -1, 0] x [0, 1; -1, 0] = [(0)(0)+(1)(-1), (0)(1)+(1)(0); (-1)(0)+(0)(-1), (-1)(1)+(0)(0)] = [-1, 0; 0, -1] = -I, which is not equal to A

Step 3: A^3 = A^2 x A = -A = [0, -1; 1, 0], which is not equal to A

Step 4: A^4 = A^2 x A^2 = (-I)(-I) = I

Step 5: A^5 = A^4 x A = I x A = A, so A^(k+1) = A first holds with k + 1 = 5

Final answer: The period of A is 4.

Question: Q8(ii) Find the period of the periodic matrix [1, -2, -6; -3, 2, 9; 2, 0, -3].

Answer: Step 1: Let A = [1, -2, -6; -3, 2, 9; 2, 0, -3]; the period is the smallest k with A^(k+1) = A

Step 2: Compute A^2 entry by entry: row 1 gives (1+6-12), (-2-4-0), (-6-18+18) = -5, -6, -6; row 2 gives (-3-6+18), (6+4+0), (18+18-27) = 9, 10, 9; row 3 gives (2+0-6), (-4+0+0), (-12+0+9) = -4, -4, -3, so A^2 = [-5, -6, -6; 9, 10, 9; -4, -4, -3]

Step 3: A^2 is not equal to A, so the period is more than 1

Step 4: Compute A^3 = A^2 x A: row 1 gives (-5+18-12), (10-12-0), (30-54+18) = 1, -2, -6; row 2 gives (-9-30+18), (18+20+0), (-54+90-27) = -3, 2, 9; row 3 gives (-4+12-6), (8-8-0), (24-36+9) = 2, 0, -3, so A^3 = [1, -2, -6; -3, 2, 9; 2, 0, -3] = A

Step 5: A^(k+1) = A first holds with k + 1 = 3, that is k = 2

Final answer: The period of A is 2.

Question: Q8(iii) Find the period of the periodic matrix [2, -3, -5; -1, 4, 5; 1, -3, -4].

Answer: Step 1: Let A = [2, -3, -5; -1, 4, 5; 1, -3, -4]; the period is the smallest k with A^(k+1) = A

Step 2: Compute A^2: the (1,1) entry is (2)(2) + (-3)(-1) + (-5)(1) = 4 + 3 - 5 = 2, the (1,2) entry is (2)(-3) + (-3)(4) + (-5)(-3) = -6 - 12 + 15 = -3, and the (1,3) entry is (2)(-5) + (-3)(5) + (-5)(-4) = -10 - 15 + 20 = -5, so row 1 of A^2 is the same as row 1 of A

Step 3: Row 2 of A^2 is (-2-4+5), (3+16-15), (5+20-20) = -1, 4, 5 and row 3 is (2+3-4), (-3-12+12), (-5-15+16) = 1, -3, -4, so A^2 = A exactly

Step 4: A^(k+1) = A already holds with k + 1 = 2, that is k = 1, which means A is in fact idempotent

Final answer: The period of A is 1.

Question: Q9(i) State whether the matrix [4, 3, 3; -1, 0, -1; -4, -4, -3] is idempotent or involutory.

Answer: Step 1: Recall the tests: A is idempotent if A^2 = A and involutory if A^2 = I, so compute A^2

Step 2: Row 1 of A^2: (16-3-12), (12+0-12), (12-3-9) = 1, 0, 0

Step 3: Row 2 of A^2: (-4+0+4), (-3+0+4), (-3+0+3) = 0, 1, 0

Step 4: Row 3 of A^2: (-16+4+12), (-12+0+12), (-12+4+9) = 0, 0, 1

Step 5: So A^2 = [1, 0, 0; 0, 1, 0; 0, 0, 1] = I, while A^2 is not equal to A

Final answer: The matrix is involutory; it is not idempotent.

Question: Q9(ii) State whether the matrix [-1, 2, 4; 1, -2, -4; -1, 2, 4] is idempotent or involutory.

Answer: Step 1: Compute A^2 and compare it with both A and I

Step 2: Row 1 of A^2: (1+2-4), (-2-4-8) with signs worked carefully: (1,1) entry = (-1)(-1) + (2)(1) + (4)(-1) = 1 + 2 - 4 = -1; (1,2) entry = (-1)(2) + (2)(-2) + (4)(2) = -2 - 4 + 8 = 2; (1,3) entry = (-1)(4) + (2)(-4) + (4)(4) =

-4 - 8 + 16 = 4

Step 3: Row 2 of A^2: (-1-2+4), (2+4+8) computed as (1)(2) + (-2)(-2) + (-4)(2) = 2 + 4 - 8 = -2 for the middle entry; in full row 2 is 1, -2, -4 and row 3 is -1, 2, 4

Step 4: So A^2 = [-1, 2, 4; 1, -2, -4; -1, 2, 4] = A, and A^2 is not equal to I

Final answer: The matrix is idempotent; it is not involutory.

Question: Q9(iii) State whether the matrix [2, 1, 1; -1, 0, -1; -2, -2, -1] is idempotent or involutory.

Answer: Step 1: Compute A^2 and compare it with both A and I

Step 2: Row 1 of A^2: (4-1-2), (2+0-2), (2-1-1) = 1, 0, 0

Step 3: Row 2 of A^2: (-2+0+2), (-1+0+2), (-1+0+1) = 0, 1, 0

Step 4: Row 3 of A^2: (-4+2+2), (-2+0+2), (-2+2+1) = 0, 0, 1

Step 5: So A^2 = I, while A^2 is not equal to A

Final answer: The matrix is involutory; it is not idempotent.

Question: Q10(i) State whether the matrix [3, 1 - 2i, 4 + 7i; 1 + 2i, -4, -2i; 4 - 7i, 2i, 5] is Hermitian, skew-Hermitian, or neither.

Answer: Step 1: A matrix is Hermitian if (Abar)^t = A and skew-Hermitian if (Abar)^t = -A; first write the conjugate Abar = [3, 1 + 2i, 4 - 7i; 1 - 2i, -4, 2i; 4 + 7i, -2i, 5]

Step 2: Transpose the conjugate: (Abar)^t = [3, 1 - 2i, 4 + 7i; 1 + 2i, -4, -2i; 4 - 7i, 2i, 5]

Step 3: This is exactly the original matrix A; the diagonal entries 3, -4, 5 are real and each symmetric pair are conjugates, for example a13 = 4 + 7i and a31 = 4 - 7i

Final answer: The matrix is Hermitian; it is not skew-Hermitian.

Question: Q10(ii) State whether the matrix [-i, 3i, i; 3i, 7i, -5i; i, -5i, -i] is Hermitian, skew-Hermitian, or neither.

Answer: Step 1: Write the conjugate: Abar = [i, -3i, -i; -3i, -7i, 5i; -i, 5i, i]

Step 2: Transpose the conjugate: (Abar)^t = [i, -3i, -i; -3i, -7i, 5i; -i, 5i, i]

Step 3: Compare with -A: -A = [i, -3i, -i; -3i, -7i, 5i; -i, 5i, i], which is the same, so (Abar)^t = -A

Step 4: This fits the skew-Hermitian pattern: the diagonal entries -i, 7i, -i are purely imaginary (or zero), as required

Final answer: The matrix is skew-Hermitian; it is not Hermitian.

Question: Q10(iii) State whether the matrix [1, -i, -1 + i; i, 1, 1 + i; 1 + i, -1 + i, 0] is Hermitian, skew-Hermitian, or neither.

Answer: Step 1: Write the conjugate: Abar = [1, i, -1 - i; -i, 1, 1 - i; 1 - i, -1 - i, 0]

Step 2: Transpose the conjugate: (Abar)^t = [1, -i, 1 - i; i, 1, -1 - i; -1 - i, 1 - i, 0]

Step 3: Compare with A: already the (1,3) entries differ, since (Abar)^t has 1 - i there while A has -1 + i, so (Abar)^t is not equal to A and the matrix is not Hermitian

Step 4: Compare with -A = [-1, i, 1 - i; -i, -1, -1 - i; -1 - i, 1 - i, 0]: the (1,1) entries differ (1 against -1), so (Abar)^t is not equal to -A either

Final answer: The matrix is neither Hermitian nor skew-Hermitian.

Question: Q11(i) Find x, y and z given that A = [3, x + 2i, yi; 3 - 2i, 0, 1 + zi; yi, 1 - xi, -1] is Hermitian.

Answer: Step 1: For a Hermitian matrix the diagonal entries must be real (3, 0, -1 already are) and each off-diagonal pair must be conjugates: aij = conjugate of aji

Step 2: Use the (1,2) and (2,1) pair: x + 2i = conjugate of (3 - 2i) = 3 + 2i, so comparing real parts gives x = 3

Step 3: Use the (1,3) and (3,1) pair: yi = conjugate of yi = -yi, so 2yi = 0 and y = 0

Step 4: Use the (2,3) and (3,2) pair: 1 + zi = conjugate of (1 - xi) = 1 + xi, so z = x

Step 5: Since x = 3 from Step 2, z = 3

Final answer: x = 3, y = 0, z = 3.

Question: Q11(ii) Find x, y and z given that B = [1 + xi, y + 2i; 3 + zi, 2] is Hermitian.

Answer: Step 1: For a Hermitian matrix both diagonal entries must be real numbers

Step 2: The (1,1) entry 1 + xi is real only if its imaginary part is zero, so x = 0; the (2,2) entry 2 is already real

Step 3: The off-diagonal entries must be conjugates: y + 2i = conjugate of (3 + zi) = 3 - zi

Step 4: Compare real parts: y = 3; compare imaginary parts: 2 = -z, so z = -2

Final answer: x = 0, y = 3, z = -2.

Question: Q11(iii) Find x, y and z given that C = [7, x, y + i; 5i, -1, 7 - 2i; 3 - i, 7 - 3i, -4] is Hermitian.

Answer: Step 1: Apply the Hermitian conditions pair by pair; the diagonal entries 7, -1, -4 are real, as required

Step 2: The (1,2) and (2,1) pair gives x = conjugate of 5i = -5i, so x = -5i

Step 3: The (1,3) and (3,1) pair gives y + i = conjugate of (3 - i) = 3 + i, so comparing real parts gives y = 3

Step 4: Now check the remaining pair, which contains no unknown: the condition needs a23 = conjugate of a32, that is 7 - 2i should equal conjugate of (7 - 3i) = 7 + 3i, but 7 - 2i is not equal to 7 + 3i

Step 5: This last condition fails no matter what x and y are, because it does not involve them; the matrix as printed in the book therefore cannot be made Hermitian (the entry 7 - 3i appears to be a misprint, since 7 + 2i in that position would satisfy the condition)

Final answer: From the pairs containing the unknowns, x = -5i and y = 3 (C contains no z); but strictly, with the entries exactly as printed, no values make C Hermitian because the (2,3)/(3,2) pair 7 - 2i and 7 - 3i are not conjugates of each other.

Question: Q12(a)(i) Find f, g and h so that the matrix [5, 2, 1; g - 3, -1, 4; f + 1, 2 - h, 7] becomes a symmetric matrix.

Answer: Step 1: A symmetric matrix satisfies aij = aji for every pair, so equate each pair of symmetric entries

Step 2: From a21 = a12: g - 3 = 2, so g = 5

Step 3: From a31 = a13: f + 1 = 1, so f = 0

Step 4: From a32 = a23: 2 - h = 4, so h = -2

Step 5: Check: with these values the matrix is [5, 2, 1; 2, -1, 4; 1, 4, 7], which equals its own transpose

Final answer: f = 0, g = 5, h = -2.

Question: Q12(a)(ii) Find f, g and h so that the matrix [5, 2, 1; g - 3, -1, 4; f + 1, 2 - h, 7] becomes an upper triangular matrix.

Answer: Step 1: In an upper triangular matrix every entry below the main diagonal is zero: a21 = 0, a31 = 0 and a32 = 0

Step 2: From a21 = 0: g - 3 = 0, so g = 3

Step 3: From a31 = 0: f + 1 = 0, so f = -1

Step 4: From a32 = 0: 2 - h = 0, so h = 2

Step 5: Check: the matrix becomes [5, 2, 1; 0, -1, 4; 0, 0, 7], which is upper triangular

Final answer: f = -1, g = 3, h = 2.

Question: Q12(b)(i) Find f, g and h so that the matrix [3, -4, 2; 2 - f, 8, 7; h - 7, g + 4, -1] becomes a symmetric matrix.

Answer: Step 1: Equate each pair of symmetric entries, aij = aji

Step 2: From a21 = a12: 2 - f = -4, so f = 6

Step 3: From a31 = a13: h - 7 = 2, so h = 9

Step 4: From a32 = a23: g + 4 = 7, so g = 3

Step 5: Check: the matrix becomes [3, -4, 2; -4, 8, 7; 2, 7, -1], which equals its own transpose

Final answer: f = 6, g = 3, h = 9.

Question: Q12(b)(ii) Find f, g and h so that the matrix [3, -4, 2; 2 - f, 8, 7; h - 7, g + 4, -1] becomes an upper triangular matrix.

Answer: Step 1: Set every entry below the main diagonal equal to zero

Step 2: From a21 = 0: 2 - f = 0, so f = 2

Step 3: From a31 = 0: h - 7 = 0, so h = 7

Step 4: From a32 = 0: g + 4 = 0, so g = -4

Step 5: Check: the matrix becomes [3, -4, 2; 0, 8, 7; 0, 0, -1], which is upper triangular

Final answer: f = 2, g = -4, h = 7.

Exercise 2.2

Question: Q1(i) If A = [1 + i, 2, 3i; 4i, 5 - i, 2 + 3i; 0, 5, 1 - i] and B = [2 + i, 3 - i, 4; i, 0, 5 - i; 6 + i, 2, 2 + 3i], find A + B.

Answer: Step 1: Both matrices are 3 x 3, so they can be added entry by entry

Step 2: First row: (1 + i) + (2 + i) = 3 + 2i, 2 + (3 - i) = 5 - i, 3i + 4 = 4 + 3i

Step 3: Second row: 4i + i = 5i, (5 - i) + 0 = 5 - i, (2 + 3i) + (5 - i) = 7 + 2i

Step 4: Third row: 0 + (6 + i) = 6 + i, 5 + 2 = 7, (1 - i) + (2 + 3i) = 3 + 2i

Final answer: A + B = [3 + 2i, 5 - i, 4 + 3i; 5i, 5 - i, 7 + 2i; 6 + i, 7, 3 + 2i].

Question: Q1(ii) With A and B as in Q1, find A - B.

Answer: Step 1: Subtract corresponding entries of B from A

Step 2: First row: (1 + i) - (2 + i) = -1, 2 - (3 - i) = -1 + i, 3i - 4 = -4 + 3i

Step 3: Second row: 4i - i = 3i, (5 - i) - 0 = 5 - i, (2 + 3i) - (5 - i) = -3 + 4i

Step 4: Third row: 0 - (6 + i) = -6 - i, 5 - 2 = 3, (1 - i) - (2 + 3i) = -1 - 4i

Final answer: A - B = [-1, -1 + i, -4 + 3i; 3i, 5 - i, -3 + 4i; -6 - i, 3, -1 - 4i].

Question: Q1(iii) With A and B as in Q1, find 2A - B.

Answer: Step 1: First double every entry of A: 2A = [2 + 2i, 4, 6i; 8i, 10 - 2i, 4 + 6i; 0, 10, 2 - 2i]

Step 2: First row of 2A - B: (2 + 2i) - (2 + i) = i, 4 - (3 - i) = 1 + i, 6i - 4 = -4 + 6i

Step 3: Second row: 8i - i = 7i, (10 - 2i) - 0 = 10 - 2i, (4 + 6i) - (5 - i) = -1 + 7i

Step 4: Third row: 0 - (6 + i) = -6 - i, 10 - 2 = 8, (2 - 2i) - (2 + 3i) = -5i

Final answer: 2A - B = [i, 1 + i, -4 + 6i; 7i, 10 - 2i, -1 + 7i; -6 - i, 8, -5i].

Question: Q1(iv) With A and B as in Q1, find 3A + B + I, where I is the unit matrix of order 3.

Answer: Step 1: First triple every entry of A: 3A = [3 + 3i, 6, 9i; 12i, 15 - 3i, 6 + 9i; 0, 15, 3 - 3i]

Step 2: Add B entry by entry: first row (3 + 3i) + (2 + i) = 5 + 4i, 6 + (3 - i) = 9 - i, 9i + 4 = 4 + 9i; second row 12i + i = 13i, (15 - 3i) + 0 = 15 - 3i, (6 + 9i) + (5 - i) = 11 + 8i; third row 0 + (6 + i) = 6 + i, 15 + 2 = 17, (3 - 3i) + (2 + 3i) = 5

Step 3: Add I = [1, 0, 0; 0, 1, 0; 0, 0, 1], which only increases the three diagonal entries by 1: 5 + 4i becomes 6 + 4i, 15 - 3i becomes 16 - 3i, and 5 becomes 6

Final answer: 3A + B + I = [6 + 4i, 9 - i, 4 + 9i; 13i, 16 - 3i, 11 + 8i; 6 + i, 17, 6].

Question: Q2(i) Let A = [3, -4, 1; 4, 5, 7; -2, -3, 8], B = [3, 4, -1; 2, -3, 5] and C = [-5, 4; 6, -3; -2, 7]. Wherever possible compute AC.

Answer: Step 1: Check the orders: A is 3 x 3 and C is 3 x 2; the inner numbers 3 and 3 match, so AC exists and has order 3 x 2

Step 2: First row of AC: (3)(-5) + (-4)(6) + (1)(-2) = -15 - 24 - 2 = -41, and (3)(4) + (-4)(-3) + (1)(7) = 12 + 12 + 7 = 31

Step 3: Second row: (4)(-5) + (5)(6) + (7)(-2) = -20 + 30 - 14 = -4, and (4)(4) + (5)(-3) + (7)(7) = 16 - 15 + 49 = 50

Step 4: Third row: (-2)(-5) + (-3)(6) + (8)(-2) = 10 - 18 - 16 = -24, and (-2)(4) + (-3)(-3) + (8)(7) = -8 + 9 + 56 = 57

Final answer: AC = [-41, 31; -4, 50; -24, 57].

Question: Q2(ii) With A, B, C as in Q2, wherever possible compute BC.

Answer: Step 1: B is 2 x 3 and C is 3 x 2; the inner numbers match, so BC exists and has order 2 x 2

Step 2: First row: (3)(-5) + (4)(6) + (-1)(-2) = -15 + 24 + 2 = 11, and (3)(4) + (4)(-3) + (-1)(7) = 12 - 12 - 7 = -7

Step 3: Second row: (2)(-5) + (-3)(6) + (5)(-2) = -10 - 18 - 10 = -38, and (2)(4) + (-3)(-3) + (5)(7) = 8 + 9 + 35 = 52

Final answer: BC = [11, -7; -38, 52].

Question: Q2(iii) With A, B, C as in Q2, wherever possible compute AB.

Answer: Step 1: A is 3 x 3 and B is 2 x 3

Step 2: For the product AB the number of columns of A must equal the number of rows of B, but A has 3 columns while B has only 2 rows

Step 3: The inner numbers 3 and 2 do not match, so the multiplication cannot be performed

Final answer: AB is not possible (the matrices are not conformable for

multiplication in this order).

Question: Q2(iv) With A, B, C as in Q2, wherever possible compute BA.

Answer: Step 1: B is 2 x 3 and A is 3 x 3; the inner numbers match, so BA exists and has order 2 x 3

Step 2: First row: (3)(3) + (4)(4) + (-1)(-2) = 9 + 16 + 2 = 27, (3)(-4) + (4)(5) + (-1)(-3) = -12 + 20 + 3 = 11, (3)(1) + (4)(7) + (-1)(8) = 3 + 28 - 8 = 23

Step 3: Second row: (2)(3) + (-3)(4) + (5)(-2) = 6 - 12 - 10 = -16, (2)(-4) + (-3)(5) + (5)(-3) = -8 - 15 - 15 = -38, (2)(1) + (-3)(7) + (5)(8) = 2 - 21 + 40 = 21

Final answer: BA = [27, 11, 23; -16, -38, 21].

Question: Q2(v) With A, B, C as in Q2, wherever possible compute A^2.

Answer: Step 1: A^2 = A x A, and A is 3 x 3, so the product exists and is 3 x 3

Step 2: First row: (9 - 16 - 2), (-12 - 20 - 3), (3 - 28 + 8) computed entry by entry as (3)(3)+(-4)(4)+(1)(-2) = -9, (3)(-4)+(-4)(5)+(1)(-3) = -35, (3)(1)+(-4)(7)+(1)(8) = -17

Step 3: Second row: (4)(3)+(5)(4)+(7)(-2) = 18, (4)(-4)+(5)(5)+(7)(-3) = -12, (4)(1)+(5)(7)+(7)(8) = 95

Step 4: Third row: (-2)(3)+(-3)(4)+(8)(-2) = -34, (-2)(-4)+(-3)(5)+(8)(-3) = -31, (-2)(1)+(-3)(7)+(8)(8) = 41

Final answer: A^2 = [-9, -35, -17; 18, -12, 95; -34, -31, 41].

Question: Q2(vi) With A, B, C as in Q2, wherever possible compute CB.

Answer: Step 1: C is 3 x 2 and B is 2 x 3; the inner numbers match, so CB exists and has order 3 x 3

Step 2: First row: (-5)(3) + (4)(2) = -7, (-5)(4) + (4)(-3) = -32, (-5)(-1) + (4)(5) = 25

Step 3: Second row: (6)(3) + (-3)(2) = 12, (6)(4) + (-3)(-3) = 33, (6)(-1) + (-3)(5) = -21

Step 4: Third row: (-2)(3) + (7)(2) = 8, (-2)(4) + (7)(-3) = -29, (-2)(-1) + (7)(5) = 37

Final answer: CB = [-7, -32, 25; 12, 33, -21; 8, -29, 37].

Question: Q2(vii) With A, B, C as in Q2, wherever possible compute (AB)C.

Answer: Step 1: The inner product AB must be found first

Step 2: From Q2(iii), AB does not exist because A has 3 columns but B has only 2 rows

Step 3: Since AB itself cannot be formed, multiplying it by C is out of the question

Final answer: (AB)C is not possible.

Question: Q2(viii) With A, B, C as in Q2, wherever possible compute C^t B^t.

Answer: Step 1: Write the transposes: C^t = [-5, 6, -2; 4, -3, 7] of order 2 x 3 and B^t = [3, 2; 4, -3; -1, 5] of order 3 x 2

Step 2: The inner numbers 3 and 3 match, so the product exists and has order 2 x 2

Step 3: First row: (-5)(3) + (6)(4) + (-2)(-1) = -15 + 24 + 2 = 11, and (-5)(2) + (6)(-3) + (-2)(5) = -10 - 18 - 10 = -38

Step 4: Second row: (4)(3) + (-3)(4) + (7)(-1) = 12 - 12 - 7 = -7, and (4)(2) + (-3)(-3) + (7)(5) = 8 + 9 + 35 = 52

Step 5: As a check, this equals (BC)^t from Q2(ii), as the rule (BC)^t = C^t B^t requires

Final answer: C^t B^t = [11, -38; -7, 52].

Question: Q2(ix) With A, B, C as in Q2, wherever possible compute C^t A^t.

Answer: Step 1: Write the transposes: C^t = [-5, 6, -2; 4, -3, 7] of order 2 x 3 and A^t = [3, 4, -2; -4, 5, -3; 1, 7, 8] of order 3 x 3

Step 2: The inner numbers match, so the product exists and has order 2 x 3

Step 3: First row: (-5)(3) + (6)(-4) + (-2)(1) = -41, (-5)(4) + (6)(5) + (-2)(7) = -4, (-5)(-2) + (6)(-3) + (-2)(8) = -24

Step 4: Second row: (4)(3) + (-3)(-4) + (7)(1) = 31, (4)(4) + (-3)(5) + (7)(7) = 50, (4)(-2) + (-3)(-3) + (7)(8) = 57

Step 5: As a check, this equals (AC)^t from Q2(i)

Final answer: C^t A^t = [-41, -4, -24; 31, 50, 57].

Question: Q2(x) With A, B, C as in Q2, wherever possible compute (CA)B.

Answer: Step 1: The inner product CA must be found first: C is 3 x 2 and A is 3 x 3

Step 2: For CA the number of columns of C must equal the number of rows of A, but C has 2 columns while A has 3 rows

Step 3: So CA does not exist, and therefore (CA)B cannot be formed either

Final answer: (CA)B is not possible.

Question: Q3 Let X = [1, -3, 2; 2, 1, -3; 4, -3, -1], Y = [2, 1, -1; 3, -2, -1; 2, -5, -1] and Z = [1, 4, 1; 2, 1, 1; 1, -2, 1]. Show that XY = XZ.

Answer: Step 1: Compute XY row by row: row 1 is (2 - 9 + 4), (1 + 6 - 10), (-1 + 3 - 2) = -3, -3, 0, from (1)(2)+(-3)(3)+(2)(2), (1)(1)+(-3)(-2)+(2)(-5), (1)(-1)+(-3)(-1)+(2)(-1)

Step 2: Row 2 of XY is (4 + 3 - 6), (2 - 2 + 15), (-2 - 3 + 3) = 1, 15, 0

Step 3: Row 3 of XY is (8 - 9 - 2), (4 + 6 + 5), (-4 + 3 + 1) = -3, 15, 0, so XY = [-3, -3, 0; 1, 15, 0; -3, 15, 0]

Step 4: Compute XZ row 1: (1)(1)+(-3)(2)+(2)(1) = -3, (1)(4)+(-3)(1)+(2)(-2) = -3, (1)(1)+(-3)(1)+(2)(1) = 0

Step 5: Row 2 of XZ: (2)(1)+(1)(2)+(-3)(1) = 1, (2)(4)+(1)(1)+(-3)(-2) = 15, (2)(1)+(1)(1)+(-3)(1) = 0

Step 6: Row 3 of XZ: (4)(1)+(-3)(2)+(-1)(1) = -3, (4)(4)+(-3)(1)+(-1)(-2) = 15, (4)(1)+(-3)(1)+(-1)(1) = 0, so XZ = [-3, -3, 0; 1, 15, 0; -3, 15, 0]

Step 7: XY and XZ are identical, even though Y is not equal to Z; this shows the cancellation law does not hold for matrices

Final answer: XY = XZ = [-3, -3, 0; 1, 15, 0; -3, 15, 0], hence shown.

Question: Q4 If A = [-2, 1, 0; -1, 4, 3; 0, 8, 5], find A^2 - 5A + 4I.

Answer: Step 1: Compute A^2 = A x A: row 1 is (4 - 1 + 0), (-2 + 4 + 0), (0 + 3 + 0) = 3, 2, 3

Step 2: Row 2 of A^2 is (2 - 4 + 0), (-1 + 16 + 24), (0 + 12 + 15) = -2, 39, 27

Step 3: Row 3 of A^2 is (0 - 8 + 0), (0 + 32 + 40), (0 + 24 + 25) = -8, 72, 49, so A^2 = [3, 2, 3; -2, 39, 27; -8, 72, 49]

Step 4: Compute 5A = [-10, 5, 0; -5, 20, 15; 0, 40, 25] and 4I = [4, 0, 0; 0, 4, 0; 0, 0, 4]

Step 5: Subtract and add entry by entry: first row (3 + 10 + 4), (2 - 5), (3 - 0) = 17, -3, 3; second row (-2 + 5), (39 - 20 + 4), (27 - 15) = 3, 23, 12; third row (-8 - 0), (72 - 40), (49 - 25 + 4) = -8, 32, 28

Final answer: A^2 - 5A + 4I = [17, -3, 3; 3, 23, 12; -8, 32, 28].

Question: Q5 Prove the identity: { [1, w, w^2; w, w^2, 1; w^2, 1, w] + [w, w^2, 1; w^2, 1, w; w, w^2, 1] } x [1; w; w^2] = [0; 0; 0], where w is a complex cube root of unity.

Answer: Step 1: Recall the facts about a complex cube root of unity: w^3 = 1 and 1 + w + w^2 = 0, so 1 + w = -w^2, w + w^2 = -1 and 1 + w^2 = -w

Step 2: Add the two matrices entry by entry: the sum S has first row (1 + w), (w + w^2), (w^2 + 1) = -w^2, -1, -w

Step 3: The second row of S is (w + w^2), (w^2 + 1), (1 + w) = -1, -w, -w^2, and the third row is (w^2 + w), (1 + w^2), (w + 1) = -1, -w, -w^2, so S = [-w^2, -1, -w; -1, -w, -w^2; -1, -w, -w^2]

Step 4: Multiply S by the column [1; w; w^2]: the first entry is (-w^2)(1) + (-1)(w) + (-w)(w^2) = -w^2 - w - w^3 = -(w^2 + w + 1) = 0, using w^3 = 1

Step 5: The second entry is (-1)(1) + (-w)(w) + (-w^2)(w^2) = -1 - w^2 - w^4 = -1 - w^2 - w = 0, using w^4 = w^3 x w = w

Step 6: The third entry is the same combination (-1)(1) + (-w)(w) + (-w^2)(w^2) = -1 - w^2 - w = 0

Final answer: The product is [0; 0; 0], hence the identity is proved.

Question: Q6(i) If A = [3, 2, 1; 4, -2, 5; -1, 0, -7] and B = [9, 4, 6; 7, -8, 5; -1, 0, 3], verify the commutative property under addition.

Answer: Step 1: Compute A + B entry by entry: [3 + 9, 2 + 4, 1 + 6; 4 + 7, -2 - 8, 5 + 5; -1 - 1, 0 + 0, -7 + 3] = [12, 6, 7; 11, -10, 10; -2, 0, -4]

Step 2: Compute B + A entry by entry: [9 + 3, 4 + 2, 6 + 1; 7 + 4, -8 - 2, 5 + 5; -1 - 1, 0 + 0, 3 - 7] = [12, 6, 7; 11, -10, 10; -2, 0, -4]

Step 3: The two results are identical entry by entry, because ordinary addition of numbers is commutative in each position

Final answer: A + B = B + A = [12, 6, 7; 11, -10, 10; -2, 0, -4], so the commutative property under addition is verified.

Question: Q6(ii) If A = [3, 2, 1; 4, -2, 5; -1, 0, -7] and B = [9, 4, 6; 7, -8, 5; -1, 0, 3], verify that (AB)^t = B^t A^t.

Answer: Step 1: Compute AB: row 1 is (27 + 14 - 1), (12 - 16 + 0), (18 + 10 + 3) = 40, -4, 31

Step 2: Row 2 of AB is (36 - 14 - 5), (16 + 16 + 0), (24 - 10 + 15) = 17, 32, 29, and row 3 is (-9 + 0 + 7), (-4 + 0 + 0), (-6 + 0 - 21) = -2, -4, -27, so AB = [40, -4, 31; 17, 32, 29; -2, -4, -27]

Step 3: Transpose the product: (AB)^t = [40, 17, -2; -4, 32, -4; 31, 29, -27]

Step 4: Write B^t = [9, 7, -1; 4, -8, 0; 6, 5, 3] and A^t = [3, 4, -1; 2, -2, 0; 1, 5, -7]

Step 5: Compute B^t A^t: row 1 is (27 + 14 - 1), (36 - 14 + 0)... working entry by entry: (1,1) = (9)(3) + (7)(2) + (-1)(1) = 40, (1,2) = (9)(4) + (7)(-2) + (-1)(5) = 17, (1,3) = (9)(-1) + (7)(0) + (-1)(-7) = -2

Step 6: Row 2 of B^t A^t: (4)(3) + (-8)(2) + (0)(1) = -4, (4)(4) + (-8)(-2) + (0)(5) = 32, (4)(-1) + (-8)(0) + (0)(-7) = -4; row 3: (6)(3) + (5)(2) + (3)(1) = 31, (6)(4) + (5)(-2) + (3)(5) = 29, (6)(-1) + (5)(0) + (3)(-7) = -27, so B^t A^t = [40, 17, -2; -4, 32, -4; 31, 29, -27]

Step 7: The results of Step 3 and Step 6 are identical

Final answer: (AB)^t = B^t A^t = [40, 17, -2; -4, 32, -4; 31, 29, -27], hence verified.

Exercise 2.3

Question: Q1(i) Evaluate the determinant |5, 4, 3; 3, -4, 0; 2, 3, 1|.

Answer: Step 1: Expand along the first row: det = 5 x M11 - 4 x M12 + 3 x M13, where the M are the 2 x 2 minors

Step 2: M11 = (-4)(1) - (0)(3) = -4, so the first term is 5 x (-4) = -20

Step 3: M12 = (3)(1) - (0)(2) = 3, so the second term is -4 x 3 = -12

Step 4: M13 = (3)(3) - (-4)(2) = 9 + 8 = 17, so the third term is 3 x 17 = 51

Step 5: Add the three terms: -20 - 12 + 51 = 19

Final answer: The determinant is 19.

Question: Q1(ii) Evaluate the determinant |-2, 4, 3; 5, 4, -2; 2, 7, 3|.

Answer: Step 1: Expand along the first row: det = -2 x M11 - 4 x M12 + 3 x M13

Step 2: M11 = (4)(3) - (-2)(7) = 12 + 14 = 26, so the first term is -2 x 26 = -52

Step 3: M12 = (5)(3) - (-2)(2) = 15 + 4 = 19, so the second term is -4 x 19 = -76

Step 4: M13 = (5)(7) - (4)(2) = 35 - 8 = 27, so the third term is 3 x 27 = 81

Step 5: Add the terms: -52 - 76 + 81 = -47

Final answer: The determinant is -47.

Question: Q1(iii) Evaluate the determinant |2c, c, c; a, a, 2a; b, 2b, b|.

Answer: Step 1: Take the common factor a out of the second row and the common factor b out of the third row (Property 5): det = ab x |2c, c, c; 1, 1, 2; 1, 2, 1|

Step 2: Expand the remaining determinant along its first row: 2c x [(1)(1) - (2)(2)] - c x [(1)(1) - (2)(1)] + c x [(1)(2) - (1)(1)]

Step 3: Evaluate the three minors: (1 - 4) = -3, (1 - 2) = -1, (2 - 1) = 1

Step 4: So the bracket becomes 2c(-3) - c(-1) + c(1) = -6c + c + c = -4c

Step 5: Multiply back by ab: det = ab x (-4c) = -4abc

Final answer: The determinant is -4abc.

Question: Q1(iv) Evaluate the determinant |1, 0, 1 - i; 0, 1, i; 1 + i, i, 1|.

Answer: Step 1: Expand along the first row: det = 1 x M11 - 0 x M12 + (1 - i) x M13

Step 2: M11 = (1)(1) - (i)(i) = 1 - i^2 = 1 + 1 = 2, so the first term is 2

Step 3: The middle term is 0 because the entry a12 itself is 0

Step 4: M13 = (0)(i) - (1)(1 + i) = -(1 + i), so the third term is (1 - i) x (-(1 + i)) = -[(1)(1) - i^2] = -(1 + 1) = -2

Step 5: Add the terms: 2 + 0 - 2 = 0

Final answer: The determinant is 0.

Question: Q1(v) Evaluate the determinant |7, -2, 1; 2, 2, 4; 4, 3, 7|.

Answer: Step 1: Expand along the first row: det = 7 x M11 - (-2) x M12 + 1

x M13 = 7 x M11 + 2 x M12 + M13

Step 2: M11 = (2)(7) - (4)(3) = 14 - 12 = 2, so the first term is 7 x 2 = 14

Step 3: M12 = (2)(7) - (4)(4) = 14 - 16 = -2, so the second term is 2 x (-2) = -4

Step 4: M13 = (2)(3) - (2)(4) = 6 - 8 = -2, so the third term is -2

Step 5: Add the terms: 14 - 4 - 2 = 8

Final answer: The determinant is 8.

Question: Q1(vi) Evaluate the determinant |1, 1, 1; 1, w, w^2; w^2, w, 1|, where w is a complex cube root of unity.

Answer: Step 1: Expand along the first row: det = 1 x M11 - 1 x M12 + 1 x M13

Step 2: M11 = (w)(1) - (w^2)(w) = w - w^3 = w - 1, using w^3 = 1

Step 3: M12 = (1)(1) - (w^2)(w^2) = 1 - w^4 = 1 - w, using w^4 = w^3 x w = w

Step 4: M13 = (1)(w) - (w)(w^2) = w - w^3 = w - 1

Step 5: Add with the expansion signs: (w - 1) - (1 - w) + (w - 1) = 3w - 3

Final answer: The determinant is 3w - 3 = 3(w - 1).

Question: Q2(i) Identify whether the matrix [4, 0, 1; 7, 5, 5; -12, -6, -7] is singular or non-singular.

Answer: Step 1: A square matrix is singular if its determinant is 0 and non-singular if its determinant is not 0, so evaluate the determinant by expanding along the first row

Step 2: The minors are M11 = (5)(-7) - (5)(-6) = -35 + 30 = -5, M12 = (7)(-7) - (5)(-12) = -49 + 60 = 11, M13 = (7)(-6) - (5)(-12) = -42 + 60 = 18

Step 3: det = 4 x (-5) - 0 x 11 + 1 x 18 = -20 + 0 + 18 = -2

Step 4: The determinant is not zero

Final answer: Since det = -2 (not 0), the matrix is non-singular.

Question: Q2(ii) Identify whether the matrix [4, 2, 0; 3, 0, 1; -1, 1, -1] is singular or non-singular.

Answer: Step 1: Expand the determinant along the first row

Step 2: The minors are M11 = (0)(-1) - (1)(1) = -1, M12 = (3)(-1) - (1)(-1) = -3 + 1 = -2, M13 = (3)(1) - (0)(-1) = 3

Step 3: det = 4 x (-1) - 2 x (-2) + 0 x 3 = -4 + 4 + 0 = 0

Step 4: The determinant is zero

Final answer: Since det = 0, the matrix is singular.

Question: Q2(iii) Identify whether the matrix [13, -5, 4; 8, 1, 3; 7, -1, 2] is singular or non-singular.

Answer: Step 1: Expand the determinant along the first row

Step 2: The minors are M11 = (1)(2) - (3)(-1) = 2 + 3 = 5, M12 = (8)(2) - (3)(7) = 16 - 21 = -5, M13 = (8)(-1) - (1)(7) = -8 - 7 = -15

Step 3: det = 13 x 5 - (-5) x (-5) + 4 x (-15) = 65 - 25 - 60 = -20

Step 4: The determinant is not zero

Final answer: Since det = -20 (not 0), the matrix is non-singular.

Question: Q2(iv) Identify whether the matrix [1, 6, 3; -2, 1, 0; 6, 4, 2] is singular or non-singular.

Answer: Step 1: Expand the determinant along the first row

Step 2: The minors are M11 = (1)(2) - (0)(4) = 2, M12 = (-2)(2) - (0)(6) =

-4, M13 = (-2)(4) - (1)(6) = -8 - 6 = -14

Step 3: det = 1 x 2 - 6 x (-4) + 3 x (-14) = 2 + 24 - 42 = -16

Step 4: The determinant is not zero

Final answer: Since det = -16 (not 0), the matrix is non-singular.

Question: Q2(v) Identify whether the matrix [4, 0, 0; 0, 3, 0; 0, 0, 2] is singular or non-singular.

Answer: Step 1: The matrix is diagonal, and the determinant of a diagonal matrix is the product of its main diagonal elements

Step 2: det = 4 x 3 x 2 = 24

Step 3: The determinant is not zero

Final answer: Since det = 24 (not 0), the matrix is non-singular.

Question: Q2(vi) Identify whether the matrix [20, 10, 30; 2, 1, 3; 0, 20, 0] is singular or non-singular.

Answer: Step 1: Look at the rows before expanding: the first row 20, 10, 30 is exactly 10 times the second row 2, 1, 3

Step 2: By Property 7, if one row is a constant multiple of another row, the determinant is zero

Step 3: Confirm by expansion along the first row if wished: M11 = (1)(0) - (3)(20) = -60, M12 = (2)(0) - (3)(0) = 0, M13 = (2)(20) - (1)(0) = 40, giving det = 20(-60) - 10(0) + 30(40) = -1200 + 1200 = 0

Final answer: Since det = 0, the matrix is singular.

Question: Q3(i) Find the value of x for which the matrix [10, 5; 6, x] is singular.

Answer: Step 1: The matrix is singular exactly when its determinant is 0

Step 2: For a 2 x 2 matrix the determinant is (10)(x) - (5)(6) = 10x - 30

Step 3: Set the determinant equal to zero: 10x - 30 = 0

Step 4: Solve: 10x = 30, so x = 3

Final answer: x = 3.

Question: Q3(ii) Find the value of x for which the matrix [5, 4; x, 8] is singular.

Answer: Step 1: Set the determinant equal to zero

Step 2: The determinant is (5)(8) - (4)(x) = 40 - 4x

Step 3: 40 - 4x = 0 gives 4x = 40

Step 4: Solve: x = 10

Final answer: x = 10.

Question: Q3(iii) Find the value of x for which the matrix [6, 3, 7; 3, -4, 2; 5, x, 1] is singular.

Answer: Step 1: Expand the determinant along the first row: the minors are M11 = (-4)(1) - (2)(x) = -4 - 2x, M12 = (3)(1) - (2)(5) = -7, M13 = (3)(x) - (-4)(5) = 3x + 20

Step 2: det = 6(-4 - 2x) - 3(-7) + 7(3x + 20) = -24 - 12x + 21 + 21x + 140 = 9x + 137

Step 3: For a singular matrix set det = 0: 9x + 137 = 0

Step 4: Solve: 9x = -137, so x = -137/9

Final answer: x = -137/9.

Question: Q3(iv) Find the value of x for which the matrix [x, -2, 1; 2, -3, 4; x, -2, -1] is singular.

Answer: Step 1: Expand along the first row: the minors are M11 = (-3)(-1) - (4)(-2) = 3 + 8 = 11, M12 = (2)(-1) - (4)(x) = -2 - 4x, M13 = (2)(-2) - (-3)(x) = -4 + 3x

Step 2: det = x(11) - (-2)(-2 - 4x) + 1(-4 + 3x) = 11x - (4 + 8x) - 4 + 3x = 6x - 8

Step 3: Set det = 0: 6x - 8 = 0

Step 4: Solve: 6x = 8, so x = 4/3

Final answer: x = 4/3.

Question: Q4(i) Find the adjoint of the matrix [3, 4; -1, 2].

Answer: Step 1: For a 2 x 2 matrix [a, b; c, d] the cofactors are A11 = d, A12 = -b, A21 = -c, A22 = a

Step 2: Here A11 = 2, A12 = -4, A21 = 1, A22 = 3, so the cofactor matrix is [2, -4; 1, 3]

Step 3: The adjoint is the transpose of the cofactor matrix: adj = [2, 1; -4, 3]^t form, that is, interchange the diagonal entries and change the signs of the off-diagonal entries of the original matrix

Final answer: adj A = [2, -4; 1, 3].

Question: Q4(ii) Find the adjoint of the matrix [-0.3, 0.5; 1, 2].

Answer: Step 1: For a 2 x 2 matrix, swap the diagonal entries and change the signs of the off-diagonal entries to get the adjoint

Step 2: The diagonal entries -0.3 and 2 swap places, giving 2 in the (1,1) position and -0.3 in the (2,2) position

Step 3: The off-diagonal entries change sign: 0.5 becomes -0.5 and 1 becomes -1

Final answer: adj A = [2, -0.5; -1, -0.3].

Question: Q4(iii) Find the adjoint of the matrix [4, 6, 8; 1, 3, 2; 2, 7, 5].

Answer: Step 1: Find all nine cofactors Aij = (-1)^(i+j) Mij, first row: A11 = +[(3)(5) - (2)(7)] = 15 - 14 = 1, A12 = -[(1)(5) - (2)(2)] = -(5 - 4) = -1, A13 = +[(1)(7) - (3)(2)] = 7 - 6 = 1

Step 2: Second row of cofactors: A21 = -[(6)(5) - (8)(7)] = -(30 - 56) = 26, A22 = +[(4)(5) - (8)(2)] = 20 - 16 = 4, A23 = -[(4)(7) - (6)(2)] = -(28 - 12) = -16

Step 3: Third row of cofactors: A31 = +[(6)(2) - (8)(3)] = 12 - 24 = -12, A32 = -[(4)(2) - (8)(1)] = -(8 - 8) = 0, A33 = +[(4)(3) - (6)(1)] = 12 - 6 = 6

Step 4: The cofactor matrix is [1, -1, 1; 26, 4, -16; -12, 0, 6]

Step 5: The adjoint is the transpose of the cofactor matrix

Final answer: adj A = [1, 26, -12; -1, 4, 0; 1, -16, 6].

Question: Q4(iv) Find the adjoint of the matrix [1, 0, 1 - i; 1 + i, i, 1; 0, 1, -i].

Answer: Step 1: Compute the cofactors one row at a time: A11 = +[(i)(-i) - (1)(1)] = 1 - 1 = 0, since (i)(-i) = 1

Step 2: A12 = -[(1 + i)(-i) - (1)(0)] = -[(-i - i^2)] = -[(1 - i)] = -1 + i, and A13 = +[(1 + i)(1) - (i)(0)] = 1 + i

Step 3: Second row: A21 = -[(0)(-i) - (1 - i)(1)] = -[-(1 - i)] = 1 - i, A22 = +[(1)(-i) - (1 - i)(0)] = -i, A23 = -[(1)(1) - (0)(0)] = -1

Step 4: Third row: A31 = +[(0)(1) - (1 - i)(i)] = -[i - i^2] = -(1 + i) = -1 - i, A32 = -[(1)(1) - (1 - i)(1 + i)] = -[1 - 2] = 1, A33 = +[(1)(i) - (0)(1 + i)] = i

Step 5: The cofactor matrix is [0, -1 + i, 1 + i; 1 - i, -i, -1; -1 - i, 1, i], and

the adjoint is its transpose

Final answer: adj A = [0, 1 - i, -1 - i; -1 + i, -i, 1; 1 + i, -1, i].

Question: Q4(v) Find the adjoint of the diagonal matrix [5, 0, 0; 0, 6, 0; 0, 0, 7].

Answer: Step 1: For a diagonal matrix every off-diagonal cofactor is 0, because each such minor contains a row or column of zeros

Step 2: The diagonal cofactors are the products of the other two diagonal entries: A11 = (6)(7) = 42, A22 = (5)(7) = 35, A33 = (5)(6) = 30

Step 3: The cofactor matrix is therefore [42, 0, 0; 0, 35, 0; 0, 0, 30], which is already symmetric, so its transpose is itself

Final answer: adj A = [42, 0, 0; 0, 35, 0; 0, 0, 30].

Question: Q5(i) If A = [11, 10; 7, 5] and B = [1, 3; 2, 1], verify that (A^-1)^-1 = A.

Answer: Step 1: Compute det A = (11)(5) - (10)(7) = 55 - 70 = -15, which is not zero, so A^-1 exists

Step 2: adj A = [5, -10; -7, 11], so A^-1 = (1/-15) x [5, -10; -7, 11] = [-1/3, 2/3; 7/15, -11/15]

Step 3: Now invert A^-1: its determinant is (-1/3)(-11/15) - (2/3)(7/15) = 11/45 - 14/45 = -3/45 = -1/15

Step 4: adj(A^-1) = [-11/15, -2/3; -7/15, -1/3], so (A^-1)^-1 = (1/(-1/15)) x adj(A^-1) = -15 x [-11/15, -2/3; -7/15, -1/3] = [11, 10; 7, 5]

Step 5: This is exactly the original matrix A

Final answer: (A^-1)^-1 = [11, 10; 7, 5] = A, hence verified.

Question: Q5(ii) If A = [11, 10; 7, 5] and B = [1, 3; 2, 1], verify that

(AB)^-1 = B^-1 A^-1.

Answer: Step 1: Compute AB = [(11)(1) + (10)(2), (11)(3) + (10)(1); (7)(1) + (5)(2), (7)(3) + (5)(1)] = [31, 43; 17, 26]

Step 2: det(AB) = (31)(26) - (43)(17) = 806 - 731 = 75, and adj(AB) = [26, -43; -17, 31], so (AB)^-1 = (1/75) x [26, -43; -17, 31]

Step 3: From Q5(i), A^-1 = [-1/3, 2/3; 7/15, -11/15]; also det B = (1)(1) - (3)(2) = -5 and adj B = [1, -3; -2, 1], so B^-1 = (1/-5) x [1, -3; -2, 1] = [-1/5, 3/5; 2/5, -1/5]

Step 4: Multiply B^-1 A^-1: the (1,1) entry is (-1/5)(-1/3) + (3/5)(7/15) = 1/15 + 21/75 = 26/75; the (1,2) entry is (-1/5)(2/3) + (3/5)(-11/15) = -2/15 - 33/75 = -43/75; the (2,1) entry is (2/5)(-1/3) + (-1/5)(7/15) = -2/15 - 7/75 = -17/75; the (2,2) entry is (2/5)(2/3) + (-1/5)(-11/15) = 4/15 + 11/75 = 31/75

Step 5: So B^-1 A^-1 = (1/75) x [26, -43; -17, 31], identical to the left side

Final answer: (AB)^-1 = B^-1 A^-1 = [26/75, -43/75; -17/75, 31/75], hence verified.

Question: Q5(iii) If A = [11, 10; 7, 5] and B = [1, 3; 2, 1], verify that adj(AB) = (adj B)(adj A).

Answer: Step 1: From Q5(ii), AB = [31, 43; 17, 26], so adj(AB) is obtained by swapping the diagonal entries and changing the signs of the off-diagonal entries: adj(AB) = [26, -43; -17, 31]

Step 2: adj A = [5, -10; -7, 11] and adj B = [1, -3; -2, 1]

Step 3: Multiply (adj B)(adj A): the (1,1) entry is (1)(5) + (-3)(-7) = 5 + 21 = 26; the (1,2) entry is (1)(-10) + (-3)(11) = -10 - 33 = -43

Step 4: The (2,1) entry is (-2)(5) + (1)(-7) = -17, and the (2,2) entry is

(-2)(-10) + (1)(11) = 31

Step 5: So (adj B)(adj A) = [26, -43; -17, 31], identical to adj(AB)

Final answer: adj(AB) = (adj B)(adj A) = [26, -43; -17, 31], hence verified.

Question: Q5(iv) If A = [11, 10; 7, 5] and B = [1, 3; 2, 1], verify that (5A)^-1 = (1/5) A^-1.

Answer: Step 1: 5A = [55, 50; 35, 25], and det(5A) = (55)(25) - (50)(35) = 1375 - 1750 = -375

Step 2: adj(5A) = [25, -50; -35, 55], so (5A)^-1 = (1/-375) x [25, -50; -35, 55] = [-1/15, 2/15; 7/75, -11/75]

Step 3: From Q5(i), A^-1 = [-1/3, 2/3; 7/15, -11/15], so (1/5) A^-1 = [-1/15, 2/15; 7/75, -11/75]

Step 4: The two results are identical

Final answer: (5A)^-1 = (1/5) A^-1 = [-1/15, 2/15; 7/75, -11/75], hence verified.

Question: Q6(i) Verify that { [2, 3; 1, -1] x [3, 1; 2, 0] }^-1 = [3, 1; 2, 0]^-1 x [2, 3; 1, -1]^-1.

Answer: Step 1: Multiply the two matrices inside the braces: [2, 3; 1, -1] x [3, 1; 2, 0] = [(6 + 6), (2 + 0); (3 - 2), (1 + 0)] = [12, 2; 1, 1]

Step 2: Its determinant is (12)(1) - (2)(1) = 10, so the left side is (1/10) x [1, -2; -1, 12] = [1/10, -1/5; -1/10, 6/5]

Step 3: Find the two inverses on the right: det[3, 1; 2, 0] = -2, so [3, 1; 2, 0]^-1 = (1/-2) x [0, -1; -2, 3] = [0, 1/2; 1, -3/2]; and det[2, 3; 1, -1] = -5, so [2, 3; 1, -1]^-1 = (1/-5) x [-1, -3; -1, 2] = [1/5, 3/5; 1/5, -2/5]

Step 4: Multiply them in the stated order: the (1,1) entry is (0)(1/5) +

(1/2)(1/5) = 1/10; the (1,2) entry is (0)(3/5) + (1/2)(-2/5) = -1/5; the (2,1) entry is (1)(1/5) + (-3/2)(1/5) = -1/10; the (2,2) entry is (1)(3/5) + (-3/2)(-2/5) = 3/5 + 3/5 = 6/5

Step 5: The right side equals [1/10, -1/5; -1/10, 6/5], the same as the left side

Final answer: Both sides equal [1/10, -1/5; -1/10, 6/5], hence verified.

Question: Q6(ii) Verify that { [1, i; i, 1] x [1, -i; -i, 1] }^-1 = [1, -i; -i, 1]^-1 x [1, i; i, 1]^-1.

Answer: Step 1: Multiply the matrices inside the braces: [1, i; i, 1] x [1, -i; -i, 1] = [(1 + 1), (-i + i); (i - i), (1 + 1)] = [2, 0; 0, 2] = 2I, using i x (-i) = 1

Step 2: The inverse of 2I is (1/2) I, so the left side is [1/2, 0; 0, 1/2]

Step 3: Find the separate inverses: det[1, i; i, 1] = 1 - i^2 = 2, so [1, i; i, 1]^-1 = (1/2) x [1, -i; -i, 1]; and det[1, -i; -i, 1] = 1 - (-i)^2 = 1 + 1 = 2, so [1, -i; -i, 1]^-1 = (1/2) x [1, i; i, 1]

Step 4: Multiply in the stated order: (1/2)[1, i; i, 1] x (1/2)[1, -i; -i, 1] = (1/4) x [(1 + 1), (-i + i); (i - i), (1 + 1)] = (1/4) x [2, 0; 0, 2] = [1/2, 0; 0, 1/2]

Step 5: The right side equals the left side

Final answer: Both sides equal [1/2, 0; 0, 1/2], hence verified.

Question: Q7(i) Use the adjoint method to calculate the inverse of [5, 6; 3, 4], if possible.

Answer: Step 1: Compute the determinant: det = (5)(4) - (6)(3) = 20 - 18 = 2, which is not zero, so the inverse exists

Step 2: Find the adjoint by swapping the diagonal entries and changing the signs of the off-diagonal entries: adj A = [4, -6; -3, 5]

Step 3: Apply A^-1 = (1/det A) x adj A = (1/2) x [4, -6; -3, 5]

Final answer: A^-1 = [2, -3; -3/2, 5/2].

Question: Q7(ii) Use the adjoint method to calculate the inverse of [7, 3; 9, 6], if possible.

Answer: Step 1: Compute the determinant: det = (7)(6) - (3)(9) = 42 - 27 = 15, not zero, so the inverse exists

Step 2: Find the adjoint: adj A = [6, -3; -9, 7]

Step 3: Apply A^-1 = (1/15) x [6, -3; -9, 7]

Final answer: A^-1 = [2/5, -1/5; -3/5, 7/15].

Question: Q7(iii) Use the adjoint method to calculate the inverse of [1, 2, 3; 2, 4, 6; 3, 0, 9], if possible.

Answer: Step 1: Compute the determinant by expanding along the first row: the minors are M11 = (4)(9) - (6)(0) = 36, M12 = (2)(9) - (6)(3) = 0, M13 = (2)(0) - (4)(3) = -12

Step 2: det = 1(36) - 2(0) + 3(-12) = 36 - 0 - 36 = 0

Step 3: Notice why: the second row 2, 4, 6 is exactly twice the first row 1, 2, 3, which always makes a determinant zero

Step 4: Since det A = 0, the matrix is singular and the formula A^-1 = adj A / det A cannot be used

Final answer: The inverse does not exist because the matrix is singular (det A = 0).

Question: Q7(iv) Use the adjoint method to calculate the inverse of [2, 3, 4; 3, 1, 5; -5, 1, 0], if possible.

Answer: Step 1: Compute the determinant along the first row: the minors

are M11 = (1)(0) - (5)(1) = -5, M12 = (3)(0) - (5)(-5) = 25, M13 = (3)(1) - (1)(-5) = 8, so det = 2(-5) - 3(25) + 4(8) = -10 - 75 + 32 = -53, not zero, so the inverse exists

Step 2: First row of cofactors: A11 = -5, A12 = -25, A13 = 8

Step 3: Second row of cofactors: A21 = -[(3)(0) - (4)(1)] = 4, A22 = +[(2)(0) - (4)(-5)] = 20, A23 = -[(2)(1) - (3)(-5)] = -17

Step 4: Third row of cofactors: A31 = +[(3)(5) - (4)(1)] = 11, A32 = -[(2)(5) - (4)(3)] = 2, A33 = +[(2)(1) - (3)(3)] = -7

Step 5: The adjoint is the transpose of the cofactor matrix [-5, -25, 8; 4, 20, -17; 11, 2, -7], giving adj A = [-5, 4, 11; -25, 20, 2; 8, -17, -7]

Step 6: Apply A^-1 = (1/-53) x adj A

Final answer: A^-1 = [5/53, -4/53, -11/53; 25/53, -20/53, -2/53; -8/53, 17/53, 7/53].

Question: Q7(v) Use the adjoint method to calculate the inverse of [1, 0, 2; 0, 2, 1; 1, -1, 1], if possible.

Answer: Step 1: Compute the determinant along the first row: the minors are M11 = (2)(1) - (1)(-1) = 3, M12 = (0)(1) - (1)(1) = -1, M13 = (0)(-1) - (2)(1) = -2, so det = 1(3) - 0(-1) + 2(-2) = -1, not zero, so the inverse exists

Step 2: The nine cofactors are: A11 = 3, A12 = 1, A13 = -2; A21 = -2, A22 = -1, A23 = 1; A31 = -4, A32 = -1, A33 = 2

Step 3: The adjoint is the transpose of this cofactor matrix: adj A = [3, -2, -4; 1, -1, -1; -2, 1, 2]

Step 4: Apply A^-1 = (1/-1) x adj A = -adj A

Final answer: A^-1 = [-3, 2, 4; -1, 1, 1; 2, -1, -2].

Question: Q7(vi) Use the adjoint method to calculate the inverse of [2, 1, 0; 1, 1, 0; 2, -3, 5], if possible.

Answer: Step 1: Compute the determinant by expanding along the third column, which has two zeros: det = 5 x [(2)(1) - (1)(1)] = 5 x 1 = 5, not zero, so the inverse exists

Step 2: The nine cofactors are: A11 = (1)(5) - (0)(-3) = 5, A12 = -[(1)(5) - (0)(2)] = -5, A13 = (1)(-3) - (1)(2) = -5; A21 = -[(1)(5) - (0)(-3)] = -5, A22 = (2)(5) - (0)(2) = 10, A23 = -[(2)(-3) - (1)(2)] = 8; A31 = (1)(0) - (0)(1) = 0, A32 = -[(2)(0) - (0)(1)] = 0, A33 = (2)(1) - (1)(1) = 1

Step 3: The adjoint is the transpose of the cofactor matrix: adj A = [5, -5, 0; -5, 10, 0; -5, 8, 1]

Step 4: Apply A^-1 = (1/5) x adj A

Final answer: A^-1 = [1, -1, 0; -1, 2, 0; -1, 8/5, 1/5].

Exercise 2.4

Question: Q1 Let A = [2, 5, 0; 3, 4, 6; 1, -5, 1]. Verify that |A| = |A^t|.

Answer: Step 1: Expand |A| along the first row: the minors are M11 = (4)(1) - (6)(-5) = 34, M12 = (3)(1) - (6)(1) = -3, M13 = (3)(-5) - (4)(1) = -19

Step 2: |A| = 2(34) - 5(-3) + 0(-19) = 68 + 15 + 0 = 83

Step 3: Write the transpose: A^t = [2, 3, 1; 5, 4, -5; 0, 6, 1]

Step 4: Expand |A^t| along its first row: the minors are (4)(1) - (-5)(6) = 34, (5)(1) - (-5)(0) = 5, (5)(6) - (4)(0) = 30

Step 5: |A^t| = 2(34) - 3(5) + 1(30) = 68 - 15 + 30 = 83

Step 6: Both determinants come out the same, as Property 1 states

Final answer: |A| = |A^t| = 83, hence verified.

Question: Q2(i) Without expanding, prove that |x + y, y + z, z + x; z, x, y; 1, 1, 1| = 0.

Answer: Step 1: Apply the row operation R1 -> R1 + R2, which does not change the value of a determinant (Property 8 type operation)

Step 2: The new first row is (x + y + z), (y + z + x), (z + x + y), that is, every entry becomes x + y + z

Step 3: Take the common factor (x + y + z) out of the first row (Property 5): the first row becomes 1, 1, 1

Step 4: The first row is now identical to the third row 1, 1, 1

Step 5: By Property 3, a determinant with two identical rows is zero

Final answer: The determinant is 0, hence proved without expansion.

Question: Q2(ii) Without expanding, prove that |1, x, y + z; 1, y, z + x; 1, z, x + y| = 0.

Answer: Step 1: Apply the column operation C3 -> C3 + C2, which leaves the determinant unchanged

Step 2: The new third column is (y + z + x), (z + x + y), (x + y + z), so every entry is x + y + z

Step 3: Take the common factor (x + y + z) out of the third column: it becomes 1, 1, 1

Step 4: The third column is now identical to the first column 1, 1, 1

Step 5: By Property 3, a determinant with two identical columns is zero

Final answer: The determinant is 0, hence proved without expansion.

Question: Q2(iii) Without expanding, prove that |k, b, a, c + d; k, b, c, a + d; k, d, c, b + a; k, a, d, b + c| = 0.

Answer: Step 1: Apply the column operation C4 -> C4 + C2 + C3, which leaves the determinant unchanged

Step 2: The new fourth column entries are: row 1: (c + d) + b + a = a + b + c + d; row 2: (a + d) + b + c = a + b + c + d; row 3: (b + a) + d + c = a + b + c + d; row 4: (b + c) + a + d = a + b + c + d

Step 3: So the fourth column is (a + b + c + d) times 1, 1, 1, 1

Step 4: The first column is k times 1, 1, 1, 1, so the first and fourth columns are proportional to each other

Step 5: By Property 7, a determinant with two proportional columns is zero

Final answer: The determinant is 0, hence proved without expansion.

Question: Q3(i) Without expanding, prove that |1, a, b + c; 1, b, c + a; 1, c, a + b| = 0.

Answer: Step 1: Apply the column operation C3 -> C3 + C2

Step 2: The new third column is (b + c + a), (c + a + b), (a + b + c), so every entry is a + b + c

Step 3: Take the common factor (a + b + c) out of the third column: it becomes 1, 1, 1

Step 4: The third column is now identical to the first column

Step 5: By Property 3, the determinant is zero

Final answer: The determinant is 0, hence proved.

Question: Q3(ii) Without expanding, prove that |x + 1, x + 3, x + 5; x + 4,

x + 6, x + 8; x + 7, x + 9, x + 11| = 0.

Answer: Step 1: Notice the pattern: each row increases by 3 down the columns and each column increases down the rows in steps of 3

Step 2: Apply R2 -> R2 - R1: the second row becomes (3, 3, 3)

Step 3: Apply R3 -> R3 - R1: the third row becomes (6, 6, 6)

Step 4: The third row is now exactly 2 times the second row, so two rows are proportional

Step 5: By Property 7, the determinant is zero

Final answer: The determinant is 0, hence proved.

Question: Q3(iii) Without expanding, prove that |x + 1, x + 3, x + 5; x + 4, x + 6, x + 8; x + 7, x + 9, x + 11| = 0. (Note: as printed in the book, part (iii) is the same determinant as part (ii).)

Answer: Step 1: As printed, this determinant is identical to the one in part (ii), so the same proof applies

Step 2: Apply R2 -> R2 - R1: the second row becomes (3, 3, 3)

Step 3: Apply R3 -> R3 - R1: the third row becomes (6, 6, 6), which is 2 times the new second row

Step 4: Two rows are proportional, so by Property 7 the determinant is zero

Final answer: The determinant is 0, hence proved.

Question: Q3(iv) Without expanding, prove that |b + c, a, a; b, c + a, b; c, c, a + b| = 4abc.

Answer: Step 1: Apply the row operation R1 -> R1 - R2 - R3, which does not change the determinant

Step 2: The new first row is: (b + c - b - c), (a - (c + a) - c), (a - b - (a + b))

= 0, -2c, -2b

Step 3: Now expand along this first row, which has only two non-zero entries: det = -(-2c) x |(b, b); (c, a + b)| with the cofactor sign worked in, that is det = 2c x [b(a + b) - bc] - 2b x [b x c - c(c + a)] with signs as per the expansion

Step 4: Evaluate the two 2 x 2 determinants: b(a + b) - bc = ab + b^2 - bc, and bc - c(c + a) = bc - c^2 - ca

Step 5: Combine: det = 2c(ab + b^2 - bc) - 2b(bc - c^2 - ca) = 2abc + 2b^2 c - 2bc^2 - 2b^2 c + 2bc^2 + 2abc

Step 6: The terms 2b^2 c and 2bc^2 cancel in pairs, leaving 2abc + 2abc

Final answer: The determinant is 4abc, hence proved.

Question: Q3(v) Without expanding, prove that |1 + a, 1, 1; 1, 1 + b, 1; 1, 1, 1 + c| = abc(1 + 1/a + 1/b + 1/c) = abc + bc + ca + ab.

Answer: Step 1: Take the factor a out of the first row, b out of the second row and c out of the third row (Property 5): the determinant equals abc x |1 + 1/a, 1/a, 1/a; 1/b, 1 + 1/b, 1/b; 1/c, 1/c, 1 + 1/c|

Step 2: Apply C2 -> C2 - C1: the second column becomes (1/a - 1 - 1/a), (1 + 1/b - 1/b), (1/c - 1/c) = -1, 1, 0

Step 3: Apply C3 -> C3 - C1: the third column becomes -1, 0, 1

Step 4: The determinant inside is now |(1 + 1/a), -1, -1; 1/b, 1, 0; 1/c, 0, 1|; expand it along its structure: it equals (1 + 1/a)(1)(1) + 1 x (1/b)(1) + 1 x (1/c)(1), collecting the three diagonal-type contributions (1 + 1/a) + 1/b + 1/c

Step 5: So the whole determinant is abc(1 + 1/a + 1/b + 1/c)

Step 6: Multiply out: abc + abc/a + abc/b + abc/c = abc + bc + ca + ab

Final answer: The determinant is abc(1 + 1/a + 1/b + 1/c) = abc + bc + ca + ab, hence proved.

Question: Q3(vi) Without expanding, prove that |alpha, beta gamma, alpha beta gamma; beta, gamma alpha, alpha beta gamma; gamma, alpha beta, alpha beta gamma| = |alpha, alpha^2, alpha^3; beta, beta^2, beta^3; gamma, gamma^2, gamma^3| (writing the Greek letters in words as the determinants contain alpha, beta, gamma).

Answer: Step 1: Work on the left determinant: interchange C2 and C3, which changes the sign: L = -|alpha, alpha beta gamma, beta gamma; beta, alpha beta gamma, gamma alpha; gamma, alpha beta gamma, alpha beta|

Step 2: Take the common factor alpha beta gamma out of the new second column: L = -(alpha beta gamma) x |alpha, 1, beta gamma; beta, 1, gamma alpha; gamma, 1, alpha beta|

Step 3: Interchange C1 and C2, changing the sign again: L = (alpha beta gamma) x |1, alpha, beta gamma; 1, beta, gamma alpha; 1, gamma, alpha beta|

Step 4: Multiply R1 by alpha, R2 by beta and R3 by gamma inside the determinant and divide by alpha beta gamma to compensate: the rows become (alpha, alpha^2, alpha beta gamma), (beta, beta^2, alpha beta gamma), (gamma, gamma^2, alpha beta gamma), so L = |alpha, alpha^2, alpha beta gamma; beta, beta^2, alpha beta gamma; gamma, gamma^2, alpha beta gamma| with the factors cancelling

Step 5: Take alpha beta gamma out of the third column and then shift the columns cyclically (two interchanges, so the sign is unchanged): L =

(alpha beta gamma) x |1, alpha, alpha^2; 1, beta, beta^2; 1, gamma, gamma^2|

Step 6: For the right side, take alpha out of its first row, beta out of its second row and gamma out of its third row: R = (alpha beta gamma) x |1, alpha, alpha^2; 1, beta, beta^2; 1, gamma, gamma^2|

Step 7: The two sides are now identical

Final answer: Both determinants are equal, hence proved.

Question: Q4 Without expanding, prove that |a, b, c; x, y, z; p, q, r| = |y, b, q; x, a, p; z, c, r| = |x, y, z; p, q, r; a, b, c|.

Answer: Step 1: Prove the second equality first: starting from D = |a, b, c; x, y, z; p, q, r|, interchange R1 and R2 (sign changes), getting -|x, y, z; a, b, c; p, q, r|

Step 2: Interchange R1 and R3 of the result (sign changes again), getting +|x, y, z; p, q, r; a, b, c|; two sign changes cancel, so D = |x, y, z; p, q, r; a, b, c|, which is the third determinant; this is just a cyclic shift of the rows

Step 3: For the first equality, use the transpose property: D equals its transpose |a, x, p; b, y, q; c, z, r|

Step 4: Interchange C1 and C2 of the transpose (sign changes): -|x, a, p; y, b, q; z, c, r|

Step 5: Interchange R1 and R2 (sign changes again): +|y, b, q; x, a, p; z, c, r|, which is the second determinant

Step 6: The two sign changes cancel, so all three determinants are equal

Final answer: |a, b, c; x, y, z; p, q, r| = |y, b, q; x, a, p; z, c, r| = |x, y, z; p, q, r; a, b, c|, hence proved.

Question: Q5 If a, b, c are different and Delta = |a, a^2, 1 + a^3; b, b^2, 1 + b^3; c, c^2, 1 + c^3| = 0, then show that 1 + abc = 0.

Answer: Step 1: Split the third column using Property 6 (each entry is a sum 1 + a^3 etc.): Delta = |a, a^2, 1; b, b^2, 1; c, c^2, 1| + |a, a^2, a^3; b, b^2, b^3; c, c^2, c^3| = Delta1 + Delta2

Step 2: In Delta2 take a out of the first row, b out of the second row and c out of the third row: Delta2 = abc x |1, a, a^2; 1, b, b^2; 1, c, c^2|

Step 3: In Delta1 interchange C1 and C3 (one sign change) and then C2 and C3 (a second sign change): the two changes cancel and Delta1 = |1, a, a^2; 1, b, b^2; 1, c, c^2|, the same determinant as in Step 2

Step 4: So Delta = (1 + abc) x |1, a, a^2; 1, b, b^2; 1, c, c^2|

Step 5: The remaining determinant is the Vandermonde determinant, equal to (b - a)(c - a)(c - b) up to sign; since a, b, c are all different, none of these factors is zero, so this determinant is not zero

Step 6: But Delta is given to be 0, and the product in Step 4 can only be zero through its first factor

Final answer: Therefore 1 + abc = 0, hence shown.

Exercise 2.5

Question: Q1(i) Reduce the matrix [5, 9, 3; -3, 5, 6; -1, -5, -3] into echelon form using elementary row operations.

Answer: Step 1: Start with A = [5, 9, 3; -3, 5, 6; -1, -5, -3]; bring the simplest row to the top: R1 <-> R3 gives [-1, -5, -3; -3, 5, 6; 5, 9, 3]

Step 2: R1 -> (-1) R1 gives [1, 5, 3; -3, 5, 6; 5, 9, 3], so the first leading

entry is 1

Step 3: R2 -> R2 + 3R1 gives [1, 5, 3; 0, 20, 15; 5, 9, 3], clearing the entry below the first pivot

Step 4: R3 -> R3 - 5R1 gives [1, 5, 3; 0, 20, 15; 0, -16, -12]

Step 5: R2 -> (1/20) R2 gives [1, 5, 3; 0, 1, 3/4; 0, -16, -12], making the second leading entry 1

Step 6: R3 -> R3 + 16R2 gives [1, 5, 3; 0, 1, 3/4; 0, 0, 0]

Step 7: The result has leading 1s moving to the right, zeros below each leading entry, and the zero row at the bottom, which is exactly echelon form

Final answer: The echelon form is [1, 5, 3; 0, 1, 3/4; 0, 0, 0].

Question: Q1(ii) Reduce the matrix [1, 2, -1; 2, 4, 1; 3, 6, 2] into echelon form using elementary row operations.

Answer: Step 1: Start with A = [1, 2, -1; 2, 4, 1; 3, 6, 2]; the first pivot is already 1

Step 2: R2 -> R2 - 2R1 gives [1, 2, -1; 0, 0, 3; 3, 6, 2]

Step 3: R3 -> R3 - 3R1 gives [1, 2, -1; 0, 0, 3; 0, 0, 5]; note the second column has become all zeros below the first row, so the next pivot is in the third column

Step 4: R3 -> R3 - (5/3)R2 gives [1, 2, -1; 0, 0, 3; 0, 0, 0]

Step 5: R2 -> (1/3) R2 gives [1, 2, -1; 0, 0, 1; 0, 0, 0], making the second leading entry 1

Final answer: The echelon form is [1, 2, -1; 0, 0, 1; 0, 0, 0].

Question: Q1(iii) Reduce the matrix [3, -4, 0, 9; 2, 4, -1, 0; 10, 0, -2, -4]

into echelon form using elementary row operations.

Answer: Step 1: Start with A = [3, -4, 0, 9; 2, 4, -1, 0; 10, 0, -2, -4]; R1 -> (1/3) R1 gives [1, -4/3, 0, 3; 2, 4, -1, 0; 10, 0, -2, -4]

Step 2: R2 -> R2 - 2R1 gives [1, -4/3, 0, 3; 0, 20/3, -1, -6; 10, 0, -2, -4]

Step 3: R3 -> R3 - 10R1 gives [1, -4/3, 0, 3; 0, 20/3, -1, -6; 0, 40/3, -2, -34]

Step 4: R2 -> (3/20) R2 gives [1, -4/3, 0, 3; 0, 1, -3/20, -9/10; 0, 40/3, -2, -34]

Step 5: R3 -> R3 - (40/3)R2 gives [1, -4/3, 0, 3; 0, 1, -3/20, -9/10; 0, 0, 0, -22], because -2 - (40/3)(-3/20) = 0 and -34 - (40/3)(-9/10) = -22

Step 6: R3 -> (-1/22) R3 gives [1, -4/3, 0, 3; 0, 1, -3/20, -9/10; 0, 0, 0, 1]

Final answer: The echelon form is [1, -4/3, 0, 3; 0, 1, -3/20, -9/10; 0, 0, 0, 1].

Question: Q2(i) Reduce the matrix [3, 5, 4; 4, 1, 5; 7, 6, 3] into reduced echelon form using elementary row operations.

Answer: Step 1: Start with A = [3, 5, 4; 4, 1, 5; 7, 6, 3]; R1 -> (1/3) R1 gives [1, 5/3, 4/3; 4, 1, 5; 7, 6, 3]

Step 2: R2 -> R2 - 4R1 gives [1, 5/3, 4/3; 0, -17/3, -1/3; 7, 6, 3]

Step 3: R3 -> R3 - 7R1 gives [1, 5/3, 4/3; 0, -17/3, -1/3; 0, -17/3, -19/3], which is echelon form

Step 4: R2 -> (-3/17) R2 gives [1, 5/3, 4/3; 0, 1, 1/17; 0, -17/3, -19/3]

Step 5: R3 -> R3 + (17/3)R2 gives [1, 5/3, 4/3; 0, 1, 1/17; 0, 0, -6], because -19/3 + (17/3)(1/17) = -6

Step 6: R3 -> (-1/6) R3 gives [1, 5/3, 4/3; 0, 1, 1/17; 0, 0, 1]

Step 7: Now clear above the pivots: R1 -> R1 - (4/3)R3 gives [1, 5/3, 0; 0,

1, 1/17; 0, 0, 1]

Step 8: R2 -> R2 - (1/17)R3 gives [1, 5/3, 0; 0, 1, 0; 0, 0, 1]

Step 9: R1 -> R1 - (5/3)R2 gives [1, 0, 0; 0, 1, 0; 0, 0, 1]

Final answer: The reduced echelon form is the identity matrix [1, 0, 0; 0, 1, 0; 0, 0, 1].

Question: Q2(ii) Reduce the matrix [0, 3, -2; 2, -4, 6; 2, 3, -1] into reduced echelon form using elementary row operations.

Answer: Step 1: Start with A = [0, 3, -2; 2, -4, 6; 2, 3, -1]; the (1,1) entry is 0, so swap rows: R1 <-> R2 gives [2, -4, 6; 0, 3, -2; 2, 3, -1]

Step 2: R1 -> (1/2) R1 gives [1, -2, 3; 0, 3, -2; 2, 3, -1]

Step 3: R3 -> R3 - 2R1 gives [1, -2, 3; 0, 3, -2; 0, 7, -7]

Step 4: R2 -> (1/3) R2 gives [1, -2, 3; 0, 1, -2/3; 0, 7, -7]

Step 5: R3 -> R3 - 7R2 gives [1, -2, 3; 0, 1, -2/3; 0, 0, -7/3], because -7 - 7(-2/3) = -7/3

Step 6: R3 -> (-3/7) R3 gives [1, -2, 3; 0, 1, -2/3; 0, 0, 1], which is echelon form

Step 7: Clear above the third pivot: R1 -> R1 - 3R3 gives [1, -2, 0; 0, 1, -2/3; 0, 0, 1]

Step 8: R2 -> R2 + (2/3)R3 gives [1, -2, 0; 0, 1, 0; 0, 0, 1]

Step 9: R1 -> R1 + 2R2 gives [1, 0, 0; 0, 1, 0; 0, 0, 1]

Final answer: The reduced echelon form is the identity matrix [1, 0, 0; 0, 1, 0; 0, 0, 1].

Question: Q2(iii) Reduce the matrix [1, 1, 1, -1; 2, 2, 1, -3; -1, -1, 1, -3] into

reduced echelon form using elementary row operations.

Answer: Step 1: Start with A = [1, 1, 1, -1; 2, 2, 1, -3; -1, -1, 1, -3]; R2 -> R2 - 2R1 gives [1, 1, 1, -1; 0, 0, -1, -1; -1, -1, 1, -3]

Step 2: R3 -> R3 + R1 gives [1, 1, 1, -1; 0, 0, -1, -1; 0, 0, 2, -4]; the second column has no pivot, so the second pivot is in the third column

Step 3: R2 -> (-1) R2 gives [1, 1, 1, -1; 0, 0, 1, 1; 0, 0, 2, -4]

Step 4: R3 -> R3 - 2R2 gives [1, 1, 1, -1; 0, 0, 1, 1; 0, 0, 0, -6]

Step 5: R3 -> (-1/6) R3 gives [1, 1, 1, -1; 0, 0, 1, 1; 0, 0, 0, 1], which is echelon form

Step 6: Clear above the pivots: R1 -> R1 + R3 gives [1, 1, 1, 0; 0, 0, 1, 1; 0, 0, 0, 1]

Step 7: R2 -> R2 - R3 gives [1, 1, 1, 0; 0, 0, 1, 0; 0, 0, 0, 1]

Step 8: R1 -> R1 - R2 gives [1, 1, 0, 0; 0, 0, 1, 0; 0, 0, 0, 1]; the 1 in position (1,2) stays because column 2 contains no pivot

Final answer: The reduced echelon form is [1, 1, 0, 0; 0, 0, 1, 0; 0, 0, 0, 1].

Question: Q3(i) Find the rank of the matrix [1, 2, 3; 2, 3, 4; 0, 2, 2] using elementary row operations.

Answer: Step 1: The rank is the number of non-zero rows in the echelon form, so reduce the matrix

Step 2: R2 -> R2 - 2R1 gives [1, 2, 3; 0, -1, -2; 0, 2, 2]

Step 3: R2 -> (-1) R2 gives [1, 2, 3; 0, 1, 2; 0, 2, 2]

Step 4: R3 -> R3 - 2R2 gives [1, 2, 3; 0, 1, 2; 0, 0, -2]

Step 5: R3 -> (-1/2) R3 gives [1, 2, 3; 0, 1, 2; 0, 0, 1], an echelon form with no zero row

Step 6: There are 3 non-zero rows (this agrees with det A = 2, which is not zero)

Final answer: The rank of the matrix is 3.

Question: Q3(ii) Find the rank of the matrix [2, 3, 4; 3, 1, 2; -1, 2, 2] using elementary row operations.

Answer: Step 1: Reduce to echelon form: R1 <-> R3 gives [-1, 2, 2; 3, 1, 2; 2, 3, 4]

Step 2: R1 -> (-1) R1 gives [1, -2, -2; 3, 1, 2; 2, 3, 4]

Step 3: R2 -> R2 - 3R1 gives [1, -2, -2; 0, 7, 8; 2, 3, 4]

Step 4: R3 -> R3 - 2R1 gives [1, -2, -2; 0, 7, 8; 0, 7, 8]

Step 5: R3 -> R3 - R2 gives [1, -2, -2; 0, 7, 8; 0, 0, 0], because rows 2 and 3 had become identical

Step 6: R2 -> (1/7) R2 gives the echelon form [1, -2, -2; 0, 1, 8/7; 0, 0, 0]

Step 7: There are only 2 non-zero rows (this agrees with det A = 0)

Final answer: The rank of the matrix is 2.

Question: Q3(iii) Find the rank of the matrix [0, -3, -6, 4, 9; -1, -2, -1, 3, 1; -2, -3, 0, 3, -7; 1, 4, 5, -9, -7] using elementary row operations.

Answer: Step 1: R1 <-> R4 gives [1, 4, 5, -9, -7; -1, -2, -1, 3, 1; -2, -3, 0, 3, -7; 0, -3, -6, 4, 9]

Step 2: R2 -> R2 + R1 gives [1, 4, 5, -9, -7; 0, 2, 4, -6, -6; -2, -3, 0, 3, -7; 0, -3, -6, 4, 9]

Step 3: R3 -> R3 + 2R1 gives [1, 4, 5, -9, -7; 0, 2, 4, -6, -6; 0, 5, 10, -15, -21; 0, -3, -6, 4, 9]

Step 4: R2 -> (1/2) R2 gives [1, 4, 5, -9, -7; 0, 1, 2, -3, -3; 0, 5, 10, -15, -21;

0, -3, -6, 4, 9]

Step 5: R3 -> R3 - 5R2 gives [1, 4, 5, -9, -7; 0, 1, 2, -3, -3; 0, 0, 0, 0, -6; 0, -3, -6, 4, 9]

Step 6: R4 -> R4 + 3R2 gives [1, 4, 5, -9, -7; 0, 1, 2, -3, -3; 0, 0, 0, 0, -6; 0, 0, 0, -5, 0]

Step 7: R3 <-> R4 gives [1, 4, 5, -9, -7; 0, 1, 2, -3, -3; 0, 0, 0, -5, 0; 0, 0, 0, 0, -6], putting the rows in echelon order

Step 8: R3 -> (-1/5) R3 and R4 -> (-1/6) R4 give the echelon form [1, 4, 5, -9, -7; 0, 1, 2, -3, -3; 0, 0, 0, 1, 0; 0, 0, 0, 0, 1]

Step 9: All 4 rows are non-zero

Final answer: The rank of the matrix is 4.

Question: Q4(i) Find the inverse of the matrix [1, 2, -3; 0, -2, 0; -2, -2, 2] using elementary row operations.

Answer: Step 1: Write the augmented matrix [A : I] = [1, 2, -3 : 1, 0, 0; 0, -2, 0 : 0, 1, 0; -2, -2, 2 : 0, 0, 1] and reduce the left side to I

Step 2: R3 -> R3 + 2R1 gives [1, 2, -3 : 1, 0, 0; 0, -2, 0 : 0, 1, 0; 0, 2, -4 : 2, 0, 1]

Step 3: R2 -> (-1/2) R2 gives [1, 2, -3 : 1, 0, 0; 0, 1, 0 : 0, -1/2, 0; 0, 2, -4 : 2, 0, 1]

Step 4: R3 -> R3 - 2R2 gives [1, 2, -3 : 1, 0, 0; 0, 1, 0 : 0, -1/2, 0; 0, 0, -4 : 2, 1, 1]

Step 5: R3 -> (-1/4) R3 gives [1, 2, -3 : 1, 0, 0; 0, 1, 0 : 0, -1/2, 0; 0, 0, 1 : -1/2, -1/4, -1/4]

Step 6: R1 -> R1 + 3R3 gives [1, 2, 0 : -1/2, -3/4, -3/4; 0, 1, 0 : 0, -1/2, 0;

0, 0, 1 : -1/2, -1/4, -1/4]

Step 7: R1 -> R1 - 2R2 gives [1, 0, 0 : -1/2, 1/4, -3/4; 0, 1, 0 : 0, -1/2, 0; 0, 0, 1 : -1/2, -1/4, -1/4]; the left side is now I, so the right side is A^-1

Final answer: A^-1 = [-1/2, 1/4, -3/4; 0, -1/2, 0; -1/2, -1/4, -1/4].

Question: Q4(ii) Find the inverse of the matrix [1, 2, -1; 0, -1, 3; 1, 0, 2] using elementary row operations.

Answer: Step 1: Write [A : I] = [1, 2, -1 : 1, 0, 0; 0, -1, 3 : 0, 1, 0; 1, 0, 2 : 0, 0, 1]

Step 2: R3 -> R3 - R1 gives [1, 2, -1 : 1, 0, 0; 0, -1, 3 : 0, 1, 0; 0, -2, 3 : -1, 0, 1]

Step 3: R2 -> (-1) R2 gives [1, 2, -1 : 1, 0, 0; 0, 1, -3 : 0, -1, 0; 0, -2, 3 : -1, 0, 1]

Step 4: R3 -> R3 + 2R2 gives [1, 2, -1 : 1, 0, 0; 0, 1, -3 : 0, -1, 0; 0, 0, -3 : -1, -2, 1]

Step 5: R3 -> (-1/3) R3 gives [1, 2, -1 : 1, 0, 0; 0, 1, -3 : 0, -1, 0; 0, 0, 1 : 1/3, 2/3, -1/3]

Step 6: R1 -> R1 + R3 gives [1, 2, 0 : 4/3, 2/3, -1/3; 0, 1, -3 : 0, -1, 0; 0, 0, 1 : 1/3, 2/3, -1/3]

Step 7: R2 -> R2 + 3R3 gives [1, 2, 0 : 4/3, 2/3, -1/3; 0, 1, 0 : 1, 1, -1; 0, 0, 1 : 1/3, 2/3, -1/3]

Step 8: R1 -> R1 - 2R2 gives [1, 0, 0 : -2/3, -4/3, 5/3; 0, 1, 0 : 1, 1, -1; 0, 0, 1 : 1/3, 2/3, -1/3]

Final answer: A^-1 = [-2/3, -4/3, 5/3; 1, 1, -1; 1/3, 2/3, -1/3].

Question: Q4(iii) Find the inverse of the matrix [1, -3, 2; 2, 1, 0; 0, -1, 1]

using elementary row operations.

Answer: Step 1: Write [A : I] = [1, -3, 2 : 1, 0, 0; 2, 1, 0 : 0, 1, 0; 0, -1, 1 : 0, 0, 1]

Step 2: R2 -> R2 - 2R1 gives [1, -3, 2 : 1, 0, 0; 0, 7, -4 : -2, 1, 0; 0, -1, 1 : 0, 0, 1]

Step 3: R3 -> (-1) R3 gives [1, -3, 2 : 1, 0, 0; 0, 7, -4 : -2, 1, 0; 0, 1, -1 : 0, 0, -1], and then R2 <-> R3 gives [1, -3, 2 : 1, 0, 0; 0, 1, -1 : 0, 0, -1; 0, 7, -4 : -2, 1, 0]

Step 4: R3 -> R3 - 7R2 gives [1, -3, 2 : 1, 0, 0; 0, 1, -1 : 0, 0, -1; 0, 0, 3 : -2, 1, 7]

Step 5: R3 -> (1/3) R3 gives [1, -3, 2 : 1, 0, 0; 0, 1, -1 : 0, 0, -1; 0, 0, 1 : -2/3, 1/3, 7/3]

Step 6: R1 -> R1 - 2R3 gives [1, -3, 0 : 7/3, -2/3, -14/3; 0, 1, -1 : 0, 0, -1; 0, 0, 1 : -2/3, 1/3, 7/3]

Step 7: R2 -> R2 + R3 gives [1, -3, 0 : 7/3, -2/3, -14/3; 0, 1, 0 : -2/3, 1/3, 4/3; 0, 0, 1 : -2/3, 1/3, 7/3]

Step 8: R1 -> R1 + 3R2 gives [1, 0, 0 : 1/3, 1/3, -2/3; 0, 1, 0 : -2/3, 1/3, 4/3; 0, 0, 1 : -2/3, 1/3, 7/3]

Final answer: A^-1 = [1/3, 1/3, -2/3; -2/3, 1/3, 4/3; -2/3, 1/3, 7/3].

Exercise 2.6

Question: Q1(i) Solve the homogeneous system x + 2y - 2z = 0, 2x + y + 5z = 0, 5x + 4y + 8z = 0 for non-trivial solutions, if possible.

Answer: Step 1: Write the system as AX = 0 with A = [1, 2, -2; 2, 1, 5; 5, 4, 8]; a homogeneous system has non-trivial solutions exactly when rank(A) is less than the number of unknowns, 3

Step 2: R2 -> R2 - 2R1 gives [1, 2, -2; 0, -3, 9; 5, 4, 8]

Step 3: R3 -> R3 - 5R1 gives [1, 2, -2; 0, -3, 9; 0, -6, 18]

Step 4: R3 -> R3 - 2R2 gives [1, 2, -2; 0, -3, 9; 0, 0, 0]; R2 -> (-1/3) R2 gives the echelon form [1, 2, -2; 0, 1, -3; 0, 0, 0], so rank(A) = 2 < 3 and non-trivial solutions exist

Step 5: The reduced system is x + 2y - 2z = 0 and y - 3z = 0, that is y = 3z

Step 6: Put y = 3z in the first equation: x + 6z - 2z = 0, so x = -4z

Step 7: Take z as the free variable, z = k, where k is any non-zero real number for a non-trivial solution

Final answer: The non-trivial solutions are x = -4k, y = 3k, z = k for k not equal to 0; for example k = 1 gives (x, y, z) = (-4, 3, 1).

Question: Q1(ii) Solve the homogeneous system x + 4y + 2z = 0, 2x + y - 3z = 0, 3x + 2y - 4z = 0 for non-trivial solutions, if possible.

Answer: Step 1: Write A = [1, 4, 2; 2, 1, -3; 3, 2, -4] and reduce it to echelon form

Step 2: R2 -> R2 - 2R1 gives [1, 4, 2; 0, -7, -7; 3, 2, -4]

Step 3: R3 -> R3 - 3R1 gives [1, 4, 2; 0, -7, -7; 0, -10, -10]

Step 4: R2 -> (-1/7) R2 gives [1, 4, 2; 0, 1, 1; 0, -10, -10]

Step 5: R3 -> R3 + 10R2 gives [1, 4, 2; 0, 1, 1; 0, 0, 0], so rank(A) = 2 < 3 and non-trivial solutions exist

Step 6: The reduced system is x + 4y + 2z = 0 and y + z = 0, that is y = -z

Step 7: Put y = -z in the first equation: x - 4z + 2z = 0, so x = 2z

Step 8: Take z = k as the free variable, k not equal to 0 for a non-trivial solution

Final answer: The non-trivial solutions are x = 2k, y = -k, z = k for k not equal to 0; for example k = 1 gives (x, y, z) = (2, -1, 1).

Question: Q2(i) Determine the consistency of the non-homogeneous system x - 2y - 2z = -1, 2x + 3y + z = 1, 5x - 4y - 3z = 1.

Answer: Step 1: Form the augmented matrix Ab = [1, -2, -2, -1; 2, 3, 1, 1; 5, -4, -3, 1]; the system is consistent if rank(A) = rank(Ab)

Step 2: R2 -> R2 - 2R1 gives [1, -2, -2, -1; 0, 7, 5, 3; 5, -4, -3, 1]

Step 3: R3 -> R3 - 5R1 gives [1, -2, -2, -1; 0, 7, 5, 3; 0, 6, 7, 6]

Step 4: R3 -> R3 - (6/7)R2 gives [1, -2, -2, -1; 0, 7, 5, 3; 0, 0, 19/7, 24/7], because 7 - (6/7)(5) = 19/7 and 6 - (6/7)(3) = 24/7

Step 5: The echelon form has 3 non-zero rows in both A and Ab, so rank(A) = rank(Ab) = 3, which also equals the number of unknowns

Final answer: The system is consistent, and in fact it has the unique solution x = 11/19, y = -9/19, z = 24/19.

Question: Q2(ii) Determine the consistency of the non-homogeneous system x + 2y + z = 2, 2x + y + 2z = -1, 2x + 3y - z = 9.

Answer: Step 1: Form the augmented matrix Ab = [1, 2, 1, 2; 2, 1, 2, -1; 2, 3, -1, 9]

Step 2: R2 -> R2 - 2R1 gives [1, 2, 1, 2; 0, -3, 0, -5; 2, 3, -1, 9]

Step 3: R3 -> R3 - 2R1 gives [1, 2, 1, 2; 0, -3, 0, -5; 0, -1, -3, 5]

Step 4: R3 -> R3 - (1/3)R2 gives [1, 2, 1, 2; 0, -3, 0, -5; 0, 0, -3, 20/3], because 5 - (1/3)(-5) = 20/3

Step 5: There are 3 non-zero rows in both parts, so rank(A) = rank(Ab) = 3 = number of unknowns

Final answer: The system is consistent, and in fact it has the unique solution x = 8/9, y = 5/3, z = -20/9.

Question: Q3(i) Solve the non-homogeneous system x + 2y + z = 8, 2x - y + z = 3, x + y - z = 0 using the matrix inversion method.

Answer: Step 1: Write the system as AX = B with A = [1, 2, 1; 2, -1, 1; 1, 1, -1], X = [x; y; z] and B = [8; 3; 0]

Step 2: Compute det A along the first row: the minors are (-1)(-1) - (1)(1) = 0, (2)(-1) - (1)(1) = -3, (2)(1) - (-1)(1) = 3, so det A = 1(0) - 2(-3) + 1(3) = 9, not zero, so A^-1 exists

Step 3: The cofactor matrix of A is [0, 3, 3; 3, -2, 1; 3, 1, -5], which is symmetric, so adj A is the same matrix [0, 3, 3; 3, -2, 1; 3, 1, -5]

Step 4: A^-1 = (1/9) x [0, 3, 3; 3, -2, 1; 3, 1, -5]

Step 5: Compute X = A^-1 B = (1/9) x [(0)(8) + (3)(3) + (3)(0); (3)(8) + (-2)(3) + (1)(0); (3)(8) + (1)(3) + (-5)(0)] = (1/9) x [9; 18; 27]

Final answer: x = 1, y = 2, z = 3.

Question: Q3(ii) Solve the non-homogeneous system 2x - y + 2z = 4, x + 10y - 3z = 10, -x + y + z = -6 using the matrix inversion method.

Answer: Step 1: Write A = [2, -1, 2; 1, 10, -3; -1, 1, 1] and B = [4; 10; -6]

Step 2: Compute det A: the first-row minors are (10)(1) - (-3)(1) = 13,

(1)(1) - (-3)(-1) = -2, (1)(1) - (10)(-1) = 11, so det A = 2(13) - (-1)(-2) + 2(11) = 26 - 2 + 22 = 46, not zero

Step 3: The nine cofactors are A11 = 13, A12 = 2, A13 = 11; A21 = 3, A22 = 4, A23 = -1; A31 = -17, A32 = 8, A33 = 21

Step 4: adj A is the transpose of the cofactor matrix: adj A = [13, 3, -17; 2, 4, 8; 11, -1, 21], so A^-1 = (1/46) x adj A

Step 5: Compute X = A^-1 B = (1/46) x [(13)(4) + (3)(10) + (-17)(-6); (2)(4) + (4)(10) + (8)(-6); (11)(4) + (-1)(10) + (21)(-6)] = (1/46) x [184; 0; -92]

Final answer: x = 4, y = 0, z = -2.

Question: Q4(i) Solve the non-homogeneous system -x + y + z = 0, x + 2y = 5, -3x + 2y - z = -2 using the Gauss elimination method.

Answer: Step 1: Write the augmented matrix, putting 0 for the missing z term in the second equation: Ab = [-1, 1, 1, 0; 1, 2, 0, 5; -3, 2, -1, -2]

Step 2: R1 -> (-1) R1 gives [1, -1, -1, 0; 1, 2, 0, 5; -3, 2, -1, -2]

Step 3: R2 -> R2 - R1 gives [1, -1, -1, 0; 0, 3, 1, 5; -3, 2, -1, -2]

Step 4: R3 -> R3 + 3R1 gives [1, -1, -1, 0; 0, 3, 1, 5; 0, -1, -4, -2]

Step 5: R3 -> R3 + (1/3)R2 gives [1, -1, -1, 0; 0, 3, 1, 5; 0, 0, -11/3, -1/3], because -4 + (1/3)(1) = -11/3 and -2 + (1/3)(5) = -1/3; this is echelon form

Step 6: Back-substitute from the third row: (-11/3) z = -1/3, so z = 1/11

Step 7: From the second row: 3y + z = 5, so 3y = 5 - 1/11 = 54/11 and y = 18/11

Step 8: From the first row: x - y - z = 0, so x = y + z = 18/11 + 1/11 = 19/11

Step 9: Check in the third original equation: -3(19/11) + 2(18/11) - 1/11 = (-57 + 36 - 1)/11 = -2, which is correct

Final answer: x = 19/11, y = 18/11, z = 1/11.

Question: Q4(ii) Solve the non-homogeneous system x + 2y + z = 8, 2x - y + z = 3, x + y - z = 0 using the Gauss elimination method.

Answer: Step 1: Write the augmented matrix Ab = [1, 2, 1, 8; 2, -1, 1, 3; 1, 1, -1, 0]

Step 2: R2 -> R2 - 2R1 gives [1, 2, 1, 8; 0, -5, -1, -13; 1, 1, -1, 0]

Step 3: R3 -> R3 - R1 gives [1, 2, 1, 8; 0, -5, -1, -13; 0, -1, -2, -8]

Step 4: R2 <-> R3 gives [1, 2, 1, 8; 0, -1, -2, -8; 0, -5, -1, -13] (swapping makes the arithmetic easier and does not affect the solution)

Step 5: R3 -> R3 - 5R2 gives [1, 2, 1, 8; 0, -1, -2, -8; 0, 0, 9, 27], which is echelon form

Step 6: Back-substitute: from the third row, 9z = 27, so z = 3

Step 7: From the second row, -y - 2z = -8, so -y = -8 + 6 = -2 and y = 2

Step 8: From the first row, x + 2(2) + 3 = 8, so x = 1

Final answer: x = 1, y = 2, z = 3.

Question: Q5(i) Solve the non-homogeneous system x - y + 4z = 4, 2x + 2y - z = 2, 3x - 2y + 3z = -3 using the Gauss-Jordan method.

Answer: Step 1: Write the augmented matrix Ab = [1, -1, 4, 4; 2, 2, -1, 2; 3, -2, 3, -3]; in the Gauss-Jordan method we reduce it all the way to reduced echelon form

Step 2: R2 -> R2 - 2R1 gives [1, -1, 4, 4; 0, 4, -9, -6; 3, -2, 3, -3]

Step 3: R3 -> R3 - 3R1 gives [1, -1, 4, 4; 0, 4, -9, -6; 0, 1, -9, -15]

Step 4: R2 <-> R3 gives [1, -1, 4, 4; 0, 1, -9, -15; 0, 4, -9, -6]

Step 5: R3 -> R3 - 4R2 gives [1, -1, 4, 4; 0, 1, -9, -15; 0, 0, 27, 54]

Step 6: R3 -> (1/27) R3 gives [1, -1, 4, 4; 0, 1, -9, -15; 0, 0, 1, 2]

Step 7: R1 -> R1 - 4R3 gives [1, -1, 0, -4; 0, 1, -9, -15; 0, 0, 1, 2]

Step 8: R2 -> R2 + 9R3 gives [1, -1, 0, -4; 0, 1, 0, 3; 0, 0, 1, 2]

Step 9: R1 -> R1 + R2 gives [1, 0, 0, -1; 0, 1, 0, 3; 0, 0, 1, 2]; the left side is I, so the solution can be read off directly

Final answer: x = -1, y = 3, z = 2.

Question: Q5(ii) Solve the non-homogeneous system 2x + 2y - z = 4, x - 2y + z = 2, x + y = 0 using the Gauss-Jordan method.

Answer: Step 1: Write the augmented matrix, with 0 for the missing z in the third equation: Ab = [2, 2, -1, 4; 1, -2, 1, 2; 1, 1, 0, 0]

Step 2: R1 <-> R2 gives [1, -2, 1, 2; 2, 2, -1, 4; 1, 1, 0, 0]

Step 3: R2 -> R2 - 2R1 gives [1, -2, 1, 2; 0, 6, -3, 0; 1, 1, 0, 0]

Step 4: R3 -> R3 - R1 gives [1, -2, 1, 2; 0, 6, -3, 0; 0, 3, -1, -2]

Step 5: R2 -> (1/6) R2 gives [1, -2, 1, 2; 0, 1, -1/2, 0; 0, 3, -1, -2]

Step 6: R3 -> R3 - 3R2 gives [1, -2, 1, 2; 0, 1, -1/2, 0; 0, 0, 1/2, -2]

Step 7: R3 -> 2R3 gives [1, -2, 1, 2; 0, 1, -1/2, 0; 0, 0, 1, -4]

Step 8: R1 -> R1 - R3 gives [1, -2, 0, 6; 0, 1, -1/2, 0; 0, 0, 1, -4]

Step 9: R2 -> R2 + (1/2)R3 gives [1, -2, 0, 6; 0, 1, 0, -2; 0, 0, 1, -4]

Step 10: R1 -> R1 + 2R2 gives [1, 0, 0, 2; 0, 1, 0, -2; 0, 0, 1, -4]

Final answer: x = 2, y = -2, z = -4.

Question: Q6(i) Solve the non-homogeneous system x - 2y + z = 2, 2x + 2y - z = 4, x + y = 0 using Cramer's Rule.

Answer: Step 1: The coefficient matrix is A = [1, -2, 1; 2, 2, -1; 1, 1, 0] and B = [2; 4; 0]; compute |A| by expanding along the first row: |A| = 1[(2)(0) - (-1)(1)] - (-2)[(2)(0) - (-1)(1)] + 1[(2)(1) - (2)(1)] = 1(1) + 2(1) + 1(0) = 3, not zero, so Cramer's rule applies

Step 2: Replace the first column by B: |A1| = |2, -2, 1; 4, 2, -1; 0, 1, 0| = 2[(2)(0) - (-1)(1)] - (-2)[(4)(0) - (-1)(0)] + 1[(4)(1) - (2)(0)] = 2(1) + 2(0) + 4 = 6

Step 3: Replace the second column by B: |A2| = |1, 2, 1; 2, 4, -1; 1, 0, 0|; expanding along the third row gives 1 x [(2)(-1) - (1)(4)] = -6 (the other two terms contain the factor 0)

Step 4: Replace the third column by B: |A3| = |1, -2, 2; 2, 2, 4; 1, 1, 0| = 1[(2)(0) - (4)(1)] - (-2)[(2)(0) - (4)(1)] + 2[(2)(1) - (2)(1)] = -4 - 8 + 0 = -12

Step 5: Apply Cramer's rule: x = |A1| / |A| = 6/3 = 2, y = |A2| / |A| = -6/3 = -2, z = |A3| / |A| = -12/3 = -4

Final answer: x = 2, y = -2, z = -4.

Question: Q6(ii) Solve the non-homogeneous system x - 2y + 0.z = -4, 3x + y + 0.z = -5, 2x + 0.y + z = -1 using Cramer's Rule.

Answer: Step 1: The coefficient matrix is A = [1, -2, 0; 3, 1, 0; 2, 0, 1] and B = [-4; -5; -1]; expand |A| along the third column: |A| = 1 x [(1)(1) - (-2)(3)] = 1 + 6 = 7, not zero, so Cramer's rule applies

Step 2: |A1| = |-4, -2, 0; -5, 1, 0; -1, 0, 1|; expanding along the third column: |A1| = 1 x [(-4)(1) - (-2)(-5)] = -4 - 10 = -14

Step 3: |A2| = |1, -4, 0; 3, -5, 0; 2, -1, 1|; expanding along the third

column: |A2| = 1 x [(1)(-5) - (-4)(3)] = -5 + 12 = 7

Step 4: |A3| = |1, -2, -4; 3, 1, -5; 2, 0, -1| = 1[(1)(-1) - (-5)(0)] - (-2)[(3)(-1) - (-5)(2)] + (-4)[(3)(0) - (1)(2)] = 1(-1) + 2(7) - 4(-2) = -1 + 14 + 8 = 21

Step 5: Apply Cramer's rule: x = -14/7 = -2, y = 7/7 = 1, z = 21/7 = 3

Final answer: x = -2, y = 1, z = 3.

Review Exercise 2

Question: MCQ (i) If a matrix A has m rows and n columns, then the order of A is: (a) m x n (b) n x m (c) mn (d) m^n.

Answer: Step 1: The order of a matrix is written as (number of rows) x (number of columns)

Step 2: Here the number of rows is m and the number of columns is n

Final answer: (a) m x n.

Question: MCQ (ii) Any matrix of order m x 1 is called: (a) Row matrix (b) Column matrix (c) Square matrix (d) Zero matrix.

Answer: Step 1: An order of m x 1 means m rows but only 1 column

Step 2: A matrix with a single column is by definition a column matrix

Final answer: (b) Column matrix.

Question: MCQ (iii) For the square matrix A = [aij], if all aij = 0 for i not equal to j and all aij = k (non-zero) for i = j, then A is called: (a) Diagonal matrix (b) Scalar matrix (c) Identity matrix (d) Null matrix.

Answer: Step 1: All off-diagonal elements are zero, so A is diagonal

Step 2: All diagonal elements are equal to the same non-zero scalar k,

which is the extra condition that makes a diagonal matrix a scalar matrix; it would be the identity matrix only in the special case k = 1

Final answer: (b) Scalar matrix.

Question: MCQ (iv) The matrix [7] is: (a) Square matrix (b) Row matrix (c) Column matrix (d) All of these.

Answer: Step 1: The matrix [7] has 1 row and 1 column

Step 2: Equal numbers of rows and columns make it a square matrix; a single row makes it a row matrix; a single column makes it a column matrix, so all three descriptions apply at once

Final answer: (d) All of these.

Question: MCQ (v) (k A B C)^t, where k is a non-zero scalar, equals: (a) k A^t B^t C^t (b) k C^t B^t A^t (c) (1/k) A^t B^t C^t (d) (1/k) C^t B^t A^t.

Answer: Step 1: The transpose of a product reverses the order of the factors: (ABC)^t = C^t B^t A^t

Step 2: A scalar factor is unchanged by transposition, so the k stays as k in front

Final answer: (b) k C^t B^t A^t.

Question: MCQ (vi) |1, 0; 5, 6| - |1, 2; 3, 4| = _____: (a) 4 (b) 8 (c) -2 (d) 10.

Answer: Step 1: Evaluate the first determinant: (1)(6) - (0)(5) = 6

Step 2: Evaluate the second determinant: (1)(4) - (2)(3) = 4 - 6 = -2

Step 3: Subtract: 6 - (-2) = 8

Final answer: (b) 8.

Question: MCQ (vii) If AB = BA, then which one is true, where A and B are square matrices: (a) A and B are multiplicative inverses of each other (b) One of A or B is null matrix (c) One of A or B is identity matrix (d) All of these.

Answer: Step 1: Test each statement as a condition that guarantees AB = BA: if B = A^-1 then AB = BA = I, so (a) gives AB = BA

Step 2: If one matrix is the null matrix then both products are the null matrix, so AB = BA, and (b) also gives AB = BA

Step 3: If one matrix is the identity matrix then AB = BA = the other matrix, so (c) also gives AB = BA

Step 4: Since each of the three listed situations makes AB = BA true, the combined option is the intended answer (note AB = BA can also hold in other cases, for example when A = B)

Final answer: (d) All of these.

Question: MCQ (viii) If A = [-7], then |A| =: (a) 7 (b) -7 (c) 0 (d) Not possible.

Answer: Step 1: The determinant of a 1 x 1 matrix is the element itself

Step 2: So |A| = |-7| as a determinant = -7 (this is a determinant, not an absolute value)

Final answer: (b) -7.

Question: MCQ (ix) Let A = [aij] be a square matrix. Then the cofactor of aij is equal to: (a) Mij (b) (-1)^(i+j) Mij (c) (-1)^(ij) Mij (d) (-1)^(i+j) aij.

Answer: Step 1: By definition the cofactor is the minor with the sign factor (-1)^(i+j) attached

Step 2: That is, Aij = (-1)^(i+j) Mij, where Mij is the minor of aij

Final answer: (b) (-1)^(i+j) Mij.

Question: MCQ (x) For any triangular matrix A, |A| is equal to: (a) Product of leading diagonal elements (b) Sum of leading diagonal elements (c) Sum of squares of diagonal elements (d) All of these.

Answer: Step 1: Expanding a triangular determinant step by step leaves only the product of the main diagonal entries; all other terms contain a zero factor

Step 2: This is the standard note given with the determinant properties in this unit

Final answer: (a) Product of leading diagonal elements.

Question: MCQ (xi) A square matrix A = [aij] for which all aij = 0 when i < j is called: (a) Upper triangular (b) Lower triangular (c) Symmetric (d) Hermitian.

Answer: Step 1: The condition i < j describes the positions above the main diagonal

Step 2: If all elements above the main diagonal are zero, the non-zero part lies on and below the diagonal, which is a lower triangular matrix

Final answer: (b) Lower triangular.

Question: MCQ (xii) A triangular matrix is always a: (a) Diagonal matrix (b) Scalar matrix (c) Square matrix (d) All of these.

Answer: Step 1: Triangular (upper or lower) is defined only when the numbers of rows and columns are equal, because it refers to elements above or below the main diagonal

Step 2: A triangular matrix need not be diagonal (its off-diagonal entries on one side may be non-zero) and need not be scalar

Final answer: (c) Square matrix.

Question: MCQ (xiii) A square matrix A is skew-symmetric if: (a) A^t = A (b) A^t = -A (c) (Abar)^t = A (d) None.

Answer: Step 1: By definition a square matrix is skew-symmetric when its transpose equals its own negative

Step 2: Option (a) is the symmetric condition and option (c) is the Hermitian condition

Final answer: (b) A^t = -A.

Question: MCQ (xiv) A square matrix A is a Hermitian matrix if: (a) A^t = A (b) A^t = -A (c) (Abar)^t = A (d) (Abar)^t = -A.

Answer: Step 1: By definition a Hermitian matrix equals its own conjugate transpose

Step 2: Option (d) is the skew-Hermitian condition

Final answer: (c) (Abar)^t = A.

Question: MCQ (xv) In a skew-Hermitian matrix, elements of the main diagonal are _____ numbers: (a) Purely real (b) Complex (c) Either zero or purely imaginary (d) None.

Answer: Step 1: For a skew-Hermitian matrix each diagonal element a satisfies conjugate of a = -a

Step 2: A number equal to the negative of its own conjugate has zero real part, so it is either 0 or purely imaginary

Final answer: (c) Either zero or purely imaginary.

Question: MCQ (xvi) If |a, b; 0, 7| = |2, 3; 1, -9|, then: (a) a = -3 (b) a = b (c) a = 1/3 (d) a = -1/3.

Answer: Step 1: Evaluate the left determinant: (a)(7) - (b)(0) = 7a

Step 2: Evaluate the right determinant: (2)(-9) - (3)(1) = -18 - 3 = -21

Step 3: Set them equal: 7a = -21, so a = -3

Final answer: (a) a = -3.

Question: MCQ (xvii) The number of non-zero rows in the echelon form of a matrix is called: (a) Order of a matrix (b) Rank of a matrix (c) Leading column (d) Leading row.

Answer: Step 1: By definition, after reducing a matrix to echelon form, the count of its non-zero rows is a named quantity

Step 2: That quantity is the rank of the matrix

Final answer: (b) Rank of a matrix.

Question: MCQ (xviii) If A is any square matrix and A = -A^t, then A is a: (a) Symmetric matrix (b) Skew-symmetric matrix (c) Hermitian matrix (d) Skew-Hermitian matrix.

Answer: Step 1: Rewrite the condition as A^t = -A

Step 2: This is exactly the definition of a skew-symmetric matrix

Final answer: (b) Skew-symmetric matrix.

Question: MCQ (xix) If A is an idempotent matrix, then: (a) A^2 = I (b) A^2 = 0 (c) A^2 = A (d) A^2 = A^t.

Answer: Step 1: By definition a square matrix is idempotent when squaring it gives the matrix itself

Final answer: (c) A^2 = A.

Question: MCQ (xx) The cofactor A22 of the matrix [1, 2, 4; -1, 2, 5; 0, 1, -1] is: (a) 0 (b) -1 (c) 1 (d) 2.

Answer: Step 1: Delete row 2 and column 2 to get the minor: M22 = |1, 4;

0, -1| = (1)(-1) - (4)(0) = -1

Step 2: Apply the sign: A22 = (-1)^(2+2) M22 = (+1)(-1) = -1

Final answer: (b) -1.

Question: Q2 Find the values of x and y if [x, 5; 7, y - 3] + [3, -4; 1, 2] = [7, 1; 8, 14].

Answer: Step 1: Add the left side entry by entry: [x + 3, 5 - 4; 7 + 1, (y - 3) + 2] = [x + 3, 1; 8, y - 1]

Step 2: Equate with the right side [7, 1; 8, 14]: the (1,2) and (2,1) entries agree automatically (1 = 1 and 8 = 8), which confirms the set-up

Step 3: From the (1,1) entries: x + 3 = 7, so x = 4

Step 4: From the (2,2) entries: y - 1 = 14, so y = 15

Final answer: x = 4, y = 15.

Question: Q3(a) For A = [0, 6, 7; -6, 0, 8; 7, -8, 0], B = [0, 1, 1; 1, 0, 2; 1, 2, 0] and C = [2; -2; 3], calculate AC.

Answer: Step 1: A is 3 x 3 and C is 3 x 1, so AC exists and is 3 x 1

Step 2: First entry: (0)(2) + (6)(-2) + (7)(3) = 0 - 12 + 21 = 9

Step 3: Second entry: (-6)(2) + (0)(-2) + (8)(3) = -12 + 0 + 24 = 12

Step 4: Third entry: (7)(2) + (-8)(-2) + (0)(3) = 14 + 16 + 0 = 30

Final answer: AC = [9; 12; 30].

Question: Q3(b) For A, B, C as in Q3, calculate BC.

Answer: Step 1: B is 3 x 3 and C is 3 x 1, so BC exists and is 3 x 1

Step 2: First entry: (0)(2) + (1)(-2) + (1)(3) = 0 - 2 + 3 = 1

Step 3: Second entry: (1)(2) + (0)(-2) + (2)(3) = 2 + 0 + 6 = 8

Step 4: Third entry: (1)(2) + (2)(-2) + (0)(3) = 2 - 4 + 0 = -2

Final answer: BC = [1; 8; -2].

Question: Q3(c) For A, B, C as in Q3, calculate (A + B)C and verify that (A + B)C = AC + BC.

Answer: Step 1: Add A and B entry by entry: A + B = [0, 7, 8; -5, 0, 10; 8, -6, 0]

Step 2: Multiply by C: first entry (0)(2) + (7)(-2) + (8)(3) = 0 - 14 + 24 = 10

Step 3: Second entry: (-5)(2) + (0)(-2) + (10)(3) = -10 + 0 + 30 = 20

Step 4: Third entry: (8)(2) + (-6)(-2) + (0)(3) = 16 + 12 + 0 = 28, so (A + B)C = [10; 20; 28]

Step 5: Add the results of Q3(a) and Q3(b): AC + BC = [9 + 1; 12 + 8; 30 - 2] = [10; 20; 28]

Step 6: The two sides are identical

Final answer: (A + B)C = [10; 20; 28] = AC + BC, hence verified (this is the distributive law of matrix multiplication over addition).

Question: Q4 If A = [3, 1; -1, 2], show that A^2 - 5A + 7I2 = 0.

Answer: Step 1: Compute A^2 = [3, 1; -1, 2] x [3, 1; -1, 2]: the entries are (9 - 1), (3 + 2), (-3 - 2), (-1 + 4) = 8, 5, -5, 3, so A^2 = [8, 5; -5, 3]

Step 2: Compute 5A = [15, 5; -5, 10] and 7I2 = [7, 0; 0, 7]

Step 3: Subtract and add entry by entry: (8 - 15 + 7), (5 - 5 + 0), (-5 + 5 + 0), (3 - 10 + 7) = 0, 0, 0, 0

Final answer: A^2 - 5A + 7I2 = [0, 0; 0, 0] = 0, hence shown.

Question: Q5 Find the degree or index of the nilpotent matrix [1, -3, -4; -1,

3, 4; 1, -3, -4].

Answer: Step 1: Let A = [1, -3, -4; -1, 3, 4; 1, -3, -4]; A is not zero, so the index is at least 2; note that row 3 equals row 1 and row 2 is the negative of row 1

Step 2: Compute A^2: the (1,1) entry is (1)(1) + (-3)(-1) + (-4)(1) = 1 + 3 - 4 = 0

Step 3: The (1,2) entry is (1)(-3) + (-3)(3) + (-4)(-3) = -3 - 9 + 12 = 0, and the (1,3) entry is (1)(-4) + (-3)(4) + (-4)(-4) = -4 - 12 + 16 = 0

Step 4: Every other entry of A^2 works out to 0 in the same way, because each row of A is a multiple of the first row, so A^2 = [0, 0, 0; 0, 0, 0; 0, 0, 0] = 0

Final answer: A is nilpotent of index (degree) 2.

Question: Q6 Show that the matrix [1, -2, -6; -3, 2, 9; 2, 0, -3] is a periodic matrix of period 2.

Answer: Step 1: A matrix is periodic of period 2 when A^3 = A but A^2 is not equal to A; let A = [1, -2, -6; -3, 2, 9; 2, 0, -3]

Step 2: Compute A^2: row 1 is (1 + 6 - 12), (-2 - 4 + 0), (-6 - 18 + 18) = -5, -6, -6; row 2 is (-3 - 6 + 18), (6 + 4 + 0), (18 + 18 - 27) = 9, 10, 9; row 3 is (2 + 0 - 6), (-4 + 0 + 0), (-12 + 0 + 9) = -4, -4, -3, so A^2 = [-5, -6, -6; 9, 10, 9; -4, -4, -3], which is not equal to A

Step 3: Compute A^3 = A^2 x A: row 1 is (-5 + 18 - 12), (10 - 12 + 0), (30 - 54 + 18) = 1, -2, -6

Step 4: Row 2 of A^3 is (-9 - 30 + 18), (18 + 20 + 0), (-54 + 90 - 27) = -3, 2, 9, and row 3 is (-4 + 12 - 6), (8 - 8 + 0), (24 - 36 + 9) = 2, 0, -3

Step 5: So A^3 = [1, -2, -6; -3, 2, 9; 2, 0, -3] = A, that is A^(2+1) = A with

k = 2

Final answer: A is a periodic matrix of period 2, hence shown.

Question: Q7(i) State whether the matrix [1, -1, 0; 0, -1, 0; 0, 0, 1] is idempotent or involutory.

Answer: Step 1: Compute A^2 and compare it with A and with I

Step 2: Row 1 of A^2: (1)(1) + (-1)(0) + (0)(0) = 1, (1)(-1) + (-1)(-1) + (0)(0) = -1 + 1 = 0, (1)(0) + (-1)(0) + (0)(1) = 0

Step 3: Row 2 of A^2: 0, (-1)(-1) = 1, 0; row 3 of A^2: 0, 0, 1

Step 4: So A^2 = [1, 0, 0; 0, 1, 0; 0, 0, 1] = I, while A^2 is not equal to A

Final answer: The matrix is involutory; it is not idempotent.

Question: Q7(ii) State whether the matrix [0, 1, -1; 4, -3, 4; 3, -3, 4] is idempotent or involutory.

Answer: Step 1: Compute A^2 and compare it with A and with I

Step 2: Row 1 of A^2: (0 + 4 - 3), (0 - 3 + 3), (0 + 4 - 4) = 1, 0, 0

Step 3: Row 2 of A^2: (0 - 12 + 12), (4 + 9 - 12), (-4 - 12 + 16) = 0, 1, 0

Step 4: Row 3 of A^2: (0 - 12 + 12), (3 + 9 - 12), (-3 - 12 + 16) = 0, 0, 1

Step 5: So A^2 = I, while A^2 is not equal to A

Final answer: The matrix is involutory; it is not idempotent.

Question: Q8(i) State whether the matrix [1, 1 - i, 2; -1 - i, 3i, i; -2, i, 0] is Hermitian or skew-Hermitian.

Answer: Step 1: Write the conjugate: Abar = [1, 1 + i, 2; -1 + i, -3i, -i; -2, -i, 0]

Step 2: Transpose the conjugate: (Abar)^t = [1, -1 + i, -2; 1 + i, -3i, -i; 2,

-i, 0]

Step 3: Compare with A: the (1,3) entries differ (A has 2, the conjugate transpose has -2), so (Abar)^t is not equal to A and the matrix is not Hermitian

Step 4: Compare with -A = [-1, -1 + i, -2; 1 + i, -3i, -i; 2, -i, 0]: everything matches except the (1,1) entry, where the conjugate transpose has 1 but -A has -1; this is because the diagonal entry 1 is a non-zero real number, whereas a skew-Hermitian matrix must have zero or purely imaginary diagonal entries

Final answer: The matrix is neither Hermitian nor skew-Hermitian.

Question: Q8(ii) State whether the matrix [4, 1 - i, 7; 1 + i, 6, -i; 7, i, 5] is Hermitian or skew-Hermitian.

Answer: Step 1: Write the conjugate: Abar = [4, 1 + i, 7; 1 - i, 6, i; 7, -i, 5]

Step 2: Transpose the conjugate: (Abar)^t = [4, 1 - i, 7; 1 + i, 6, -i; 7, i, 5]

Step 3: This is exactly the original matrix A; the diagonal entries 4, 6, 5 are real, and the symmetric pairs are conjugates of each other, for example a12 = 1 - i with a21 = 1 + i, and a23 = -i with a32 = i

Final answer: The matrix is Hermitian; it is not skew-Hermitian.

Question: Q9(a) Let A be a square matrix. Show that A + (Abar)^t is Hermitian.

Answer: Step 1: Let B = A + (Abar)^t; to prove B is Hermitian we must show (Bbar)^t = B

Step 2: Use the rules (X + Y)bar = Xbar + Ybar and ((Xbar)^t)bar = X^t, and also (X^t)bar = (Xbar)^t: Bbar = Abar + ((Abar)^t)bar = Abar + A^t

Step 3: Transpose the result: (Bbar)^t = (Abar)^t + (A^t)^t = (Abar)^t +

A

Step 4: This is exactly B, since addition is commutative: (Bbar)^t = A + (Abar)^t = B

Final answer: (Bbar)^t = B, so A + (Abar)^t is Hermitian, hence shown.

Question: Q9(b) Let A be a square matrix. Show that A - (Abar)^t is skew-Hermitian.

Answer: Step 1: Let C = A - (Abar)^t; to prove C is skew-Hermitian we must show (Cbar)^t = -C

Step 2: Take the conjugate: Cbar = Abar - ((Abar)^t)bar = Abar - A^t, using ((Abar)^t)bar = A^t

Step 3: Transpose: (Cbar)^t = (Abar)^t - (A^t)^t = (Abar)^t - A

Step 4: Factor out the minus sign: (Abar)^t - A = -(A - (Abar)^t) = -C

Final answer: (Cbar)^t = -C, so A - (Abar)^t is skew-Hermitian, hence shown.

Question: Q10 Show that the determinant |a, b, c; a + 2x, b + 2y, c + 2z; x, y, z| = 0.

Answer: Step 1: Split the second row using Property 6 (each of its entries is a sum): the determinant equals |a, b, c; a, b, c; x, y, z| + |a, b, c; 2x, 2y, 2z; x, y, z|

Step 2: In the first of these, rows 1 and 2 are identical, so by Property 3 it is 0

Step 3: In the second, take the factor 2 out of the second row (Property 5): it becomes 2 x |a, b, c; x, y, z; x, y, z|, and now rows 2 and 3 are identical, so by Property 3 it is also 0

Step 4: Add the two parts: 0 + 0 = 0

Final answer: The determinant is 0, hence shown.

Question: Q11 Show that the determinant |x + y + 2z, x, y; z, y + z + 2x, y; z, x, z + x + 2y| = 2(x + y + z)^3.

Answer: Step 1: Let s = x + y + z; apply the column operation C1 -> C1 + C2 + C3, which does not change the determinant

Step 2: The new first column entries are: row 1: (x + y + 2z) + x + y = 2x + 2y + 2z = 2s; row 2: z + (y + z + 2x) + y = 2s; row 3: z + x + (z + x + 2y) = 2s

Step 3: Take the common factor 2s out of the first column: the determinant is 2s x |1, x, y; 1, y + z + 2x, y; 1, x, z + x + 2y|

Step 4: Apply R2 -> R2 - R1: the second row becomes (0), (y + z + 2x - x), (y - y) = 0, s, 0

Step 5: Apply R3 -> R3 - R1: the third row becomes (0), (x - x), (z + x + 2y - y) = 0, 0, s

Step 6: The determinant is now upper triangular in effect: |1, x, y; 0, s, 0; 0, 0, s|, whose value is the product of the diagonal entries 1 x s x s = s^2

Step 7: Multiply by the factor taken out in Step 3: total = 2s x s^2 = 2 s^3

Final answer: The determinant is 2(x + y + z)^3, hence shown.

Question: Q12 Solve the system 2x + 4y - z = 0, x - 2y - 2z = 2, -5x - 8y + 3z = -2 by Cramer's rule, Gauss elimination, Gauss-Jordan and matrix inversion methods.

Answer: Step 1: Write A = [2, 4, -1; 1, -2, -2; -5, -8, 3] and B = [0; 2; -2]; first compute |A| along the first row: the minors are (-2)(3) - (-2)(-8) = -22, (1)(3) - (-2)(-5) = -7, (1)(-8) - (-2)(-5) = -18, so |A| = 2(-22) - 4(-7) +

(-1)(-18) = -44 + 28 + 18 = 2, not zero, so the system has a unique solution and all four methods apply

Step 2 (Cramer's rule): |A1| = |0, 4, -1; 2, -2, -2; -2, -8, 3| = 0 - 4[(2)(3) - (-2)(-2)] + (-1)[(2)(-8) - (-2)(-2)] = -4(2) - 1(-20) = 12, so x = 12/2 = 6

Step 3 (Cramer's rule): |A2| = |2, 0, -1; 1, 2, -2; -5, -2, 3| = 2[(2)(3) - (-2)(-2)] - 0 + (-1)[(1)(-2) - (2)(-5)] = 2(2) - 1(8) = -4, so y = -4/2 = -2

Step 4 (Cramer's rule): |A3| = |2, 4, 0; 1, -2, 2; -5, -8, -2| = 2[(-2)(-2) - (2)(-8)] - 4[(1)(-2) - (2)(-5)] + 0 = 2(20) - 4(8) = 8, so z = 8/2 = 4

Step 5 (Gauss elimination): the augmented matrix [2, 4, -1, 0; 1, -2, -2, 2; -5, -8, 3, -2] under R1 <-> R2, then R2 -> R2 - 2R1 and R3 -> R3 + 5R1 becomes [1, -2, -2, 2; 0, 8, 3, -4; 0, -18, -7, 8]; then R3 -> R3 + (9/4)R2 becomes [1, -2, -2, 2; 0, 8, 3, -4; 0, 0, -1/4, -1], giving z = 4 by back-substitution, then 8y + 12 = -4 gives y = -2, then x + 4 - 8 = 2 gives x = 6

Step 6 (Gauss-Jordan): continue from the echelon form: R3 -> -4R3 gives [1, -2, -2, 2; 0, 8, 3, -4; 0, 0, 1, 4]; R2 -> R2 - 3R3 and R1 -> R1 + 2R3 give [1, -2, 0, 10; 0, 8, 0, -16; 0, 0, 1, 4]; R2 -> (1/8)R2 gives [1, -2, 0, 10; 0, 1, 0, -2; 0, 0, 1, 4]; R1 -> R1 + 2R2 gives [1, 0, 0, 6; 0, 1, 0, -2; 0, 0, 1, 4], so x = 6, y = -2, z = 4 directly

Step 7 (Matrix inversion): the cofactor matrix of A is [-22, 7, -18; -4, 1, -4; -10, 3, -8], so adj A = [-22, -4, -10; 7, 1, 3; -18, -4, -8] and A^-1 = (1/2) x adj A

Step 8: X = A^-1 B = (1/2) x [(-22)(0) + (-4)(2) + (-10)(-2); (7)(0) + (1)(2) + (3)(-2); (-18)(0) + (-4)(2) + (-8)(-2)] = (1/2) x [12; -4; 8] = [6; -2; 4]

Step 9: All four methods agree, and the check 2(6) + 4(-2) - 4 = 0, 6 -

2(-2) - 2(4) = 2, -5(6) - 8(-2) + 3(4) = -2 confirms the answer

Final answer: By all four methods, x = 6, y = -2, z = 4.

Unit 3: Vectors

Exercise 3.1

Question: Exercise 3.1 Q1(i): Classify the quantity: Force - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector needs magnitude and direction

Step 2: Force has a size in newtons, for example a push of 10 N

Step 3: A push of 10 N is incomplete until we also say in which direction it acts, for example 10 N to the east

Step 4: Since both magnitude and direction are needed to specify force completely, it is a vector quantity

Final answer: Force is a vector.

Question: Exercise 3.1 Q1(ii): Classify the quantity: Speed - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector needs magnitude and direction

Step 2: Speed is stated as a single number with a unit, for example 60 km/h

Step 3: No direction is needed to say how fast an object is moving; 60 km/h fully describes its speed

Step 4: Since only magnitude is required, speed is not a vector

Final answer: Speed is a scalar.

Question: Exercise 3.1 Q1(iii): Classify the quantity: Velocity - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector needs magnitude and direction

Step 2: Velocity tells how fast an object moves, which is its magnitude

Step 3: Velocity also tells the direction of motion, for example 60 km/h towards the north

Step 4: Changing the direction while keeping the speed same changes the velocity, so direction is an essential part of it

Final answer: Velocity is a vector.

Question: Exercise 3.1 Q1(iv): Classify the quantity: Area - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector needs magnitude and direction

Step 2: Area is measured as a number of square units, for example 24 m^2

Step 3: The statement 24 m^2 is complete by itself and no direction can be attached to it

Step 4: Since only magnitude is required, area is not a vector

Final answer: Area is a scalar.

Question: Exercise 3.1 Q1(v): Classify the quantity: Acceleration - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector

needs magnitude and direction

Step 2: Acceleration is the rate of change of velocity, for example 5 m/s^2

Step 3: Velocity itself has direction, so its change also has a direction; an acceleration of 5 m/s^2 downwards is different from 5 m/s^2 upwards

Step 4: Since magnitude and direction are both needed, acceleration is a vector

Final answer: Acceleration is a vector.

Question: Exercise 3.1 Q1(vi): Classify the quantity: Work - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector needs magnitude and direction

Step 2: Work is measured in joules as a single number, for example 200 J of work is done

Step 3: Although work is calculated from force and displacement, the result itself carries no direction; it can only be positive, negative or zero

Step 4: Since only magnitude (with sign) is required, work is not a vector

Final answer: Work is a scalar.

Question: Exercise 3.1 Q1(vii): Classify the quantity: Volume - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector needs magnitude and direction

Step 2: Volume is stated as a number of cubic units, for example 5 litres or 125 cm^3

Step 3: This number completely describes the space occupied and no

direction is involved

Step 4: Since only magnitude is required, volume is not a vector

Final answer: Volume is a scalar.

Question: Exercise 3.1 Q1(viii): Classify the quantity: Weight - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector needs magnitude and direction

Step 2: Weight is the force of gravity on a body, measured in newtons

Step 3: This force always acts in a definite direction, towards the centre of the earth, that is vertically downwards

Step 4: Since weight is a force with magnitude and direction, it is a vector

Final answer: Weight is a vector.

Question: Exercise 3.1 Q1(ix): Classify the quantity: Mass - scalar or vector?

Answer: Step 1: Recall the test: a scalar needs only magnitude, a vector needs magnitude and direction

Step 2: Mass is stated as a single number with a unit, for example 10 kg

Step 3: The value 10 kg completely describes the mass of the body and no direction can be attached to it

Step 4: Since only magnitude is required, mass is not a vector

Final answer: Mass is a scalar.

Question: Exercise 3.1 Q2(i): Write the vector PQ in the form xi + yj, where P(4, 6) and Q(-4, 5).

Answer: Step 1: Use the rule PQ = (position vector of Q) - (position vector

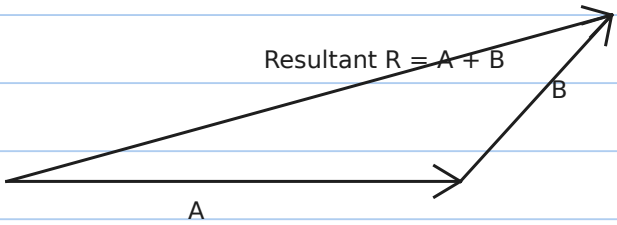
of P), that is subtract the coordinates of the initial point from the coordinates of the terminal point

Step 2: The x-component is x = -4 - 4 = -8

Step 3: The y-component is y = 5 - 6 = -1

Step 4: Write the components with the unit vectors i and j: PQ = -8i - 1j

Final answer: PQ = -8i - j.



Question: Exercise 3.1 Q2(ii): Write the vector PQ in the form xi + yj, where P(-5/2, 9) and Q(10, 8).

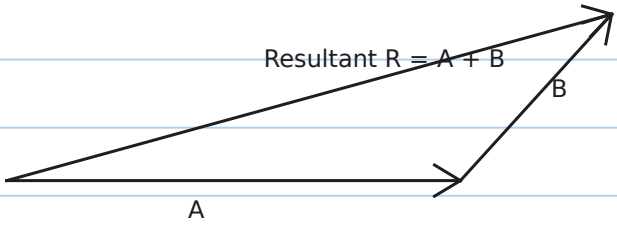
Answer: Step 1: Use PQ = (coordinates of Q) - (coordinates of P), component by component

Step 2: The x-component is x = 10 - (-5/2) = 10 + 5/2 = 25/2

Step 3: The y-component is y = 8 - 9 = -1

Step 4: Combine the components: PQ = (25/2)i - 1j

Final answer: PQ = (25/2)i - j.



Question: Exercise 3.1 Q2(iii): Write the vector PQ in the form xi + yj, where P(-3, 6) and Q(5, 9).

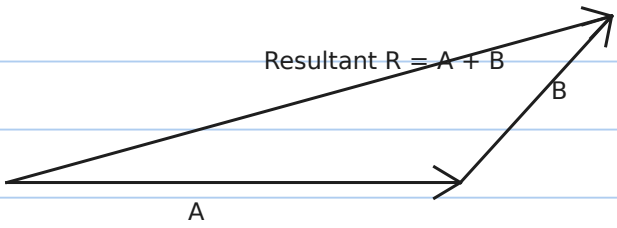
Answer: Step 1: Use PQ = (coordinates of Q) - (coordinates of P), component by component

Step 2: The x-component is x = 5 - (-3) = 5 + 3 = 8

Step 3: The y-component is y = 9 - 6 = 3

Step 4: Combine the components: PQ = 8i + 3j

Final answer: PQ = 8i + 3j.



Question: Exercise 3.1 Q2(iv): Write the vector PQ in the form xi + yj, where P(-6, 2) and Q(0, -5).

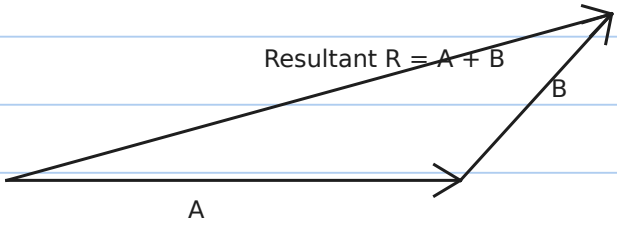
Answer: Step 1: Use PQ = (coordinates of Q) - (coordinates of P), component by component

Step 2: The x-component is x = 0 - (-6) = 6

Step 3: The y-component is y = -5 - 2 = -7

Step 4: Combine the components: PQ = 6i - 7j

Final answer: PQ = 6i - 7j.



Question: Exercise 3.1 Q3(i): If p = -5i + 4j, q = -3i + 5j and r = -6i + 7j, find 2p + 3q.

Answer: Step 1: Multiply p by 2 component-wise: 2p = 2(-5i + 4j) = -10i + 8j

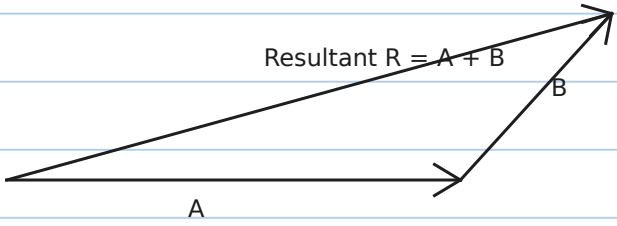
Step 2: Multiply q by 3 component-wise: 3q = 3(-3i + 5j) = -9i + 15j

Step 3: Add the i-components: -10 + (-9) = -19

Step 4: Add the j-components: 8 + 15 = 23

Step 5: Combine: 2p + 3q = -19i + 23j

Final answer: 2p + 3q = -19i + 23j.



Question: Exercise 3.1 Q3(ii): If p = -5i + 4j, q = -3i + 5j and r = -6i + 7j, find 3p + (1/2)q + r.

Answer: Step 1: Multiply p by 3: 3p = -15i + 12j

Step 2: Multiply q by 1/2: (1/2)q = -(3/2)i + (5/2)j

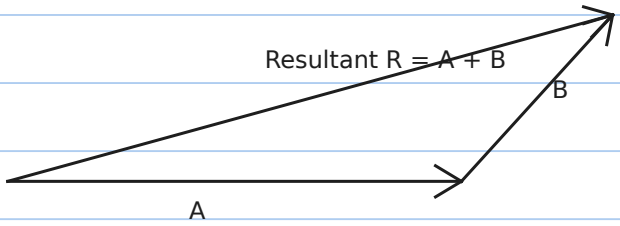
Step 3: Keep r as it is: r = -6i + 7j

Step 4: Add the i-components: -15 - 3/2 - 6 = -45/2

Step 5: Add the j-components: 12 + 5/2 + 7 = 43/2

Step 6: Combine: 3p + (1/2)q + r = -(45/2)i + (43/2)j

Final answer: 3p + (1/2)q + r = -(45/2)i + (43/2)j.



Question: Exercise 3.1 Q3(iii): If p = -5i + 4j, q = -3i + 5j and r = -6i + 7j, find 3p + 5q - 4r.

Answer: Step 1: Multiply p by 3: 3p = -15i + 12j

Step 2: Multiply q by 5: 5q = -15i + 25j

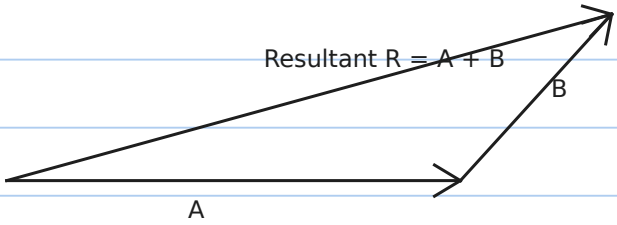
Step 3: Multiply r by -4: -4r = -4(-6i + 7j) = 24i - 28j

Step 4: Add the i-components: -15 - 15 + 24 = -6

Step 5: Add the j-components: 12 + 25 - 28 = 9

Step 6: Combine: 3p + 5q - 4r = -6i + 9j

Final answer: 3p + 5q - 4r = -6i + 9j.



Question: Exercise 3.1 Q3(iv): If p = -5i + 4j, q = -3i + 5j and r = -6i + 7j, find (1/2)p - 2r + 4q.

Answer: Step 1: Multiply p by 1/2: (1/2)p = -(5/2)i + 2j

Step 2: Multiply r by -2: -2r = -2(-6i + 7j) = 12i - 14j

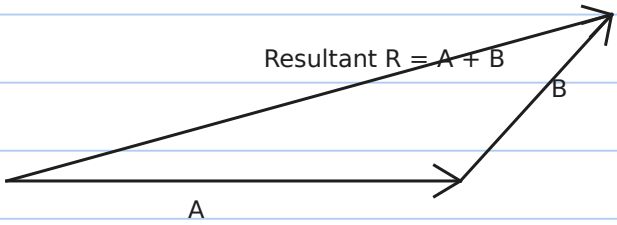
Step 3: Multiply q by 4: 4q = -12i + 20j

Step 4: Add the i-components: -5/2 + 12 - 12 = -5/2

Step 5: Add the j-components: 2 - 14 + 20 = 8

Step 6: Combine: (1/2)p - 2r + 4q = -(5/2)i + 8j

Final answer: (1/2)p - 2r + 4q = -(5/2)i + 8j.



Question: Exercise 3.1 Q4: P(-2, 5), Q(8, 0), R(7/2, 6) and S(-10, 7) are given points. Find the sum of the vectors PQ and RS.

Answer: Step 1: Find PQ by subtracting the coordinates of P from Q: PQ = (8 - (-2))i + (0 - 5)j = 10i - 5j

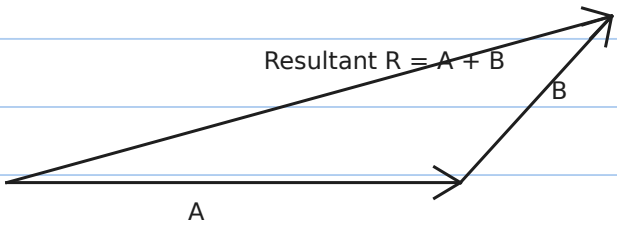
Step 2: Find RS by subtracting the coordinates of R from S: RS = (-10 - 7/2)i + (7 - 6)j = -(27/2)i + 1j

Step 3: Add the i-components of PQ and RS: 10 - 27/2 = -7/2

Step 4: Add the j-components of PQ and RS: -5 + 1 = -4

Step 5: Combine: PQ + RS = -(7/2)i - 4j

Final answer: PQ + RS = -(7/2)i - 4j.



Question: Exercise 3.1 Q5(i): Find the unit vector in the direction of p = -3i + 8j.

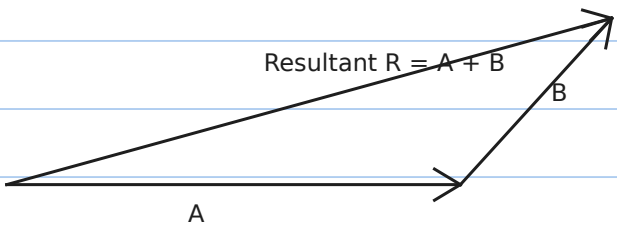
Answer: Step 1: A unit vector in the direction of p is p / |p|, so first find the magnitude

Step 2: |p| = sqrt((-3)^2 + (8)^2) = sqrt(9 + 64) = sqrt(73)

Step 3: Divide each component of p by sqrt(73): the i-component is -3/sqrt(73) and the j-component is 8/sqrt(73)

Step 4: Write the result as a single vector: (-3i + 8j)/sqrt(73), whose magnitude is sqrt(9 + 64)/sqrt(73) = 1, as a unit vector must have

Final answer: The unit vector is (-3i + 8j)/sqrt(73).



Question: Exercise 3.1 Q5(ii): Find the unit vector in the direction of q = 5i - 9j.

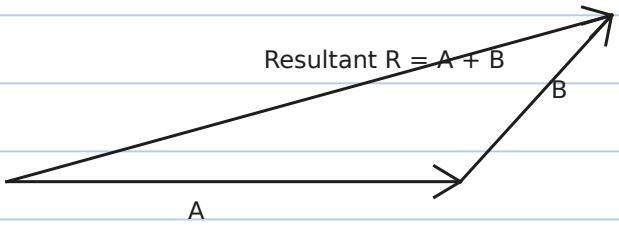
Answer: Step 1: A unit vector in the direction of q is q / |q|, so first find the magnitude

Step 2: |q| = sqrt((5)^2 + (-9)^2) = sqrt(25 + 81) = sqrt(106)

Step 3: Divide each component of q by sqrt(106): the i-component is 5/sqrt(106) and the j-component is -9/sqrt(106)

Step 4: Write the result as a single vector: (5i - 9j)/sqrt(106), and check sqrt(25 + 81)/sqrt(106) = 1

Final answer: The unit vector is (5i - 9j)/sqrt(106).



Question: Exercise 3.1 Q5(iii): Find the unit vector in the direction of r = 10i - 12j.

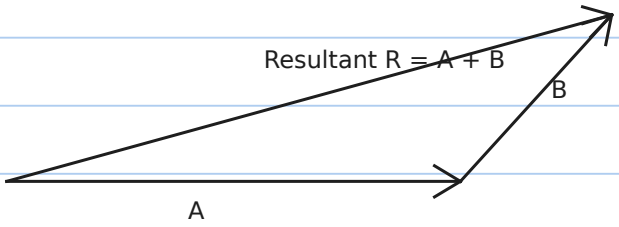
Answer: Step 1: A unit vector in the direction of r is r / |r|, so first find the magnitude

Step 2: |r| = sqrt((10)^2 + (-12)^2) = sqrt(100 + 144) = sqrt(244) = 2sqrt(61)

Step 3: Divide each component by 2sqrt(61): the i-component is 10/(2sqrt(61)) = 5/sqrt(61) and the j-component is -12/(2sqrt(61)) = -6/sqrt(61)

Step 4: Write the simplified result: (5i - 6j)/sqrt(61)

Final answer: The unit vector is (5i - 6j)/sqrt(61).



Question: Exercise 3.1 Q5(iv): Find the unit vector in the direction of s = 7i + 5j.

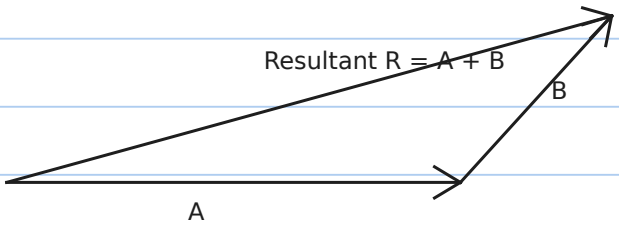
Answer: Step 1: A unit vector in the direction of s is s / |s|, so first find the magnitude

Step 2: |s| = sqrt((7)^2 + (5)^2) = sqrt(49 + 25) = sqrt(74)

Step 3: Divide each component of s by sqrt(74): the i-component is 7/sqrt(74) and the j-component is 5/sqrt(74)

Step 4: Write the result as a single vector: (7i + 5j)/sqrt(74), with magnitude sqrt(49 + 25)/sqrt(74) = 1

Final answer: The unit vector is (7i + 5j)/sqrt(74).



Question: Exercise 3.1 Q6(i): Find the vector in the direction of a = 2i - 3j having magnitude 6 units.

Answer: Step 1: The required vector is (given magnitude) x (unit vector in the direction of a), so first find |a|

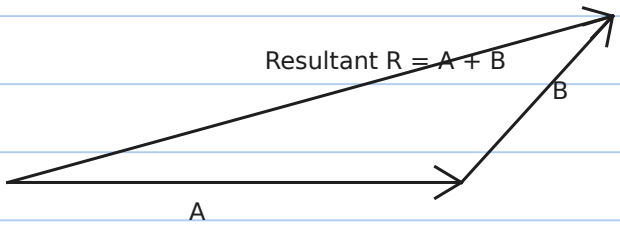
Step 2: |a| = sqrt((2)^2 + (-3)^2) = sqrt(4 + 9) = sqrt(13)

Step 3: The unit vector in the direction of a is a/|a| = (2i - 3j)/sqrt(13)

Step 4: Multiply the unit vector by the required magnitude 6: 6(2i - 3j)/sqrt(13) = (12i - 18j)/sqrt(13)

Step 5: Check its length: sqrt(144 + 324)/sqrt(13) = sqrt(468)/sqrt(13) = sqrt(36) = 6, as required

Final answer: The required vector is (12i - 18j)/sqrt(13).



Question: Exercise 3.1 Q6(ii): Find the vector in the direction of b = 10i - 9j having magnitude 7 units.

Answer: Step 1: The required vector is (given magnitude) x (unit vector in the direction of b), so first find |b|

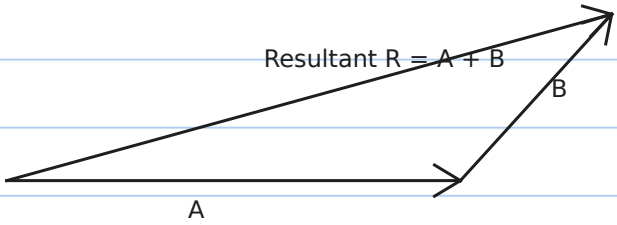
Step 2: |b| = sqrt((10)^2 + (-9)^2) = sqrt(100 + 81) = sqrt(181)

Step 3: The unit vector in the direction of b is b/|b| = (10i - 9j)/sqrt(181)

Step 4: Multiply the unit vector by the required magnitude 7: 7(10i - 9j)/sqrt(181) = (70i - 63j)/sqrt(181)

Step 5: Check its length: sqrt(4900 + 3969)/sqrt(181) = sqrt(8869)/sqrt(181) = sqrt(49) = 7, as required

Final answer: The required vector is (70i - 63j)/sqrt(181).



Question: Exercise 3.1 Q6(iii): Find the vector in the direction of c = 5i + 6j having magnitude 11 units.

Answer: Step 1: The required vector is (given magnitude) x (unit vector in the direction of c), so first find |c|

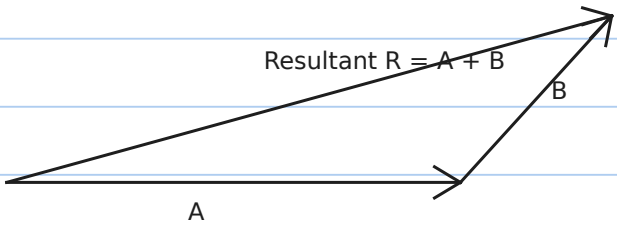
Step 2: |c| = sqrt((5)^2 + (6)^2) = sqrt(25 + 36) = sqrt(61)

Step 3: The unit vector in the direction of c is c/|c| = (5i + 6j)/sqrt(61)

Step 4: Multiply the unit vector by the required magnitude 11: 11(5i + 6j)/sqrt(61) = (55i + 66j)/sqrt(61)

Step 5: Check its length: sqrt(3025 + 4356)/sqrt(61) = sqrt(7381)/sqrt(61) = sqrt(121) = 11, as required

Final answer: The required vector is (55i + 66j)/sqrt(61).



Question: Exercise 3.1 Q7(i): Find the value of the unknown if 4i + 5j = pi + 5j.

Answer: Step 1: Two vectors are equal only when their corresponding components are equal

Step 2: Compare the i-components of the two sides: 4 = p

Step 3: Compare the j-components as a check: 5 = 5, which is already true and gives no new information

Step 4: Hence the single unknown is fixed by the i-components alone

Final answer: p = 4.

Question: Exercise 3.1 Q7(ii): Find the value of the unknown if (5, 7) = (5, -q).

Answer: Step 1: Two vectors written as ordered pairs are equal only when their corresponding components are equal

Step 2: Compare the first components: 5 = 5, which is already true

Step 3: Compare the second components: 7 = -q

Step 4: Multiply both sides by -1 to isolate q: q = -7

Final answer: q = -7.

Question: Exercise 3.1 Q7(iii): Find the value of the unknown if (7/2)i - (3/5)j = pi - (3/sqrt(25))j.

Answer: Step 1: First simplify the known coefficient on the right: sqrt(25) = 5, so 3/sqrt(25) = 3/5

Step 2: Two vectors are equal only when their corresponding components are equal

Step 3: Compare the i-components: 7/2 = p

Step 4: Compare the j-components as a check: -3/5 = -3/5, which is true

Final answer: p = 7/2.

Question: Exercise 3.1 Q7(iv): Find the value of the unknown if the column

vector (5/2, -7) equals the column vector (p, -7).

Answer: Step 1: Two column vectors are equal only when their corresponding entries are equal

Step 2: Compare the top entries: 5/2 = p

Step 3: Compare the bottom entries as a check: -7 = -7, which is already true

Step 4: Hence the unknown is fixed by the top entry alone

Final answer: p = 5/2.

Question: Exercise 3.1 Q8(i): Find the position vector of the point dividing the join of A and B in the ratio 3:5, where the position vectors of A and B are 4i - 5j and 2i + 7j respectively.

Answer: Step 1: For internal division in the ratio m:n, the position vector is r = (m x b + n x a)/(m + n), where a and b are the position vectors of A and B; here m = 3 and n = 5

Step 2: Compute m x b = 3(2i + 7j) = 6i + 21j

Step 3: Compute n x a = 5(4i - 5j) = 20i - 25j

Step 4: Add them: (6 + 20)i + (21 - 25)j = 26i - 4j

Step 5: Divide by m + n = 8: r = (26/8)i - (4/8)j = (13/4)i - (1/2)j

Final answer: The position vector is (13/4)i - (1/2)j.

Question: Exercise 3.1 Q8(ii): Find the position vector of the point dividing the join of A and B in the ratio 3:2, where the position vectors of A and B are 3i + 5j and 11i + 8j respectively.

Answer: Step 1: For internal division in the ratio m:n, r = (m x b + n x a)/(m + n); here m = 3 and n = 2

Step 2: Compute m x b = 3(11i + 8j) = 33i + 24j

Step 3: Compute n x a = 2(3i + 5j) = 6i + 10j

Step 4: Add them: (33 + 6)i + (24 + 10)j = 39i + 34j

Step 5: Divide by m + n = 5: r = (39/5)i + (34/5)j

Final answer: The position vector is (39/5)i + (34/5)j.

Question: Exercise 3.1 Q8(iii): Find the position vector of the point dividing the join of A and B in the ratio 4:3, where the position vectors of A and B are 2i - 3j and 2i + 6j respectively.

Answer: Step 1: For internal division in the ratio m:n, r = (m x b + n x a)/(m + n); here m = 4 and n = 3

Step 2: Compute m x b = 4(2i + 6j) = 8i + 24j

Step 3: Compute n x a = 3(2i - 3j) = 6i - 9j

Step 4: Add them: (8 + 6)i + (24 - 9)j = 14i + 15j

Step 5: Divide by m + n = 7: r = (14/7)i + (15/7)j = 2i + (15/7)j

Final answer: The position vector is 2i + (15/7)j.

Question: Exercise 3.1 Q9: Prove the hexagon law of vector addition.

Answer: Step 1: State the law: if five vectors are represented in magnitude and direction by five consecutive sides of a hexagon taken in the same order, then their resultant is represented by the sixth (closing) side taken in the opposite order

Step 2: Let the hexagon be ABCDEF and let the five sides AB, BC, CD, DE and EF be represented by vectors a, b, c, d and e respectively; we prove that AF = a + b + c + d + e, where AF is the closing side

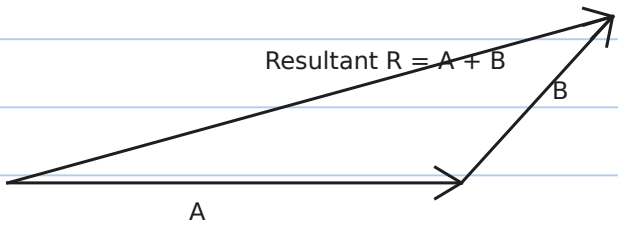
Step 3: In triangle ABC, by the triangle law, AC = AB + BC = a + b

Step 4: In triangle ACD, by the triangle law, AD = AC + CD = (a + b) + c = a + b + c

Step 5: In triangle ADE, by the triangle law, AE = AD + DE = (a + b + c) + d = a + b + c + d

Step 6: In triangle AEF, by the triangle law, AF = AE + EF = (a + b + c + d) + e = a + b + c + d + e, and AF is exactly the closing side of the hexagon taken in the opposite order

Final answer: AF = a + b + c + d + e, which proves the hexagon law of vector addition.



Question: Exercise 3.1 Q10(a): A(3, -5), B(6, 0) and C(2, 4) are vertices of a parallelogram. Using vectors, find the coordinates of point D if ABCD is the parallelogram.

Answer: Step 1: In parallelogram ABCD the vertices go in order, so the diagonal AC and the diagonal BD have the same midpoint

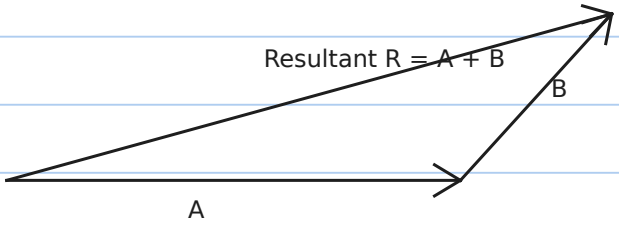
Step 2: Let D = (x, y); position vectors are a = 3i - 5j, b = 6i + 0j and c = 2i + 4j

Step 3: For a parallelogram, d = a + c - b, because OA + OC = OB + OD from the equal-midpoint condition

Step 4: Compute component-wise: d = (3 + 2 - 6)i + (-5 + 4 - 0)j = -1i - 1j

Step 5: Read off the coordinates from the position vector -i - j

Final answer: D = (-1, -1).



Question: Exercise 3.1 Q10(b): A(3, -5), B(6, 0) and C(2, 4) are given points. Using vectors, find the coordinates of point D if ADBC is the parallelogram.

Answer: Step 1: In parallelogram ADBC the vertices go in the order A-D-B-C, so the diagonals are AB and DC, and they have the same midpoint

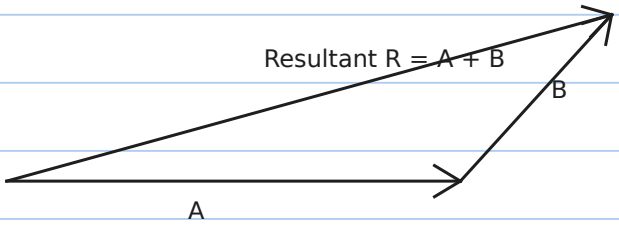
Step 2: Equal midpoints give (a + b)/2 = (d + c)/2, so d = a + b - c

Step 3: Substitute a = 3i - 5j, b = 6i + 0j and c = 2i + 4j

Step 4: Compute component-wise: d = (3 + 6 - 2)i + (-5 + 0 - 4)j = 7i - 9j

Step 5: Read off the coordinates from the position vector 7i - 9j

Final answer: D = (7, -9).



Question: Exercise 3.1 Q10(c): A(3, -5), B(6, 0) and C(2, 4) are given points. Using vectors, find the coordinates of point D if ABDC is the parallelogram.

Answer: Step 1: In parallelogram ABDC the vertices go in the order A-B-D-C, so the diagonals are AD and BC, and they have the same midpoint

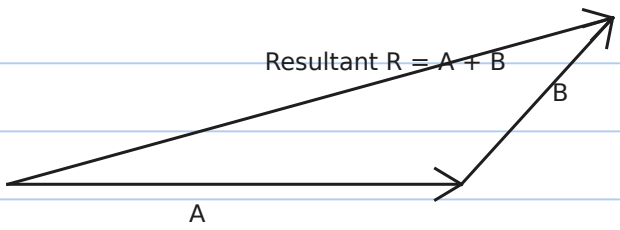
Step 2: Equal midpoints give (a + d)/2 = (b + c)/2, so d = b + c - a

Step 3: Substitute a = 3i - 5j, b = 6i + 0j and c = 2i + 4j

Step 4: Compute component-wise: d = (6 + 2 - 3)i + (0 + 4 - (-5))j = 5i + 9j

Step 5: Read off the coordinates from the position vector 5i + 9j

Final answer: D = (5, 9).



Question: Exercise 3.1 Q11: Given that A is the point (1, 3), AB is the column vector (4, 1) and AD is the column vector (2, 3). Find the coordinates of vertices B, C and D of the parallelogram ABCD.

Answer: Step 1: The position vector of A is a = (1, 3); a vertex reached by a side vector is found by adding that side vector to a

Step 2: Find B: b = a + AB = (1, 3) + (4, 1) = (5, 4)

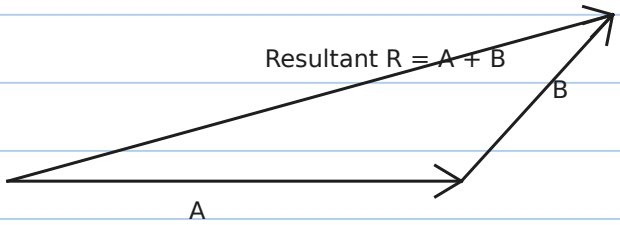
Step 3: Find D: d = a + AD = (1, 3) + (2, 3) = (3, 6)

Step 4: In a parallelogram the fourth vertex is reached by both side vectors together: c = a + AB + AD = (1, 3) + (4, 1) + (2, 3)

Step 5: Add component-wise: c = (1 + 4 + 2, 3 + 1 + 3) = (7, 7)

Step 6: Check with the midpoint rule: midpoint of AC = (4, 5) and midpoint of BD = ((5 + 3)/2, (4 + 6)/2) = (4, 5), which agree

Final answer: B = (5, 4), C = (7, 7) and D = (3, 6).



Question: Exercise 3.1 Q12(i): Using vectors, prove that if one pair of opposite sides of a quadrilateral are congruent and parallel, then it is a parallelogram.

Answer: Step 1: Let the quadrilateral be ABCD with position vectors a, b, c and d, and suppose the side AB is parallel and congruent (equal in length and direction as a vector) to the side DC, so AB = DC as vectors

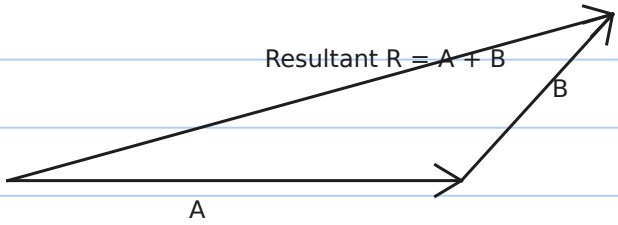
Step 2: Write the two sides as vectors: AB = b - a and DC = c - d, so the hypothesis gives b - a = c - d

Step 3: Rearrange the equation by moving a to the right and d to the left: b + d = c + a, equivalently d - a = c - b after grouping the terms for the other pair of sides

Step 4: But d - a is the vector AD and c - b is the vector BC, so AD = BC as vectors, which means AD is also parallel and congruent to BC

Step 5: A quadrilateral in which both pairs of opposite sides are parallel (and equal) is by definition a parallelogram

Final answer: Hence ABCD is a parallelogram.



Question: Exercise 3.1 Q12(ii): Using vectors, prove that the line segments joining the midpoints of the opposite sides of a quadrilateral bisect each other.

Answer: Step 1: Let the quadrilateral have vertices with position vectors a, b, c and d

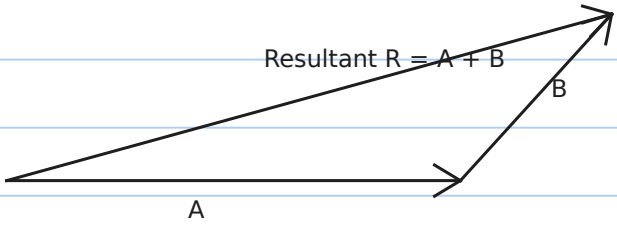
Step 2: The midpoints of the four sides are: midpoint of AB = (a + b)/2, midpoint of BC = (b + c)/2, midpoint of CD = (c + d)/2 and midpoint of DA = (d + a)/2

Step 3: The segment joining the midpoints of AB and CD has midpoint ((a + b)/2 + (c + d)/2)/2 = (a + b + c + d)/4

Step 4: The segment joining the midpoints of BC and DA has midpoint ((b + c)/2 + (d + a)/2)/2 = (a + b + c + d)/4

Step 5: The two segments have the same midpoint (a + b + c + d)/4, so each segment passes through the midpoint of the other, which is exactly what bisecting each other means

Final answer: Hence the two segments bisect each other.



Question: Exercise 3.1 Q12(iii): Using vectors, prove that the medians of a triangle are concurrent.

Answer: Step 1: Let the triangle have vertices with position vectors a, b and c; a median joins a vertex to the midpoint of the opposite side

Step 2: The midpoint of BC is (b + c)/2, so points on the median from A have the form a + t((b + c)/2 - a) for a parameter t

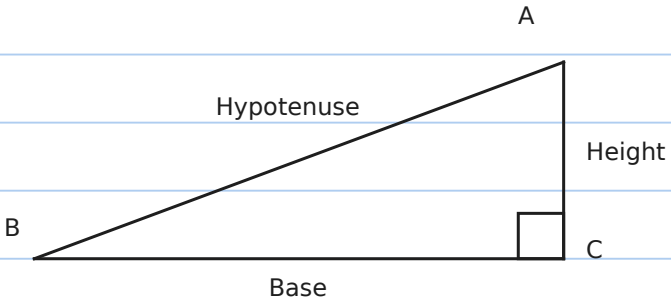
Step 3: Consider the point g = (a + b + c)/3, the average of the three position vectors

Step 4: Check that g lies on the median from A: setting t = 2/3 gives a + (2/3)((b + c)/2 - a) = a(1 - 2/3) + (b + c)/3 = (a + b + c)/3 = g, so g divides the median from A in the ratio 2:1

Step 5: By the same calculation with the vertices relabelled, g also lies on the median from B (dividing it 2:1) and on the median from C (dividing it 2:1)

Step 6: Since one and the same point g lies on all three medians, the three medians meet at a single point

Final answer: Hence the medians of a triangle are concurrent at the centroid (a + b + c)/3.



Question: Exercise 3.1 Q12(iv): Using vectors, prove that the midpoint of the hypotenuse of a right-angled triangle is equidistant from its vertices.

Answer: Step 1: Place the right angle at the origin O, and let the other two vertices be A and B with position vectors u and v; because the angle at O is 90 deg, u and v are perpendicular, so u.v = 0

Step 2: The hypotenuse is AB, and its midpoint M has position vector m = (u + v)/2

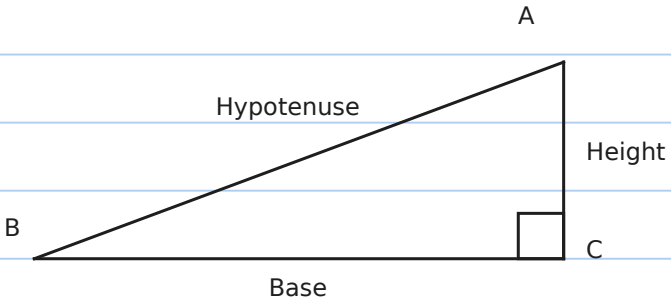
Step 3: Distance from M to O: |m| = |u + v|/2

Step 4: Distance from M to A: |u - m| = |u - (u + v)/2| = |(u - v)/2| = |u - v|/2, and distance from M to B: |v - m| = |v - u|/2 = |u - v|/2

Step 5: Compare |u + v| and |u - v| by squaring: |u + v|^2 = |u|^2 + 2u.v + |v|^2 = |u|^2 + |v|^2 (since u.v = 0), and |u - v|^2 = |u|^2 - 2u.v + |v|^2 = |u|^2 + |v|^2; so |u + v| = |u - v|

Step 6: Therefore |m| = |u - m| = |v - m|, that is M is at the same distance from O, A and B

Final answer: Hence the midpoint of the hypotenuse is equidistant from all three vertices.



Question: Exercise 3.1 Q13: The points (1, 1), (5, 4), (8, 9) and (0, 3) represent the position vectors of A, B, C and D respectively. Show that ABCD is a trapezium.

Answer: Step 1: A trapezium has one pair of opposite sides parallel but not equal; find the four side vectors first

Step 2: AB = (5 - 1, 4 - 1) = (4, 3), and its length is |AB| = sqrt(16 + 9) = 5

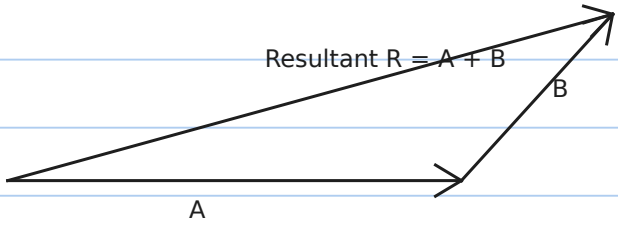
Step 3: BC = (8 - 5, 9 - 4) = (3, 5)

Step 4: CD = (0 - 8, 3 - 9) = (-8, -6), and its length is |CD| = sqrt(64 + 36) = 10

Step 5: Compare AB and DC: DC = -CD = (8, 6) = 2(4, 3) = 2 AB, so AB and DC are parallel (one is a scalar multiple of the other)

Step 6: But |DC| = 10 is not equal to |AB| = 5, so the parallel sides are not congruent, and ABCD is not a parallelogram; also BC = (3, 5) is not a multiple of AD = D - A = (-1, 2), so the other pair is not parallel

Final answer: Exactly one pair of opposite sides is parallel and unequal, hence ABCD is a trapezium.



Question: Exercise 3.1 Q14: A, B, C are the points with position vectors a, b and 2a - b respectively. D divides AC in the ratio 2:3 and E divides BD in the ratio 4:1. Find the position vector of E.

Answer: Step 1: Use the internal division formula: a point dividing the join of points with position vectors p and q in the ratio m:n has position vector (m x (second vector) + n x (first vector))/(m + n)

Step 2: Find D on AC with AD:DC = 2:3: d = (3a + 2(2a - b))/(2 + 3)

Step 3: Expand the numerator: 3a + 4a - 2b = 7a - 2b, so d = (7a - 2b)/5

Step 4: Find E on BD with BE:ED = 4:1: e = (1 x b + 4 x d)/(4 + 1) = (b + 4d)/5

Step 5: Substitute d: b + 4(7a - 2b)/5 = (5b + 28a - 8b)/5 = (28a - 3b)/5

Step 6: Divide by 5: e = (28a - 3b)/25

Final answer: The position vector of E is (28a - 3b)/25.

Question: Exercise 3.1 Q15: In a parallelogram ABCD, X is the midpoint of AB and Y divides BC in the ratio 1:2. Show that if Z divides DX in the ratio 6:1, then it also divides AY in the ratio 3:4.

Answer: Step 1: Let the position vectors of A, B, C, D be a, b, c, d; for a parallelogram the diagonals have the same midpoint, so a + c = b + d, that is c = b + d - a

Step 2: X is the midpoint of AB, so x = (a + b)/2

Step 3: Y divides BC in the ratio 1:2, so by the section formula y = (2b + 1c)/3; substituting c = b + d - a gives y = (2b + b + d - a)/3 = (3b + d - a)/3

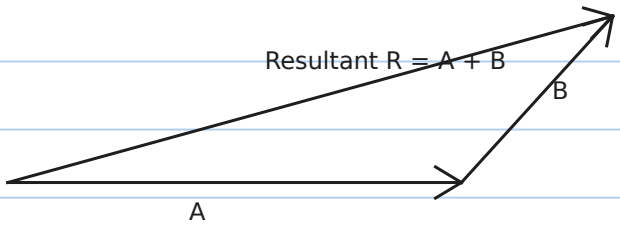
Step 4: Z divides DX in the ratio 6:1, so z = (1 x d + 6 x x)/(6 + 1) = (d + 6(a + b)/2)/7 = (d + 3a + 3b)/7

Step 5: Now compute the point that divides AY in the ratio 3:4: it is (4a + 3y)/(3 + 4) = (4a + 3y)/7

Step 6: Substitute 3y = 3b + d - a from Step 3: 4a + (3b + d - a) = 3a + 3b + d, so the point is (3a + 3b + d)/7

Step 7: This is exactly the same vector as z found in Step 4

Final answer: Hence the point Z that divides DX in the ratio 6:1 is the same point as the one dividing AY in the ratio 3:4, so Z also divides AY in the ratio 3:4.



Exercise 3.2

Question: Exercise 3.2 Q1(i): Let u = i + 2j + k, v = 3i - 5j and w = -8i + 7j - 2k. Find 3u + 2v.

Answer: Step 1: Note that v = 3i - 5j means v = 3i - 5j + 0k, so all three components are available

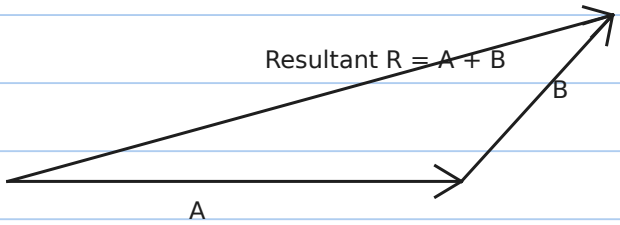
Step 2: Multiply u by 3 component-wise: 3u = 3i + 6j + 3k

Step 3: Multiply v by 2 component-wise: 2v = 6i - 10j + 0k

Step 4: Add the i-components: 3 + 6 = 9; add the j-components: 6 - 10 = -4; add the k-components: 3 + 0 = 3

Step 5: Combine: 3u + 2v = 9i - 4j + 3k

Final answer: 3u + 2v = 9i - 4j + 3k.



Question: Exercise 3.2 Q1(ii): Let u = i + 2j + k, v = 3i - 5j and w = -8i + 7j - 2k. Find v - 2u.

Answer: Step 1: Write v with all components: v = 3i - 5j + 0k

Step 2: Multiply u by 2 component-wise: 2u = 2i + 4j + 2k

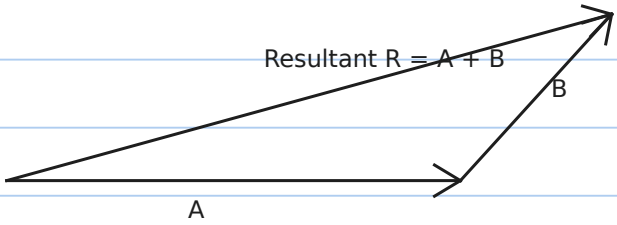
Step 3: Subtract component-wise, i-components: 3 - 2 = 1

Step 4: Subtract the j-components: -5 - 4 = -9

Step 5: Subtract the k-components: 0 - 2 = -2

Step 6: Combine: v - 2u = 1i - 9j - 2k

Final answer: v - 2u = i - 9j - 2k.



Question: Exercise 3.2 Q1(iii): Let u = i + 2j + k, v = 3i - 5j and w = -8i + 7j - 2k. Find u - 3v + 2w.

Answer: Step 1: Write u and v with all components: u = 1i + 2j + 1k and v = 3i - 5j + 0k

Step 2: Multiply v by -3: -3v = -9i + 15j + 0k

Step 3: Multiply w by 2: 2w = -16i + 14j - 4k

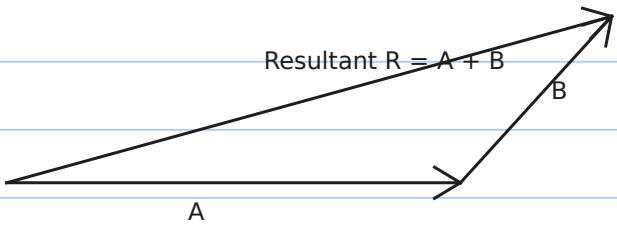
Step 4: Add the i-components: 1 - 9 - 16 = -24

Step 5: Add the j-components: 2 + 15 + 14 = 31

Step 6: Add the k-components: 1 + 0 - 4 = -3

Step 7: Combine: u - 3v + 2w = -24i + 31j - 3k

Final answer: u - 3v + 2w = -24i + 31j - 3k.



Question: Exercise 3.2 Q1(iv): Let u = i + 2j + k, v = 3i - 5j and w = -8i + 7j - 2k. Find 3u + v - w.

Answer: Step 1: Multiply u by 3: 3u = 3i + 6j + 3k

Step 2: Write v with all components: v = 3i - 5j + 0k

Step 3: Change the sign of w: -w = 8i - 7j + 2k

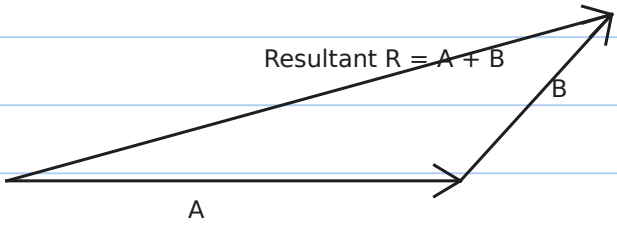
Step 4: Add the i-components: 3 + 3 + 8 = 14

Step 5: Add the j-components: 6 - 5 - 7 = -6

Step 6: Add the k-components: 3 + 0 + 2 = 5

Step 7: Combine: 3u + v - w = 14i - 6j + 5k

Final answer: 3u + v - w = 14i - 6j + 5k.



Question: Exercise 3.2 Q1(v): Let u = i + 2j + k, v = 3i - 5j and w = -8i + 7j - 2k. Find -2u + (1/2)v - 3w.

Answer: Step 1: Multiply u by -2: -2u = -2i - 4j - 2k

Step 2: Multiply v = 3i - 5j + 0k by 1/2: (1/2)v = (3/2)i - (5/2)j + 0k

Step 3: Multiply w by -3: -3w = 24i - 21j + 6k

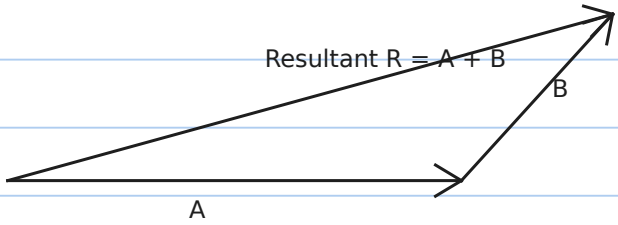
Step 4: Add the i-components: -2 + 3/2 + 24 = 47/2

Step 5: Add the j-components: -4 - 5/2 - 21 = -55/2

Step 6: Add the k-components: -2 + 0 + 6 = 4

Step 7: Combine: -2u + (1/2)v - 3w = (47/2)i - (55/2)j + 4k

Final answer: -2u + (1/2)v - 3w = (47/2)i - (55/2)j + 4k.



Question: Exercise 3.2 Q2(i): If a = -5i + 3j - 4k, find -2a.

Answer: Step 1: Scalar multiplication multiplies every component by the scalar

Step 2: Multiply the i-component by -2: -2 x (-5) = 10

Step 3: Multiply the j-component by -2: -2 x 3 = -6

Step 4: Multiply the k-component by -2: -2 x (-4) = 8

Step 5: Combine: -2a = 10i - 6j + 8k; the negative scalar also reverses the direction of a

Final answer: -2a = 10i - 6j + 8k.

Question: Exercise 3.2 Q2(ii): If a = -5i + 3j - 4k, find 3|a|.

Answer: Step 1: First find the magnitude |a| = sqrt((-5)^2 + (3)^2 + (-4)^2)

Step 2: Square each component: (-5)^2 = 25, (3)^2 = 9 and (-4)^2 = 16

Step 3: Add them: 25 + 9 + 16 = 50, so |a| = sqrt(50) = 5sqrt(2)

Step 4: Multiply the magnitude by 3: 3|a| = 3 x 5sqrt(2) = 15sqrt(2)

Final answer: 3|a| = 15sqrt(2).

Question: Exercise 3.2 Q2(iii): If a = -5i + 3j - 4k, find (4/5)a.

Answer: Step 1: Scalar multiplication multiplies every component by 4/5

Step 2: Multiply the i-component: (4/5) x (-5) = -4

Step 3: Multiply the j-component: (4/5) x 3 = 12/5

Step 4: Multiply the k-component: (4/5) x (-4) = -16/5

Step 5: Combine: (4/5)a = -4i + (12/5)j - (16/5)k

Final answer: (4/5)a = -4i + (12/5)j - (16/5)k.

Question: Exercise 3.2 Q2(iv): If a = -5i + 3j - 4k, find -(1/2)a.

Answer: Step 1: Scalar multiplication multiplies every component by -1/2

Step 2: Multiply the i-component: (-1/2) x (-5) = 5/2

Step 3: Multiply the j-component: (-1/2) x 3 = -3/2

Step 4: Multiply the k-component: (-1/2) x (-4) = 2

Step 5: Combine: -(1/2)a = (5/2)i - (3/2)j + 2k

Final answer: -(1/2)a = (5/2)i - (3/2)j + 2k.

Question: Exercise 3.2 Q3(i): Let A = i + 2j - 3k, B = 2i - j + 2k and C = 3i - j + 5k. Find A - 2B.

Answer: Step 1: Multiply B by 2 component-wise: 2B = 4i - 2j + 4k

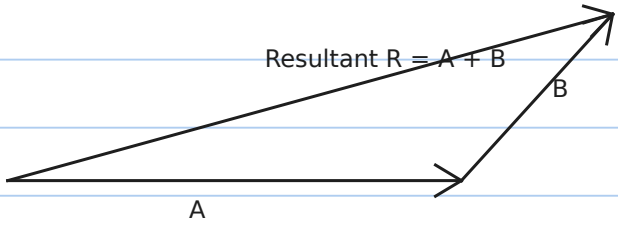
Step 2: Subtract the i-components: 1 - 4 = -3

Step 3: Subtract the j-components: 2 - (-2) = 4

Step 4: Subtract the k-components: -3 - 4 = -7

Step 5: Combine: A - 2B = -3i + 4j - 7k

Final answer: A - 2B = -3i + 4j - 7k.



Question: Exercise 3.2 Q3(ii): Let A = i + 2j - 3k, B = 2i - j + 2k and C = 3i - j + 5k. Find 3B + 2C.

Answer: Step 1: Multiply B by 3 component-wise: 3B = 6i - 3j + 6k

Step 2: Multiply C by 2 component-wise: 2C = 6i - 2j + 10k

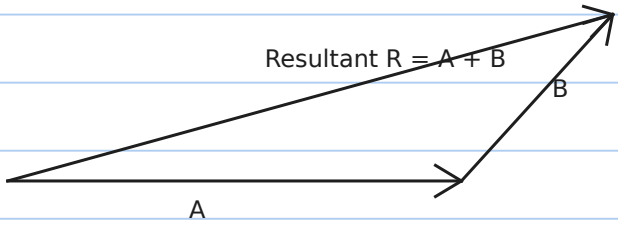
Step 3: Add the i-components: 6 + 6 = 12

Step 4: Add the j-components: -3 - 2 = -5

Step 5: Add the k-components: 6 + 10 = 16

Step 6: Combine: 3B + 2C = 12i - 5j + 16k

Final answer: 3B + 2C = 12i - 5j + 16k.



Question: Exercise 3.2 Q3(iii): Let A = i + 2j - 3k, B = 2i - j + 2k and C = 3i - j + 5k. Find 3A - (2B + C).

Answer: Step 1: First work out the bracket: 2B = 4i - 2j + 4k, so 2B + C = (4 + 3)i + (-2 - 1)j + (4 + 5)k = 7i - 3j + 9k

Step 2: Multiply A by 3: 3A = 3i + 6j - 9k

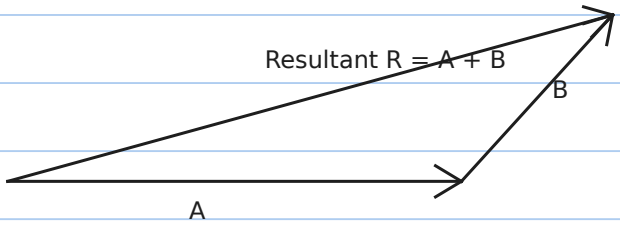
Step 3: Subtract the i-components: 3 - 7 = -4

Step 4: Subtract the j-components: 6 - (-3) = 9

Step 5: Subtract the k-components: -9 - 9 = -18

Step 6: Combine: 3A - (2B + C) = -4i + 9j - 18k

Final answer: 3A - (2B + C) = -4i + 9j - 18k.



Question: Exercise 3.2 Q4(i): Let u = i - 3j + 2k, v = i + j and w = 2i + 2j - 4k. Find |u + v - w|.

Answer: Step 1: Write v with all components: v = 1i + 1j + 0k

Step 2: Add the i-components of u + v - w: 1 + 1 - 2 = 0

Step 3: Add the j-components: -3 + 1 - 2 = -4

Step 4: Add the k-components: 2 + 0 - (-4) = 6

Step 5: So u + v - w = 0i - 4j + 6k = -4j + 6k

Step 6: Find its magnitude: |u + v - w| = sqrt(0^2 + (-4)^2 + (6)^2) = sqrt(0 + 16 + 36) = sqrt(52) = 2sqrt(13)

Final answer: |u + v - w| = 2sqrt(13).

Question: Exercise 3.2 Q4(ii): Let u = i - 3j + 2k, v = i + j and w = 2i + 2j - 4k. Find |u| + |v|.

Answer: Step 1: Find |u| by the magnitude formula: |u| = sqrt((1)^2 + (-3)^2 + (2)^2) = sqrt(1 + 9 + 4) = sqrt(14)

Step 2: Write v with all components: v = 1i + 1j + 0k

Step 3: Find |v|: |v| = sqrt((1)^2 + (1)^2 + (0)^2) = sqrt(2)

Step 4: Add the two scalar magnitudes: |u| + |v| = sqrt(14) + sqrt(2)

Step 5: Note this is a sum of two lengths, not the magnitude of a sum, so it cannot be simplified further

Final answer: |u| + |v| = sqrt(14) + sqrt(2).

Question: Exercise 3.2 Q4(iii): Let u = i - 3j + 2k, v = i + j and w = 2i + 2j - 4k. Find |(1/|w|) w|.

Answer: Step 1: Find |w| first: |w| = sqrt((2)^2 + (2)^2 + (-4)^2) = sqrt(4 + 4 + 16) = sqrt(24) = 2sqrt(6)

Step 2: Form the scaled vector (1/|w|) w = (1/sqrt(24))(2i + 2j - 4k); this is the unit vector in the direction of w

Step 3: The magnitude of a scalar multiple satisfies |c w| = |c| |w| for any scalar c

Step 4: Apply this with c = 1/|w|: |(1/|w|) w| = (1/|w|) x |w| = (1/sqrt(24)) x sqrt(24) = 1

Step 5: This confirms the general fact that dividing any non-zero vector by its own magnitude always gives a vector of magnitude 1

Final answer: |(1/|w|) w| = 1.

Question: Exercise 3.2 Q5(a): The position vector of point P is the column vector (5, -3, 2) and the position vector of point Q is the column vector (-2, 4, -6). Find 4PQ and |5PQ|.

Answer: Step 1: The vector PQ is (position vector of Q) - (position vector of P), entry by entry

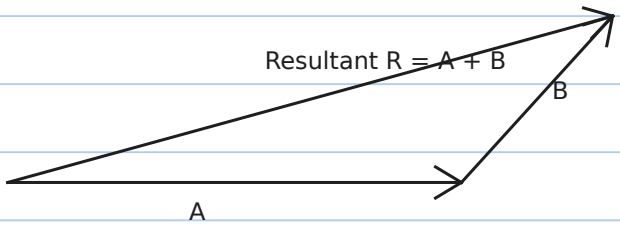
Step 2: PQ = (-2 - 5, 4 - (-3), -6 - 2) = (-7, 7, -8)

Step 3: Multiply every entry by 4: 4PQ = (-28, 28, -32), that is 4PQ = -28i + 28j - 32k

Step 4: For the second part, |PQ| = sqrt((-7)^2 + (7)^2 + (-8)^2) = sqrt(49 + 49 + 64) = sqrt(162) = 9sqrt(2)

Step 5: Use |5PQ| = 5|PQ| = 5 x 9sqrt(2) = 45sqrt(2)

Final answer: 4PQ = (-28, 28, -32) and |5PQ| = 45sqrt(2).



Question: Exercise 3.2 Q5(b)(i): If a = 3i - 5j + 2k and b = -i - 2j + 3k, find the unit vector a-hat in the direction of a.

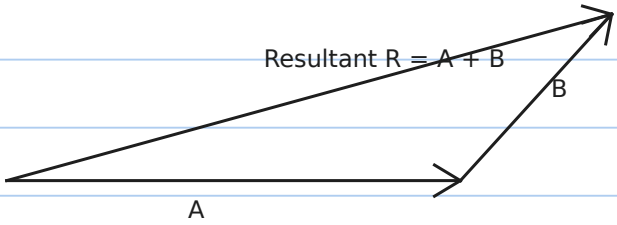
Answer: Step 1: The unit vector in the direction of a is a-hat = a / |a|

Step 2: Find |a| = sqrt((3)^2 + (-5)^2 + (2)^2) = sqrt(9 + 25 + 4) = sqrt(38)

Step 3: Divide each component of a by sqrt(38): i-component 3/sqrt(38), j-component -5/sqrt(38), k-component 2/sqrt(38)

Step 4: Combine into one vector: a-hat = (3i - 5j + 2k)/sqrt(38)

Final answer: a-hat = (3i - 5j + 2k)/sqrt(38).



Question: Exercise 3.2 Q5(b)(ii): If a = 3i - 5j + 2k and b = -i - 2j + 3k, find the unit vector b-hat in the direction of b.

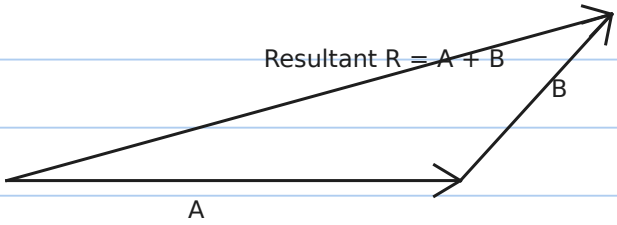
Answer: Step 1: The unit vector in the direction of b is b-hat = b / |b|

Step 2: Find |b| = sqrt((-1)^2 + (-2)^2 + (3)^2) = sqrt(1 + 4 + 9) = sqrt(14)

Step 3: Divide each component of b by sqrt(14): i-component -1/sqrt(14), j-component -2/sqrt(14), k-component 3/sqrt(14)

Step 4: Combine into one vector: b-hat = (-i - 2j + 3k)/sqrt(14)

Final answer: b-hat = (-i - 2j + 3k)/sqrt(14).



Question: Exercise 3.2 Q5(b)(iii): If a = 3i - 5j + 2k and b = -i - 2j + 3k, find |a + b|.

Answer: Step 1: Add the vectors component-wise: i-components: 3 + (-1) = 2

Step 2: Add the j-components: -5 + (-2) = -7

Step 3: Add the k-components: 2 + 3 = 5, so a + b = 2i - 7j + 5k

Step 4: Apply the magnitude formula: |a + b| = sqrt((2)^2 + (-7)^2 + (5)^2)

Step 5: Square and add: 4 + 49 + 25 = 78

Final answer: |a + b| = sqrt(78).

Question: Exercise 3.2 Q5(b)(iv): If a = 3i - 5j + 2k and b = -i - 2j + 3k, find |a - b|.

Answer: Step 1: Subtract the vectors component-wise: i-components: 3 - (-1) = 4

Step 2: Subtract the j-components: -5 - (-2) = -3

Step 3: Subtract the k-components: 2 - 3 = -1, so a - b = 4i - 3j - k

Step 4: Apply the magnitude formula: |a - b| = sqrt((4)^2 + (-3)^2 + (-1)^2)

Step 5: Square and add: 16 + 9 + 1 = 26

Final answer: |a - b| = sqrt(26).

Question: Exercise 3.2 Q6(i): Find the vector A1A2 when A1(0, 0, 0) and A2(-2, 5, 1).

Answer: Step 1: The vector A1A2 is (coordinates of A2) - (coordinates of A1), component by component

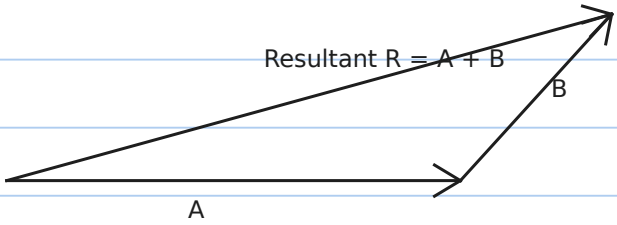
Step 2: x-component: -2 - 0 = -2

Step 3: y-component: 5 - 0 = 5

Step 4: z-component: 1 - 0 = 1

Step 5: Combine: A1A2 = -2i + 5j + k

Final answer: A1A2 = -2i + 5j + k.



Question: Exercise 3.2 Q6(ii): Find the vector A1A2 when A1(2, 1, -3) and A2(7, 1, -3).

Answer: Step 1: The vector A1A2 is (coordinates of A2) - (coordinates of A1), component by component

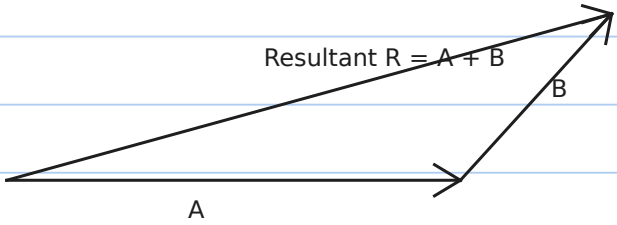
Step 2: x-component: 7 - 2 = 5

Step 3: y-component: 1 - 1 = 0

Step 4: z-component: -3 - (-3) = 0

Step 5: Combine: A1A2 = 5i + 0j + 0k = 5i, a vector parallel to the x-axis

Final answer: A1A2 = 5i.



Question: Exercise 3.2 Q6(iii): Find the vector A1A2 when A1(5, -2, 1) and A2(2, 4, 2).

Answer: Step 1: The vector A1A2 is (coordinates of A2) - (coordinates of A1), component by component

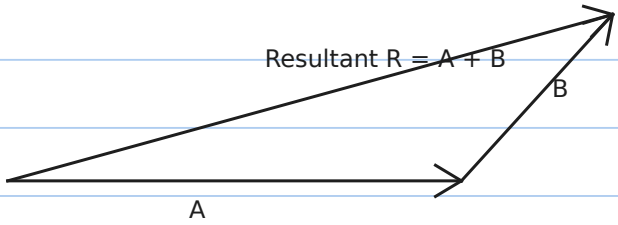
Step 2: x-component: 2 - 5 = -3

Step 3: y-component: 4 - (-2) = 6

Step 4: z-component: 2 - 1 = 1

Step 5: Combine: A1A2 = -3i + 6j + k

Final answer: A1A2 = -3i + 6j + k.



Question: Exercise 3.2 Q7(i): Find the initial point of the vector a = (-2, 1, 2) if its terminal point is (4, 0, -1).

Answer: Step 1: For any vector, vector = (terminal point) - (initial point), so initial point = (terminal point) - (vector)

Step 2: x-coordinate: 4 - (-2) = 6

Step 3: y-coordinate: 0 - 1 = -1

Step 4: z-coordinate: -1 - 2 = -3

Step 5: Check by adding: (6, -1, -3) + (-2, 1, 2) = (4, 0, -1), which is the given terminal point

Final answer: The initial point is (6, -1, -3).

Question: Exercise 3.2 Q7(ii): Find the terminal point of the vector A1A2 = (1, 3, -3) if its initial point is (-2, 1, 4).

Answer: Step 1: For any vector, terminal point = (initial point) + (vector), component by component

Step 2: x-coordinate: -2 + 1 = -1

Step 3: y-coordinate: 1 + 3 = 4

Step 4: z-coordinate: 4 + (-3) = 1

Step 5: Check by subtracting: (-1, 4, 1) - (-2, 1, 4) = (1, 3, -3), which is the given vector

Final answer: The terminal point is (-1, 4, 1).

Question: Exercise 3.2 Q8(a): The initial point of a vector a of magnitude 5 is (1, -sqrt(3), -5). Find k if the terminal point is (3, sqrt(3), k).

Answer: Step 1: The vector is (terminal) - (initial): a = (3 - 1, sqrt(3) - (-sqrt(3)), k - (-5)) = (2, 2sqrt(3), k + 5)

Step 2: Its magnitude is 5, so |a|^2 = 25: (2)^2 + (2sqrt(3))^2 + (k + 5)^2 = 25

Step 3: Simplify the known squares: 4 + 12 + (k + 5)^2 = 25, so (k + 5)^2 = 9

Step 4: Take square roots: k + 5 = 3 or k + 5 = -3

Step 5: Solve the two cases: k = 3 - 5 = -2, or k = -3 - 5 = -8

Final answer: k = -2 or k = -8.

Question: Exercise 3.2 Q8(b): The initial point of a vector a of magnitude 5 is (1, -sqrt(3), -5). Find k if the terminal point is (2, -3sqrt(3), -10k).

Answer: Step 1: The vector is (terminal) - (initial): a = (2 - 1, -3sqrt(3) - (-sqrt(3)), -10k - (-5)) = (1, -2sqrt(3), 5 - 10k)

Step 2: Its magnitude is 5, so |a|^2 = 25: (1)^2 + (-2sqrt(3))^2 + (5 - 10k)^2 = 25

Step 3: Simplify the known squares: 1 + 12 + (5 - 10k)^2 = 25, so (5 - 10k)^2 = 12

Step 4: Take square roots: 5 - 10k = 2sqrt(3) or 5 - 10k = -2sqrt(3)

Step 5: Solve: 10k = 5 - 2sqrt(3) giving k = (5 - 2sqrt(3))/10 = 1/2 - sqrt(3)/5, or 10k = 5 + 2sqrt(3) giving k = (5 + 2sqrt(3))/10 = 1/2 + sqrt(3)/5

Final answer: k = 1/2 - sqrt(3)/5 or k = 1/2 + sqrt(3)/5.

Question: Exercise 3.2 Q8(c): The initial point of a vector a of magnitude 5 is (1, -sqrt(3), -5). Find k if the terminal point is (-2, k, -3).

Answer: Step 1: The vector is (terminal) - (initial): a = (-2 - 1, k - (-sqrt(3)), -3 - (-5)) = (-3, k + sqrt(3), 2)

Step 2: Its magnitude is 5, so |a|^2 = 25: (-3)^2 + (k + sqrt(3))^2 + (2)^2 = 25

Step 3: Simplify the known squares: 9 + (k + sqrt(3))^2 + 4 = 25, so (k + sqrt(3))^2 = 12

Step 4: Take square roots: k + sqrt(3) = 2sqrt(3) or k + sqrt(3) = -2sqrt(3)

Step 5: Solve: k = 2sqrt(3) - sqrt(3) = sqrt(3), or k = -2sqrt(3) - sqrt(3) = -3sqrt(3)

Final answer: k = sqrt(3) or k = -3sqrt(3).

Question: Exercise 3.2 Q9: Find the coordinates of A, if OA is of length 6 units in the direction of OB, where B is the point (2, -1, 4).

Answer: Step 1: A lies on the ray from O through B at distance 6, so OA = 6 x (unit vector in the direction of OB) and the position vector of A is exactly OA

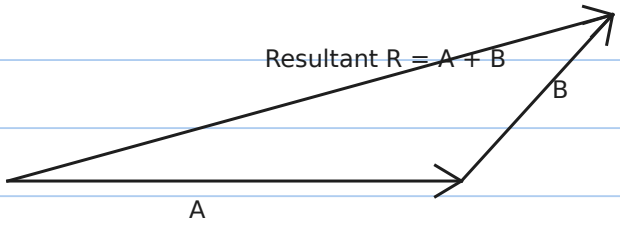
Step 2: OB = 2i - j + 4k, and |OB| = sqrt((2)^2 + (-1)^2 + (4)^2) = sqrt(4 + 1 + 16) = sqrt(21)

Step 3: The unit vector in the direction of OB is (2i - j + 4k)/sqrt(21)

Step 4: Multiply by 6: OA = 6(2i - j + 4k)/sqrt(21) = (12i - 6j + 24k)/sqrt(21); rationalising, 6/sqrt(21) = 2sqrt(21)/7, so OA = (4sqrt(21)/7)i - (2sqrt(21)/7)j + (8sqrt(21)/7)k

Step 5: The coordinates of A are the components of its position vector

Final answer: A = (12/sqrt(21), -6/sqrt(21), 24/sqrt(21)), that is (4sqrt(21)/7, -2sqrt(21)/7, 8sqrt(21)/7).



Question: Exercise 3.2 Q10(i): P, Q, R, S are points with position vectors (1, 1, -1), (1, -1, 2), (0, 1, 1) and (2, 1, 0) respectively. Find |PQ| and |QS|.

Answer: Step 1: PQ = (position vector of Q) - (position vector of P) = (1 - 1, -1 - 1, 2 - (-1)) = (0, -2, 3)

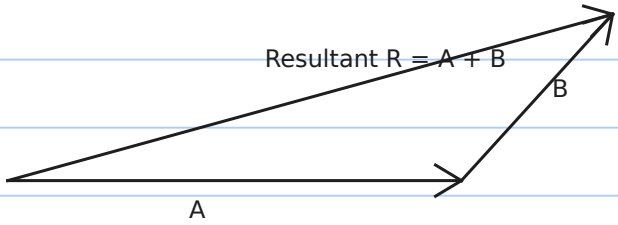
Step 2: |PQ| = sqrt(0^2 + (-2)^2 + (3)^2) = sqrt(0 + 4 + 9) = sqrt(13)

Step 3: QS = (position vector of S) - (position vector of Q) = (2 - 1, 1 - (-1), 0 - 2) = (1, 2, -2)

Step 4: |QS| = sqrt((1)^2 + (2)^2 + (-2)^2) = sqrt(1 + 4 + 4) = sqrt(9) = 3

Step 5: Both results are lengths, so both are non-negative scalars

Final answer: |PQ| = sqrt(13) and |QS| = 3.



Question: Exercise 3.2 Q10(ii)(a): P, Q, R, S are points with position vectors (1, 1, -1), (1, -1, 2), (0, 1, 1) and (2, 1, 0) respectively. Find the position vector of the point which divides QR internally in the ratio 3:2.

Answer: Step 1: For internal division of QR in the ratio m:n = 3:2, the position vector is r = (m x (position vector of R) + n x (position vector of Q))/(m + n)

Step 2: Compute 3 x (position vector of R) = 3(0, 1, 1) = (0, 3, 3)

Step 3: Compute 2 x (position vector of Q) = 2(1, -1, 2) = (2, -2, 4)

Step 4: Add them component-wise: (0 + 2, 3 - 2, 3 + 4) = (2, 1, 7)

Step 5: Divide by m + n = 5: r = (2/5, 1/5, 7/5)

Final answer: The position vector is (2/5, 1/5, 7/5), that is (2/5)i + (1/5)j + (7/5)k.

Question: Exercise 3.2 Q10(ii)(b): P, Q, R, S are points with position vectors (1, 1, -1), (1, -1, 2), (0, 1, 1) and (2, 1, 0) respectively. Find the position vector of the point which divides PR externally in the ratio 3:2.

Answer: Step 1: For external division of PR in the ratio m:n = 3:2, the position vector is r = (m x (position vector of R) - n x (position vector of P))/(m - n)

Step 2: Compute 3 x (position vector of R) = 3(0, 1, 1) = (0, 3, 3)

Step 3: Compute 2 x (position vector of P) = 2(1, 1, -1) = (2, 2, -2)

Step 4: Subtract component-wise: (0 - 2, 3 - 2, 3 - (-2)) = (-2, 1, 5)

Step 5: Divide by m - n = 1, so the vector is unchanged: r = (-2, 1, 5)

Final answer: The position vector is (-2, 1, 5), that is -2i + j + 5k.

Question: Exercise 3.2 Q10(iii): P, Q, R, S are points with position vectors (1, 1, -1), (1, -1, 2), (0, 1, 1) and (2, 1, 0) respectively. If X and Y are the midpoints of PR and RS, show that XY = (1/2)PS.

Answer: Step 1: The midpoint of a segment is the average of the position vectors of its endpoints

Step 2: X is the midpoint of PR: x = ((1, 1, -1) + (0, 1, 1))/2 = (1/2, 1, 0)

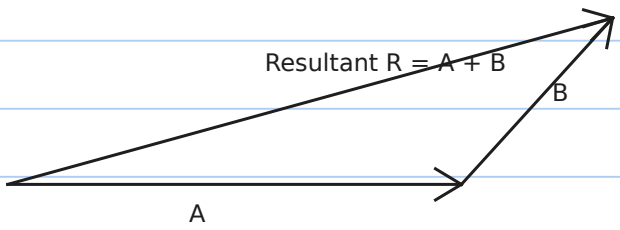
Step 3: Y is the midpoint of RS: y = ((0, 1, 1) + (2, 1, 0))/2 = (1, 1, 1/2)

Step 4: XY = y - x = (1 - 1/2, 1 - 1, 1/2 - 0) = (1/2, 0, 1/2)

Step 5: PS = (position vector of S) - (position vector of P) = (2 - 1, 1 - 1, 0 - (-1)) = (1, 0, 1), so (1/2)PS = (1/2, 0, 1/2)

Step 6: The two results in Steps 4 and 5 are identical

Final answer: Hence XY = (1/2)PS, as required.



Question: Exercise 3.2 Q11(i): Find the vector OA, where |OA| = 6 and OA is in the direction of the vector (2, -3, 6).

Answer: Step 1: OA is a scalar multiple of the given direction vector: OA = |OA| x (unit vector in that direction)

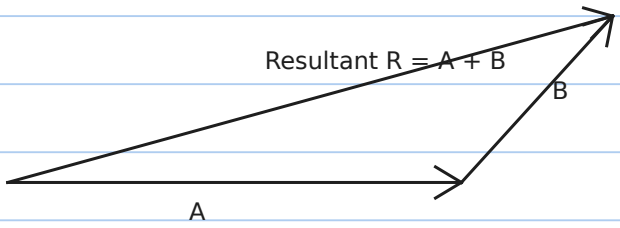
Step 2: The magnitude of the direction vector (2, -3, 6) is sqrt((2)^2 + (-3)^2 + (6)^2) = sqrt(4 + 9 + 36) = sqrt(49) = 7

Step 3: The unit vector in that direction is (2i - 3j + 6k)/7

Step 4: Multiply by the required magnitude 6: OA = 6(2i - 3j + 6k)/7 = (12i - 18j + 36k)/7

Step 5: Check: |OA| = (6/7) x 7 = 6, as required

Final answer: OA = (12/7)i - (18/7)j + (36/7)k.



Question: Exercise 3.2 Q11(ii): Find the vector OA, where |OA| = 2 and OA is in the opposite direction of the vector (8, 1, -4).

Answer: Step 1: Because the direction is opposite, OA = -|OA| x (unit vector in the direction of (8, 1, -4))

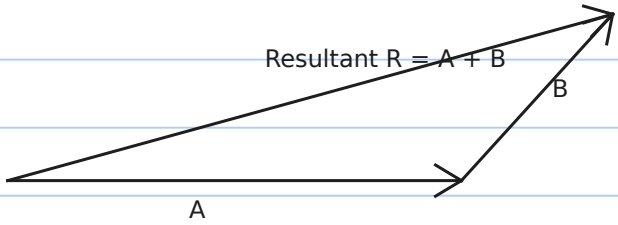
Step 2: The magnitude of (8, 1, -4) is sqrt((8)^2 + (1)^2 + (-4)^2) = sqrt(64 + 1 + 16) = sqrt(81) = 9

Step 3: The unit vector in the direction of (8, 1, -4) is (8i + j - 4k)/9

Step 4: Multiply by -2: OA = -2(8i + j - 4k)/9 = (-16i - 2j + 8k)/9

Step 5: Check: |OA| = (2/9) x 9 = 2, and every component has the opposite sign to the given vector, as required for the opposite direction

Final answer: OA = -(16/9)i - (2/9)j + (8/9)k.



Question: Exercise 3.2 Q12(a): Find the magnitude of the vector a = -2i + 3j - 4k.

Answer: Step 1: Use the magnitude formula |a| = sqrt(a1^2 + a2^2 + a3^2) for a = a1 i + a2 j + a3 k

Step 2: Identify the components: a1 = -2, a2 = 3, a3 = -4

Step 3: Square each component: (-2)^2 = 4, (3)^2 = 9, (-4)^2 = 16

Step 4: Add the squares: 4 + 9 + 16 = 29

Step 5: Take the square root: |a| = sqrt(29)

Final answer: |a| = sqrt(29).

Question: Exercise 3.2 Q12(b): Find the value of t if the vectors -3i + 5j - 6k and 2i - 3j + tk have the same magnitude.

Answer: Step 1: Equal magnitudes means the squares of the magnitudes are equal

Step 2: Square of the magnitude of the first vector: (-3)^2 + (5)^2 + (-6)^2 = 9 + 25 + 36 = 70

Step 3: Square of the magnitude of the second vector: (2)^2 + (-3)^2 + t^2 = 4 + 9 + t^2 = 13 + t^2

Step 4: Set them equal: 13 + t^2 = 70, so t^2 = 57

Step 5: Take square roots, keeping both signs: t = sqrt(57) or t = -sqrt(57)

Final answer: t = +-sqrt(57).

Question: Exercise 3.2 Q12(c): Find the vector whose magnitude is 5 times the length of a and is in the opposite direction of a = 4i - 5j + 6k.

Answer: Step 1: A vector of magnitude m in the opposite direction of a is -m x (a/|a|); here m = 5|a|, so the required vector is -5|a| x (a/|a|)

Step 2: The factor |a| cancels: -5|a| x (a/|a|) = -5a

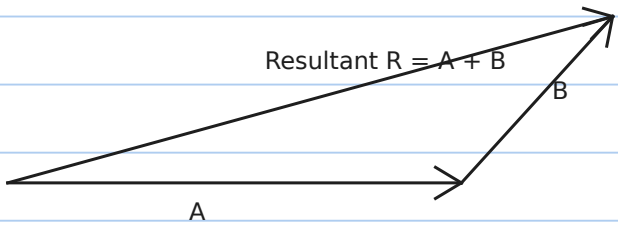
Step 3: Multiply a by -5 component-wise: i-component: -5 x 4 = -20

Step 4: j-component: -5 x (-5) = 25

Step 5: k-component: -5 x 6 = -30

Step 6: Combine, and note |a| = sqrt(16 + 25 + 36) = sqrt(77), so the result has magnitude 5sqrt(77), exactly 5 times the length of a, with all signs reversed

Final answer: The required vector is -20i + 25j - 30k.



Question: Exercise 3.2 Q12(d): Express a = -2i + 4j - 6k as a product of its magnitude and direction.

Answer: Step 1: Any vector can be written as a = |a| x (a-hat), the product of its magnitude and the unit vector (direction) along it

Step 2: Find the magnitude: |a| = sqrt((-2)^2 + (4)^2 + (-6)^2) = sqrt(4 + 16 + 36) = sqrt(56) = 2sqrt(14)

Step 3: Find the direction (unit vector): a-hat = a/|a| = (-2i + 4j - 6k)/(2sqrt(14)) = (-i + 2j - 3k)/sqrt(14)

Step 4: Write a as the product: a = 2sqrt(14) x [(-i + 2j - 3k)/sqrt(14)]

Step 5: Check by multiplying out: 2sqrt(14)/sqrt(14) = 2, and 2(-i + 2j - 3k) = -2i + 4j - 6k = a, which is correct

Final answer: a = 2sqrt(14) x [(-i + 2j - 3k)/sqrt(14)], a magnitude of 2sqrt(14) times the unit direction (-i + 2j - 3k)/sqrt(14).

Question: Exercise 3.2 Q13: Find the length of the median through point O of triangle OCD, if C and D are (2, 7, -1) and (4, 1, 2) respectively.

Answer: Step 1: The median through O goes from O to the midpoint of the opposite side CD

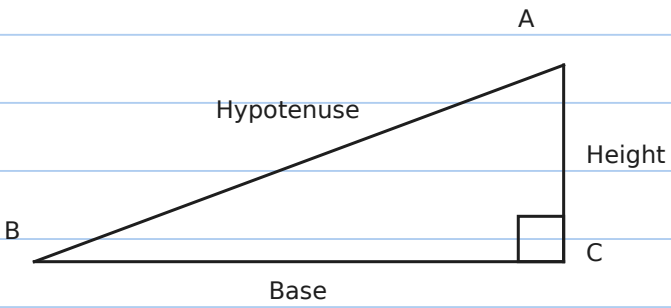
Step 2: Midpoint of CD = ((2 + 4)/2, (7 + 1)/2, (-1 + 2)/2) = (3, 4, 1/2)

Step 3: The median is the vector from O(0, 0, 0) to this midpoint, namely 3i + 4j + (1/2)k

Step 4: Its length is sqrt((3)^2 + (4)^2 + (1/2)^2) = sqrt(9 + 16 + 1/4) = sqrt(101/4)

Step 5: Simplify: sqrt(101/4) = sqrt(101)/2 (about 5.02 units)

Final answer: The length of the median is sqrt(101)/2 units.



Question: Exercise 3.2 Q14: If S and T are the midpoints of PR and QR

respectively, show that ST = (1/2)PQ. (P, Q, R are points with position vectors p, q, r.)

Answer: Step 1: Let the position vectors of P, Q, R be p, q, r; a midpoint is the average of the position vectors of the endpoints

Step 2: S is the midpoint of PR, so s = (p + r)/2

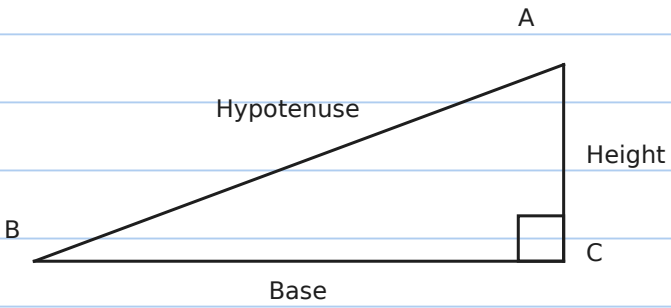
Step 3: T is the midpoint of QR, so t = (q + r)/2

Step 4: ST = t - s = (q + r)/2 - (p + r)/2 = (q + r - p - r)/2 = (q - p)/2

Step 5: But q - p is exactly the vector PQ, so ST = (1/2)(q - p) = (1/2)PQ

Step 6: This also shows ST is parallel to PQ and half its length, the midpoint theorem for a triangle

Final answer: Hence ST = (1/2)PQ.



Exercise 3.3

Question: Exercise 3.3 Q1(i): If a = -5i + 3j - 4k and b = 2i + 4j + 5k, verify that 0 + a = a + 0 = a, where 0 is the zero vector.

Answer: Step 1: The zero vector is 0 = 0i + 0j + 0k; vector addition is done component-wise

Step 2: Compute 0 + a component-wise: i: 0 + (-5) = -5; j: 0 + 3 = 3; k: 0

+ (-4) = -4

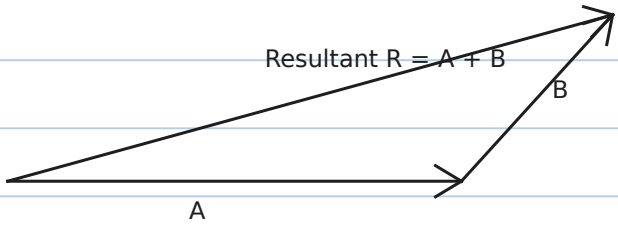
Step 3: So 0 + a = -5i + 3j - 4k = a

Step 4: Compute a + 0 component-wise: i: -5 + 0 = -5; j: 3 + 0 = 3; k: -4 + 0 = -4

Step 5: So a + 0 = -5i + 3j - 4k = a

Step 6: Both orders give a itself, which shows 0 behaves as the identity for vector addition

Final answer: 0 + a = a + 0 = a = -5i + 3j - 4k, verified.



Question: Exercise 3.3 Q1(ii): If a = -5i + 3j - 4k and b = 2i + 4j + 5k, find the additive inverses of a, b and 2a + 3b.

Answer: Step 1: The additive inverse of a vector is its negative: every component changes sign

Step 2: Inverse of a: -a = -(-5i + 3j - 4k) = 5i - 3j + 4k, and indeed a + (-a) = 0

Step 3: Inverse of b: -b = -(2i + 4j + 5k) = -2i - 4j - 5k

Step 4: First compute 2a + 3b: 2a = -10i + 6j - 8k and 3b = 6i + 12j + 15k, so 2a + 3b = (-10 + 6)i + (6 + 12)j + (-8 + 15)k = -4i + 18j + 7k

Step 5: Its additive inverse is found by changing every sign: -(2a + 3b) = 4i - 18j - 7k

Step 6: Check: (-4i + 18j + 7k) + (4i - 18j - 7k) = 0, as an inverse must satisfy

Final answer: The additive inverses are: of a: 5i - 3j + 4k; of b: -2i - 4j - 5k; of 2a + 3b: 4i - 18j - 7k.

Question: Exercise 3.3 Q1(iii): If a = -5i + 3j - 4k and b = 2i + 4j + 5k, verify the commutative law for vector addition.

Answer: Step 1: The commutative law claims a + b = b + a; compute both sides component-wise

Step 2: a + b, i-components: -5 + 2 = -3

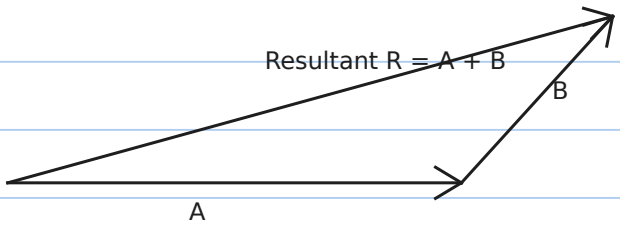
Step 3: a + b, j-components: 3 + 4 = 7

Step 4: a + b, k-components: -4 + 5 = 1, so a + b = -3i + 7j + k

Step 5: b + a, i-components: 2 - 5 = -3; j-components: 4 + 3 = 7; k-components: 5 - 4 = 1, so b + a = -3i + 7j + k

Step 6: The two sides are identical because ordinary addition of numbers is commutative in each component

Final answer: a + b = b + a = -3i + 7j + k, so the commutative law is verified.



Question: Exercise 3.3 Q2: Let u = 3i + j + 5k, v = i - 2j and w = -i + 3j - k. Verify the associative law of vector addition.

Answer: Step 1: The associative law claims u + (v + w) = (u + v) + w;

write v with all components: v = 1i - 2j + 0k

Step 2: Compute v + w: i: 1 - 1 = 0; j: -2 + 3 = 1; k: 0 - 1 = -1, so v + w = 0i + 1j - 1k

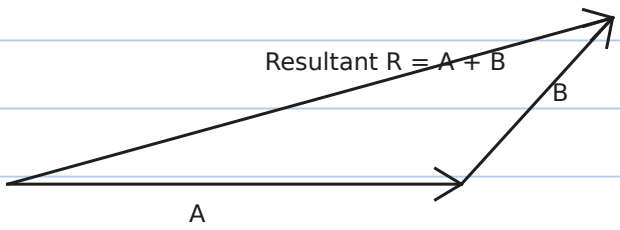
Step 3: Left side: u + (v + w): i: 3 + 0 = 3; j: 1 + 1 = 2; k: 5 - 1 = 4, so u + (v + w) = 3i + 2j + 4k

Step 4: Compute u + v: i: 3 + 1 = 4; j: 1 - 2 = -1; k: 5 + 0 = 5, so u + v = 4i - j + 5k

Step 5: Right side: (u + v) + w: i: 4 - 1 = 3; j: -1 + 3 = 2; k: 5 - 1 = 4, so (u + v) + w = 3i + 2j + 4k

Step 6: The left and right sides are equal

Final answer: u + (v + w) = (u + v) + w = 3i + 2j + 4k, so the associative law is verified.



Question: Exercise 3.3 Q3: Let a = 3i + 2j + 5k be a vector and m = 5 be a scalar. Verify the commutative law of scalar multiplication.

Answer: Step 1: The commutative law for scalar multiplication claims m(a) = (a)m, that is the scalar may be written on either side

Step 2: Compute m(a) = 5(3i + 2j + 5k) by multiplying each component by 5

Step 3: i-component: 5 x 3 = 15; j-component: 5 x 2 = 10; k-component: 5 x 5 = 25

Step 4: So m(a) = 15i + 10j + 25k

Step 5: Compute (a)m = (3i + 2j + 5k) x 5 in exactly the same way: each component is again multiplied by 5, giving 15i + 10j + 25k

Step 6: Both sides agree, because multiplying components by a number does not depend on the side on which the scalar is written

Final answer: m(a) = (a)m = 15i + 10j + 25k, so the commutative law of scalar multiplication is verified.

Question: Exercise 3.3 Q4: Let a = 5i + 2j - 3k be a vector and m = 2, n = 5 be scalars. Verify the associative law of scalar multiplication.

Answer: Step 1: The associative law claims m(n(a)) = (mn)(a); compute both sides

Step 2: First n(a) = 5(5i + 2j - 3k) = 25i + 10j - 15k

Step 3: Then m(n(a)) = 2(25i + 10j - 15k) = 50i + 20j - 30k

Step 4: On the other side, mn = 2 x 5 = 10

Step 5: (mn)(a) = 10(5i + 2j - 3k) = 50i + 20j - 30k

Step 6: Both sides are equal, since each component ends up multiplied by 2 x 5 = 10 either way

Final answer: m(n(a)) = (mn)(a) = 50i + 20j - 30k, so the associative law of scalar multiplication is verified.

Question: Exercise 3.3 Q5(i): Let a = 3i + j + 5k and b = -5i + 3j - 4k be two vectors, and m = 3, n = 7 be scalars. Prove that m(a + b) = ma + mb.

Answer: Step 1: First compute a + b component-wise: i: 3 - 5 = -2; j: 1 + 3 = 4; k: 5 - 4 = 1, so a + b = -2i + 4j + k

Step 2: Left side: m(a + b) = 3(-2i + 4j + k) = -6i + 12j + 3k

Step 3: Compute ma = 3(3i + j + 5k) = 9i + 3j + 15k

Step 4: Compute mb = 3(-5i + 3j - 4k) = -15i + 9j - 12k

Step 5: Right side: ma + mb: i: 9 - 15 = -6; j: 3 + 9 = 12; k: 15 - 12 = 3, so ma + mb = -6i + 12j + 3k

Step 6: The left side of Step 2 and the right side of Step 5 are identical

Final answer: m(a + b) = ma + mb = -6i + 12j + 3k, hence proved.

Question: Exercise 3.3 Q5(ii): Let a = 3i + j + 5k and b = -5i + 3j - 4k be two vectors, and m = 3, n = 7 be scalars. Prove that (m + n)a = ma + na.

Answer: Step 1: Left side: m + n = 3 + 7 = 10, so (m + n)a = 10(3i + j + 5k)

Step 2: Multiply each component by 10: (m + n)a = 30i + 10j + 50k

Step 3: Compute ma = 3(3i + j + 5k) = 9i + 3j + 15k

Step 4: Compute na = 7(3i + j + 5k) = 21i + 7j + 35k

Step 5: Right side: ma + na: i: 9 + 21 = 30; j: 3 + 7 = 10; k: 15 + 35 = 50, so ma + na = 30i + 10j + 50k

Step 6: This matches the left side of Step 2 exactly

Final answer: (m + n)a = ma + na = 30i + 10j + 50k, hence proved.

Exercise 3.4

Question: Exercise 3.4 Q1(i): If a = 2i + 3j + 5k, b = i - 2j - 3k and c = 5i + j - k, find a.b.

Answer: Step 1: The dot product multiplies corresponding components and

adds the products: a.b = a1b1 + a2b2 + a3b3

Step 2: i-components product: 2 x 1 = 2

Step 3: j-components product: 3 x (-2) = -6

Step 4: k-components product: 5 x (-3) = -15

Step 5: Add the three products term by term: 2 - 6 - 15 = -19

Step 6: The dot product is a scalar, so the answer is a single number with no i, j, k parts

Final answer: a.b = -19.

Question: Exercise 3.4 Q1(ii): If a = 2i + 3j + 5k, b = i - 2j - 3k and c = 5i + j - k, find a.(b + c).

Answer: Step 1: First add b and c component-wise: i: 1 + 5 = 6; j: -2 + 1 = -1; k: -3 - 1 = -4, so b + c = 6i - j - 4k

Step 2: Now take the dot product a.(b + c) = a1(b+c)1 + a2(b+c)2 + a3(b+c)3

Step 3: i-components product: 2 x 6 = 12

Step 4: j-components product: 3 x (-1) = -3

Step 5: k-components product: 5 x (-4) = -20

Step 6: Add the products: 12 - 3 - 20 = -11

Final answer: a.(b + c) = -11.

Question: Exercise 3.4 Q1(iii): If a = 2i + 3j + 5k, b = i - 2j - 3k and c = 5i + j - k, find (2a + 3b).(a - 2b).

Answer: Step 1: Compute 2a = 4i + 6j + 10k and 3b = 3i - 6j - 9k

Step 2: Add them: 2a + 3b = (4 + 3)i + (6 - 6)j + (10 - 9)k = 7i + 0j + 1k

= 7i + k

Step 3: Compute a - 2b: 2b = 2i - 4j - 6k, so a - 2b = (2 - 2)i + (3 - (-4))j + (5 - (-6))k = 0i + 7j + 11k = 7j + 11k

Step 4: Take the dot product term by term: i: 7 x 0 = 0

Step 5: j: 0 x 7 = 0

Step 6: k: 1 x 11 = 11

Step 7: Add the products: 0 + 0 + 11 = 11

Final answer: (2a + 3b).(a - 2b) = 11.

Question: Exercise 3.4 Q2(i): If a = 5i - j - 3k and b = i + 3j - 5k, show that a.b = b.a.

Answer: Step 1: Compute a.b by multiplying corresponding components: i: 5 x 1 = 5

Step 2: j: (-1) x 3 = -3

Step 3: k: (-3) x (-5) = 15

Step 4: Add: a.b = 5 - 3 + 15 = 17

Step 5: Compute b.a in the reverse order: i: 1 x 5 = 5; j: 3 x (-1) = -3; k: (-5) x (-3) = 15

Step 6: Add: b.a = 5 - 3 + 15 = 17

Step 7: Both orders give the same three products because multiplication of numbers is commutative

Final answer: a.b = b.a = 17, hence shown.

Question: Exercise 3.4 Q2(ii): If a = 5i - j - 3k and b = i + 3j - 5k, show that the vectors a + b and a - b are perpendicular.

Answer: Step 1: Two vectors are perpendicular (orthogonal) exactly when their dot product is zero, so compute a + b and a - b first

Step 2: a + b: i: 5 + 1 = 6; j: -1 + 3 = 2; k: -3 - 5 = -8, so a + b = 6i + 2j - 8k

Step 3: a - b: i: 5 - 1 = 4; j: -1 - 3 = -4; k: -3 - (-5) = 2, so a - b = 4i - 4j + 2k

Step 4: Dot product, i-term: 6 x 4 = 24

Step 5: Dot product, j-term: 2 x (-4) = -8

Step 6: Dot product, k-term: (-8) x 2 = -16

Step 7: Add: 24 - 8 - 16 = 0

Final answer: (a + b).(a - b) = 0, hence a + b and a - b are perpendicular.

Question: Exercise 3.4 Q3(i): Check whether a = 2i - 3j + 2k and b = 5i + 8j + 7k are orthogonal or not.

Answer: Step 1: Two non-zero vectors are orthogonal exactly when their dot product is zero

Step 2: i-term of the dot product: 2 x 5 = 10

Step 3: j-term: (-3) x 8 = -24

Step 4: k-term: 2 x 7 = 14

Step 5: Add term by term: 10 - 24 + 14 = 0

Step 6: Since the dot product is zero, the angle between the vectors is 90 deg

Final answer: a.b = 0, so a and b are orthogonal (perpendicular).

Question: Exercise 3.4 Q3(ii): Check whether a = i + 3j - 4k and b = -2i + j

+ 3k are orthogonal or not.

Answer: Step 1: Two non-zero vectors are orthogonal exactly when their dot product is zero

Step 2: i-term of the dot product: 1 x (-2) = -2

Step 3: j-term: 3 x 1 = 3

Step 4: k-term: (-4) x 3 = -12

Step 5: Add term by term: -2 + 3 - 12 = -11

Step 6: Since the dot product is -11, which is not zero, the angle between the vectors is not 90 deg

Final answer: a.b = -11, not equal to 0, so a and b are not orthogonal.

Question: Exercise 3.4 Q4(i): Verify the distributive law for the dot product for a = i - j + 2k, b = 2i + 3j - k and c = 5i - j + 2k.

Answer: Step 1: The distributive law claims a.(b + c) = a.b + a.c; first compute b + c = (2 + 5)i + (3 - 1)j + (-1 + 2)k = 7i + 2j + k

Step 2: Left side: a.(b + c): i: 1 x 7 = 7; j: (-1) x 2 = -2; k: 2 x 1 = 2; total 7 - 2 + 2 = 7

Step 3: Compute a.b: i: 1 x 2 = 2; j: (-1) x 3 = -3; k: 2 x (-1) = -2; total 2 - 3 - 2 = -3

Step 4: Compute a.c: i: 1 x 5 = 5; j: (-1) x (-1) = 1; k: 2 x 2 = 4; total 5 + 1 + 4 = 10

Step 5: Right side: a.b + a.c = -3 + 10 = 7

Step 6: Left side (7) equals right side (7)

Final answer: a.(b + c) = a.b + a.c = 7, so the distributive law is verified.

Question: Exercise 3.4 Q4(ii): Verify the distributive law for the dot product for a = 3i - 5j + 7k, b = -4i + j + k and c = 6i + j - 2k.

Answer: Step 1: The distributive law claims a.(b + c) = a.b + a.c; first compute b + c = (-4 + 6)i + (1 + 1)j + (1 - 2)k = 2i + 2j - k

Step 2: Left side: a.(b + c): i: 3 x 2 = 6; j: (-5) x 2 = -10; k: 7 x (-1) = -7; total 6 - 10 - 7 = -11

Step 3: Compute a.b: i: 3 x (-4) = -12; j: (-5) x 1 = -5; k: 7 x 1 = 7; total -12 - 5 + 7 = -10

Step 4: Compute a.c: i: 3 x 6 = 18; j: (-5) x 1 = -5; k: 7 x (-2) = -14; total 18 - 5 - 14 = -1

Step 5: Right side: a.b + a.c = -10 + (-1) = -11

Step 6: Left side (-11) equals right side (-11)

Final answer: a.(b + c) = a.b + a.c = -11, so the distributive law is verified.

Question: Exercise 3.4 Q5(i): Find the angle between the vectors i - 2j + 3k and 3i - 2j + k.

Answer: Step 1: Use cos(theta) = (a.b)/(|a||b|), so compute the dot product and both magnitudes

Step 2: Dot product term by term: i: 1 x 3 = 3; j: (-2) x (-2) = 4; k: 3 x 1 = 3; so a.b = 3 + 4 + 3 = 10

Step 3: |a| = sqrt(1 + 4 + 9) = sqrt(14)

Step 4: |b| = sqrt(9 + 4 + 1) = sqrt(14)

Step 5: cos(theta) = 10/(sqrt(14) x sqrt(14)) = 10/14 = 5/7

Step 6: theta = cos^-1(5/7), which is approximately 44.4 deg

Final answer: The angle between the vectors is cos^-1(5/7), approximately

44.4 deg.

Question: Exercise 3.4 Q5(ii): Find the angle between the vectors 3i + 5j - 3k and i - 4j + k.

Answer: Step 1: Use cos(theta) = (a.b)/(|a||b|), so compute the dot product and both magnitudes

Step 2: Dot product term by term: i: 3 x 1 = 3; j: 5 x (-4) = -20; k: (-3) x 1 = -3; so a.b = 3 - 20 - 3 = -20

Step 3: |a| = sqrt(9 + 25 + 9) = sqrt(43)

Step 4: |b| = sqrt(1 + 16 + 1) = sqrt(18) = 3sqrt(2)

Step 5: cos(theta) = -20/(sqrt(43) x sqrt(18)) = -20/sqrt(774), approximately -0.719

Step 6: The negative cosine shows the angle is obtuse; theta = cos^-1(-20/sqrt(774)), approximately 136.0 deg

Final answer: The angle between the vectors is cos^-1(-20/sqrt(774)), approximately 136.0 deg.

Question: Exercise 3.4 Q6(i): Find the direction cosines and direction angles of the vector 2i + j - 3k.

Answer: Step 1: The direction cosines are the components divided by the magnitude: cos(alpha) = a1/|a|, cos(beta) = a2/|a|, cos(gamma) = a3/|a|

Step 2: |a| = sqrt((2)^2 + (1)^2 + (-3)^2) = sqrt(4 + 1 + 9) = sqrt(14)

Step 3: cos(alpha) = 2/sqrt(14), so alpha = cos^-1(2/sqrt(14)) which is approximately 57.7 deg

Step 4: cos(beta) = 1/sqrt(14), so beta = cos^-1(1/sqrt(14)) which is approximately 74.5 deg

Step 5: cos(gamma) = -3/sqrt(14), so gamma = cos^-1(-3/sqrt(14)) which is approximately 143.3 deg

Step 6: Check: cos^2(alpha) + cos^2(beta) + cos^2(gamma) = (4 + 1 + 9)/14 = 1, as it must be

Final answer: Direction cosines are 2/sqrt(14), 1/sqrt(14), -3/sqrt(14); direction angles are approximately alpha = 57.7 deg, beta = 74.5 deg, gamma = 143.3 deg.

Question: Exercise 3.4 Q6(ii): Find the direction cosines and direction angles of the vector 2i - 4j + 5k.

Answer: Step 1: The direction cosines are the components divided by the magnitude: cos(alpha) = a1/|a|, cos(beta) = a2/|a|, cos(gamma) = a3/|a|

Step 2: |a| = sqrt((2)^2 + (-4)^2 + (5)^2) = sqrt(4 + 16 + 25) = sqrt(45) = 3sqrt(5)

Step 3: cos(alpha) = 2/sqrt(45), so alpha = cos^-1(2/sqrt(45)) which is approximately 72.7 deg

Step 4: cos(beta) = -4/sqrt(45), so beta = cos^-1(-4/sqrt(45)) which is approximately 126.6 deg

Step 5: cos(gamma) = 5/sqrt(45), so gamma = cos^-1(5/sqrt(45)) which is approximately 41.8 deg

Step 6: Check: cos^2(alpha) + cos^2(beta) + cos^2(gamma) = (4 + 16 + 25)/45 = 1, as it must be

Final answer: Direction cosines are 2/sqrt(45), -4/sqrt(45), 5/sqrt(45); direction angles are approximately alpha = 72.7 deg, beta = 126.6 deg, gamma = 41.8 deg.

Question: Exercise 3.4 Q6(iii): Find the direction cosines and direction

angles of the vector -3i + 4j + 5k.

Answer: Step 1: The direction cosines are the components divided by the magnitude: cos(alpha) = a1/|a|, cos(beta) = a2/|a|, cos(gamma) = a3/|a|

Step 2: |a| = sqrt((-3)^2 + (4)^2 + (5)^2) = sqrt(9 + 16 + 25) = sqrt(50) = 5sqrt(2)

Step 3: cos(alpha) = -3/sqrt(50), so alpha = cos^-1(-3/sqrt(50)) which is approximately 115.1 deg

Step 4: cos(beta) = 4/sqrt(50), so beta = cos^-1(4/sqrt(50)) which is approximately 55.6 deg

Step 5: cos(gamma) = 5/sqrt(50) = 1/sqrt(2), so gamma = cos^-1(1/sqrt(2)) = 45 deg exactly

Step 6: Check: cos^2(alpha) + cos^2(beta) + cos^2(gamma) = (9 + 16 + 25)/50 = 1, as it must be

Final answer: Direction cosines are -3/sqrt(50), 4/sqrt(50), 5/sqrt(50); direction angles are approximately alpha = 115.1 deg, beta = 55.6 deg, gamma = 45 deg.

Question: Exercise 3.4 Q7: If alpha, beta, gamma are the direction angles of a vector, show that sin^2(alpha) + sin^2(beta) + sin^2(gamma) = 2.

Answer: Step 1: Start from the standard identity for direction cosines: cos^2(alpha) + cos^2(beta) + cos^2(gamma) = 1, which holds because the direction cosines are the components of a unit vector

Step 2: For each angle use sin^2(x) = 1 - cos^2(x)

Step 3: Add the three relations: sin^2(alpha) + sin^2(beta) + sin^2(gamma) = (1 - cos^2(alpha)) + (1 - cos^2(beta)) + (1 - cos^2(gamma))

Step 4: Group the terms: = 3 - (cos^2(alpha) + cos^2(beta) + cos^2(gamma))

Step 5: Substitute the identity of Step 1: = 3 - 1 = 2

Final answer: Hence sin^2(alpha) + sin^2(beta) + sin^2(gamma) = 2, as required.

Question: Exercise 3.4 Q8: If the measures of two of the direction angles of a vector are 45 deg and 60 deg, find the measure of the third angle.

Answer: Step 1: Let the third direction angle be gamma; use cos^2(alpha) + cos^2(beta) + cos^2(gamma) = 1 with alpha = 45 deg and beta = 60 deg

Step 2: cos(45 deg) = 1/sqrt(2), so cos^2(45 deg) = 1/2

Step 3: cos(60 deg) = 1/2, so cos^2(60 deg) = 1/4

Step 4: Substitute: 1/2 + 1/4 + cos^2(gamma) = 1, so cos^2(gamma) = 1 - 3/4 = 1/4

Step 5: Take square roots: cos(gamma) = 1/2 or cos(gamma) = -1/2

Step 6: Direction angles lie between 0 deg and 180 deg, so gamma = 60 deg (from cos = 1/2) or gamma = 120 deg (from cos = -1/2); both are possible because the squares alone cannot fix the sign of the third component

Final answer: The third direction angle is 60 deg or 120 deg.

Question: Exercise 3.4 Q9: Find the work done by the force F = 7i + 9j - 11k in moving an object along a straight line from (4, 2, 7) to (6, 4, 9).

Answer: Step 1: Work done by a constant force is W = F.d, where d is the displacement vector

Step 2: Find the displacement: d = (6 - 4)i + (4 - 2)j + (9 - 7)k = 2i + 2j + 2k

Step 3: Take the dot product term by term: i: 7 x 2 = 14

Step 4: j: 9 x 2 = 18

Step 5: k: (-11) x 2 = -22

Step 6: Add the products: W = 14 + 18 - 22 = 10

Final answer: The work done is 10 Joules (units of work).

Question: Exercise 3.4 Q10(i): Calculate the projection of a along b and the projection of b along a when a = i - 2j + 3k and b = 3i - j - 5k.

Answer: Step 1: The projection of a along b is (a.b)/|b| and the projection of b along a is (a.b)/|a|, so first find a.b

Step 2: a.b term by term: i: 1 x 3 = 3; j: (-2) x (-1) = 2; k: 3 x (-5) = -15; so a.b = 3 + 2 - 15 = -10

Step 3: |b| = sqrt((3)^2 + (-1)^2 + (-5)^2) = sqrt(9 + 1 + 25) = sqrt(35)

Step 4: Projection of a along b = -10/sqrt(35), approximately -1.69; the negative sign means the component of a along b points opposite to b

Step 5: |a| = sqrt((1)^2 + (-2)^2 + (3)^2) = sqrt(1 + 4 + 9) = sqrt(14)

Step 6: Projection of b along a = -10/sqrt(14), approximately -2.67

Final answer: Projection of a along b is -10/sqrt(35) (approx. -1.69) and projection of b along a is -10/sqrt(14) (approx. -2.67).

Question: Exercise 3.4 Q10(ii): Calculate the projection of a along b and the projection of b along a when a = 2i + 3j + 2k and b = i + 2j + k.

Answer: Step 1: The projection of a along b is (a.b)/|b| and the projection of b along a is (a.b)/|a|, so first find a.b

Step 2: a.b term by term: i: 2 x 1 = 2; j: 3 x 2 = 6; k: 2 x 1 = 2; so a.b = 2 + 6 + 2 = 10

Step 3: |b| = sqrt((1)^2 + (2)^2 + (1)^2) = sqrt(1 + 4 + 1) = sqrt(6)

Step 4: Projection of a along b = 10/sqrt(6), approximately 4.08

Step 5: |a| = sqrt((2)^2 + (3)^2 + (2)^2) = sqrt(4 + 9 + 4) = sqrt(17)

Step 6: Projection of b along a = 10/sqrt(17), approximately 2.43

Final answer: Projection of a along b is 10/sqrt(6) (approx. 4.08) and projection of b along a is 10/sqrt(17) (approx. 2.43).

Question: Exercise 3.4 Q11: Calculate the work done by a force F = 3i + 4j + 5k in displacing a body from position B to position A along a straight path, where the position vectors of A and B are rA = 2i + 5j - 2k and rB = 7i + 3j - 5k respectively.

Answer: Step 1: Work done by a constant force is W = F.d, where d is the displacement from the starting point B to the finishing point A

Step 2: Displacement d = rA - rB = (2 - 7)i + (5 - 3)j + (-2 - (-5))k = -5i + 2j + 3k

Step 3: Take the dot product F.d term by term: i: 3 x (-5) = -15

Step 4: j: 4 x 2 = 8

Step 5: k: 5 x 3 = 15

Step 6: Add the products: W = -15 + 8 + 15 = 8

Final answer: The work done is 8 Joules (units of work).

Exercise 3.5

Question: Exercise 3.5 Q1(i): Find a x b and b x a if a = 4i + j + k and b = i + 2j + 3k.

Answer: Step 1: Use the determinant form: a x b = i(a2b3 - a3b2) - j(a1b3 - a3b1) + k(a1b2 - a2b1), with (a1, a2, a3) = (4, 1, 1) and (b1, b2, b3) = (1, 2, 3)

Step 2: i-component: a2b3 - a3b2 = (1)(3) - (1)(2) = 3 - 2 = 1

Step 3: j-component: -(a1b3 - a3b1) = -(4 x 3 - 1 x 1) = -(12 - 1) = -11

Step 4: k-component: a1b2 - a2b1 = (4)(2) - (1)(1) = 8 - 1 = 7

Step 5: Combine: a x b = 1i - 11j + 7k

Step 6: The reverse product only changes the sign: b x a = -(a x b) = -1i + 11j - 7k

Final answer: a x b = i - 11j + 7k and b x a = -i + 11j - 7k.

Question: Exercise 3.5 Q1(ii): Find a x b and b x a if a = -2i + 5j - 3k and b = 3i + 2j + k.

Answer: Step 1: Use the determinant form: a x b = i(a2b3 - a3b2) - j(a1b3 - a3b1) + k(a1b2 - a2b1), with (a1, a2, a3) = (-2, 5, -3) and (b1, b2, b3) = (3, 2, 1)

Step 2: i-component: a2b3 - a3b2 = (5)(1) - (-3)(2) = 5 + 6 = 11

Step 3: j-component: -(a1b3 - a3b1) = -((-2)(1) - (-3)(3)) = -(-2 + 9) = -7

Step 4: k-component: a1b2 - a2b1 = (-2)(2) - (5)(3) = -4 - 15 = -19

Step 5: Combine: a x b = 11i - 7j - 19k

Step 6: The reverse product only changes the sign: b x a = -(a x b) = -11i + 7j + 19k

Final answer: a x b = 11i - 7j - 19k and b x a = -11i + 7j + 19k.

Question: Exercise 3.5 Q2: Verify that a x b = -(b x a) for a = i - 7j + 7k and b = 3i - 2j + 2k.

Answer: Step 1: Compute a x b by the determinant form with (a1, a2, a3) = (1, -7, 7) and (b1, b2, b3) = (3, -2, 2)

Step 2: i-component: a2b3 - a3b2 = (-7)(2) - (7)(-2) = -14 + 14 = 0

Step 3: j-component: -(a1b3 - a3b1) = -(1 x 2 - 7 x 3) = -(2 - 21) = 19

Step 4: k-component: a1b2 - a2b1 = (1)(-2) - (-7)(3) = -2 + 21 = 19

Step 5: So a x b = 0i + 19j + 19k = 19j + 19k

Step 6: Compute b x a: i-component: b2a3 - b3a2 = (-2)(7) - (2)(-7) = -14 + 14 = 0; j-component: -(b1a3 - b3a1) = -(3 x 7 - 2 x 1) = -(21 - 2) = -19; k-component: b1a2 - b2a1 = (3)(-7) - (-2)(1) = -21 + 2 = -19

Step 7: So b x a = -19j - 19k = -(19j + 19k) = -(a x b)

Final answer: a x b = 19j + 19k and b x a = -19j - 19k, hence a x b = -(b x a) is verified.

Question: Exercise 3.5 Q3: Find a unit vector which is orthogonal to both the vectors a = i - 2j + 3k and b = 3i - 2j + k.

Answer: Step 1: A vector orthogonal to both a and b is given by the cross product a x b; a unit vector in that direction is then (a x b)/|a x b|

Step 2: i-component of a x b: a2b3 - a3b2 = (-2)(1) - (3)(-2) = -2 + 6 = 4

Step 3: j-component: -(a1b3 - a3b1) = -(1 x 1 - 3 x 3) = -(1 - 9) = 8

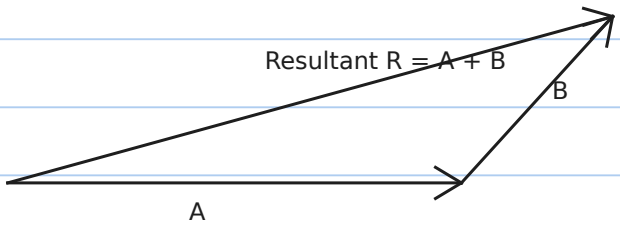
Step 4: k-component: a1b2 - a2b1 = (1)(-2) - (-2)(3) = -2 + 6 = 4

Step 5: So a x b = 4i + 8j + 4k, and |a x b| = sqrt(16 + 64 + 16) = sqrt(96) = 4sqrt(6)

Step 6: Divide by the magnitude: (4i + 8j + 4k)/(4sqrt(6)) = (i + 2j + k)/sqrt(6)

Step 7: The opposite vector -(i + 2j + k)/sqrt(6) is also a unit vector orthogonal to both, since reversing direction keeps it perpendicular to the same plane

Final answer: A unit vector orthogonal to both is (i + 2j + k)/sqrt(6) (and also its negative).



Question: Exercise 3.5 Q4: Find a unit vector perpendicular to each of the vectors a + b and a - b, where a = 3i + 5j - k and b = i + 2j - 5k.

Answer: Step 1: First compute the two vectors: a + b = (3 + 1)i + (5 + 2)j + (-1 - 5)k = 4i + 7j - 6k

Step 2: a - b = (3 - 1)i + (5 - 2)j + (-1 - (-5))k = 2i + 3j + 4k

Step 3: A vector perpendicular to both is their cross product (a + b) x (a - b); compute its i-component: (7)(4) - (-6)(3) = 28 + 18 = 46

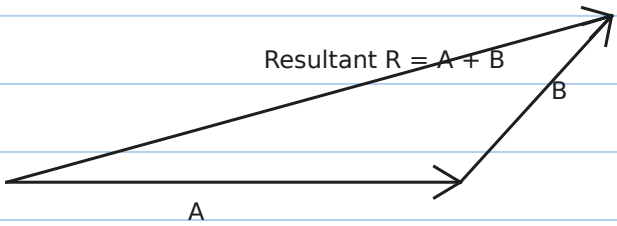
Step 4: j-component: -((4)(4) - (-6)(2)) = -(16 + 12) = -28

Step 5: k-component: (4)(3) - (7)(2) = 12 - 14 = -2

Step 6: So (a + b) x (a - b) = 46i - 28j - 2k, with magnitude sqrt(46^2 + (-28)^2 + (-2)^2) = sqrt(2116 + 784 + 4) = sqrt(2904) = 2sqrt(726)

Step 7: Divide by the magnitude and simplify by 2: (46i - 28j - 2k)/(2sqrt(726)) = (23i - 14j - k)/sqrt(726)

Final answer: A unit vector perpendicular to both a + b and a - b is (23i - 14j - k)/sqrt(726) (and also its negative).



Question: Exercise 3.5 Q5: Find the area of the parallelogram whose adjacent sides are represented by a = 5i - j + 3k and b = i + k.

Answer: Step 1: The area of the parallelogram with adjacent sides a and b is |a x b|

Step 2: Write b with all components: b = 1i + 0j + 1k, so (a1, a2, a3) = (5, -1, 3) and (b1, b2, b3) = (1, 0, 1)

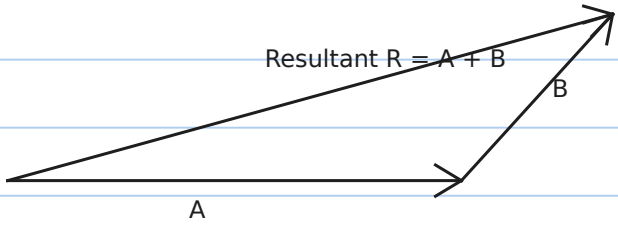
Step 3: i-component of a x b: a2b3 - a3b2 = (-1)(1) - (3)(0) = -1 - 0 = -1

Step 4: j-component: -(a1b3 - a3b1) = -(5 x 1 - 3 x 1) = -(5 - 3) = -2

Step 5: k-component: a1b2 - a2b1 = (5)(0) - (-1)(1) = 0 + 1 = 1

Step 6: So a x b = -i - 2j + k, and the area is |a x b| = sqrt((-1)^2 + (-2)^2 + (1)^2) = sqrt(1 + 4 + 1) = sqrt(6)

Final answer: The area of the parallelogram is sqrt(6) square units (approx. 2.45 square units).



Question: Exercise 3.5 Q6: Find the area of the triangle with vertices A(1, 1, 2), B(2, 3, 5) and C(1, 5, 5).

Answer: Step 1: Form two side vectors from the same vertex A: AB = B - A and AC = C - A

Step 2: AB = (2 - 1, 3 - 1, 5 - 2) = (1, 2, 3), that is AB = i + 2j + 3k

Step 3: AC = (1 - 1, 5 - 1, 5 - 2) = (0, 4, 3), that is AC = 0i + 4j + 3k

Step 4: The area of the triangle is (1/2)|AB x AC|; compute the cross product: i-component: (2)(3) - (3)(4) = 6 - 12 = -6

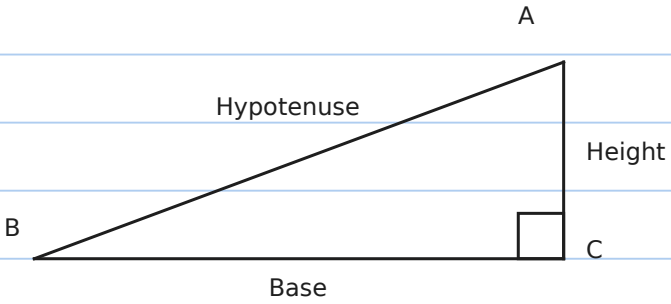
Step 5: j-component: -((1)(3) - (3)(0)) = -(3 - 0) = -3

Step 6: k-component: (1)(4) - (2)(0) = 4 - 0 = 4

Step 7: So AB x AC = -6i - 3j + 4k, with magnitude sqrt(36 + 9 + 16) = sqrt(61)

Step 8: Area = (1/2)sqrt(61)

Final answer: The area of the triangle is (1/2)sqrt(61) square units (approx. 3.91 square units).



Question: Exercise 3.5 Q7: If a = 2i + j + 7k, b = i - 3j + 5k and c = -i + 5j + k, verify that a x (b + c) = (a x b) + (a x c).

Answer: Step 1: First compute b + c = (1 - 1)i + (-3 + 5)j + (5 + 1)k = 0i + 2j + 6k

Step 2: Left side, a x (b + c): i-component: (1)(6) - (7)(2) = 6 - 14 = -8

Step 3: j-component: -((2)(6) - (7)(0)) = -(12 - 0) = -12

Step 4: k-component: (2)(2) - (1)(0) = 4 - 0 = 4, so a x (b + c) = -8i - 12j + 4k

Step 5: Compute a x b: i: (1)(5) - (7)(-3) = 5 + 21 = 26; j: -((2)(5) - (7)(1)) = -(10 - 7) = -3; k: (2)(-3) - (1)(1) = -6 - 1 = -7, so a x b = 26i - 3j - 7k

Step 6: Compute a x c: i: (1)(1) - (7)(5) = 1 - 35 = -34; j: -((2)(1) - (7)(-1)) = -(2 + 7) = -9; k: (2)(5) - (1)(-1) = 10 + 1 = 11, so a x c = -34i - 9j + 11k

Step 7: Right side: (a x b) + (a x c) = (26 - 34)i + (-3 - 9)j + (-7 + 11)k = -8i - 12j + 4k

Step 8: The right side equals the left side found in Step 4

Final answer: a x (b + c) = (a x b) + (a x c) = -8i - 12j + 4k, hence verified.

Question: Exercise 3.5 Q8: Using the cross product, find the angle between i - 7j + 7k and 3i - 2j + 2k.

Answer: Step 1: From the definition a x b = |a||b| sin(theta) n, we get sin(theta) = |a x b|/(|a||b|); call the vectors a = i - 7j + 7k and b = 3i - 2j +

2k

Step 2: Compute a x b: i-component: (-7)(2) - (7)(-2) = -14 + 14 = 0

Step 3: j-component: -((1)(2) - (7)(3)) = -(2 - 21) = 19

Step 4: k-component: (1)(-2) - (-7)(3) = -2 + 21 = 19, so a x b = 19j + 19k

Step 5: |a x b| = sqrt(0 + 361 + 361) = sqrt(722) = 19sqrt(2)

Step 6: |a| = sqrt(1 + 49 + 49) = sqrt(99) = 3sqrt(11), and |b| = sqrt(9 + 4 + 4) = sqrt(17)

Step 7: sin(theta) = 19sqrt(2)/(3sqrt(11) x sqrt(17)) = 19sqrt(2)/sqrt(1683), approximately 0.655

Step 8: sin alone leaves two possible angles (theta and 180 deg - theta); decide with the dot product: a.b = (1)(3) + (-7)(-2) + (7)(2) = 3 + 14 + 14 = 31 > 0, so theta is acute

Step 9: theta = sin^-1(0.655), approximately 40.9 deg (the same value is given by cos(theta) = 31/sqrt(1683))

Final answer: The angle between the vectors is approximately 40.9 deg.

Question: Exercise 3.5 Q9: Find the moment about the point A(1, 3, 5) of the force F = i + 2j - k applied at B(3, -2, 5).

Answer: Step 1: The moment about A is M = r x F, where r is the vector from the reference point A to the point of application B

Step 2: r = AB = B - A = (3 - 1, -2 - 3, 5 - 5) = (2, -5, 0), that is r = 2i - 5j + 0k

Step 3: Compute r x F: i-component: (-5)(-1) - (0)(2) = 5 - 0 = 5

Step 4: j-component: -((2)(-1) - (0)(1)) = -(-2 - 0) = 2

Step 5: k-component: (2)(2) - (-5)(1) = 4 + 5 = 9

Step 6: So the moment vector is M = 5i + 2j + 9k

Step 7: Its magnitude is |M| = sqrt(25 + 4 + 81) = sqrt(110), approximately 10.49

Final answer: The moment about A is M = 5i + 2j + 9k, with magnitude sqrt(110) (approx. 10.49) units.

Question: Exercise 3.5 Q10: If you apply a 40 N force to the end of a 0.3 meter wrench at an angle of 110 degrees, calculate the magnitude of the moment Mo produced by the applied force.

Answer: Step 1: The magnitude of the moment (torque) is |Mo| = |r| |F| sin(theta), where r is the radius vector from the axis of rotation to the point of application and theta is the angle between r and F

Step 2: Here |r| = 0.3 m, |F| = 40 N and theta = 110 deg

Step 3: Substitute: |Mo| = (0.3)(40) sin(110 deg) = 12 sin(110 deg)

Step 4: Use sin(110 deg) = sin(70 deg) approximately 0.9397 (since sin(180 deg - x) = sin(x))

Step 5: Multiply: 12 x 0.9397 = 11.28

Step 6: The unit is newton-meter, because a length in meters is multiplied by a force in newtons

Final answer: The magnitude of the moment is approximately 11.28 Nm.

Exercise 3.6

Question: Exercise 3.6 Q1: Find the scalar triple product of the vectors i +

2j + 3k, -i - j + k and i + j + k.

Answer: Step 1: The scalar triple product a.(b x c) equals the determinant whose rows are the components of a, b and c in order: rows (1, 2, 3), (-1, -1, 1) and (1, 1, 1)

Step 2: Expand along the first row: 1 x [(-1)(1) - (1)(1)] - 2 x [(-1)(1) - (1)(1)] + 3 x [(-1)(1) - (-1)(1)]

Step 3: Evaluate the first minor: (-1)(1) - (1)(1) = -1 - 1 = -2, giving the term 1 x (-2) = -2

Step 4: Evaluate the second minor: (-1)(1) - (1)(1) = -2, giving the term -2 x (-2) = +4

Step 5: Evaluate the third minor: (-1)(1) - (-1)(1) = -1 + 1 = 0, giving the term 3 x 0 = 0

Step 6: Add the terms: -2 + 4 + 0 = 2

Final answer: The scalar triple product is 2.

Question: Exercise 3.6 Q2: Compute [i + j, i, i - j + k], the scalar triple product of the vectors i + j, i and i - j + k.

Answer: Step 1: The scalar triple product [a, b, c] = a.(b x c) equals the determinant with rows (1, 1, 0) for i + j, (1, 0, 0) for i, and (1, -1, 1) for i - j + k

Step 2: Expand along the first row: 1 x [(0)(1) - (0)(-1)] - 1 x [(1)(1) - (0)(1)] + 0 x [(1)(-1) - (0)(1)]

Step 3: Evaluate the first minor: (0)(1) - (0)(-1) = 0, giving the term 1 x 0 = 0

Step 4: Evaluate the second minor: (1)(1) - (0)(1) = 1, giving the term -1 x 1 = -1

Step 5: The third term is 0 because the first-row third entry is 0

Step 6: Add the terms: 0 - 1 + 0 = -1

Final answer: [i + j, i, i - j + k] = -1.

Question: Exercise 3.6 Q3(i): Find the volume of the parallelepiped whose edges are represented by the vectors 3i + 2j - k, i - 2j + k and i + 2j - 4k.

Answer: Step 1: The volume of the parallelepiped is the absolute value of the scalar triple product of its three edge vectors

Step 2: Set up the determinant with rows (3, 2, -1), (1, -2, 1) and (1, 2, -4)

Step 3: Expand along the first row: 3 x [(-2)(-4) - (1)(2)] - 2 x [(1)(-4) - (1)(1)] + (-1) x [(1)(2) - (-2)(1)]

Step 4: First minor: (-2)(-4) - (1)(2) = 8 - 2 = 6, term = 3 x 6 = 18

Step 5: Second minor: (1)(-4) - (1)(1) = -4 - 1 = -5, term = -2 x (-5) = +10

Step 6: Third minor: (1)(2) - (-2)(1) = 2 + 2 = 4, term = (-1) x 4 = -4

Step 7: Add: 18 + 10 - 4 = 24; volume is positive, so take the absolute value

Final answer: The volume of the parallelepiped is 24 cubic units.

Question: Exercise 3.6 Q3(ii): Find the volume of the parallelepiped whose edges are represented by the vectors i - 2j + 3k, 2i - j - k and i + 2j - 4k.

Answer: Step 1: The volume of the parallelepiped is the absolute value of the scalar triple product of its three edge vectors

Step 2: Set up the determinant with rows (1, -2, 3), (2, -1, -1) and (1, 2, -4)

Step 3: Expand along the first row: 1 x [(-1)(-4) - (-1)(2)] - (-2) x [(2)(-4) - (-1)(1)] + 3 x [(2)(2) - (-1)(1)]

Step 4: First minor: (-1)(-4) - (-1)(2) = 4 + 2 = 6, term = 1 x 6 = 6

Step 5: Second minor: (2)(-4) - (-1)(1) = -8 + 1 = -7, term = +2 x (-7) = -14

Step 6: Third minor: (2)(2) - (-1)(1) = 4 + 1 = 5, term = 3 x 5 = 15

Step 7: Add: 6 - 14 + 15 = 7; volume is positive, so take the absolute value

Final answer: The volume of the parallelepiped is 7 cubic units.

Question: Exercise 3.6 Q4: Find the volume of the tetrahedron whose vertices are A(2, 1, 8), B(3, 2, 9), C(2, 1, 4) and D(3, 3, 10).

Answer: Step 1: The volume of a tetrahedron is (1/6) of the absolute scalar triple product of the three edge vectors from one vertex

Step 2: From vertex A: AB = B - A = (3 - 2, 2 - 1, 9 - 8) = (1, 1, 1)

Step 3: AC = C - A = (2 - 2, 1 - 1, 4 - 8) = (0, 0, -4)

Step 4: AD = D - A = (3 - 2, 3 - 1, 10 - 8) = (1, 2, 2)

Step 5: Form the determinant with rows (1, 1, 1), (0, 0, -4) and (1, 2, 2) and expand along the first row: 1 x [(0)(2) - (-4)(2)] - 1 x [(0)(2) - (-4)(1)] + 1 x [(0)(2) - (0)(1)]

Step 6: Minors: first: 0 + 8 = 8, term = 8; second: 0 + 4 = 4, term = -4; third: 0 - 0 = 0, term = 0

Step 7: Scalar triple product = 8 - 4 + 0 = 4, so volume = (1/6) x |4| = 4/6 = 2/3

Final answer: The volume of the tetrahedron is 2/3 cubic units.

Question: Exercise 3.6 Q5: Show that the vectors 4i - j + k, 3i - 2j - k and i + j + 2k are coplanar.

Answer: Step 1: Three vectors are coplanar exactly when their scalar triple product is zero, because then the parallelepiped they form has zero volume

Step 2: Set up the determinant with rows (4, -1, 1), (3, -2, -1) and (1, 1, 2)

Step 3: Expand along the first row: 4 x [(-2)(2) - (-1)(1)] - (-1) x [(3)(2) - (-1)(1)] + 1 x [(3)(1) - (-2)(1)]

Step 4: First minor: (-2)(2) - (-1)(1) = -4 + 1 = -3, term = 4 x (-3) = -12

Step 5: Second minor: (3)(2) - (-1)(1) = 6 + 1 = 7, term = +1 x 7 = 7

Step 6: Third minor: (3)(1) - (-2)(1) = 3 + 2 = 5, term = 1 x 5 = 5

Step 7: Add: -12 + 7 + 5 = 0

Final answer: The scalar triple product is 0, hence the three vectors are coplanar.

Question: Exercise 3.6 Q6: Find lambda if the vectors i + j + 2k, lambda i - j + k and 3i - 2j - k are coplanar.

Answer: Step 1: Coplanar vectors have scalar triple product zero, so set the determinant with rows (1, 1, 2), (lambda, -1, 1) and (3, -2, -1) equal to zero

Step 2: Expand along the first row: 1 x [(-1)(-1) - (1)(-2)] - 1 x [(lambda)(-1) - (1)(3)] + 2 x [(lambda)(-2) - (-1)(3)] = 0

Step 3: First minor: (-1)(-1) - (1)(-2) = 1 + 2 = 3, term = 3

Step 4: Second minor: -lambda - 3, term = -1 x (-lambda - 3) = lambda + 3

Step 5: Third minor: -2lambda + 3, term = 2 x (-2lambda + 3) = -4lambda + 6

Step 6: Add all terms and set to zero: 3 + (lambda + 3) + (-4lambda + 6) = 12 - 3lambda = 0

Step 7: Solve: 3lambda = 12, so lambda = 4

Final answer: lambda = 4.

Question: Exercise 3.6 Q7: Show that i + j + k, i - 2j + 3k and 2i - j + 3k are not coplanar vectors.

Answer: Step 1: Three vectors are not coplanar when their scalar triple product is not zero

Step 2: Set up the determinant with rows (1, 1, 1), (1, -2, 3) and (2, -1, 3)

Step 3: Expand along the first row: 1 x [(-2)(3) - (3)(-1)] - 1 x [(1)(3) - (3)(2)] + 1 x [(1)(-1) - (-2)(2)]

Step 4: First minor: (-2)(3) - (3)(-1) = -6 + 3 = -3, term = -3

Step 5: Second minor: (1)(3) - (3)(2) = 3 - 6 = -3, term = -1 x (-3) = +3

Step 6: Third minor: (1)(-1) - (-2)(2) = -1 + 4 = 3, term = 3

Step 7: Add: -3 + 3 + 3 = 3, which is not zero

Final answer: The scalar triple product is 3, not equal to 0, hence the three vectors are not coplanar (they form a parallelepiped of volume 3).

Review Exercise 3

Question: Review Exercise 3 Q1(i) (multiple choice): The unit vector in the direction of r = 2i + j - k is: (a) (1/sqrt(6))(2i + j - k) (b) sqrt(6)(2i + j - k) (c) 6(2i + j - k) (d) (1/6)(2i + j - k).

Answer: Step 1: The unit vector is r/|r|, so find |r| = sqrt((2)^2 + (1)^2 +

(-1)^2)

Step 2: Square and add the components: 4 + 1 + 1 = 6, so |r| = sqrt(6)

Step 3: Divide the vector by sqrt(6): the unit vector is (2i + j - k)/sqrt(6), which is (1/sqrt(6))(2i + j - k)

Step 4: Option (b) would have magnitude 6, option (c) magnitude 6sqrt(6) and option (d) magnitude sqrt(6)/6, so none of them is a unit vector

Final answer: Option (a), (1/sqrt(6))(2i + j - k).

Question: Review Exercise 3 Q1(ii) (multiple choice): The magnitude of a x b represents the _____ of a parallelogram with adjacent sides a and b: (a) opposite sides (b) diagonal (c) area (d) volume.

Answer: Step 1: By the geometric interpretation of the cross product, a x b is a vector whose length equals |a||b| sin(theta)

Step 2: For a parallelogram with adjacent sides a and b, the area is base x height = |a| x (|b| sin(theta)) = |a||b| sin(theta)

Step 3: This is exactly the magnitude |a x b|

Step 4: Volume belongs to three vectors (scalar triple product), and a diagonal is a vector sum, so those options do not fit

Final answer: Option (c), area.

Question: Review Exercise 3 Q1(iii) (multiple choice): The volume of a tetrahedron determined by vectors a, b and c is: (a) [a, b, c] (b) (1/3)[a, b, c] (c) (1/6)[a, b, c] (d) None.

Answer: Step 1: The scalar triple product [a, b, c] gives the volume of the parallelepiped formed by the three vectors

Step 2: The volume of a tetrahedron is (1/3) x (area of its triangular base)

x (height); its base is half of the parallelogram on two of the edges, so the volume is (1/3) x (1/2)|b x c| x (the component of a perpendicular to that base) = (1/6) a.(b x c) = (1/6)[a, b, c]

Step 3: Option (a) is the parallelepiped volume and option (b) would be the volume of a pyramid on the full parallelogram base, so neither is the tetrahedron volume

Final answer: Option (c), (1/6)[a, b, c] (in absolute value).

Question: Review Exercise 3 Q1(iv) (multiple choice): a.(b x c) = _____: (a) a.b.c (b) a x b x c (c) b.(a x c) (d) (a x b).c.

Answer: Step 1: a.(b x c) is the scalar triple product [a, b, c]

Step 2: In a scalar triple product the dot and cross may be interchanged, and the vectors may be cycled, without changing the value: a.(b x c) = (a x b).c = b.(c x a)

Step 3: Option (c) has the wrong order: b.(a x c) = -[a, b, c], because swapping a and c changes the sign

Step 4: Options (a) and (b) are not valid vector expressions for this product at all

Final answer: Option (d), (a x b).c.

Question: Review Exercise 3 Q1(v) (multiple choice): If a and b are parallel, then a.b = _____: (a) 1 (b) -1 (c) 0 (d) ab (where a = |a| and b = |b|).

Answer: Step 1: By definition a.b = |a||b| cos(theta), where theta is the angle between the vectors

Step 2: For parallel vectors in the same direction, theta = 0 deg and cos(0 deg) = 1

Step 3: Substitute: a.b = |a||b| x 1 = ab in the notation of the options

Step 4: The value 0 in option (c) would be for perpendicular vectors, not parallel ones

Final answer: Option (d), ab.

Question: Review Exercise 3 Q1(vi) (multiple choice): If three vectors a, b and c are coplanar, then [a, b, c] = _____: (a) 1 (b) 0 (c) c (d) a.

Answer: Step 1: [a, b, c] is the scalar triple product, equal to the signed volume of the parallelepiped formed by the three vectors

Step 2: If the three vectors are coplanar, they all lie in one plane, so the parallelepiped is flat and its height is zero

Step 3: A flat parallelepiped has volume zero, so the determinant giving the triple product is also zero

Step 4: Options (c) and (d) are vectors, while the triple product is a scalar, so they cannot be correct

Final answer: Option (b), 0.

Question: Review Exercise 3 Q1(vii) (multiple choice): Direction cosines of the vector i + j - k are: (a) 1/sqrt(3), -1/sqrt(3), 1/sqrt(3) (b) -1/sqrt(3), 1/sqrt(3), 1/sqrt(3) (c) 1, 1, -1 (d) 1/sqrt(3), 1/sqrt(3), -1/sqrt(3).

Answer: Step 1: Direction cosines are the components divided by the magnitude of the vector

Step 2: The magnitude is sqrt((1)^2 + (1)^2 + (-1)^2) = sqrt(3)

Step 3: Divide each component by sqrt(3): cos(alpha) = 1/sqrt(3), cos(beta) = 1/sqrt(3), cos(gamma) = -1/sqrt(3)

Step 4: The signs must match the components (+, +, -); options (a) and (b) have a wrong sign, and option (c) is just the components themselves, whose squares sum to 3, not 1

Final answer: Option (d), 1/sqrt(3), 1/sqrt(3), -1/sqrt(3).

Question: Review Exercise 3 Q1(viii) (multiple choice): a x b is _____ to the plane of a and b: (a) Parallel (b) Perpendicular (c) Opposite (d) None of these.

Answer: Step 1: By the definition of the cross product, a x b = |a||b| sin(theta) n, where n is a unit vector given by the right-hand rule

Step 2: The vector n is perpendicular to both a and b

Step 3: Anything perpendicular to two intersecting vectors of a plane is perpendicular to the plane itself

Final answer: Option (b), Perpendicular.

Question: Review Exercise 3 Q1(ix) (multiple choice): If a = P1P2, where P1(0, 0, 1) and P2(-3, 1, 2), then |a| =: (a) sqrt(12) (b) sqrt(10) (c) sqrt(13) (d) sqrt(11).

Answer: Step 1: The vector is terminal minus initial: a = P1P2 = (-3 - 0, 1 - 0, 2 - 1) = (-3, 1, 1)

Step 2: Apply the magnitude formula: |a| = sqrt((-3)^2 + (1)^2 + (1)^2)

Step 3: Square and add the components: 9 + 1 + 1 = 11

Step 4: So |a| = sqrt(11)

Final answer: Option (d), sqrt(11).

Question: Review Exercise 3 Q1(x) (multiple choice): If p = 4i + 6k and q = 6i - 4j, then |p - q| is: (a) sqrt(8) (b) 2sqrt(14) (c) 2sqrt(26) (d) 2sqrt(3).

Answer: Step 1: Write both vectors with all components: p = 4i + 0j + 6k and q = 6i - 4j + 0k

Step 2: Subtract component-wise: p - q = (4 - 6)i + (0 + 4)j + (6 - 0)k = -2i

+ 4j + 6k

Step 3: Apply the magnitude formula: |p - q| = sqrt((-2)^2 + (4)^2 + (6)^2) = sqrt(4 + 16 + 36) = sqrt(56)

Step 4: Simplify the surd: sqrt(56) = sqrt(4 x 14) = 2sqrt(14)

Final answer: Option (b), 2sqrt(14).

Question: Review Exercise 3 Q1(xi) (multiple choice): For non-zero vectors a and b, a x b is a unit vector and |a| = |b| = sqrt(2); then the angle theta between a and b is: (a) pi/2 (b) pi/3 (c) pi/6 (d) -pi/2.

Answer: Step 1: Take magnitudes in the definition of the cross product: |a x b| = |a||b| sin(theta)

Step 2: Substitute the given values: 1 = (sqrt(2))(sqrt(2)) sin(theta) = 2 sin(theta)

Step 3: Solve: sin(theta) = 1/2

Step 4: The angle between two vectors lies between 0 and pi, and sin(theta) = 1/2 in this range gives theta = pi/6 (30 deg) or theta = 5pi/6 (150 deg); only pi/6 appears among the options

Final answer: Option (c), pi/6.

Question: Review Exercise 3 Q1(xii) (multiple choice): Magnitude of the vector a = 3i - j + 2k is: (a) 13 (b) sqrt(12) (c) sqrt(14) (d) sqrt(11).

Answer: Step 1: Write the components: a1 = 3, a2 = -1, a3 = 2

Step 2: Apply the magnitude formula: |a| = sqrt((3)^2 + (-1)^2 + (2)^2)

Step 3: Square and add the components: 9 + 1 + 4 = 14

Step 4: So |a| = sqrt(14)

Final answer: Option (c), sqrt(14).

Question: Review Exercise 3 Q1(xiii) (multiple choice): The position vector of the point (1, 0, 2) is: (a) i + j + 2k (b) i + 2j (c) 2i + 3k (d) i + 2k.

Answer: Step 1: The position vector of a point (x, y, z) is xi + yj + zk, measured from the origin

Step 2: Substitute x = 1, y = 0 and z = 2: the position vector is 1i + 0j + 2k

Step 3: The zero j-component is simply dropped: i + 2k

Step 4: Option (a) would be the point (1, 1, 2) and option (b) the point (1, 2, 0), so they describe different points

Final answer: Option (d), i + 2k.

Question: Review Exercise 3 Q1(xiv) (multiple choice): If O is the origin, OP = 2i + 3j - 4k and OQ = 5i + 4j - 3k, then PQ is equal to: (a) 7i + 7j - 7k (b) -3i + j - k (c) 3i + j + k (d) -7i - 7j + 7k.

Answer: Step 1: Use the rule PQ = OQ - OP, subtracting position vectors component-wise

Step 2: i-components: 5 - 2 = 3

Step 3: j-components: 4 - 3 = 1

Step 4: k-components: -3 - (-4) = 1

Step 5: Combine: PQ = 3i + j + k

Step 6: Option (a) is the sum OP + OQ with wrong signs, and option (b) is OP - OQ, the reverse vector

Final answer: Option (c), 3i + j + k.

Question: Review Exercise 3 Q1(xv) (multiple choice): k x j = _____: (a) i (b) k (c) j (d) -i.

Answer: Step 1: Use the cyclic order of the basic unit vectors: i x j = k, j x k = i and k x i = j

Step 2: Reversing the order of a cross product changes its sign, so j x k = i gives k x j = -i

Step 3: Check by definition: k and j are perpendicular unit vectors, and the right-hand rule from k to j points in the -i direction

Final answer: Option (d), -i.

Question: Review Exercise 3 Q1(xvi) (multiple choice): If |a x b| = |a.b|, then the angle between a and b is: (a) 0 (b) pi/2 (c) pi/4 (d) pi.

Answer: Step 1: Write both sides using the angle theta between the vectors: |a x b| = |a||b| sin(theta) and |a.b| = |a||b| |cos(theta)|

Step 2: Set them equal (for non-zero a and b, divide both sides by |a||b|): sin(theta) = |cos(theta)|

Step 3: This gives tan(theta) = 1 in magnitude, so theta = pi/4 or theta = 3pi/4

Step 4: Of the given options, only pi/4 (45 deg) satisfies the equation

Final answer: Option (c), pi/4.

Question: Review Exercise 3 Q1(xvii) (multiple choice): The distance of the point (-3, 4, 5) from the origin is: (a) 50 (b) 5sqrt(2) (c) 6 (d) None of these.

Answer: Step 1: The distance from the origin is the magnitude of the position vector -3i + 4j + 5k

Step 2: Apply the magnitude formula: sqrt((-3)^2 + (4)^2 + (5)^2)

Step 3: Square and add the components: 9 + 16 + 25 = 50

Step 4: Simplify: sqrt(50) = sqrt(25 x 2) = 5sqrt(2)

Step 5: Option (a) is the squared distance, not the distance itself

Final answer: Option (b), 5sqrt(2).

Question: Review Exercise 3 Q1(xviii) (multiple choice): The vector having initial point (2, 5, 0) and terminal point (-3, 7, 4) is: (a) -i + 12j + 4k (b) -5i + 2j - 4k (c) -5i + 2j + 4k (d) -i + j + k.

Answer: Step 1: A vector is (terminal point) - (initial point), component by component

Step 2: x-component: -3 - 2 = -5

Step 3: y-component: 7 - 5 = 2

Step 4: z-component: 4 - 0 = 4

Step 5: Combine: -5i + 2j + 4k

Step 6: Option (b) has the wrong sign in the k-component, and options (a) and (d) match none of the correct components

Final answer: Option (c), -5i + 2j + 4k.

Question: Review Exercise 3 Q1(xix) (multiple choice): If |a| = 10, |b| = 2 and a.b = 0, then the value of |a x b| is: (a) 10 (b) 16 (c) 14 (d) 20.

Answer: Step 1: a.b = 0 for non-zero vectors means the vectors are perpendicular, so the angle between them is theta = 90 deg

Step 2: Use |a x b| = |a||b| sin(theta)

Step 3: Substitute: |a x b| = (10)(2) sin(90 deg) = 20 x 1

Step 4: The result is the full product of the magnitudes because sin(90 deg) = 1

Final answer: Option (d), 20.

Question: Review Exercise 3 Q1(xx) (multiple choice): The number of vectors of unit length perpendicular to the plane of the vectors a = 2i + j + 2k and b = j + k is: (a) One (b) Two (c) Three (d) Infinite.

Answer: Step 1: Every vector perpendicular to the plane of a and b is parallel to the single direction of a x b (the normal direction to the plane)

Step 2: Along that one line through the origin there are exactly two vectors of unit length: the unit normal n and its opposite -n

Step 3: There are infinitely many perpendicular vectors of all lengths, but the unit-length condition leaves exactly the two opposite choices

Step 4: Options (a), (c) and (d) therefore do not fit

Final answer: Option (b), Two.

Question: Review Exercise 3 Q2(i): Write PQ in the form xi + yj, where P(4, 6) and Q(-4, 5).

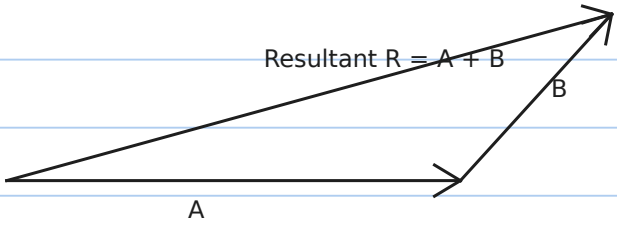
Answer: Step 1: Use PQ = (coordinates of Q) - (coordinates of P), component by component

Step 2: The x-component is x = -4 - 4 = -8

Step 3: The y-component is y = 5 - 6 = -1

Step 4: Write the components with the unit vectors: PQ = -8i - 1j

Final answer: PQ = -8i - j.



Question: Review Exercise 3 Q2(ii): Write PQ in the form xi + yj, where P(-5/2, 9) and Q(10, 8).

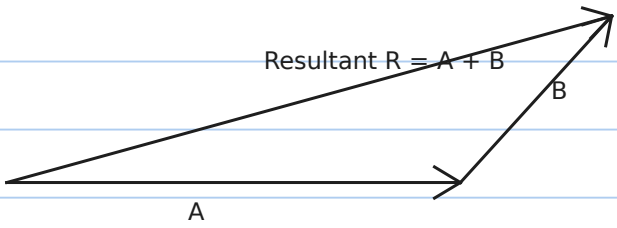
Answer: Step 1: Use PQ = (coordinates of Q) - (coordinates of P), component by component

Step 2: The x-component is x = 10 - (-5/2) = 10 + 5/2 = 25/2

Step 3: The y-component is y = 8 - 9 = -1

Step 4: Write the components with the unit vectors: PQ = (25/2)i - 1j

Final answer: PQ = (25/2)i - j.



Question: Review Exercise 3 Q3: Find the vector PQ joining the points P(3, 4, 2) and Q(-2, 3, 4), and also find the direction cosines of PQ.

Answer: Step 1: PQ = (coordinates of Q) - (coordinates of P), component by component

Step 2: x-component: -2 - 3 = -5; y-component: 3 - 4 = -1; z-component: 4 - 2 = 2, so PQ = -5i - j + 2k

Step 3: Find its magnitude: |PQ| = sqrt((-5)^2 + (-1)^2 + (2)^2) = sqrt(25

+ 1 + 4) = sqrt(30)

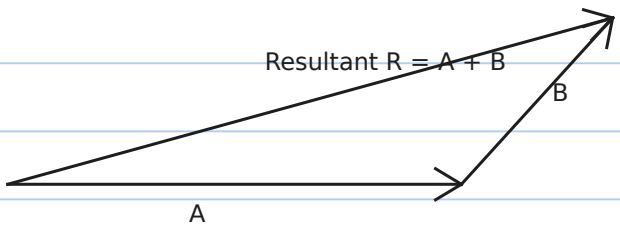
Step 4: The direction cosines are the components divided by the magnitude: cos(alpha) = -5/sqrt(30)

Step 5: cos(beta) = -1/sqrt(30)

Step 6: cos(gamma) = 2/sqrt(30)

Step 7: Check: (25 + 1 + 4)/30 = 1, as the sum of the squares of direction cosines must be

Final answer: PQ = -5i - j + 2k, with direction cosines -5/sqrt(30), -1/sqrt(30) and 2/sqrt(30).



Question: Review Exercise 3 Q4: Show that P(2, 3, 6), Q(3, 7, 4) and R(4, 11, -2) are collinear. (Note: worked exactly as printed, the book's claim does not hold; see the final step.)

Answer: Step 1: Three points are collinear exactly when the vectors along them, such as PQ and QR, are parallel (scalar multiples of each other)

Step 2: PQ = Q - P = (3 - 2, 7 - 3, 4 - 6) = (1, 4, -2), that is PQ = i + 4j - 2k

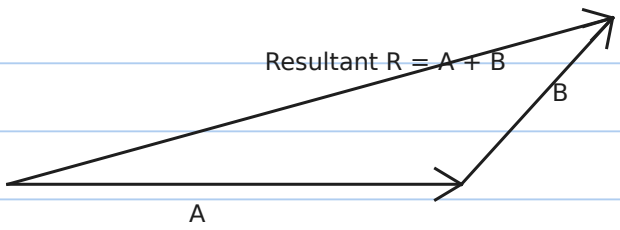
Step 3: QR = R - Q = (4 - 3, 11 - 7, -2 - 4) = (1, 4, -6), that is QR = i + 4j - 6k

Step 4: For parallelism every component ratio must be the same: the i- and j-ratios are 1/1 = 1 and 4/4 = 1, but the k-ratio is (-6)/(-2) = 3, which is different

Step 5: Confirm with the cross product: PQ x QR = i((4)(-6) - (-2)(4)) - j((1)(-6) - (-2)(1)) + k((1)(4) - (4)(1)) = (-24 + 8)i - (-6 + 2)j + 0k = -16i + 4j, which is not the zero vector

Step 6: Also |PQ| = sqrt(21) and |QR| = sqrt(53) give |PQ| + |QR| approximately 11.86, while |PR| = sqrt((2)^2 + (8)^2 + (-8)^2) = sqrt(132) approximately 11.49, so Q does not even lie on segment PR

Final answer: As printed, P, Q and R are NOT collinear: PQ = i + 4j - 2k is not parallel to QR = i + 4j - 6k. This appears to be a misprint in the book - if R were (4, 11, 2), then QR = i + 4j - 2k = PQ and the points would indeed be collinear.



Question: Review Exercise 3 Q5: Show that the vectors (1/7)i + j + k, 2i - 3j + 3k and 4i + j - 5k are mutually perpendicular. (Note: worked exactly as printed, the book's claim does not hold; see the final step.)

Answer: Step 1: Vectors are mutually perpendicular exactly when every pair has dot product zero; call the vectors a = (1/7)i + j + k, b = 2i - 3j + 3k and c = 4i + j - 5k

Step 2: Compute a.b term by term: i: (1/7)(2) = 2/7; j: (1)(-3) = -3; k: (1)(3) = 3

Step 3: Add: a.b = 2/7 - 3 + 3 = 2/7, which is not zero

Step 4: Compute a.c term by term: i: (1/7)(4) = 4/7; j: (1)(1) = 1; k: (1)(-5) = -5; total a.c = 4/7 + 1 - 5 = -24/7, which is not zero

Step 5: Compute b.c term by term: i: (2)(4) = 8; j: (-3)(1) = -3; k: (3)(-5) = -15; total b.c = 8 - 3 - 15 = -10, which is not zero

Step 6: Since none of the three pairwise dot products is zero, no pair of these vectors is perpendicular

Final answer: As printed, the three vectors are NOT mutually perpendicular (a.b = 2/7, a.c = -24/7 and b.c = -10, all non-zero), so the statement in the book appears to contain a misprint in the vectors.

Question: Review Exercise 3 Q6: Find the area of the parallelogram whose adjacent sides are 3i - 5j + 6k and i + 3j - 4k. Also find the unit vector parallel to its diagonal through the common initial point.

Answer: Step 1: Call the sides a = 3i - 5j + 6k and b = i + 3j - 4k; the area is |a x b|

Step 2: i-component of a x b: (-5)(-4) - (6)(3) = 20 - 18 = 2

Step 3: j-component: -((3)(-4) - (6)(1)) = -(-12 - 6) = 18

Step 4: k-component: (3)(3) - (-5)(1) = 9 + 5 = 14

Step 5: So a x b = 2i + 18j + 14k, and the area is sqrt(2^2 + 18^2 + 14^2) = sqrt(4 + 324 + 196) = sqrt(524) = 2sqrt(131), approximately 22.89 square units

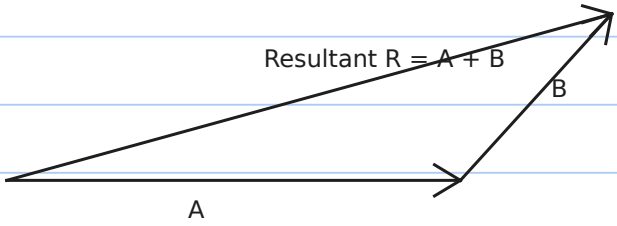
Step 6: The diagonal through the common initial point is a + b = (3 + 1)i + (-5 + 3)j + (6 - 4)k = 4i - 2j + 2k

Step 7: Its magnitude is |a + b| = sqrt(16 + 4 + 4) = sqrt(24) = 2sqrt(6)

Step 8: The unit vector along the diagonal is (4i - 2j + 2k)/(2sqrt(6)) = (2i - j + k)/sqrt(6)

Final answer: The area is 2sqrt(131) square units (approx. 22.89), and the

unit vector parallel to the diagonal is (2i - j + k)/sqrt(6).



Question: Review Exercise 3 Q7: Prove that [p + q, q + r, r + p] = 2[p, q, r].

Answer: Step 1: The left side means (p + q).((q + r) x (r + p)); first expand the cross product using distributivity: (q + r) x (r + p) = q x r + q x p + r x r + r x p

Step 2: The term r x r is the zero vector, so (q + r) x (r + p) = q x r + q x p + r x p

Step 3: Now dot with (p + q), giving six scalar triple products: p.(q x r) + p.(q x p) + p.(r x p) + q.(q x r) + q.(q x p) + q.(r x p)

Step 4: Any triple product with a repeated vector is zero, because a vector is perpendicular to the cross product of itself with anything: p.(q x p) = 0, p.(r x p) = 0, q.(q x r) = 0 and q.(q x p) = 0

Step 5: Two terms survive: p.(q x r) = [p, q, r] and q.(r x p) = [q, r, p]

Step 6: A cyclic change of order does not change a scalar triple product, so [q, r, p] = [p, q, r]

Step 7: Add the surviving terms: [p, q, r] + [p, q, r] = 2[p, q, r]

Final answer: Hence [p + q, q + r, r + p] = 2[p, q, r], as required.

Question: Review Exercise 3 Q8: For the vectors a = i + 2j + k, b = 3i - j - k and c = i + 8j + k, verify the distributive property of the cross product.

Answer: Step 1: Verify a x (b + c) = (a x b) + (a x c); first compute b + c = (3 + 1)i + (-1 + 8)j + (-1 + 1)k = 4i + 7j + 0k

Step 2: Left side, a x (b + c): i-component: (2)(0) - (1)(7) = 0 - 7 = -7

Step 3: j-component: -((1)(0) - (1)(4)) = -(0 - 4) = 4

Step 4: k-component: (1)(7) - (2)(4) = 7 - 8 = -1, so a x (b + c) = -7i + 4j - k

Step 5: Compute a x b: i: (2)(-1) - (1)(-1) = -2 + 1 = -1; j: -((1)(-1) - (1)(3)) = -(-1 - 3) = 4; k: (1)(-1) - (2)(3) = -1 - 6 = -7, so a x b = -i + 4j - 7k

Step 6: Compute a x c: i: (2)(1) - (1)(8) = 2 - 8 = -6; j: -((1)(1) - (1)(1)) = -(1 - 1) = 0; k: (1)(8) - (2)(1) = 8 - 2 = 6, so a x c = -6i + 0j + 6k

Step 7: Right side: (a x b) + (a x c) = (-1 - 6)i + (4 + 0)j + (-7 + 6)k = -7i + 4j - k

Step 8: The right side equals the left side found in Step 4

Final answer: a x (b + c) = (a x b) + (a x c) = -7i + 4j - k, so the distributive property of the cross product is verified.

Unit 4: Sequences and Series

Exercise 4.1

Question: Q1(i). Find the nth term (rule of formation) of the sequence 2, 4, 6, ...

Answer: Step 1: write the terms with their positions, a1 = 2, a2 = 4, a3 = 6

Step 2: observe each term is twice its position, a1 = 2(1), a2 = 2(2), a3 = 2(3)

Step 3: generalise the pattern by replacing the position with n

Final answer: an = 2n

Question: Q1(ii). Find the nth term (rule of formation) of the sequence 1^2, 2^2, 3^2, ...

Answer: Step 1: write the terms with their positions, a1 = 1^2, a2 = 2^2, a3 = 3^2

Step 2: observe each term is the square of its position number

Step 3: generalise by replacing the position with n

Final answer: an = n^2

Question: Q1(iii). Find the nth term (rule of formation) of the sequence 3, 9, 27, ...

Answer: Step 1: write the terms as powers of 3, a1 = 3 = 3^1, a2 = 9 = 3^2, a3 = 27 = 3^3

Step 2: observe the exponent of 3 in each term equals the position of the term

Step 3: generalise by replacing the position with n

Final answer: an = 3^n

Question: Q1(iv). Find the nth term (rule of formation) of the sequence 1/2, 2/3, 3/4, 4/5, ...

Answer: Step 1: compare numerator and denominator of each term, a1 = 1/2, a2 = 2/3, a3 = 3/4, a4 = 4/5

Step 2: observe the numerator of each term equals its position n

Step 3: observe the denominator is always one more than the numerator, that is n + 1

Step 4: combine the two observations into one fraction

Final answer: an = n/(n + 1)

Question: Q1(v). Find the nth term (rule of formation) of the sequence 1, -1/3, 1/9, -1/27, ...

Answer: Step 1: write the terms ignoring signs, 1, 1/3, 1/9, 1/27, which are 1/3^0, 1/3^1, 1/3^2, 1/3^3

Step 2: observe the denominator of the nth term is 3^(n-1)

Step 3: observe the signs alternate +, -, +, -, starting positive, which is given by the factor (-1)^(n+1), or equivalently write the terms as (-1/3)^(n-1)

Step 4: check, n = 1 gives 1, n = 2 gives -1/3, n = 3 gives 1/9, n = 4 gives -1/27

Final answer: an = (-1/3)^(n-1), i.e. an = (-1)^(n+1) / 3^(n-1)

Question: Q1(vi). Find the nth term (rule of formation) of the sequence 1, 8, 27, 64, ...

Answer: Step 1: recognise the terms as cubes, a1 = 1 = 1^3, a2 = 8 = 2^3, a3 = 27 = 3^3, a4 = 64 = 4^3

Step 2: observe each term is the cube of its position number

Step 3: generalise by replacing the position with n

Final answer: an = n^3

Question: Q2(i). Find the first five terms of the sequence with general term an = (2n/3)(n + 1).

Answer: Step 1: put n = 1, a1 = (2(1)/3)(1 + 1) = (2/3)(2) = 4/3

Step 2: put n = 2, a2 = (2(2)/3)(2 + 1) = (4/3)(3) = 4

Step 3: put n = 3, a3 = (2(3)/3)(3 + 1) = (2)(4) = 8

Step 4: put n = 4, a4 = (2(4)/3)(4 + 1) = (8/3)(5) = 40/3

Step 5: put n = 5, a5 = (2(5)/3)(5 + 1) = (10/3)(6) = 20

Final answer: the first five terms are 4/3, 4, 8, 40/3, 20

Question: Q2(ii). Find the first five terms of the sequence with general term an = (-1)^(n+1) . 3^(n-1).

Answer: Step 1: put n = 1, a1 = (-1)^2 . 3^0 = (1)(1) = 1

Step 2: put n = 2, a2 = (-1)^3 . 3^1 = (-1)(3) = -3

Step 3: put n = 3, a3 = (-1)^4 . 3^2 = (1)(9) = 9

Step 4: put n = 4, a4 = (-1)^5 . 3^3 = (-1)(27) = -27

Step 5: put n = 5, a5 = (-1)^6 . 3^4 = (1)(81) = 81

Final answer: the first five terms are 1, -3, 9, -27, 81

Question: Q2(iii). Find the first five terms of the sequence with general term an = (1/4) n^2 (n + 1)^2.

Answer: Step 1: put n = 1, a1 = (1/4)(1)(2)^2 = (1/4)(4) = 1

Step 2: put n = 2, a2 = (1/4)(4)(3)^2 = (1/4)(36) = 9

Step 3: put n = 3, a3 = (1/4)(9)(4)^2 = (1/4)(144) = 36

Step 4: put n = 4, a4 = (1/4)(16)(5)^2 = (1/4)(400) = 100

Step 5: put n = 5, a5 = (1/4)(25)(6)^2 = (1/4)(900) = 225

Final answer: the first five terms are 1, 9, 36, 100, 225

Question: Q2(iv). Find the first five terms of the sequence with general term an = n/(3n + 1).

Answer: Step 1: put n = 1, a1 = 1/(3 + 1) = 1/4

Step 2: put n = 2, a2 = 2/(6 + 1) = 2/7

Step 3: put n = 3, a3 = 3/(9 + 1) = 3/10

Step 4: put n = 4, a4 = 4/(12 + 1) = 4/13

Step 5: put n = 5, a5 = 5/(15 + 1) = 5/16

Final answer: the first five terms are 1/4, 2/7, 3/10, 4/13, 5/16

Question: Q2(v). Find the first five terms of the sequence with general term an = (1/6) n (n + 1)(n + 2).

Answer: Step 1: put n = 1, a1 = (1/6)(1)(2)(3) = 6/6 = 1

Step 2: put n = 2, a2 = (1/6)(2)(3)(4) = 24/6 = 4

Step 3: put n = 3, a3 = (1/6)(3)(4)(5) = 60/6 = 10

Step 4: put n = 4, a4 = (1/6)(4)(5)(6) = 120/6 = 20

Step 5: put n = 5, a5 = (1/6)(5)(6)(7) = 210/6 = 35

Final answer: the first five terms are 1, 4, 10, 20, 35

Question: Q2(vi). Find the first five terms of the sequence with general term an = (1/6) n (n + 1)(2n + 7).

Answer: Step 1: put n = 1, a1 = (1/6)(1)(2)(9) = 18/6 = 3

Step 2: put n = 2, a2 = (1/6)(2)(3)(11) = 66/6 = 11

Step 3: put n = 3, a3 = (1/6)(3)(4)(13) = 156/6 = 26

Step 4: put n = 4, a4 = (1/6)(4)(5)(15) = 300/6 = 50

Step 5: put n = 5, a5 = (1/6)(5)(6)(17) = 510/6 = 85

Final answer: the first five terms are 3, 11, 26, 50, 85

Question: Q2(vii). Find the first five terms of the sequence defined by a(n+1) = 6 + an, with a1 = 2.

Answer: Step 1: the given first term is a1 = 2

Step 2: put n = 1 in the rule, a2 = 6 + a1 = 6 + 2 = 8

Step 3: put n = 2, a3 = 6 + a2 = 6 + 8 = 14

Step 4: put n = 3, a4 = 6 + a3 = 6 + 14 = 20

Step 5: put n = 4, a5 = 6 + a4 = 6 + 20 = 26

Final answer: the first five terms are 2, 8, 14, 20, 26

Question: Q2(viii). Find the first five terms of the sequence defined by an = (n/2) a(n+1), with a1 = 1. (As printed in the book; the rule is used in the forward direction a(n+1) = (2/n) an.)

Answer: Step 1: rearrange the printed rule an = (n/2) a(n+1) to get a(n+1) = (2/n) an, and note a1 = 1

Step 2: put n = 1, a2 = (2/1) a1 = 2(1) = 2

Step 3: put n = 2, a3 = (2/2) a2 = (1)(2) = 2

Step 4: put n = 3, a4 = (2/3) a3 = (2/3)(2) = 4/3

Step 5: put n = 4, a5 = (2/4) a4 = (1/2)(4/3) = 2/3

Final answer: the first five terms are 1, 2, 2, 4/3, 2/3

Question: Q3. Find the sequence by using T(n+1) = (n + 1) Tn, where T1 = 2.

Answer: Step 1: the given first term is T1 = 2

Step 2: put n = 1, T2 = (1 + 1) T1 = 2(2) = 4

Step 3: put n = 2, T3 = (2 + 1) T2 = 3(4) = 12

Step 4: put n = 3, T4 = (3 + 1) T3 = 4(12) = 48

Step 5: put n = 4, T5 = (4 + 1) T4 = 5(48) = 240

Final answer: the sequence is 2, 4, 12, 48, 240, ...

Question: Q4. Find the values of the nth term of the triangular sequence when n = 7, 9, 12 and 16.

Answer: Step 1: write the formula for the nth triangular number, tn = n(n + 1)/2

Step 2: put n = 7, t7 = 7(8)/2 = 56/2 = 28

Step 3: put n = 9, t9 = 9(10)/2 = 90/2 = 45

Step 4: put n = 12, t12 = 12(13)/2 = 156/2 = 78

Step 5: put n = 16, t16 = 16(17)/2 = 272/2 = 136

Final answer: t7 = 28, t9 = 45, t12 = 78, t16 = 136

Question: Q5. Find the first five terms of the sequence with general term an = (n + 1)! / 2!.

Answer: Step 1: recall 2! = 2 and put n = 1, a1 = 2!/2! = 2/2 = 1

Step 2: put n = 2, a2 = 3!/2! = 6/2 = 3

Step 3: put n = 3, a3 = 4!/2! = 24/2 = 12

Step 4: put n = 4, a4 = 5!/2! = 120/2 = 60

Step 5: put n = 5, a5 = 6!/2! = 720/2 = 360

Final answer: the first five terms are 1, 3, 12, 60, 360

Question: Q6. Find the Pascal sequence when n = 5, 6, 7 and 8.

Answer: Step 1: write the Pascal formula for row n, term r is n! / (r! (n - r)!), for r = 0, 1, 2, ..., n

Step 2: for n = 5, the values for r = 0 to 5 are 5!/(0!5!) = 1, 5!/(1!4!) = 5, 5!/(2!3!) = 10, 5!/(3!2!) = 10, 5!/(4!1!) = 5, 5!/(5!0!) = 1, giving 1, 5, 10, 10, 5, 1

Step 3: for n = 6, the values for r = 0 to 6 are 1, 6, 15, 20, 15, 6, 1

Step 4: for n = 7, the values for r = 0 to 7 are 1, 7, 21, 35, 35, 21, 7, 1

Step 5: for n = 8, the values for r = 0 to 8 are 1, 8, 28, 56, 70, 56, 28, 8, 1 (each middle entry is the sum of the two entries above it in the previous row)

Final answer: n = 5: 1, 5, 10, 10, 5, 1; n = 6: 1, 6, 15, 20, 15, 6, 1; n = 7: 1, 7, 21, 35, 35, 21, 7, 1; n = 8: 1, 8, 28, 56, 70, 56, 28, 8, 1

Exercise 4.2

Question: Q1(i). Find the indicated term a12 of the A.P. 1, 5, 9, ...

Answer: Step 1: identify the first term a = 1

Step 2: find the common difference d = 5 - 1 = 4

Step 3: write the formula an = a + (n - 1)d

Step 4: substitute n = 12, a = 1, d = 4, a12 = 1 + (12 - 1)(4)

Step 5: simplify, a12 = 1 + (11)(4) = 1 + 44

Final answer: a12 = 45

Question: Q1(ii). Find the indicated term a10 of the A.P. -15, -9, -3, ...

Answer: Step 1: identify the first term a = -15

Step 2: find the common difference d = -9 - (-15) = 6

Step 3: write the formula an = a + (n - 1)d

Step 4: substitute n = 10, a10 = -15 + (10 - 1)(6)

Step 5: simplify, a10 = -15 + (9)(6) = -15 + 54

Final answer: a10 = 39

Question: Q1(iii). Find the indicated term a7 of the A.P. 2, 6, 10, 14, ...

Answer: Step 1: identify the first term a = 2

Step 2: find the common difference d = 6 - 2 = 4

Step 3: write the formula an = a + (n - 1)d

Step 4: substitute n = 7, a7 = 2 + (7 - 1)(4)

Step 5: simplify, a7 = 2 + (6)(4) = 2 + 24

Final answer: a7 = 26

Question: Q1(iv). Find the indicated term a30 of the A.P. -5, 4, 13, ...

Answer: Step 1: identify the first term a = -5

Step 2: find the common difference d = 4 - (-5) = 9

Step 3: write the formula an = a + (n - 1)d

Step 4: substitute n = 30, a30 = -5 + (30 - 1)(9)

Step 5: simplify, a30 = -5 + (29)(9) = -5 + 261

Final answer: a30 = 256

Question: Q1(v). Find the indicated term a14 of the A.P. 23, 26, 29, ...

Answer: Step 1: identify the first term a = 23

Step 2: find the common difference d = 26 - 23 = 3

Step 3: write the formula an = a + (n - 1)d

Step 4: substitute n = 14, a14 = 23 + (14 - 1)(3)

Step 5: simplify, a14 = 23 + (13)(3) = 23 + 39

Final answer: a14 = 62

Question: Q1(vi). Find the indicated term a6 of the A.P. 1/3, 1/2, 2/3, ...

Answer: Step 1: identify the first term a = 1/3

Step 2: find the common difference d = 1/2 - 1/3 = 3/6 - 2/6 = 1/6

Step 3: write the formula an = a + (n - 1)d

Step 4: substitute n = 6, a6 = 1/3 + (6 - 1)(1/6)

Step 5: simplify, a6 = 1/3 + 5/6 = 2/6 + 5/6 = 7/6

Final answer: a6 = 7/6

Question: Q2(i). Find the first five terms of the arithmetic sequence defined by an = 2n - 3.

Answer: Step 1: put n = 1, a1 = 2(1) - 3 = -1

Step 2: put n = 2, a2 = 2(2) - 3 = 1

Step 3: put n = 3, a3 = 2(3) - 3 = 3

Step 4: put n = 4, a4 = 2(4) - 3 = 5

Step 5: put n = 5, a5 = 2(5) - 3 = 7

Final answer: the first five terms are -1, 1, 3, 5, 7

Question: Q2(ii). Find the first five terms of the arithmetic sequence defined by a(n-2) = 3n - 5.

Answer: Step 1: replace n by n + 2 to express the rule in terms of an, an = 3(n + 2) - 5 = 3n + 1

Step 2: put n = 1, a1 = 3(1) + 1 = 4

Step 3: put n = 2, a2 = 3(2) + 1 = 7

Step 4: put n = 3, a3 = 3(3) + 1 = 10

Step 5: put n = 4, a4 = 3(4) + 1 = 13

Step 6: put n = 5, a5 = 3(5) + 1 = 16

Final answer: the first five terms are 4, 7, 10, 13, 16

Question: Q2(iii). Find the first five terms of the arithmetic sequence defined by an = n/(2n + 1).

Answer: Step 1: put n = 1, a1 = 1/(2 + 1) = 1/3

Step 2: put n = 2, a2 = 2/(4 + 1) = 2/5

Step 3: put n = 3, a3 = 3/(6 + 1) = 3/7

Step 4: put n = 4, a4 = 4/(8 + 1) = 4/9

Step 5: put n = 5, a5 = 5/(10 + 1) = 5/11

Final answer: the first five terms are 1/3, 2/5, 3/7, 4/9, 5/11 (the book groups this formula with the arithmetic-sequence exercise; the terms are computed directly from the formula)

Question: Q2(iv). Find the first five terms of the sequence defined by an = (n + 1) a(n-1), with a1 = 1.

Answer: Step 1: the given first term is a1 = 1

Step 2: put n = 2, a2 = (2 + 1) a1 = 3(1) = 3

Step 3: put n = 3, a3 = (3 + 1) a2 = 4(3) = 12

Step 4: put n = 4, a4 = (4 + 1) a3 = 5(12) = 60

Step 5: put n = 5, a5 = (5 + 1) a4 = 6(60) = 360

Final answer: the first five terms are 1, 3, 12, 60, 360

Question: Q2(v). Find the first five terms of the sequence defined by an - a(n-1) = n + 2, with a1 = 2.

Answer: Step 1: rearrange the rule as an = a(n-1) + n + 2, with a1 = 2

Step 2: put n = 2, a2 = a1 + 2 + 2 = 2 + 4 = 6

Step 3: put n = 3, a3 = a2 + 3 + 2 = 6 + 5 = 11

Step 4: put n = 4, a4 = a3 + 4 + 2 = 11 + 6 = 17

Step 5: put n = 5, a5 = a4 + 5 + 2 = 17 + 7 = 24

Final answer: the first five terms are 2, 6, 11, 17, 24

Question: Q2(vi). Find the first five terms of the arithmetic sequence whose first term is a = 5 and three of whose consecutive terms are 23, 26, 29.

Answer: Step 1: find the common difference from the given consecutive terms, d = 26 - 23 = 3 (also 29 - 26 = 3)

Step 2: the first term is a1 = a = 5

Step 3: add d repeatedly, a2 = 5 + 3 = 8

Step 4: continue, a3 = 8 + 3 = 11

Step 5: continue, a4 = 11 + 3 = 14

Step 6: continue, a5 = 14 + 3 = 17

Final answer: the first five terms are 5, 8, 11, 14, 17 (check: continuing with d = 3 gives 20, 23, 26, 29, so 23, 26, 29 do occur as consecutive terms)

Question: Q3. If a(n-3) = 2n - 5, find the nth term of the sequence.

Answer: Step 1: replace n by n + 3 in the given rule so that the subscript becomes n

Step 2: substitute, an = 2(n + 3) - 5

Step 3: expand the bracket, an = 2n + 6 - 5

Step 4: simplify the constants

Final answer: an = 2n + 1

Question: Q4. Find the 20th term of the sequence 1, 2 - x, 3 - 2x, ...

Answer: Step 1: identify the first term a = 1

Step 2: find the common difference d = (2 - x) - 1 = 1 - x (check: (3 - 2x) - (2 - x) = 1 - x, the same)

Step 3: write the formula an = a + (n - 1)d

Step 4: substitute n = 20, a20 = 1 + (20 - 1)(1 - x) = 1 + 19(1 - x)

Step 5: expand, a20 = 1 + 19 - 19x

Final answer: a20 = 20 - 19x

Question: Q5. Find the 21st term of the A.P. whose 6th term is 11 and whose 15th term is 47.

Answer: Step 1: write the formula an = a + (n - 1)d

Step 2: use the 6th term, a + 5d = 11 ...(i)

Step 3: use the 15th term, a + 14d = 47 ...(ii)

Step 4: subtract (i) from (ii), 9d = 36, so d = 4

Step 5: substitute d = 4 in (i), a = 11 - 5(4) = 11 - 20 = -9

Step 6: find the 21st term, a21 = a + 20d = -9 + 20(4) = -9 + 80

Final answer: a21 = 71

Question: Q6(a). Which term of the sequence -15, -9, -3, ... is 75?

Answer: Step 1: identify a = -15 and d = -9 - (-15) = 6

Step 2: let the required term be an = 75 and write an = a + (n - 1)d

Step 3: substitute, 75 = -15 + (n - 1)(6)

Step 4: simplify, 90 = 6(n - 1), so n - 1 = 15

Step 5: solve for n

Final answer: n = 16, so 75 is the 16th term

Question: Q6(b). Which term of the sequence 5, 2, -1, ... is -85?

Answer: Step 1: identify a = 5 and d = 2 - 5 = -3

Step 2: let the required term be an = -85 and write an = a + (n - 1)d

Step 3: substitute, -85 = 5 + (n - 1)(-3)

Step 4: simplify, -90 = -3(n - 1), so n - 1 = 30

Step 5: solve for n

Final answer: n = 31, so -85 is the 31st term

Question: Q6(c). Which term of the sequence -2, 4, 10, ... is 148?

Answer: Step 1: identify a = -2 and d = 4 - (-2) = 6

Step 2: let the required term be an = 148 and write an = a + (n - 1)d

Step 3: substitute, 148 = -2 + (n - 1)(6)

Step 4: simplify, 150 = 6(n - 1), so n - 1 = 25

Step 5: solve for n

Final answer: n = 26, so 148 is the 26th term

Question: Q7. Find the nth term of the sequence (6/7)^2, (11/7)^2, (16/7)^2, ...

Answer: Step 1: look at the numerators inside the brackets, 6, 11, 16, which form an A.P. with a = 6 and d = 5

Step 2: find the nth numerator, 6 + (n - 1)(5) = 5n + 1

Step 3: the denominator stays 7 for every term

Step 4: every term is squared, so square the general fraction

Final answer: an = ((5n + 1)/7)^2

Question: Q8(i). If a, b, c are the lth, mth and nth terms of an A.P., show that a(m - n) + b(n - l) + c(l - m) = 0.

Answer: Step 1: let the A.P. have first term A and common difference D, so a = A + (l - 1)D, b = A + (m - 1)D, c = A + (n - 1)D

Step 2: substitute these in the left-hand side, LHS = [A + (l - 1)D](m - n) + [A + (m - 1)D](n - l) + [A + (n - 1)D](l - m)

Step 3: collect the terms containing A: A[(m - n) + (n - l) + (l - m)] = A(0)

= 0

Step 4: collect the terms containing D: D[(l - 1)(m - n) + (m - 1)(n - l) + (n - 1)(l - m)]

Step 5: expand the bracket, (lm - ln - m + n) + (mn - ml - n + l) + (nl - nm - l + m) = 0, since every term cancels

Step 6: therefore LHS = 0 + D(0) = 0

Final answer: a(m - n) + b(n - l) + c(l - m) = 0, as required

Question: Q8(ii). If a, b, c are the lth, mth and nth terms of an A.P., show that l(b - c) + m(c - a) + n(a - b) = 0.

Answer: Step 1: let the A.P. have first term A and common difference D, so a = A + (l - 1)D, b = A + (m - 1)D, c = A + (n - 1)D

Step 2: compute the differences, b - c = (m - 1)D - (n - 1)D = (m - n)D, c - a = (n - l)D, a - b = (l - m)D

Step 3: substitute in the left-hand side, LHS = l(m - n)D + m(n - l)D + n(l - m)D

Step 4: take D common, LHS = D[lm - ln + mn - lm + nl - mn]

Step 5: simplify the bracket, lm - ln + mn - lm + nl - mn = 0

Final answer: l(b - c) + m(c - a) + n(a - b) = 0, as required

Question: Q9. If 1/a, 1/b, 1/c are in A.P., show that b = 2ac/(a + c).

Answer: Step 1: since 1/a, 1/b, 1/c are in A.P., the middle term is the arithmetic mean of its neighbours, 1/b = (1/2)(1/a + 1/c)

Step 2: simplify the right-hand side, 1/b = (1/2)((c + a)/(ac)) = (a + c)/(2ac)

Step 3: take reciprocals of both sides

Final answer: b = 2ac/(a + c), as required

Question: Q10. If 1/(a - b), 1/(b - c), 1/(c - a) are in A.P., then show that (a + c)/(a - b) = (a + b)/(c - a). (Statement as printed in the book; see the note at the end of the solution.)

Answer: Step 1: since the three fractions are in A.P., twice the middle term equals the sum of the first and third terms, 2/(b - c) = 1/(a - b) + 1/(c - a)

Step 2: combine the right-hand side over the common denominator (a - b)(c - a), 1/(a - b) + 1/(c - a) = [(c - a) + (a - b)] / [(a - b)(c - a)] = (c - b)/[(a - b)(c - a)]

Step 3: note c - b = -(b - c), so the condition becomes 2/(b - c) = -(b - c)/[(a - b)(c - a)]

Step 4: cross-multiply, 2(a - b)(c - a) = -(b - c)^2, that is, (b - c)^2 + 2(a - b)(c - a) = 0 ...(i)

Step 5: expand (i), b^2 - 2bc + c^2 + 2ac - 2a^2 - 2bc + 2ab = 0, i.e. b^2 + c^2 - 2a^2 + 2ab + 2ac - 4bc = 0

Step 6: on the other hand, the printed equality (a + c)/(a - b) = (a + b)/(c - a) cross-multiplies to (a + c)(c - a) = (a + b)(a - b), i.e. c^2 - a^2 = a^2 - b^2, i.e. b^2 + c^2 = 2a^2, which is a different condition from Step 5

Final answer: the A.P. condition is exactly equivalent to (b - c)^2 + 2(a - b)(c - a) = 0 (verified numerically, e.g. a - b = 1, b - c = 1 + sqrt(3) satisfies the A.P. condition and this relation); the equality printed in the book does not follow from it and appears to be a misprint in the book

Question: Q11. An object falling from rest falls 12 metres during the first second, 24 metres during the next second, 36 metres during the third second and so on. How much will it fall during the 8th second?

Answer: Step 1: the distances 12, 24, 36, ... form an A.P. with a = 12

Step 2: find the common difference d = 24 - 12 = 12

Step 3: write the formula an = a + (n - 1)d

Step 4: substitute n = 8, a8 = 12 + (8 - 1)(12)

Step 5: simplify, a8 = 12 + (7)(12) = 12 + 84

Final answer: the object falls 96 metres during the 8th second

Question: Q12. A man deposits Rs. 13,000 in a bank in the first month, Rs. 14,500 in the second month, Rs. 16,000 in the third month and so on. Find how much he has to deposit in the bank at the end of a year.

Answer: Step 1: the deposits 13000, 14500, 16000, ... form an A.P. with a = 13000

Step 2: find the common difference d = 14500 - 13000 = 1500

Step 3: one year has 12 months, so we need the 12th term, using an = a + (n - 1)d

Step 4: substitute n = 12, a12 = 13000 + (12 - 1)(1500)

Step 5: simplify, a12 = 13000 + (11)(1500) = 13000 + 16500

Final answer: he deposits Rs. 29,500 in the 12th month (at the end of the year)

Question: Q13. A boy saves Rs. 200 at the end of the first week and goes on increasing his saving by Rs. 25 weekly. After how many weeks will his weekly saving be Rs. 2000?

Answer: Step 1: the weekly savings 200, 225, 250, ... form an A.P. with a = 200

Step 2: the common difference is d = 25

Step 3: let the required saving be an = 2000 and write an = a + (n - 1)d

Step 4: substitute, 2000 = 200 + (n - 1)(25)

Step 5: simplify, 1800 = 25(n - 1), so n - 1 = 72

Step 6: solve for n

Final answer: after 73 weeks his weekly saving will be Rs. 2000

Exercise 4.3

Question: Q1(i). Find the A.M. between 18 and 26.

Answer: Step 1: write the formula for the arithmetic mean of two numbers, A.M. = (a + b)/2

Step 2: substitute a = 18 and b = 26, A.M. = (18 + 26)/2

Step 3: simplify the numerator, A.M. = 44/2

Final answer: A.M. = 22

Question: Q1(ii). Find the A.M. between 10 + 5 sqrt(3) and 4 - 5 sqrt(3).

Answer: Step 1: write the formula A.M. = (a + b)/2

Step 2: substitute, A.M. = [(10 + 5 sqrt(3)) + (4 - 5 sqrt(3))]/2

Step 3: collect like terms in the numerator, the sqrt(3) terms cancel, numerator = 10 + 4 = 14

Step 4: divide by 2

Final answer: A.M. = 7

Question: Q1(iii). Find the A.M. between 3a + 2b and 5a - 6b.

Answer: Step 1: write the formula A.M. = (a + b)/2, here the two numbers

are 3a + 2b and 5a - 6b

Step 2: substitute, A.M. = [(3a + 2b) + (5a - 6b)]/2

Step 3: collect like terms, numerator = 8a - 4b

Step 4: divide each term by 2

Final answer: A.M. = 4a - 2b

Question: Q1(iv). Find the A.M. between 5 + 7 sqrt(5) i and 8 - 7 sqrt(5) i.

Answer: Step 1: write the formula A.M. = (a + b)/2

Step 2: substitute, A.M. = [(5 + 7 sqrt(5) i) + (8 - 7 sqrt(5) i)]/2

Step 3: collect like terms, the imaginary parts cancel, numerator = 5 + 8 = 13

Step 4: divide by 2

Final answer: A.M. = 13/2

Question: Q1(v). Find the A.M. between 3a^2 - 5a + 6 and -a^2 + 7a - 4.

Answer: Step 1: write the formula A.M. = (a + b)/2

Step 2: substitute, A.M. = [(3a^2 - 5a + 6) + (-a^2 + 7a - 4)]/2

Step 3: collect like terms, numerator = 2a^2 + 2a + 2

Step 4: divide each term by 2

Final answer: A.M. = a^2 + a + 1

Question: Q1(vi). Find the A.M. between 3a - 5 and 5a + 3.

Answer: Step 1: write the formula A.M. = (a + b)/2

Step 2: substitute, A.M. = [(3a - 5) + (5a + 3)]/2

Step 3: collect like terms, numerator = 8a - 2

Step 4: divide each term by 2

Final answer: A.M. = 4a - 1

Question: Q2. Insert three A.Ms. between 3 and 11.

Answer: Step 1: let the A.P. be 3, A1, A2, A3, 11, which has 5 terms, so 11 is the 5th term

Step 2: write a5 = a + (5 - 1)d, i.e. 11 = 3 + 4d

Step 3: solve, 4d = 8, so d = 2

Step 4: compute the means one by one, A1 = 3 + 2 = 5

Step 5: A2 = 5 + 2 = 7

Step 6: A3 = 7 + 2 = 9

Final answer: the three A.Ms. are 5, 7, 9

Question: Q3. Insert four A.Ms. between sqrt(2) and 12/sqrt(2).

Answer: Step 1: simplify the second extreme, 12/sqrt(2) = 6 sqrt(2)

Step 2: the A.P. sqrt(2), A1, A2, A3, A4, 6 sqrt(2) has 6 terms, so 6 sqrt(2) = sqrt(2) + 5d

Step 3: solve, 5d = 5 sqrt(2), so d = sqrt(2)

Step 4: A1 = sqrt(2) + sqrt(2) = 2 sqrt(2)

Step 5: A2 = 2 sqrt(2) + sqrt(2) = 3 sqrt(2)

Step 6: A3 = 4 sqrt(2) and A4 = 5 sqrt(2)

Final answer: the four A.Ms. are 2 sqrt(2), 3 sqrt(2), 4 sqrt(2), 5 sqrt(2)

Question: Q4. Insert 5 A.Ms. between sqrt(5)/2 and 4 sqrt(5).

Answer: Step 1: with 5 means inserted there are 7 terms in all, so 4 sqrt(5) is the 7th term, 4 sqrt(5) = sqrt(5)/2 + 6d

Step 2: solve for d, 6d = 4 sqrt(5) - sqrt(5)/2 = 7 sqrt(5)/2, so d = 7 sqrt(5)/12

Step 3: A1 = sqrt(5)/2 + 7 sqrt(5)/12 = 6 sqrt(5)/12 + 7 sqrt(5)/12 = 13 sqrt(5)/12

Step 4: A2 = 13 sqrt(5)/12 + 7 sqrt(5)/12 = 20 sqrt(5)/12 = 5 sqrt(5)/3

Step 5: A3 = 20 sqrt(5)/12 + 7 sqrt(5)/12 = 27 sqrt(5)/12 = 9 sqrt(5)/4

Step 6: A4 = 27 sqrt(5)/12 + 7 sqrt(5)/12 = 34 sqrt(5)/12 = 17 sqrt(5)/6

Step 7: A5 = 34 sqrt(5)/12 + 7 sqrt(5)/12 = 41 sqrt(5)/12 (check: A5 + d = 48 sqrt(5)/12 = 4 sqrt(5))

Final answer: the five A.Ms. are 13 sqrt(5)/12, 5 sqrt(5)/3, 9 sqrt(5)/4, 17 sqrt(5)/6, 41 sqrt(5)/12

Question: Q5. Insert 6 A.Ms. between -30 and 30.

Answer: Step 1: with 6 means inserted there are 8 terms in all, so 30 = -30 + 7d

Step 2: solve, 7d = 60, so d = 60/7

Step 3: A1 = -30 + 60/7 = -210/7 + 60/7 = -150/7

Step 4: A2 = -150/7 + 60/7 = -90/7

Step 5: A3 = -90/7 + 60/7 = -30/7

Step 6: A4 = -30/7 + 60/7 = 30/7

Step 7: A5 = 30/7 + 60/7 = 90/7 and A6 = 90/7 + 60/7 = 150/7 (check: A6 + d = 210/7 = 30)

Final answer: the six A.Ms. are -150/7, -90/7, -30/7, 30/7, 90/7, 150/7

Question: Q6. If 5 and 8 are two A.Ms. between c and d, find c and d.

Answer: Step 1: since 5 and 8 are the two A.Ms. between c and d, the numbers c, 5, 8, d are in A.P.

Step 2: find the common difference from the two known consecutive terms, d_common = 8 - 5 = 3

Step 3: go one step backwards from 5, c = 5 - 3 = 2

Step 4: go one step forwards from 8, d = 8 + 3 = 11

Final answer: c = 2 and d = 11

Question: Q7. Find n so that [a^(n+5) + b^(n+5)] / [a^(n+4) + b^(n+4)] may be the A.M. between a and b.

Answer: Step 1: the A.M. between a and b is (a + b)/2, so set [a^(n+5) + b^(n+5)] / [a^(n+4) + b^(n+4)] = (a + b)/2

Step 2: cross-multiply, 2a^(n+5) + 2b^(n+5) = (a + b)[a^(n+4) + b^(n+4)]

Step 3: expand the right-hand side, (a + b)[a^(n+4) + b^(n+4)] = a^(n+5) + a b^(n+4) + b a^(n+4) + b^(n+5)

Step 4: cancel a^(n+5) and b^(n+5) from both sides, giving a^(n+5) + b^(n+5) = a b^(n+4) + b a^(n+4)

Step 5: rearrange, a^(n+5) - b a^(n+4) = a b^(n+4) - b^(n+5), i.e. a^(n+4)(a - b) = b^(n+4)(a - b)

Step 6: divide by (a - b) (for a not equal to b), a^(n+4) = b^(n+4), i.e. (a/b)^(n+4) = 1

Step 7: a power equals 1 only when the exponent is 0, so n + 4 = 0

Final answer: n = -4

Exercise 4.4

Question: Q1(i). Sum the series 3 + 8 + 13 + ... to 16 terms.

Answer: Step 1: identify a = 3 and d = 8 - 3 = 5

Step 2: write the formula Sn = n/2 [2a + (n - 1)d]

Step 3: substitute n = 16, a = 3, d = 5, S16 = 16/2 [2(3) + (16 - 1)(5)]

Step 4: simplify inside the bracket, 2(3) + (15)(5) = 6 + 75 = 81

Step 5: multiply, S16 = 8(81)

Final answer: S16 = 648

Question: Q1(ii). Sum the series (-3) + (-1) + 1 + 3 + 5 + ... to 18 terms.

Answer: Step 1: identify a = -3 and d = -1 - (-3) = 2

Step 2: write the formula Sn = n/2 [2a + (n - 1)d]

Step 3: substitute n = 18, S18 = 18/2 [2(-3) + (18 - 1)(2)]

Step 4: simplify inside the bracket, -6 + (17)(2) = -6 + 34 = 28

Step 5: multiply, S18 = 9(28)

Final answer: S18 = 252

Question: Q1(iii). Sum the series 4.57 + 4.87 + 5.17 + ... to 22 terms.

Answer: Step 1: identify a = 4.57 and d = 4.87 - 4.57 = 0.30

Step 2: write the formula Sn = n/2 [2a + (n - 1)d]

Step 3: substitute n = 22, S22 = 22/2 [2(4.57) + (22 - 1)(0.30)]

Step 4: simplify inside the bracket, 9.14 + (21)(0.30) = 9.14 + 6.30 = 15.44

Step 5: multiply, S22 = 11(15.44)

Final answer: S22 = 169.84

Question: Q1(iv). Sum the series 3/sqrt(2) + 2 sqrt(2) + 5/sqrt(2) + ... to n terms.

Answer: Step 1: rewrite the second term with denominator sqrt(2), 2 sqrt(2) = 4/sqrt(2), so the series is 3/sqrt(2) + 4/sqrt(2) + 5/sqrt(2) + ...

Step 2: identify a = 3/sqrt(2) and d = 4/sqrt(2) - 3/sqrt(2) = 1/sqrt(2)

Step 3: write the formula Sn = n/2 [2a + (n - 1)d]

Step 4: substitute, Sn = n/2 [6/sqrt(2) + (n - 1)/sqrt(2)] = n/2 . [(n + 5)/sqrt(2)]

Step 5: simplify, Sn = n(n + 5)/(2 sqrt(2)), and rationalising gives Sn = sqrt(2) n (n + 5)/4

Final answer: Sn = n(n + 5)/(2 sqrt(2)) = sqrt(2) n (n + 5)/4

Question: Q1(v). Sum the series 3/sqrt(2) + 2 sqrt(2) + 5/sqrt(2) + ... to 15 terms.

Answer: Step 1: from Q1(iv) this series has a = 3/sqrt(2), d = 1/sqrt(2) and sum to n terms Sn = n(n + 5)/(2 sqrt(2))

Step 2: substitute n = 15, S15 = 15(15 + 5)/(2 sqrt(2))

Step 3: simplify the numerator, 15(20) = 300, so S15 = 300/(2 sqrt(2)) = 150/sqrt(2)

Step 4: rationalise, 150/sqrt(2) = 150 sqrt(2)/2

Final answer: S15 = 75 sqrt(2) (approximately 106.07)

Question: Q2(i). Find the number of terms of the series (-7) + (-5) + (-3) + ... if the sum is 65.

Answer: Step 1: identify a = -7 and d = -5 - (-7) = 2

Step 2: write Sn = n/2 [2a + (n - 1)d] and set it equal to 65

Step 3: substitute, 65 = n/2 [2(-7) + (n - 1)(2)] = n/2 (2n - 16) = n(n - 8)

Step 4: form the quadratic, n^2 - 8n - 65 = 0

Step 5: factorise, (n - 13)(n + 5) = 0, so n = 13 or n = -5

Step 6: reject the negative value since the number of terms is positive

Final answer: n = 13 terms

Question: Q2(ii). Find the number of terms of the series -7 + (-4) + (-1) + ... if the sum is 114.

Answer: Step 1: identify a = -7 and d = -4 - (-7) = 3

Step 2: write Sn = n/2 [2a + (n - 1)d] and set it equal to 114

Step 3: substitute, 114 = n/2 [2(-7) + (n - 1)(3)] = n/2 (3n - 17)

Step 4: form the quadratic, 3n^2 - 17n - 228 = 0

Step 5: solve using the discriminant, disc = 289 + 2736 = 3025 = 55^2, so n = (17 + 55)/6 = 12 (the other root is negative)

Final answer: n = 12 terms

Question: Q2(iii). Find the number of terms of the series -9 + (-6) + (-3) + ... if the sum is 3225.

Answer: Step 1: identify a = -9 and d = -6 - (-9) = 3

Step 2: write Sn = n/2 [2a + (n - 1)d] and set it equal to 3225

Step 3: substitute, 3225 = n/2 [2(-9) + (n - 1)(3)] = n/2 (3n - 21)

Step 4: form the quadratic, 3n^2 - 21n - 6450 = 0, i.e. n^2 - 7n - 2150 = 0

Step 5: solve using the discriminant, disc = 49 + 8600 = 8649 = 93^2, so n = (7 + 93)/2 = 50 (the other root is negative)

Final answer: n = 50 terms

Question: Q2(iv). Find the number of terms of the series -9 - 6 - 3 + 0 + ... if the sum is 66.

Answer: Step 1: identify a = -9 and d = -6 - (-9) = 3

Step 2: write Sn = n/2 [2a + (n - 1)d] and set it equal to 66

Step 3: substitute, 66 = n/2 [2(-9) + (n - 1)(3)] = n/2 (3n - 21)

Step 4: form the quadratic, 3n^2 - 21n - 132 = 0, i.e. n^2 - 7n - 44 = 0

Step 5: factorise, (n - 11)(n + 4) = 0, so n = 11 or n = -4

Step 6: reject the negative value

Final answer: n = 11 terms

Question: Q3(i). Sum the series 3 + 5 - 7 + 9 + 11 - 13 + 15 + 17 - 19 + ... to 3n terms.

Answer: Step 1: group the 3n terms into n groups of three consecutive terms, (3 + 5 - 7) + (9 + 11 - 13) + (15 + 17 - 19) + ...

Step 2: evaluate the groups, 1, 7, 13, ...

Step 3: the group sums form an A.P. with first term 1 and common difference 6, and there are n groups

Step 4: write the sum of n groups, S = n/2 [2(1) + (n - 1)(6)]

Step 5: simplify, S = n/2 (6n - 4) = n(3n - 2)

Final answer: the sum to 3n terms is n(3n - 2)

Question: Q3(ii). Sum the series 1 + 4 - 7 + 10 + 13 - 16 + 19 + 22 - 25 +

... to 3n terms.

Answer: Step 1: group the 3n terms into n groups of three, (1 + 4 - 7) + (10 + 13 - 16) + (19 + 22 - 25) + ...

Step 2: evaluate the groups, -2, 7, 16, ...

Step 3: the group sums form an A.P. with first term -2 and common difference 9, and there are n groups

Step 4: write the sum, S = n/2 [2(-2) + (n - 1)(9)]

Step 5: simplify, S = n/2 (9n - 13)

Final answer: the sum to 3n terms is n(9n - 13)/2

Question: Q3(iii). Sum the series 9 + 12 - 15 + 18 + 21 - 24 + 27 + 30 - 33 + ... to 3n terms.

Answer: Step 1: group the 3n terms into n groups of three, (9 + 12 - 15) + (18 + 21 - 24) + (27 + 30 - 33) + ...

Step 2: evaluate the groups, 6, 15, 24, ...

Step 3: the group sums form an A.P. with first term 6 and common difference 9, and there are n groups

Step 4: write the sum, S = n/2 [2(6) + (n - 1)(9)]

Step 5: simplify, S = n/2 (9n + 3) = 3n(3n + 1)/2

Final answer: the sum to 3n terms is 3n(3n + 1)/2

Question: Q4. If Sn = n(2n - 1), then find the series.

Answer: Step 1: the nth term is obtained from an = Sn - S(n-1)

Step 2: compute S(n-1) = (n - 1)(2(n - 1) - 1) = (n - 1)(2n - 3) = 2n^2 - 5n + 3

Step 3: subtract, an = (2n^2 - n) - (2n^2 - 5n + 3) = 4n - 3

Step 4: put n = 1, 2, 3, 4 in an = 4n - 3, giving 1, 5, 9, 13

Final answer: the series is 1 + 5 + 9 + 13 + ... (an A.P. with a = 1, d = 4)

Question: Q5. Find the sum of the first 100 natural numbers which are neither exactly divisible by 3 nor by 7.

Answer: Step 1: find the sum of all the first 100 natural numbers, S = 100(101)/2 = 5050

Step 2: find the sum of the numbers divisible by 3, namely 3 + 6 + ... + 99, which has 33 terms, sum = 33/2 (3 + 99) = 33(51) = 1683

Step 3: find the sum of the numbers divisible by 7, namely 7 + 14 + ... + 98, which has 14 terms, sum = 14/2 (7 + 98) = 7(105) = 735

Step 4: the numbers divisible by both 3 and 7 (i.e. by 21) have been subtracted twice, so add them back once: 21 + 42 + 63 + 84 = 210

Step 5: combine, required sum = 5050 - 1683 - 735 + 210

Final answer: the required sum is 2842

Question: Q6. If S2, S3, S5 are the sums of 2n, 3n and 5n terms of an A.P., show that S5 = 5(S3 - S2).

Answer: Step 1: let the A.P. have first term a and common difference d, and write the sum of k terms as Sk_terms = k/2 [2a + (k - 1)d]

Step 2: write S3 - S2 = (3n/2)[2a + (3n - 1)d] - (2n/2)[2a + (2n - 1)d]

Step 3: expand both parts, S3 - S2 = n/2 { 3[2a + (3n - 1)d] - 2[2a + (2n - 1)d] }

Step 4: simplify inside the braces, 6a + (9n - 3)d - 4a - (4n - 2)d = 2a + (5n - 1)d, so S3 - S2 = n/2 [2a + (5n - 1)d]

Step 5: multiply by 5, 5(S3 - S2) = (5n/2)[2a + (5n - 1)d]

Step 6: recognise the right-hand side as the sum of 5n terms of the same A.P., which is S5

Final answer: S5 = 5(S3 - S2), as required

Question: Q7. The sums of n terms of two arithmetic series are in the ratio (3n + 2) : (n + 1). Find the ratio of their 8th terms.

Answer: Step 1: for two A.P.s, the ratio of their nth terms equals the ratio of the sums of (2n - 1) terms, an/a'n = S(2n-1)/S'(2n-1)

Step 2: for the 8th terms put 2n - 1 = 15, i.e. use the sums of 15 terms

Step 3: substitute n = 15 in the given ratio (3n + 2) : (n + 1), giving (3(15) + 2) : (15 + 1)

Step 4: simplify, 45 + 2 = 47 and 15 + 1 = 16

Final answer: the ratio of the 8th terms is 47 : 16

Question: Q8. The sum of three numbers in A.P. is 27 and their product is 405. Find the numbers.

Answer: Step 1: take the three numbers as a - d, a, a + d

Step 2: use the sum, (a - d) + a + (a + d) = 27, so 3a = 27 and a = 9

Step 3: use the product, (a - d)(a)(a + d) = 405, i.e. a(a^2 - d^2) = 405

Step 4: substitute a = 9, 9(81 - d^2) = 405, so 81 - d^2 = 45

Step 5: solve, d^2 = 36, so d = 6 (taking the positive value; d = -6 gives the same numbers in reverse order)

Step 6: write the numbers, a - d = 3, a = 9, a + d = 15

Final answer: the numbers are 3, 9, 15

Question: Q9. Find five numbers in A.P. whose sum is 25 and the sum of whose squares is 135.

Answer: Step 1: take the five numbers as a - 2d, a - d, a, a + d, a + 2d

Step 2: use the sum, 5a = 25, so a = 5

Step 3: use the sum of squares, (a - 2d)^2 + (a - d)^2 + a^2 + (a + d)^2 + (a + 2d)^2 = 135, which simplifies to 5a^2 + 10d^2 = 135

Step 4: substitute a = 5, 5(25) + 10d^2 = 135, so 10d^2 = 10 and d^2 = 1

Step 5: solve, d = 1 (d = -1 gives the same numbers in reverse order)

Step 6: write the numbers, 5 - 2 = 3, 5 - 1 = 4, 5, 5 + 1 = 6, 5 + 2 = 7

Final answer: the numbers are 3, 4, 5, 6, 7

Question: Q10. The sum of Rs. 42,000 is distributed among five persons so that each person after the first receives Rs. 80 less than the preceding person. How much does each person receive?

Answer: Step 1: the amounts form an A.P. with common difference d = -80 and 5 terms

Step 2: let the first person receive a, and write S5 = 5/2 [2a + (5 - 1)(-80)] = 42000

Step 3: simplify, 5/2 (2a - 320) = 42000, so 2a - 320 = 16800

Step 4: solve, 2a = 17120, so a = 8560

Step 5: subtract 80 repeatedly for the remaining persons, 8480, 8400, 8320, 8240 (check: 8560 + 8480 + 8400 + 8320 + 8240 = 42000)

Final answer: the five persons receive Rs. 8560, Rs. 8480, Rs. 8400, Rs. 8320 and Rs. 8240

Question: Q11. A well digging company charges Rs. 1,250 for the first meter, Rs. 1,500 for the second meter and Rs. 1,750 for the third meter and so on. What is the depth of a well that costs Rs. 50,000?

Answer: Step 1: the charges 1250, 1500, 1750, ... form an A.P. with a = 1250 and d = 250

Step 2: let the depth be n metres, so the total cost is Sn = 50000

Step 3: write Sn = n/2 [2a + (n - 1)d], 50000 = n/2 [2500 + (n - 1)(250)]

Step 4: simplify, 50000 = n/2 (250n + 2250), i.e. 250n^2 + 2250n - 100000 = 0, i.e. n^2 + 9n - 400 = 0

Step 5: factorise, (n + 25)(n - 16) = 0, so n = 16 or n = -25

Step 6: reject the negative value

Final answer: the depth of the well is 16 metres

Question: Q12. A man borrows Rs. 25,000 and agrees to repay with a total profit of Rs. 10,000 in 10 installments, each installment being less than the preceding by Rs. 200. What should be his first installment?

Answer: Step 1: the total repayment is 25000 + 10000 = 35000, paid in 10 installments

Step 2: the installments form an A.P. with d = -200 and n = 10

Step 3: let the first installment be a and write S10 = 10/2 [2a + (10 - 1)(-200)] = 35000

Step 4: simplify, 5(2a - 1800) = 35000, so 2a - 1800 = 7000

Step 5: solve, 2a = 8800

Final answer: the first installment should be Rs. 4400

Exercise 4.5

Question: Q1(i). Find the indicated term, the 5th term, of the G.P. 3, 6, 12, ...

Answer: Step 1: identify the first term a = 3

Step 2: find the common ratio r = 6/3 = 2

Step 3: write the formula an = a r^(n-1)

Step 4: substitute n = 5, a5 = 3(2)^(5-1) = 3(2^4)

Step 5: simplify, 2^4 = 16, so a5 = 3(16)

Final answer: a5 = 48

Question: Q1(ii). Find the indicated term, the 9th term, of the G.P. 32, 16, 8, ...

Answer: Step 1: identify the first term a = 32

Step 2: find the common ratio r = 16/32 = 1/2

Step 3: write the formula an = a r^(n-1)

Step 4: substitute n = 9, a9 = 32(1/2)^(9-1) = 32(1/2)^8

Step 5: simplify, (1/2)^8 = 1/256, so a9 = 32/256

Final answer: a9 = 1/8

Question: Q1(iii). Find the indicated term, the 12th term, of the G.P. 4, 2 sqrt(2), 2, ...

Answer: Step 1: identify the first term a = 4

Step 2: find the common ratio r = 2 sqrt(2)/4 = sqrt(2)/2 = 1/sqrt(2)

Step 3: write the formula an = a r^(n-1)

Step 4: substitute n = 12, a12 = 4(1/sqrt(2))^(12-1) = 4(1/sqrt(2))^11

Step 5: simplify, (1/sqrt(2))^11 = 1/(32 sqrt(2)), so a12 = 4/(32 sqrt(2)) = 1/(8 sqrt(2))

Final answer: a12 = 1/(8 sqrt(2)) = sqrt(2)/16

Question: Q1(iv). Find the indicated term, the 16th term, of the G.P. i, 1, -i, ...

Answer: Step 1: identify the first term a = i

Step 2: find the common ratio r = 1/i = -i (multiplying numerator and denominator by i)

Step 3: write the formula an = a r^(n-1)

Step 4: substitute n = 16, a16 = i(-i)^(16-1) = i(-i)^15

Step 5: simplify the power, (-i)^15 = (-i)^12 . (-i)^3 = (1)(i) = i, since (-i)^4 = 1 and (-i)^3 = i

Step 6: multiply, a16 = i . i = i^2 = -1 (check by the repeating pattern i, 1, -i, -1 of period 4: the 16th term is -1)

Final answer: a16 = -1

Question: Q1(v). Find the indicated term, the 9th term, of the G.P. b^2 - c^2, b + c, (b + c)/(b - c), ...

Answer: Step 1: identify the first term a = b^2 - c^2 = (b - c)(b + c)

Step 2: find the common ratio r = (b + c)/(b^2 - c^2) = (b + c)/[(b - c)(b + c)] = 1/(b - c) (check with the third term: (b + c) . 1/(b - c) = (b + c)/(b - c))

Step 3: write the formula an = a r^(n-1)

Step 4: substitute n = 9, a9 = (b^2 - c^2) [1/(b - c)]^8 = (b - c)(b + c)/(b - c)^8

Step 5: cancel one factor (b - c)

Final answer: a9 = (b + c)/(b - c)^7

Question: Q2. In a G.P., a1 = 5/9 and a6 = 15625/9. Find its nth term.

Answer: Step 1: write the formula an = a r^(n-1) with a = 5/9

Step 2: use the 6th term, a6 = (5/9) r^5 = 15625/9

Step 3: divide both sides by 5/9, r^5 = (15625/9)/(5/9) = 3125

Step 4: recognise 3125 = 5^5, so r = 5

Step 5: substitute a and r in the general term, an = (5/9)(5)^(n-1)

Step 6: simplify using the law of exponents, (5)(5)^(n-1) = 5^n

Final answer: an = 5^n / 9

Question: Q3. Find the nth term of the geometric sequence if a5/a3 = 4/9 and a2 = 4/9.

Answer: Step 1: write a5 = a r^4 and a3 = a r^2, so a5/a3 = r^2

Step 2: use the given ratio, r^2 = 4/9, so r = 2/3 or r = -2/3

Step 3: use a2 = a r = 4/9 with r = 2/3, a = (4/9)/(2/3) = (4/9)(3/2) = 2/3

Step 4: write the nth term for this case, an = (2/3)(2/3)^(n-1) = (2/3)^n

Step 5: use a2 = a r = 4/9 with r = -2/3, a = (4/9)/(-2/3) = -2/3

Step 6: write the nth term for this case, an = (-2/3)(-2/3)^(n-1) = (-2/3)^n

Final answer: an = (2/3)^n (with a = 2/3, r = 2/3) or an = (-2/3)^n (with a = -2/3, r = -2/3)

Question: Q4. Show that the reciprocals of the terms of the G.P. a1, a1 r^2, a1 r^4, ... form another G.P.

Answer: Step 1: write the terms of the given G.P., a1, a1 r^2, a1 r^4, a1

r^6, ..., whose general term is a1 (r^2)^(n-1), a G.P. with common ratio r^2

Step 2: take reciprocals of the terms, 1/a1, 1/(a1 r^2), 1/(a1 r^4), 1/(a1 r^6), ...

Step 3: find the ratio of the second reciprocal to the first, [1/(a1 r^2)] / (1/a1) = 1/r^2

Step 4: find the ratio of the third reciprocal to the second, [1/(a1 r^4)] / [1/(a1 r^2)] = 1/r^2

Step 5: the ratio is the same (constant) for every pair of consecutive terms, namely 1/r^2

Final answer: hence the reciprocals form another G.P. with first term 1/a1 and common ratio 1/r^2, as required

Question: Q5. How many terms are in the G.P. 1/3, 1/12, 1/48, ..., 1/196608?

Answer: Step 1: identify a = 1/3 and r = (1/12)/(1/3) = 1/4

Step 2: let the last term be the nth term, an = a r^(n-1) = 1/196608

Step 3: substitute, (1/3)(1/4)^(n-1) = 1/196608

Step 4: simplify, (1/4)^(n-1) = 3/196608 = 1/65536

Step 5: write 65536 as a power of 4, 65536 = 4^8, so (1/4)^(n-1) = (1/4)^8

Step 6: compare exponents, n - 1 = 8

Final answer: n = 9 terms

Question: Q6. Find three consecutive numbers in G.P. whose sum is 39 and their product is 729.

Answer: Step 1: take the three numbers as a/r, a, a r

Step 2: use the product, (a/r)(a)(a r) = a^3 = 729 = 9^3, so a = 9

Step 3: use the sum, 9/r + 9 + 9r = 39, so 9(1/r + r) = 30 and 1/r + r = 10/3

Step 4: multiply by 3r, 3 + 3r^2 = 10r, i.e. 3r^2 - 10r + 3 = 0

Step 5: factorise, (3r - 1)(r - 3) = 0, so r = 3 or r = 1/3

Step 6: for r = 3 the numbers are 9/3 = 3, 9, 9(3) = 27; for r = 1/3 they are 27, 9, 3

Final answer: the numbers are 3, 9, 27

Question: Q7. The number of bacteria in a culture increased in G.P. from 515,000 to 15,45,000 in 7 days. Find the daily rate of increase, assuming the rate of increase to be constant.

Answer: Step 1: following the book's worked example for this type of problem, take the initial number as the first term a = 515000 and the number after 7 days as the 7th term a7 = 1545000

Step 2: write a7 = a r^(7-1), i.e. 1545000 = 515000 r^6

Step 3: divide, r^6 = 1545000/515000 = 3

Step 4: take the sixth root, r = 3^(1/6) = approximately 1.201

Step 5: interpret the ratio: each day the number of bacteria becomes about 1.201 times the previous day's number, an increase of 1.201 - 1 = 0.201, i.e. about 20.1% per day

Final answer: the daily growth factor is 3^(1/6), approximately 1.201, so the daily rate of increase is about 20.1% (the population grows to about 120.1% of itself each day)

Question: Q8. Find the profit on Rs. 1000 for 5 years at 4% per annum compound profit.

Answer: Step 1: with compound profit the amounts form a G.P. with ratio r = 1 + 4/100 = 1.04

Step 2: the amount after 5 years is A = 1000 (1.04)^5

Step 3: compute the power in stages, (1.04)^2 = 1.0816, (1.04)^4 = (1.0816)^2 = 1.16986 (approximately), (1.04)^5 = 1.16986(1.04) = 1.21665 (approximately)

Step 4: multiply by the principal, A = 1000(1.21665) = Rs. 1216.65 (approximately)

Step 5: subtract the principal to get the profit, 1216.65 - 1000

Final answer: the compound profit is approximately Rs. 216.65

Exercise 4.6

Question: Q1(i). Find the G.M. between -2 and -8.

Answer: Step 1: write the formula for the geometric mean of two numbers, G = +/- sqrt(ab)

Step 2: substitute a = -2 and b = -8, G = +/- sqrt((-2)(-8))

Step 3: simplify under the root, (-2)(-8) = 16

Step 4: take the square root, sqrt(16) = 4

Final answer: G.M. = +/- 4 (check: -2, -4, -8 and -2, 4, -8 are both G.P.s)

Question: Q1(ii). Find the G.M. between 3 and 1/3.

Answer: Step 1: write the formula G = +/- sqrt(ab)

Step 2: substitute a = 3 and b = 1/3, G = +/- sqrt(3 . 1/3)

Step 3: simplify under the root, 3 . (1/3) = 1

Step 4: take the square root, sqrt(1) = 1

Final answer: G.M. = +/- 1

Question: Q1(iii). Find the G.M. between 8 sqrt(2) and 9 sqrt(2).

Answer: Step 1: write the formula G = +/- sqrt(ab)

Step 2: substitute, G = +/- sqrt(8 sqrt(2) . 9 sqrt(2))

Step 3: simplify under the root, 8 . 9 . (sqrt(2))^2 = 72 . 2 = 144

Step 4: take the square root, sqrt(144) = 12

Final answer: G.M. = +/- 12

Question: Q1(iv). Find the G.M. between -2i and 8i.

Answer: Step 1: write the formula G = +/- sqrt(ab)

Step 2: substitute, G = +/- sqrt((-2i)(8i))

Step 3: simplify under the root, (-2i)(8i) = -16 i^2 = -16(-1) = 16

Step 4: take the square root, sqrt(16) = 4

Final answer: G.M. = +/- 4

Question: Q1(v). Find the G.M. between 7 + i and 7 - i.

Answer: Step 1: write the formula G = +/- sqrt(ab)

Step 2: substitute, G = +/- sqrt((7 + i)(7 - i))

Step 3: simplify under the root using (x + y)(x - y) = x^2 + y^2 for conjugates, (7 + i)(7 - i) = 49 + 1 = 50

Step 4: take the square root, sqrt(50) = 5 sqrt(2)

Final answer: G.M. = +/- 5 sqrt(2)

Question: Q1(vi). Find the G.M. between 16/9 and 9/25.

Answer: Step 1: write the formula G = +/- sqrt(ab)

Step 2: substitute, G = +/- sqrt((16/9)(9/25))

Step 3: simplify under the root, the 9s cancel, giving 16/25

Step 4: take the square root, sqrt(16/25) = 4/5

Final answer: G.M. = +/- 4/5

Question: Q2(i). Insert two G.Ms. between 3 and 81.

Answer: Step 1: let the G.P. be 3, G1, G2, 81, so 81 is the 4th term, 81 = 3 r^3

Step 2: solve for the ratio, r^3 = 27, so r = 3

Step 3: compute the first mean, G1 = 3 . 3 = 9

Step 4: compute the second mean, G2 = 9 . 3 = 27 (check: 27 . 3 = 81)

Final answer: the two G.Ms. are 9 and 27

Question: Q2(ii). Insert three G.Ms. between 2 and 1/2.

Answer: Step 1: let the G.P. be 2, G1, G2, G3, 1/2, so 1/2 is the 5th term, 1/2 = 2 r^4

Step 2: solve for the ratio, r^4 = 1/4, so r = 1/sqrt(2) (taking the positive value; r = -1/sqrt(2) gives the alternating set -sqrt(2), 1, -1/sqrt(2))

Step 3: G1 = 2 . (1/sqrt(2)) = sqrt(2)

Step 4: G2 = sqrt(2) . (1/sqrt(2)) = 1

Step 5: G3 = 1 . (1/sqrt(2)) = 1/sqrt(2) (check: G3 . r = (1/sqrt(2))(1/sqrt(2)) = 1/2)

Final answer: the three G.Ms. are sqrt(2), 1, 1/sqrt(2)

Question: Q2(iii). Insert four G.Ms. between 3 and 96.

Answer: Step 1: let the G.P. be 3, G1, G2, G3, G4, 96, so 96 is the 6th term, 96 = 3 r^5

Step 2: solve for the ratio, r^5 = 32 = 2^5, so r = 2

Step 3: G1 = 3 . 2 = 6

Step 4: G2 = 6 . 2 = 12

Step 5: G3 = 12 . 2 = 24

Step 6: G4 = 24 . 2 = 48 (check: 48 . 2 = 96)

Final answer: the four G.Ms. are 6, 12, 24, 48

Question: Q2(iv). Insert five G.Ms. between 16 and 1024.

Answer: Step 1: let the G.P. be 16, G1, G2, G3, G4, G5, 1024, so 1024 is the 7th term, 1024 = 16 r^6

Step 2: solve for the ratio, r^6 = 64, so r = 2 (taking the positive value; r = -2 gives -32, 64, -128, 256, -512)

Step 3: G1 = 16 . 2 = 32

Step 4: G2 = 32 . 2 = 64

Step 5: G3 = 64 . 2 = 128

Step 6: G4 = 128 . 2 = 256

Step 7: G5 = 256 . 2 = 512 (check: 512 . 2 = 1024)

Final answer: the five G.Ms. are 32, 64, 128, 256, 512

Question: Q3. The A.M. between two numbers is 29 and the geometric mean is 21. Find the numbers.

Answer: Step 1: let the numbers be a and b; from the A.M., (a + b)/2 = 29, so a + b = 58

Step 2: from the G.M., sqrt(ab) = 21, so ab = 441

Step 3: a and b are the roots of t^2 - (a + b) t + ab = 0, i.e. t^2 - 58 t + 441 = 0

Step 4: compute the discriminant, 58^2 - 4(441) = 3364 - 1764 = 1600, and sqrt(1600) = 40

Step 5: solve, t = (58 + 40)/2 = 49 or t = (58 - 40)/2 = 9

Final answer: the numbers are 49 and 9 (check: (49 + 9)/2 = 29 and sqrt(49 . 9) = 21)

Question: Q4. For what value of n is [a^(n-2) + b^(n-2)] / [a^(n-3) + b^(n-3)] the G.M. between a and b?

Answer: Step 1: take the (positive) G.M. between a and b as sqrt(ab), and set [a^(n-2) + b^(n-2)] / [a^(n-3) + b^(n-3)] = sqrt(ab)

Step 2: cross-multiply, a^(n-2) + b^(n-2) = sqrt(ab) [a^(n-3) + b^(n-3)] = a^(n - 5/2) b^(1/2) + a^(1/2) b^(n - 5/2)

Step 3: bring the terms in a to one side and the terms in b to the other, a^(n-2) - a^(n - 5/2) b^(1/2) = a^(1/2) b^(n - 5/2) - b^(n-2)

Step 4: factorise both sides, a^(n - 5/2) [a^(1/2) - b^(1/2)] = b^(n - 5/2) [a^(1/2) - b^(1/2)]

Step 5: cancel the common factor [a^(1/2) - b^(1/2)] (for a not equal to b), giving a^(n - 5/2) = b^(n - 5/2), i.e. (a/b)^(n - 5/2) = 1

Step 6: a power equals 1 only when the exponent is 0, so n - 5/2 = 0

Final answer: n = 5/2

Question: Q5. Show that the nth root of the product of n geometric means between x and y is the geometric mean between x and y.

Answer: Step 1: let G1, G2, ..., Gn be the n G.Ms. between x and y, so x, G1, G2, ..., Gn, y is a G.P. with common ratio r = (y/x)^(1/(n+1))

Step 2: write the means, Gk = x r^k for k = 1, 2, ..., n

Step 3: form the product, G1 G2 ... Gn = x^n r^(1 + 2 + ... + n)

Step 4: use the sum of the first n natural numbers, 1 + 2 + ... + n = n(n + 1)/2, so the product = x^n r^[n(n + 1)/2]

Step 5: substitute r = (y/x)^(1/(n+1)), product = x^n (y/x)^(n/2) = [x^2 . (y/x)]^(n/2) = (xy)^(n/2)

Step 6: take the nth root, (product)^(1/n) = [(xy)^(n/2)]^(1/n) = (xy)^(1/2) = sqrt(xy)

Final answer: the nth root of the product of the n G.Ms. is sqrt(xy), which is the G.M. between x and y, as required

Question: Q6. The A.M. of two positive integral numbers exceeds their (positive) G.M. by 2 and their sum is 20. Find the numbers.

Answer: Step 1: let the numbers be a and b with a + b = 20, so the A.M. is (a + b)/2 = 10

Step 2: use the given condition, G.M. = A.M. - 2 = 10 - 2 = 8, so sqrt(ab) = 8 and ab = 64

Step 3: a and b are the roots of t^2 - 20 t + 64 = 0

Step 4: compute the discriminant, 400 - 256 = 144, and sqrt(144) = 12

Step 5: solve, t = (20 + 12)/2 = 16 or t = (20 - 12)/2 = 4

Final answer: the numbers are 16 and 4 (check: A.M. = 10, G.M. = sqrt(64)

= 8, and 10 exceeds 8 by 2)

Question: Q7. The A.M. between two numbers is 5 and their (positive) G.M. is 4. Find the numbers.

Answer: Step 1: let the numbers be a and b; from the A.M., (a + b)/2 = 5, so a + b = 10

Step 2: from the G.M., sqrt(ab) = 4, so ab = 16

Step 3: a and b are the roots of t^2 - 10 t + 16 = 0

Step 4: factorise, (t - 8)(t - 2) = 0, so t = 8 or t = 2

Final answer: the numbers are 8 and 2 (check: (8 + 2)/2 = 5 and sqrt(8 . 2) = 4)

Exercise 4.7

Question: Q1(i). Find the sum: 1 + 1/2 + 1/4 + ... + 1/512.

Answer: Step 1: this is a geometric series with a = 1 and r = 1/2

Step 2: find the number of terms: 1/512 = (1/2)^9 is the 10th term, so n = 10

Step 3: write the formula Sn = a(1 - r^n)/(1 - r)

Step 4: substitute, S10 = 1(1 - (1/2)^10)/(1 - 1/2) = (1 - 1/1024)/(1/2)

Step 5: simplify, S10 = 2(1023/1024) = 1023/512

Final answer: the sum is 1023/512 (approximately 1.998)

Question: Q1(ii). Find the sum: 2^6 + 2^7 + 2^8 + ... + 2^13.

Answer: Step 1: this is a geometric series with a = 2^6 = 64 and r = 2

Step 2: count the terms, from 2^6 to 2^13 there are 13 - 6 + 1 = 8 terms

Step 3: write the formula Sn = a(r^n - 1)/(r - 1)

Step 4: substitute, S8 = 64(2^8 - 1)/(2 - 1) = 64(256 - 1)

Step 5: simplify, S8 = 64(255)

Final answer: the sum is 16320

Question: Q1(iii). Find the sum: 1/x + 1/x^2 + 1/x^3 + ... + 1/x^8.

Answer: Step 1: this is a geometric series with a = 1/x and r = 1/x

Step 2: count the terms, n = 8

Step 3: write the formula Sn = a(1 - r^n)/(1 - r)

Step 4: substitute, S8 = (1/x)(1 - (1/x)^8)/(1 - 1/x)

Step 5: simplify, multiplying numerator and denominator suitably, S8 = (1 - (1/x)^8)/(x - 1) = (x^8 - 1)/(x^8 (x - 1))

Final answer: the sum is (1 - 1/x^8)/(x - 1)

Question: Q1(iv). Find the sum: 0.9 + 0.99 + 0.999 + ... to n terms.

Answer: Step 1: write each term as 1 minus a small part, (1 - 0.1) + (1 - 0.01) + (1 - 0.001) + ..., i.e. the kth term is 1 - (1/10)^k

Step 2: sum the parts separately, Sn = n - [(1/10) + (1/10)^2 + ... + (1/10)^n]

Step 3: the bracket is a geometric series with a = 1/10, r = 1/10, so it equals (1/10)(1 - (1/10)^n)/(1 - 1/10)

Step 4: simplify the bracket, (1/10)/(9/10) = 1/9, so the bracket equals (1/9)(1 - (1/10)^n)

Step 5: subtract from n

Final answer: Sn = n - (1/9)(1 - (1/10)^n)

Question: Q1(v). Find the sum: 7 + 77 + 777 + ... to n terms.

Answer: Step 1: take 7 common, Sn = 7[1 + 11 + 111 + ... to n terms]

Step 2: write each bracket term using 9, 1 = 9/9, 11 = 99/9, 111 = 999/9, i.e. the kth term of the bracket is (10^k - 1)/9

Step 3: sum, Sn = (7/9)[(10 + 10^2 + ... + 10^n) - n]

Step 4: the powers of 10 form a geometric series with sum 10(10^n - 1)/(10 - 1) = (10/9)(10^n - 1)

Step 5: substitute

Final answer: Sn = (7/9)[(10/9)(10^n - 1) - n]

Question: Q1(vi). Find the sum: 0.2 + 0.22 + 0.222 + ... to n terms.

Answer: Step 1: take 2 common and write each term with denominator 9, the kth term is (2/9)(1 - (1/10)^k), since 0.2 = (2/9)(0.9), 0.22 = (2/9)(0.99), etc.

Step 2: sum, Sn = (2/9)[n - ((1/10) + (1/10)^2 + ... + (1/10)^n)]

Step 3: the bracketed geometric series equals (1/9)(1 - (1/10)^n), as in Q1(iv)

Step 4: substitute

Final answer: Sn = (2/9) n - (2/81)(1 - (1/10)^n)

Question: Q1(vii). Find the sum: 3 + 33 + 333 + ... to n terms.

Answer: Step 1: take 3 common, Sn = 3[1 + 11 + 111 + ... to n terms]

Step 2: write the kth bracket term as (10^k - 1)/9

Step 3: sum, Sn = (3/9)[(10 + 10^2 + ... + 10^n) - n] = (1/3)[

(10/9)(10^n - 1) - n]

Final answer: Sn = (1/3)[(10/9)(10^n - 1) - n]

Question: Q2(i). Find the sum to n terms of the series 1 + (a + b) + (a^2 + ab + b^2) + (a^3 + a^2 b + a b^2 + b^3) + ..., where a > 1, b > 1 and a > b.

Answer: Step 1: recognise each term as a quotient, the mth term is (a^m - b^m)/(a - b), since (a - b)(a^(m-1) + a^(m-2) b + ... + b^(m-1)) = a^m - b^m

Step 2: write the sum, Sn = 1/(a - b) [sum from m=1 to n of a^m - sum from m=1 to n of b^m]

Step 3: sum each geometric series, sum of a^m = a(a^n - 1)/(a - 1) and sum of b^m = b(b^n - 1)/(b - 1)

Step 4: substitute

Final answer: Sn = 1/(a - b) [a(a^n - 1)/(a - 1) - b(b^n - 1)/(b - 1)]

Question: Q2(ii). Find the sum to n terms of the series r + (1 + k) r^2 + (1 + k + k^2) r^3 + ..., where k and r are proper fractions. (The book prints the third term with r^2; it is solved by the clear pattern r, r^2, r^3, ...)

Answer: Step 1: write the mth term using 1 + k + ... + k^(m-1) = (1 - k^m)/(1 - k), so Tm = r^m (1 - k^m)/(1 - k)

Step 2: split the term, Tm = 1/(1 - k) [r^m - (kr)^m]

Step 3: sum the two geometric series separately, sum of r^m (m = 1 to n) = r(1 - r^n)/(1 - r) and sum of (kr)^m = kr(1 - (kr)^n)/(1 - kr)

Step 4: substitute and keep the factor 1/(1 - k)

Final answer: Sn = 1/(1 - k) [r(1 - r^n)/(1 - r) - kr(1 - (kr)^n)/(1 - kr)]

Question: Q3. If an = (1/4)^n then find its sum up to n terms.

Answer: Step 1: the terms are a1 = 1/4, a2 = (1/4)^2, ..., so the sum is a geometric series with first term 1/4 and ratio 1/4

Step 2: write the formula Sn = a(1 - r^n)/(1 - r)

Step 3: substitute a = 1/4 and r = 1/4, Sn = (1/4)(1 - (1/4)^n)/(1 - 1/4)

Step 4: simplify, (1/4)/(3/4) = 1/3

Final answer: Sn = (1/3)(1 - (1/4)^n) = (1/3)(1 - 1/4^n)

Question: Q4(i). Find the sum of the infinite geometric series 1/5 + 1/25 + 1/125 + ...

Answer: Step 1: identify a = 1/5 and r = (1/25)/(1/5) = 1/5

Step 2: since |r| = 1/5 < 1, the series converges and S = a/(1 - r)

Step 3: substitute, S = (1/5)/(1 - 1/5) = (1/5)/(4/5)

Step 4: simplify

Final answer: S = 1/4

Question: Q4(ii). Find the sum of the infinite geometric series 1/2 + 1/4 + 1/8 + ...

Answer: Step 1: identify a = 1/2 and r = 1/2

Step 2: since |r| < 1, S = a/(1 - r)

Step 3: substitute, S = (1/2)/(1 - 1/2) = (1/2)/(1/2)

Final answer: S = 1

Question: Q4(iii). Find the sum of the infinite geometric series 9/4 + 3/2 + 1 + 2/3 + ...

Answer: Step 1: identify a = 9/4 and r = (3/2)/(9/4) = (3/2)(4/9) = 2/3

Step 2: since |r| = 2/3 < 1, S = a/(1 - r)

Step 3: substitute, S = (9/4)/(1 - 2/3) = (9/4)/(1/3)

Step 4: simplify, (9/4)(3)

Final answer: S = 27/4

Question: Q4(iv). Find the sum of the infinite geometric series 2 + sqrt(2) + 1 + ...

Answer: Step 1: identify a = 2 and r = sqrt(2)/2 = 1/sqrt(2)

Step 2: since |r| = 1/sqrt(2) < 1, S = a/(1 - r)

Step 3: substitute, S = 2/(1 - 1/sqrt(2))

Step 4: multiply numerator and denominator by sqrt(2), S = 2 sqrt(2)/(sqrt(2) - 1)

Step 5: rationalise by multiplying by (sqrt(2) + 1), S = 2 sqrt(2)(sqrt(2) + 1)/(2 - 1) = 2 sqrt(2)(sqrt(2) + 1) = 4 + 2 sqrt(2)

Final answer: S = 4 + 2 sqrt(2) (approximately 6.83)

Question: Q4(v). Find the sum of the infinite geometric series 2 + 1 + 0.5 + ...

Answer: Step 1: identify a = 2 and r = 1/2

Step 2: since |r| < 1, S = a/(1 - r)

Step 3: substitute, S = 2/(1 - 1/2) = 2/(1/2)

Final answer: S = 4

Question: Q4(vi). Find the sum of the infinite geometric series 4 + 2 sqrt(2) + 2 + sqrt(2) + 1 + ...

Answer: Step 1: identify a = 4 and r = (2 sqrt(2))/4 = 1/sqrt(2)

Step 2: since |r| = 1/sqrt(2) < 1, S = a/(1 - r)

Step 3: substitute, S = 4/(1 - 1/sqrt(2))

Step 4: multiply numerator and denominator by sqrt(2), S = 4 sqrt(2)/(sqrt(2) - 1)

Step 5: rationalise, S = 4 sqrt(2)(sqrt(2) + 1) = 8 + 4 sqrt(2)

Final answer: S = 8 + 4 sqrt(2) (approximately 13.66)

Question: Q4(vii). Find the sum of the infinite geometric series 0.1 + 0.05 + 0.025 + ...

Answer: Step 1: identify a = 0.1 and r = 0.05/0.1 = 0.5

Step 2: since |r| < 1, S = a/(1 - r)

Step 3: substitute, S = 0.1/(1 - 0.5) = 0.1/0.5

Final answer: S = 0.2

Question: Q5(i). Find the vulgar fraction equivalent to the recurring decimal 0.7 (7 recurring).

Answer: Step 1: write the decimal as an infinite geometric series, 0.777... = 0.7 + 0.07 + 0.007 + ...

Step 2: identify a = 0.7 and r = 0.1

Step 3: use S = a/(1 - r), S = 0.7/(1 - 0.1) = 0.7/0.9

Step 4: simplify, 0.7/0.9 = 7/9

Final answer: 0.777... = 7/9

Question: Q5(ii). Find the vulgar fraction equivalent to the recurring decimal 1.74 (both 7 and 4 recurring).

Answer: Step 1: separate the whole number, 1.747474... = 1 +

0.747474...

Step 2: write the recurring part as a series, 0.74 + 0.0074 + 0.000074 + ..., with a = 0.74 and r = 0.01

Step 3: sum it, 0.74/(1 - 0.01) = 0.74/0.99 = 74/99

Step 4: add the whole number, 1 + 74/99 = (99 + 74)/99

Final answer: 1.747474... = 173/99

Question: Q5(iii). Find the vulgar fraction equivalent to the recurring decimal 0.259 (2, 5 and 9 all recurring).

Answer: Step 1: write the decimal as an infinite geometric series, 0.259259... = 0.259 + 0.000259 + ..., with a = 0.259 and r = 0.001

Step 2: use S = a/(1 - r), S = 0.259/(1 - 0.001) = 0.259/0.999

Step 3: multiply numerator and denominator by 1000, S = 259/999

Step 4: simplify by dividing by 37, 259/999 = 7/27

Final answer: 0.259259... = 7/27

Question: Q5(iv). Find the vulgar fraction equivalent to the recurring decimal 1.148 (only 8 recurring).

Answer: Step 1: separate the non-recurring part, 1.14888... = 1.14 + 0.00888...

Step 2: write the recurring tail as a series, 0.008 + 0.0008 + 0.00008 + ..., with a = 0.008 and r = 0.1

Step 3: sum the tail, 0.008/(1 - 0.1) = 0.008/0.9 = 8/900 = 2/225

Step 4: convert the non-recurring part, 1.14 = 114/100 = 57/50

Step 5: add, 57/50 + 2/225 = (2565 + 20)/2250 = 2585/2250

Step 6: simplify by dividing by 5

Final answer: 1.14888... = 517/450

Question: Q5(v). Find the vulgar fraction equivalent to the recurring decimal 2.23 (both 2 and 3 recurring).

Answer: Step 1: separate the whole number, 2.232323... = 2 + 0.232323...

Step 2: write the recurring part as a series, 0.23 + 0.0023 + ..., with a = 0.23 and r = 0.01

Step 3: sum it, 0.23/(1 - 0.01) = 0.23/0.99 = 23/99

Step 4: add the whole number, 2 + 23/99 = (198 + 23)/99

Final answer: 2.232323... = 221/99

Question: Q6. If a = b/2 + b^2/4 + b^3/8 + ..., if 0 < b < 2, then prove that b = 2a/(1 + a).

Answer: Step 1: the right-hand series is geometric with first term b/2 and common ratio b/2

Step 2: since 0 < b < 2, |b/2| < 1, so the sum is a = (b/2)/(1 - b/2)

Step 3: simplify, a = (b/2)/((2 - b)/2) = b/(2 - b)

Step 4: cross-multiply, a(2 - b) = b, i.e. 2a - ab = b

Step 5: collect the terms in b, 2a = b + ab = b(1 + a)

Step 6: divide by (1 + a)

Final answer: b = 2a/(1 + a), as required

Question: Q7. The sum of an infinite geometric series is half the sum of the squares of its terms. If the sum of its first two terms is 4 1/2, find the

series.

Answer: Step 1: let the series have first term a and ratio r; its sum is S = a/(1 - r)

Step 2: the squares of the terms form a geometric series with first term a^2 and ratio r^2, whose sum is a^2/(1 - r^2)

Step 3: use the given condition, a/(1 - r) = (1/2) . a^2/(1 - r^2)

Step 4: simplify using 1 - r^2 = (1 - r)(1 + r) and cancelling a/(1 - r), 1 = a/[2(1 + r)], so a = 2(1 + r) ...(i)

Step 5: use the second condition, a + ar = 9/2, i.e. a(1 + r) = 9/2 ...(ii)

Step 6: substitute (i) in (ii), 2(1 + r)^2 = 9/2, so (1 + r)^2 = 9/4 and 1 + r = 3/2, giving r = 1/2

Step 7: from (i), a = 2(3/2) = 3

Step 8: write the series with a = 3, r = 1/2 (check: sum = 3/(1/2) = 6; sum of squares = 9/(1 - 1/4) = 12, and 6 is half of 12)

Final answer: the series is 3 + 3/2 + 3/4 + 3/8 + ...

Question: Q8. Joining the midpoints of the sides of an equilateral triangle, an equilateral triangle having half the perimeter of the original triangle is obtained. We form a sequence of nested equilateral triangles in this manner with the original triangle having perimeter 5/2. What will be the total perimeter of all the triangles formed in this way?

Answer: Step 1: the perimeters form a geometric sequence, 5/2, 5/4, 5/8, ..., since each new triangle has half the perimeter of the previous one

Step 2: identify a = 5/2 and r = 1/2

Step 3: the process continues indefinitely, so use the infinite sum S = a/(1

- r)

Step 4: substitute, S = (5/2)/(1 - 1/2) = (5/2)/(1/2)

Step 5: simplify

Final answer: the total perimeter of all the triangles is 5

Exercise 4.8

Question: Q1(i). Find the indicated term, the 9th term, of the harmonic progression 1/3, 1/7, 1/11, ...

Answer: Step 1: take reciprocals to get the corresponding A.P., 3, 7, 11, ...

Step 2: identify a = 3 and d = 7 - 3 = 4 for the A.P.

Step 3: write the A.P. formula an = a + (n - 1)d

Step 4: substitute n = 9, a9 = 3 + (9 - 1)(4) = 3 + 32 = 35

Step 5: the 9th term of the H.P. is the reciprocal of 35

Final answer: the 9th term is 1/35

Question: Q1(ii). Find the indicated term, the 12th term, of the harmonic progression 1/3, 1/5, 1/7, ...

Answer: Step 1: take reciprocals to get the corresponding A.P., 3, 5, 7, ...

Step 2: identify a = 3 and d = 2

Step 3: write an = a + (n - 1)d

Step 4: substitute n = 12, a12 = 3 + (12 - 1)(2) = 3 + 22 = 25

Step 5: take the reciprocal

Final answer: the 12th term is 1/25

Question: Q1(iii). Find the indicated term, the 8th term, of the harmonic progression 9/5, 9/13, 9/21, ...

Answer: Step 1: take reciprocals to get the corresponding A.P., 5/9, 13/9, 21/9, ...

Step 2: identify a = 5/9 and d = 13/9 - 5/9 = 8/9

Step 3: write an = a + (n - 1)d

Step 4: substitute n = 8, a8 = 5/9 + (8 - 1)(8/9) = 5/9 + 56/9 = 61/9

Step 5: take the reciprocal

Final answer: the 8th term is 9/61

Question: Q1(iv). Find the indicated term, the 15th term, of the harmonic progression -1/5, -1/3, -1, ...

Answer: Step 1: take reciprocals to get the corresponding A.P., -5, -3, -1, ...

Step 2: identify a = -5 and d = -3 - (-5) = 2

Step 3: write an = a + (n - 1)d

Step 4: substitute n = 15, a15 = -5 + (15 - 1)(2) = -5 + 28 = 23

Step 5: take the reciprocal of 23 (the terms of the A.P. increase through 1, 3, 5, ..., so the later terms of the H.P. are positive)

Final answer: the 15th term is 1/23

Question: Q1(v). Find the indicated term, the 8th term, of the harmonic progression 1/2, 1/5, 1/8, ...

Answer: Step 1: take reciprocals to get the corresponding A.P., 2, 5, 8, ...

Step 2: identify a = 2 and d = 3

Step 3: write an = a + (n - 1)d

Step 4: substitute n = 8, a8 = 2 + (8 - 1)(3) = 2 + 21 = 23

Step 5: take the reciprocal

Final answer: the 8th term is 1/23

Question: Q1(vi). Find the indicated term, the 11th term, of the harmonic progression 1/3, 1/8, 1/13, ...

Answer: Step 1: take reciprocals to get the corresponding A.P., 3, 8, 13, ...

Step 2: identify a = 3 and d = 5

Step 3: write an = a + (n - 1)d

Step 4: substitute n = 11, a11 = 3 + (11 - 1)(5) = 3 + 50 = 53

Step 5: take the reciprocal

Final answer: the 11th term is 1/53

Question: Q2. If the 7th and 10th terms of an H.P. are 1/3 and 5/21 respectively, find its 14th and 20th terms.

Answer: Step 1: take reciprocals; the corresponding A.P. has 7th term 3 and 10th term 21/5

Step 2: write the two equations, a + 6d = 3 ...(i) and a + 9d = 21/5 ...(ii)

Step 3: subtract (i) from (ii), 3d = 21/5 - 3 = 6/5, so d = 2/5

Step 4: substitute in (i), a = 3 - 6(2/5) = 3 - 12/5 = 3/5

Step 5: find the 14th term of the A.P., a + 13d = 3/5 + 13(2/5) = 3/5 + 26/5 = 29/5, so the 14th term of the H.P. is 5/29

Step 6: find the 20th term of the A.P., a + 19d = 3/5 + 19(2/5) = 3/5 + 38/5 = 41/5, so the 20th term of the H.P. is 5/41

Final answer: the 14th term is 5/29 and the 20th term is 5/41

Question: Q3. If the sum of the first and sixth terms of an H.P. is 31/116, find the harmonic sequence if the first term is 1/4.

Answer: Step 1: the first term is T1 = 1/4, so T6 = 31/116 - 1/4 = 31/116 - 29/116 = 2/116 = 1/58

Step 2: take reciprocals; the corresponding A.P. has first term 4 and sixth term 58

Step 3: write a + 5d = 58 with a = 4, so 5d = 54 and d = 54/5

Step 4: list the A.P. terms, 4, 4 + 54/5 = 74/5, 4 + 108/5 = 128/5, 4 + 162/5 = 182/5, 4 + 216/5 = 236/5, 58

Step 5: take reciprocals to return to the H.P.

Final answer: the harmonic sequence is 1/4, 5/74, 5/128, 5/182, 5/236, 1/58, ...

Question: Q4. The first term of an H.P. is -1/3 and its fifth term is 1/5. Find its 9th term.

Answer: Step 1: take reciprocals; the corresponding A.P. has first term -3 and fifth term 5

Step 2: write a + 4d = 5 with a = -3, so 4d = 8 and d = 2

Step 3: find the 9th term of the A.P., a + 8d = -3 + 8(2) = -3 + 16 = 13

Step 4: take the reciprocal

Final answer: the 9th term of the H.P. is 1/13

Question: Q5. If the pth term of an H.P. is q and the qth term is p, prove that the (p + q)th term is pq/(p + q).

Answer: Step 1: let A and D be the first term and common difference of

the corresponding A.P.; the pth term of the H.P. being q gives A + (p - 1)D = 1/q ...(i)

Step 2: the qth term of the H.P. being p gives A + (q - 1)D = 1/p ...(ii)

Step 3: subtract (ii) from (i), (p - q)D = 1/q - 1/p = (p - q)/(pq)

Step 4: cancel (p - q), giving D = 1/(pq)

Step 5: substitute D in (i), A = 1/q - (p - 1)/(pq) = [p - (p - 1)]/(pq) = 1/(pq)

Step 6: find the (p + q)th term of the A.P., A + (p + q - 1)D = 1/(pq) + (p + q - 1)/(pq) = (p + q)/(pq)

Step 7: the (p + q)th term of the H.P. is the reciprocal of this

Final answer: the (p + q)th term is pq/(p + q), as required

Exercise 4.9

Question: Q1(i). Insert the H.M. between 1/64 and 1/81.

Answer: Step 1: write the formula for the harmonic mean of two numbers, H = 2ab/(a + b)

Step 2: substitute a = 1/64 and b = 1/81, numerator = 2(1/64)(1/81) = 2/5184

Step 3: compute the denominator, 1/64 + 1/81 = (81 + 64)/5184 = 145/5184

Step 4: divide, H = (2/5184)/(145/5184) = 2/145

Final answer: the H.M. is 2/145

Question: Q1(ii). Insert the H.M. between 1/5 and 1/17.

Answer: Step 1: write the formula H = 2ab/(a + b)

Step 2: substitute a = 1/5 and b = 1/17, numerator = 2(1/5)(1/17) = 2/85

Step 3: compute the denominator, 1/5 + 1/17 = (17 + 5)/85 = 22/85

Step 4: divide, H = (2/85)/(22/85) = 2/22 = 1/11

Final answer: the H.M. is 1/11

Question: Q1(iii). Insert the H.M. between 3 + sqrt(2) and 3 - sqrt(2).

Answer: Step 1: write the formula H = 2ab/(a + b)

Step 2: compute the product, ab = (3 + sqrt(2))(3 - sqrt(2)) = 9 - 2 = 7

Step 3: compute the sum, a + b = (3 + sqrt(2)) + (3 - sqrt(2)) = 6

Step 4: substitute, H = 2(7)/6 = 14/6

Final answer: the H.M. is 7/3

Question: Q1(iv). Insert the H.M. between 2 + 3i and 2 - 3i.

Answer: Step 1: write the formula H = 2ab/(a + b)

Step 2: compute the product, ab = (2 + 3i)(2 - 3i) = 4 + 9 = 13

Step 3: compute the sum, a + b = (2 + 3i) + (2 - 3i) = 4

Step 4: substitute, H = 2(13)/4 = 26/4

Final answer: the H.M. is 13/2

Question: Q1(v). Insert the H.M. between 1/7 and 1/11.

Answer: Step 1: write the formula H = 2ab/(a + b)

Step 2: substitute, numerator = 2(1/7)(1/11) = 2/77

Step 3: compute the denominator, 1/7 + 1/11 = (11 + 7)/77 = 18/77

Step 4: divide, H = (2/77)/(18/77) = 2/18 = 1/9

Final answer: the H.M. is 1/9

Question: Q1(vi). Insert the H.M. between 3 and 7.

Answer: Step 1: write the formula H = 2ab/(a + b)

Step 2: substitute a = 3 and b = 7, H = 2(3)(7)/(3 + 7)

Step 3: simplify, H = 42/10

Final answer: the H.M. is 21/5

Question: Q2(i). Insert four H.Ms. between 4 and 20.

Answer: Step 1: insert the means through the corresponding A.P. of reciprocals, 1/4, ..., 1/20, in which 1/20 is the 6th term

Step 2: find the common difference of the A.P., d = (1/20 - 1/4)/5 = (-1/5)/5 = -1/25

Step 3: the A.P. terms after 1/4 are 1/4 - 1/25 = 21/100, 1/4 - 2/25 = 17/100, 1/4 - 3/25 = 13/100, 1/4 - 4/25 = 9/100

Step 4: take reciprocals of these four terms

Final answer: the four H.Ms. are 100/21, 100/17, 100/13, 100/9

Question: Q2(ii). Insert four H.Ms. between 1/9 and 1/81.

Answer: Step 1: use the corresponding A.P. of reciprocals, 9, ..., 81, in which 81 is the 6th term

Step 2: find the common difference, d = (81 - 9)/5 = 72/5

Step 3: the A.P. terms after 9 are 9 + 72/5 = 117/5, 9 + 144/5 = 189/5, 9 + 216/5 = 261/5, 9 + 288/5 = 333/5

Step 4: take reciprocals of these four terms

Final answer: the four H.Ms. are 5/117, 5/189, 5/261, 5/333

Question: Q3. Insert five harmonic means between 1/4 and 1/24.

Answer: Step 1: use the corresponding A.P. of reciprocals, 4, ..., 24, in which 24 is the 7th term

Step 2: find the common difference, d = (24 - 4)/6 = 20/6 = 10/3

Step 3: the A.P. terms after 4 are 4 + 10/3 = 22/3, 4 + 20/3 = 32/3, 4 + 30/3 = 14, 4 + 40/3 = 52/3, 4 + 50/3 = 62/3

Step 4: take reciprocals of these five terms

Final answer: the five H.Ms. are 3/22, 3/32, 1/14, 3/52, 3/62

Question: Q4. Prove that the square of the geometric mean of two numbers equals the product of the A.M. and the H.M. of the two numbers.

Answer: Step 1: let the two numbers be a and b, with A.M. A = (a + b)/2

Step 2: write the G.M., G = sqrt(ab), so G^2 = ab

Step 3: write the H.M., H = 2ab/(a + b)

Step 4: form the product A . H = [(a + b)/2] . [2ab/(a + b)]

Step 5: cancel (a + b) and the factor 2, A . H = ab

Step 6: compare with Step 2

Final answer: A . H = ab = G^2, i.e. the square of the G.M. equals the product of the A.M. and the H.M., as required

Question: Q5. Find n so that [x^(n-5) + y^(n-5)] / [x^(n-6) + y^(n-6)] may be the H.M. between x and y.

Answer: Step 1: the H.M. between x and y is 2xy/(x + y), so set [x^(n-5) + y^(n-5)] / [x^(n-6) + y^(n-6)] = 2xy/(x + y)

Step 2: cross-multiply, (x + y)[x^(n-5) + y^(n-5)] = 2xy [x^(n-6) + y^(n-6)]

Step 3: expand both sides, x^(n-4) + x y^(n-5) + y x^(n-5) + y^(n-4) = 2 x^(n-5) y + 2 x y^(n-5)

Step 4: rearrange, x^(n-4) - 2 x^(n-5) y + y x^(n-5) = 2 x y^(n-5) - x y^(n-5) - y^(n-4)

Step 5: factorise both sides, x^(n-5)(x - y) = y^(n-5)(x - y)

Step 6: divide by (x - y) (for x not equal to y), x^(n-5) = y^(n-5), i.e. (x/y)^(n-5) = 1

Step 7: a power equals 1 only when the exponent is 0, so n - 5 = 0

Final answer: n = 5

Question: Q6. If 5 is the harmonic mean between 2 and b, find b.

Answer: Step 1: write the H.M. formula with a = 2, H = 2(2)b/(2 + b) = 4b/(2 + b)

Step 2: set it equal to 5, 4b/(2 + b) = 5

Step 3: cross-multiply, 4b = 5(2 + b) = 10 + 5b

Step 4: solve, 4b - 5b = 10, so -b = 10

Step 5: check: H.M. of 2 and -10 is 2(2)(-10)/(2 - 10) = -40/-8 = 5

Final answer: b = -10

Question: Q7. Find the arithmetic, harmonic and geometric means between 1 and 9. Also verify that AH = G^2.

Answer: Step 1: compute the A.M., A = (1 + 9)/2 = 10/2 = 5

Step 2: compute the G.M., G = sqrt(1 . 9) = sqrt(9) = 3

Step 3: compute the H.M., H = 2(1)(9)/(1 + 9) = 18/10 = 9/5

Step 4: verify the relation, A . H = 5 . (9/5) = 9

Step 5: compare with G^2 = 3^2 = 9

Final answer: A.M. = 5, G.M. = 3, H.M. = 9/5, and AH = 9 = G^2, verified

Question: Q8. The A.M. of two numbers is 8 and the H.M. is 6. Find the numbers.

Answer: Step 1: let the numbers be a and b; from the A.M., a + b = 16

Step 2: from the H.M., 2ab/(a + b) = 6, so 2ab = 6(16) = 96 and ab = 48

Step 3: a and b are the roots of t^2 - 16 t + 48 = 0

Step 4: compute the discriminant, 256 - 192 = 64, and sqrt(64) = 8

Step 5: solve, t = (16 + 8)/2 = 12 or t = (16 - 8)/2 = 4

Final answer: the numbers are 12 and 4 (check: H.M. = 2(48)/16 = 6)

Question: Q9. The H.M. of two numbers is 24/5 and the G.M. is 6. Find the numbers.

Answer: Step 1: let the numbers be a and b; from the G.M., sqrt(ab) = 6, so ab = 36

Step 2: from the H.M., 2ab/(a + b) = 24/5, so 2(36)/(a + b) = 24/5

Step 3: solve for the sum, a + b = 72 . 5/24 = 15

Step 4: a and b are the roots of t^2 - 15 t + 36 = 0

Step 5: compute the discriminant, 225 - 144 = 81, and sqrt(81) = 9

Step 6: solve, t = (15 + 9)/2 = 12 or t = (15 - 9)/2 = 3

Final answer: the numbers are 12 and 3 (check: H.M. = 2(36)/15 = 72/15 = 24/5)

Review Exercise 4

Question: Q1(i). Select the correct option: A sequence is a function whose domain is the set of: (a) integers (b) rational numbers (c) natural numbers (d) real numbers.

Answer: Step 1: by definition a sequence is a function a : N -> R (or C), i.e. its terms are indexed by the counting numbers 1, 2, 3, ...

Step 2: the domain is therefore the set of natural numbers

Final answer: (c) natural numbers

Question: Q1(ii). Select the correct option: If H is the harmonic mean between x and y then H is: (a) 2(x + y)/(xy) (b) (x + y)/(2xy) (c) 2xy/(x + y) (d) xy/(x + y).

Answer: Step 1: the harmonic mean of two numbers is the reciprocal of the A.M. of their reciprocals, H = 2xy/(x + y)

Step 2: compare with the options

Final answer: (c) 2xy/(x + y)

Question: Q1(iii). Select the correct option: If an - a(n-1) = n + 1 and a4 = 14 then a5 = ...: (a) 3 (b) 5 (c) 14 (d) 20.

Answer: Step 1: put n = 5 in the given rule, a5 - a4 = 5 + 1 = 6

Step 2: substitute a4 = 14, a5 = 14 + 6

Step 3: simplify

Final answer: (d) 20

Question: Q1(iv). Select the correct option: A sequence {an} in which a(n+1) - an is the same number for all n in N is called: (a) A.P (b) G.P (c)

H.P (d) None of these.

Answer: Step 1: a constant difference between consecutive terms is the defining property of an arithmetic progression

Final answer: (a) A.P

Question: Q1(v). Select the correct option: If a(n-1), an, a(n+1) are in A.P., then an is called: (a) A.M (b) G.M (c) H.M (d) Mid-point.

Answer: Step 1: in an A.P. the middle term of three consecutive terms is the arithmetic mean of its two neighbours

Final answer: (a) A.M

Question: Q1(vi). Select the correct option: The arithmetic mean between c and d is: (a) (c + d)/2 (b) (c + d)/(2cd) (c) 2cd/(c + d) (d) 2/(c + d).

Answer: Step 1: the A.M. of two numbers is half their sum, (c + d)/2

Final answer: (a) (c + d)/2

Question: Q1(vii). Select the correct option: The harmonic mean between sqrt(2) and 3 sqrt(2) is: (a) 4 sqrt(2) (b) 4/sqrt(2) (c) (3/2) sqrt(2) (d) None of these.

Answer: Step 1: write H = 2xy/(x + y) with x = sqrt(2), y = 3 sqrt(2)

Step 2: compute the numerator, 2 . sqrt(2) . 3 sqrt(2) = 2 . 3 . 2 = 12

Step 3: compute the denominator, sqrt(2) + 3 sqrt(2) = 4 sqrt(2)

Step 4: divide, H = 12/(4 sqrt(2)) = 3/sqrt(2) = 3 sqrt(2)/2

Final answer: (c) (3/2) sqrt(2)

Question: Q1(viii). Select the correct option: For any G.P. the common ratio r is equal to: (a) an/a(n+1) (b) a(n-1)/an (c) an/a(n-1) (d) a(n+1) - an, for n in N, n > 1.

Answer: Step 1: the common ratio is obtained by dividing any term by the term immediately before it

Final answer: (c) an/a(n-1)

Question: Q1(ix). Select the correct option: No term of a G.P. is: (a) 0 (b) 1 (c) Negative (d) Imaginary number.

Answer: Step 1: in a G.P. with non-zero first term and non-zero ratio, every term an = a r^(n-1) is non-zero

Final answer: (a) 0

Question: Q1(x). Select the correct option: The sum of an infinite geometric series is a finite number if: (a) |r| > 1 (b) |r| = 1 (c) |r| >= 1 (d) |r| < 1.

Answer: Step 1: an infinite geometric series converges to the finite sum a/(1 - r) exactly when the terms shrink to zero, i.e. |r| < 1

Final answer: (d) |r| < 1

Question: Q1(xi). Select the correct option: If the reciprocals of the terms of a sequence form an A.P., then it is: (a) Harmonic sequence (b) Arithmetic sequence (c) Reciprocal sequence (d) Series.

Answer: Step 1: this is the definition of a harmonic progression

Final answer: (a) Harmonic sequence

Question: Q1(xii). Select the correct option: If [a^(n+1) + b^(n+1)] / [a^n + b^n] is the A.M. between a and b, then n is equal to: (a) 0 (b) -1 (c) 1 (d) 1/2.

Answer: Step 1: set the expression equal to (a + b)/2 and cross-multiply, 2a^(n+1) + 2b^(n+1) = (a + b)(a^n + b^n)

Step 2: expand the right-hand side and cancel, giving a^(n+1) + b^(n+1)

= a b^n + b a^n

Step 3: factorise, a^n (a - b) = b^n (a - b), so (a/b)^n = 1

Step 4: a power equals 1 only when the exponent is 0

Final answer: (a) 0

Question: Q1(xiii). Select the correct option: If 1, x - 1, 3 are in A.P., then x = ...: (a) 0 (b) 1 (c) 2 (d) 3.

Answer: Step 1: the middle term is the A.M. of its neighbours, x - 1 = (1 + 3)/2 = 2

Step 2: solve, x = 2 + 1

Final answer: (d) 3

Question: Q1(xiv). Select the correct option: The G.M. between -2 and 8 is: (a) 4i or -4i (b) 4 or -4 (c) 16 or -16 (d) 3 or -5.

Answer: Step 1: write G = +/- sqrt(ab) = +/- sqrt((-2)(8))

Step 2: simplify, (-2)(8) = -16, and sqrt(-16) = 4i

Final answer: (a) 4i or -4i

Question: Q1(xv). Select the correct option: The general term of a sequence is (-1)^n n^2. Its 4th term is: (a) -4 (b) -16 (c) 16 (d) 4.

Answer: Step 1: put n = 4, a4 = (-1)^4 . 4^2

Step 2: simplify, (-1)^4 = 1 and 4^2 = 16, so a4 = 16

Final answer: (c) 16

Question: Q1(xvi). Select the correct option: If |r| > 1 then an infinite geometric series is .....: (a) Convergent (b) Divergent (c) Undefined (d) both a and b.

Answer: Step 1: when |r| > 1 the terms grow in magnitude and the partial

sums do not approach a finite limit

Final answer: (b) Divergent

Question: Q1(xvii). Select the correct option: The harmonic mean of 1/3 and 2/5 is: (a) 4/11 (b) 3/4 (c) 5/11 (d) 11/4.

Answer: Step 1: write H = 2xy/(x + y) with x = 1/3, y = 2/5

Step 2: compute the numerator, 2(1/3)(2/5) = 4/15

Step 3: compute the denominator, 1/3 + 2/5 = 5/15 + 6/15 = 11/15

Step 4: divide, H = (4/15)/(11/15) = 4/11

Final answer: (a) 4/11

Question: Q1(xviii). Select the correct option: If (a - b)/(b - c) = a/b, then a, b and c are in: (a) A.P (b) G.P (c) H.P (d) None of these.

Answer: Step 1: cross-multiply, b(a - b) = a(b - c)

Step 2: expand, ab - b^2 = ab - ac

Step 3: cancel ab, giving b^2 = ac

Step 4: b^2 = ac is the condition for a, b, c to be in geometric progression

Final answer: (b) G.P

Question: Q2. The 5th term of an arithmetic sequence is 60 and the 8th term is 90. Find the 12th term.

Answer: Step 1: write the two equations, a + 4d = 60 ...(i) and a + 7d = 90 ...(ii)

Step 2: subtract (i) from (ii), 3d = 30, so d = 10

Step 3: substitute in (i), a = 60 - 4(10) = 20

Step 4: find the 12th term, a12 = a + 11d = 20 + 11(10) = 20 + 110

Final answer: a12 = 130

Question: Q3. There are n A.Ms. between 8 and 32 such that the ratio of the third and the seventh means is 3 : 5. Find the value of n.

Answer: Step 1: with n means between 8 and 32, the common difference is d = (32 - 8)/(n + 1) = 24/(n + 1)

Step 2: write the third mean, A3 = 8 + 3d = 8 + 72/(n + 1)

Step 3: write the seventh mean, A7 = 8 + 7d = 8 + 168/(n + 1)

Step 4: use the ratio, [8 + 72/(n + 1)] / [8 + 168/(n + 1)] = 3/5

Step 5: cross-multiply, 5[8 + 72/(n + 1)] = 3[8 + 168/(n + 1)], i.e. 40 + 360/(n + 1) = 24 + 504/(n + 1)

Step 6: simplify, 16 = 144/(n + 1), so n + 1 = 9

Final answer: n = 8

Question: Q4. Find the sum of all the integers between 280 and 350 which are exactly divisible by 9.

Answer: Step 1: the first integer divisible by 9 after 280 is 288 and the last before 350 is 342

Step 2: these form an A.P. with a = 288, d = 9, last term 342

Step 3: count the terms, n = (342 - 288)/9 + 1 = 6 + 1 = 7

Step 4: write the sum, Sn = n/2 (a + l) = 7/2 (288 + 342)

Step 5: simplify, 7/2 (630) = 7(315)

Final answer: the sum is 2205

Question: Q5. How many terms are there in a G.P. if a = 8, an = 1/512 and r = 1/2?

Answer: Step 1: write an = a r^(n-1), 1/512 = 8(1/2)^(n-1)

Step 2: divide by 8, (1/2)^(n-1) = 1/(512 . 8) = 1/4096

Step 3: write 4096 as a power of 2, 4096 = 2^12, so (1/2)^(n-1) = (1/2)^12

Step 4: compare exponents, n - 1 = 12

Final answer: n = 13 terms

Question: Q6. The yearly depreciation of a certain machine is 20% of the value at the beginning of the year. If the original cost of the machine is Rs. 50,000, find its value after 5 years.

Answer: Step 1: each year the machine keeps 100% - 20% = 80% of its value, so the values form a G.P. with ratio r = 0.8

Step 2: the value after 5 years is V = 50000 (0.8)^5

Step 3: compute the power in stages, (0.8)^2 = 0.64, (0.8)^4 = (0.64)^2 = 0.4096, (0.8)^5 = 0.4096(0.8) = 0.32768

Step 4: multiply, V = 50000(0.32768)

Final answer: the value after 5 years is Rs. 16,384

Question: Q7. Find r, if S10 = 244 S5 in a G.P.

Answer: Step 1: write the ratio using Sn = a(r^n - 1)/(r - 1), S10/S5 = (r^10 - 1)/(r^5 - 1) = 244

Step 2: factorise the numerator, r^10 - 1 = (r^5 - 1)(r^5 + 1)

Step 3: cancel (r^5 - 1), giving r^5 + 1 = 244

Step 4: solve, r^5 = 243 = 3^5

Final answer: r = 3

Question: Q8. If the A.M. and H.M. between two numbers are 5 and 4 1/5 respectively, find the numbers.

Answer: Step 1: let the numbers be a and b; from the A.M., a + b = 10

Step 2: from the H.M., 2ab/(a + b) = 21/5, so 2ab = (21/5)(10) = 42 and ab = 21

Step 3: a and b are the roots of t^2 - 10 t + 21 = 0

Step 4: factorise, (t - 7)(t - 3) = 0, so t = 7 or t = 3

Final answer: the numbers are 7 and 3

Question: Q9. If the G.M. and H.M. between two numbers are 15 and 5 10/13 respectively, find the numbers.

Answer: Step 1: note 5 10/13 = 75/13; let the numbers be a and b; from the G.M., ab = 15^2 = 225

Step 2: from the H.M., 2ab/(a + b) = 75/13, so 2(225)/(a + b) = 75/13

Step 3: solve for the sum, a + b = 450 . 13/75 = 78

Step 4: a and b are the roots of t^2 - 78 t + 225 = 0

Step 5: factorise, (t - 75)(t - 3) = 0, so t = 75 or t = 3

Final answer: the numbers are 75 and 3

Question: Q10. The second term of an H.P. is 1/2 and the fifth term is -1/4. Find the 12th term.

Answer: Step 1: take reciprocals; the corresponding A.P. has second term 2 and fifth term -4

Step 2: write the equations, a + d = 2 ...(i) and a + 4d = -4 ...(ii)

Step 3: subtract (i) from (ii), 3d = -6, so d = -2

Step 4: substitute in (i), a = 2 - (-2) = 4

Step 5: find the 12th term of the A.P., a + 11d = 4 + 11(-2) = 4 - 22 = -18

Step 6: take the reciprocal

Final answer: the 12th term of the H.P. is -1/18

Unit 5: Miscellaneous Series

Exercise 5.1

Question: Sum the series up to the indicated terms: 1^2 + 2^2 + 3^2 + ... + 10^2

Answer: Step 1: The series is the sum of the squares of the first 10 natural numbers, so we use the formula sum of n^2 = n(n+1)(2n+1)/6

Step 2: Here the number of terms is n = 10

Step 3: Substitute n = 10 in the formula, sum = 10(10+1)(2(10)+1)/6

Step 4: Simplify the factors, sum = 10 x 11 x 21 / 6

Step 5: Multiply the numerator, 10 x 11 x 21 = 2310

Step 6: Divide by 6, 2310/6 = 385

Final answer: 385

Question: Sum the series up to the indicated terms: 1^3 + 2^3 + 3^3 + ... + 15^3

Answer: Step 1: The series is the sum of the cubes of the first 15 natural numbers, so we use the formula sum of n^3 = [n(n+1)/2]^2

Step 2: Here the number of terms is n = 15

Step 3: Substitute n = 15 in the formula, sum = [15(15+1)/2]^2

Step 4: Simplify inside the bracket, 15 x 16 / 2 = 120

Step 5: Square the result, 120^2 = 14400

Final answer: 14400

Question: Sum the series up to n terms: 1^2 + 3^2 + 5^2 + ... up to n terms

Answer: Step 1: The terms are the squares of the odd numbers 1, 3, 5, ..., so the kth term is (2k-1)^2

Step 2: Expand the kth term, (2k-1)^2 = 4k^2 - 4k + 1

Step 3: Take the sum from k = 1 to n, S_n = 4 x sum of k^2 - 4 x sum of k + sum of 1

Step 4: Substitute the standard sums, S_n = 4 x n(n+1)(2n+1)/6 - 4 x n(n+1)/2 + n

Step 5: Simplify each part, S_n = (2n(n+1)(2n+1))/3 - 2n(n+1) + n

Step 6: Take n common and use the common denominator 3, S_n = n[2(n+1)(2n+1) - 6(n+1) + 3]/3

Step 7: Expand inside the bracket, 2(2n^2+3n+1) - 6n - 6 + 3 = 4n^2 - 1

Final answer: S_n = n(4n^2 - 1)/3 = n(2n-1)(2n+1)/3

Question: Sum the series up to n terms: 2^2 + 4^2 + 6^2 + 8^2 + ... up to n terms

Answer: Step 1: The terms are the squares of the even numbers 2, 4, 6, ..., so the kth term is (2k)^2

Step 2: Simplify the kth term, (2k)^2 = 4k^2

Step 3: Take the sum from k = 1 to n, S_n = 4 x sum of k^2

Step 4: Substitute the formula for the sum of squares, S_n = 4 x n(n+1)(2n+1)/6

Step 5: Cancel 4 with 6, S_n = 2n(n+1)(2n+1)/3

Final answer: S_n = 2n(n+1)(2n+1)/3

Question: Sum the series up to n terms: 1^3 + 3^3 + 5^3 + ... up to n terms

Answer: Step 1: The terms are the cubes of the odd numbers 1, 3, 5, ..., so the kth term is (2k-1)^3

Step 2: Expand the kth term, (2k-1)^3 = 8k^3 - 12k^2 + 6k - 1

Step 3: Take the sum from k = 1 to n, S_n = 8 x sum of k^3 - 12 x sum of k^2 + 6 x sum of k - sum of 1

Step 4: Substitute the standard sums, S_n = 8[n(n+1)/2]^2 - 12 x n(n+1)(2n+1)/6 + 6 x n(n+1)/2 - n

Step 5: Simplify each part, S_n = 2n^2(n+1)^2 - 2n(n+1)(2n+1) + 3n(n+1) - n

Step 6: Take n common, S_n = n[2n(n+1)^2 - 2(n+1)(2n+1) + 3(n+1) - 1]

Step 7: Expand and collect inside the bracket, the bracket becomes 2n^3 - n

Step 8: Factor n once more, S_n = n x n(2n^2 - 1)

Final answer: S_n = n^2(2n^2 - 1)

Question: Sum the series up to n terms: 2 x 1^2 + 5 x 2^2 + 8 x 3^2 + ... up to n terms

Answer: Step 1: In the kth term the first factor 2, 5, 8, ... is in arithmetic progression with kth value 3k-1, and the second factor is k^2

Step 2: So the kth term is (3k-1)k^2 = 3k^3 - k^2

Step 3: Take the sum from k = 1 to n, S_n = 3 x sum of k^3 - sum of k^2

Step 4: Substitute the standard sums, S_n = 3[n(n+1)/2]^2 -

n(n+1)(2n+1)/6

Step 5: Write with denominator 12, S_n = n(n+1)[9n(n+1) - 2(2n+1)]/12

Step 6: Expand inside the bracket, 9n^2 + 9n - 4n - 2 = 9n^2 + 5n - 2

Step 7: Factor 9n^2 + 5n - 2 = (n+1)(9n-2), so S_n = n(n+1)(n+1)(9n-2)/12

Final answer: S_n = n(n+1)^2(9n-2)/12, which is the same as n(9n^3 + 14n^2 + 3n - 2)/12

Question: Sum the series up to n terms: 1 + (1+2) + (1+2+3) + ... up to n terms

Answer: Step 1: The kth bracket is 1+2+...+k, which is the sum of the first k natural numbers

Step 2: So the kth term is T_k = k(k+1)/2 = (k^2 + k)/2

Step 3: Take the sum from k = 1 to n, S_n = (1/2)[sum of k^2 + sum of k]

Step 4: Substitute the standard sums, S_n = (1/2)[n(n+1)(2n+1)/6 + n(n+1)/2]

Step 5: Take n(n+1) common, S_n = (n(n+1)/2)[(2n+1)/6 + 1/2]

Step 6: Simplify the bracket, (2n+1)/6 + 3/6 = (2n+4)/6 = (n+2)/3

Step 7: Multiply, S_n = n(n+1)(n+2)/6

Final answer: S_n = n(n+1)(n+2)/6

Question: Sum the series up to n terms: 2 + (2+5) + (2+5+8) + ... up to n terms

Answer: Step 1: The kth bracket is 2+5+8+... up to k terms, an arithmetic series with first term 2 and common difference 3

Step 2: So the kth term is T_k = (k/2)[2(2) + (k-1)(3)] = (k/2)(3k+1) = (3k^2 + k)/2

Step 3: Take the sum from k = 1 to n, S_n = (1/2)[3 x sum of k^2 + sum of k]

Step 4: Substitute the standard sums, S_n = (1/2)[3 x n(n+1)(2n+1)/6 + n(n+1)/2]

Step 5: Simplify, S_n = (n(n+1)/4)[(2n+1) + 1]

Step 6: Simplify the bracket, (2n+1) + 1 = 2n + 2 = 2(n+1)

Step 7: Multiply, S_n = (n(n+1)/4) x 2(n+1) = n(n+1)^2/2

Final answer: S_n = n(n+1)^2/2

Question: Find the sum of n terms of the series whose nth term is (i) 3n^2 + n + 1

Answer: Step 1: The kth term is T_k = 3k^2 + k + 1

Step 2: Take the sum from k = 1 to n, S_n = 3 x sum of k^2 + sum of k + sum of 1

Step 3: Substitute the standard sums, S_n = 3 x n(n+1)(2n+1)/6 + n(n+1)/2 + n

Step 4: Simplify the first two parts, S_n = n(n+1)(2n+1)/2 + n(n+1)/2 + n

Step 5: Take n common, S_n = n[(n+1)(2n+1)/2 + (n+1)/2 + 1]

Step 6: Combine the fractions, (n+1)(2n+1) + (n+1) = (n+1)(2n+2) = 2(n+1)^2, so the bracket is (n+1)^2 + 1

Step 7: Expand (n+1)^2 + 1 = n^2 + 2n + 2

Final answer: S_n = n(n^2 + 2n + 2)

Question: Find the sum of n terms of the series whose nth term is (ii) n^2 + 4n + 1

Answer: Step 1: The kth term is T_k = k^2 + 4k + 1

Step 2: Take the sum from k = 1 to n, S_n = sum of k^2 + 4 x sum of k + sum of 1

Step 3: Substitute the standard sums, S_n = n(n+1)(2n+1)/6 + 4 x n(n+1)/2 + n

Step 4: Simplify, S_n = n(n+1)(2n+1)/6 + 2n(n+1) + n

Step 5: Take n common with denominator 6, S_n = n[(n+1)(2n+1) + 12(n+1) + 6]/6

Step 6: Expand inside the bracket, 2n^2 + 3n + 1 + 12n + 12 + 6 = 2n^2 + 15n + 19

Final answer: S_n = n(2n^2 + 15n + 19)/6

Question: Find the sum of n terms of the series whose nth term is (iii) n^3 + 3n^2 + 2n + 1

Answer: Step 1: The kth term is T_k = k^3 + 3k^2 + 2k + 1

Step 2: Take the sum from k = 1 to n, S_n = sum of k^3 + 3 x sum of k^2 + 2 x sum of k + sum of 1

Step 3: Substitute the standard sums, S_n = [n(n+1)/2]^2 + 3 x n(n+1)(2n+1)/6 + 2 x n(n+1)/2 + n

Step 4: Simplify each part, S_n = n^2(n+1)^2/4 + n(n+1)(2n+1)/2 + n(n+1) + n

Step 5: Take n common with denominator 4, S_n = n[n(n+1)^2 + 2(n+1)(2n+1) + 4(n+1) + 4]/4

Step 6: Expand inside the bracket, n(n^2+2n+1) = n^3 + 2n^2 + n, 2(n+1)(2n+1) = 4n^2 + 6n + 2, and 4(n+1) + 4 = 4n + 8

Step 7: Add all parts, the bracket becomes n^3 + 6n^2 + 11n + 10

Final answer: S_n = n(n^3 + 6n^2 + 11n + 10)/4

Question: (i) Find the sum of first forty six natural numbers starting from 10

Answer: Step 1: Forty six numbers starting from 10 run from 10 to 10 + 45 = 55

Step 2: The required sum is the sum of the first 55 natural numbers minus the sum of the first 9 natural numbers

Step 3: Use sum of first n natural numbers = n(n+1)/2 for both parts

Step 4: Sum up to 55 is 55 x 56 / 2 = 1540

Step 5: Sum up to 9 is 9 x 10 / 2 = 45

Step 6: Subtract, 1540 - 45 = 1495

Final answer: 1495

Question: (ii) Find the sum of last 12 terms of the series: 1 + 2 + 3 + ... + 39

Answer: Step 1: The series has 39 terms, so the last 12 terms are the terms from 28 to 39

Step 2: The required sum is the sum of the first 39 natural numbers minus the sum of the first 27 natural numbers

Step 3: Use sum of first n natural numbers = n(n+1)/2 for both parts

Step 4: Sum up to 39 is 39 x 40 / 2 = 780

Step 5: Sum up to 27 is 27 x 28 / 2 = 378

Step 6: Subtract, 780 - 378 = 402

Final answer: 402

Question: Find the sum of 9th to 21st terms of the series of squares of first n natural numbers

Answer: Step 1: The required sum is 9^2 + 10^2 + ... + 21^2, the sum of squares up to 21 minus the sum of squares up to 8

Step 2: Use sum of n^2 = n(n+1)(2n+1)/6 for both parts

Step 3: Sum of squares up to 21 is 21 x 22 x 43 / 6 = 3311

Step 4: Sum of squares up to 8 is 8 x 9 x 17 / 6 = 204

Step 5: Subtract, 3311 - 204 = 3107

Final answer: 3107

Question: Find the sum of last 12 terms of the series of cubes of first 30 natural numbers

Answer: Step 1: The last 12 terms are the cubes from 19^3 to 30^3, so the sum is the sum of cubes up to 30 minus the sum of cubes up to 18

Step 2: Use sum of n^3 = [n(n+1)/2]^2 for both parts

Step 3: Sum of cubes up to 30 is [30 x 31 / 2]^2 = 465^2 = 216225

Step 4: Sum of cubes up to 18 is [18 x 19 / 2]^2 = 171^2 = 29241

Step 5: Subtract, 216225 - 29241 = 186984

Final answer: 186984

Exercise 5.2

Question: Sum the series up to 50 terms: 1 x 2 + 2 x 2^2 + 3 x 2^3 + 4 x 2^4 + ... to 50 terms

Answer: Step 1: The series is arithmetico-geometric with kth term k x 2^k, so here a = 1, d = 1 and r = 2

Step 2: Let S be the sum of n terms, S = 1x2 + 2x2^2 + 3x2^3 + ... + n x 2^n

Step 3: Multiply both sides by r = 2, 2S = 1x2^2 + 2x2^3 + ... + (n-1) x 2^n + n x 2^(n+1)

Step 4: Subtract the second equation from the first, -S = 2 + 2^2 + 2^3 + ... + 2^n - n x 2^(n+1)

Step 5: The middle part is a geometric series with sum 2(2^n - 1)/(2 - 1) = 2^(n+1) - 2

Step 6: So -S = 2^(n+1) - 2 - n x 2^(n+1) = -(n-1) x 2^(n+1) - 2

Step 7: Multiply by -1, S = (n-1) x 2^(n+1) + 2

Step 8: Put n = 50, S = 49 x 2^51 + 2

Final answer: 49 x 2^51 + 2

Question: Sum the series up to 30 terms: 1 + 2 x 5 + 3 x 5^2 + 4 x 5^3 + ... to 30 terms

Answer: Step 1: The kth term is k x 5^(k-1), an arithmetico-geometric series with a = 1, d = 1 and r = 5

Step 2: Use the formula S_n = a/(1-r) + dr/(1-r)^2 - dr^n/(1-r)^2 - [a + (n-1)d] r^n/(1-r)

Step 3: Substitute a = 1, d = 1, r = 5 and n = 30

Step 4: Compute the constant parts, a/(1-r) = 1/(1-5) = -1/4 and dr/(1-r)^2 = 5/16

Step 5: Compute the terms containing 5^30, dr^n/(1-r)^2 = 5^30/16 and [a + (n-1)d] r^n/(1-r) = 30 x 5^30/(1-5) = -15 x 5^30/2

Step 6: Combine the constant parts, -1/4 + 5/16 = 1/16

Step 7: Combine the 5^30 parts with denominator 16, -5^30/16 + 15 x 5^30/2 = (-1 + 120) x 5^30/16 = 119 x 5^30/16

Step 8: Add both groups, S_30 = [1 + 119 x 5^30]/16

Final answer: (1 + 119 x 5^30)/16

Question: Sum the series up to 25 terms: 1 + 3/2 + 5/2^2 + 7/2^3 + ... to 25 terms

Answer: Step 1: The numerators 1, 3, 5, 7, ... are in arithmetic progression and the denominators are powers of 2, so a = 1, d = 2 and r = 1/2

Step 2: Use the formula S_n = a/(1-r) + dr/(1-r)^2 - dr^n/(1-r)^2 - [a + (n-1)d] r^n/(1-r)

Step 3: Substitute a = 1, d = 2, r = 1/2 and n = 25

Step 4: Compute the constant parts, a/(1-r) = 1/(1/2) = 2 and dr/(1-r)^2 = (2 x 1/2)/(1/2)^2 = 1/(1/4) = 4, giving 6 in total

Step 5: Compute dr^n/(1-r)^2 = 2 x (1/2)^25 / (1/4) = 8/2^25 = 1/2^22

Step 6: Compute [a + (n-1)d] r^n/(1-r) = (1 + 24 x 2) x (1/2)^25 / (1/2) = 49/2^24

Step 7: Combine the two correction terms, 1/2^22 + 49/2^24 = (4 + 49)/2^24 = 53/2^24

Step 8: Subtract from 6, S_25 = 6 - 53/2^24

Final answer: 6 - 53/2^24

Question: Sum the series up to n terms: 1 + 8/5 + 15/5^2 + 22/5^3 + ... to n terms

Answer: Step 1: The numerators 1, 8, 15, 22, ... are in arithmetic progression with d = 7 and the ratio is r = 1/5, so a = 1, d = 7, r = 1/5

Step 2: Use the formula S_n = a/(1-r) + dr/(1-r)^2 - dr^n/(1-r)^2 - [a + (n-1)d] r^n/(1-r)

Step 3: Compute the constant parts, a/(1-r) = 1/(4/5) = 5/4 and dr/(1-r)^2 = (7/5)/(16/25) = 35/16

Step 4: Add the constant parts, 5/4 + 35/16 = 20/16 + 35/16 = 55/16

Step 5: Compute dr^n/(1-r)^2 = 7 x (1/5)^n / (16/25) = 175/(16 x 5^n)

Step 6: Compute [a + (n-1)d] r^n/(1-r) = (7n - 6) x (1/5)^n / (4/5) = (35n - 30)/(4 x 5^n) = (140n - 120)/(16 x 5^n)

Step 7: Add the two correction terms, (175 + 140n - 120)/(16 x 5^n) = (140n + 55)/(16 x 5^n)

Final answer: S_n = 55/16 - (140n + 55)/(16 x 5^n)

Question: Find the sum to infinity of the series: 1 + 4/3 + 9/3^2 + 14/3^3 + 19/3^4 + ...

Answer: Step 1: Let S = 1 + 4/3 + 9/3^2 + 14/3^3 + 19/3^4 + ... where the ratio is r = 1/3, which is less than 1 so the sum to infinity exists

Step 2: Multiply both sides by 1/3, (1/3)S = 1/3 + 4/3^2 + 9/3^3 + 14/3^4 + ...

Step 3: Subtract the second equation from the first, (2/3)S = 1 + (4/3 -

1/3) + (9-4)/3^2 + (14-9)/3^3 + (19-14)/3^4 + ...

Step 4: Simplify the differences, (2/3)S = 1 + 1 + 5/3^2 + 5/3^3 + 5/3^4 + ...

Step 5: The tail 5/3^2 + 5/3^3 + ... is geometric with first term 5/9 and ratio 1/3, so its sum is (5/9)/(1 - 1/3) = (5/9)(3/2) = 5/6

Step 6: So (2/3)S = 2 + 5/6 = 17/6

Step 7: Multiply both sides by 3/2, S = (3/2)(17/6) = 17/4

Final answer: 17/4

Question: Find the sum to infinity of the series: 1 + 9a + 17a^2 + 25a^3 + ... where |a| < 1

Answer: Step 1: The numerators 1, 9, 17, 25, ... are in arithmetic progression with a = 1 and d = 8, and the geometric ratio is r = a with |a| < 1

Step 2: For an arithmetico-geometric series the sum to infinity is S = a/(1-r) + dr/(1-r)^2

Step 3: Substitute the values, S = 1/(1-a) + 8a/(1-a)^2

Step 4: Write with the common denominator (1-a)^2, S = [(1-a) + 8a]/(1-a)^2

Step 5: Simplify the numerator, (1-a) + 8a = 1 + 7a

Final answer: S = (1 + 7a)/(1-a)^2

Question: Find the sum to infinity of the series: 1 + 4b + 7b^2 + 10b^3 + ... where |b| < 1

Answer: Step 1: The numerators 1, 4, 7, 10, ... are in arithmetic progression with a = 1 and d = 3, and the geometric ratio is r = b with |b|

< 1

Step 2: For an arithmetico-geometric series the sum to infinity is S = a/(1-r) + dr/(1-r)^2

Step 3: Substitute the values, S = 1/(1-b) + 3b/(1-b)^2

Step 4: Write with the common denominator (1-b)^2, S = [(1-b) + 3b]/(1-b)^2

Step 5: Simplify the numerator, (1-b) + 3b = 1 + 2b

Final answer: S = (1 + 2b)/(1-b)^2

Question: Find the sum to infinity of the series: 1 + 4/4 + 7/4^2 + 10/4^3 + ...

Answer: Step 1: The numerators 1, 4, 7, 10, ... are in arithmetic progression with a = 1 and d = 3, and the geometric ratio is r = 1/4

Step 2: For an arithmetico-geometric series the sum to infinity is S = a/(1-r) + dr/(1-r)^2

Step 3: Substitute the values, S = 1/(1 - 1/4) + (3 x 1/4)/(1 - 1/4)^2

Step 4: Compute the first part, 1/(3/4) = 4/3

Step 5: Compute the second part, (3/4)/(9/16) = (3/4)(16/9) = 4/3

Step 6: Add, S = 4/3 + 4/3 = 8/3

Final answer: 8/3

Question: Find the sum to infinity of the series: 1 + 5/3 + 9/3^2 + 13/3^3 + ...

Answer: Step 1: The numerators 1, 5, 9, 13, ... are in arithmetic progression with a = 1 and d = 4, and the geometric ratio is r = 1/3

Step 2: For an arithmetico-geometric series the sum to infinity is S = a/(1-r) + dr/(1-r)^2

Step 3: Substitute the values, S = 1/(1 - 1/3) + (4 x 1/3)/(1 - 1/3)^2

Step 4: Compute the first part, 1/(2/3) = 3/2

Step 5: Compute the second part, (4/3)/(4/9) = (4/3)(9/4) = 3

Step 6: Add, S = 3/2 + 3 = 9/2

Final answer: 9/2

Question: Find the sum to infinity of the series: 1 + 5a + 9a^2 + 13a^3 + ... where |a| < 1

Answer: Step 1: The numerators 1, 5, 9, 13, ... are in arithmetic progression with a = 1 and d = 4, and the geometric ratio is r = a with |a| < 1

Step 2: For an arithmetico-geometric series the sum to infinity is S = a/(1-r) + dr/(1-r)^2

Step 3: Substitute the values, S = 1/(1-a) + 4a/(1-a)^2

Step 4: Write with the common denominator (1-a)^2, S = [(1-a) + 4a]/(1-a)^2

Step 5: Simplify the numerator, (1-a) + 4a = 1 + 3a

Final answer: S = (1 + 3a)/(1-a)^2

Exercise 5.3

Question: Find the sum of the series to n terms: 3 + 6 + 12 + 21 + 33 + ...

to n terms

Answer: Step 1: The differences of consecutive terms are 3, 6, 9, 12, ..., which are in arithmetic progression, so we use the method of differences

Step 2: Let the nth term be a quadratic An^2 + Bn + C, using the first three terms, A + B + C = 3, 4A + 2B + C = 6 and 9A + 3B + C = 12

Step 3: Subtract consecutive equations, 3A + B = 3 and 5A + B = 6

Step 4: Subtract again, 2A = 3 so A = 3/2, then B = 3 - 9/2 = -3/2 and C = 3

Step 5: So the kth term is T_k = (3k^2 - 3k + 6)/2 = (3/2)(k^2 - k + 2)

Step 6: Take the sum, S_n = (3/2)[sum of k^2 - sum of k + 2n] = (3/2)[n(n+1)(2n+1)/6 - n(n+1)/2 + 2n]

Step 7: Take n common inside, the bracket becomes n[(n+1)(2n+1) - 3(n+1) + 12]/6 = n(2n^2 + 10)/6 = n(n^2 + 5)/3

Step 8: Multiply by 3/2, S_n = (3/2) x n(n^2 + 5)/3

Final answer: S_n = n(n^2 + 5)/2

Question: Find the sum of the series to n terms: 5 + 10 + 17 + 26 + 37 + ... to n terms

Answer: Step 1: The differences of consecutive terms are 5, 7, 9, 11, ..., which are in arithmetic progression, so we use the method of differences

Step 2: Let the nth term be An^2 + Bn + C, using the first three terms, A + B + C = 5, 4A + 2B + C = 10 and 9A + 3B + C = 17

Step 3: Subtract consecutive equations, 3A + B = 5 and 5A + B = 7

Step 4: Subtract again, 2A = 2 so A = 1, then B = 2 and C = 2

Step 5: So the kth term is T_k = k^2 + 2k + 2

Step 6: Take the sum, S_n = sum of k^2 + 2 x sum of k + 2n = n(n+1)(2n+1)/6 + n(n+1) + 2n

Step 7: Take n common with denominator 6, S_n = n[(n+1)(2n+1) + 6(n+1) + 12]/6

Step 8: Expand inside the bracket, 2n^2 + 3n + 1 + 6n + 6 + 12 = 2n^2 + 9n + 19

Final answer: S_n = n(2n^2 + 9n + 19)/6

Question: Find the sum of the series to n terms: 7 + 14 + 26 + 43 + 65 + ... to n terms

Answer: Step 1: The differences of consecutive terms are 7, 12, 17, 22, ..., which are in arithmetic progression with difference 5, so we use the method of differences

Step 2: Let the nth term be An^2 + Bn + C, using the first three terms, A + B + C = 7, 4A + 2B + C = 14 and 9A + 3B + C = 26

Step 3: Subtract consecutive equations, 3A + B = 7 and 5A + B = 12

Step 4: Subtract again, 2A = 5 so A = 5/2, then B = 7 - 15/2 = -1/2 and C = 5

Step 5: So the kth term is T_k = (5k^2 - k + 10)/2

Step 6: Take the sum, S_n = (1/2)[5 x sum of k^2 - sum of k + 10n] = (1/2)[5n(n+1)(2n+1)/6 - n(n+1)/2 + 10n]

Step 7: Take n common inside with denominator 6, the bracket becomes n[5(n+1)(2n+1) - 3(n+1) + 60]/6 = n(10n^2 + 12n + 62)/6

Step 8: Multiply by 1/2 and simplify, S_n = n(5n^2 + 6n + 31)/6

Final answer: S_n = n(5n^2 + 6n + 31)/6

Question: Find the sum of the series to n terms: 3 + 6 + 15 + 42 + 123 + ... to n terms

Answer: Step 1: The differences of consecutive terms are 3, 9, 27, 81, ..., which are in geometric progression with ratio 3, so we use the method of differences

Step 2: Write S_n = 3 + 6 + 15 + 42 + ... + a_n, and the same sum shifted by one term

Step 3: Subtract the shifted sum, 0 = 3 + {3 + 9 + 27 + ... to (n-1) terms} - a_n

Step 4: The bracket is geometric with first term 3, ratio 3 and (n-1) terms, so its sum is 3(3^(n-1) - 1)/(3 - 1) = (3^n - 3)/2

Step 5: So a_n = 3 + (3^n - 3)/2 = (3^n + 3)/2

Step 6: Take the sum of the nth term formula, S_n = (1/2)[sum of 3^k + sum of 3] from k = 1 to n

Step 7: The geometric sum is 3(3^n - 1)/(3 - 1) = (3^(n+1) - 3)/2 and the sum of the constant 3 is 3n

Step 8: Combine, S_n = (1/2)[(3^(n+1) - 3)/2 + 3n] = (3^(n+1) - 3 + 6n)/4

Final answer: S_n = (3^(n+1) + 6n - 3)/4

Question: Find the sum of the series to n terms: 7 + 14 + 45 + 100 + 179 + ... to n terms

Answer: Step 1: The differences of consecutive terms are 7, 31, 55, 79, ..., and their differences are all 24, so the nth term is a quadratic and we use the method of differences

Step 2: Let the nth term be An^2 + Bn + C, using the first three terms, A

+ B + C = 7, 4A + 2B + C = 14 and 9A + 3B + C = 45

Step 3: Subtract consecutive equations, 3A + B = 7 and 5A + B = 31

Step 4: Subtract again, 2A = 24 so A = 12, then B = 7 - 36 = -29 and C = 7 - 12 + 29 = 24

Step 5: Check with the fourth term, 12(16) - 29(4) + 24 = 192 - 116 + 24 = 100, which is correct

Step 6: So the kth term is T_k = 12k^2 - 29k + 24

Step 7: Take the sum, S_n = 12 x sum of k^2 - 29 x sum of k + 24n = 12 x n(n+1)(2n+1)/6 - 29 x n(n+1)/2 + 24n

Step 8: Take n common, S_n = n[2(n+1)(2n+1) - 29(n+1)/2 + 24] = n(8n^2 - 17n + 23)/2

Final answer: S_n = n(8n^2 - 17n + 23)/2

Question: Find the sum of the series to n terms: 3 + 6 + 21 + 96 + 471 + ... to n terms

Answer: Step 1: The differences of consecutive terms are 3, 15, 75, 375, ..., which are in geometric progression with ratio 5, so we use the method of differences

Step 2: Write S_n = 3 + 6 + 21 + 96 + ... + a_n, and the same sum shifted by one term

Step 3: Subtract the shifted sum, 0 = 3 + {3 + 15 + 75 + ... to (n-1) terms} - a_n

Step 4: The bracket is geometric with first term 3, ratio 5 and (n-1) terms, so its sum is 3(5^(n-1) - 1)/(5 - 1) = 3(5^(n-1) - 1)/4

Step 5: So a_n = 3 + 3(5^(n-1) - 1)/4 = (9 + 3 x 5^(n-1))/4 = (3/4)(3 +

5^(n-1))

Step 6: Check with the fourth term, (3/4)(3 + 125) = 96, which is correct

Step 7: Take the sum, S_n = (3/4)[3n + sum of 5^(k-1)] from k = 1 to n, and the geometric sum is (5^n - 1)/(5 - 1) = (5^n - 1)/4

Step 8: Combine, S_n = (3/4)[3n + (5^n - 1)/4] = 9n/4 + 3(5^n - 1)/16

Step 9: Write with denominator 16, S_n = (36n + 3 x 5^n - 3)/16

Final answer: S_n = (3 x 5^n + 36n - 3)/16

Exercise 5.4

Question: Find the sum of the series to n terms: 1/(1x3) + 1/(3x5) + 1/(5x7) + ... to n terms

Answer: Step 1: The kth term is 1/((2k-1)(2k+1)), a rational term, so we use partial fractions

Step 2: Write 1/((2k-1)(2k+1)) = (1/2)[1/(2k-1) - 1/(2k+1)], since the difference of the two fractions gives 2/((2k-1)(2k+1))

Step 3: Take the sum from k = 1 to n, S_n = (1/2)[(1 - 1/3) + (1/3 - 1/5) + (1/5 - 1/7) + ... + (1/(2n-1) - 1/(2n+1))]

Step 4: All middle terms cancel in pairs (telescoping), leaving only the first and last terms

Step 5: So S_n = (1/2)[1 - 1/(2n+1)]

Step 6: Simplify, 1 - 1/(2n+1) = 2n/(2n+1), so S_n = (1/2)(2n/(2n+1))

Final answer: S_n = n/(2n+1)

Question: Find the sum of the series to n terms: 1/(1x5) + 1/(5x9) + 1/(9x13) + ... to n terms

Answer: Step 1: The kth term is 1/((4k-3)(4k+1)), a rational term, so we use partial fractions

Step 2: Write 1/((4k-3)(4k+1)) = (1/4)[1/(4k-3) - 1/(4k+1)], since the difference of the two fractions gives 4/((4k-3)(4k+1))

Step 3: Take the sum from k = 1 to n, S_n = (1/4)[(1 - 1/5) + (1/5 - 1/9) + (1/9 - 1/13) + ... + (1/(4n-3) - 1/(4n+1))]

Step 4: All middle terms cancel in pairs (telescoping), leaving only the first and last terms

Step 5: So S_n = (1/4)[1 - 1/(4n+1)]

Step 6: Simplify, 1 - 1/(4n+1) = 4n/(4n+1), so S_n = (1/4)(4n/(4n+1))

Final answer: S_n = n/(4n+1)

Question: Find the sum of the series to infinity: 1/(1x5x9) + 1/(5x9x13) + 1/(9x13x17) + ... to infinity

Answer: Step 1: The kth term is 1/((4k-3)(4k+1)(4k+5)), so we look for a difference of two simpler fractions

Step 2: Consider f(k) = 1/((4k-3)(4k+1)), then f(k) - f(k+1) = 1/((4k-3)(4k+1)) - 1/((4k+1)(4k+5))

Step 3: Put over the common denominator, the numerator is (4k+5) - (4k-3) = 8, so f(k) - f(k+1) = 8/((4k-3)(4k+1)(4k+5))

Step 4: Hence the kth term is (1/8)[f(k) - f(k+1)], the method of differences form

Step 5: Sum to N terms, the sum telescopes to (1/8)[f(1) - f(N+1)]

Step 6: Compute f(1) = 1/(1 x 5) = 1/5, and as N tends to infinity f(N+1) tends to 0

Step 7: So the sum to infinity is (1/8)(1/5 - 0) = 1/40

Final answer: 1/40

Question: Find the sum of the series to infinity: 1/(1x4x7) + 1/(4x7x10) + 1/(7x10x13) + ... to infinity

Answer: Step 1: The kth term is 1/((3k-2)(3k+1)(3k+4)), so we look for a difference of two simpler fractions

Step 2: Consider f(k) = 1/((3k-2)(3k+1)), then f(k) - f(k+1) = 1/((3k-2)(3k+1)) - 1/((3k+1)(3k+4))

Step 3: Put over the common denominator, the numerator is (3k+4) - (3k-2) = 6, so f(k) - f(k+1) = 6/((3k-2)(3k+1)(3k+4))

Step 4: Hence the kth term is (1/6)[f(k) - f(k+1)], the method of differences form

Step 5: Sum to N terms, the sum telescopes to (1/6)[f(1) - f(N+1)]

Step 6: Compute f(1) = 1/(1 x 4) = 1/4, and as N tends to infinity f(N+1) tends to 0

Step 7: So the sum to infinity is (1/6)(1/4 - 0) = 1/24

Final answer: 1/24

Question: Find the sum of the series: sum from k=1 to infinity of 1/(4k^2 - 1)

Answer: Step 1: Factor the denominator, 4k^2 - 1 = (2k-1)(2k+1), so the kth term is 1/((2k-1)(2k+1))

Step 2: Use partial fractions, 1/((2k-1)(2k+1)) = (1/2)[1/(2k-1) - 1/(2k+1)]

Step 3: Take the sum from k = 1 to N, the sum is (1/2)[(1 - 1/3) + (1/3 - 1/5) + ... + (1/(2N-1) - 1/(2N+1))]

Step 4: All middle terms cancel in pairs (telescoping), leaving (1/2)[1 - 1/(2N+1)]

Step 5: As N tends to infinity, 1/(2N+1) tends to 0

Step 6: So the sum to infinity is (1/2)(1 - 0) = 1/2

Final answer: 1/2

Question: Find the sum of the series: sum from k=1 to n of 1/(9k^2 + 3k - 2)

Answer: Step 1: Factor the denominator, 9k^2 + 3k - 2 = (3k-1)(3k+2), since (3k-1)(3k+2) = 9k^2 + 6k - 3k - 2

Step 2: Use partial fractions, 1/((3k-1)(3k+2)) = (1/3)[1/(3k-1) - 1/(3k+2)], since the difference gives 3/((3k-1)(3k+2))

Step 3: Take the sum from k = 1 to n, S_n = (1/3)[(1/2 - 1/5) + (1/5 - 1/8) + ... + (1/(3n-1) - 1/(3n+2))]

Step 4: All middle terms cancel in pairs (telescoping), leaving S_n = (1/3)[1/2 - 1/(3n+2)]

Step 5: Simplify the bracket, 1/2 - 1/(3n+2) = (3n+2-2)/(2(3n+2)) = 3n/(2(3n+2))

Step 6: Multiply by 1/3, S_n = n/(2(3n+2))

Final answer: S_n = n/(6n + 4)

Question: Find the sum of the series: sum from k=2 to infinity of 1/(k^2 - 1)

Answer: Step 1: Factor the denominator, k^2 - 1 = (k-1)(k+1), so the kth

term is 1/((k-1)(k+1))

Step 2: Use partial fractions, 1/((k-1)(k+1)) = (1/2)[1/(k-1) - 1/(k+1)]

Step 3: Take the sum from k = 2 to N, write the terms out: (1/2)[(1/1 - 1/3) + (1/2 - 1/4) + (1/3 - 1/5) + ...]

Step 4: Because the two fractions are two steps apart, the cancelling leaves the first two positive terms and the last two negative terms, giving (1/2)[1 + 1/2 - 1/N - 1/(N+1)]

Step 5: As N tends to infinity, 1/N and 1/(N+1) both tend to 0

Step 6: So the sum to infinity is (1/2)(3/2) = 3/4

Final answer: 3/4

Question: Find the sum of the series: sum from k=2 to n of 1/(k^2 - k)

Answer: Step 1: Factor the denominator, k^2 - k = k(k-1), so the kth term is 1/(k(k-1))

Step 2: Use partial fractions, 1/(k(k-1)) = 1/(k-1) - 1/k

Step 3: Take the sum from k = 2 to n, S_n = (1/1 - 1/2) + (1/2 - 1/3) + (1/3 - 1/4) + ... + (1/(n-1) - 1/n)

Step 4: All middle terms cancel in pairs (telescoping), leaving only the first and last terms

Step 5: So S_n = 1 - 1/n

Step 6: Write as a single fraction, 1 - 1/n = (n-1)/n

Final answer: S_n = (n-1)/n

Review Exercise 5

Question: Select the correct option: (i) If a_n = 5 - 3n + 2n^2, then a_(2n) = ... (a) 5 - 6n + 2n^2 (b) 5 - 6n + 4n^2 (c) 5 + 6n + 4n^2 (d) 5 - 6n + 8n^2

Answer: Step 1: To find a_(2n) we replace n by 2n everywhere in a_n = 5 - 3n + 2n^2

Step 2: The constant term stays 5

Step 3: The linear term becomes -3(2n) = -6n

Step 4: The squared term becomes 2(2n)^2 = 2 x 4n^2 = 8n^2

Step 5: Collect the three parts, a_(2n) = 5 - 6n + 8n^2

Final answer: option (d) 5 - 6n + 8n^2

Question: Select the correct option: (ii) If a_(n-2) = 3n - 11, then a_n = ... (a) 3n + 5 (b) 3n - 5 (c) 3n - 9 (d) 3n - 13

Answer: Step 1: Put m = n - 2, so that n = m + 2

Step 2: Then a_m = 3(m + 2) - 11

Step 3: Expand, a_m = 3m + 6 - 11

Step 4: Simplify, a_m = 3m - 5

Step 5: Rename the variable m back to n, a_n = 3n - 5

Final answer: option (b) 3n - 5

Question: Select the correct option: (iii) sum from n=1 to 50 of n = ... (a) 1274 (b) 1275 (c) 1280 (d) 1285

Answer: Step 1: The sum is 1 + 2 + 3 + ... + 50, the sum of the first 50

natural numbers

Step 2: Use the formula sum of n = n(n+1)/2 with n = 50

Step 3: Substitute, sum = 50 x 51 / 2

Step 4: Simplify, 50/2 = 25, so sum = 25 x 51

Step 5: Multiply, 25 x 51 = 1275

Final answer: option (b) 1275

Question: Select the correct option: (iv) 1/(n(n+1)) = ... (a) 1/n + 1/(n+1) (b) 1/n - 1/(n+1) (c) 2/n + 1/(n+1) (d) 1/n - 3/(n+1)

Answer: Step 1: Split 1/(n(n+1)) into partial fractions A/n + B/(n+1)

Step 2: Multiply both sides by n(n+1), giving 1 = A(n+1) + Bn

Step 3: Put n = 0, giving 1 = A

Step 4: Put n = -1, giving 1 = -B, so B = -1

Step 5: Hence 1/(n(n+1)) = 1/n - 1/(n+1), which can be checked because the difference has numerator (n+1) - n = 1

Final answer: option (b) 1/n - 1/(n+1)

Question: Select the correct option: (v) The nth term of 1/2, 1/5, 1/8, ... is: (a) 1/(3n-1) (b) 3n - 1 (c) 2n + 1 (d) 1/(3n+1)

Answer: Step 1: Look at the denominators 2, 5, 8, ...

Step 2: They form an arithmetic progression with first term 2 and common difference 3

Step 3: The nth denominator is 2 + (n-1)(3) = 3n - 1

Step 4: Check: for n = 1, 2 and 3 this gives 2, 5 and 8, which is correct

Step 5: So the nth term of the sequence is the reciprocal, 1/(3n-1)

Final answer: option (a) 1/(3n-1)

Question: Select the correct option: (vi) sum of n is equal to: (a) n(n+1)/2 (b) n(n+1)(2n+1)/6 (c) n^2(n+1)^2/2 (d) n^2

Answer: Step 1: Sum of n means 1 + 2 + 3 + ... + n, the sum of the first n natural numbers

Step 2: Pair the first and last terms, 1 + n, 2 + (n-1), ..., each pair adding to n+1

Step 3: There are n/2 such pairs

Step 4: So the sum is (n/2)(n+1) = n(n+1)/2

Step 5: Option (b) is the sum of squares and must not be confused with this

Final answer: option (a) n(n+1)/2

Question: Select the correct option: (vii) sum of n^2 is equal to: (a) n(n+1)/2 (b) n(n+1)(2n+1)/6 (c) n^2(n+1)^2/2 (d) n^2

Answer: Step 1: Sum of n^2 means 1^2 + 2^2 + 3^2 + ... + n^2

Step 2: It is obtained from the identity sum of [k^3 - (k-1)^3] = n^3 by expanding the cubes

Step 3: Expanding gives 3 x sum of k^2 - 3 x sum of k + n = n^3

Step 4: Substituting sum of k = n(n+1)/2 and solving for sum of k^2 gives n(n+1)(2n+1)/6

Step 5: Option (a) is the sum of first powers, so it is not this sum

Final answer: option (b) n(n+1)(2n+1)/6

Question: Select the correct option: (viii) sum of n^3 is equal to: (a) n(n+1)/2 (b) n(n+1)(2n+1)/6 (c) n^2(n+1)^2/4 (d) n(n+1)^2/4

Answer: Step 1: Sum of n^3 means 1^3 + 2^3 + 3^3 + ... + n^3

Step 2: It is obtained from the identity sum of [k^4 - (k-1)^4] = n^4 by expanding the fourth powers and using the sums of k and k^2

Step 3: Solving gives sum of k^3 = [n(n+1)/2]^2

Step 4: Square the bracket, [n(n+1)/2]^2 = n^2(n+1)^2/4

Step 5: Note this is the square of the sum of the first n natural numbers

Final answer: option (c) n^2(n+1)^2/4

Question: Select the correct option: (ix) If S_n = (n+1)^2, then S_(2n) is equal to: (a) 2n + 1 (b) 4n^2 + 4n + 1 (c) (2n-1)^2 (d) Cannot be determined

Answer: Step 1: S_(2n) is found by replacing n by 2n in S_n = (n+1)^2

Step 2: Substitute, S_(2n) = (2n + 1)^2

Step 3: Expand the square, (2n+1)^2 = (2n)^2 + 2(2n)(1) + 1^2

Step 4: Simplify each part, 4n^2 + 4n + 1

Final answer: option (b) 4n^2 + 4n + 1

Question: Select the correct option: (x) sum from n=3 to 20 of n^0 = ... (a) 1 (b) 19 (c) 20 (d) 18

Answer: Step 1: For every value of n, n^0 = 1

Step 2: So the sum is simply the number of terms, each contributing 1

Step 3: Count the terms from n = 3 to n = 20 inclusive, 20 - 3 + 1 = 18 terms

Step 4: Hence the sum is 18 x 1 = 18

Final answer: option (d) 18

Question: Sum the series up to n terms: 2 x 1^2 + 4 x 2^2 + 6 x 3^2 + ... up to n terms

Answer: Step 1: In the kth term the first factor 2, 4, 6, ... has kth value 2k, and the second factor is k^2

Step 2: So the kth term is 2k x k^2 = 2k^3

Step 3: Take the sum from k = 1 to n, S_n = 2 x sum of k^3

Step 4: Substitute the formula for the sum of cubes, S_n = 2[n(n+1)/2]^2

Step 5: Square the bracket, [n(n+1)/2]^2 = n^2(n+1)^2/4

Step 6: Multiply by 2, S_n = n^2(n+1)^2/2

Final answer: S_n = n^2(n+1)^2/2

Question: Sum the series to infinity: 1/(1x4) + 1/(4x7) + 1/(7x10) + ... to infinity

Answer: Step 1: The kth term is 1/((3k-2)(3k+1)), a rational term, so we use partial fractions

Step 2: Write 1/((3k-2)(3k+1)) = (1/3)[1/(3k-2) - 1/(3k+1)], since the difference of the two fractions gives 3/((3k-2)(3k+1))

Step 3: Take the sum from k = 1 to N, the sum is (1/3)[(1 - 1/4) + (1/4 - 1/7) + (1/7 - 1/10) + ... + (1/(3N-2) - 1/(3N+1))]

Step 4: All middle terms cancel in pairs (telescoping), leaving (1/3)[1 - 1/(3N+1)]

Step 5: As N tends to infinity, 1/(3N+1) tends to 0

Step 6: So the sum to infinity is (1/3)(1 - 0) = 1/3

Final answer: 1/3

Unit 6: Permutation, Combination and Probability

Exercise 6.1

Question: Evaluate (i) 4!

Answer: Step 1: by definition n! = n(n - 1)(n - 2)...3.2.1

Step 2: 4! = 4 x 3 x 2 x 1

Step 3: 4 x 3 = 12, 12 x 2 = 24, 24 x 1 = 24

Final answer: 4! = 24

Question: Evaluate (ii) 6!

Answer: Step 1: by definition n! = n(n - 1)(n - 2)...3.2.1

Step 2: 6! = 6 x 5 x 4 x 3 x 2 x 1

Step 3: 6 x 5 = 30, 30 x 4 = 120, 120 x 3 = 360, 360 x 2 = 720, 720 x 1 = 720

Final answer: 6! = 720

Question: Evaluate (iii) 8! / 5!

Answer: Step 1: expand the larger factorial down to the smaller one: 8! = 8 x 7 x 6 x 5!

Step 2: 8! / 5! = (8 x 7 x 6 x 5!) / 5!

Step 3: cancel 5! from numerator and denominator, giving 8 x 7 x 6

Step 4: 8 x 7 = 56, 56 x 6 = 336

Final answer: 8! / 5! = 336

Question: Evaluate (iv) 10! / 7!

Answer: Step 1: expand 10! down to 7!: 10! = 10 x 9 x 8 x 7!

Step 2: 10! / 7! = (10 x 9 x 8 x 7!) / 7!

Step 3: cancel 7!, giving 10 x 9 x 8

Step 4: 10 x 9 = 90, 90 x 8 = 720

Final answer: 10! / 7! = 720

Question: Evaluate (v) 5! x 7! x 3!

Answer: Step 1: find each factorial: 5! = 5 x 4 x 3 x 2 x 1 = 120, 7! = 7 x 6 x 5 x 4 x 3 x 2 x 1 = 5040, 3! = 3 x 2 x 1 = 6

Step 2: multiply the first two values: 120 x 5040 = 604800

Step 3: multiply by the third value: 604800 x 6 = 3628800

Final answer: 5! x 7! x 3! = 3628800

Question: Evaluate (vi) 10! / 8! + 5!

Answer: Step 1: division is done before addition, so first compute 10! / 8! = (10 x 9 x 8!) / 8! = 10 x 9 = 90

Step 2: compute 5! = 5 x 4 x 3 x 2 x 1 = 120

Step 3: add the two results: 90 + 120 = 210

Final answer: 10! / 8! + 5! = 210

Question: Evaluate (vii) 16! / (8! x 12!)

Answer: Step 1: expand 16! down to 12!: 16! = 16 x 15 x 14 x 13 x 12!

Step 2: cancel 12!: 16! / (8! x 12!) = (16 x 15 x 14 x 13) / 8! = 43680 / 40320

Step 3: divide numerator and denominator by 3360: 43680 / 40320 = 13 / 12

Final answer: 16! / (8! x 12!) = 13/12

Question: Evaluate (viii) (9!)^2 / ((5!)^3 x 7!)

Answer: Step 1: find the factorials: 9! = 362880, 5! = 120, 7! = 5040

Step 2: numerator (9!)^2 = 362880^2 = 131681894400

Step 3: denominator (5!)^3 x 7! = 120^3 x 5040 = 1728000 x 5040 = 8709120000

Step 4: divide: 131681894400 / 8709120000 = 378/25 = 15.12

Final answer: (9!)^2 / ((5!)^3 x 7!) = 378/25

Question: Evaluate (ix) 9! / (2! x (9 - 2)!)

Answer: Step 1: simplify the brackets: (9 - 2)! = 7! and 2! = 2

Step 2: expand 9! down to 7!: 9! = 9 x 8 x 7!

Step 3: 9! / (2! x 7!) = (9 x 8 x 7!) / (2 x 7!) = (9 x 8) / 2

Step 4: 72 / 2 = 36

Final answer: 9! / (2! x (9 - 2)!) = 36

Question: Write in factorial form (i) 6 x 5 x 4

Answer: Step 1: a descending product is completed to a factorial by multiplying and dividing by the missing lower factors

Step 2: 6 x 5 x 4 = (6 x 5 x 4 x 3 x 2 x 1) / (3 x 2 x 1)

Step 3: the numerator is 6! and the denominator is 3!

Final answer: 6 x 5 x 4 = 6! / 3!

Question: Write in factorial form (ii) 12 x 11 x 10

Answer: Step 1: complete the product down to 1 by multiplying and dividing by 9!

Step 2: 12 x 11 x 10 = (12 x 11 x 10 x 9 x 8 x ... x 1) / (9 x 8 x ... x 1)

Step 3: the numerator is 12! and the denominator is 9!

Final answer: 12 x 11 x 10 = 12! / 9!

Question: Write in factorial form (iii) n(n + 1)(n + 2) / (6 x 5)

Answer: Step 1: write the numerator as a ratio of factorials: n(n + 1)(n + 2) = (n + 2)! / (n - 1)!, because (n + 2)! = (n + 2)(n + 1)n(n - 1)!

Step 2: write the denominator as a ratio of factorials: 6 x 5 = 6! / 4!, because 6! = 6 x 5 x 4!

Step 3: divide the two ratios: [(n + 2)! / (n - 1)!] / [6! / 4!] = (n + 2)! x 4! / ((n - 1)! x 6!)

Final answer: n(n + 1)(n + 2) / (6 x 5) = 4!(n + 2)! / (6!(n - 1)!)

Question: Write in factorial form (iv) 20 x 19 x 18 x 17

Answer: Step 1: complete the product down to 1 by multiplying and dividing by 16!

Step 2: 20 x 19 x 18 x 17 = (20 x 19 x 18 x 17 x 16 x ... x 1) / (16 x 15 x ... x 1)

Step 3: the numerator is 20! and the denominator is 16!

Final answer: 20 x 19 x 18 x 17 = 20! / 16!

Question: Express as a single fraction (i) (n + 1)! / (r + 1)! + n! / r!

Answer: Step 1: expand (n + 1)! = (n + 1)n! and (r + 1)! = (r + 1)r! so

both terms can be written over the common denominator (r + 1)!

Step 2: first term = (n + 1)n! / (r + 1)! and second term = n!(r + 1) / (r + 1)!, because n! / r! = n!(r + 1) / ((r + 1) r!) = n!(r + 1) / (r + 1)!

Step 3: add the numerators: [(n + 1)n! + (r + 1)n!] / (r + 1)!

Step 4: take n! common: n![(n + 1) + (r + 1)] / (r + 1)! = n!(n + r + 2) / (r + 1)!

Final answer: (n + 1)! / (r + 1)! + n! / r! = n!(n + r + 2) / (r + 1)!

Question: Express as a single fraction (ii) (n + 1)! / r! + n! / (r + 1)!

Answer: Step 1: use the common denominator (r + 1)!: multiply the first term above and below by (r + 1)

Step 2: first term = (n + 1)!(r + 1) / (r + 1)! and the second term is already n! / (r + 1)!

Step 3: add the numerators: [(r + 1)(n + 1)! + n!] / (r + 1)!

Step 4: take n! common, using (n + 1)! = (n + 1)n!: n![(r + 1)(n + 1) + 1] / (r + 1)!

Step 5: expand the bracket: (r + 1)(n + 1) + 1 = rn + r + n + 1 + 1 = rn + r + n + 2

Final answer: (n + 1)! / r! + n! / (r + 1)! = n!(rn + r + n + 2) / (r + 1)!

Exercise 6.2

Question: Evaluate (i) 20P3

Answer: Step 1: use nPr = n! / (n - r)!, so 20P3 = 20! / 17!

Step 2: expand 20! down to 17!: 20! = 20 x 19 x 18 x 17!

Step 3: cancel 17!: 20P3 = 20 x 19 x 18

Step 4: 20 x 19 = 380, 380 x 18 = 6840

Final answer: 20P3 = 6840

Question: Evaluate (ii) 16P4

Answer: Step 1: use nPr = n! / (n - r)!, so 16P4 = 16! / 12!

Step 2: expand 16! down to 12!: 16! = 16 x 15 x 14 x 13 x 12!

Step 3: cancel 12!: 16P4 = 16 x 15 x 14 x 13

Step 4: 16 x 15 = 240, 240 x 14 = 3360, 3360 x 13 = 43680

Final answer: 16P4 = 43680

Question: Evaluate (iii) 12P5

Answer: Step 1: use nPr = n! / (n - r)!, so 12P5 = 12! / 7!

Step 2: expand 12! down to 7!: 12! = 12 x 11 x 10 x 9 x 8 x 7!

Step 3: cancel 7!: 12P5 = 12 x 11 x 10 x 9 x 8

Step 4: 12 x 11 = 132, 132 x 10 = 1320, 1320 x 9 = 11880, 11880 x 8 = 95040

Final answer: 12P5 = 95040

Question: Evaluate (iv) the multinomial permutation (8; 3, 2), i.e. the number of permutations of 8 objects in which 3 are alike of one kind, 2 are alike of another kind and the remaining 3 are all different

Answer: Step 1: for repeated objects use n! / (r! s!), where r and s are the numbers of alike objects

Step 2: here n = 8, r = 3, s = 2, so the value is 8! / (3! x 2!)

Step 3: expand: 8! = 8 x 7 x 6 x 5 x 4 x 3 x 2 x 1 = 40320, 3! = 6, 2! = 2

Step 4: 40320 / (6 x 2) = 40320 / 12 = 3360

Final answer: (8; 3, 2) = 3360

Question: Evaluate (v) the multinomial permutation (9; 2, 3, 4), i.e. the number of permutations of 9 objects in which 2, 3 and 4 are alike of three different kinds

Answer: Step 1: use n! / (r! s! t!) with n = 9, r = 2, s = 3, t = 4

Step 2: the value is 9! / (2! x 3! x 4!)

Step 3: expand: 9! = 362880, 2! = 2, 3! = 6, 4! = 24

Step 4: 362880 / (2 x 6 x 24) = 362880 / 288 = 1260

Final answer: (9; 2, 3, 4) = 1260

Question: Find the value of n (i) if nP2 = 30

Answer: Step 1: expand nP2 = n(n - 1), so n(n - 1) = 30

Step 2: expand: n^2 - n - 30 = 0

Step 3: factorise: (n - 6)(n + 5) = 0, so n = 6 or n = -5

Step 4: n must be a positive integer, so n = -5 is rejected

Final answer: n = 6

Question: Find the value of n (ii) if nP3 = 60

Answer: Step 1: expand nP3 = n(n - 1)(n - 2), so n(n - 1)(n - 2) = 60

Step 2: look for three consecutive integers whose product is 60: 5 x 4 x 3 = 60

Step 3: compare: n = 5, n - 1 = 4, n - 2 = 3

Final answer: n = 5

Question: Find the value of n (iii) if 8Pn = 8 x 7 x 6

Answer: Step 1: 8Pn = 8! / (8 - n)! is a product of n descending factors starting from 8

Step 2: the right side 8 x 7 x 6 is a product of exactly 3 such factors

Step 3: therefore the number of factors taken is 3

Final answer: n = 3

Question: Find the value of n (iv) if 11Pn = 11 x 10 x 9

Answer: Step 1: 11Pn = 11! / (11 - n)! is a product of n descending factors starting from 11

Step 2: the right side 11 x 10 x 9 is a product of exactly 3 such factors

Step 3: therefore the number of factors taken is 3

Final answer: n = 3

Question: Find the value of n (v) if (n + 2)P4 : (n + 1)P3 = 5 : 1

Answer: Step 1: expand both permutations: (n + 2)P4 = (n + 2)(n + 1)n(n - 1) and (n + 1)P3 = (n + 1)n(n - 1)

Step 2: form the ratio and cancel the common factors (n + 1)n(n - 1): ratio = (n + 2) / 1

Step 3: set the ratio equal to 5 : 1, so n + 2 = 5

Step 4: solve: n = 5 - 2 = 3

Final answer: n = 3

Question: Find the value of n (vi) if nP4 : (n - 1)P3 = 9 : 1

Answer: Step 1: expand both permutations: nP4 = n(n - 1)(n - 2)(n - 3) and (n - 1)P3 = (n - 1)(n - 2)(n - 3)

Step 2: form the ratio and cancel the common factors (n - 1)(n - 2)(n - 3): ratio = n / 1

Step 3: set the ratio equal to 9 : 1, so n = 9

Final answer: n = 9

Question: In how many ways are different batting orders possible for a cricket team consisting of 11 players?

Answer: Step 1: a batting order is an arrangement of all 11 players, and order matters, so this is a permutation of 11 different players taken all at a time

Step 2: number of orders = 11P11 = 11!

Step 3: expand: 11! = 11 x 10 x 9 x 8 x 7 x 6 x 5 x 4 x 3 x 2 x 1

Step 4: multiply step by step: 10! = 3628800 and 11 x 3628800 = 39916800

Final answer: 39916800 different batting orders

Question: In how many ways can two English books, three Chemistry books and four Physics books be arranged on a shelf so that all the books of the same subject are together?

Answer: Step 1: treat each subject group as one block; the 3 blocks can be arranged in 3! = 6 ways

Step 2: inside the English block the 2 books can be arranged in 2! = 2 ways

Step 3: inside the Chemistry block the 3 books can be arranged in 3! = 6 ways

Step 4: inside the Physics block the 4 books can be arranged in 4! = 24

ways

Step 5: by the fundamental principle multiply: 3! x 2! x 3! x 4! = 6 x 2 x 6 x 24 = 1728

Final answer: 1728 ways

Question: How many different arrangements can be formed of the word "MATRIX" using all its letters?

Answer: Step 1: MATRIX has 6 letters M, A, T, R, I, X, all different, and all are used

Step 2: the number of arrangements is the permutation of 6 different letters taken all at a time = 6!

Step 3: expand: 6! = 6 x 5 x 4 x 3 x 2 x 1 = 720

Final answer: 720 arrangements

Question: How many arrangements of the word "MATRIX" start with "R"?

Answer: Step 1: fix R in the first place, so only the remaining 5 letters M, A, T, I, X are to be arranged

Step 2: these 5 different letters can be arranged in 5! ways

Step 3: expand: 5! = 5 x 4 x 3 x 2 x 1 = 120

Final answer: 120 arrangements

Question: How many signals can be given by 8 flags of different colours when 5 of them are used at a time?

Answer: Step 1: a signal depends on which 5 flags are used and the order in which they are hoisted, so this is a permutation of 8 flags taken 5 at a time

Step 2: number of signals = 8P5 = 8! / (8 - 5)! = 8! / 3!

Step 3: expand and cancel: 8 x 7 x 6 x 5 x 4

Step 4: 8 x 7 = 56, 56 x 6 = 336, 336 x 5 = 1680, 1680 x 4 = 6720

Final answer: 6720 signals

Question: A fair coin is tossed three times. How many outcomes are possible? Find by the fundamental principle and illustrate using a tree diagram.

Answer: Step 1: each toss has 2 possible outcomes, head (H) or tail (T)

Step 2: by the fundamental principle of counting, the three tosses together have 2 x 2 x 2 = 8 outcomes

Step 3: list them from the tree diagram: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

Final answer: 8 outcomes

[diagram: A tree diagram with three levels of branching: the first toss splits into H and T branches, eac

Question: How many 5-digit numbers can be formed, without repeating any digit, from (i) the digits 5, 6, 7, 8, 9?

Answer: Step 1: all 5 digits are used and no digit is zero, so every arrangement gives a genuine 5-digit number

Step 2: the number of arrangements is 5P5 = 5!

Step 3: expand: 5! = 5 x 4 x 3 x 2 x 1 = 120

Final answer: 120 numbers

Question: How many 5-digit numbers can be formed, without repeating any digit, from (ii) the digits 1, 2, 3, 5, 7, 9?

Answer: Step 1: 5 digits are to be chosen and arranged from 6 given digits, none of which is zero

Step 2: the number of arrangements is 6P5 = 6! / (6 - 5)! = 6! / 1!

Step 3: expand: 6 x 5 x 4 x 3 x 2 = 720

Final answer: 720 numbers

Question: How many 5-digit numbers can be formed, without repeating any digit, from (iii) the digits 2, 4, 6, 8, 0?

Answer: Step 1: the first digit cannot be 0, so the first place can be filled by any of the 4 digits 2, 4, 6, 8: 4 ways

Step 2: the remaining 4 places are filled by the remaining 4 digits (including 0) in 4! = 24 ways

Step 3: by the fundamental principle: 4 x 24 = 96

Final answer: 96 numbers

Question: Prove that (i) nPr = (n - 1)Pr + r x (n - 1)P(r - 1)

Answer: Step 1: use nPr = n! / (n - r)! and write the right side with the common denominator (n - r)!: (n - 1)Pr = (n - 1)! / (n - 1 - r)! = (n - 1)!(n - r) / (n - r)! and r x (n - 1)P(r - 1) = r(n - 1)! / (n - r)!

Step 2: add the numerators: (n - 1)![(n - r) + r] / (n - r)!

Step 3: simplify the bracket: (n - r) + r = n, giving n(n - 1)! / (n - r)!

Step 4: n(n - 1)! = n!, so the right side becomes n! / (n - r)! = nPr

Final answer: nPr = (n - 1)Pr + r x (n - 1)P(r - 1), proved

Question: Prove that (ii) nPr = n x (n - 1)P(r - 1)

Answer: Step 1: start from the right side: (n - 1)P(r - 1) = (n - 1)! / ((n - 1) - (r - 1))! = (n - 1)! / (n - r)!

Step 2: multiply by n: n x (n - 1)! / (n - r)!

Step 3: n x (n - 1)! = n!, so the expression becomes n! / (n - r)!

Step 4: by definition n! / (n - r)! = nPr

Final answer: nPr = n x (n - 1)P(r - 1), proved

Question: How many arrangements can be formed using all the letters of the word (i) STATISTICS?

Answer: Step 1: STATISTICS has 10 letters: S appears 3 times, T appears 3 times, I appears 2 times, and A and C appear once each

Step 2: for repeated letters use n! / (r! s! t!): arrangements = 10! / (3! x 3! x 2!)

Step 3: expand: 10! = 3628800, 3! = 6, 2! = 2

Step 4: 3628800 / (6 x 6 x 2) = 3628800 / 72 = 50400

Final answer: 50400 arrangements

Question: How many arrangements can be formed using all the letters of the word (ii) PLANE?

Answer: Step 1: PLANE has 5 letters, all different, and all are used

Step 2: arrangements = 5!

Step 3: expand: 5! = 5 x 4 x 3 x 2 x 1 = 120

Final answer: 120 arrangements

Question: How many arrangements can be formed using all the letters of the word (iii) OBJECT?

Answer: Step 1: OBJECT has 6 letters, all different, and all are used

Step 2: arrangements = 6!

Step 3: expand: 6! = 6 x 5 x 4 x 3 x 2 x 1 = 720

Final answer: 720 arrangements

Question: How many arrangements can be formed using all the letters of the word (iv) FASTENS?

Answer: Step 1: FASTENS has 7 letters, of which S appears 2 times and the rest are all different

Step 2: arrangements = 7! / 2!

Step 3: expand and cancel: (7 x 6 x 5 x 4 x 3 x 2!) / 2! = 7 x 6 x 5 x 4 x 3

Step 4: 7 x 6 = 42, 42 x 5 = 210, 210 x 4 = 840, 840 x 3 = 2520

Final answer: 2520 arrangements

Question: How many arrangements can be formed using all the letters of the word (v) MATHEMATICS?

Answer: Step 1: MATHEMATICS has 11 letters: M appears 2 times, A appears 2 times, T appears 2 times, and H, E, I, C, S appear once each

Step 2: arrangements = 11! / (2! x 2! x 2!)

Step 3: expand: 11! = 39916800 and 2! x 2! x 2! = 8

Step 4: 39916800 / 8 = 4989600

Final answer: 4989600 arrangements

Question: How many arrangements can be formed using all the letters of the word (vi) ASSISTANCE?

Answer: Step 1: ASSISTANCE has 10 letters: S appears 3 times, A appears

2 times, and I, T, N, C, E appear once each

Step 2: arrangements = 10! / (3! x 2!)

Step 3: expand: 10! = 3628800, 3! = 6, 2! = 2

Step 4: 3628800 / (6 x 2) = 3628800 / 12 = 302400

Final answer: 302400 arrangements

Question: How many 8-digit numbers can be formed from the digits 2, 2, 2, 3, 3, 3, 5, 6?

Answer: Step 1: there are 8 digits in total; 2 appears 3 times and 3 appears 3 times, while 5 and 6 appear once each; no digit is 0, so every arrangement is an 8-digit number

Step 2: arrangements = 8! / (3! x 3!)

Step 3: expand: 8! = 40320, 3! = 6

Step 4: 40320 / (6 x 6) = 40320 / 36 = 1120

Final answer: 1120 numbers

Question: Of the 8-digit numbers formed from the digits 2, 2, 2, 3, 3, 3, 5, 6, how many are greater than 60,000,000?

Answer: Step 1: an 8-digit number exceeds 60,000,000 only if its first digit is greater than 6, or equal to 6 with the rest positive; the only available first digit that works is 6, since the next largest digit is 5 and any number starting with 5 is below 60,000,000

Step 2: fix 6 in the first place; the remaining digits are 2, 2, 2, 3, 3, 3, 5

Step 3: these 7 digits can be arranged in 7! / (3! x 3!) ways

Step 4: expand: 7! = 5040, so 5040 / (6 x 6) = 5040 / 36 = 140

Final answer: 140 numbers

Question: How many distinct permutations of the letters of the word "ESSENTIAL" are possible?

Answer: Step 1: ESSENTIAL has 9 letters: S appears 2 times, E appears 2 times, and N, T, I, A, L appear once each

Step 2: permutations = 9! / (2! x 2!)

Step 3: expand: 9! = 362880 and 2! x 2! = 4

Step 4: 362880 / 4 = 90720

Final answer: 90720 permutations

Question: In how many permutations of the word "ESSENTIAL" will the two S's be (i) together?

Answer: Step 1: treat the two S's as a single block (SS); then we have 8 objects: (SS), E, E, N, T, I, A, L, with E repeated twice

Step 2: permutations = 8! / 2!

Step 3: expand: 8! = 40320, so 40320 / 2 = 20160

Step 4: the two S's are identical, so the block has no internal arrangements to add

Final answer: 20160 permutations

Question: In how many permutations of the word "ESSENTIAL" will the two S's be (ii) separate?

Answer: Step 1: count all permutations: 9! / (2! x 2!) = 90720 (from the total count of the word ESSENTIAL)

Step 2: count the permutations in which the S's are together: 8! / 2! = 20160

Step 3: subtract: 90720 - 20160 = 70560

Final answer: 70560 permutations

Question: In how many ways can 8 persons be seated around a round table?

Answer: Step 1: in a circular arrangement only relative positions matter, so fix one person's seat and arrange the rest

Step 2: the number of circular permutations of 8 persons is (8 - 1)! = 7!

Step 3: expand: 7! = 7 x 6 x 5 x 4 x 3 x 2 x 1 = 5040

Final answer: 5040 ways

Question: In how many ways can 7 keys be arranged in a circular key ring?

Answer: Step 1: for a ring, clockwise and anticlockwise arrangements are identical, so the number is (n - 1)! / 2 with n = 7

Step 2: substitute: (7 - 1)! / 2 = 6! / 2

Step 3: expand: 6! = 720, so 720 / 2 = 360

Final answer: 360 ways

Question: How many 3-digit numbers can be formed by using each one of the digits 2, 3, 5, 7, 9 only once?

Answer: Step 1: 3 digits are chosen and arranged from the 5 given digits, each used at most once, and order matters

Step 2: the number is 5P3 = 5! / (5 - 3)! = 5! / 2!

Step 3: expand and cancel: 5 x 4 x 3 = 60

Final answer: 60 numbers

Question: How many 5-digit numbers greater than 23,000 can be formed

from the digits 1, 2, 3, 5, 6 if digits may repeat?

Answer: Step 1: the number has 5 digits and repetition is allowed, so each place normally has 5 choices; we count directly by the first two digits

Step 2: case 1, first digit is 3, 5 or 6 (3 choices): the number is automatically greater than 23,000, and the remaining 4 places are free, giving 3 x 5 x 5 x 5 x 5 = 3 x 625 = 1875 numbers

Step 3: case 2, first digit is 2: the second digit must be 3 or more, i.e. 3, 5 or 6 (3 choices), and the remaining 3 places are free, giving 1 x 3 x 5 x 5 x 5 = 375 numbers

Step 4: case 3, first digit is 1: impossible, since the number would be below 23,000

Step 5: add the cases: 1875 + 375 = 2250

Final answer: 2250 numbers

Question: Find the number of 5-digit numbers that can be formed from the digits 1, 2, 4, 6, 8 (when no digit is repeated), but (i) the digits 2 and 8 are next to each other.

Answer: Step 1: total digits available are 5 and all are used in each number

Step 2: treat 2 and 8 as one block; then there are 4 objects: (2,8), 1, 4, 6, which can be arranged in 4! = 24 ways

Step 3: inside the block, 2 and 8 can be ordered in 2 ways: 28 or 82

Step 4: multiply: 24 x 2 = 48

Final answer: 48 numbers

Question: Find the number of 5-digit numbers that can be formed from the

digits 1, 2, 4, 6, 8 (when no digit is repeated), but (ii) the digits 2 and 8 are not next to each other.

Answer: Step 1: count all 5-digit numbers using all 5 digits without repetition: 5! = 120

Step 2: count those in which 2 and 8 are next to each other: 48 (from part (i))

Step 3: subtract: 120 - 48 = 72

Final answer: 72 numbers

Question: How many 6-digit numbers can be formed without repeating any digit from the digits 0, 1, 2, 3, 4, 5?

Answer: Step 1: the first digit cannot be 0, so it can be filled in 5 ways (1, 2, 3, 4 or 5)

Step 2: the remaining 5 places are filled from the remaining 5 digits (including 0) in 5! = 120 ways

Step 3: by the fundamental principle: 5 x 120 = 600

Final answer: 600 numbers

Question: From the digits 0, 1, 2, 3, 4, 5, how many 6-digit numbers (no digit repeated) will have 0 at the tens place?

Answer: Step 1: fix 0 at the tens place

Step 2: the first digit can be any of the 5 nonzero digits: 5 ways

Step 3: the remaining 4 places (hundreds, thousands, ten-thousands... i.e. the other four positions) are filled by the remaining 4 digits in 4! = 24 ways

Step 4: by the fundamental principle: 5 x 24 = 120

Final answer: 120 numbers

Question: How many 5-digit multiples of 5 can be formed from the digits 2, 3, 5, 7, 9 when digits may repeat?

Answer: Step 1: a multiple of 5 must end in 0 or 5; of the given digits only 5 is available, so the last place has exactly 1 choice

Step 2: each of the first 4 places can be filled by any of the 5 digits, since repetition is allowed: 5 choices each

Step 3: by the fundamental principle: 5 x 5 x 5 x 5 x 1 = 625

Final answer: 625 numbers

Exercise 6.3

Question: Evaluate (i) 12C10

Answer: Step 1: use nCr = n! / (r!(n - r)!), so 12C10 = 12! / (10! x 2!)

Step 2: use the complementary form to make the calculation smaller: 12C10 = 12C2, since 12 - 10 = 2

Step 3: 12C2 = (12 x 11) / 2! = 132 / 2

Step 4: 132 / 2 = 66

Final answer: 12C10 = 66

Question: Evaluate (ii) 12C3

Answer: Step 1: use nCr = n! / (r!(n - r)!), so 12C3 = 12! / (3! x 9!)

Step 2: expand 12! down to 9!: 12! = 12 x 11 x 10 x 9!

Step 3: cancel 9!: 12C3 = (12 x 11 x 10) / 3! = 1320 / 6

Step 4: 1320 / 6 = 220

Final answer: 12C3 = 220

Question: Evaluate (iii) C(20, 17)

Answer: Step 1: use the complementary form: C(20, 17) = C(20, 3), since 20 - 17 = 3

Step 2: C(20, 3) = 20! / (3! x 17!) = (20 x 19 x 18) / 3!

Step 3: expand: 20 x 19 x 18 = 6840 and 3! = 6

Step 4: 6840 / 6 = 1140

Final answer: C(20, 17) = 1140

Question: Evaluate (iv) C(7, 4)

Answer: Step 1: use nCr = n! / (r!(n - r)!), so C(7, 4) = 7! / (4! x 3!)

Step 2: expand 7! down to 4!: 7! = 7 x 6 x 5 x 4!

Step 3: cancel 4!: C(7, 4) = (7 x 6 x 5) / 3! = 210 / 6

Step 4: 210 / 6 = 35

Final answer: C(7, 4) = 35

Question: Find the value of n when (i) nC2 = 21

Answer: Step 1: expand nC2 = n(n - 1) / 2!, so n(n - 1) / 2 = 21

Step 2: multiply by 2: n(n - 1) = 42, i.e. n^2 - n - 42 = 0

Step 3: factorise: (n - 7)(n + 6) = 0, so n = 7 or n = -6

Step 4: n must be positive, so n = -6 is rejected

Final answer: n = 7

Question: Find the value of n when (ii) nC3 = 4

Answer: Step 1: expand nC3 = n(n - 1)(n - 2) / 3!, so n(n - 1)(n - 2) = 4 x 6 = 24

Step 2: look for three consecutive integers with product 24: 4 x 3 x 2 = 24

Step 3: compare: n = 4

Step 4: check: 4C3 = 4, which is correct

Final answer: n = 4

Question: Find the value of n when (iii) nC12 = nC6

Answer: Step 1: if nCa = nCb with a not equal to b, then a + b = n (complementary combinations are equal)

Step 2: here a = 12 and b = 6, and 12 is not equal to 6

Step 3: therefore n = 12 + 6 = 18

Final answer: n = 18

Question: Find the value of n when (iv) nC15 = (17 x 16) / 2!

Answer: Step 1: simplify the right side: (17 x 16) / 2! = 272 / 2 = 136

Step 2: note that 136 = 17C2 = 17C15, because 17C15 = 17! / (15! x 2!) = (17 x 16) / 2!

Step 3: so nC15 = 17C15, which gives n = 17

Final answer: n = 17

Question: Find the value of n when (v) nC5 = nC4

Answer: Step 1: if nCa = nCb with a not equal to b, then a + b = n

Step 2: here a = 5 and b = 4, and 5 is not equal to 4

Step 3: therefore n = 5 + 4 = 9

Final answer: n = 9

Question: Find the value of n when (vi) nC10 = (12 x 11) / 2!

Answer: Step 1: simplify the right side: (12 x 11) / 2! = 132 / 2 = 66

Step 2: note that 66 = 12C2 = 12C10, because 12C10 = 12! / (10! x 2!) = (12 x 11) / 2!

Step 3: so nC10 = 12C10, which gives n = 12

Final answer: n = 12

Question: Find the values of n and r when (i) nPr = 210 and nCr = 35

Answer: Step 1: divide the two given values: nPr / nCr = 210 / 35 = 6

Step 2: but nPr = nCr x r!, so nPr / nCr = r!, giving r! = 6 = 3!, hence r = 3

Step 3: use nC3 = 35: n(n - 1)(n - 2) / 3! = 35, so n(n - 1)(n - 2) = 35 x 6 = 210

Step 4: look for three consecutive integers with product 210: 7 x 6 x 5 = 210, so n = 7

Step 5: check: 7P3 = 210 and 7C3 = 35, both correct

Final answer: n = 7 and r = 3

Question: Find the values of n and r when (ii) (n - 1)C(r - 1) : nCr : (n + 1)C(r + 1) = 3 : 6 : 11

Answer: Step 1: use the ratio relations (n - 1)C(r - 1) / nCr = r / n and (n + 1)C(r + 1) / nCr = (n + 1) / (r + 1)

Step 2: from the given ratio, r / n = 3 / 6 = 1 / 2, so n = 2r

Step 3: also (n + 1) / (r + 1) = 11 / 6, so 6(n + 1) = 11(r + 1), i.e. 6n + 6 = 11r + 11

Step 4: substitute n = 2r: 12r + 6 = 11r + 11, giving r = 5

Step 5: then n = 2 x 5 = 10

Step 6: check: 9C4 = 126, 10C5 = 252, 11C6 = 462, and 126 : 252 : 462 = 3 : 6 : 11, correct

Final answer: n = 10 and r = 5

Question: If (i) 12Cr = 12C(r + 4), then find 6Cr

Answer: Step 1: if nCa = nCb and a is not equal to b, then a + b = n; here r is not equal to r + 4, so r + (r + 4) = 12

Step 2: solve: 2r + 4 = 12, so 2r = 8 and r = 4

Step 3: now find 6C4 = 6! / (4! x 2!) = (6 x 5) / 2 = 30 / 2 = 15

Final answer: r = 4 and 6Cr = 15

Question: If (ii) nC12 = nC10, then find 25Cn

Answer: Step 1: since 12 is not equal to 10, use a + b = n: n = 12 + 10 = 22

Step 2: now find 25C22 = 25C3, using the complementary form (25 - 22 = 3)

Step 3: 25C3 = (25 x 24 x 23) / 3! = 13800 / 6 = 2300

Final answer: n = 22 and 25Cn = 2300

Question: The members of a club are 12 boys and 8 girls. In how many ways can (i) a committee of 5 members be formed?

Answer: Step 1: the committee is formed from all 12 + 8 = 20 members, with no restriction

Step 2: a committee is a selection in which order does not matter, so use combinations: 20C5 = 20! / (5! x 15!)

Step 3: expand and cancel: (20 x 19 x 18 x 17 x 16) / (5 x 4 x 3 x 2 x 1) =

1860480 / 120

Step 4: 1860480 / 120 = 15504

Final answer: 15504 ways

Question: The members of a club are 12 boys and 8 girls. In how many ways can (ii) a committee of 4 boys and 3 girls be formed?

Answer: Step 1: choose the 4 boys from 12 boys: 12C4 = (12 x 11 x 10 x 9) / 4! = 11880 / 24 = 495

Step 2: choose the 3 girls from 8 girls: 8C3 = (8 x 7 x 6) / 3! = 336 / 6 = 56

Step 3: by the product principle, multiply the two independent choices: 495 x 56 = 27720

Final answer: 27720 ways

Question: A box contains 7 red balls and 6 black balls. In how many ways can 5 balls be selected such that exactly 3 are red?

Answer: Step 1: exactly 3 red balls means the other 5 - 3 = 2 balls must be black

Step 2: choose 3 red balls from 7: 7C3 = (7 x 6 x 5) / 3! = 210 / 6 = 35

Step 3: choose 2 black balls from 6: 6C2 = (6 x 5) / 2! = 30 / 2 = 15

Step 4: by the product principle: 35 x 15 = 525

Final answer: 525 ways

Question: How many (a) diagonals can be drawn in a plane by joining the vertices of a polygon having (i) 6 sides?

Answer: Step 1: joining any 2 of the 6 vertices gives 6C2 = (6 x 5) / 2! = 15 line segments

Step 2: of these segments, 6 are the sides of the polygon itself, not diagonals

Step 3: subtract: 15 - 6 = 9

Final answer: 9 diagonals

Question: How many (b) triangles can be drawn in a plane by joining the vertices of a polygon having (i) 6 sides?

Answer: Step 1: a triangle is formed by choosing any 3 vertices of the polygon, and order does not matter

Step 2: number of triangles = 6C3 = (6 x 5 x 4) / 3! = 120 / 6

Step 3: 120 / 6 = 20

Final answer: 20 triangles

Question: How many (a) diagonals can be drawn in a plane by joining the vertices of a polygon having (ii) 8 sides?

Answer: Step 1: joining any 2 of the 8 vertices gives 8C2 = (8 x 7) / 2! = 28 line segments

Step 2: of these, 8 are the sides of the polygon

Step 3: subtract: 28 - 8 = 20

Final answer: 20 diagonals

Question: How many (b) triangles can be drawn in a plane by joining the vertices of a polygon having (ii) 8 sides?

Answer: Step 1: a triangle is formed by choosing any 3 vertices

Step 2: number of triangles = 8C3 = (8 x 7 x 6) / 3! = 336 / 6

Step 3: 336 / 6 = 56

Final answer: 56 triangles

Question: How many (a) diagonals can be drawn in a plane by joining the vertices of a polygon having (iii) 10 sides?

Answer: Step 1: joining any 2 of the 10 vertices gives 10C2 = (10 x 9) / 2! = 45 line segments

Step 2: of these, 10 are the sides of the polygon

Step 3: subtract: 45 - 10 = 35

Final answer: 35 diagonals

Question: How many (b) triangles can be drawn in a plane by joining the vertices of a polygon having (iii) 10 sides?

Answer: Step 1: a triangle is formed by choosing any 3 vertices

Step 2: number of triangles = 10C3 = (10 x 9 x 8) / 3! = 720 / 6

Step 3: 720 / 6 = 120

Final answer: 120 triangles

Question: Show that 9C6 + 9C5 = 10C6

Answer: Step 1: this is Pascal's rule nC(r - 1) + nCr = (n + 1)Cr with n = 9 and r = 6

Step 2: compute the left side: 9C6 = 9C3 = (9 x 8 x 7) / 3! = 504 / 6 = 84, and 9C5 = 9C4 = (9 x 8 x 7 x 6) / 4! = 3024 / 24 = 126

Step 3: add: 84 + 126 = 210

Step 4: compute the right side: 10C6 = 10C4 = (10 x 9 x 8 x 7) / 4! = 5040 / 24 = 210

Step 5: left side = right side = 210

Final answer: 9C6 + 9C5 = 10C6, shown

Question: There are 11 men and 9 women members of a club. How many committees of 8 members can be formed having (i) exactly five men?

Answer: Step 1: exactly 5 men means the committee has 8 - 5 = 3 women

Step 2: choose 5 men from 11: 11C5 = (11 x 10 x 9 x 8 x 7) / 5! = 55440 / 120 = 462

Step 3: choose 3 women from 9: 9C3 = (9 x 8 x 7) / 3! = 504 / 6 = 84

Step 4: by the product principle: 462 x 84 = 38808

Final answer: 38808 committees

Question: There are 11 men and 9 women members of a club. How many committees of 8 members can be formed having (ii) at most five women?

Answer: Step 1: at most 5 women means the committee may have 0, 1, 2, 3, 4 or 5 women, with the remaining members men

Step 2: list the cases (men, women): (8, 0): 11C8 x 9C0 = 165 x 1 = 165; (7, 1): 11C7 x 9C1 = 330 x 9 = 2970; (6, 2): 11C6 x 9C2 = 462 x 36 = 16632; (5, 3): 11C5 x 9C3 = 462 x 84 = 38808; (4, 4): 11C4 x 9C4 = 330 x 126 = 41580; (3, 5): 11C3 x 9C5 = 165 x 126 = 20790

Step 3: these cases are mutually exclusive, so add them by the sum principle: 165 + 2970 + 16632 + 38808 + 41580 + 20790 = 120945

Final answer: 120945 committees

Question: There are 11 men and 9 women members of a club. How many committees of 8 members can be formed having (iii) at least five women?

Answer: Step 1: at least 5 women means the committee may have 5, 6, 7 or 8 women, with the remaining members men

Step 2: list the cases (men, women): (3, 5): 11C3 x 9C5 = 165 x 126 = 20790; (2, 6): 11C2 x 9C6 = 55 x 84 = 4620; (1, 7): 11C1 x 9C7 = 11 x 36 = 396; (0, 8): 11C0 x 9C8 = 1 x 9 = 9

Step 3: add the mutually exclusive cases: 20790 + 4620 + 396 + 9 = 25815

Final answer: 25815 committees

Question: Find the number of combinations of the letters of the word "Question" taken 5 at a time.

Answer: Step 1: the word Question has 8 letters Q, u, e, s, t, i, o, n, all different

Step 2: combinations taken 5 at a time = 8C5 = 8C3 (using the complementary form)

Step 3: 8C3 = (8 x 7 x 6) / 3! = 336 / 6 = 56

Final answer: 56 combinations

Question: How many words can be formed by 3 vowels and 4 consonants out of 5 vowels and 7 consonants?

Answer: Step 1: choose 3 vowels from 5: 5C3 = (5 x 4 x 3) / 3! = 60 / 6 = 10

Step 2: choose 4 consonants from 7: 7C4 = 7C3 = (7 x 6 x 5) / 3! = 210 / 6 = 35

Step 3: the chosen 7 letters can be arranged to form words in 7! = 5040 ways

Step 4: by the product principle: 10 x 35 x 5040 = 350 x 5040 = 1764000

Final answer: 1764000 words

Exercise 6.4

Question: Two fair coins are tossed. Find the probability of getting (i) same faces.

Answer: Step 1: the sample space is S = {HH, HT, TH, TT}, so n(S) = 4

Step 2: the event of same faces is E = {HH, TT}, so n(E) = 2

Step 3: P(E) = n(E) / n(S) = 2 / 4

Step 4: simplify: 2 / 4 = 1 / 2

Final answer: 1/2

Question: Two fair coins are tossed. Find the probability of getting (ii) all heads.

Answer: Step 1: the sample space is S = {HH, HT, TH, TT}, so n(S) = 4

Step 2: the event of all heads is E = {HH}, so n(E) = 1

Step 3: P(E) = n(E) / n(S) = 1 / 4

Final answer: 1/4

Question: Two fair coins are tossed. Find the probability of getting (iii) at most one head.

Answer: Step 1: the sample space is S = {HH, HT, TH, TT}, so n(S) = 4

Step 2: at most one head means 0 or 1 head: E = {HT, TH, TT}, so n(E) = 3

Step 3: P(E) = n(E) / n(S) = 3 / 4

Final answer: 3/4

Question: Two fair coins are tossed. Find the probability of getting (iv) at

least two tails.

Answer: Step 1: the sample space is S = {HH, HT, TH, TT}, so n(S) = 4

Step 2: at least two tails with two coins means both tails: E = {TT}, so n(E) = 1

Step 3: P(E) = n(E) / n(S) = 1 / 4

Final answer: 1/4

Question: Two fair coins are tossed. Find the probability of getting (v) at least one tail.

Answer: Step 1: the sample space is S = {HH, HT, TH, TT}, so n(S) = 4

Step 2: at least one tail: E = {HT, TH, TT}, so n(E) = 3

Step 3: P(E) = n(E) / n(S) = 3 / 4

Final answer: 3/4

Question: Two dice, one red and the other green, are rolled simultaneously. The numbers of dots on the tops are added. Find the probability of getting a sum of (i) 8.

Answer: Step 1: two dice give n(S) = 6 x 6 = 36 equally likely outcomes

Step 2: the pairs giving sum 8 are (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), so n(E) = 5

Step 3: P(E) = n(E) / n(S) = 5 / 36

Final answer: 5/36

Question: Two dice, one red and the other green, are rolled simultaneously. The numbers of dots on the tops are added. Find the probability of getting a sum of (ii) 10.

Answer: Step 1: two dice give n(S) = 36 equally likely outcomes

Step 2: the pairs giving sum 10 are (4, 6), (5, 5), (6, 4), so n(E) = 3

Step 3: P(E) = n(E) / n(S) = 3 / 36

Step 4: simplify: 3 / 36 = 1 / 12

Final answer: 1/12

Question: Two dice, one red and the other green, are rolled simultaneously. The numbers of dots on the tops are added. Find the probability of getting a sum of (iii) 12.

Answer: Step 1: two dice give n(S) = 36 equally likely outcomes

Step 2: the only pair giving sum 12 is (6, 6), so n(E) = 1

Step 3: P(E) = n(E) / n(S) = 1 / 36

Final answer: 1/36

Question: A bag contains 40 balls out of which 5 are green, 15 are red and the remaining are black. A ball is drawn out of the bag. Find the probability of getting (i) the ball is green.

Answer: Step 1: total outcomes n(S) = 40

Step 2: favourable outcomes: green balls = 5, so n(E) = 5

Step 3: P(E) = n(E) / n(S) = 5 / 40

Step 4: simplify: 5 / 40 = 1 / 8

Final answer: 1/8

Question: A bag contains 40 balls out of which 5 are green, 15 are red and the remaining are black. A ball is drawn out of the bag. Find the probability of getting (ii) the ball is black.

Answer: Step 1: black balls = 40 - 5 - 15 = 20

Step 2: total outcomes n(S) = 40 and favourable outcomes n(E) = 20

Step 3: P(E) = n(E) / n(S) = 20 / 40

Step 4: simplify: 20 / 40 = 1 / 2

Final answer: 1/2

Question: A bag contains 40 balls out of which 5 are green, 15 are red and the remaining are black. A ball is drawn out of the bag. Find the probability of getting (iii) the ball is not green.

Answer: Step 1: balls that are not green = 40 - 5 = 35

Step 2: total outcomes n(S) = 40 and favourable outcomes n(E) = 35

Step 3: P(E) = n(E) / n(S) = 35 / 40

Step 4: simplify: 35 / 40 = 7 / 8 (equivalently, 1 - P(green) = 1 - 1/8 = 7/8)

Final answer: 7/8

Question: A card is drawn from a well shuffled deck of cards. Find the probability of (i) getting a King.

Answer: Step 1: total outcomes n(S) = 52 cards

Step 2: there are 4 Kings, so n(E) = 4

Step 3: P(E) = n(E) / n(S) = 4 / 52

Step 4: simplify: 4 / 52 = 1 / 13

Final answer: 1/13

Question: A card is drawn from a well shuffled deck of cards. Find the probability of (ii) getting a club.

Answer: Step 1: total outcomes n(S) = 52 cards

Step 2: there are 13 clubs, so n(E) = 13

Step 3: P(E) = n(E) / n(S) = 13 / 52

Step 4: simplify: 13 / 52 = 1 / 4

Final answer: 1/4

Question: A card is drawn from a well shuffled deck of cards. Find the probability of (iii) getting a face card.

Answer: Step 1: total outcomes n(S) = 52 cards

Step 2: face cards are the Jacks, Queens and Kings: 4 suits x 3 = 12, so n(E) = 12

Step 3: P(E) = n(E) / n(S) = 12 / 52

Step 4: simplify: 12 / 52 = 3 / 13

Final answer: 3/13

Question: A die is rolled. Find the probability using a Venn diagram of getting (i) an even number.

Answer: Step 1: the sample space is S = {1, 2, 3, 4, 5, 6}, so n(S) = 6

Step 2: let A be the event of an even number: A = {2, 4, 6}, so n(A) = 3

Step 3: P(A) = n(A) / n(S) = 3 / 6

Step 4: simplify: 3 / 6 = 1 / 2

Final answer: 1/2

[diagram: A Venn diagram with a rectangle for S = {1, 2, 3, 4, 5, 6}, one circle A holding the even num

Question: A die is rolled. Find the probability using a Venn diagram of getting (ii) a number greater than 4.

Answer: Step 1: the sample space is S = {1, 2, 3, 4, 5, 6}, so n(S) = 6

Step 2: let B be the event of a number greater than 4: B = {5, 6}, so n(B) = 2

Step 3: P(B) = n(B) / n(S) = 2 / 6

Step 4: simplify: 2 / 6 = 1 / 3

Final answer: 1/3

Question: A die is rolled. Find the probability using a Venn diagram of getting (iii) a number which is even and greater than 4.

Answer: Step 1: the sample space is S = {1, 2, 3, 4, 5, 6}, so n(S) = 6

Step 2: with A = {2, 4, 6} (even) and B = {5, 6} (greater than 4), the overlapping part of the Venn diagram is A and B together = {6}, so n = 1

Step 3: P = 1 / 6

Final answer: 1/6

Question: In a three child family, by using a tree diagram, find the probability of having (i) three boys.

Answer: Step 1: each child is a boy (B) or a girl (G), so the sample space has 2 x 2 x 2 = 8 outcomes: BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG

Step 2: the event of three boys is {BBB}, so n(E) = 1

Step 3: P(E) = n(E) / n(S) = 1 / 8

Final answer: 1/8

[diagram: A tree diagram with three levels: the first child branches into B and G, each branch splits aga

Question: In a three child family, by using a tree diagram, find the probability of having (ii) exactly two boys.

Answer: Step 1: the sample space has n(S) = 8 outcomes: BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG

Step 2: the outcomes with exactly two boys are BBG, BGB, GBB, so n(E) = 3

Step 3: P(E) = n(E) / n(S) = 3 / 8

Final answer: 3/8

Question: In a three child family, by using a tree diagram, find the probability of having (iii) at least one girl.

Answer: Step 1: the sample space has n(S) = 8 outcomes

Step 2: use the complement: the only outcome with no girl is BBB, so the outcomes with at least one girl number 8 - 1 = 7

Step 3: P(E) = n(E) / n(S) = 7 / 8

Final answer: 7/8

Question: In a single throw of two fair dice, find the probability that the product of the numbers on the dice is (i) between 2 and 10 (both inclusive).

Answer: Step 1: two dice give n(S) = 36 equally likely outcomes

Step 2: count the pairs (a, b) with 2 <= ab <= 10: for a = 1: b = 2, 3, 4, 5,

6 (5 pairs); a = 2: b = 1, 2, 3, 4, 5 (5 pairs); a = 3: b = 1, 2, 3 (3 pairs); a = 4: b = 1, 2 (2 pairs); a = 5: b = 1, 2 (2 pairs); a = 6: b = 1 (1 pair)

Step 3: total favourable outcomes n(E) = 5 + 5 + 3 + 2 + 2 + 1 = 18

Step 4: P(E) = 18 / 36 = 1 / 2

Final answer: 1/2

Question: In a single throw of two fair dice, find the probability that the product of the numbers on the dice is (ii) divisible by 5.

Answer: Step 1: two dice give n(S) = 36 equally likely outcomes

Step 2: the product is divisible by 5 only when at least one die shows 5: the pairs are (1, 5), (2, 5), (3, 5), (4, 5), (5, 5), (6, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 6), so n(E) = 11

Step 3: P(E) = n(E) / n(S) = 11 / 36

Final answer: 11/36

Question: A marble is drawn at random from a box containing 20 red, 10 white, 25 orange and 15 blue marbles. Find the probability that it is (i) orange or red.

Answer: Step 1: total marbles n(S) = 20 + 10 + 25 + 15 = 70

Step 2: favourable marbles: orange or red = 25 + 20 = 45 (the events are mutually exclusive, so their counts add)

Step 3: P = 45 / 70

Step 4: simplify: 45 / 70 = 9 / 14

Final answer: 9/14

Question: A marble is drawn at random from a box containing 20 red, 10 white, 25 orange and 15 blue marbles. Find the probability that it is (ii) not

blue or red, i.e. neither blue nor red.

Answer: Step 1: total marbles n(S) = 70

Step 2: marbles that are neither blue nor red are the white and orange ones: 10 + 25 = 35

Step 3: P = 35 / 70

Step 4: simplify: 35 / 70 = 1 / 2

Final answer: 1/2

Question: A marble is drawn at random from a box containing 20 red, 10 white, 25 orange and 15 blue marbles. Find the probability that it is (iii) red, white or blue.

Answer: Step 1: total marbles n(S) = 70

Step 2: favourable marbles: red + white + blue = 20 + 10 + 15 = 45

Step 3: P = 45 / 70

Step 4: simplify: 45 / 70 = 9 / 14

Final answer: 9/14

Question: The king, queen and jack of clubs are removed from a deck of 52 playing cards and the rest are shuffled. A card is drawn from the remaining cards. Find the probability of getting (i) a heart.

Answer: Step 1: after removing 3 cards, the remaining cards number n(S) = 52 - 3 = 49

Step 2: no heart was removed, so hearts = 13

Step 3: P = 13 / 49

Final answer: 13/49

Question: The king, queen and jack of clubs are removed from a deck of 52 playing cards and the rest are shuffled. A card is drawn from the remaining cards. Find the probability of getting (ii) a queen.

Answer: Step 1: the remaining cards number n(S) = 49

Step 2: the queen of clubs was removed, so queens left = 4 - 1 = 3

Step 3: P = 3 / 49

Final answer: 3/49

Question: The king, queen and jack of clubs are removed from a deck of 52 playing cards and the rest are shuffled. A card is drawn from the remaining cards. Find the probability of getting (iii) a club.

Answer: Step 1: the remaining cards number n(S) = 49

Step 2: three clubs were removed, so clubs left = 13 - 3 = 10

Step 3: P = 10 / 49

Final answer: 10/49

Question: The king, queen and jack of clubs are removed from a deck of 52 playing cards and the rest are shuffled. A card is drawn from the remaining cards. Find the probability of getting (iv) '9' of red colour.

Answer: Step 1: the remaining cards number n(S) = 49

Step 2: the red 9's are the 9 of hearts and the 9 of diamonds, and neither was removed, so there are 2 favourable cards

Step 3: P = 2 / 49

Final answer: 2/49

Question: A pair of fair dice is thrown. If the two numbers appearing are different, find the probability that (i) the sum is 10.

Answer: Step 1: this is a conditional probability: the reduced sample space is the outcomes where the two numbers differ, which number 36 - 6 = 30 (excluding the 6 doubles)

Step 2: the pairs with sum 10 are (4, 6), (5, 5), (6, 4); of these, (5, 5) has equal numbers, leaving (4, 6) and (6, 4): 2 favourable outcomes

Step 3: P = 2 / 30

Step 4: simplify: 2 / 30 = 1 / 15

Final answer: 1/15

Question: A pair of fair dice is thrown. If the two numbers appearing are different, find the probability that (ii) the sum is six or less.

Answer: Step 1: the reduced sample space (different numbers) has 36 - 6 = 30 outcomes

Step 2: count the pairs with different numbers and sum at most 6: (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1), (4, 2), (5, 1): 12 favourable outcomes (the doubles (1, 1), (2, 2), (3, 3) are excluded)

Step 3: P = 12 / 30

Step 4: simplify: 12 / 30 = 2 / 5

Final answer: 2/5

Question: Given that P(A) = 0.3, P(B) = 0.7 and P(A and B) = 0.21, find (i) P(A/B).

Answer: Step 1: use the conditional probability formula P(A/B) = P(A and B) / P(B)

Step 2: substitute: P(A/B) = 0.21 / 0.7

Step 3: divide: 0.21 / 0.7 = 0.3

Final answer: P(A/B) = 0.3

Question: Given that P(A) = 0.3, P(B) = 0.7 and P(A and B) = 0.21, find (ii) P(B/A).

Answer: Step 1: use the conditional probability formula P(B/A) = P(A and B) / P(A)

Step 2: substitute: P(B/A) = 0.21 / 0.3

Step 3: divide: 0.21 / 0.3 = 0.7

Final answer: P(B/A) = 0.7

Question: If one card is selected at random from a deck of 52 playing cards, what is the probability that the card is a club or a face card or both?

Answer: Step 1: let A = the card is a club, P(A) = 13 / 52, and B = the card is a face card, P(B) = 12 / 52

Step 2: the cards that are both a club and a face card are the jack, queen and king of clubs: 3 cards, so P(A and B) = 3 / 52

Step 3: by the addition theorem: P(A or B) = P(A) + P(B) - P(A and B) = 13/52 + 12/52 - 3/52

Step 4: = 22 / 52 = 11 / 26

Final answer: 11/26

Question: For two events A and B, P(A) = 0.5, P(B) = 0.2 and P(A or B) = 0.4. Find P(A and B).

Answer: Step 1: use the addition theorem: P(A or B) = P(A) + P(B) - P(A and B)

Step 2: substitute the given values: 0.4 = 0.5 + 0.2 - P(A and B)

Step 3: rearrange: P(A and B) = 0.5 + 0.2 - 0.4

Step 4: = 0.7 - 0.4 = 0.3

Final answer: P(A and B) = 0.3

Question: A natural number is chosen out of the first 35 natural numbers. What is the probability that the chosen number is divisible by 8 or 9?

Answer: Step 1: total outcomes n(S) = 35

Step 2: numbers divisible by 8: 8, 16, 24, 32: 4 numbers, so P(A) = 4 / 35

Step 3: numbers divisible by 9: 9, 18, 27: 3 numbers, so P(B) = 3 / 35

Step 4: a number divisible by both 8 and 9 must be divisible by 72, and there is none within the first 35, so A and B are mutually exclusive and P(A and B) = 0

Step 5: by the addition theorem for mutually exclusive events: P(A or B) = 4/35 + 3/35 = 7 / 35 = 1 / 5

Final answer: 1/5

Question: A bag contains 15 black, 25 red and 10 white balls. A ball is drawn at random. Find the probability that it is either red or white.

Answer: Step 1: total balls n(S) = 15 + 25 + 10 = 50

Step 2: favourable balls: red or white = 25 + 10 = 35 (mutually exclusive colours, so the counts add)

Step 3: P = 35 / 50

Step 4: simplify: 35 / 50 = 7 / 10

Final answer: 7/10

Question: Two events A and B are such that P(A) = 1/4, P(A/B) = 1/2 and P(B/A) = 2/3. Find P(A and B) and P(B).

Answer: Step 1: use P(B/A) = P(A and B) / P(A), so P(A and B) = P(B/A) x

P(A) = (2/3) x (1/4) = 2 / 12 = 1 / 6

Step 2: use P(A/B) = P(A and B) / P(B), so P(B) = P(A and B) / P(A/B) = (1/6) / (1/2)

Step 3: divide: (1/6) x (2/1) = 2 / 6 = 1 / 3

Final answer: P(A and B) = 1/6 and P(B) = 1/3

Question: A fair die is thrown twice. Find the probability that an even number of dots appear in the first throw and the number of dots in the second throw is less than 4.

Answer: Step 1: let A = even number on the first throw = {2, 4, 6}, so P(A) = 3 / 6 = 1 / 2

Step 2: let B = number less than 4 on the second throw = {1, 2, 3}, so P(B) = 3 / 6 = 1 / 2

Step 3: the two throws are independent, so by the multiplication theorem for independent events: P(A and B) = P(A) x P(B) = (1/2) x (1/2)

Step 4: = 1 / 4 (equivalently, 9 favourable pairs out of 36)

Final answer: 1/4

Question: Three missiles are fired at a target. If the probabilities of hitting the target are 0.5, 0.3 and 0.6 respectively, and the missiles are fired independently, what is the probability that all the missiles hit the target?

Answer: Step 1: the three missiles are fired independently, so the probability that all three events happen together is the product of their probabilities (multiplication theorem for independent events)

Step 2: substitute: P(all hit) = 0.5 x 0.3 x 0.6

Step 3: multiply step by step: 0.5 x 0.3 = 0.15, and 0.15 x 0.6 = 0.09

Final answer: 0.09

Review Exercise 6

Question: Select the correct answer: (i) If n = 0, then (n + 1)! / n! = ____ (a) 0 (b) 1 (c) n (d) infinity

Answer: Step 1: substitute n = 0: (0 + 1)! / 0! = 1! / 0!

Step 2: use 1! = 1 and the definition 0! = 1

Step 3: the value is 1 / 1 = 1

Final answer: (b) 1

Question: Select the correct answer: (ii) The probability of getting 7 in throwing a dice is: (a) 0 (b) 1 (c) -1 (d) not defined

Answer: Step 1: a die has faces 1 to 6 only, so getting 7 is an impossible event

Step 2: the number of favourable outcomes is 0, so P = 0 / 6 = 0

Final answer: (a) 0

Question: Select the correct answer: (iii) The factorial form of 12 x 11 x 10 is: (a) 12! / 9! (b) 12! (c) C(12, 9)! (d) (12!) x (9!)

Answer: Step 1: complete the product down to 1: 12 x 11 x 10 = (12 x 11 x 10 x 9 x ... x 1) / (9 x ... x 1)

Step 2: the numerator is 12! and the denominator is 9!

Final answer: (a) 12! / 9!

Question: Select the correct answer: (iv) If two independent events A and B occur in p and q ways respectively, then the number of ways that both

events can occur is: (a) p + q ways (b) p.q ways (c) (pq)^r ways (d) rp + qr ways

Answer: Step 1: by the fundamental principle of counting, when two independent operations are performed together, their numbers of ways are multiplied

Step 2: total ways = p x q

Final answer: (b) p.q ways

Question: Select the correct answer: (v) An arrangement of n objects according to some definite order is called: (a) combination (b) permutation (c) factorial (d) none of these

Answer: Step 1: an arrangement in which order matters and follows a definite order is by definition a permutation

Final answer: (b) Permutation

Question: Select the correct answer: (vi) An arrangement of n objects without any order is called: (a) combination (b) permutation (c) factorial (d) none of these

Answer: Step 1: a selection in which order does not matter is by definition a combination

Final answer: (a) Combination

Question: Select the correct answer: (vii) The number of permutations of the letters of the word COMMITTEE is: (a) (9; 2, 2, 2) (b) (6; 1, 2, 2) (c) (9; 2, 2, 1) (d) (9; 2, 3, 1)

Answer: Step 1: COMMITTEE has 9 letters

Step 2: count the repetitions: M appears 2 times, T appears 2 times and E appears 2 times; C, O and I appear once each

Step 3: the multinomial symbol is therefore (9; 2, 2, 2), i.e. 9! / (2! x 2! x 2!)

Final answer: (a) (9; 2, 2, 2)

Question: Select the correct answer: (viii) 8 x 7 x 6 is equal to: (a) 8P3 (b) 8C3 (c) 8P5 (d) 8C5

Answer: Step 1: 8 x 7 x 6 is a product of 3 descending factors starting from 8

Step 2: this is exactly nPr with n = 8 and r = 3, since 8P3 = 8! / 5! = 8 x 7 x 6 = 336

Final answer: (a) 8P3

Question: Select the correct answer: (ix) If r = n, then nPr is equal to: (a) r! (b) (n - r)! (c) 1 (d) 0

Answer: Step 1: put r = n in nPr = n! / (n - r)!: nPn = n! / 0! = n!

Step 2: since r = n, n! is the same as r!

Final answer: (a) r!

Question: Select the correct answer: (x) The number of ways that a necklace of n beads of different colours can be made is: (a) (n - 1)! (b) n! / 2 (c) (n! - 1) / 2 (d) (n - 1)! / 2

Answer: Step 1: beads on a necklace form a circular permutation, giving (n - 1)! arrangements

Step 2: a necklace can be turned over, so clockwise and anticlockwise orders are identical and the count is halved

Step 3: number of ways = (n - 1)! / 2

Final answer: (d) (n - 1)! / 2

Question: Select the correct answer: (xi) Any subset of a sample space is called: (a) sample space (b) an event (c) a trial (d) random variable

Answer: Step 1: by definition, any subset of the sample space S is called an event

Final answer: (b) an event

Question: Select the correct answer: (xii) For two events A and B, if A and B have empty intersection, then events A and B are called: (a) mutually exclusive (b) not mutually exclusive (c) overlapping (d) dependent events

Answer: Step 1: two events that cannot occur together, i.e. have no common outcome, are mutually exclusive

Final answer: (a) Mutually exclusive

Question: Select the correct answer: (xiii) When a dice is rolled and a coin is tossed, all possible outcomes are: (a) 6 (b) 12 (c) 18 (d) 24

Answer: Step 1: the dice has 6 outcomes and the coin has 2 outcomes

Step 2: by the fundamental principle: 6 x 2 = 12

Final answer: (b) 12

Question: Select the correct answer: (xiv) If two events A and B have equal chance of occurrence, then the events are: (a) equally likely (b) not equally likely (c) dependent (d) not mutually exclusive

Answer: Step 1: events with the same chance of occurrence are by definition equally likely events

Final answer: (a) Equally likely

Question: Select the correct answer: (xv) If E is an event of a sample space S, then: (a) $P(E) = n(S) / n(E)$ (b) $0 <= P(E) <= 1$ (c) $0 < P(E) < 1$ (d) all of these

Answer: Step 1: probability is defined as P(E) = n(E) / n(S), and since n(E) lies between 0 and n(S), P(E) always lies between 0 and 1 inclusive

Step 2: option (a) has the ratio inverted and option (c) excludes the end values 0 and 1, so both are wrong

Final answer: (b) 0 <= P(E) <= 1

Question: Select the correct answer: (xvi) The probability of getting the tail in a single toss of a coin is: (a) 1/3 (b) 2/3 (c) 1/2 (d) 1/4

Answer: Step 1: the sample space is {H, T}, so n(S) = 2

Step 2: the favourable outcome is {T}, so n(E) = 1

Step 3: P = n(E) / n(S) = 1 / 2

Final answer: (c) 1/2

Question: Select the correct answer: (xvii) Three dice are rolled simultaneously, then n(S) is equal to: (a) 36 (b) 18 (c) 216 (d) 6

Answer: Step 1: each die has 6 outcomes

Step 2: by the fundamental principle: n(S) = 6 x 6 x 6 = 216

Final answer: (c) 216

Question: Select the correct answer: (xviii) Two teams A and B are playing a match; the probability that team A does not lose is: (a) 1/3 (b) 2/3 (c) 1/2 (d) 0

Answer: Step 1: the equally likely results of the match are: A wins, draw, A loses, so n(S) = 3

Step 2: A does not lose in 2 of these cases: A wins or the match is drawn

Step 3: P = 2 / 3

Final answer: (b) 2/3

Question: Select the correct answer: (xix) If nC6 = nC12, then n equals: (a) 18 (b) 12 (c) 6 (d) 20

Answer: Step 1: if nCa = nCb with a not equal to b, then a + b = n

Step 2: here n = 6 + 12 = 18

Final answer: (a) 18

Question: Evaluate (i) (6C3 x 4C3) / 10C4

Answer: Step 1: compute each combination: 6C3 = (6 x 5 x 4) / 3! = 120 / 6 = 20

Step 2: 4C3 = 4C1 = 4

Step 3: 10C4 = (10 x 9 x 8 x 7) / 4! = 5040 / 24 = 210

Step 4: substitute: (20 x 4) / 210 = 80 / 210

Step 5: simplify by dividing by 10: 80 / 210 = 8 / 21

Final answer: 8/21

Question: Evaluate (ii) (6P3 x 4P3) / 10P4

Answer: Step 1: compute each permutation: 6P3 = 6 x 5 x 4 = 120

Step 2: 4P3 = 4 x 3 x 2 = 24

Step 3: 10P4 = 10 x 9 x 8 x 7 = 5040

Step 4: substitute: (120 x 24) / 5040 = 2880 / 5040

Step 5: simplify by dividing by 720: 2880 / 5040 = 4 / 7

Final answer: 4/7

Question: Evaluate (iii) the sum of 3Cr for r = 1 to 3, i.e. 3C1 + 3C2 + 3C3

Answer: Step 1: compute each term: 3C1 = 3

Step 2: 3C2 = 3C1 = 3

Step 3: 3C3 = 1

Step 4: add: 3 + 3 + 1 = 7

Final answer: 7

Question: There are seven seats available in a compartment. In how many ways can seven persons be seated?

Answer: Step 1: seating 7 different persons in 7 seats is an arrangement in which order matters, i.e. a permutation of 7 persons

Step 2: number of ways = 7!

Step 3: expand: 7! = 7 x 6 x 5 x 4 x 3 x 2 x 1 = 5040

Final answer: 5040 ways

Question: A room has 3 lamps. From a collection of 10 light bulbs of which 6 are no good, a person selects 3 at random and puts them in the sockets. What is the probability that he will have light?

Answer: Step 1: good bulbs = 10 - 6 = 4; the room will have light if at least one of the 3 selected bulbs is good

Step 2: use the complement: total ways of selecting 3 bulbs from 10 = 10C3 = (10 x 9 x 8) / 3! = 720 / 6 = 120

Step 3: ways of selecting 3 bulbs that are all no good = 6C3 = (6 x 5 x 4) / 3! = 120 / 6 = 20, so P(no light) = 20 / 120 = 1 / 6

Step 4: P(light) = 1 - 1/6 = 5 / 6

Final answer: 5/6

Question: If two dice are thrown simultaneously, what is the probability of obtaining a sum of 7 or a sum of 11?

Answer: Step 1: two dice give n(S) = 36 outcomes

Step 2: pairs with sum 7: (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1): 6 outcomes, so P(sum 7) = 6 / 36

Step 3: pairs with sum 11: (5, 6), (6, 5): 2 outcomes, so P(sum 11) = 2 / 36

Step 4: the two events are mutually exclusive, so add: P = (6 + 2) / 36 = 8 / 36

Step 5: simplify: 8 / 36 = 2 / 9

Final answer: 2/9

Question: A bag contains 8 white, 10 black and 12 red balls. Three balls are drawn from the bag, each ball being replaced before the next draw. What is the probability that the first ball is white, the second ball is black and the third ball is red?

Answer: Step 1: total balls = 8 + 10 + 12 = 30; because each ball is replaced, the draws are independent and the total stays 30

Step 2: P(first white) = 8 / 30

Step 3: P(second black) = 10 / 30

Step 4: P(third red) = 12 / 30

Step 5: by the multiplication theorem for independent events: P = (8/30) x (10/30) x (12/30) = 960 / 27000

Step 6: simplify by dividing by 120: 960 / 27000 = 8 / 225

Final answer: 8/225

Question: In how many ways can a football team of 11 players be selected out of 15 players?

Answer: Step 1: selecting a team is a combination, since order does not matter

Step 2: number of ways = 15C11 = 15C4 (complementary form)

Step 3: 15C4 = (15 x 14 x 13 x 12) / 4! = 32760 / 24 = 1365

Final answer: 1365 ways

Question: How many of the football teams of 11 players selected out of 15 players will include a particular player?

Answer: Step 1: fix the particular player in the team

Step 2: the remaining 10 players are chosen from the other 14 players: 14C10 = 14C4

Step 3: 14C4 = (14 x 13 x 12 x 11) / 4! = 24024 / 24 = 1001

Final answer: 1001 teams

Unit 7: Mathematical Induction and Binomial Theorem

Exercise 7.1

Question: Prove by mathematical induction, for every positive integer n: (i) 1 + 3 + 5 + ... + (2n - 1) = n^2

Answer: Step 1 (statement): let P(n) be the statement 1 + 3 + 5 + ... + (2n - 1) = n^2

Step 2 (base case, n = 1): LHS = 1 and RHS = 1^2 = 1, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true for some positive integer k, i.e. 1 + 3 + 5 + ... + (2k - 1) = k^2

Step 4 (inductive step): add the (k + 1)th term (2(k + 1) - 1) = (2k + 1) to both sides: 1 + 3 + ... + (2k - 1) + (2k + 1) = k^2 + (2k + 1)

Step 5: simplify the right side: k^2 + 2k + 1 = (k + 1)^2, which is exactly P(k + 1)

Step 6 (conclusion): therefore P(k) true implies P(k + 1) true, so by the principle of mathematical induction P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (ii) 3 + 6 + 9 + ... + 3n = (3/2)n(n + 1)

Answer: Step 1 (statement): let P(n) be the statement 3 + 6 + 9 + ... + 3n = (3/2)n(n + 1)

Step 2 (base case, n = 1): LHS = 3 and RHS = (3/2)(1)(2) = 3, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: 3 + 6 + ... + 3k = (3/2)k(k + 1)

Step 4 (inductive step): add the (k + 1)th term 3(k + 1) to both sides: 3 + 6 + ... + 3k + 3(k + 1) = (3/2)k(k + 1) + 3(k + 1)

Step 5: take (3(k + 1)) common on the right: 3(k + 1)[k/2 + 1] = 3(k + 1)(k + 2)/2 = (3/2)(k + 1)(k + 2), which is P(k + 1)

Step 6 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (iii) 1 + 4 + 7 + ... + (3n - 2) = n(3n - 1)/2

Answer: Step 1 (statement): let P(n) be the statement 1 + 4 + 7 + ... + (3n - 2) = n(3n - 1)/2

Step 2 (base case, n = 1): LHS = 1 and RHS = 1(3 - 1)/2 = 1, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: 1 + 4 + ... + (3k - 2) = k(3k - 1)/2

Step 4 (inductive step): add the (k + 1)th term (3(k + 1) - 2) = (3k + 1) to both sides: LHS sum up to (3k + 1) = k(3k - 1)/2 + (3k + 1)

Step 5: simplify: [k(3k - 1) + 2(3k + 1)] / 2 = (3k^2 - k + 6k + 2)/2 =

(3k^2 + 5k + 2)/2 = (k + 1)(3k + 2)/2

Step 6: note (k + 1)(3k + 2)/2 = (k + 1)(3(k + 1) - 1)/2, which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n:

(iv) 1 + 1/2 + 1/4 + ... + 1/2^(n - 1) = 2(1 - 1/2^n)

Answer: Step 1 (statement): let P(n) be the given statement

Step 2 (base case, n = 1): LHS = 1 and RHS = 2(1 - 1/2) = 2(1/2) = 1, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: 1 + 1/2 + ... + 1/2^(k - 1) = 2(1 - 1/2^k)

Step 4 (inductive step): add the (k + 1)th term 1/2^k to both sides: sum = 2(1 - 1/2^k) + 1/2^k = 2 - 2/2^k + 1/2^k

Step 5: since 2/2^k = 1/2^(k - 1), we get 2 - 1/2^(k - 1) + 1/2^k; combine the last two terms over the denominator 2^k: 2 - (2 - 1)/2^k = 2 - 1/2^k

Step 6: and 2(1 - 1/2^(k + 1)) = 2 - 2/2^(k + 1) = 2 - 1/2^k, the same value, so P(k + 1) is true

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n:

(v) 2 + 6 + 18 + ... + 2 x 3^(n - 1) = 3^n - 1

Answer: Step 1 (statement): let P(n) be the given statement

Step 2 (base case, n = 1): LHS = 2 and RHS = 3^1 - 1 = 2, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: 2 + 6 + ... + 2 x 3^(k - 1) = 3^k - 1

Step 4 (inductive step): add the (k + 1)th term 2 x 3^k to both sides: sum = (3^k - 1) + 2 x 3^k

Step 5: simplify: 3^k + 2 x 3^k - 1 = 3 x 3^k - 1 = 3^(k + 1) - 1, which is P(k + 1)

Step 6 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (vi) 2 + 6 + 12 + ... + n(n + 1) = (1/3)n(n + 1)(n + 2)

Answer: Step 1 (statement): let P(n) be the given statement (the general term is j(j + 1))

Step 2 (base case, n = 1): LHS = 1 x 2 = 2 and RHS = (1/3)(1)(2)(3) = 2, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: 2 + 6 + ... + k(k + 1) = (1/3)k(k + 1)(k + 2)

Step 4 (inductive step): add the (k + 1)th term (k + 1)(k + 2) to both sides: sum = (1/3)k(k + 1)(k + 2) + (k + 1)(k + 2)

Step 5: take (k + 1)(k + 2) common: (k + 1)(k + 2)[k/3 + 1] = (k + 1)(k + 2)(k + 3)/3 = (1/3)(k + 1)(k + 2)(k + 3), which is P(k + 1)

Step 6 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n:

(vii) 2^2 + 4^2 + 6^2 + ... + (2n)^2 = (2/3)n(n + 1)(2n + 1)

Answer: Step 1 (statement): let P(n) be the given statement

Step 2 (base case, n = 1): LHS = 2^2 = 4 and RHS = (2/3)(1)(2)(3) = 4, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: 2^2 + 4^2 + ... + (2k)^2 = (2/3)k(k + 1)(2k + 1)

Step 4 (inductive step): add the (k + 1)th term (2(k + 1))^2 = 4(k + 1)^2 to both sides: sum = (2/3)k(k + 1)(2k + 1) + 4(k + 1)^2

Step 5: take (k + 1) common: (k + 1)[(2/3)k(2k + 1) + 4(k + 1)] = (k + 1)[(4k^2 + 2k)/3 + (12k + 12)/3] = (k + 1)(4k^2 + 14k + 12)/3

Step 6: factorise: 4k^2 + 14k + 12 = 2(k + 2)(2k + 3), so the sum = (2/3)(k + 1)(k + 2)(2k + 3) = (2/3)(k + 1)(k + 2)(2(k + 1) + 1), which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n:

(viii) 1 x 3 + 2 x 4 + 3 x 5 + ... + n(n + 2) = (1/6)n(n + 1)(2n + 7)

Answer: Step 1 (statement): let P(n) be the given statement

Step 2 (base case, n = 1): LHS = 1 x 3 = 3 and RHS = (1/6)(1)(2)(9) = 3, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: 1 x 3 + ... + k(k + 2) =

(1/6)k(k + 1)(2k + 7)

Step 4 (inductive step): add the (k + 1)th term (k + 1)(k + 3) to both sides: sum = (1/6)k(k + 1)(2k + 7) + (k + 1)(k + 3)

Step 5: take (k + 1) common and use denominator 6: (k + 1)[k(2k + 7) + 6(k + 3)] / 6 = (k + 1)(2k^2 + 7k + 6k + 18)/6 = (k + 1)(2k^2 + 13k + 18)/6

Step 6: factorise: 2k^2 + 13k + 18 = (k + 2)(2k + 9), so the sum = (1/6)(k + 1)(k + 2)(2k + 9) = (1/6)(k + 1)(k + 2)(2(k + 1) + 7), which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (ix) 1/(1 x 2) + 1/(2 x 3) + 1/(3 x 4) + ... + 1/(n(n + 1)) = n/(n + 1)

Answer: Step 1 (statement): let P(n) be the given statement

Step 2 (base case, n = 1): LHS = 1/(1 x 2) = 1/2 and RHS = 1/(1 + 1) = 1/2, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: sum up to 1/(k(k + 1)) = k/(k + 1)

Step 4 (inductive step): add the (k + 1)th term 1/((k + 1)(k + 2)) to both sides: sum = k/(k + 1) + 1/((k + 1)(k + 2))

Step 5: take the common denominator (k + 1)(k + 2): [k(k + 2) + 1] / ((k + 1)(k + 2)) = (k^2 + 2k + 1) / ((k + 1)(k + 2))

Step 6: factorise the numerator: (k + 1)^2 / ((k + 1)(k + 2)) = (k + 1)/(k + 2), which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (x) 1 x 2 + 2 x 2^2 + 3 x 2^3 + 4 x 2^4 + ... + n x 2^n = (n - 1) x 2^(n + 1) + 2

Answer: Step 1 (statement): let P(n) be the given statement (the general term is j x 2^j)

Step 2 (base case, n = 1): LHS = 1 x 2 = 2 and RHS = (1 - 1) x 2^2 + 2 = 2, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: sum up to k x 2^k = (k - 1) x 2^(k + 1) + 2

Step 4 (inductive step): add the (k + 1)th term (k + 1) x 2^(k + 1) to both sides: sum = (k - 1) x 2^(k + 1) + 2 + (k + 1) x 2^(k + 1)

Step 5: combine the powers of 2: [(k - 1) + (k + 1)] x 2^(k + 1) + 2 = 2k x 2^(k + 1) + 2

Step 6: rewrite: 2k x 2^(k + 1) = k x 2^(k + 2) = ((k + 1) - 1) x 2^((k + 1) + 1), so the sum = ((k + 1) - 1) x 2^((k + 1) + 1) + 2, which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (xi) 1! x 1 + 2! x 2 + 3! x 3 + ... + n! x n = (n + 1)! - 1

Answer: Step 1 (statement): let P(n) be the given statement

Step 2 (base case, n = 1): LHS = 1! x 1 = 1 and RHS = 2! - 1 = 1, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: sum up to k! x k = (k + 1)! - 1

Step 4 (inductive step): add the (k + 1)th term (k + 1)! x (k + 1) to both sides: sum = (k + 1)! - 1 + (k + 1)(k + 1)!

Step 5: take (k + 1)! common: (k + 1)![1 + (k + 1)] - 1 = (k + 1)!(k + 2) - 1

Step 6: (k + 1)!(k + 2) = (k + 2)!, so the sum = (k + 2)! - 1 = ((k + 1) + 1)! - 1, which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (xii) 1/(a(a + 1)) + 1/((a + 1)(a + 2)) + ... + 1/((a + n - 1)(a + n)) = n/(a(a + n))

Answer: Step 1 (statement): let P(n) be the given statement

Step 2 (base case, n = 1): LHS = 1/(a(a + 1)) and RHS = 1/(a(a + 1)), so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: sum up to 1/((a + k - 1)(a + k)) = k/(a(a + k))

Step 4 (inductive step): add the (k + 1)th term 1/((a + k)(a + k + 1)) to both sides: sum = k/(a(a + k)) + 1/((a + k)(a + k + 1))

Step 5: take the common denominator a(a + k)(a + k + 1): [k(a + k + 1) +

a] / (a(a + k)(a + k + 1))

Step 6: expand the numerator: ka + k^2 + k + a = k(a + k) + (k + a)... grouping as (k + 1)(a + k): indeed (k + 1)(a + k) = ka + k^2 + a + k, the same expression

Step 7: cancel (a + k): the sum = (k + 1) / (a(a + k + 1)), which is P(k + 1)

Step 8 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (xiii) 1^2/(1 x 3) + 2^2/(3 x 5) + ... + n^2/((2n - 1)(2n + 1)) = n(n + 1)/(2(2n + 1))

Answer: Step 1 (statement): let P(n) be the given statement

Step 2 (base case, n = 1): LHS = 1/(1 x 3) = 1/3 and RHS = 1 x 2 / (2 x 3) = 1/3, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: sum up to k^2/((2k - 1)(2k + 1)) = k(k + 1)/(2(2k + 1))

Step 4 (inductive step): add the (k + 1)th term (k + 1)^2/((2k + 1)(2k + 3)) to both sides: sum = k(k + 1)/(2(2k + 1)) + (k + 1)^2/((2k + 1)(2k + 3))

Step 5: take (k + 1)/(2k + 1) common: (k + 1)/(2k + 1) x [k/2 + (k + 1)/(2k + 3)] = (k + 1)/(2k + 1) x [k(2k + 3) + 2(k + 1)] / (2(2k + 3))

Step 6: simplify the bracket numerator: 2k^2 + 3k + 2k + 2 = 2k^2 + 5k + 2 = (2k + 1)(k + 2)

Step 7: cancel (2k + 1): the sum = (k + 1)(k + 2) / (2(2k + 3)) = (k + 1)(k

+ 2) / (2(2(k + 1) + 1)), which is P(k + 1)

Step 8 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (xiv)(i) a + ar + ar^2 + ... + ar^(n - 1) = a(1 - r^n)/(1 - r), where r is not equal to 1

Answer: Step 1 (statement): let P(n) be the geometric series formula

Step 2 (base case, n = 1): LHS = a and RHS = a(1 - r)/(1 - r) = a, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: a + ar + ... + ar^(k - 1) = a(1 - r^k)/(1 - r)

Step 4 (inductive step): add the (k + 1)th term ar^k to both sides: sum = a(1 - r^k)/(1 - r) + ar^k

Step 5: write over the common denominator (1 - r): a[(1 - r^k) + r^k(1 - r)] / (1 - r) = a[1 - r^k + r^k - r^(k + 1)] / (1 - r)

Step 6: the terms -r^k and +r^k cancel, giving a(1 - r^(k + 1))/(1 - r), which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, the formula is true for every positive integer n

Final answer: proved

Question: Prove by mathematical induction, for every positive integer n: (xiv)(ii) C(n, 0) + C(n, 1) + ... + C(n, n) = 2^n

Answer: Step 1 (statement): let P(n) be the statement that the sum of all

binomial coefficients C(n, 0) + C(n, 1) + ... + C(n, n) equals 2^n

Step 2 (base case, n = 1): LHS = C(1, 0) + C(1, 1) = 1 + 1 = 2 and RHS = 2^1 = 2, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: C(k, 0) + C(k, 1) + ... + C(k, k) = 2^k

Step 4 (inductive step): use Pascal's rule C(k + 1, r) = C(k, r - 1) + C(k, r); then the sum for n = k + 1 splits into two sums: [C(k, 0) + C(k, 1) + ... + C(k, k)] + [C(k, 0) + C(k, 1) + ... + C(k, k)], because each interior coefficient is formed once as C(k, r) and once as C(k, r - 1), while the end terms C(k + 1, 0) = C(k, 0) = 1 and C(k + 1, k + 1) = C(k, k) = 1 each appear once in total across the two sums

Step 5: by the hypothesis each of the two sums equals 2^k, so the total is 2^k + 2^k = 2 x 2^k = 2^(k + 1), which is P(k + 1)

Step 6 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Prove, for n >= r: (xiv)(iii) C(n, r - 1) + C(n, r) = C(n + 1, r) (Pascal's rule)

Answer: Step 1: write both terms on the left in factorial form: C(n, r - 1) = n! / ((r - 1)!(n - r + 1)!) and C(n, r) = n! / (r!(n - r)!)

Step 2: express both over the common denominator r!(n - r + 1)!: the first term is multiplied above and below by r, giving r x n! / (r!(n - r + 1)!), and the second term is multiplied above and below by (n - r + 1), giving (n - r + 1) x n! / (r!(n - r + 1)!)

Step 3: add the numerators: [r + (n - r + 1)] x n! / (r!(n - r + 1)!) = (n + 1)

x n! / (r!(n - r + 1)!)

Step 4: (n + 1) x n! = (n + 1)!, and (n - r + 1)! = ((n + 1) - r)!, so the sum = (n + 1)! / (r!((n + 1) - r)!)

Step 5: by the definition of combinations this is C(n + 1, r)

Final answer: C(n, r - 1) + C(n, r) = C(n + 1, r), proved

Question: Prove by mathematical induction that (i) 2^(3n + 2) - 28n - 4 is divisible by 49 for all n in N

Answer: Step 1 (statement): let P(n) be the statement that f(n) = 2^(3n + 2) - 28n - 4 is divisible by 49

Step 2 (base case, n = 1): f(1) = 2^5 - 28 - 4 = 32 - 32 = 0, and 0 is divisible by 49, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true, i.e. f(k) = 2^(3k + 2) - 28k - 4 is divisible by 49

Step 4 (inductive step): form f(k + 1) = 2^(3k + 5) - 28(k + 1) - 4 = 8 x 2^(3k + 2) - 28k - 32

Step 5: from the hypothesis, 2^(3k + 2) = f(k) + 28k + 4; substitute: f(k + 1) = 8[f(k) + 28k + 4] - 28k - 32 = 8f(k) + 224k + 32 - 28k - 32

Step 6: simplify: f(k + 1) = 8f(k) + 196k = 8f(k) + 49 x 4k; the first term is divisible by 49 by the hypothesis and 49 x 4k is a multiple of 49, so f(k + 1) is divisible by 49

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for all n in N

Final answer: proved

Question: Prove by mathematical induction that (ii) 3^(2n + 2) - 8n - 9 is

divisible by 64 for all n in N

Answer: Step 1 (statement): let P(n) be the statement that f(n) = 3^(2n + 2) - 8n - 9 is divisible by 64

Step 2 (base case, n = 1): f(1) = 3^4 - 8 - 9 = 81 - 17 = 64, which is divisible by 64, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true, i.e. f(k) = 3^(2k + 2) - 8k - 9 is divisible by 64

Step 4 (inductive step): form f(k + 1) = 3^(2k + 4) - 8(k + 1) - 9 = 9 x 3^(2k + 2) - 8k - 17

Step 5: from the hypothesis, 3^(2k + 2) = f(k) + 8k + 9; substitute: f(k + 1) = 9[f(k) + 8k + 9] - 8k - 17 = 9f(k) + 72k + 81 - 8k - 17

Step 6: simplify: f(k + 1) = 9f(k) + 64k + 64 = 9f(k) + 64(k + 1); both terms are divisible by 64, so f(k + 1) is divisible by 64

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for all n in N

Final answer: proved

Question: Prove by mathematical induction that (iii) 7^n - 4^n is divisible by 3 for all n in N

Answer: Step 1 (statement): let P(n) be the statement that 7^n - 4^n is divisible by 3

Step 2 (base case, n = 1): 7 - 4 = 3, which is divisible by 3, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true, i.e. 7^k - 4^k is divisible by 3

Step 4 (inductive step): form 7^(k + 1) - 4^(k + 1) = 7 x 7^k - 4 x 4^k

Step 5: add and subtract 7 x 4^k: = 7(7^k - 4^k) + 7 x 4^k - 4 x 4^k = 7(7^k - 4^k) + 3 x 4^k

Step 6: the first term is divisible by 3 by the hypothesis, and 3 x 4^k is a multiple of 3, so 7^(k + 1) - 4^(k + 1) is divisible by 3

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for all n in N

Final answer: proved

Question: Prove by mathematical induction that (i) 2^(n + 1) > (2n + 3) for all integral values of n >= 2

Answer: Step 1 (statement): let P(n) be the inequality 2^(n + 1) > 2n + 3, which is false for n = 1, so we use the modified principle starting at n = 2

Step 2 (base case, n = 2): LHS = 2^3 = 8 and RHS = 2(2) + 3 = 7, and 8 > 7, so P(2) is true

Step 3 (inductive hypothesis): assume P(k) is true for some integer k >= 2: 2^(k + 1) > 2k + 3

Step 4 (inductive step): multiply both sides of the hypothesis by 2: 2^(k + 2) > 2(2k + 3) = 4k + 6

Step 5: compare with the target: 4k + 6 = (2k + 5) + (2k + 1) > 2k + 5 = 2(k + 1) + 3, since 2k + 1 > 0

Step 6: therefore 2^(k + 2) > 2(k + 1) + 3, which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for all integers n >= 2

Final answer: proved

Question: Prove by mathematical induction that (ii) 3^(n - 1) > 2^n for all

integral values of n >= 3

Answer: Step 1 (statement): let P(n) be the inequality 3^(n - 1) > 2^n, starting at n = 3

Step 2 (base case, n = 3): LHS = 3^2 = 9 and RHS = 2^3 = 8, and 9 > 8, so P(3) is true

Step 3 (inductive hypothesis): assume P(k) is true for some integer k >= 3: 3^(k - 1) > 2^k

Step 4 (inductive step): multiply both sides by 3: 3^k > 3 x 2^k

Step 5: compare with the target 2^(k + 1): 3 x 2^k = 2^(k + 1) + 2^k > 2^(k + 1), since 2^k > 0

Step 6: therefore 3^((k + 1) - 1) = 3^k > 2^(k + 1), which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for all integers n >= 3

Final answer: proved

Question: Prove by mathematical induction that (iii) n! > 3^(n - 1) for all integral values of n >= 5

Answer: Step 1 (statement): let P(n) be the inequality n! > 3^(n - 1), starting at n = 5

Step 2 (base case, n = 5): LHS = 5! = 120 and RHS = 3^4 = 81, and 120 > 81, so P(5) is true

Step 3 (inductive hypothesis): assume P(k) is true for some integer k >= 5: k! > 3^(k - 1)

Step 4 (inductive step): (k + 1)! = (k + 1) x k! > (k + 1) x 3^(k - 1), using the hypothesis

Step 5: since k >= 5, we have k + 1 >= 6 > 3, so (k + 1) x 3^(k - 1) > 3 x 3^(k - 1) = 3^k

Step 6: therefore (k + 1)! > 3^((k + 1) - 1) = 3^k, which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for all integers n >= 5

Final answer: proved

Exercise 7.2

Question: Expand by means of the binomial theorem: (i) (a + b)^8

Answer: Step 1: the general term is Tr+1 = C(8, r) a^(8 - r) b^r, and the binomial coefficients for n = 8 are 1, 8, 28, 56, 70, 56, 28, 8, 1

Step 2: write the terms one by one: T1 = a^8, T2 = 8a^7 b, T3 = 28a^6 b^2, T4 = 56a^5 b^3, T5 = 70a^4 b^4, T6 = 56a^3 b^5, T7 = 28a^2 b^6, T8 = 8a b^7, T9 = b^8

Step 3: in each term the powers of a decrease by 1 and the powers of b increase by 1, the sum of the powers always being 8

Final answer: (a + b)^8 = a^8 + 8a^7 b + 28a^6 b^2 + 56a^5 b^3 + 70a^4 b^4 + 56a^3 b^5 + 28a^2 b^6 + 8a b^7 + b^8

Question: Expand by means of the binomial theorem: (ii) (2x - 3y)^4

Answer: Step 1: write the binomial as (2x + (-3y))^4; the general term is Tr+1 = C(4, r) (2x)^(4 - r) (-3y)^r, with coefficients 1, 4, 6, 4, 1

Step 2: T1 = (2x)^4 = 16x^4

Step 3: T2 = 4(2x)^3 (-3y) = 4 x 8x^3 x (-3y) = -96x^3 y

Step 4: T3 = 6(2x)^2 (-3y)^2 = 6 x 4x^2 x 9y^2 = 216x^2 y^2

Step 5: T4 = 4(2x)(-3y)^3 = 4 x 2x x (-27y^3) = -216x y^3

Step 6: T5 = (-3y)^4 = 81y^4

Final answer: (2x - 3y)^4 = 16x^4 - 96x^3 y + 216x^2 y^2 - 216x y^3 + 81y^4

Question: Expand by means of the binomial theorem: (iii) (sqrt(x/y) - sqrt(y/x))^4

Answer: Step 1: let A = sqrt(x/y) and B = sqrt(y/x); note that AB = sqrt((x/y)(y/x)) = 1, A^2 = x/y and B^2 = y/x

Step 2: the general term is Tr+1 = C(4, r) A^(4 - r) (-B)^r, with coefficients 1, 4, 6, 4, 1

Step 3: T1 = A^4 = (x/y)^2 = x^2/y^2

Step 4: T2 = -4 A^3 B = -4 A^2 (AB) = -4(x/y) x 1 = -4x/y

Step 5: T3 = 6 A^2 B^2 = 6(AB)^2 = 6 x 1 = 6

Step 6: T4 = -4 A B^3 = -4 (AB) B^2 = -4y/x

Step 7: T5 = B^4 = (y/x)^2 = y^2/x^2

Final answer: (sqrt(x/y) - sqrt(y/x))^4 = x^2/y^2 - 4x/y + 6 - 4y/x + y^2/x^2

Question: Expand by means of the binomial theorem: (iv) (x^2/2 - 2/x)^6

Answer: Step 1: the general term is Tr+1 = C(6, r) (x^2/2)^(6 - r) (-2/x)^r, with coefficients 1, 6, 15, 20, 15, 6, 1

Step 2: T1 = (x^2/2)^6 = x^12 / 64

Step 3: T2 = 6(x^2/2)^5 (-2/x) = 6 x (x^10/32) x (-2/x) = -(3/8)x^9

Step 4: T3 = 15(x^2/2)^4 (-2/x)^2 = 15 x (x^8/16) x (4/x^2) = (15/4)x^6

Step 5: T4 = 20(x^2/2)^3 (-2/x)^3 = 20 x (x^6/8) x (-8/x^3) = -20x^3

Step 6: T5 = 15(x^2/2)^2 (-2/x)^4 = 15 x (x^4/4) x (16/x^4) = 60

Step 7: T6 = 6(x^2/2)(-2/x)^5 = 6 x (x^2/2) x (-32/x^5) = -96/x^3

Step 8: T7 = (-2/x)^6 = 64/x^6

Final answer: (x^2/2 - 2/x)^6 = x^12/64 - (3/8)x^9 + (15/4)x^6 - 20x^3 + 60 - 96/x^3 + 64/x^6

Question: Expand by means of the binomial theorem: (v) (3x^2 - 2y^3)^5

Answer: Step 1: the general term is Tr+1 = C(5, r) (3x^2)^(5 - r) (-2y^3)^r, with coefficients 1, 5, 10, 10, 5, 1

Step 2: T1 = (3x^2)^5 = 243x^10

Step 3: T2 = 5(3x^2)^4 (-2y^3) = 5 x 81x^8 x (-2y^3) = -810x^8 y^3

Step 4: T3 = 10(3x^2)^3 (-2y^3)^2 = 10 x 27x^6 x 4y^6 = 1080x^6 y^6

Step 5: T4 = 10(3x^2)^2 (-2y^3)^3 = 10 x 9x^4 x (-8y^9) = -720x^4 y^9

Step 6: T5 = 5(3x^2)(-2y^3)^4 = 5 x 3x^2 x 16y^12 = 240x^2 y^12

Step 7: T6 = (-2y^3)^5 = -32y^15

Final answer: (3x^2 - 2y^3)^5 = 243x^10 - 810x^8 y^3 + 1080x^6 y^6 - 720x^4 y^9 + 240x^2 y^12 - 32y^15

Question: Use the binomial theorem to find the value of (i) (11)^5

Answer: Step 1: write 11 = 10 + 1, so (11)^5 = (10 + 1)^5

Step 2: expand with coefficients 1, 5, 10, 10, 5, 1: (10 + 1)^5 = 10^5 + 5

x 10^4 + 10 x 10^3 + 10 x 10^2 + 5 x 10 + 1

Step 3: evaluate each term: 100000 + 50000 + 10000 + 1000 + 50 + 1

Step 4: add step by step: 100000 + 50000 = 150000; 150000 + 10000 = 160000; 160000 + 1000 = 161000; 161000 + 50 = 161050; 161050 + 1 = 161051

Final answer: (11)^5 = 161051

Question: Use the binomial theorem to find the value of (ii) (19)^6

Answer: Step 1: write 19 = 20 - 1, so (19)^6 = (20 - 1)^6

Step 2: expand with coefficients 1, 6, 15, 20, 15, 6, 1 and alternating signs: 20^6 - 6 x 20^5 + 15 x 20^4 - 20 x 20^3 + 15 x 20^2 - 6 x 20 + 1

Step 3: evaluate each term: 64000000 - 19200000 + 2400000 - 160000 + 6000 - 120 + 1

Step 4: add step by step: 64000000 - 19200000 = 44800000; 44800000 + 2400000 = 47200000; 47200000 - 160000 = 47040000; 47040000 + 6000 = 47046000; 47046000 - 120 = 47045880; 47045880 + 1 = 47045881

Final answer: (19)^6 = 47045881

Question: Use the binomial theorem to find the value of (iii) (99)^4

Answer: Step 1: write 99 = 100 - 1, so (99)^4 = (100 - 1)^4

Step 2: expand with coefficients 1, 4, 6, 4, 1 and alternating signs: 100^4 - 4 x 100^3 + 6 x 100^2 - 4 x 100 + 1

Step 3: evaluate each term: 100000000 - 4000000 + 60000 - 400 + 1

Step 4: add step by step: 100000000 - 4000000 = 96000000; 96000000 + 60000 = 96060000; 96060000 - 400 = 96059600; 96059600 + 1 =

96059601

Final answer: (99)^4 = 96059601

Question: Use the binomial theorem to compute the value correct to four places of decimal: (i) (1.01)^6

Answer: Step 1: write (1.01)^6 = (1 + 0.01)^6 and expand: 1 + 6(0.01) + 15(0.01)^2 + 20(0.01)^3 + 15(0.01)^4 + 6(0.01)^5 + (0.01)^6

Step 2: evaluate the terms: 1 + 0.06 + 0.0015 + 0.00002 + 0.00000015 + 0.0000000006 + ...

Step 3: the terms decrease very fast, so terms beyond the fourth do not affect the fourth decimal place

Step 4: add: 1 + 0.06 = 1.06; 1.06 + 0.0015 = 1.0615; 1.0615 + 0.00002 = 1.06152

Final answer: (1.01)^6 = 1.0615 (correct to four decimal places)

Question: Use the binomial theorem to compute the value correct to four places of decimal: (ii) (1.02)^7

Answer: Step 1: write (1.02)^7 = (1 + 0.02)^7 and expand: 1 + 7(0.02) + 21(0.02)^2 + 35(0.02)^3 + 35(0.02)^4 + 21(0.02)^5 + ...

Step 2: evaluate the terms: 1 + 0.14 + 0.0084 + 0.00028 + 0.0000056 + 0.0000000672 + ...

Step 3: add the significant terms step by step: 1 + 0.14 = 1.14; 1.14 + 0.0084 = 1.1484; 1.1484 + 0.00028 = 1.14868; 1.14868 + 0.0000056 = 1.1486856

Step 4: round to four decimal places: 1.1486856 becomes 1.1487

Final answer: (1.02)^7 = 1.1487 (correct to four decimal places)

Question: Use the binomial theorem to compute the value correct to four places of decimal: (iii) (2.03)^5

Answer: Step 1: write 2.03 = 2(1 + 0.015), so (2.03)^5 = 2^5 (1 + 0.015)^5 = 32(1 + 0.015)^5

Step 2: expand (1 + 0.015)^5: 1 + 5(0.015) + 10(0.015)^2 + 10(0.015)^3 + 5(0.015)^4 + (0.015)^5

Step 3: evaluate the terms: 1 + 0.075 + 0.00225 + 0.00003375 + 0.0000002531 + 0.0000000008 = 1.0772840039

Step 4: multiply by 32: 32 x 1.0772840039 = 34.4730881

Step 5: round to four decimal places

Final answer: (2.03)^5 = 34.4731 (correct to four decimal places)

Question: Evaluate (i) (1 + 2 sqrt(a))^4 - (1 - 2 sqrt(a))^4

Answer: Step 1: expand both brackets by the binomial theorem with coefficients 1, 4, 6, 4, 1: (1 + 2 sqrt(a))^4 = 1 + 4(2 sqrt(a)) + 6(2 sqrt(a))^2 + 4(2 sqrt(a))^3 + (2 sqrt(a))^4 and (1 - 2 sqrt(a))^4 is the same with alternating signs

Step 2: on subtraction, the even-power terms cancel and the odd-power terms double: difference = 2[4(2 sqrt(a)) + 4(2 sqrt(a))^3]

Step 3: simplify the powers: (2 sqrt(a)) = 2 sqrt(a) and (2 sqrt(a))^3 = 8 a sqrt(a)

Step 4: difference = 2[8 sqrt(a) + 32 a sqrt(a)] = 16 sqrt(a) + 64 a sqrt(a)

Step 5: factorise: 16 sqrt(a) (1 + 4a)

Final answer: (1 + 2 sqrt(a))^4 - (1 - 2 sqrt(a))^4 = 16 sqrt(a) (1 + 4a)

Question: Evaluate (ii) (2 - sqrt(1 - a))^5 + (2 + sqrt(1 - a))^5

Answer: Step 1: let t = sqrt(1 - a), so t^2 = 1 - a; expand both brackets with coefficients 1, 5, 10, 10, 5, 1

Step 2: on addition, the odd-power terms in t cancel and the even-power terms double: sum = 2[2^5 + 10 x 2^3 t^2 + 5 x 2 t^4]

Step 3: substitute t^2 = 1 - a and t^4 = (1 - a)^2: sum = 2[32 + 80(1 - a) + 10(1 - a)^2]

Step 4: expand: (1 - a)^2 = 1 - 2a + a^2, so the bracket = 32 + 80 - 80a + 10 - 20a + 10a^2 = 122 - 100a + 10a^2

Step 5: multiply by 2: 244 - 200a + 20a^2

Final answer: (2 - sqrt(1 - a))^5 + (2 + sqrt(1 - a))^5 = 244 - 200a + 20a^2

Question: Evaluate (iii) (2 + sqrt(3))^5 - (2 - sqrt(3))^5

Answer: Step 1: expand both brackets with coefficients 1, 5, 10, 10, 5, 1

Step 2: on subtraction, the even-power terms of sqrt(3) cancel and the odd-power terms double: difference = 2[5 x 2^4 x sqrt(3) + 10 x 2^2 x (sqrt(3))^3 + (sqrt(3))^5]

Step 3: simplify the powers: (sqrt(3))^3 = 3 sqrt(3) and (sqrt(3))^5 = 9 sqrt(3)

Step 4: each term: 5 x 16 x sqrt(3) = 80 sqrt(3); 10 x 4 x 3 sqrt(3) = 120 sqrt(3); and 9 sqrt(3)

Step 5: difference = 2[80 sqrt(3) + 120 sqrt(3) + 9 sqrt(3)] = 2 x 209 sqrt(3) = 418 sqrt(3)

Final answer: (2 + sqrt(3))^5 - (2 - sqrt(3))^5 = 418 sqrt(3)

Question: Find the indicated term in the expansion: (i) ((2/3)x - 3/2)^10,

the 8th term

Answer: Step 1: the general term is Tr+1 = C(10, r) ((2/3)x)^(10 - r) (-3/2)^r

Step 2: the 8th term is T8 = T(7 + 1), so put r = 7: T8 = C(10, 7) ((2/3)x)^3 (-3/2)^7

Step 3: evaluate the coefficient C(10, 7) = C(10, 3) = (10 x 9 x 8) / 3! = 720 / 6 = 120

Step 4: evaluate the powers: ((2/3)x)^3 = 8x^3/27 and (-3/2)^7 = -2187/128

Step 5: multiply: 120 x (8/27) x (-2187/128) = 120 x 8 x (-2187) / (27 x 128); simplify 2187/27 = 81 and 8/128 = 1/16, giving 120 x 81 x (-1/16) x x^3 = -9720/16 x^3

Step 6: -9720 / 16 = -1215/2

Final answer: the 8th term is -(1215/2)x^3

Question: Find the indicated term in the expansion: (ii) (a/b - b/a)^8, the 7th term

Answer: Step 1: the general term is Tr+1 = C(8, r) (a/b)^(8 - r) (-b/a)^r

Step 2: the 7th term is T7 = T(6 + 1), so put r = 6: T7 = C(8, 6) (a/b)^2 (-b/a)^6

Step 3: evaluate C(8, 6) = C(8, 2) = (8 x 7) / 2! = 28

Step 4: the sign is positive since 6 is even, and (a/b)^2 x (b/a)^6 = a^2 b^6 / (b^2 a^6) = b^4 / a^4

Step 5: multiply: 28 x b^4/a^4

Final answer: the 7th term is 28 b^4 / a^4

Question: Find the indicated term in the expansion: (iii) (2x - 1/y)^10, the last term

Answer: Step 1: the expansion of (2x - 1/y)^10 has 10 + 1 = 11 terms, so the last term is T11 = T(10 + 1)

Step 2: the general term is Tr+1 = C(10, r) (2x)^(10 - r) (-1/y)^r; put r = 10

Step 3: T11 = C(10, 10) (2x)^0 (-1/y)^10 = 1 x 1 x (1/y^10), since 10 is even

Final answer: the last term is 1/y^10

Question: Find the middle term in the expansion of: (i) (b - 2c/3)^10

Answer: Step 1: n = 10 is even, so there is one middle term, the (10/2 + 1) = 6th term

Step 2: the general term is Tr+1 = C(10, r) b^(10 - r) (-2c/3)^r; the 6th term has r = 5

Step 3: T6 = C(10, 5) b^5 (-2c/3)^5, with C(10, 5) = 10! / (5! x 5!) = 3628800 / (120 x 120) = 252

Step 4: (-2c/3)^5 = -32 c^5 / 243

Step 5: multiply: 252 x (-32/243) b^5 c^5 = -(8064/243) b^5 c^5, and 8064 / 243 simplifies (dividing by 9) to 896/27

Final answer: the middle term is -(896/27) b^5 c^5

Question: Find the middle term in the expansion of: (ii) (1 - (1/2)a^2)^14

Answer: Step 1: n = 14 is even, so there is one middle term, the (14/2 + 1) = 8th term

Step 2: the general term is Tr+1 = C(14, r) (1)^(14 - r) (-a^2/2)^r; the 8th

term has r = 7

Step 3: T8 = C(14, 7) (-a^2/2)^7, with C(14, 7) = 14! / (7! x 7!) = 3432

Step 4: (-a^2/2)^7 = -a^14 / 128 (negative because 7 is odd)

Step 5: multiply: -3432/128 a^14, and 3432 / 128 simplifies (dividing by 8) to 429/16

Final answer: the middle term is -(429/16) a^14

Question: Find the middle term in the expansion of: (iii) (a/b - b/a)^18

Answer: Step 1: n = 18 is even, so there is one middle term, the (18/2 + 1) = 10th term

Step 2: the general term is Tr+1 = C(18, r) (a/b)^(18 - r) (-b/a)^r; the 10th term has r = 9

Step 3: T10 = C(18, 9) (a/b)^9 (-b/a)^9, with C(18, 9) = 18! / (9! x 9!) = 48620

Step 4: (a/b)^9 x (-b/a)^9 = -(a^9 b^9) / (b^9 a^9) = -1 (negative because 9 is odd)

Step 5: multiply: 48620 x (-1) = -48620

Final answer: the middle term is -48620

Question: Find the middle term in the expansion of: (iv) (a - 1/a)^(2n)

Answer: Step 1: the index is 2n, an even number, so there is one middle term, the (2n/2 + 1) = (n + 1)th term

Step 2: the general term is Tr+1 = C(2n, r) a^(2n - r) (-1/a)^r; the (n + 1)th term has r = n

Step 3: T(n + 1) = C(2n, n) a^(2n - n) (-1/a)^n = C(2n, n) a^n x (-1)^n / a^n

Step 4: the powers of a cancel: a^n / a^n = 1

Final answer: the middle term is (-1)^n C(2n, n)

Question: Find the two middle terms of: (i) (x^3 + 1/x^2)^7

Answer: Step 1: n = 7 is odd, so there are two middle terms, the ((7 + 1)/2) = 4th and ((7 + 3)/2) = 5th terms

Step 2: the general term is Tr+1 = C(7, r) (x^3)^(7 - r) (1/x^2)^r = C(7, r) x^(21 - 3r) x^(-2r) = C(7, r) x^(21 - 5r)

Step 3: the 4th term (r = 3): T4 = C(7, 3) x^(21 - 15) = 35x^6, since C(7, 3) = (7 x 6 x 5) / 3! = 210 / 6 = 35

Step 4: the 5th term (r = 4): T5 = C(7, 4) x^(21 - 20) = 35x, since C(7, 4) = C(7, 3) = 35

Final answer: the two middle terms are 35x^6 and 35x

Question: Find the two middle terms of: (ii) (2b - b^2/4)^9

Answer: Step 1: n = 9 is odd, so there are two middle terms, the ((9 + 1)/2) = 5th and ((9 + 3)/2) = 6th terms

Step 2: the general term is Tr+1 = C(9, r) (2b)^(9 - r) (-b^2/4)^r

Step 3: the 5th term (r = 4): T5 = C(9, 4) (2b)^5 (-b^2/4)^4 = 126 x 32b^5 x (b^8/256) = 126 x (32/256) b^13 = (63/4) b^13, since C(9, 4) = (9 x 8 x 7 x 6) / 4! = 3024 / 24 = 126

Step 4: the 6th term (r = 5): T6 = C(9, 5) (2b)^4 (-b^2/4)^5 = 126 x 16b^4 x (-b^10/1024) = -(2016/1024) b^14 = -(63/32) b^14

Final answer: the two middle terms are (63/4) b^13 and -(63/32) b^14

Question: Write, in the simplified form, the term independent of x in (i) (2x + 1/x^2)^9

Answer: Step 1: the general term is Tr+1 = C(9, r) (2x)^(9 - r) (1/x^2)^r = C(9, r) 2^(9 - r) x^(9 - r) x^(-2r) = C(9, r) 2^(9 - r) x^(9 - 3r)

Step 2: for the term to be independent of x, the power of x must be zero: 9 - 3r = 0, so r = 3

Step 3: the required term is T4 = C(9, 3) 2^(9 - 3) = C(9, 3) x 2^6

Step 4: C(9, 3) = (9 x 8 x 7) / 3! = 504 / 6 = 84 and 2^6 = 64

Step 5: multiply: 84 x 64 = 5376

Final answer: the term independent of x is 5376

Question: Write, in the simplified form, the term independent of x in (ii) (2x + 5/x)^6

Answer: Step 1: the general term is Tr+1 = C(6, r) (2x)^(6 - r) (5/x)^r = C(6, r) 2^(6 - r) 5^r x^(6 - r) x^(-r) = C(6, r) 2^(6 - r) 5^r x^(6 - 2r)

Step 2: for independence of x: 6 - 2r = 0, so r = 3

Step 3: the required term is T4 = C(6, 3) 2^3 5^3

Step 4: C(6, 3) = (6 x 5 x 4) / 3! = 120 / 6 = 20, 2^3 = 8 and 5^3 = 125

Step 5: multiply: 20 x 8 x 125 = 20000

Final answer: the term independent of x is 20000

Question: Obtain in the simplified form: (i) the term involving x^6 in the expansion of (2x^3 - 1/x^2)^7

Answer: Step 1: the general term is Tr+1 = C(7, r) (2x^3)^(7 - r) (-1/x^2)^r = C(7, r) 2^(7 - r) (-1)^r x^(3(7 - r)) x^(-2r) = C(7, r) 2^(7 - r) (-1)^r x^(21 - 5r)

Step 2: set the power of x equal to 6: 21 - 5r = 6, so 5r = 15 and r = 3

Step 3: the required term is T4 = C(7, 3) 2^4 (-1)^3 x^6

Step 4: C(7, 3) = 35, 2^4 = 16 and (-1)^3 = -1, so the term = 35 x 16 x (-1) x^6 = -560x^6

Final answer: the term involving x^6 is -560x^6

Question: Obtain in the simplified form: (ii) the term involving a^8 in the expansion of (a^2 - 1/a^2)^12

Answer: Step 1: the general term is Tr+1 = C(12, r) (a^2)^(12 - r) (-1/a^2)^r = C(12, r) (-1)^r a^(2(12 - r)) a^(-2r) = C(12, r) (-1)^r a^(24 - 4r)

Step 2: set the power of a equal to 8: 24 - 4r = 8, so 4r = 16 and r = 4

Step 3: the required term is T5 = C(12, 4) (-1)^4 a^8

Step 4: C(12, 4) = (12 x 11 x 10 x 9) / 4! = 11880 / 24 = 495 and (-1)^4 = 1

Final answer: the term involving a^8 is 495a^8

Question: Obtain in the simplified form: (iii) the coefficient of x in the expansion of (x^2 + a^2/x)^5

Answer: Step 1: the general term is Tr+1 = C(5, r) (x^2)^(5 - r) (a^2/x)^r = C(5, r) a^(2r) x^(2(5 - r)) x^(-r) = C(5, r) a^(2r) x^(10 - 3r)

Step 2: set the power of x equal to 1: 10 - 3r = 1, so 3r = 9 and r = 3

Step 3: the required term is T4 = C(5, 3) a^6 x

Step 4: C(5, 3) = C(5, 2) = (5 x 4) / 2! = 10

Step 5: the term is 10a^6 x, so the coefficient of x is 10a^6

Final answer: the coefficient of x is 10a^6 (the term is 10a^6 x)

Question: Does the expansion of ((3a/2) - 1/(3a^3))^9 contain a term (i) independent of a?

Answer: Step 1: the general term is Tr+1 = C(9, r) (3a/2)^(9 - r) (-1/(3a^3))^r; the power of a in this term is (9 - r) from the first part and -3r from the second part, i.e. a^(9 - r - 3r) = a^(9 - 4r)

Step 2: for a term independent of a, the power must be zero: 9 - 4r = 0, so r = 9/4

Step 3: r = 9/4 is not a whole number, and r must be an integer between 0 and 9 for a term of the expansion

Final answer: No, the expansion contains no term independent of a

Question: Does the expansion of ((3a/2) - 1/(3a^3))^9 contain a term (ii) involving a^10?

Answer: Step 1: from the general term, the power of a in Tr+1 is 9 - 4r

Step 2: for a term involving a^10: 9 - 4r = 10, so -4r = 1 and r = -1/4

Step 3: r = -1/4 is negative and not a whole number, so no term of the expansion has this value of r

Final answer: No, the expansion contains no term involving a^10

Question: The coefficients of the fifth, sixth and seventh terms of the expansion of (1 + x)^n form an A.P. Find n.

Answer: Step 1: in (1 + x)^n the coefficients of the 5th, 6th and 7th terms are C(n, 4), C(n, 5) and C(n, 6)

Step 2: for an arithmetic progression, the middle term is the average of its neighbours: 2 C(n, 5) = C(n, 4) + C(n, 6)

Step 3: use the ratio relations C(n, 5)/C(n, 4) = (n - 4)/5 and C(n, 6)/C(n, 5) = (n - 5)/6; dividing the A.P. condition by C(n, 5) gives 2 = 5/(n - 4) + (n -

5)/6

Step 4: multiply by 6(n - 4): 12(n - 4) = 30 + (n - 5)(n - 4), i.e. 12n - 48 = 30 + n^2 - 9n + 20

Step 5: simplify: 0 = n^2 - 21n + 98, i.e. n^2 - 21n + 98 = 0

Step 6: factorise: (n - 7)(n - 14) = 0, so n = 7 or n = 14

Step 7: check n = 7: coefficients are C(7, 4) = 35, C(7, 5) = 21, C(7, 6) = 7, an A.P. with common difference -14; check n = 14: C(14, 4) = 1001, C(14, 5) = 2002, C(14, 6) = 3003, an A.P. with common difference 1001

Final answer: n = 7 or n = 14

Exercise 7.3

Question: Find the first four terms in the expansion of (i) (8 - 16x)^(1/3) (the cube root of (8 - 16x)), where |x| < 1/2

Answer: Step 1: make the first term 1: (8 - 16x)^(1/3) = 8^(1/3) (1 - 2x)^(1/3) = 2(1 - 2x)^(1/3)

Step 2: use the binomial series (1 + z)^n = 1 + nz + n(n - 1)/2! z^2 + n(n - 1)(n - 2)/3! z^3 + ... with n = 1/3 and z = -2x

Step 3: the terms are: 1; (1/3)(-2x) = -(2/3)x; (1/3)(-2/3)/2 x (4x^2) = -(4/9)x^2; (1/3)(-2/3)(-5/3)/6 x (-8x^3) = -(40/81)x^3

Step 4: so (1 - 2x)^(1/3) = 1 - (2/3)x - (4/9)x^2 - (40/81)x^3 - ...

Step 5: multiply by 2

Final answer: (8 - 16x)^(1/3) = 2 - (4/3)x - (8/9)x^2 - (80/81)x^3 - ...

Question: Find the first four terms in the expansion of (ii) 3/(3 + x)

Answer: Step 1: make the first term of the binomial 1: 3/(3 + x) = 1 / (1 + x/3) = (1 + x/3)^(-1)

Step 2: use the binomial series with n = -1 and z = x/3: (1 + z)^(-1) = 1 - z + z^2 - z^3 + ...

Step 3: substitute z = x/3 term by term: 1 - x/3 + (x/3)^2 - (x/3)^3

Step 4: simplify the powers: (x/3)^2 = x^2/9 and (x/3)^3 = x^3/27

Final answer: 3/(3 + x) = 1 - x/3 + x^2/9 - x^3/27 + ..., valid for |x| < 3

Question: Find the first four terms in the expansion of (iii) (1 + 3x)^(-1/3)

Answer: Step 1: use the binomial series (1 + z)^n with n = -1/3 and z = 3x, valid for |3x| < 1

Step 2: first terms of the series: 1 + nz + n(n - 1)/2! z^2 + n(n - 1)(n - 2)/3! z^3

Step 3: nz = (-1/3)(3x) = -x

Step 4: n(n - 1)/2! z^2 = [(-1/3)(-4/3)/2] (3x)^2 = (2/9)(9x^2) = 2x^2

Step 5: n(n - 1)(n - 2)/3! z^3 = [(-1/3)(-4/3)(-7/3)/6] (3x)^3 = (-28/162)(27x^3) = -(14/3)x^3

Final answer: (1 + 3x)^(-1/3) = 1 - x + 2x^2 - (14/3)x^3 + ...

Question: Find the first four terms in the expansion of (iv) sqrt(1 + 2x)/(1 - x)

Answer: Step 1: write the expression as (1 + 2x)^(1/2) (1 - x)^(-1)

Step 2: expand the first factor with n = 1/2, z = 2x: (1 + 2x)^(1/2) = 1 + x - (1/2)x^2 + (1/2)x^3 + ...

Step 3: expand the second factor: (1 - x)^(-1) = 1 + x + x^2 + x^3 + ...

Step 4: multiply the two series, keeping terms up to x^3: constant term = 1

Step 5: x term = 1 + 1 = 2, giving 2x

Step 6: x^2 term = 1 (from 1 x x^2) + 1 (from x x x) - 1/2 (from -(1/2)x^2 x 1) = 3/2, giving (3/2)x^2

Step 7: x^3 term = 1 + 1 - 1/2 + 1/2 = 2, giving 2x^3

Final answer: sqrt(1 + 2x)/(1 - x) = 1 + 2x + (3/2)x^2 + 2x^3 + ...

Question: Find the first negative term in the expansion of (i) (1 + y)^(4/3)

Answer: Step 1: the general term of (1 + y)^n is Tr+1 = [n(n - 1)(n - 2)...(n - r + 1) / r!] y^r, with n = 4/3

Step 2: the factors in the numerator are (4/3), (1/3), (-2/3), ...; the term becomes negative as soon as one negative factor appears, i.e. when (n - r + 1) = (7/3 - r) first becomes negative

Step 3: 7/3 - r < 0 when r > 7/3, so the first such value is r = 3, which gives the 4th term

Step 4: compute T4 = [(4/3)(1/3)(-2/3) / 3!] y^3 = [(-8/27) / 6] y^3 = -(8/162) y^3

Step 5: simplify: -(8/162) = -(4/81)

Final answer: the first negative term is the 4th term, -(4/81)y^3

Question: Find the first negative term in the expansion of (ii) (1 + 2x)^(5/2)

Answer: Step 1: the general term is Tr+1 = [n(n - 1)...(n - r + 1) / r!] (2x)^r, with n = 5/2

Step 2: the numerator factors are (5/2), (3/2), (1/2), (-1/2), ...; the first

negative factor appears when (n - r + 1) = (7/2 - r) becomes negative

Step 3: 7/2 - r < 0 when r > 7/2, so the first such value is r = 4, which gives the 5th term

Step 4: compute T5 = [(5/2)(3/2)(1/2)(-1/2) / 4!] (2x)^4 = [(-15/16) / 24] x 16x^4

Step 5: (-15/16)/24 = -15/384 = -5/128, and (-5/128) x 16 = -80/128 = -5/8

Final answer: the first negative term is the 5th term, -(5/8)x^4

Question: If x is so small that its square and higher powers may be neglected, show that (i) sqrt(1 + (1/4)x) is approximately 1 + (1/8)x

Answer: Step 1: write the expression as (1 + x/4)^(1/2)

Step 2: expand by the binomial series, keeping only the first two terms (first approximation): (1 + z)^n is approximately 1 + nz, with n = 1/2 and z = x/4

Step 3: (1 + x/4)^(1/2) is approximately 1 + (1/2)(x/4)

Step 4: the neglected terms contain x^2 and higher powers, which are negligible by the given condition

Step 5: (1/2)(x/4) = x/8

Final answer: sqrt(1 + (1/4)x) is approximately 1 + (1/8)x, shown

Question: If x is so small that its square and higher powers may be neglected, show that (ii) (1 + 8x)^(1/4) is approximately 1 + 2x

Answer: Step 1: expand by the binomial series, keeping only the first two terms: (1 + z)^n is approximately 1 + nz, with n = 1/4 and z = 8x

Step 2: (1 + 8x)^(1/4) is approximately 1 + (1/4)(8x)

Step 3: the neglected terms contain x^2 and higher powers, which are negligible by the given condition

Step 4: (1/4)(8x) = 2x

Final answer: (1 + 8x)^(1/4) is approximately 1 + 2x, shown

Question: Using the binomial series, find the value up to three places of decimals of (i) sqrt(24)

Answer: Step 1: write 24 = 25 - 1, so sqrt(24) = (25 - 1)^(1/2) = 5(1 - 1/25)^(1/2)

Step 2: expand (1 - 1/25)^(1/2) with n = 1/2 and z = -1/25: 1 + (1/2)(-1/25) + (1/2)(-1/2)/2 (-1/25)^2 + (1/2)(-1/2)(-3/2)/6 (-1/25)^3 + ...

Step 3: evaluate the terms: 1 - 0.02 - 0.0002 - 0.000004 - ... = 0.979796 (the terms decrease rapidly since 1/25 is small)

Step 4: multiply by 5: 5 x 0.979796 = 4.89898

Step 5: round to three decimal places

Final answer: sqrt(24) = 4.899 (approximately)

Question: Using the binomial series, find the value up to three places of decimals of (ii) the cube root of 28

Answer: Step 1: write 28 = 27 + 1, so 28^(1/3) = (27(1 + 1/27))^(1/3) = 3(1 + 1/27)^(1/3)

Step 2: expand with n = 1/3 and z = 1/27: 1 + (1/3)(1/27) + (1/3)(-2/3)/2 (1/27)^2 + (1/3)(-2/3)(-5/3)/6 (1/27)^3 + ...

Step 3: evaluate the terms: 1 + 0.0123457 - 0.0001524 + 0.0000031 + ... = 1.0121964

Step 4: multiply by 3: 3 x 1.0121964 = 3.0365892

Step 5: round to three decimal places

Final answer: the cube root of 28 = 3.037 (approximately)

Question: Using the binomial series, find the value up to three places of decimals of (iii) the fifth root of 241

Answer: Step 1: write 241 = 243 - 2 = 3^5 - 2, so 241^(1/5) = 3(1 - 2/243)^(1/5)

Step 2: expand with n = 1/5 and z = -2/243: 1 + (1/5)(-2/243) + (1/5)(-4/5)/2 (-2/243)^2 + ...

Step 3: evaluate the terms: 1 - 0.0016461 - 0.0000054 - ... = 0.9983485 (later terms are below 0.0000001)

Step 4: multiply by 3: 3 x 0.9983485 = 2.9950455

Step 5: round to three decimal places

Final answer: the fifth root of 241 = 2.995 (approximately)

Question: Using the binomial series, find the value up to three places of decimals of (iv) (1280)^(1/4)

Answer: Step 1: write 1280 = 1296 - 16 = 6^4 - 16, so (1280)^(1/4) = 6(1 - 16/1296)^(1/4) = 6(1 - 1/81)^(1/4)

Step 2: expand with n = 1/4 and z = -1/81: 1 + (1/4)(-1/81) + (1/4)(-3/4)/2 (-1/81)^2 + ...

Step 3: evaluate the terms: 1 - 0.0030864 - 0.0000143 - ... = 0.9968993

Step 4: multiply by 6: 6 x 0.9968993 = 5.9813958

Step 5: round to three decimal places

Final answer: (1280)^(1/4) = 5.981 (approximately)

Question: Using the binomial series, find the value up to three places of decimals of (v) 1/(the fifth root of 252), i.e. 252^(-1/5)

Answer: Step 1: write 252 = 243 + 9 = 3^5 (1 + 9/243) = 3^5 (1 + 1/27), so 252^(-1/5) = (1/3)(1 + 1/27)^(-1/5)

Step 2: expand with n = -1/5 and z = 1/27: 1 + (-1/5)(1/27) + (-1/5)(-6/5)/2 (1/27)^2 + (-1/5)(-6/5)(-11/5)/6 (1/27)^3 + ...

Step 3: evaluate the terms: 1 - 0.0074074 + 0.0001646 - 0.0000045 + ... = 0.9927527

Step 4: divide by 3: 0.9927527 / 3 = 0.3309176

Step 5: round to three decimal places

Final answer: 1/(fifth root of 252) = 0.331 (approximately)

Question: Identify the series as a binomial expansion and find its sum: (i) 1 + 3/4 + (3 x 5)/(4 x 8) + (3 x 5 x 7)/(4 x 8 x 12) + ...

Answer: Step 1: compare with (1 + x)^n = 1 + nx + n(n - 1)/2! x^2 + n(n - 1)(n - 2)/3! x^3 + ...

Step 2: from the second term: nx = 3/4

Step 3: from the third term: n(n - 1) x^2 / 2 = (3 x 5)/(4 x 8) = 15/32, so n(n - 1) x^2 = 15/16

Step 4: square the equation of Step 2: n^2 x^2 = 9/16; subtract from Step 3: n^2 x^2 - n(n - 1) x^2 = (nx)(x) = 9/16 - 15/16 = -3/8, so (3/4) x = -3/8, giving x = -1/2

Step 5: then n = (3/4) / (-1/2) = -3/2; check the fourth term: [(-3/2)(-5/2)(-7/2)/6] (-1/2)^3 = 35/128, and the given term is (3 x 5 x 7)/(4 x 8 x 12) = 105/384 = 35/128, correct

Step 6: the series is (1 - 1/2)^(-3/2) = (1/2)^(-3/2) = 2^(3/2)

Final answer: the sum is 2 sqrt(2)

Question: Identify the series as a binomial expansion and find its sum: (ii) 1 + (1/2)(1/5) - (1/2!)(1/4)(1/5^2) + ...

Answer: Step 1: compare with (1 + x)^n = 1 + nx + n(n - 1)/2! x^2 + ... (rename the variable z here to avoid confusion, so compare with (1 + z)^n)

Step 2: from the second term: nz = (1/2)(1/5) = 1/10

Step 3: from the third term: n(n - 1) z^2 / 2 = -(1/2)(1/4)(1/25) = -1/200, so n(n - 1) z^2 = -1/100

Step 4: square Step 2: n^2 z^2 = 1/100; subtract Step 3: (nz)(z) = 1/100 + 1/100 = 1/50, so (1/10) z = 1/50, giving z = 1/5

Step 5: then n = (1/10) / (1/5) = 1/2

Step 6: the series is (1 + 1/5)^(1/2) = (6/5)^(1/2)

Final answer: the sum is sqrt(6/5)

Question: Identify the series as a binomial expansion and find its sum: (iii) 1 - (1/2)(1/4) + (1 x 3)/(2! x 4) (1/4)^2 - (1 x 3 x 5)/(3! x 8) (1/4)^3 + ...

Answer: Step 1: compare with (1 + z)^n = 1 + nz + n(n - 1)/2! z^2 + n(n - 1)(n - 2)/3! z^3 + ...; the powers show z = 1/4

Step 2: from the second term: n(1/4) = -(1/2)(1/4), so n = -1/2

Step 3: check the third term: n(n - 1)/2 = (-1/2)(-3/2)/2 = 3/8 = (1 x 3)/(2! x 4) = 3/8, correct

Step 4: check the fourth term: n(n - 1)(n - 2)/6 = (-1/2)(-3/2)(-5/2)/6 = -5/16, and the given coefficient is -(1 x 3 x 5)/(3! x 8) = -15/48 = -5/16,

correct

Step 5: the series is (1 + 1/4)^(-1/2) = (5/4)^(-1/2) = sqrt(4/5)

Final answer: the sum is 2/sqrt(5)

Question: Identify the series as a binomial expansion and find its sum: (iv) 1 - (1/2)(1/2) + (1 x 3)/(2 x 4) (1/2)^2 - (1 x 3 x 5)/(2 x 4 x 6) (1/2)^3 + ...

Answer: Step 1: compare with (1 + z)^n; the powers show z = 1/2

Step 2: from the second term: n(1/2) = -(1/2)(1/2), so n = -1/2

Step 3: check the third term: n(n - 1)/2 = (-1/2)(-3/2)/2 = 3/8 = (1 x 3)/(2 x 4), correct; the fourth term check gives -(1 x 3 x 5)/(2 x 4 x 6) = -15/48 = -5/16 = (-1/2)(-3/2)(-5/2)/6, correct

Step 4: the series is (1 + 1/2)^(-1/2) = (3/2)^(-1/2) = sqrt(2/3)

Final answer: the sum is sqrt(2/3)

Question: Identify the series as a binomial expansion and find its sum: (v) 1 - (1/2)(1/3) + (1 x 3)/(2 x 4) (1/3)^2 - (1 x 3 x 5)/(2 x 4 x 6) (1/3)^3 + ...

Answer: Step 1: compare with (1 + z)^n; the powers show z = 1/3

Step 2: from the second term: n(1/3) = -(1/2)(1/3), so n = -1/2

Step 3: check the third term: n(n - 1)/2 = 3/8 = (1 x 3)/(2 x 4), correct, and the fourth term matches -(1 x 3 x 5)/(2 x 4 x 6) in the same way as the previous parts

Step 4: the series is (1 + 1/3)^(-1/2) = (4/3)^(-1/2) = sqrt(3/4)

Final answer: the sum is sqrt(3)/2

Question: Use the binomial theorem to show that (i) 1 + 1/4 + (1 x 3)/(4 x 8) + (1 x 3 x 5)/(4 x 8 x 12) + ... = sqrt(2)

Answer: Step 1: compare the series with (1 + x)^n = 1 + nx + n(n - 1)/2!

x^2 + ...

Step 2: from the second term: nx = 1/4

Step 3: from the third term: n(n - 1) x^2 / 2 = 3/32, so n(n - 1) x^2 = 3/16

Step 4: square Step 2: n^2 x^2 = 1/16; subtract Step 3: (nx)(x) = 1/16 - 3/16 = -1/8, so (1/4) x = -1/8, giving x = -1/2

Step 5: then n = (1/4) / (-1/2) = -1/2; the fourth term checks: [(-1/2)(-3/2)(-5/2)/6] (-1/2)^3 = 5/128 = (1 x 3 x 5)/(4 x 8 x 12), correct

Step 6: the series is (1 - 1/2)^(-1/2) = (1/2)^(-1/2) = sqrt(2)

Final answer: the sum is sqrt(2), shown

Question: Use the binomial theorem to show that (ii) 1 + 1/4 + (1 x 3)/(4 x 6) + (1 x 3 x 5)/(4 x 6 x 8) + ... = the cube root of 4

Answer: Step 1: compare the series with (1 + x)^n = 1 + nx + n(n - 1)/2! x^2 + ...

Step 2: from the second term: nx = 1/4

Step 3: from the third term: n(n - 1) x^2 / 2 = (1 x 3)/(4 x 6) = 1/8, so n(n - 1) x^2 = 1/4

Step 4: square Step 2: n^2 x^2 = 1/16; subtract Step 3: (nx)(x) = 1/16 - 1/4 = -3/16, so (1/4) x = -3/16, giving x = -3/4

Step 5: then n = (1/4) / (-3/4) = -1/3

Step 6: the series is (1 - 3/4)^(-1/3) = (1/4)^(-1/3) = 4^(1/3)

Final answer: the sum is the cube root of 4, shown

Question: If y = (1/2)(1/16) - (1/(2! x 4))(1/16^2) + (1/3!)((1 x 3)/8)(1/16^3) - ..., then show that 16y(y + 2) = 1

Answer: Step 1: add 1 to both sides and compare 1 + y with the binomial series (1 + z)^n = 1 + nz + n(n - 1)/2! z^2 + n(n - 1)(n - 2)/3! z^3 + ...

Step 2: from the second term: nz = (1/2)(1/16) = 1/32

Step 3: from the third term: n(n - 1) z^2 / 2 = -(1/(2 x 4))(1/256) = -1/2048, so n(n - 1) z^2 = -1/1024

Step 4: square Step 2: n^2 z^2 = 1/1024; subtract Step 3: (nz)(z) = 1/1024 + 1/1024 = 1/512, so (1/32) z = 1/512, giving z = 1/16

Step 5: then n = (1/32) / (1/16) = 1/2; the fourth term checks: [(1/2)(-1/2)(-3/2)/6] (1/16)^3 = (1/16)(1/16^3) = (1/3!) x ((1 x 3)/8) x (1/16^3), correct

Step 6: therefore 1 + y = (1 + 1/16)^(1/2) = sqrt(17/16), so (1 + y)^2 = 17/16

Step 7: expand: 1 + 2y + y^2 = 17/16, so y^2 + 2y = 1/16

Step 8: multiply by 16: 16y^2 + 32y = 1, i.e. 16y(y + 2) = 1

Final answer: shown

Question: If 1/x = 2/5 + ((1 x 3)/2!) (2/5)^2 + ((1 x 3 x 5)/3!) (2/5)^3 + ..., then show that 4x^2 - 2x - 1 = 0

Answer: Step 1: add 1 to both sides and compare 1 + 1/x with (1 + z)^n = 1 + nz + n(n - 1)/2! z^2 + ...

Step 2: from the second term: nz = 2/5

Step 3: from the third term: n(n - 1) z^2 / 2 = ((1 x 3)/2) (2/5)^2 = (3/2)(4/25) = 6/25, so n(n - 1) z^2 = 12/25

Step 4: square Step 2: n^2 z^2 = 4/25; subtract Step 3: (nz)(z) = 4/25 - 12/25 = -8/25, so (2/5) z = -8/25, giving z = -4/5

Step 5: then n = (2/5) / (-4/5) = -1/2

Step 6: therefore 1 + 1/x = (1 - 4/5)^(-1/2) = (1/5)^(-1/2) = sqrt(5)

Step 7: so sqrt(5) = 1 + 1/x; square both sides: 5 = 1 + 2/x + 1/x^2

Step 8: multiply by x^2: 5x^2 = x^2 + 2x + 1, giving 4x^2 - 2x - 1 = 0

Final answer: shown

Question: If y/2 = (1/2)(4/9) + ((1 x 3)/(2^2 x 2!)) (4/9)^2 + ((1 x 3 x 5)/(2^3 x 3!)) (4/9)^3 + ..., then show that 5y^2 + 20y - 16 = 0

Answer: Step 1: the coefficients (1 x 3 x 5 x ... x (2k - 1)) / (2^k k!) are exactly the coefficients of the binomial series of (1 - t)^(-1/2), with t = 4/9

Step 2: therefore 1 + y/2 = (1 - 4/9)^(-1/2) = (5/9)^(-1/2) = 3/sqrt(5)

Step 3: square both sides: (1 + y/2)^2 = 9/5

Step 4: expand the left side: 1 + y + y^2/4 = 9/5

Step 5: multiply throughout by 20: 20 + 20y + 5y^2 = 36

Step 6: rearrange: 5y^2 + 20y + 20 - 36 = 0, i.e. 5y^2 + 20y - 16 = 0

Final answer: shown

Review Exercise 7

Question: Select the correct answer: (i) The general term of the binomial expansion (a + x)^n, where n is in N, is: (a) C(n, r) a^n x (b) C(n, r) a^(n - r) x^r (c) C(n, r) a^r x^(n - r) (d) C(n, r) (ax)^(n - r)

Answer: Step 1: by the binomial theorem, the (r + 1)th term of (a + x)^n is Tr+1 = C(n, r) a^(n - r) x^r, where the powers of a fall from n to 0 and

the powers of x rise from 0 to n

Final answer: (b) C(n, r) a^(n - r) x^r

Question: Select the correct answer: (ii) The number of terms in the expansion of (a + b)^(2n) are: (a) n (b) 2n + 1 (c) 2^n (d) 2^(n - 1)

Answer: Step 1: the expansion of (a + b)^m has m + 1 terms

Step 2: here m = 2n, so the number of terms is 2n + 1

Final answer: (b) 2n + 1

Question: Select the correct answer: (iii) In the expansion (a + x)^n, the exponent of x: (a) decreases from n to 0 (b) increases from 0 to n (c) remains n everywhere (d) becomes 0 at the end

Answer: Step 1: in the general term C(n, r) a^(n - r) x^r, as r runs from 0 to n the exponent of x runs from 0 up to n

Final answer: (b) increases from 0 to n

Question: Select the correct answer: (iv) The middle term in the expansion of (a + b)^(2n) is: (a) nth term (b) (n + 1)th term (c) (2n + 1)th term (d) (2n - 1)th term

Answer: Step 1: the expansion has 2n + 1 terms, an odd number, so there is a single middle term

Step 2: the middle term is the ((2n + 2)/2) = (n + 1)th term

Final answer: (b) (n + 1)th term

Question: Select the correct answer: (v) The term independent of x in the expansion of (a + 2x)^n is: (a) first term (b) middle term (c) last term (d) 2nd last term

Answer: Step 1: the general term is C(n, r) a^(n - r) (2x)^r; every term with r >= 1 contains x

Step 2: the only term free of x is the one with r = 0, namely the first term a^n

Final answer: (a) First term

Question: Select the correct answer: (vi) The coefficient of the last term in the expansion of (2 - x)^7 is: (a) 1 (b) -1 (c) 7 (d) -7

Answer: Step 1: the last term of (2 - x)^7 is C(7, 7) 2^0 (-x)^7 = -x^7, because 7 is odd

Step 2: its coefficient is -1

Final answer: (b) -1

Question: Select the correct answer: (vii) In the expansion (a + 1/2)^7, the number of middle terms is: (a) one (b) two (c) three (d) four

Answer: Step 1: the index 7 is odd, so the expansion has 8 terms, an even number

Step 2: an even number of terms gives two middle terms (the 4th and 5th)

Final answer: (b) two

Question: Select the correct answer: (viii) The sum of the odd binomial coefficients in the expansion of (a + x)^n is: (a) 2^n (b) 2^(n - 1) (c) 2^(n + 1) (d) n + 1

Answer: Step 1: put a = 1 and x = 1 in (a + x)^n: the sum of all binomial coefficients is 2^n

Step 2: put a = 1 and x = -1: the alternating sum is 0, which shows the sum of the odd-position coefficients equals the sum of the even-position coefficients

Step 3: hence each sum is half of 2^n, i.e. 2^(n - 1)

Final answer: (b) 2^(n - 1)

Question: Select the correct answer: (ix) C(n + 1, 0) + C(n + 1, 1) + C(n + 1, 2) + ... + C(n + 1, n + 1) is equal to: (a) 2^n (b) 2^(n + 1) (c) 2^(n - 1) (d) cannot be determined

Answer: Step 1: the sum of all binomial coefficients C(m, 0) + C(m, 1) + ... + C(m, m) equals 2^m

Step 2: here m = n + 1, so the sum is 2^(n + 1)

Final answer: (b) 2^(n + 1)

Question: Select the correct answer: (x) The number of terms in the expansion of (1 + x)^(1/3) is: (a) 4/3 (b) 4 (c) infinity (d) 2

Answer: Step 1: for a fractional (or negative) index the binomial expansion does not terminate; it is an infinite series valid for |x| < 1

Final answer: (c) infinity

Question: Select the correct answer: (xi) 1 - x + x^2 - x^3 + ... is equal to: (a) (1 + x)^(-1) (b) (1 - x)^(-1) (c) (1 + x)^(-2) (d) (1 - x)^(-2)

Answer: Step 1: the binomial series (1 + z)^(-1) = 1 - z + z^2 - z^3 + ...

Step 2: putting z = x gives exactly 1 - x + x^2 - x^3 + ...

Final answer: (a) (1 + x)^(-1)

Question: Select the correct answer: (xii) The expansion of (1 - 2x)^(-2) is valid if: (a) |x| < 0 (b) |x| < 1/2 (c) |x| < 2 (d) |x| < 1

Answer: Step 1: the binomial series for (1 + z)^n with fractional or negative n is valid for |z| < 1

Step 2: here z = -2x, so we need |2x| < 1, i.e. |x| < 1/2

Final answer: (b) |x| < 1/2

Question: Select the correct answer: (xiii) The middle term in the expansion of (a + b)^n is the (n/2 + 1)th term; then n is: (a) odd (b) even (c) prime (d) none of these

Answer: Step 1: a single middle term at position (n/2 + 1) requires n/2 to be a whole number and the number of terms n + 1 to be odd

Step 2: both conditions hold exactly when n is even

Final answer: (b) Even

Question: Select the correct answer: (xiv) In (a + 1/a)^8, the sum of the binomial coefficients is: (a) 64 (b) 128 (c) 256 (d) 512

Answer: Step 1: the sum of the binomial coefficients in (a + b)^n is 2^n, found by putting a = b = 1

Step 2: here n = 8, so the sum is 2^8 = 256

Final answer: (c) 256

Question: Prove by the principle of mathematical induction that 8 x 10^n - 2 is divisible by 6 for all n in N

Answer: Step 1 (statement): let P(n) be the statement that f(n) = 8 x 10^n - 2 is divisible by 6

Step 2 (base case, n = 1): f(1) = 80 - 2 = 78 = 6 x 13, which is divisible by 6, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true, i.e. 8 x 10^k - 2 is divisible by 6

Step 4 (inductive step): form f(k + 1) = 8 x 10^(k + 1) - 2 = 10(8 x 10^k) - 2

Step 5: write 8 x 10^k = f(k) + 2 from the hypothesis form: f(k + 1) = 10[f(k) + 2] - 2 = 10 f(k) + 20 - 2 = 10 f(k) + 18

Step 6: 10 f(k) is divisible by 6 by the hypothesis, and 18 = 6 x 3 is a multiple of 6, so f(k + 1) is divisible by 6

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for all n in N

Final answer: proved

Question: Prove by mathematical induction that 1^2 + 3^2 + 5^2 + ... + (2n - 1)^2 = (1/3)n(2n - 1)(2n + 1)

Answer: Step 1 (statement): let P(n) be the given statement (sum of the squares of the first n odd numbers)

Step 2 (base case, n = 1): LHS = 1^2 = 1 and RHS = (1/3)(1)(1)(3) = 1, so P(1) is true

Step 3 (inductive hypothesis): assume P(k) is true: 1^2 + 3^2 + ... + (2k - 1)^2 = (1/3)k(2k - 1)(2k + 1)

Step 4 (inductive step): add the (k + 1)th term (2(k + 1) - 1)^2 = (2k + 1)^2 to both sides: sum = (1/3)k(2k - 1)(2k + 1) + (2k + 1)^2

Step 5: take (2k + 1) common: (2k + 1)[k(2k - 1)/3 + (2k + 1)] = (2k + 1)[(2k^2 - k + 6k + 3)/3] = (2k + 1)(2k^2 + 5k + 3)/3

Step 6: factorise: 2k^2 + 5k + 3 = (k + 1)(2k + 3), so the sum = (1/3)(k + 1)(2k + 1)(2k + 3) = (1/3)(k + 1)(2(k + 1) - 1)(2(k + 1) + 1), which is P(k + 1)

Step 7 (conclusion): by the principle of mathematical induction, P(n) is true for every positive integer n

Final answer: proved

Question: Given that the coefficient of y^2 in the expansion of 1/(1 +

my)^2 is 96, find m

Answer: Step 1: write the expression as (1 + my)^(-2) and use the binomial series: (1 + z)^(-2) = 1 - 2z + 3z^2 - 4z^3 + ..., with z = my

Step 2: the general coefficient pattern comes from (-2)(-3)/2! = 3 for the z^2 term

Step 3: so (1 + my)^(-2) = 1 - 2my + 3m^2 y^2 - 4m^3 y^3 + ..., and the coefficient of y^2 is 3m^2

Step 4: set 3m^2 = 96, giving m^2 = 32

Step 5: take square roots: m = sqrt(32) or m = -sqrt(32)

Final answer: m = 4 sqrt(2) or m = -4 sqrt(2)

Question: Given that the terms involving m^4 and higher powers may be neglected, and that 1/(a + bm)^3 - 1/(1 + 3m)^4 = c m^2 + d m^3, find the values of a, b, c and d

Answer: Step 1: expand the first term: 1/(a + bm)^3 = a^(-3) (1 + (b/a) m)^(-3) = a^(-3) [1 - 3(b/a) m + 6(b/a)^2 m^2 - 10(b/a)^3 m^3 + ...]

Step 2: expand the second term: 1/(1 + 3m)^4 = (1 + 3m)^(-4) = 1 - 12m + 90m^2 - 540m^3 + ..., since (-4)(3) = -12, [(-4)(-5)/2](3^2) = 90 and [(-4)(-5)(-6)/6](3^3) = -540

Step 3: the right side has no constant term, so the constant terms on the left must cancel: a^(-3) - 1 = 0, giving a^3 = 1 and a = 1

Step 4: with a = 1, the first expansion becomes 1 - 3b m + 6b^2 m^2 - 10b^3 m^3; the right side has no m term, so the m coefficients must cancel: -3b - (-12) = 0, giving 3b = 12 and b = 4

Step 5: compare the m^2 coefficients: c = 6b^2 - 90 = 6(16) - 90 = 96 - 90 = 6

Step 6: compare the m^3 coefficients: d = -10b^3 - (-540) = -10(64) + 540 = -640 + 540 = -100

Final answer: a = 1, b = 4, c = 6, d = -100

Unit 8: Functions and Graphs

Exercise 8.1

Question: Q1(i). Let X = {a, b, c} and Y = {1, 2, 3}. In the mapping diagram for f, a -> 1 and c -> 2, and b has no arrow. Is f a function from X to Y?

Answer: Step 1: A relation is a function only if every element of the domain X has exactly one image in Y

Step 2: Here a has one image 1 and c has one image 2, which is fine

Step 3: But b has no image at all, so one element of X is left unmapped

Final answer: f is NOT a function from X to Y, because b has no image.

Question: Q1(ii). Let X = {a, b, c} and Y = {1, 2, 3}. In the mapping diagram for g, a -> 2, b -> 1 and b -> 3, and c -> 2. Is g a function from X to Y?

Answer: Step 1: A function must give each element of X one and only one image

Step 2: Here a -> 2 and c -> 2, which is allowed, because different elements may share an image

Step 3: But b has two different images, 1 and 3, from the two arrows starting at b

Final answer: g is NOT a function from X to Y, because b has two images.

Question: Q1(iii). Let X = {a, b, c} and Y = {1, 2, 3}. In the mapping

diagram for h, a -> 2, b -> 3 and c -> 2. Is h a function from X to Y?

Answer: Step 1: Check each element of X one by one

Step 2: a has the single image 2, b has the single image 3, and c has the single image 2

Step 3: Every element of X has exactly one image, even though 2 is the image of both a and c

Final answer: h IS a function from X to Y.

Question: Q2(i). Which of f1, with X = {2, 5, 8}, Y = {1, 2, 9}, 2 -> 1, 5 -> 2, 8 -> 9, is injective, surjective or bijective? Give a reason.

Answer: Step 1: The images 1, 2, 9 are all distinct, so distinct elements have distinct images, hence f1 is injective

Step 2: The range is {1, 2, 9}, which equals the co-domain Y, so every element of Y has a pre-image, hence f1 is surjective

Step 3: A function that is both injective and surjective is bijective

Final answer: f1 is injective, surjective and therefore bijective.

Question: Q2(ii). Which of f2, with X = {2, 5, 8}, Y = {-3, 1, 2, 9}, 2 -> -3, 5 -> 1, 8 -> 9, is injective, surjective or bijective? Give a reason.

Answer: Step 1: The images -3, 1, 9 are all distinct, so f2 is injective

Step 2: The range is {-3, 1, 9}, but the co-domain is {-3, 1, 2, 9}, and 2 in Y has no pre-image

Step 3: Since the range is not equal to the co-domain, f2 is not surjective, and so it cannot be bijective

Final answer: f2 is injective only, not surjective and not bijective.

Question: Q2(iii). Which of f3, with X = {1, 2, 3}, Y = {3, 4}, 1 -> 3, 2 ->

4, 3 -> 3, is injective, surjective or bijective? Give a reason.

Answer: Step 1: The elements 1 and 3 both have the same image 3, so distinct elements do not have distinct images, hence f3 is not injective

Step 2: The range is {3, 4}, which equals the co-domain Y, so f3 is surjective

Step 3: Since it is not injective, it cannot be bijective

Final answer: f3 is surjective only, not injective and not bijective.

Question: Q2(iv). Which of f4, with X = {2, 4, 5}, Y = {7, 8, 9}, 2 -> 7, 4 -> 9, 5 -> 8, is injective, surjective or bijective? Give a reason.

Answer: Step 1: The images 7, 9, 8 are all distinct, so f4 is injective

Step 2: The range is {7, 8, 9}, which equals the co-domain Y, so f4 is surjective

Step 3: Being both injective and surjective, it is bijective

Final answer: f4 is injective, surjective and therefore bijective.

Question: Q2(v). Which of f5, with X = {a, b, c}, Y = {1, 2, 3, 4}, a -> 2, b -> 1, c -> 3, is injective, surjective or bijective? Give a reason.

Answer: Step 1: The images 2, 1, 3 are all distinct, so f5 is injective

Step 2: The range is {1, 2, 3}, but 4 in the co-domain Y = {1, 2, 3, 4} has no pre-image

Step 3: So f5 is not surjective and cannot be bijective

Final answer: f5 is injective only, not surjective and not bijective.

Question: Q2(vi). Which of f6, with X = {a, b, c, d}, Y = {1, 2, 3}, a -> 1, b -> 2, c -> 3, d -> 3, is injective, surjective or bijective? Give a reason.

Answer: Step 1: The elements c and d both have the image 3, so f6 is not

injective

Step 2: The range is {1, 2, 3}, which equals the co-domain Y, so f6 is surjective

Step 3: Since it is not injective, it cannot be bijective

Final answer: f6 is surjective only, not injective and not bijective.

Question: Q3(i). If f : Z -> R is defined by f(x) = x^3 + 1, find f(0), if possible.

Answer: Step 1: 0 belongs to Z, the domain, so f(0) can be found

Step 2: Substitute x = 0, so f(0) = (0)^3 + 1

Step 3: (0)^3 = 0, so f(0) = 0 + 1

Final answer: f(0) = 1.

Question: Q3(ii). If f : Z -> R is defined by f(x) = x^3 + 1, find f(5), if possible.

Answer: Step 1: 5 belongs to Z, so f(5) can be found

Step 2: Substitute x = 5, so f(5) = (5)^3 + 1

Step 3: (5)^3 = 125, so f(5) = 125 + 1

Final answer: f(5) = 126.

Question: Q3(iii). If f : Z -> R is defined by f(x) = x^3 + 1, find f(3/4), if possible.

Answer: Step 1: The domain of f is Z, the set of integers

Step 2: 3/4 is a fraction, not an integer, so 3/4 does not belong to the domain

Step 3: A function value can only be found for inputs that lie in the domain

Final answer: f(3/4) is not possible, because 3/4 is not in Z.

Question: Q3(iv). If f : Z -> R is defined by f(x) = x^3 + 1, find f(-2), if possible.

Answer: Step 1: -2 belongs to Z, so f(-2) can be found

Step 2: Substitute x = -2, so f(-2) = (-2)^3 + 1

Step 3: (-2)^3 = -8, so f(-2) = -8 + 1

Final answer: f(-2) = -7.

Question: Q4(i). If f(x) = x^3, find f(2).

Answer: Step 1: Substitute x = 2 in f(x) = x^3

Step 2: f(2) = (2)^3

Step 3: (2)^3 = 2 x 2 x 2 = 8

Final answer: f(2) = 8.

Question: Q4(ii). If f(x) = x^3, find f(-10).

Answer: Step 1: Substitute x = -10

Step 2: f(-10) = (-10)^3

Step 3: The cube of a negative number is negative, and 10^3 = 1000, so (-10)^3 = -1000

Final answer: f(-10) = -1000.

Question: Q4(iii). If f(x) = x^3, find f(1/2).

Answer: Step 1: Substitute x = 1/2

Step 2: f(1/2) = (1/2)^3

Step 3: (1/2)^3 = 1/8

Final answer: f(1/2) = 1/8.

Question: Q4(iv). If f(x) = x^3, find f(5a).

Answer: Step 1: Substitute x = 5a

Step 2: f(5a) = (5a)^3

Step 3: Cube each factor, (5)^3 = 125 and (a)^3 = a^3, so (5a)^3 = 125a^3

Final answer: f(5a) = 125a^3.

Question: Q4(v). If f(x) = x^3, find f(a/3).

Answer: Step 1: Substitute x = a/3

Step 2: f(a/3) = (a/3)^3

Step 3: (a/3)^3 = a^3 / 27

Final answer: f(a/3) = a^3/27.

Question: Q4(vi). If f(x) = x^3, find f(a + h).

Answer: Step 1: Substitute x = a + h, so f(a + h) = (a + h)^3

Step 2: Use the identity (a + h)^3 = a^3 + 3a^2h + 3ah^2 + h^3

Step 3: Write the terms in order of powers of h

Final answer: f(a + h) = a^3 + 3a^2h + 3ah^2 + h^3.

Question: Q4(vii). If f(x) = x^3, find [f(a + h) - f(a - h)] / (2h), where h != 0.

Answer: Step 1: f(a + h) = (a + h)^3 = a^3 + 3a^2h + 3ah^2 + h^3

Step 2: f(a - h) = (a - h)^3 = a^3 - 3a^2h + 3ah^2 - h^3

Step 3: Subtract, f(a + h) - f(a - h) = 6a^2h + 2h^3, because the a^3 and 3ah^2 terms cancel

Step 4: Divide by 2h, giving (6a^2h + 2h^3) / (2h) = 3a^2 + h^2

Final answer: [f(a + h) - f(a - h)] / (2h) = 3a^2 + h^2.

Question: Q4(viii). If f(x) = x^3, find f(a + h) - f(a - h).

Answer: Step 1: f(a + h) = a^3 + 3a^2h + 3ah^2 + h^3

Step 2: f(a - h) = a^3 - 3a^2h + 3ah^2 - h^3

Step 3: Subtract the second expansion from the first, term by term

Step 4: a^3 - a^3 = 0, 3a^2h - (-3a^2h) = 6a^2h, 3ah^2 - 3ah^2 = 0, h^3 - (-h^3) = 2h^3

Final answer: f(a + h) - f(a - h) = 6a^2h + 2h^3.

Question: Q5(i). Given f : R -> R defined by f(x) = x^2, find [f(5 + h) - f(5)] / h, where h != 0.

Answer: Step 1: f(5 + h) = (5 + h)^2 = 25 + 10h + h^2

Step 2: f(5) = (5)^2 = 25

Step 3: Subtract, f(5 + h) - f(5) = 10h + h^2 = h(10 + h)

Step 4: Divide by h, which is allowed because h != 0, giving 10 + h

Final answer: [f(5 + h) - f(5)] / h = 10 + h.

Question: Q5(ii). Given f : R -> R defined by f(x) = x^2, find [f(a + h) - f(a)] / h, where h != 0.

Answer: Step 1: f(a + h) = (a + h)^2 = a^2 + 2ah + h^2

Step 2: f(a) = a^2

Step 3: Subtract, f(a + h) - f(a) = 2ah + h^2 = h(2a + h)

Step 4: Divide by h, giving 2a + h

Final answer: [f(a + h) - f(a)] / h = 2a + h.

Question: Q6. The domain of the function f(x) = 5x + 1 is {0, 1, 2, 3, 4, 5}. Find its range.

Answer: Step 1: Put each domain value in f(x) = 5x + 1 one by one

Step 2: f(0) = 1, f(1) = 6, f(2) = 11

Step 3: f(3) = 16, f(4) = 21, f(5) = 26

Step 4: The range is the set of all these images, written in increasing order

Final answer: Range of f = {1, 6, 11, 16, 21, 26}.

Question: Q7(i). Find the domain of f(x) = x^2 + 3.

Answer: Step 1: f(x) = x^2 + 3 is a polynomial, and a polynomial is defined for every real x

Step 2: There is no denominator that could become zero and no square root that could become negative

Step 3: So no value of x has to be excluded

Final answer: Domain = R, all real numbers.

Question: Q7(ii). Find the domain of f(x) = sqrt(x + 3).

Answer: Step 1: For a real square root, the expression inside the root must be >= 0

Step 2: So x + 3 >= 0

Step 3: Subtract 3 from both sides, giving x >= -3

Final answer: Domain = {x in R : x >= -3} = [-3, infinity).

Question: Q7(iii). Find the domain of f(x) = 1/x.

Answer: Step 1: A fraction is undefined when its denominator is zero

Step 2: Here the denominator is x, which is zero only when x = 0

Step 3: Exclude x = 0 and keep every other real number

Final answer: Domain = R - {0}.

Question: Q7(iv). Find the domain of f(x) = sqrt(x - 5).

Answer: Step 1: The expression inside the square root must be >= 0

Step 2: So x - 5 >= 0

Step 3: Add 5 to both sides, giving x >= 5

Final answer: Domain = {x in R : x >= 5} = [5, infinity).

Question: Q7(v). Find the domain of f(x) = (x + 1) / (x - 2).

Answer: Step 1: The function is undefined when the denominator x - 2 is zero

Step 2: x - 2 = 0 gives x = 2

Step 3: Exclude x = 2 only, since the numerator causes no problem

Final answer: Domain = R - {2}.

Question: Q7(vi) [printed as (iv) again in the book]. Find the domain of f(x) = x^2 / (x^2 - 9).

Answer: Step 1: The function is undefined when the denominator x^2 - 9 is zero

Step 2: Factorise, x^2 - 9 = (x - 3)(x + 3) = 0, giving x = 3 or x = -3

Step 3: Exclude both of these values

Final answer: Domain = R - {-3, 3}.

Question: Q8(i). The domain of f(x) is {1, 2, 3, 4, 5}. Find the range of f(x) = 5x^2 + 3.

Answer: Step 1: Substitute each domain value in 5x^2 + 3

Step 2: f(1) = 5(1) + 3 = 8 and f(2) = 5(4) + 3 = 23

Step 3: f(3) = 5(9) + 3 = 48 and f(4) = 5(16) + 3 = 83

Step 4: f(5) = 5(25) + 3 = 128

Final answer: Range = {8, 23, 48, 83, 128}.

Question: Q8(ii). The domain of f(x) is {1, 2, 3, 4, 5}. Find the range of f(x) = x / (x + 1).

Answer: Step 1: Substitute each domain value in x / (x + 1)

Step 2: f(1) = 1/2 and f(2) = 2/3

Step 3: f(3) = 3/4 and f(4) = 4/5

Step 4: f(5) = 5/6

Final answer: Range = {1/2, 2/3, 3/4, 4/5, 5/6}.

Question: Q9. Find the range of the function f(x) = sqrt(x^2 + 1).

Answer: Step 1: For every real x, x^2 >= 0

Step 2: So x^2 + 1 >= 1, with equality when x = 0

Step 3: Taking square roots, which keeps the inequality direction, sqrt(x^2 + 1) >= sqrt(1) = 1

Step 4: As |x| grows, sqrt(x^2 + 1) grows without bound, so every value >= 1 is attained

Final answer: Range = {y in R : y >= 1} = [1, infinity).

Question: Q10. Find the domain and range of the function f(x) = 1 / (1 + x^2).

Answer: Step 1: The denominator 1 + x^2 is always >= 1, so it is never zero, and f is defined for all real x, giving domain = R

Step 2: Since x^2 >= 0, we have 1 + x^2 >= 1, so 1 / (1 + x^2) <= 1, with the value 1 attained at x = 0

Step 3: The fraction is always positive, and it gets close to 0 as |x| grows, but never reaches 0

Step 4: Every value between 0 and 1 is attained for a suitable x

Final answer: Domain = R and Range = {y in R : 0 < y <= 1} = (0, 1].

Question: Q11. Find the domain and range of f(x) = 1 / sqrt(25 - x).

Answer: Step 1: The expression under the root must be positive, because it is also in a denominator and cannot be zero or negative

Step 2: So 25 - x > 0, which gives x < 25, and this is the domain

Step 3: sqrt(25 - x) takes every positive value, large when x is very negative and close to 0 when x is close to 25

Step 4: Therefore 1 / sqrt(25 - x) takes every positive value, from close to 0 up to arbitrarily large values

Final answer: Domain = {x in R : x < 25} = (-infinity, 25) and Range = {y in R : y > 0} = (0, infinity).

Question: Q12(i). A function is defined by f(x) = ax + b. The images of 1 and 5 are -2 and 10 respectively. Find the values of a and b.

Answer: Step 1: The image of 1 is -2, so f(1) = a + b = -2

Step 2: The image of 5 is 10, so f(5) = 5a + b = 10

Step 3: Subtract the first equation from the second, giving 4a = 12, so a = 3

Step 4: Put a = 3 in a + b = -2, giving b = -2 - 3 = -5

Final answer: a = 3 and b = -5, so f(x) = 3x - 5.

Question: Q12(ii). For f(x) = ax + b with a and b as in part (i), the domain of f is {1, 3, 5}. Find the range of f.

Answer: Step 1: From part (i), f(x) = 3x - 5

Step 2: f(1) = 3 - 5 = -2 and f(3) = 9 - 5 = 4

Step 3: f(5) = 15 - 5 = 10

Final answer: Range = {-2, 4, 10}.

Question: Q13. If f(x) = x^3 - ax^2 + bx + 1, find the values of a and b, where f(2) = -3 and f(-1) = 0.

Answer: Step 1: f(2) = 8 - 4a + 2b + 1 = -3, which simplifies to -4a + 2b = -12, or -2a + b = -6

Step 2: f(-1) = -1 - a - b + 1 = 0, which simplifies to -a - b = 0, so b = -a

Step 3: Substitute b = -a in -2a + b = -6, giving -2a - a = -6, so -3a = -6 and a = 2

Step 4: Then b = -a = -2

Step 5: Check, f(2) = 8 - 8 - 4 + 1 = -3 and f(-1) = -1 - 2 + 2 + 1 = 0, both correct

Final answer: a = 2 and b = -2.

Exercise 8.2

Question: Q1(i). If f(x) = -5x + 1, find f^-1(36).

Answer: Step 1: First find the inverse, put y = -5x + 1

Step 2: Rearrange, 5x = 1 - y, so x = (1 - y) / 5, hence f^-1(x) = (1 - x) / 5

Step 3: Substitute x = 36, giving f^-1(36) = (1 - 36) / 5 = -35 / 5

Final answer: f^-1(36) = -7.

Question: Q1(ii). If f(x) = -5x + 1, find f^-1(-14).

Answer: Step 1: Use f^-1(x) = (1 - x) / 5 from part (i)

Step 2: Substitute x = -14, giving f^-1(-14) = (1 - (-14)) / 5 = 15 / 5

Final answer: f^-1(-14) = 3.

Question: Q1(iii). If f(x) = -5x + 1, find f^-1(0).

Answer: Step 1: Use f^-1(x) = (1 - x) / 5

Step 2: Substitute x = 0, giving f^-1(0) = (1 - 0) / 5

Final answer: f^-1(0) = 1/5.

Question: Q1(iv). If f(x) = -5x + 1, find f^-1(a).

Answer: Step 1: Use f^-1(x) = (1 - x) / 5

Step 2: Replace x by the literal a, giving f^-1(a) = (1 - a) / 5

Final answer: f^-1(a) = (1 - a) / 5.

Question: Q2(i). Given g(t) = 1 / (t - 5), t != 5, find g^-1(1/2).

Answer: Step 1: First find the inverse, put y = 1 / (t - 5)

Step 2: Take reciprocals, t - 5 = 1 / y, so t = 5 + 1 / y, hence g^-1(x) = 5 + 1 / x for x != 0

Step 3: Substitute x = 1/2, giving g^-1(1/2) = 5 + 1 / (1/2) = 5 + 2

Final answer: g^-1(1/2) = 7.

Question: Q2(ii). Given g(t) = 1 / (t - 5), t != 5, find g^-1(2).

Answer: Step 1: Use g^-1(x) = 5 + 1 / x

Step 2: Substitute x = 2, giving g^-1(2) = 5 + 1/2

Final answer: g^-1(2) = 11/2 = 5.5.

Question: Q2(iii). Given g(t) = 1 / (t - 5), t != 5, find g^-1(-1).

Answer: Step 1: Use g^-1(x) = 5 + 1 / x

Step 2: Substitute x = -1, giving g^-1(-1) = 5 + 1 / (-1) = 5 - 1

Final answer: g^-1(-1) = 4.

Question: Q2(iv). Given g(t) = 1 / (t - 5), t != 5, find g^-1(a).

Answer: Step 1: Use g^-1(x) = 5 + 1 / x, which needs a != 0

Step 2: Replace x by a, giving g^-1(a) = 5 + 1 / a

Final answer: g^-1(a) = 5 + 1/a, for a != 0.

Question: Q3(i). Find the inverse of f(x) = 12 - (1/2)x.

Answer: Step 1: Put y = 12 - x/2

Step 2: Rearrange, x/2 = 12 - y, so x = 24 - 2y

Step 3: Replace y by x to write the inverse as a function of x

Final answer: f^-1(x) = 24 - 2x.

Question: Q3(ii). Find the inverse of f(x) = (1/2)(x - 3).

Answer: Step 1: Put y = (x - 3) / 2

Step 2: Multiply by 2, 2y = x - 3, so x = 2y + 3

Step 3: Replace y by x

Final answer: f^-1(x) = 2x + 3.

Question: Q3(iii). Find the inverse of f(x) = (2x + 1) / 5.

Answer: Step 1: Put y = (2x + 1) / 5

Step 2: Multiply by 5, 5y = 2x + 1, so 2x = 5y - 1 and x = (5y - 1) / 2

Step 3: Replace y by x

Final answer: f^-1(x) = (5x - 1) / 2.

Question: Q3(iv). Find the inverse of f(x) = (5/9)(x - 32).

Answer: Step 1: Put y = (5/9)(x - 32)

Step 2: Multiply by 9/5, (9y) / 5 = x - 32, so x = (9y) / 5 + 32

Step 3: Replace y by x

Final answer: f^-1(x) = (9x) / 5 + 32.

Question: Q3(v). Find the inverse of g(x) = 180(x - 2).

Answer: Step 1: Put y = 180(x - 2)

Step 2: Divide by 180, y / 180 = x - 2, so x = y / 180 + 2

Step 3: Replace y by x

Final answer: g^-1(x) = x / 180 + 2.

Question: Q3(vi). Find the inverse of h(x) = 2 pi x.

Answer: Step 1: Put y = 2 pi x

Step 2: Divide by 2 pi, giving x = y / (2 pi)

Step 3: Replace y by x

Final answer: h^-1(x) = x / (2 pi).

Question: Q3(vii). Find the inverse of f(t) = t^2 + 5, for t >= 0.

Answer: Step 1: Put y = t^2 + 5, where y >= 5 because t >= 0

Step 2: Rearrange, t^2 = y - 5, so t = sqrt(y - 5), taking the positive root because t >= 0

Step 3: Replace y by x

Final answer: f^-1(x) = sqrt(x - 5), for x >= 5.

Question: Q3(viii). Find the inverse of f(t) = 5 sqrt(t), for t >= 0.

Answer: Step 1: Put y = 5 sqrt(t), where y >= 0

Step 2: Divide by 5, sqrt(t) = y / 5, then square both sides, t = y^2 / 25

Step 3: Replace y by x

Final answer: f^-1(x) = x^2 / 25, for x >= 0.

Question: Q3(ix). Find the inverse of f(t) = (t - 5)^3.

Answer: Step 1: Put y = (t - 5)^3

Step 2: Take the cube root of both sides, cube_root(y) = t - 5, so t = cube_root(y) + 5

Step 3: Replace y by x

Final answer: f^-1(x) = cube_root(x) + 5.

Question: Q3(x). Find the inverse of f(t) = cube_root(t + 1).

Answer: Step 1: Put y = cube_root(t + 1)

Step 2: Cube both sides, y^3 = t + 1, so t = y^3 - 1

Step 3: Replace y by x

Final answer: f^-1(x) = x^3 - 1.

Question: Q3(xi). Find the inverse of g(x) = 1 / (x - 3), for x != 3.

Answer: Step 1: Put y = 1 / (x - 3)

Step 2: Take reciprocals, x - 3 = 1 / y, so x = 1 / y + 3, with y != 0

Step 3: Replace y by x

Final answer: g^-1(x) = 1/x + 3, for x != 0.

Question: Q3(xii). Find the inverse of g(x) = 1 / (2x + 1), for x != -1/2.

Answer: Step 1: Put y = 1 / (2x + 1)

Step 2: Take reciprocals, 2x + 1 = 1 / y, so 2x = 1 / y - 1 and x = (1 - y) / (2y)

Step 3: Replace y by x

Final answer: g^-1(x) = (1 - x) / (2x), for x != 0.

Question: Q4(i). Find the inverse of f(x) = (x - 1) / (x - 3), x != 3, and determine the domain and range of f^-1.

Answer: Step 1: Put y = (x - 1) / (x - 3) and cross-multiply, y(x - 3) = x - 1, so xy - 3y = x - 1

Step 2: Collect the x terms, xy - x = 3y - 1, so x(y - 1) = 3y - 1 and x = (3y - 1) / (y - 1)

Step 3: Hence f^-1(x) = (3x - 1) / (x - 1), which is undefined at x = 1

Step 4: The domain of f^-1 is the range of f, and its range is the domain of f

Final answer: f^-1(x) = (3x - 1) / (x - 1), domain of f^-1 = R - {1} and range of f^-1 = R - {3}.

Question: Q4(ii). Find the inverse of f(x) = 5x + 7, and determine the domain and range of f^-1.

Answer: Step 1: Put y = 5x + 7

Step 2: Rearrange, 5x = y - 7, so x = (y - 7) / 5

Step 3: Hence f^-1(x) = (x - 7) / 5, a linear function defined for all real x

Final answer: f^-1(x) = (x - 7) / 5, domain of f^-1 = R and range of f^-1 = R.

Question: Q4(iii). Find the inverse of f(x) = 1 / (x + 5), x != -5, and determine the domain and range of f^-1.

Answer: Step 1: Put y = 1 / (x + 5)

Step 2: Take reciprocals, x + 5 = 1 / y, so x = 1 / y - 5, with y != 0

Step 3: Hence f^-1(x) = 1/x - 5

Step 4: Its domain excludes 0 and its range excludes -5, the value excluded from the domain of f

Final answer: f^-1(x) = 1/x - 5, domain of f^-1 = R - {0} and range of f^-1 = R - {-5}.

Question: Q4(iv). Find the inverse of f(x) = (x - 5)^2, x >= 5, and determine the domain and range of f^-1.

Answer: Step 1: Put y = (x - 5)^2, where x >= 5 so y >= 0

Step 2: Take square roots, sqrt(y) = x - 5, taking the positive root because x - 5 >= 0, so x = sqrt(y) + 5

Step 3: Hence f^-1(x) = sqrt(x) + 5

Step 4: The inverse needs x >= 0, and its values are >= 5

Final answer: f^-1(x) = sqrt(x) + 5, domain of f^-1 = {x : x >= 0} and range of f^-1 = {y : y >= 5}.

Question: Q4(v). Find the inverse of f(x) = sqrt(x + 7), x >= -7, and determine the domain and range of f^-1.

Answer: Step 1: Put y = sqrt(x + 7), where y >= 0

Step 2: Square both sides, y^2 = x + 7, so x = y^2 - 7

Step 3: Hence f^-1(x) = x^2 - 7, but only for x >= 0, the range of f

Step 4: Its values start at -7 when x = 0 and increase from there

Final answer: f^-1(x) = x^2 - 7 for x >= 0, domain of f^-1 = {x : x >= 0} and range of f^-1 = {y : y >= -7}.

Question: Q4(vi). Find the inverse of f(x) = (x - 1) / (x - 4), x != 4, and determine the domain and range of f^-1.

Answer: Step 1: Put y = (x - 1) / (x - 4) and cross-multiply, y(x - 4) = x - 1, so xy - 4y = x - 1

Step 2: Collect the x terms, xy - x = 4y - 1, so x(y - 1) = 4y - 1 and x = (4y - 1) / (y - 1)

Step 3: Hence f^-1(x) = (4x - 1) / (x - 1), undefined at x = 1

Final answer: f^-1(x) = (4x - 1) / (x - 1), domain of f^-1 = R - {1} and range of f^-1 = R - {4}.

Exercise 8.3

Question: Q1(i). Sketch the graph of y = 5x + 2.

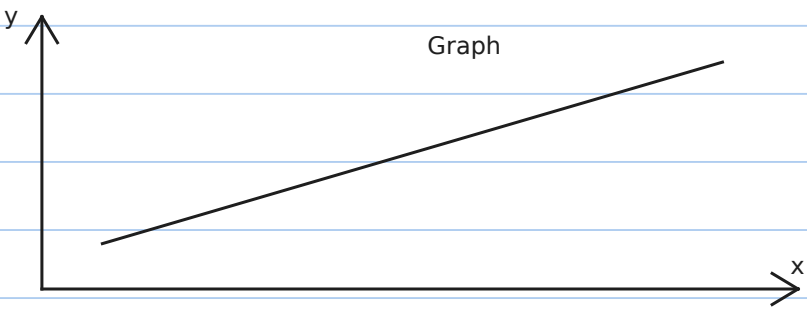
Answer: Step 1: This is a linear function, so its graph is a straight line with slope 5 and y-intercept 2

Step 2: Put x = 0, giving the y-intercept point (0, 2)

Step 3: Put y = 0, giving 5x = -2, so x = -2/5, the x-intercept point (-2/5, 0)

Step 4: Make a short table, x = -1 gives y = -3, x = 1 gives y = 7, and join the points with a straight line rising steeply to the right

Final answer: A straight line through (-2/5, 0) and (0, 2) with slope 5.



Question: Q1(ii). Sketch the graph of y = x^2 + 5.

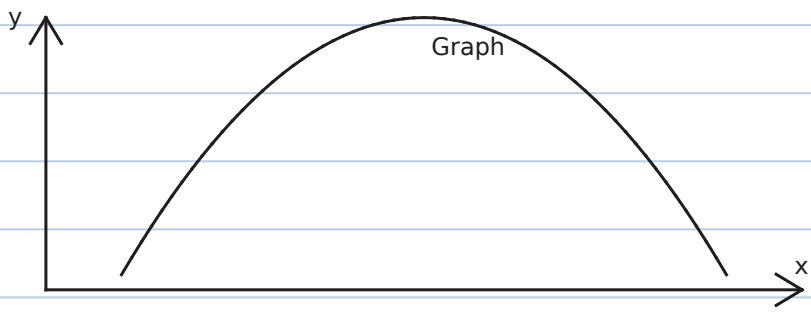
Answer: Step 1: This is a quadratic, so the graph is a parabola opening upward because the coefficient of x^2 is positive

Step 2: Complete the form, y = (x - 0)^2 + 5, so the vertex is (0, 5)

Step 3: The y-intercept is (0, 5), and there is no x-intercept because x^2 + 5 = 0 has no real solution

Step 4: Extra points, x = 1 and x = -1 give y = 6, x = 2 and x = -2 give y = 9, and the graph is symmetric about the y-axis

Final answer: An upward parabola with vertex (0, 5), axis x = 0 and range y >= 5.



Question: Q1(iii). Sketch the graph of y = x^5.

Answer: Step 1: x^5 is an odd power, so the graph passes through the origin and is symmetric about the origin

Step 2: Make a table, x = -2, -1, 0, 1, 2 give y = -32, -1, 0, 1, 32

Step 3: For 0 < x < 1 the values are small, so the curve is flat near the origin, and it rises very steeply for x > 1

Step 4: Join the points with a smooth increasing curve from the third quadrant to the first quadrant

Final answer: An increasing curve through (-1, -1), (0, 0), (1, 1), flat near the origin and steep for large |x|.

Question: Q2(a). Sketch the graph of the square root function obtained from x^2 + y^2 = 25, namely y = sqrt(25 - x^2).

Answer: Step 1: From x^2 + y^2 = 25, y^2 = 25 - x^2 and taking the positive root gives y = sqrt(25 - x^2) >= 0

Step 2: The radicand needs 25 - x^2 >= 0, so the domain is -5 <= x <= 5

Step 3: Key points are (-5, 0), (0, 5), (5, 0), and also (-3, 4) and (3, 4)

Step 4: Join them with the upper half of the circle of radius 5 centred at the origin

Final answer: The upper semicircle of radius 5, from (-5, 0) up to (0, 5) and down to (5, 0).

Question: Q2(b). Sketch the graph of the square root function obtained from x^2 + y^2 = 25, namely y = -sqrt(25 - x^2).

Answer: Step 1: Taking the negative root in y^2 = 25 - x^2 gives y = -sqrt(25 - x^2) <= 0

Step 2: The domain is again -5 <= x <= 5

Step 3: Key points are (-5, 0), (0, -5), (5, 0), and also (-3, -4) and (3, -4)

Step 4: Join them with the lower half of the circle of radius 5 centred at the origin

Final answer: The lower semicircle of radius 5, from (-5, 0) down to (0, -5) and up to (5, 0).

Question: Q3(i). Sketch the graph of the quadratic function y = x^2 - 2x + 1.

Answer: Step 1: Factorise or complete the square, y = (x - 1)^2

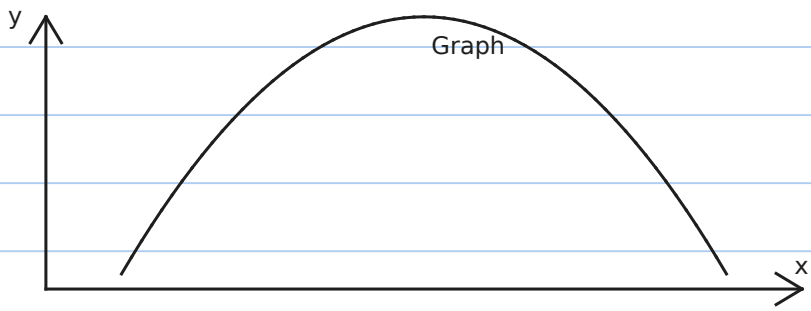
Step 2: The vertex is (1, 0) and the parabola opens upward because the

coefficient of x^2 is positive

Step 3: The x-intercept is the single repeated point (1, 0) and the y-intercept is (0, 1)

Step 4: Extra points, x = -1 and x = 3 give y = 4, x = -2 and x = 4 give y = 9

Final answer: An upward parabola touching the x-axis at its vertex (1, 0), through (0, 1).



Question: Q3(ii). Sketch the graph of the quadratic function y = -3x^2 + 6x.

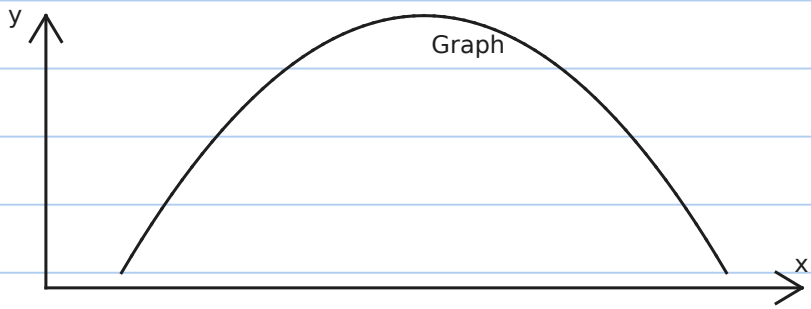
Answer: Step 1: Complete the square, y = -3(x^2 - 2x) = -3(x - 1)^2 + 3

Step 2: The vertex is (1, 3) and the parabola opens downward because the coefficient of x^2 is negative

Step 3: Factorise for the x-intercepts, y = -3x(x - 2), so the graph cuts the x-axis at (0, 0) and (2, 0), and the y-intercept is (0, 0)

Step 4: Extra points, x = -1 and x = 3 give y = -9

Final answer: A downward parabola with vertex (1, 3), through (0, 0) and (2, 0).



Question: Q4(i). Sketch the graph of y = x^2 + 3x + 2 using factors.

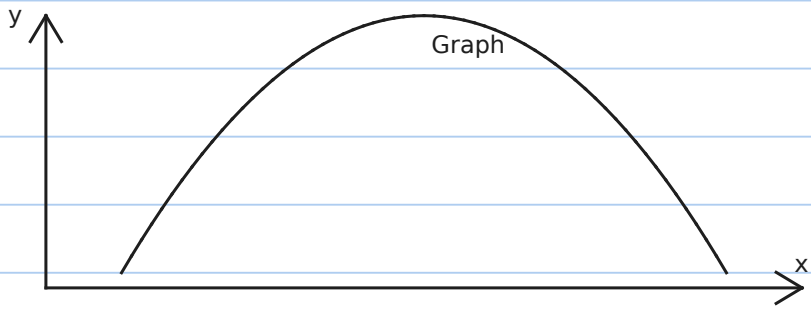
Answer: Step 1: Write in factor form, y = (x + 1)(x + 2)

Step 2: Put y = 0, giving x = -1 and x = -2, so the x-intercepts are (-1, 0) and (-2, 0)

Step 3: Put x = 0, giving y = 2, so the y-intercept is (0, 2)

Step 4: The coefficient of x^2 is positive, so the parabola opens upward, and its vertex is midway between the intercepts at x = -3/2, where y = (-1/2)(1/2) = -1/4, giving vertex (-3/2, -1/4)

Final answer: An upward parabola through (-2, 0), (-1, 0), (0, 2) with vertex (-3/2, -1/4).



Question: Q4(ii). Sketch the graph of y = -x^2 - 5x using factors.

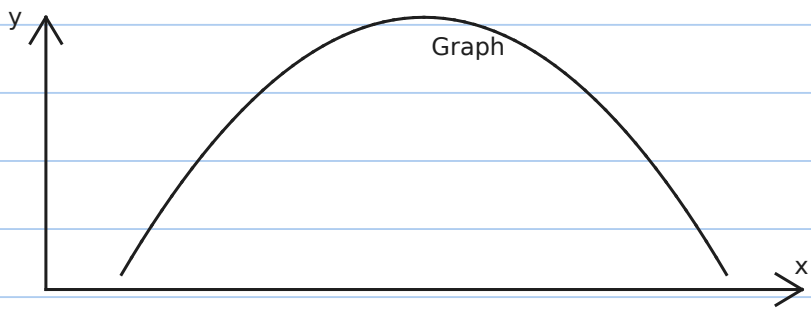
Answer: Step 1: Write in factor form, y = -x(x + 5)

Step 2: Put y = 0, giving x = 0 and x = -5, so the x-intercepts are (0, 0) and (-5, 0)

Step 3: Put x = 0, giving y = 0, so the y-intercept is also (0, 0)

Step 4: The coefficient of x^2 is negative, so the parabola opens downward, and its vertex is midway between the intercepts at x = -5/2, where y = -(-5/2)(-5/2 + 5) = 25/4, giving vertex (-5/2, 25/4)

Final answer: A downward parabola through (-5, 0), (0, 0) with vertex (-5/2, 25/4).



Question: Q5. Find the equation of the function of the type y = f(x) = ax^2 + bx + c which cuts the x-axis at (-4, 0) and (3, 0) and also passes through the point (2, -4).

Answer: Step 1: Use the factor form with intercepts p = -4 and q = 3, so y = a(x + 4)(x - 3)

Step 2: The extra point (2, -4) lies on the curve, so substitute x = 2, y = -4

Step 3: -4 = a(2 + 4)(2 - 3) = a(6)(-1) = -6a, so a = 2/3

Step 4: Expand, y = (2/3)(x^2 + x - 12) = (2/3)x^2 + (2/3)x - 8

Final answer: y = (2/3)x^2 + (2/3)x - 8.

Question: Q6. Find the equation of the graph of the function of the type y = ax^2 + bx + c which crosses the x-axis at (-8, 0) and (9, 0) and also passes through (5, 10).

Answer: Step 1: Use the factor form with intercepts p = -8 and q = 9, so y = a(x + 8)(x - 9)

Step 2: Substitute the extra point (5, 10), giving 10 = a(5 + 8)(5 - 9) =

a(13)(-4) = -52a

Step 3: So a = -10/52 = -5/26

Step 4: Expand, y = (-5/26)(x^2 - x - 72) = (-5/26)x^2 + (5/26)x + 180/13

Final answer: y = (-5/26)x^2 + (5/26)x + 180/13.

Question: Q7. Find the equation of the function of the type y = 3x^2 + bx + c of the parabola which cuts the x-axis at (-3, 0) and (5, 0).

Answer: Step 1: Use the factor form with the given leading coefficient 3, so y = 3(x + 3)(x - 5)

Step 2: Expand the brackets, (x + 3)(x - 5) = x^2 - 2x - 15

Step 3: Multiply by 3, giving y = 3x^2 - 6x - 45

Step 4: Compare with y = 3x^2 + bx + c to read off b and c

Final answer: y = 3x^2 - 6x - 45, so b = -6 and c = -45.

Question: Q8. Find the equation of the graph of the function of the type y = 2x^2 + bx + c which crosses the x-axis at (-4, 0) and (5, 0).

Answer: Step 1: Use the factor form with the given leading coefficient 2, so y = 2(x + 4)(x - 5)

Step 2: Expand the brackets, (x + 4)(x - 5) = x^2 - x - 20

Step 3: Multiply by 2, giving y = 2x^2 - 2x - 40

Step 4: Compare with y = 2x^2 + bx + c

Final answer: y = 2x^2 - 2x - 40, so b = -2 and c = -40.

Exercise 8.4

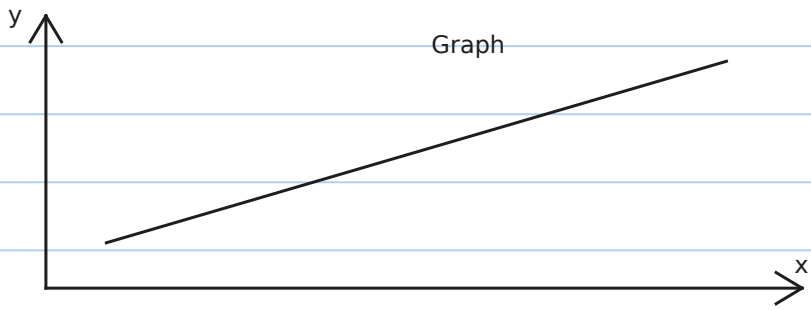
Question: Q1(i). Graphically find the points of intersection of the linear function f(x) = x + 7 with the co-ordinate axes.

Answer: Step 1: For the y-axis intersection put x = 0, giving y = 7, the point (0, 7)

Step 2: For the x-axis intersection put y = 0, giving 0 = x + 7, so x = -7, the point (-7, 0)

Step 3: Plot these two points and join them with a straight line of slope 1, which is the graph read off in the graphical method

Final answer: The line meets the axes at (-7, 0) and (0, 7).



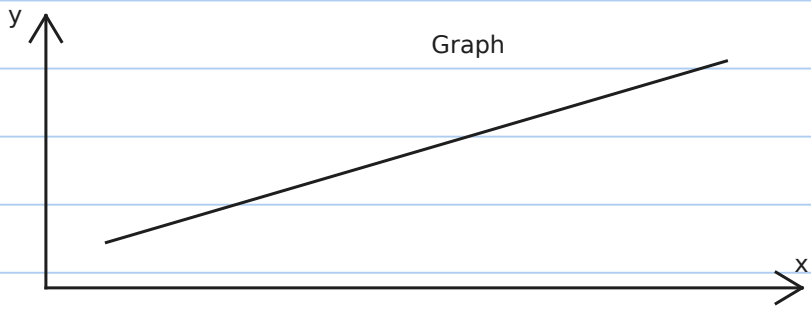
Question: Q1(ii). Graphically find the points of intersection of the linear function f(x) = -x + 5 with the co-ordinate axes.

Answer: Step 1: Put x = 0, giving y = 5, the point (0, 5)

Step 2: Put y = 0, giving 0 = -x + 5, so x = 5, the point (5, 0)

Step 3: Join the two points with a straight line of slope -1

Final answer: The line meets the axes at (5, 0) and (0, 5).



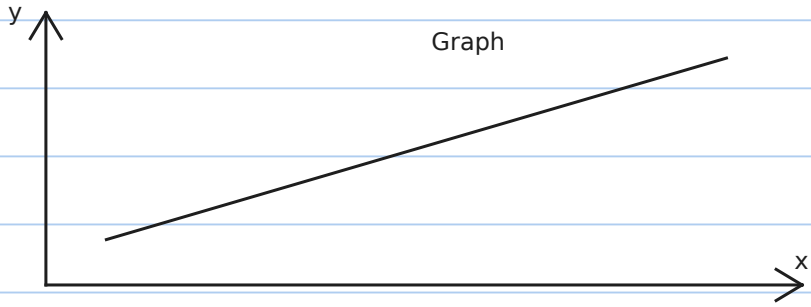
Question: Q1(iii). Graphically find the points of intersection of the linear function f(x) = 3x + 2 with the co-ordinate axes.

Answer: Step 1: Put x = 0, giving y = 2, the point (0, 2)

Step 2: Put y = 0, giving 0 = 3x + 2, so x = -2/3, the point (-2/3, 0)

Step 3: Join the points with a straight line of slope 3

Final answer: The line meets the axes at (-2/3, 0) and (0, 2).



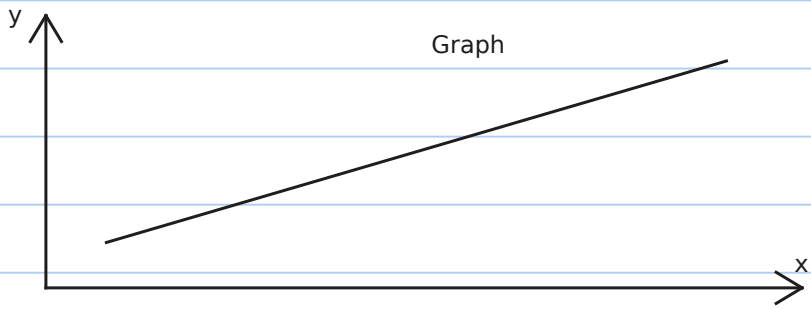
Question: Q1(iv). Graphically find the points of intersection of the linear function f(x) = -2x + 4 with the co-ordinate axes.

Answer: Step 1: Put x = 0, giving y = 4, the point (0, 4)

Step 2: Put y = 0, giving 0 = -2x + 4, so x = 2, the point (2, 0)

Step 3: Join the points with a straight line of slope -2

Final answer: The line meets the axes at (2, 0) and (0, 4).



Question: Q2(i). Find the point of intersection of 2x + y = 5 and x + y = 4 graphically, where y is a function of x.

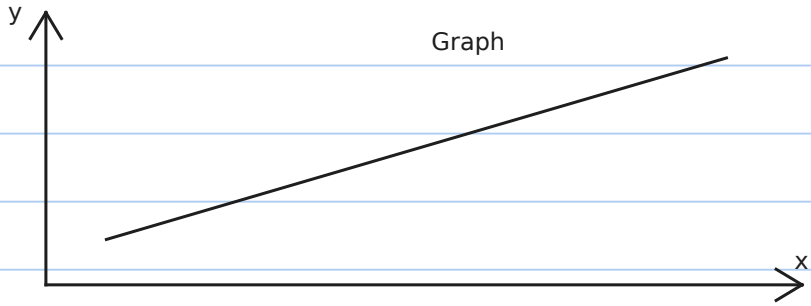
Answer: Step 1: Write both as functions, y = 5 - 2x and y = 4 - x, and draw both lines

Step 2: At the intersection the y values are equal, so 5 - 2x = 4 - x

Step 3: Rearrange, 5 - 4 = 2x - x, giving x = 1

Step 4: Put x = 1 in y = 4 - x, giving y = 3, which is the point where the two drawn lines cross

Final answer: The point of intersection is (1, 3).



Question: Q2(ii). Find the point of intersection of 3x - 2y = 4 and x + 4y = 6 graphically, where y is a function of x.

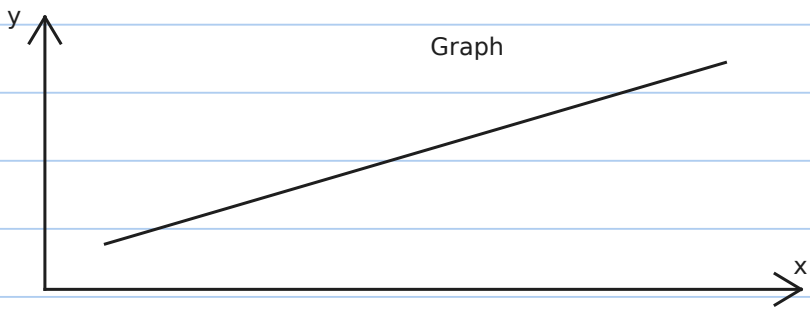
Answer: Step 1: Write both as functions, y = (3x - 4) / 2 and y = (6 - x) / 4, and draw both lines

Step 2: At the intersection, (3x - 4) / 2 = (6 - x) / 4, so 2(3x - 4) = 6 - x

Step 3: Expand and solve, 6x - 8 = 6 - x, so 7x = 14 and x = 2

Step 4: Put x = 2 in x + 4y = 6, giving 4y = 4, so y = 1

Final answer: The point of intersection is (2, 1).



Question: Q3(i). Find the points of intersection of f(x) = x + 2 and g(x) = x^2 - 4x + 6 graphically.

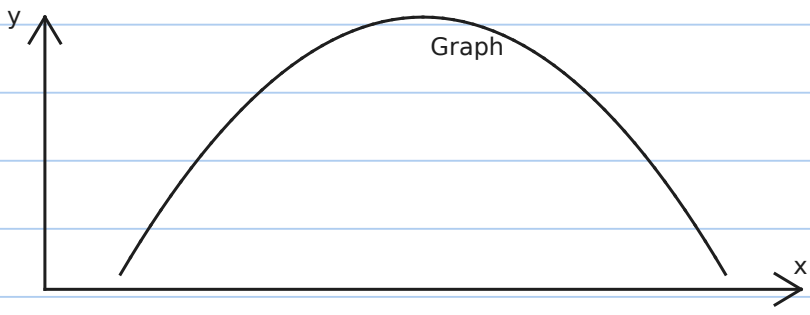
Answer: Step 1: Draw the line y = x + 2 and the upward parabola y = x^2 - 4x + 6, whose vertex is (2, 2)

Step 2: At an intersection x + 2 = x^2 - 4x + 6

Step 3: Rearrange, x^2 - 5x + 4 = 0, and factorise, (x - 1)(x - 4) = 0, so x = 1 or x = 4

Step 4: From the line, x = 1 gives y = 3 and x = 4 gives y = 6

Final answer: The points of intersection are (1, 3) and (4, 6).



Question: Q3(ii). Find the points of intersection of f(x) = x + 4 and g(x) = x^2 - 6x + 10 graphically.

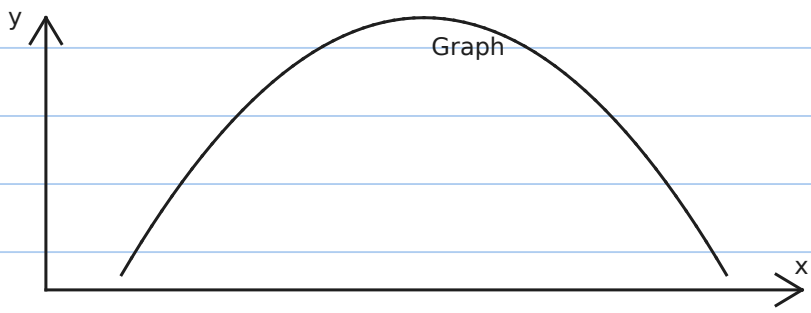
Answer: Step 1: Draw the line y = x + 4 and the upward parabola y = x^2 - 6x + 10, whose vertex is (3, 1)

Step 2: At an intersection x + 4 = x^2 - 6x + 10

Step 3: Rearrange, x^2 - 7x + 6 = 0, and factorise, (x - 1)(x - 6) = 0, so x = 1 or x = 6

Step 4: From the line, x = 1 gives y = 5 and x = 6 gives y = 10

Final answer: The points of intersection are (1, 5) and (6, 10).



Question: Q4. The paths of two aeroplanes A and B in the plane are determined by the straight lines 2x - y = 6 and 3x + y = 4 respectively. Graphically find the point where the two paths cross each other.

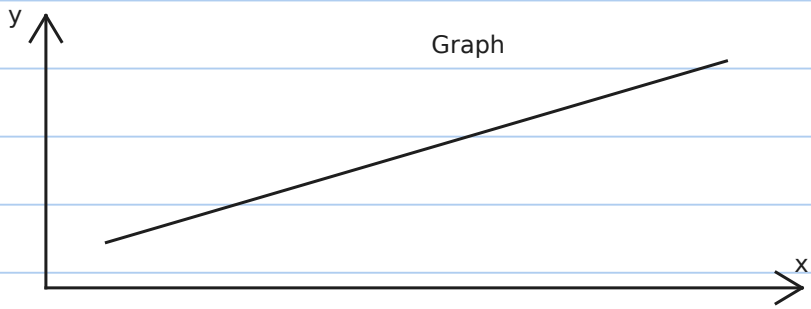
Answer: Step 1: Write the paths as y = 2x - 6 and y = 4 - 3x and draw both lines

Step 2: At the crossing point 2x - 6 = 4 - 3x

Step 3: Solve, 5x = 10, so x = 2

Step 4: Put x = 2 in y = 2x - 6, giving y = -2, and check in the second line, 3(2) + (-2) = 4, which is correct

Final answer: The two paths cross at the point (2, -2).



Question: Q5. If the sum of two numbers is 6 and the square of the first number is greater than 6 by the second number, solve graphically and find the numbers.

Answer: Step 1: Let the first number be x and the second be y, then the sum condition gives x + y = 6, so y = 6 - x, a straight line

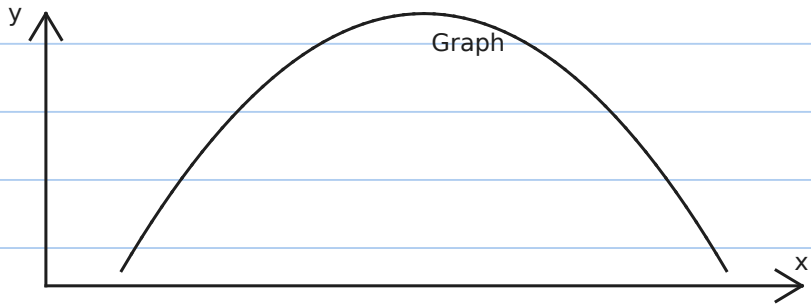
Step 2: The second condition says x^2 = y + 6, so y = x^2 - 6, an upward parabola with vertex (0, -6)

Step 3: Draw both graphs and set them equal at the intersections, x^2 - 6 = 6 - x, so x^2 + x - 12 = 0

Step 4: Factorise, (x + 4)(x - 3) = 0, giving x = 3 or x = -4

Step 5: For x = 3, y = 6 - 3 = 3, and for x = -4, y = 6 - (-4) = 10, and indeed (-4)^2 = 16 = 10 + 6

Final answer: The graphs meet at (3, 3) and (-4, 10), so the numbers are 3 and 3, or -4 and 10.



Review Exercise 8

Question: Q1(i). Select the correct answer: If f : A -> B is a function, then it is an onto function if: (a) Range = B (b) Range is a proper subset of B (c) Image is not repeated (d) Domain != A.

Answer: Step 1: An onto, or surjective, function is one in which every element of the co-domain B is the image of some element of A

Step 2: That is exactly the statement that the set of all images, the range, fills the whole of B

Final answer: (a) Range = B.

Question: Q1(ii). Select the correct answer: The graph of y = x^6 is symmetric to: (a) x-axis (b) y-axis (c) origin (d) none.

Answer: Step 1: Replace x by -x, giving (-x)^6 = x^6, so the equation is unchanged

Step 2: Unchanged under x -> -x means symmetry about the y-axis

Final answer: (b) y-axis.

Question: Q1(iii). Select the correct answer: A one-to-one function is also called a: (a) Injective (b) Surjective (c) Bijective (d) Inverse function.

Answer: Step 1: One-to-one means distinct inputs give distinct outputs

Step 2: That is the definition of an injective function

Final answer: (a) Injective.

Question: Q1(iv). Select the correct answer: The inverse of a function exists only if it is: (a) Injective (b) Bijective (c) Surjective (d) all of these.

Answer: Step 1: For the reversed mapping to be a function, each output

must come from exactly one input, so the function must be one-to-one

Step 2: It must also cover the whole co-domain, so it must be onto, and one-to-one plus onto is bijective

Final answer: (b) Bijective.

Question: Q1(v). Select the correct answer: The function f = {(x, y) | y = mx + c}, where m and c are real numbers, is: (a) Linear (b) Quadratic (c) A circle (d) A point.

Answer: Step 1: The equation y = mx + c is the standard form of a straight line

Step 2: A function whose graph is a straight line is a linear function

Final answer: (a) Linear.

Question: Q1(vi). Select the correct answer: The function f = {(x, y) | y = ax^2 + bx + c, a != 0} is: (a) Linear (b) Quadratic (c) A circle (d) A point.

Answer: Step 1: The highest power of x in y = ax^2 + bx + c with a != 0 is 2

Step 2: A function of degree 2 is a quadratic function, with a parabola as its graph

Final answer: (b) Quadratic.

Question: Q1(vii). Select the correct answer: The range of f(x) = x^2 + 3 for all x in R is: (a) R (b) R- (c) (3, infinity) (d) [3, infinity).

Answer: Step 1: x^2 >= 0 for all real x, so x^2 + 3 >= 3

Step 2: The value 3 itself is attained at x = 0, so 3 is included in the range

Final answer: (d) [3, infinity).

Question: Q1(viii). Select the correct answer: The graph of y = x^n is

symmetric to the ___ if n is an odd integer: (a) x-axis (b) y-axis (c) origin (d) none.

Answer: Step 1: For odd n, (-x)^n = -x^n, so the point (-x, -y) lies on the graph whenever (x, y) does

Step 2: That is symmetry about the origin

Final answer: (c) origin.

Question: Q1(ix). Select the correct answer: The graph of a linear function is: (a) Circle (b) Straight line (c) Parabola (d) Triangle.

Answer: Step 1: A linear function has the form y = mx + c

Step 2: Its graph is by definition a straight line

Final answer: (b) Straight line.

Question: Q1(x). Select the correct answer: If f is a function from A to B, the domain of f is equal to: (a) Any subset of A (b) A x B (c) A (d) B.

Answer: Step 1: In f : A -> B, the set A is the set of all inputs

Step 2: Every element of A must have an image, so the domain is the whole of A, not merely a subset

Final answer: (c) A.

Question: Q1(xi). Select the correct answer: Every function is a: (a) Relation (b) Inverse function (c) One-to-one (d) None of these.

Answer: Step 1: A function is a special kind of relation, one in which each input has exactly one output

Step 2: A function need not be one-to-one and need not have an inverse, so the other options fail in general

Final answer: (a) Relation.

Question: Q1(xii). Select the correct answer: If f and g are equal, where f(x) = 7x - 4 and g(x) = x, then x =: (a) 1/3 (b) 2/3 (c) 1 (d) 4/3.

Answer: Step 1: Equal functions give equal values, so set 7x - 4 = x

Step 2: Rearrange, 6x = 4, so x = 4/6 = 2/3

Final answer: (b) 2/3.

Question: Q1(xiii). Select the correct answer: The domain of the function f(x) = (3x + 1) / (x - 1) is the set of all: (a) real numbers (b) rational numbers (c) complex numbers (d) real numbers except 1.

Answer: Step 1: The function fails only when the denominator x - 1 is zero, that is at x = 1

Step 2: Every other real number is allowed

Final answer: (d) real numbers except 1.

Question: Q1(xiv). Select the correct answer: The function f(x) = |x| - x^2 is: (a) odd (b) Linear (c) Even (d) Neither even nor odd.

Answer: Step 1: Compute f(-x) = |-x| - (-x)^2 = |x| - x^2

Step 2: This equals f(x), which is the test for an even function

Final answer: (c) Even.

Question: Q2(i). Find the domain of f(x) = (2x - 1) / (x + 4).

Answer: Step 1: The function is undefined when the denominator x + 4 is zero

Step 2: x + 4 = 0 gives x = -4

Step 3: Exclude this single value

Final answer: Domain = R - {-4}.

Question: Q2(ii). Find the domain of f(x) = sqrt(x + 3).

Answer: Step 1: The expression inside the square root must be >= 0

Step 2: So x + 3 >= 0, giving x >= -3

Final answer: Domain = {x in R : x >= -3} = [-3, infinity).

Question: Q3. Graphically, find the points of intersection of the functions f(x) = x and g(x) = x^2 - x.

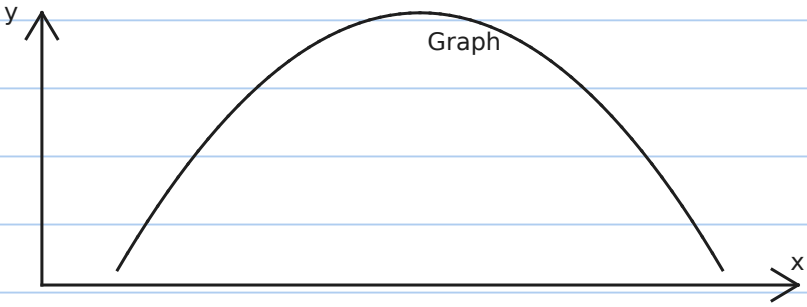
Answer: Step 1: Draw the line y = x and the upward parabola y = x^2 - x = x(x - 1), which meets the x-axis at 0 and 1

Step 2: At an intersection, x = x^2 - x

Step 3: Rearrange, x^2 - 2x = 0, so x(x - 2) = 0, giving x = 0 or x = 2

Step 4: On the line y = x, these give the points (0, 0) and (2, 2)

Final answer: The points of intersection are (0, 0) and (2, 2).



Question: Q4. Find the equation, in the form y = ax^2 + bx + c, of the parabola which cuts the x-axis at (1, 0) and (5, 0) and cuts the y-axis at (0, 15).

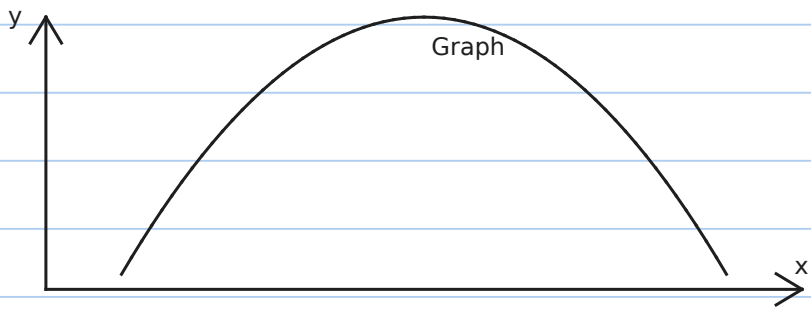
Answer: Step 1: Use the factor form with intercepts 1 and 5, so y = a(x - 1)(x - 5)

Step 2: The y-intercept is 15, so put x = 0, y = 15, giving 15 = a(-1)(-5) = 5a

Step 3: So a = 3

Step 4: Expand, y = 3(x^2 - 6x + 5) = 3x^2 - 18x + 15

Final answer: y = 3x^2 - 18x + 15.



Question: Q5. The function f is defined by f(x) = ax^2 + bx + c. Given that f(0) = 5, f(-1) = 15 and f(1) = 1, find the values of a, b and c.

Answer: Step 1: f(0) = c = 5

Step 2: f(-1) = a - b + c = 15, and with c = 5 this gives a - b = 10

Step 3: f(1) = a + b + c = 1, which gives a + b = -4

Step 4: Add the two equations, 2a = 6, so a = 3, and then b = -4 - 3 = -7

Step 5: Check, f(-1) = 3 + 7 + 5 = 15 and f(1) = 3 - 7 + 5 = 1, both correct

Final answer: a = 3, b = -7 and c = 5, so f(x) = 3x^2 - 7x + 5.

Question: Q6. The function f is defined by f(x) = x^2 - 3x + 5. Find the values of x for which f(x) = f(3).

Answer: Step 1: First compute f(3) = (3)^2 - 3(3) + 5 = 9 - 9 + 5 = 5

Step 2: Set f(x) = 5, so x^2 - 3x + 5 = 5

Step 3: Simplify, x^2 - 3x = 0, and factorise, x(x - 3) = 0

Step 4: So x = 0 or x = 3

Final answer: f(x) = f(3) when x = 0 or x = 3.

Unit 9: Linear Programming

Exercise 9.1

Question: Q1(i). Solve the inequality 2x + 5 < 9, for all x in R, and represent the solution on a number line.

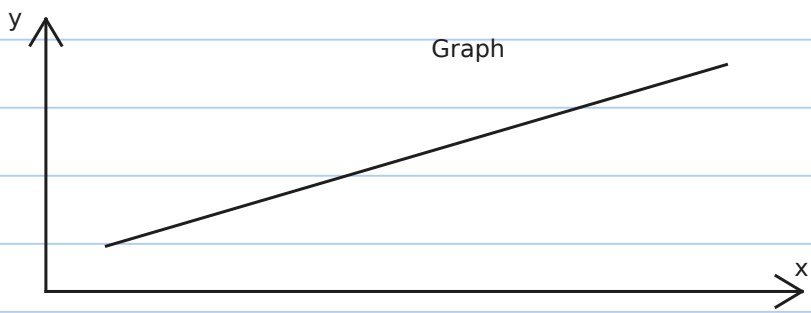
Answer: Step 1: subtract 5 from both sides, 2x + 5 - 5 < 9 - 5, so 2x < 4

Step 2: divide both sides by the positive number 2 (the inequality sign does not change), x < 2

Step 3: write the solution set, {x | x in R and x < 2}, in interval notation (-infinity, 2)

Step 4: on the number line mark an open parenthesis at 2 (2 is not included) and shade everything to the left of 2

Final answer: x < 2



Question: Q1(ii). Solve the inequality 3x - 1 > 10, for all x in R, and represent the solution on a number line.

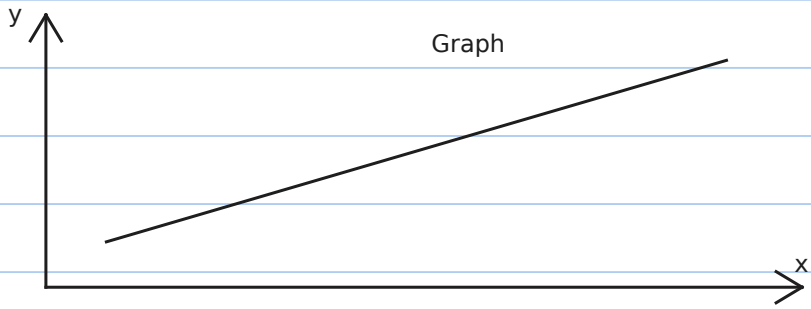
Answer: Step 1: add 1 to both sides, 3x - 1 + 1 > 10 + 1, so 3x > 11

Step 2: divide both sides by the positive number 3, x > 11/3

Step 3: write the solution set, {x | x in R and x > 11/3}, in interval notation (11/3, infinity)

Step 4: on the number line mark an open parenthesis at 11/3 (11/3 is not included) and shade everything to the right

Final answer: x > 11/3



Question: Q1(iii). Solve the inequality 4x - 5 <= 3 + 2x, for all x in R, and represent the solution on a number line.

Answer: Step 1: collect the x terms on the left and the constants on the right, 4x - 2x <= 3 + 5

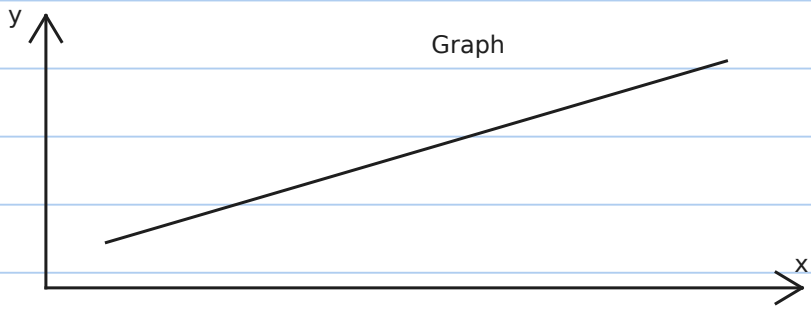
Step 2: simplify, 2x <= 8

Step 3: divide both sides by 2, x <= 4

Step 4: write the solution set, {x | x in R and x <= 4}, in interval notation (-infinity, 4]

Step 5: on the number line mark a square bracket at 4 (4 is included) and shade everything to the left

Final answer: x <= 4



Question: Q1(iv). Solve the inequality 6 < 3(x + 2) < 21, for all x in R, and represent the solution on a number line.

Answer: Step 1: divide all three parts by the positive number 3, 2 < x + 2 < 7

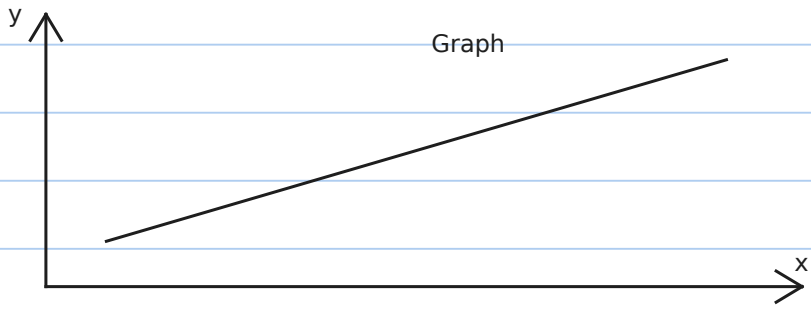
Step 2: subtract 2 from all three parts, 2 - 2 < x < 7 - 2

Step 3: simplify, 0 < x < 5

Step 4: write the solution set, {x | x in R and 0 < x < 5}, in interval notation (0, 5)

Step 5: on the number line mark open parentheses at 0 and 5 and shade the part between them

Final answer: 0 < x < 5



Question: Q2(i). Draw the graph of the linear inequality 2x - y <= 7.

Answer: Step 1: consider the corresponding equation 2x - y = 7

Step 2: find the x-intercept by putting y = 0, x = 7/2, giving the point (7/2, 0)

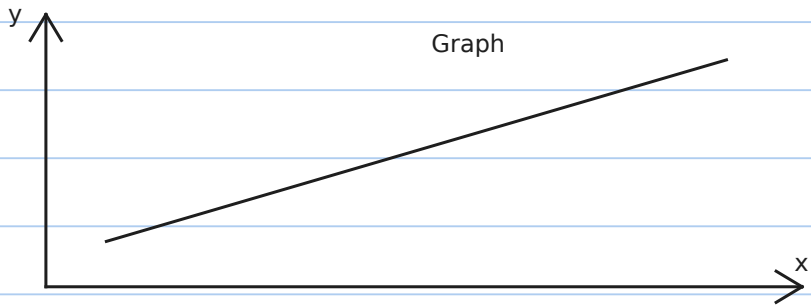
Step 3: find the y-intercept by putting x = 0, y = -7, giving the point (0, -7)

Step 4: draw the straight (solid) line through (7/2, 0) and (0, -7); the line is solid because the inequality includes equality

Step 5: test the origin in the inequality, 2(0) - 0 = 0 <= 7, which is true, so the origin is in the solution region

Step 6: shade the half-plane on the side of the line that contains the origin

Final answer: the solution is the half-plane below-right of the line 2x - y = 7 containing the origin, including the line itself



Question: Q2(ii). Draw the graph of the linear inequality 3x + 4y >= 10.

Answer: Step 1: consider the corresponding equation 3x + 4y = 10

Step 2: find the x-intercept by putting y = 0, x = 10/3, giving the point (10/3, 0)

Step 3: find the y-intercept by putting x = 0, y = 10/4 = 5/2, giving the point (0, 5/2)

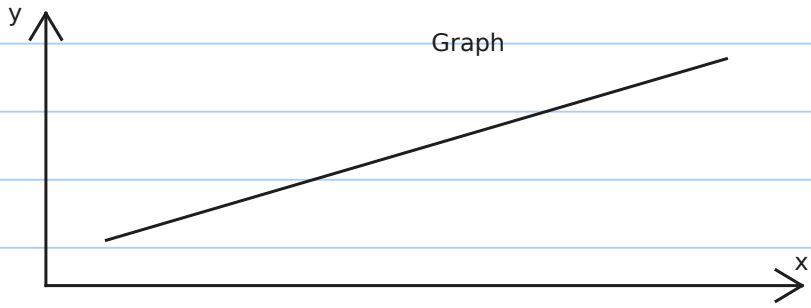
Step 4: draw the solid line through (10/3, 0) and (0, 5/2)

Step 5: test the origin, 3(0) + 4(0) = 0 >= 10, which is false, so the origin is not in the solution region

Step 6: shade the half-plane on the side of the line away from the origin

Final answer: the solution is the half-plane above the line 3x + 4y = 10 not

containing the origin, including the line itself



Question: Q2(iii). Draw the graph of the linear inequality x - 2y >= 5.

Answer: Step 1: consider the corresponding equation x - 2y = 5

Step 2: find the x-intercept by putting y = 0, x = 5, giving the point (5, 0)

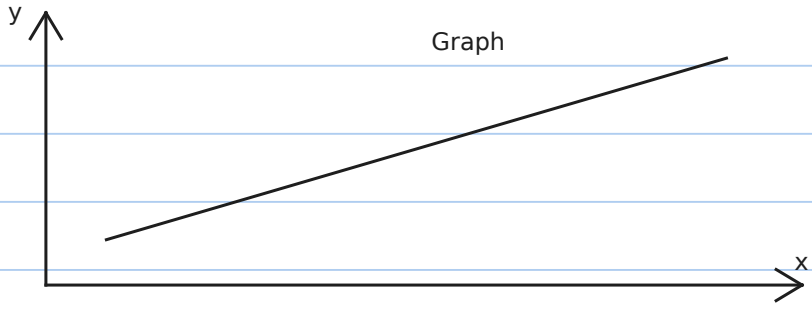
Step 3: find the y-intercept by putting x = 0, y = -5/2, giving the point (0, -5/2)

Step 4: draw the solid line through (5, 0) and (0, -5/2)

Step 5: test the origin, 0 - 0 = 0 >= 5, which is false, so the origin is not in the solution region

Step 6: shade the half-plane on the side of the line away from the origin

Final answer: the solution is the half-plane below-right of the line x - 2y = 5 not containing the origin, including the line itself



Question: Q2(iv). Draw the graph of the linear inequality 2x + y <= 4.

Answer: Step 1: consider the corresponding equation 2x + y = 4

Step 2: find the x-intercept by putting y = 0, x = 2, giving the point (2, 0)

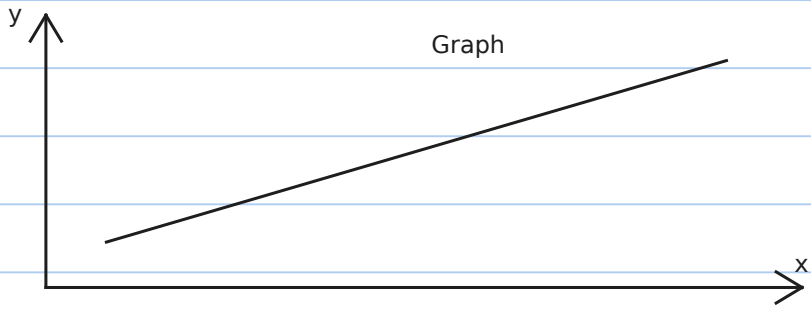
Step 3: find the y-intercept by putting x = 0, y = 4, giving the point (0, 4)

Step 4: draw the solid line through (2, 0) and (0, 4)

Step 5: test the origin, 2(0) + 0 = 0 <= 4, which is true, so the origin is in the solution region

Step 6: shade the half-plane on the side of the line that contains the origin

Final answer: the solution is the half-plane below the line 2x + y = 4 containing the origin, including the line itself



Question: Q3(i). Solve the following system of linear inequalities graphically: 3x - y >= 8; 2x + 3y >= 5.

Answer: Step 1: draw the line 3x - y = 8 through its intercepts (8/3, 0) and (0, -8); testing the origin gives 0 >= 8, false, so shade the side away from the origin

Step 2: draw the line 2x + 3y = 5 through its intercepts (5/2, 0) and (0, 5/3); testing the origin gives 0 >= 5, false, so shade the side away from the origin

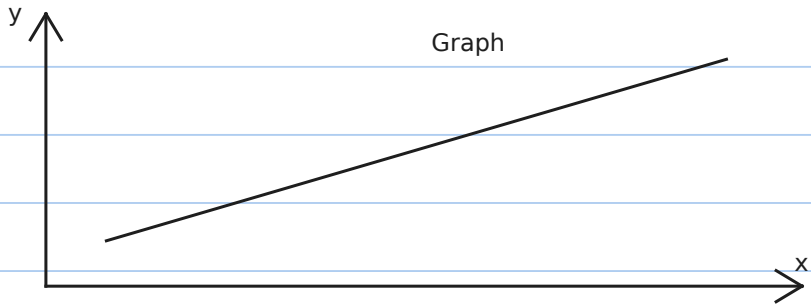
Step 3: the solution of the system is the common (overlapping) shaded region

Step 4: find the corner point where the two lines meet by solving 3x - y = 8 and 2x + 3y = 5 together: from the first, y = 3x - 8; substituting, 2x +

3(3x - 8) = 5, so 11x = 29, x = 29/11 and y = 3(29/11) - 8 = -1/11

Step 5: observe the common region extends without limit

Final answer: the solution region is the unbounded region common to both half-planes, with corner point (29/11, -1/11)



Question: Q3(ii). Solve the following system of linear inequalities graphically: 3x - 2y <= 6; x - y <= 4.

Answer: Step 1: draw the line 3x - 2y = 6 through its intercepts (2, 0) and (0, -3); testing the origin gives 0 <= 6, true, so shade the side containing the origin

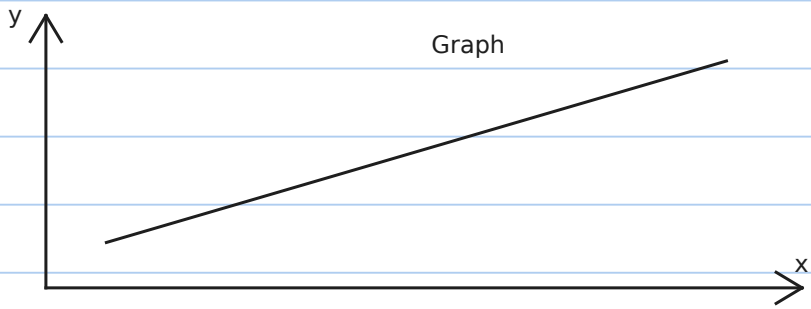
Step 2: draw the line x - y = 4 through its intercepts (4, 0) and (0, -4); testing the origin gives 0 <= 4, true, so shade the side containing the origin

Step 3: the solution of the system is the common (overlapping) shaded region

Step 4: find the corner point where the two lines meet by solving 3x - 2y = 6 and x - y = 4 together: from the second, x = y + 4; substituting, 3(y + 4) - 2y = 6, so y = -6 and x = -2

Step 5: observe the common region extends without limit

Final answer: the solution region is the unbounded region common to both half-planes, with corner point (-2, -6)



Question: Q4(i). Draw the graph of the solution region of the system of linear inequalities 2x + y <= 5; 3x - 2y <= 7; x >= 0. Also check whether the graph is bounded or not.

Answer: Step 1: draw the line 2x + y = 5 through (5/2, 0) and (0, 5) and shade the origin side, since 0 <= 5 is true

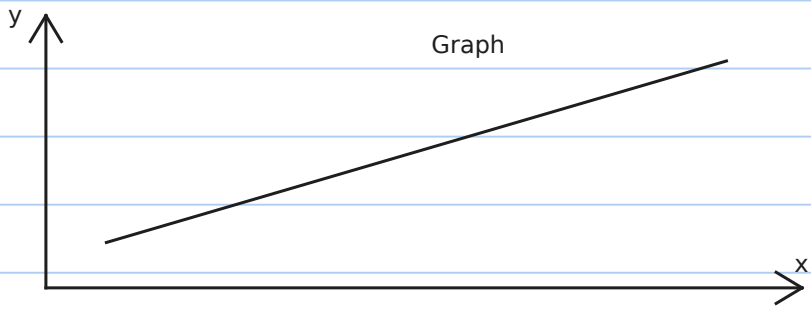
Step 2: draw the line 3x - 2y = 7 through (7/3, 0) and (0, -7/2) and shade the origin side, since 0 <= 7 is true

Step 3: apply x >= 0, keeping only the part on and to the right of the y-axis

Step 4: find the corner points: on the y-axis the two lines give (0, 5) and (0, -7/2); the two lines meet where 2x + y = 5 and 3x - 2y = 7, giving y = 5 - 2x and 3x - 2(5 - 2x) = 7, so 7x = 17, x = 17/7 and y = 1/7, the point (17/7, 1/7)

Step 5: the common region is the triangle with these three corner points, which is enclosed on all sides

Final answer: the solution region is the bounded triangular region with corner points (0, -7/2), (0, 5) and (17/7, 1/7); it is bounded



Question: Q4(ii). Draw the graph of the solution region of the system of linear inequalities x + 2y <= 6; 2x - 3y >= 8; x + 2y <= 4. Also check whether the graph is bounded or not.

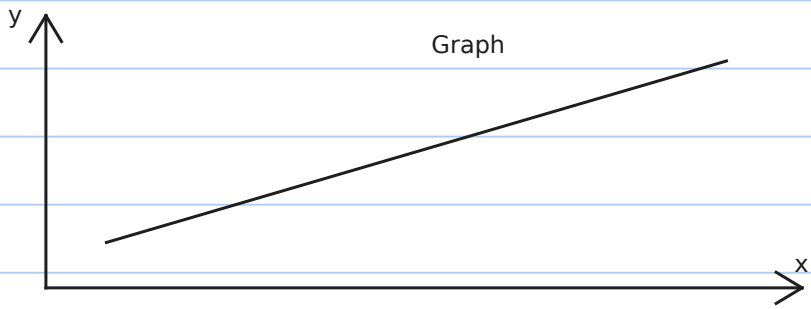
Answer: Step 1: note that x + 2y <= 4 is stronger than x + 2y <= 6, so draw the line x + 2y = 4 through (4, 0) and (0, 2) and shade the origin side (the region x + 2y <= 6 is automatically satisfied there)

Step 2: draw the line 2x - 3y = 8 through (4, 0) and (0, -8/3); testing the origin gives 0 >= 8, false, so shade the side away from the origin

Step 3: find where the two lines meet by solving x + 2y = 4 and 2x - 3y = 8 together: from the first, x = 4 - 2y; substituting, 2(4 - 2y) - 3y = 8, so -7y = 0, y = 0 and x = 4, the point (4, 0)

Step 4: the common region is the wedge with corner (4, 0) lying below the line x + 2y = 4 and on the far side of the line 2x - 3y = 8, and it extends without limit (for example as y decreases)

Final answer: the solution region has the single corner point (4, 0) and is unbounded (not bounded)



Exercise 9.2

Question: Q1(i). Draw the graph of the system of linear inequalities 2x - 3y <= 8; 3x + 2y <= 10; x >= 0, y >= 0, and identify the feasible region and the corner points.

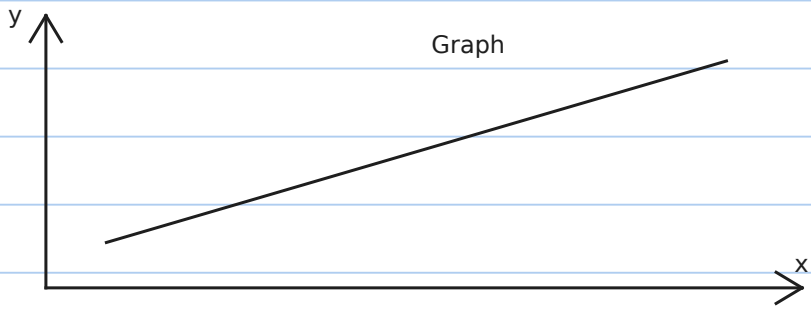
Answer: Step 1: draw the line 2x - 3y = 8 through (4, 0) and (0, -8/3); the origin satisfies 0 <= 8, so the feasible side contains the origin

Step 2: draw the line 3x + 2y = 10 through (10/3, 0) and (0, 5); the origin satisfies 0 <= 10, so the feasible side contains the origin

Step 3: restrict to the first quadrant because x >= 0 and y >= 0

Step 4: read off the corner points of the common region: (0, 0) on the axes, (0, 5) where 3x + 2y = 10 meets the y-axis, and (10/3, 0) where 3x + 2y = 10 meets the x-axis (the two lines cross at a point with negative y, outside the first quadrant, so it is not a corner of the feasible region)

Final answer: the feasible region is the triangle with corner points (0, 0), (0, 5) and (10/3, 0)



Question: Q1(ii). Draw the graph of the system of linear inequalities -5x + 4y >= 10; 2x + y <= 8; x >= 0, y >= 0, and identify the feasible region and the corner points.

Answer: Step 1: draw the line -5x + 4y = 10 through (-2, 0) and (0, 5/2); the origin gives 0 >= 10, false, so the feasible side is away from the origin (above the line)

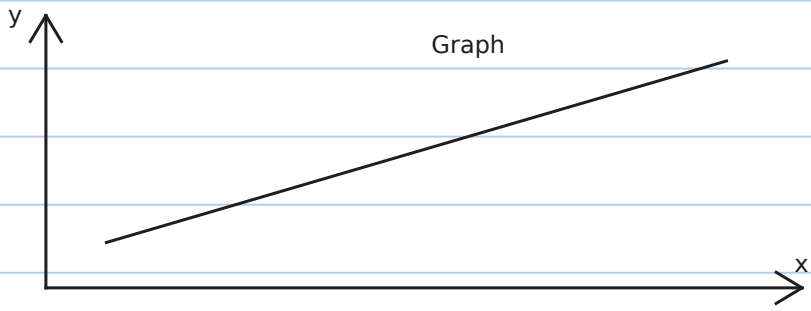
Step 2: draw the line 2x + y = 8 through (4, 0) and (0, 8); the origin satisfies 0 <= 8, so the feasible side contains the origin (below the line)

Step 3: restrict to the first quadrant

Step 4: find where the two lines meet by solving -5x + 4y = 10 and 2x + y = 8 together: from the second, y = 8 - 2x; substituting, -5x + 4(8 - 2x) = 10, so -13x = -22, x = 22/13 and y = 8 - 44/13 = 60/13

Step 5: collect the corner points: (0, 5/2) on the y-axis from the first line, (0, 8) on the y-axis from the second line, and the intersection (22/13, 60/13)

Final answer: the feasible region is the bounded region with corner points (0, 5/2), (0, 8) and (22/13, 60/13)



Question: Q1(iii). Draw the graph of the system of linear inequalities 4x + 3y <= 12; 3x - 2y <= 9; x >= 0, y >= 0, and identify the feasible region and the corner points.

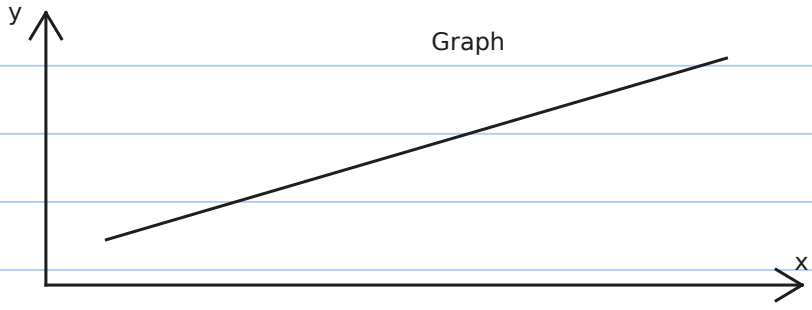
Answer: Step 1: draw the line 4x + 3y = 12 through (3, 0) and (0, 4); the origin satisfies 0 <= 12, so shade the origin side

Step 2: draw the line 3x - 2y = 9 through (3, 0) and (0, -9/2); the origin satisfies 0 <= 9, so shade the origin side

Step 3: restrict to the first quadrant; note both lines pass through (3, 0)

Step 4: read off the corner points: the origin (0, 0), the point (0, 4) where 4x + 3y = 12 meets the y-axis, and the common point (3, 0) of the two lines on the x-axis

Final answer: the feasible region is the triangle with corner points (0, 0), (0, 4) and (3, 0)



Question: Q2(i). Draw the graph of the inequalities 5x - y <= 16; 3x + 2y <= 6; x + 2y <= 5; x >= 0, y >= 0, and identify the solution space and the corner points.

Answer: Step 1: draw the line 3x + 2y = 6 through (2, 0) and (0, 3) and shade the origin side

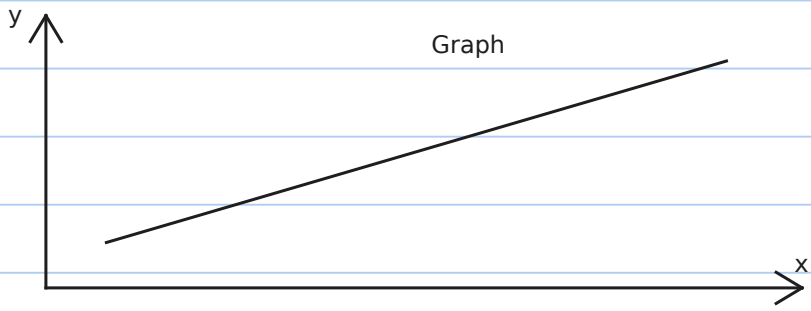
Step 2: draw the line x + 2y = 5 through (5, 0) and (0, 5/2) and shade the origin side

Step 3: draw the line 5x - y = 16 through (16/5, 0) and (0, -16); the origin satisfies it, and it does not cut the region formed by the other two lines in the first quadrant (its largest value on that region is 10, at (2, 0)), so it is a redundant constraint

Step 4: find where the two effective lines meet: 3x + 2y = 6 and x + 2y = 5 give 2x = 1, so x = 1/2 and y = 9/4

Step 5: collect the corner points: (0, 0), (0, 5/2) from x + 2y = 5 on the y-axis, the intersection (1/2, 9/4), and (2, 0) from 3x + 2y = 6 on the x-axis

Final answer: the solution space is the quadrilateral with corner points (0, 0), (0, 5/2), (1/2, 9/4) and (2, 0)



Question: Q2(ii). Draw the graph of the inequalities 5x + 3y <= 12; 2x - 3y <= 9; x + 2y <= 4; x >= 0, y >= 0, and identify the solution space and the corner points.

Answer: Step 1: draw the line 5x + 3y = 12 through (12/5, 0) and (0, 4) and shade the origin side

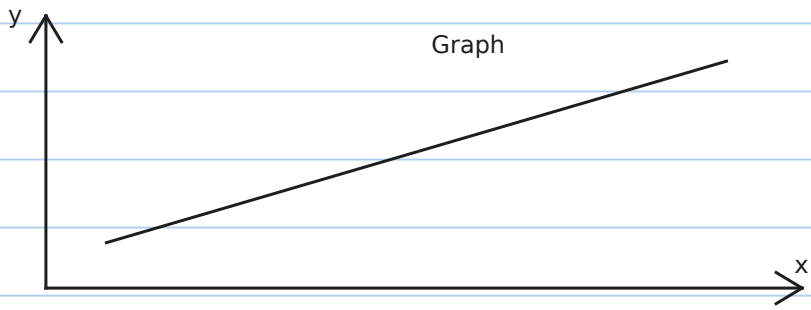
Step 2: draw the line x + 2y = 4 through (4, 0) and (0, 2) and shade the origin side

Step 3: the line 2x - 3y = 9 passes through (9/2, 0) and (0, -3); the origin satisfies 2x - 3y <= 9 and the constraint is not binding anywhere on the region formed by the other lines, so it is redundant

Step 4: find where the two effective lines meet by solving 5x + 3y = 12 and x + 2y = 4 together: from the second, x = 4 - 2y; substituting, 5(4 - 2y) + 3y = 12, so -7y = -8, y = 8/7 and x = 4 - 16/7 = 12/7

Step 5: collect the corner points: (0, 0), (0, 2) from x + 2y = 4 on the y-axis, the intersection (12/7, 8/7), and (12/5, 0) from 5x + 3y = 12 on the x-axis

Final answer: the solution space is the quadrilateral with corner points (0, 0), (0, 2), (12/7, 8/7) and (12/5, 0)



Question: Q2(iii). Draw the graph of the inequalities 6x + 5y >= 14; 4x + 2y <= 10; 3x - 2y <= 8; x >= 0, y >= 0, and identify the solution space and the corner points.

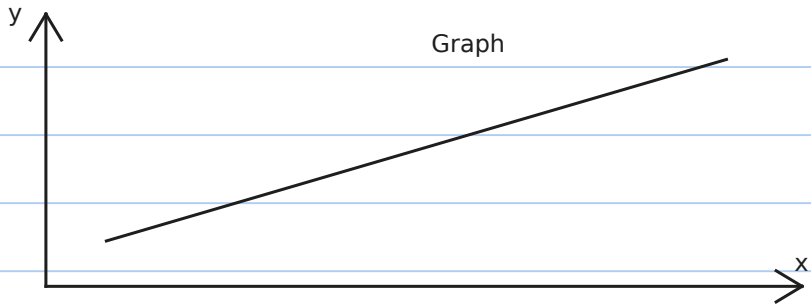
Answer: Step 1: draw the line 6x + 5y = 14 through (7/3, 0) and (0, 14/5); the origin gives 0 >= 14, false, so the feasible side is away from the origin (above the line)

Step 2: draw the line 4x + 2y = 10 through (5/2, 0) and (0, 5) and shade the origin side (below the line)

Step 3: the constraint 3x - 2y <= 8 is satisfied throughout the region formed by the other constraints in the first quadrant, so it is redundant

Step 4: collect the corner points: (0, 14/5) where 6x + 5y = 14 meets the y-axis, (0, 5) where 4x + 2y = 10 meets the y-axis, (7/3, 0) where 6x + 5y = 14 meets the x-axis, and (5/2, 0) where 4x + 2y = 10 meets the x-axis

Final answer: the solution space is the quadrilateral with corner points (0, 14/5), (0, 5), (5/2, 0) and (7/3, 0)



Exercise 9.3

Question: Q1(i). Solve by the graphical method, when x >= 0, y >= 0: Maximize Z(x, y) = 10x + 11y subject to 2x + 3y <= 8; 6x + 3y <= 10.

Answer: Step 1: draw the line 2x + 3y = 8 through (4, 0) and (0, 8/3) and shade the origin side

Step 2: draw the line 6x + 3y = 10 through (5/3, 0) and (0, 10/3) and shade the origin side, keeping the first quadrant

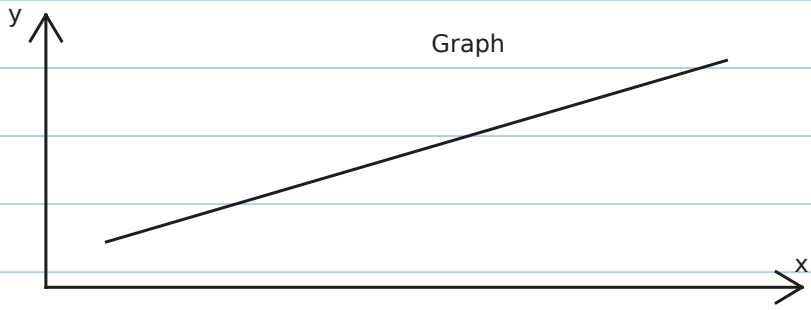
Step 3: find the intersection of the two lines: subtracting, (6x + 3y) - (2x + 3y) = 10 - 8 gives 4x = 2, so x = 1/2 and y = (8 - 1)/3 = 7/3

Step 4: list the corner points of the feasible region: (0, 0), (0, 8/3), (1/2, 7/3), (5/3, 0)

Step 5: evaluate Z = 10x + 11y at each corner: Z(0, 0) = 0; Z(0, 8/3) = 88/3 = 29.33; Z(1/2, 7/3) = 5 + 77/3 = 92/3 = 30.67; Z(5/3, 0) = 50/3 = 16.67

Step 6: the largest value is at (1/2, 7/3)

Final answer: Z maximum = 92/3 (approximately 30.67) at x = 1/2, y = 7/3



Question: Q1(ii). Solve by the graphical method, when x >= 0, y >= 0: Maximize Z(x, y) = 30x + 36y subject to 4x + 2y <= 12; 6x + 5y <= 20.

Answer: Step 1: draw the line 4x + 2y = 12 through (3, 0) and (0, 6) and shade the origin side

Step 2: draw the line 6x + 5y = 20 through (10/3, 0) and (0, 4) and shade the origin side, keeping the first quadrant

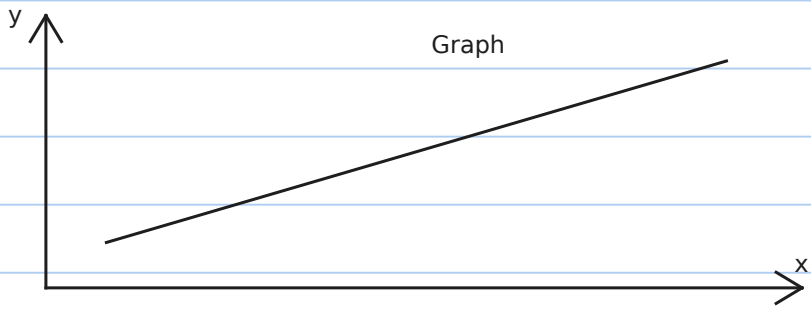
Step 3: find the intersection of the two lines: from the first, y = 6 - 2x; substituting, 6x + 5(6 - 2x) = 20 gives -4x = -10, so x = 5/2 and y = 1

Step 4: list the corner points: (0, 0), (3, 0), (5/2, 1), (0, 4)

Step 5: evaluate Z = 30x + 36y at each corner: Z(0, 0) = 0; Z(3, 0) = 90; Z(5/2, 1) = 75 + 36 = 111; Z(0, 4) = 144

Step 6: the largest value is at (0, 4)

Final answer: Z maximum = 144 at x = 0, y = 4



Question: Q1(iii). Solve by the graphical method, when x >= 0, y >= 0: Maximize Z(x, y) = 41x + 38y subject to 4x + 5y <= 26; 8x + 5y <= 22; 5x + 2y <= 10.

Answer: Step 1: draw the three lines: 4x + 5y = 26 through (13/2, 0) and (0, 26/5); 8x + 5y = 22 through (11/4, 0) and (0, 22/5); 5x + 2y = 10 through (2, 0) and (0, 5); shade the origin side of each, keeping the first quadrant

Step 2: the effective boundary is formed by 5x + 2y = 10 (near the x-axis) and 8x + 5y = 22 (near the y-axis); the first constraint is satisfied throughout this region

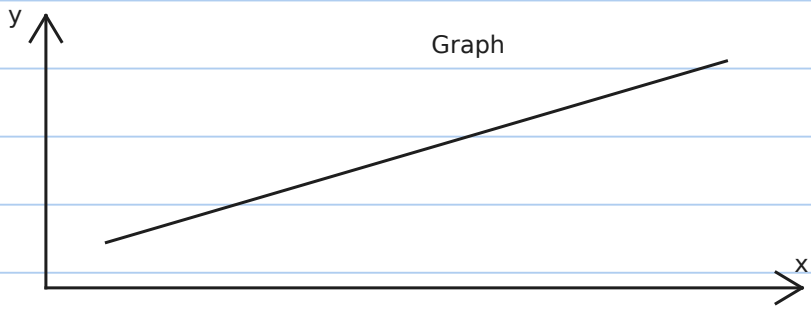
Step 3: find the intersection of 8x + 5y = 22 and 5x + 2y = 10: from the second, y = (10 - 5x)/2; substituting, 8x + 5(10 - 5x)/2 = 22 gives 16x + 50 - 25x = 44, so x = 2/3 and y = 10/3

Step 4: list the corner points: (0, 0), (2, 0), (2/3, 10/3), (0, 22/5)

Step 5: evaluate Z = 41x + 38y at each corner: Z(0, 0) = 0; Z(2, 0) = 82; Z(2/3, 10/3) = 82/3 + 380/3 = 462/3 = 154; Z(0, 22/5) = 38 . 22/5 = 836/5 = 167.2

Step 6: the largest value is at (0, 22/5)

Final answer: Z maximum = 836/5 = 167.2 at x = 0, y = 22/5



Question: Q1(iv). Solve by the graphical method, when x >= 0, y >= 0: Maximize Z = 4x + y subject to x + y <= 50; 3x + y <= 90; x - y <= 40.

Answer: Step 1: draw the lines: x + y = 50 through (50, 0) and (0, 50); 3x + y = 90 through (30, 0) and (0, 90); x - y = 40 through (40, 0) and (0, -40); shade the origin side of each, keeping the first quadrant

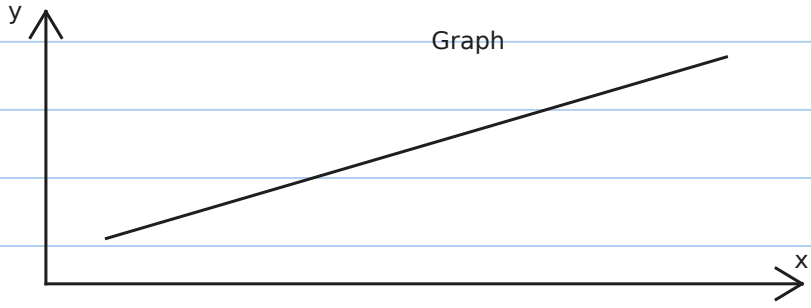
Step 2: find the intersection of x + y = 50 and 3x + y = 90: subtracting gives 2x = 40, so x = 20 and y = 30 (this point also satisfies x - y = -10 <= 40)

Step 3: list the corner points: (0, 0), (0, 50), (20, 30), (30, 0) from 3x + y = 90 on the x-axis

Step 4: evaluate Z = 4x + y at each corner: Z(0, 0) = 0; Z(0, 50) = 50; Z(20, 30) = 80 + 30 = 110; Z(30, 0) = 120

Step 5: the largest value is at (30, 0)

Final answer: Z maximum = 120 at x = 30, y = 0



Question: Q1(v). Solve by the graphical method, when x >= 0, y >= 0:

Minimize Z = 200x + 500y subject to the constraints x + 2y >= 10; 3x + 4y <= 24.

Answer: Step 1: draw the line x + 2y = 10 through (10, 0) and (0, 5); the origin fails >=, so the feasible side is away from the origin

Step 2: draw the line 3x + 4y = 24 through (8, 0) and (0, 6) and shade the origin side, keeping the first quadrant

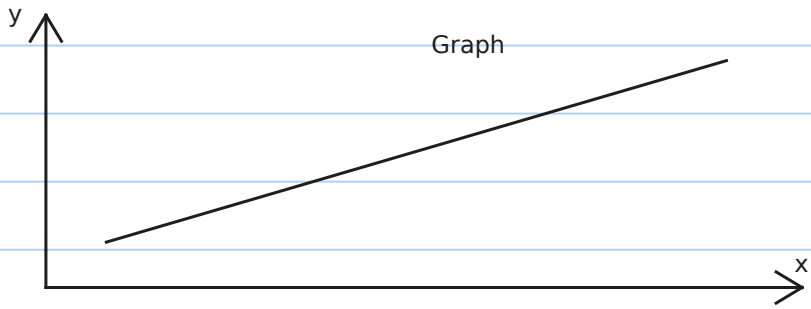
Step 3: find the intersection of the two lines: from the first, x = 10 - 2y; substituting, 3(10 - 2y) + 4y = 24 gives -2y = -6, so y = 3 and x = 4

Step 4: list the corner points of the bounded feasible region: (0, 5), (0, 6), (4, 3)

Step 5: evaluate Z = 200x + 500y at each corner: Z(0, 5) = 2500; Z(0, 6) = 3000; Z(4, 3) = 800 + 1500 = 2300

Step 6: the smallest value is at (4, 3)

Final answer: Z minimum = 2300 at x = 4, y = 3



Question: Q1(vi). Solve by the graphical method, when x >= 0, y >= 0: Minimize Z = 3x + 2y subject to the constraints 2x + y >= 6; x + y >= 4.

Answer: Step 1: draw the line 2x + y = 6 through (3, 0) and (0, 6); the feasible side is away from the origin

Step 2: draw the line x + y = 4 through (4, 0) and (0, 4); the feasible side is away from the origin, keeping the first quadrant

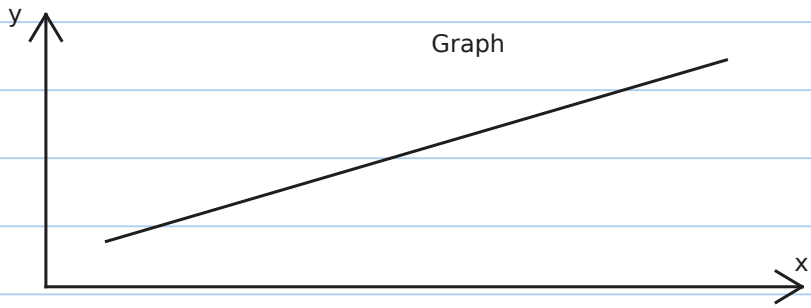
Step 3: find the intersection of the two lines: subtracting, (2x + y) - (x + y) = 6 - 4 gives x = 2 and y = 2

Step 4: list the corner points of the (unbounded) feasible region: (0, 6), (2, 2), (4, 0)

Step 5: evaluate Z = 3x + 2y at each corner: Z(0, 6) = 12; Z(2, 2) = 6 + 4 = 10; Z(4, 0) = 12

Step 6: the smallest value is at (2, 2)

Final answer: Z minimum = 10 at x = 2, y = 2



Question: Q1(vii). Solve by the graphical method, when x >= 0, y >= 0: Minimize Z = 0.12x + 0.15y subject to the constraints x + y >= 5; 2x + y >= 6; x + 3y >= 9.

Answer: Step 1: draw the three lines: x + y = 5 through (5, 0) and (0, 5); 2x + y = 6 through (3, 0) and (0, 6); x + 3y = 9 through (9, 0) and (0, 3); the feasible side of each is away from the origin, keeping the first quadrant

Step 2: find the corner points on the axes: (0, 6) from 2x + y = 6 and (9, 0) from x + 3y = 9

Step 3: find the intersection of 2x + y = 6 and x + y = 5: subtracting gives x = 1, y = 4, the point (1, 4)

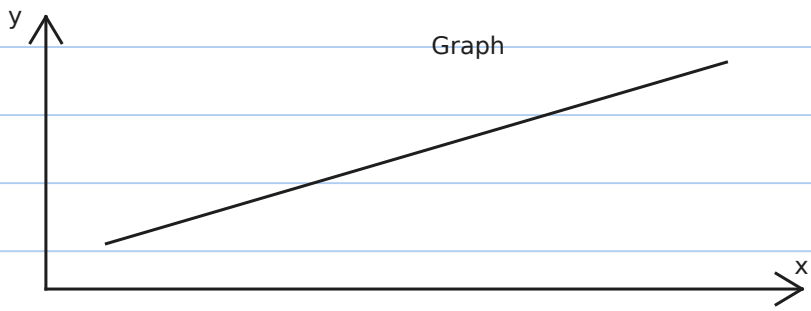
Step 4: find the intersection of x + y = 5 and x + 3y = 9: subtracting gives

2y = 4, so y = 2 and x = 3, the point (3, 2) (the intersection of 2x + y = 6 with x + 3y = 9 is (9/5, 12/5), which fails x + y >= 5 since 21/5 < 5, so it is not a corner)

Step 5: evaluate Z = 0.12x + 0.15y at each corner: Z(0, 6) = 0.90; Z(1, 4) = 0.12 + 0.60 = 0.72; Z(3, 2) = 0.36 + 0.30 = 0.66; Z(9, 0) = 1.08

Step 6: the smallest value is at (3, 2)

Final answer: Z minimum = 0.66 at x = 3, y = 2



Question: Q2. A landlord has 50 hectares of land to grow two crops X and Y, with profits per hectare estimated as Rs. 10,500 and Rs. 9,000 respectively. To control weeds, a liquid herbicide, no more than 800 litres, is to be used for crops X and Y at rates of 20 litres and 10 litres per hectare respectively. How much land should be allocated to each crop so as to maximize the total profit?

Answer: Step 1: let x hectares be allocated to crop X and y hectares to crop Y

Step 2: formulate the LP problem: Maximize Z = 10500x + 9000y subject to x + y <= 50 (total land), 20x + 10y <= 800 (herbicide), x >= 0, y >= 0

Step 3: draw the lines x + y = 50 through (50, 0) and (0, 50), and 20x + 10y = 800, i.e. 2x + y = 80, through (40, 0) and (0, 80); shade the origin side of each

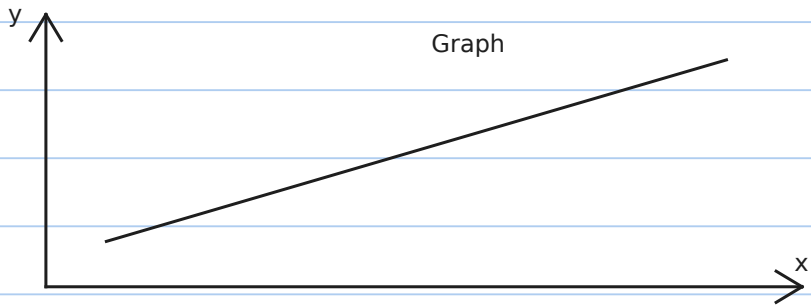
Step 4: find the intersection: from x + y = 50, y = 50 - x; substituting in 2x + y = 80 gives x = 30 and y = 20

Step 5: list the corner points: (0, 0), (40, 0), (30, 20), (0, 50)

Step 6: evaluate Z at each corner: Z(0, 0) = 0; Z(40, 0) = 420000; Z(30, 20) = 315000 + 180000 = 495000; Z(0, 50) = 450000

Step 7: the largest profit is at (30, 20)

Final answer: allocate 30 hectares to crop X and 20 hectares to crop Y for a maximum profit of Rs. 495,000



Question: Q3. A toy factory manufactures two types of toys A and B and sells them for Rs. 25 and Rs. 20 respectively. Toys A and B require 20 units and 12 units daily out of 2000 available resource units. Both require a production time of 5 minutes and the total working hours are 9 per day. What should be the quantity for each of the toys to maximize the selling amount?

Answer: Step 1: let x toys of type A and y toys of type B be made daily

Step 2: formulate the LP problem: Maximize Z = 25x + 20y subject to 20x + 12y <= 2000 (resource units) and 5x + 5y <= 9 . 60 = 540 minutes, i.e. x + y <= 108 (production time), x >= 0, y >= 0

Step 3: draw the lines 20x + 12y = 2000 through (100, 0) and (0, 500/3), and x + y = 108 through (108, 0) and (0, 108); shade the origin side of

each

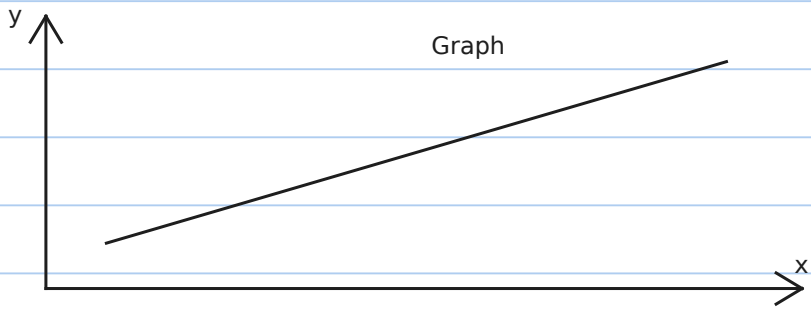
Step 4: find the intersection: from x + y = 108, y = 108 - x; substituting, 20x + 12(108 - x) = 2000 gives 8x = 704, so x = 88 and y = 20

Step 5: list the corner points: (0, 0), (100, 0), (88, 20), (0, 108)

Step 6: evaluate Z at each corner: Z(0, 0) = 0; Z(100, 0) = 2500; Z(88, 20) = 2200 + 400 = 2600; Z(0, 108) = 2160

Step 7: the largest selling amount is at (88, 20)

Final answer: make 88 toys of type A and 20 toys of type B daily for a maximum selling amount of Rs. 2600



Question: Q4. At a university, Professor Abdul Sattar wishes to employ two people, Farhan and Sarfaraz, to grade papers for his classes. Farhan is a graduate student and can grade 20 papers per hour; Farhan earns $15 per hour for grading papers. Sarfaraz is a post-doctoral associate and can grade 30 papers per hour; Sarfaraz earns $25 per hour for grading papers. Each must be employed at least one hour a week to justify their employment. If Prof. Abdul Sattar has at least 110 papers to be graded each week, how many hours per week should he employ each person to minimize the cost?

Answer: Step 1: let x be the hours per week Farhan is employed and y the hours per week Sarfaraz is employed

Step 2: formulate the LP problem: Minimize C = 15x + 25y subject to 20x + 30y >= 110 (papers to be graded), x >= 1, y >= 1 (each employed at least one hour)

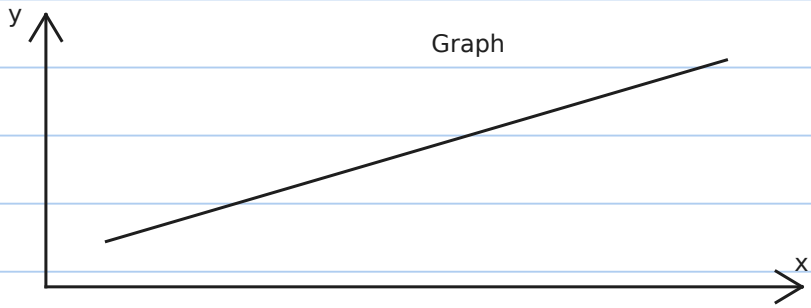
Step 3: draw the line 20x + 30y = 110, i.e. 2x + 3y = 11, through (11/2, 0) and (0, 11/3); the feasible side is away from the origin, cut by the lines x = 1 and y = 1

Step 4: find the corner points: where y = 1 meets the line, 2x + 3 = 11 gives x = 4, the point (4, 1); where x = 1 meets the line, 2 + 3y = 11 gives y = 3, the point (1, 3)

Step 5: evaluate C at each corner: C(4, 1) = 60 + 25 = 85; C(1, 3) = 15 + 75 = 90

Step 6: the smaller cost is at (4, 1) (check: 20(4) + 30(1) = 110 papers)

Final answer: employ Farhan for 4 hours per week and Sarfaraz for 1 hour per week, for a minimum cost of $85 per week



Review Exercise 9

Question: Q1(i). Select the correct answer: If 2x - 5 >= 3 then its solution set is ....., where x in Z. (a) {2, 3, 4, ...} (b) {4, 5, ...} (c) {2, 1, 0, ...} (d) {1, 0, -1, ...}.

Answer: Step 1: add 5 to both sides, 2x >= 8

Step 2: divide by 2, x >= 4

Step 3: since x is an integer, the solution set is 4, 5, 6, ...

Final answer: (b) {4, 5, ...}

Question: Q1(ii). Select the correct answer: If 4x - 3y = 8 is the corresponding equation of an inequality 4x - 3y <= 8, then its x-intercept is ..... (a) 3/2 (b) 2/3 (c) 2 (d) 1/2.

Answer: Step 1: the x-intercept is found by putting y = 0 in the equation

Step 2: substitute, 4x - 3(0) = 8, so 4x = 8

Step 3: solve, x = 2

Final answer: (c) 2

Question: Q1(iii). Select the correct answer: {x | x in R and x > 5} = ... (a) [5, infinity) (b) (-infinity, 5) (c) (5, infinity) (d) (-infinity, 5].

Answer: Step 1: the set contains all real numbers greater than 5, not including 5 itself

Step 2: in interval notation an excluded endpoint takes a round bracket, giving (5, infinity)

Final answer: (c) (5, infinity)

Question: Q1(iv). Select the correct answer: Linear Programming (LP) is used to obtain ..... solution. (a) Feasible (b) Trivial (c) Optimal (d) Infeasible.

Answer: Step 1: LP is the technique of finding the best (maximum or minimum) value of an objective function subject to constraints, i.e. the optimal solution

Final answer: (c) Optimal

Question: Q1(v). Select the correct answer: The solution space of the linear inequality 2x + 3y <= 6, for all x, y in R, includes ..... (a) All points above the line (b) All points on and below the line (c) All points below the line (d) All points on and above the line.

Answer: Step 1: test the origin in the inequality, 0 <= 6 is true, so the origin side of the line is included

Step 2: because the inequality is of the <= type, the points on the line are also included

Final answer: (b) All points on and below the line

Question: Q1(vi). Select the correct answer: If f(x, y) = 5x + 4y is an objective function and the corner points of the feasible region are (5, 4), (0, 0), (4, 0) and (0, 6), then the function is maximum at: (a) (4, 0) (b) (0, 6) (c) (0, 0) (d) (5, 4).

Answer: Step 1: evaluate f at (5, 4), f = 5(5) + 4(4) = 25 + 16 = 41

Step 2: evaluate f at (0, 0), f = 0

Step 3: evaluate f at (4, 0), f = 20

Step 4: evaluate f at (0, 6), f = 24

Step 5: the largest value is 41 at (5, 4)

Final answer: (d) (5, 4)

Question: Q1(vii). Select the correct answer: ..... are the entities whose values are to be determined in the solution of the LP problem. (a) Objective function (b) Decision Variables (c) Constraints (d) Opportunity cost.

Answer: Step 1: the unknowns of an LP problem, whose values the

decision maker must choose, are called decision variables

Final answer: (b) Decision Variables

Question: Q1(viii). Select the correct answer: ..... specifies the objective or goal of solving the LP problem. (a) Objective function (b) Decision Variables (c) Constraints (d) Opportunity cost.

Answer: Step 1: the linear function that is to be maximized or minimized expresses the goal of the problem

Final answer: (a) Objective function

Question: Q1(ix). Select the correct answer: ..... are the restrictions or limitations imposed on the LP problem. (a) Variables (b) Costs (c) Profits (d) Constraints.

Answer: Step 1: the linear inequalities that limit the values of the decision variables are the constraints of the problem

Final answer: (d) Constraints

Question: Q1(x). Select the correct answer: The region of solution in an LP problem is called ..... (a) Infeasible region (b) Unbounded region (c) Infinite region (d) Feasible region.

Answer: Step 1: the set of all points satisfying all the constraints is, by definition, the feasible region

Final answer: (d) Feasible region

Question: Q1(xi). Select the correct answer: In case of an ..... constraint, the feasible region is a straight line. (a) Less than or equal to (b) Greater than or equal to (c) Mixed (d) Equal to.

Answer: Step 1: an equality constraint $a\,x + b\,y = c$ is satisfied only by the points on the straight line itself, not by a half-plane

Final answer: (d) Equal to

Question: Q1(xii). Select the correct answer: ax + b <= 0, a not equal to 0, a, b in R, is called ..... (a) Non-linear inequality (b) Linear inequality (c) Complex inequality (d) Linear equality.

Answer: Step 1: the expression contains the variable x only to the first power and uses an inequality sign, so it is a linear inequality in one variable

Final answer: (b) Linear inequality

Question: Q1(xiii). Select the correct answer: A point of a solution region where two of its boundary lines intersect is called ..... (a) Middle point (b) Origin (c) Corner point (d) Feasible point.

Answer: Step 1: by definition, the intersection point of two boundary lines of the feasible region is a corner (vertex) point

Final answer: (c) Corner point

Question: Q1(xiv). Select the correct answer: The solution region of an inequality restricted to the first quadrant is called ..... region. (a) Combined (b) Unbounded (c) Infeasible (d) Feasible.

Answer: Step 1: any region formed by points satisfying the given constraints (here in the first quadrant, x >= 0, y >= 0) is a feasible region

Final answer: (d) Feasible

Question: Q1(xv). Select the correct answer: A function which is to be maximized or minimized is called ..... (a) Linear function (b) Equal function (c) Objective function (d) Non linear function.

Answer: Step 1: the function whose optimum value is sought in an LP problem is by definition the objective function

Final answer: (c) Objective function

Question: Q1(xvi). Select the correct answer: The feasible solution which maximizes or minimizes the objective function is called the ..... (a) Optimal solution (b) Corner solution (c) Initial solution (d) Complex solution.

Answer: Step 1: a feasible solution that gives the best possible value of the objective function is called the optimal solution

Final answer: (a) Optimal solution

Question: Q2. Draw the graph of the following system of linear inequalities and find the feasible region: 2x + 3y >= 10; 3x + 2y >= 12; x >= 0, y >= 0.

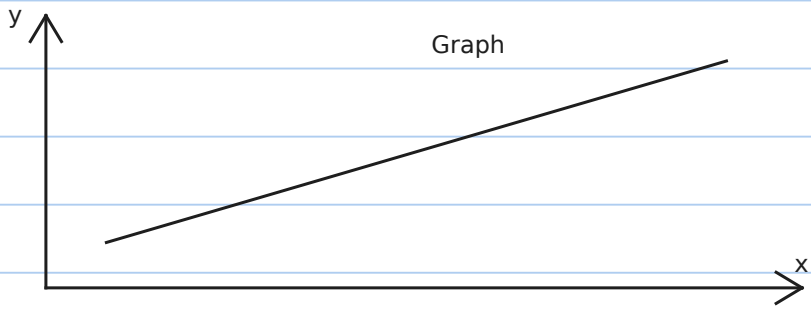
Answer: Step 1: draw the line 2x + 3y = 10 through (5, 0) and (0, 10/3); the origin fails >=, so shade the side away from the origin

Step 2: draw the line 3x + 2y = 12 through (4, 0) and (0, 6); the origin fails >=, so shade the side away from the origin, keeping the first quadrant

Step 3: find the intersection of the two lines: multiply the first by 3 and the second by 2, 6x + 9y = 30 and 6x + 4y = 24; subtracting gives 5y = 6, so y = 6/5 and x = (10 - 18/5)/2 = 16/5

Step 4: collect the corner points of the feasible region: (5, 0), (16/5, 6/5) and (0, 6)

Final answer: the feasible region is the unbounded region in the first quadrant with corner points (5, 0), (16/5, 6/5) and (0, 6)



Question: Q3. Find the feasible region graphically subject to the constraints 4x + y >= 9; 2x + 3y <= 14; x - y <= 5; x >= 0, y >= 0, and also find its corner points.

Answer: Step 1: draw the line 4x + y = 9 through (9/4, 0) and (0, 9) and shade the side away from the origin

Step 2: draw the line 2x + 3y = 14 through (7, 0) and (0, 14/3) and shade the origin side

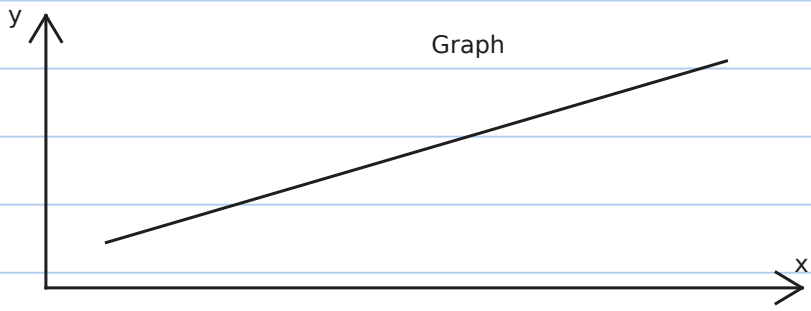
Step 3: draw the line x - y = 5 through (5, 0) and (0, -5) and shade the origin side, keeping the first quadrant

Step 4: find the intersection of 4x + y = 9 and 2x + 3y = 14: from the first, y = 9 - 4x; substituting, 2x + 3(9 - 4x) = 14 gives -10x = -13, so x = 13/10 and y = 19/5

Step 5: find the intersection of 2x + 3y = 14 and x - y = 5: from the second, x = y + 5; substituting, 2(y + 5) + 3y = 14 gives y = 4/5 and x = 29/5

Step 6: the remaining corners on the x-axis are (9/4, 0) from 4x + y = 9 and (5, 0) from x - y = 5

Final answer: the feasible region is the quadrilateral with corner points (9/4, 0), (13/10, 19/5), (29/5, 4/5) and (5, 0)



Question: Q4. Find the maximum and minimum values of the function f(x, y) = 4x - 3y subject to the constraints 2x + 3y >= 5; 3x + 2y <= 12; x >= 0, y >= 0.

Answer: Step 1: draw the line 2x + 3y = 5 through (5/2, 0) and (0, 5/3) and shade the side away from the origin

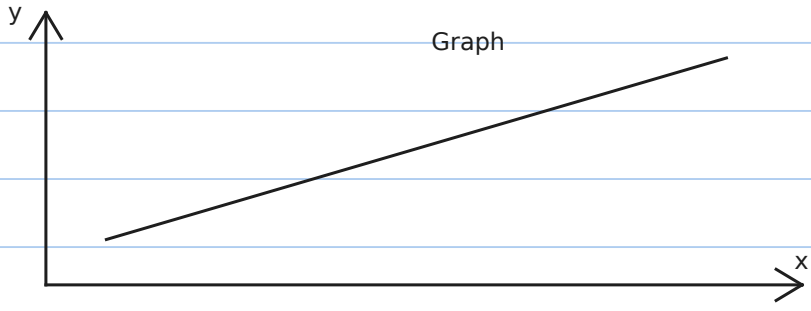
Step 2: draw the line 3x + 2y = 12 through (4, 0) and (0, 6) and shade the origin side, keeping the first quadrant (the two lines cross outside the first quadrant, so they form no corner there)

Step 3: list the corner points of the feasible region: (5/2, 0), (4, 0), (0, 6), (0, 5/3)

Step 4: evaluate f = 4x - 3y at each corner: f(5/2, 0) = 10; f(4, 0) = 16; f(0, 6) = -18; f(0, 5/3) = -5

Step 5: pick the largest and smallest values

Final answer: the maximum value is 16 at (4, 0) and the minimum value is -18 at (0, 6)



Question: Q5. A dealer wishes to purchase some fans and sewing machines. He has only Rs. 5760 to invest and space for at most 20 items. The costs of a fan and a sewing machine are Rs. 360 and Rs. 240 respectively. He wants profit on a fan and a sewing machine of Rs. 22 and Rs. 18 respectively. Assuming that he can sell all the items that he can buy, how should he invest his money in order to maximize his profit?

Answer: Step 1: let x be the number of fans and y the number of sewing machines purchased

Step 2: formulate the LP problem: Maximize P = 22x + 18y subject to 360x + 240y <= 5760 (money), x + y <= 20 (space), x >= 0, y >= 0

Step 3: simplify the money constraint by dividing by 120, 3x + 2y <= 48, and draw the lines 3x + 2y = 48 through (16, 0) and (0, 24), and x + y = 20 through (20, 0) and (0, 20), shading the origin side of each

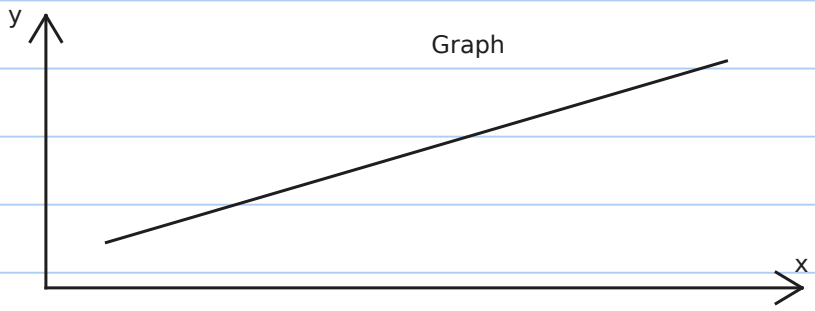
Step 4: find the intersection of the two lines: from x + y = 20, y = 20 - x; substituting, 3x + 2(20 - x) = 48 gives x = 8 and y = 12

Step 5: list the corner points: (0, 0), (16, 0), (8, 12), (0, 20)

Step 6: evaluate P at each corner: P(0, 0) = 0; P(16, 0) = 352; P(8, 12) = 176 + 216 = 392; P(0, 20) = 360

Step 7: the largest profit is at (8, 12) (check: cost 8 . 360 + 12 . 240 = 2880 + 2880 = 5760, and 8 + 12 = 20 items)

Final answer: he should buy 8 fans and 12 sewing machines for a maximum profit of Rs. 392


y
Graph
x

Unit 10: Trigonometric Identities of Sum and Difference of Angles

Exercise 10.1

Question: Q1(i). Prove that cos(360 deg + theta) = cos(theta).

Answer: Step 1: Use the sum formula cos(alpha + beta) = cos(alpha)cos(beta) - sin(alpha)sin(beta) with alpha = 360 deg and beta = theta

Step 2: LHS = cos(360 deg)cos(theta) - sin(360 deg)sin(theta)

Step 3: Use cos(360 deg) = 1 and sin(360 deg) = 0, giving LHS = (1)cos(theta) - (0)sin(theta)

Step 4: LHS = cos(theta)

Final answer: LHS = RHS, so cos(360 deg + theta) = cos(theta) is proved.

Question: Q1(ii). Prove that sin(180 deg + theta) = -sin(theta).

Answer: Step 1: Use sin(alpha + beta) = sin(alpha)cos(beta) + cos(alpha)sin(beta) with alpha = 180 deg

Step 2: LHS = sin(180 deg)cos(theta) + cos(180 deg)sin(theta)

Step 3: Use sin(180 deg) = 0 and cos(180 deg) = -1, giving LHS = (0)cos(theta) + (-1)sin(theta)

Step 4: LHS = -sin(theta)

Final answer: LHS = RHS, so sin(180 deg + theta) = -sin(theta) is proved.

Question: Q1(iii). Prove that tan(180 deg - theta) = -tan(theta).

Answer: Step 1: Use tan(alpha - beta) = [tan(alpha) - tan(beta)] / [1 + tan(alpha)tan(beta)] with alpha = 180 deg

Step 2: LHS = [tan(180 deg) - tan(theta)] / [1 + tan(180 deg)tan(theta)]

Step 3: Use tan(180 deg) = 0, giving LHS = [0 - tan(theta)] / [1 + 0]

Step 4: LHS = -tan(theta)

Final answer: LHS = RHS, so tan(180 deg - theta) = -tan(theta) is proved.

Question: Q1(iv). Prove that sin(270 deg - theta) = -cos(theta).

Answer: Step 1: Use sin(alpha - beta) = sin(alpha)cos(beta) - cos(alpha)sin(beta) with alpha = 270 deg

Step 2: LHS = sin(270 deg)cos(theta) - cos(270 deg)sin(theta)

Step 3: Use sin(270 deg) = -1 and cos(270 deg) = 0, giving LHS = (-1)cos(theta) - (0)sin(theta)

Step 4: LHS = -cos(theta)

Final answer: LHS = RHS, so sin(270 deg - theta) = -cos(theta) is proved.

Question: Q1(v). Prove that cot(270 deg + theta) = -tan(theta).

Answer: Step 1: Write cot as cos / sin, so LHS = cos(270 deg + theta) / sin(270 deg + theta)

Step 2: Expand the numerator, cos(270 deg)cos(theta) - sin(270 deg)sin(theta) = (0)cos(theta) - (-1)sin(theta) = sin(theta)

Step 3: Expand the denominator, sin(270 deg)cos(theta) + cos(270 deg)sin(theta) = (-1)cos(theta) + (0)sin(theta) = -cos(theta)

Step 4: Divide, LHS = sin(theta) / (-cos(theta)) = -tan(theta)

Final answer: LHS = RHS, so cot(270 deg + theta) = -tan(theta) is proved.

Question: Q1(vi). Prove that cot(270 deg - theta) = tan(theta).

Answer: Step 1: Write cot as cos / sin, so LHS = cos(270 deg - theta) / sin(270 deg - theta)

Step 2: Expand the numerator, cos(270 deg)cos(theta) + sin(270 deg)sin(theta) = (0)cos(theta) + (-1)sin(theta) = -sin(theta)

Step 3: Expand the denominator, sin(270 deg)cos(theta) - cos(270 deg)sin(theta) = (-1)cos(theta) - (0)sin(theta) = -cos(theta)

Step 4: Divide, LHS = (-sin(theta)) / (-cos(theta)) = tan(theta)

Final answer: LHS = RHS, so cot(270 deg - theta) = tan(theta) is proved.

Question: Q1(vii). Prove that cosec(360 deg - theta) = -cosec(theta).

Answer: Step 1: Write cosec as 1 / sin, so LHS = 1 / sin(360 deg - theta)

Step 2: Expand sin(360 deg - theta) = sin(360 deg)cos(theta) - cos(360 deg)sin(theta)

Step 3: Use sin(360 deg) = 0 and cos(360 deg) = 1, giving sin(360 deg - theta) = -sin(theta)

Step 4: Hence LHS = 1 / (-sin(theta)) = -cosec(theta)

Final answer: LHS = RHS, so cosec(360 deg - theta) = -cosec(theta) is proved.

Question: Q1(viii). Prove that sec(360 deg - theta) = sec(theta).

Answer: Step 1: Write sec as 1 / cos, so LHS = 1 / cos(360 deg - theta)

Step 2: Expand cos(360 deg - theta) = cos(360 deg)cos(theta) + sin(360 deg)sin(theta)

Step 3: Use cos(360 deg) = 1 and sin(360 deg) = 0, giving cos(360 deg - theta) = cos(theta)

Step 4: Hence LHS = 1 / cos(theta) = sec(theta)

Final answer: LHS = RHS, so sec(360 deg - theta) = sec(theta) is proved.

Question: Q2(i). Evaluate sin(150 deg)cos(300 deg) + sin(300 deg)cos(150 deg).

Answer: Step 1: The expression has the pattern sin(A)cos(B) + sin(B)cos(A), which is sin(A + B)

Step 2: Take A = 150 deg and B = 300 deg, so the expression = sin(150 deg + 300 deg) = sin(450 deg)

Step 3: Reduce, sin(450 deg) = sin(450 deg - 360 deg) = sin(90 deg)

Step 4: sin(90 deg) = 1

Final answer: The value is 1.

Question: Q2(ii). Evaluate cos(19 deg)cos(11 deg) - sin(19 deg)sin(11 deg).

Answer: Step 1: The expression has the pattern cos(A)cos(B) - sin(A)sin(B), which is cos(A + B)

Step 2: Take A = 19 deg and B = 11 deg, so the expression = cos(19 deg + 11 deg) = cos(30 deg)

Step 3: Use the standard value cos(30 deg) = sqrt(3) / 2

Final answer: The value is sqrt(3)/2.

Question: Q3(i). Verify that cos(75 deg) = (sqrt(3) - 1) / (2 sqrt(2)).

Answer: Step 1: Write 75 deg = 45 deg + 30 deg and use cos(alpha + beta) = cos(alpha)cos(beta) - sin(alpha)sin(beta)

Step 2: cos(75 deg) = cos(45 deg)cos(30 deg) - sin(45 deg)sin(30 deg)

Step 3: Substitute the standard values, = (1 / sqrt(2))(sqrt(3) / 2) - (1 / sqrt(2))(1 / 2)

Step 4: Combine over the common denominator 2 sqrt(2), giving (sqrt(3) - 1) / (2 sqrt(2))

Final answer: cos(75 deg) = (sqrt(3) - 1) / (2 sqrt(2)), as required.

Question: Q3(ii). Verify that cos(180 deg + 45 deg)cos(180 deg - 45 deg) = 1/2.

Answer: Step 1: Expand the first factor, cos(180 deg + 45 deg) = cos(180 deg)cos(45 deg) - sin(180 deg)sin(45 deg) = (-1)(1 / sqrt(2)) - 0 = -1 / sqrt(2)

Step 2: Expand the second factor, cos(180 deg - 45 deg) = cos(180 deg)cos(45 deg) + sin(180 deg)sin(45 deg) = (-1)(1 / sqrt(2)) + 0 = -1 / sqrt(2)

Step 3: Multiply the two factors, (-1 / sqrt(2)) x (-1 / sqrt(2)) = 1 / 2

Final answer: The product is 1/2, as required.

Question: Q4(i). Without using tables or calculator, find sin(75 deg).

Answer: Step 1: Write 75 deg = 45 deg + 30 deg and use sin(alpha + beta) = sin(alpha)cos(beta) + cos(alpha)sin(beta)

Step 2: sin(75 deg) = sin(45 deg)cos(30 deg) + cos(45 deg)sin(30 deg)

Step 3: Substitute values, = (1 / sqrt(2))(sqrt(3) / 2) + (1 / sqrt(2))(1 / 2)

Step 4: Combine over 2 sqrt(2), giving (sqrt(3) + 1) / (2 sqrt(2))

Final answer: sin(75 deg) = (sqrt(3) + 1) / (2 sqrt(2)).

Question: Q4(ii). Without using tables or calculator, find tan(105 deg).

Answer: Step 1: Write 105 deg = 60 deg + 45 deg and use tan(alpha + beta) = [tan(alpha) + tan(beta)] / [1 - tan(alpha)tan(beta)]

Step 2: tan(105 deg) = [tan(60 deg) + tan(45 deg)] / [1 - tan(60 deg)tan(45 deg)] = (sqrt(3) + 1) / (1 - sqrt(3))

Step 3: Rationalise by multiplying top and bottom by (1 + sqrt(3)), giving (sqrt(3) + 1)(1 + sqrt(3)) / (1 - 3) = (4 + 2 sqrt(3)) / (-2)

Step 4: Simplify, = -(2 + sqrt(3)), which is negative as expected for an angle in the second quadrant

Final answer: tan(105 deg) = -(2 + sqrt(3)).

Question: Q4(iii). Without using tables or calculator, find cos(165 deg).

Answer: Step 1: Write 165 deg = 180 deg - 15 deg, so cos(165 deg) = -cos(15 deg)

Step 2: Find cos(15 deg) from 15 deg = 45 deg - 30 deg, cos(15 deg) = cos(45 deg)cos(30 deg) + sin(45 deg)sin(30 deg)

Step 3: Substitute values, = (1 / sqrt(2))(sqrt(3) / 2) + (1 / sqrt(2))(1 / 2) = (sqrt(3) + 1) / (2 sqrt(2))

Step 4: Apply the negative sign from Step 1

Final answer: cos(165 deg) = -(sqrt(3) + 1) / (2 sqrt(2)).

Question: Q4(iv). Without using tables or calculator, find sin(255 deg).

Answer: Step 1: Write 255 deg = 180 deg + 75 deg, so sin(255 deg) = -sin(75 deg), using sin(180 deg + theta) = -sin(theta)

Step 2: From Q4(i), sin(75 deg) = (sqrt(3) + 1) / (2 sqrt(2)), obtained from sin(45 deg + 30 deg)

Step 3: Apply the negative sign

Final answer: sin(255 deg) = -(sqrt(3) + 1) / (2 sqrt(2)).

Question: Q4(v). Without using tables or calculator, find tan(5 pi / 12).

Answer: Step 1: Convert, 5 pi / 12 = 75 deg, and write 75 deg = 45 deg + 30 deg

Step 2: Use tan(alpha + beta) = [tan(alpha) + tan(beta)] / [1 - tan(alpha)tan(beta)] = [1 + 1 / sqrt(3)] / [1 - 1 / sqrt(3)]

Step 3: Multiply top and bottom by sqrt(3), giving (sqrt(3) + 1) / (sqrt(3) - 1)

Step 4: Rationalise, (sqrt(3) + 1)^2 / (3 - 1) = (4 + 2 sqrt(3)) / 2 = 2 + sqrt(3)

Final answer: tan(5 pi / 12) = 2 + sqrt(3).

Question: Q4(vi). Without using tables or calculator, find cos(13 pi / 12).

Answer: Step 1: Convert, 13 pi / 12 = 195 deg = 180 deg + 15 deg, so cos(13 pi / 12) = -cos(15 deg)

Step 2: cos(15 deg) = cos(45 deg - 30 deg) = cos(45 deg)cos(30 deg) + sin(45 deg)sin(30 deg) = (sqrt(3) + 1) / (2 sqrt(2))

Step 3: Apply the negative sign from Step 1

Final answer: cos(13 pi / 12) = -(sqrt(3) + 1) / (2 sqrt(2)).

Question: Q4(vii). Without using tables or calculator, find sin(23 pi / 12).

Answer: Step 1: Convert, 23 pi / 12 = 345 deg = 360 deg - 15 deg, so sin(23 pi / 12) = -sin(15 deg)

Step 2: sin(15 deg) = sin(45 deg - 30 deg) = sin(45 deg)cos(30 deg) - cos(45 deg)sin(30 deg) = (sqrt(3) - 1) / (2 sqrt(2))

Step 3: Apply the negative sign from Step 1

Final answer: sin(23 pi / 12) = -(sqrt(3) - 1) / (2 sqrt(2)).

Question: Q4(viii). Without using tables or calculator, find cos(25 pi / 12).

Answer: Step 1: Convert, 25 pi / 12 = 375 deg = 360 deg + 15 deg, so cos(25 pi / 12) = cos(15 deg)

Step 2: cos(15 deg) = cos(45 deg - 30 deg) = cos(45 deg)cos(30 deg) + sin(45 deg)sin(30 deg)

Step 3: Substitute values, = (1 / sqrt(2))(sqrt(3) / 2) + (1 / sqrt(2))(1 / 2) = (sqrt(3) + 1) / (2 sqrt(2))

Final answer: cos(25 pi / 12) = (sqrt(3) + 1) / (2 sqrt(2)).

Question: Q5(i). Prove that [sin(alpha + beta) + sin(alpha - beta)] / [cos(alpha + beta) + cos(alpha - beta)] = tan(alpha).

Answer: Step 1: Expand the numerator using the sin sum and difference formulas, (sin(alpha)cos(beta) + cos(alpha)sin(beta)) + (sin(alpha)cos(beta) - cos(alpha)sin(beta)) = 2 sin(alpha)cos(beta), because the cos(alpha)sin(beta) terms cancel

Step 2: Expand the denominator, (cos(alpha)cos(beta) - sin(alpha)sin(beta)) + (cos(alpha)cos(beta) + sin(alpha)sin(beta)) = 2 cos(alpha)cos(beta), because the sin(alpha)sin(beta) terms cancel

Step 3: Divide, LHS = [2 sin(alpha)cos(beta)] / [2 cos(alpha)cos(beta)], and cancel 2 cos(beta)

Step 4: LHS = sin(alpha) / cos(alpha) = tan(alpha)

Final answer: LHS = RHS, so the identity is proved.

Question: Q5(ii). Prove that [cos(alpha + beta) + cos(alpha - beta)] / [sin(alpha + beta) + sin(alpha - beta)] = cot(alpha).

Answer: Step 1: Expand the numerator, (cos(alpha)cos(beta) -

sin(alpha)sin(beta)) + (cos(alpha)cos(beta) + sin(alpha)sin(beta)) = 2 cos(alpha)cos(beta)

Step 2: Expand the denominator, (sin(alpha)cos(beta) + cos(alpha)sin(beta)) + (sin(alpha)cos(beta) - cos(alpha)sin(beta)) = 2 sin(alpha)cos(beta)

Step 3: Divide, LHS = [2 cos(alpha)cos(beta)] / [2 sin(alpha)cos(beta)], and cancel 2 cos(beta)

Step 4: LHS = cos(alpha) / sin(alpha) = cot(alpha)

Final answer: LHS = RHS, so the identity is proved.

Question: Q6(i). Prove that sin(alpha + beta)sin(alpha - beta) = sin^2(alpha) - sin^2(beta).

Answer: Step 1: Expand both factors, LHS = [sin(alpha)cos(beta) + cos(alpha)sin(beta)] x [sin(alpha)cos(beta) - cos(alpha)sin(beta)], which is a difference of squares

Step 2: So LHS = sin^2(alpha)cos^2(beta) - cos^2(alpha)sin^2(beta)

Step 3: Replace cos^2(beta) by 1 - sin^2(beta) and cos^2(alpha) by 1 - sin^2(alpha), using sin^2 + cos^2 = 1, giving LHS = sin^2(alpha)(1 - sin^2(beta)) - (1 - sin^2(alpha))sin^2(beta)

Step 4: Expand, = sin^2(alpha) - sin^2(alpha)sin^2(beta) - sin^2(beta) + sin^2(alpha)sin^2(beta), and the middle terms cancel

Step 5: LHS = sin^2(alpha) - sin^2(beta)

Final answer: LHS = RHS, so the identity is proved.

Question: Q6(ii). Prove that cos(alpha + beta)cos(alpha - beta) = cos^2(alpha) - sin^2(beta).

Answer: Step 1: Expand both factors, LHS = [cos(alpha)cos(beta) - sin(alpha)sin(beta)] x [cos(alpha)cos(beta) + sin(alpha)sin(beta)], a difference of squares

Step 2: So LHS = cos^2(alpha)cos^2(beta) - sin^2(alpha)sin^2(beta)

Step 3: Replace cos^2(beta) by 1 - sin^2(beta) and sin^2(alpha) by 1 - cos^2(alpha), giving LHS = cos^2(alpha)(1 - sin^2(beta)) - (1 - cos^2(alpha))sin^2(beta)

Step 4: Expand, = cos^2(alpha) - cos^2(alpha)sin^2(beta) - sin^2(beta) + cos^2(alpha)sin^2(beta), and the middle terms cancel

Step 5: LHS = cos^2(alpha) - sin^2(beta)

Final answer: LHS = RHS, so the identity is proved.

Question: Q7(i). Prove that cos(alpha + beta) + cos(alpha - beta) = 2 cos(alpha)cos(beta).

Answer: Step 1: Expand the first term, cos(alpha + beta) = cos(alpha)cos(beta) - sin(alpha)sin(beta)

Step 2: Expand the second term, cos(alpha - beta) = cos(alpha)cos(beta) + sin(alpha)sin(beta)

Step 3: Add them, and the sin(alpha)sin(beta) terms cancel, one positive and one negative

Step 4: LHS = 2 cos(alpha)cos(beta)

Final answer: LHS = RHS, so the identity is proved.

Question: Q7(ii). Prove that cos(alpha)cos(beta - alpha) - sin(alpha)sin(beta - alpha) = cos(beta).

Answer: Step 1: Notice the LHS has the pattern cos(A)cos(B) - sin(A)sin(B)

with A = alpha and B = beta - alpha, which is cos(A + B)

Step 2: So LHS = cos(alpha + (beta - alpha))

Step 3: Simplify the angle, alpha + beta - alpha = beta

Step 4: LHS = cos(beta)

Final answer: LHS = RHS, so the identity is proved.

Question: Q8 (preliminary values). If sin(alpha) = 12/13 and sin(beta) = 3/5, where 0 < alpha < pi/2 and 0 < beta < pi/2, find cos(alpha), cos(beta), tan(alpha), tan(beta), cot(alpha) and cot(beta) for use in parts (i) to (viii).

Answer: Step 1: Both angles are in the first quadrant, so all their ratios are positive

Step 2: From sin^2 + cos^2 = 1, cos(alpha) = sqrt(1 - (12/13)^2) = sqrt(25/169) = 5/13

Step 3: Similarly cos(beta) = sqrt(1 - (3/5)^2) = sqrt(16/25) = 4/5

Step 4: tan(alpha) = sin(alpha) / cos(alpha) = (12/13) / (5/13) = 12/5 and tan(beta) = (3/5) / (4/5) = 3/4

Step 5: Take reciprocals for cot, cot(alpha) = 5/12 and cot(beta) = 4/3

Final answer: cos(alpha) = 5/13, cos(beta) = 4/5, tan(alpha) = 12/5, tan(beta) = 3/4, cot(alpha) = 5/12, cot(beta) = 4/3.

Question: Q8(i). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, find cos(alpha + beta).

Answer: Step 1: Use cos(alpha + beta) = cos(alpha)cos(beta) - sin(alpha)sin(beta)

Step 2: Substitute cos(alpha) = 5/13, cos(beta) = 4/5, sin(alpha) = 12/13,

sin(beta) = 3/5

Step 3: = (5/13)(4/5) - (12/13)(3/5) = 20/65 - 36/65

Final answer: cos(alpha + beta) = -16/65.

Question: Q8(ii). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, find tan(alpha + beta).

Answer: Step 1: Use tan(alpha + beta) = [tan(alpha) + tan(beta)] / [1 - tan(alpha)tan(beta)]

Step 2: Substitute tan(alpha) = 12/5 and tan(beta) = 3/4, giving numerator 12/5 + 3/4 = 63/20

Step 3: The denominator is 1 - (12/5)(3/4) = 1 - 36/20 = -16/20

Step 4: Divide, (63/20) / (-16/20) = -63/16

Final answer: tan(alpha + beta) = -63/16.

Question: Q8(iii). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, find cot(alpha - beta).

Answer: Step 1: Use cot(alpha - beta) = [cot(alpha)cot(beta) + 1] / [cot(beta) - cot(alpha)]

Step 2: Substitute cot(alpha) = 5/12 and cot(beta) = 4/3, giving numerator (5/12)(4/3) + 1 = 20/36 + 1 = 14/9

Step 3: The denominator is 4/3 - 5/12 = 11/12

Step 4: Divide, (14/9) / (11/12) = (14/9) x (12/11) = 56/33

Final answer: cot(alpha - beta) = 56/33.

Question: Q8(iv). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, find cos(alpha - beta).

Answer: Step 1: Use cos(alpha - beta) = cos(alpha)cos(beta) +

sin(alpha)sin(beta)

Step 2: Substitute the values, = (5/13)(4/5) + (12/13)(3/5)

Step 3: = 20/65 + 36/65

Final answer: cos(alpha - beta) = 56/65.

Question: Q8(v). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, find sin(alpha + beta).

Answer: Step 1: Use sin(alpha + beta) = sin(alpha)cos(beta) + cos(alpha)sin(beta)

Step 2: Substitute the values, = (12/13)(4/5) + (5/13)(3/5)

Step 3: = 48/65 + 15/65

Final answer: sin(alpha + beta) = 63/65.

Question: Q8(vi). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, find sin(alpha - beta).

Answer: Step 1: Use sin(alpha - beta) = sin(alpha)cos(beta) - cos(alpha)sin(beta)

Step 2: Substitute the values, = (12/13)(4/5) - (5/13)(3/5)

Step 3: = 48/65 - 15/65

Final answer: sin(alpha - beta) = 33/65.

Question: Q8(vii). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, find cot(alpha + beta).

Answer: Step 1: Use cot(alpha + beta) = [cot(alpha)cot(beta) - 1] / [cot(alpha) + cot(beta)]

Step 2: Substitute cot(alpha) = 5/12 and cot(beta) = 4/3, giving numerator (5/12)(4/3) - 1 = 20/36 - 1 = -16/36 = -4/9

Step 3: The denominator is 5/12 + 4/3 = 21/12 = 7/4

Step 4: Divide, (-4/9) / (7/4) = -16/63, which agrees with 1 / tan(alpha + beta) = -16/63

Final answer: cot(alpha + beta) = -16/63.

Question: Q8(viii). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, find tan(alpha - beta).

Answer: Step 1: Use tan(alpha - beta) = [tan(alpha) - tan(beta)] / [1 + tan(alpha)tan(beta)]

Step 2: Substitute tan(alpha) = 12/5 and tan(beta) = 3/4, giving numerator 12/5 - 3/4 = 33/20

Step 3: The denominator is 1 + (12/5)(3/4) = 1 + 36/20 = 56/20

Step 4: Divide, (33/20) / (56/20) = 33/56

Final answer: tan(alpha - beta) = 33/56.

Question: Q8 (quadrants). With sin(alpha) = 12/13 and sin(beta) = 3/5, both angles acute, also find the quadrants of the angles (alpha + beta) and (alpha - beta).

Answer: Step 1: For alpha + beta, sin(alpha + beta) = 63/65 is positive but cos(alpha + beta) = -16/65 is negative

Step 2: Sine positive and cosine negative happens only in the second quadrant

Step 3: For alpha - beta, sin(alpha - beta) = 33/65 and cos(alpha - beta) = 56/65 are both positive, and alpha > beta because sin(alpha) > sin(beta) for acute angles, so alpha - beta is a positive acute angle

Final answer: (alpha + beta) is in the second quadrant (Quadrant II) and

(alpha - beta) is in the first quadrant (Quadrant I).

Exercise 10.2

Question: Q1(i). Generate all allied angles if theta = 60 deg.

Answer: Step 1: Allied angles are given by n(90 deg) +- theta for integer n, so here they are n(90 deg) +- 60 deg

Step 2: For n = 0, we get +- 60 deg

Step 3: For n = 1, 90 deg +- 60 deg gives 150 deg and 30 deg, and for n = -1, -90 deg +- 60 deg gives -30 deg and -150 deg

Step 4: For n = 2, 180 deg +- 60 deg gives 240 deg and 120 deg, and for n = -2, -180 deg +- 60 deg gives -120 deg and -240 deg, and so on

Final answer: The allied angles of 60 deg are +- 30 deg, +- 60 deg, +- 120 deg, +- 150 deg, +- 240 deg, ... generated by n(90 deg) +- 60 deg.

Question: Q1(ii). Generate all allied angles if theta = 50 deg.

Answer: Step 1: Use the general form n(90 deg) +- theta with theta = 50 deg

Step 2: For n = 0, we get +- 50 deg

Step 3: For n = 1, 90 deg +- 50 deg gives 140 deg and 40 deg, and for n = -1, -90 deg +- 50 deg gives -40 deg and -140 deg

Step 4: For n = 2, 180 deg +- 50 deg gives 230 deg and 130 deg, and continuing in this way gives all the rest

Final answer: The allied angles of 50 deg are +- 40 deg, +- 50 deg, +- 130 deg, +- 140 deg, +- 230 deg, ... generated by n(90 deg) +- 50 deg.

Question: Q2(i). Show that sin(90 deg - theta) / cosec(90 deg - theta) + cos(90 deg - theta) / sec(90 deg - theta) = 1.

Answer: Step 1: Use the co-ratio results sin(90 deg - theta) = cos(theta) and cosec(90 deg - theta) = sec(theta), so the first fraction is cos(theta) / sec(theta)

Step 2: Since sec(theta) = 1 / cos(theta), cos(theta) / sec(theta) = cos(theta) x cos(theta) = cos^2(theta)

Step 3: Similarly cos(90 deg - theta) = sin(theta) and sec(90 deg - theta) = cosec(theta), so the second fraction is sin(theta) / cosec(theta) = sin^2(theta)

Step 4: Add, LHS = cos^2(theta) + sin^2(theta) = 1, using sin^2 + cos^2 = 1

Final answer: LHS = 1 = RHS, as required.

Question: Q2(ii). Show that sin^2(2 deg) + sin^2(3 deg) + sin^2(87 deg) + sin^2(88 deg) = 2.

Answer: Step 1: Pair each angle with its complement, 87 deg = 90 deg - 3 deg and 88 deg = 90 deg - 2 deg

Step 2: So sin(87 deg) = sin(90 deg - 3 deg) = cos(3 deg) and sin(88 deg) = cos(2 deg)

Step 3: The sum becomes [sin^2(2 deg) + cos^2(2 deg)] + [sin^2(3 deg) + cos^2(3 deg)]

Step 4: Each bracket equals 1 by sin^2 + cos^2 = 1, so the total is 1 + 1

Final answer: The sum is 2, as required.

Question: Q3(i). Evaluate without using a calculator: sin(675 deg).

Answer: Step 1: Remove full turns, 675 deg = 360 deg + 315 deg, so

sin(675 deg) = sin(315 deg)

Step 2: Write 315 deg = 360 deg - 45 deg, so sin(315 deg) = -sin(45 deg)

Step 3: Use sin(45 deg) = 1 / sqrt(2)

Final answer: sin(675 deg) = -1 / sqrt(2).

Question: Q3(ii). Evaluate without using a calculator: tan(25 pi / 6).

Answer: Step 1: Reduce the angle, 25 pi / 6 = 4 pi + pi / 6, and tan has period pi, so tan(25 pi / 6) = tan(pi / 6)

Step 2: pi / 6 is 30 deg

Step 3: Use the standard value tan(30 deg) = 1 / sqrt(3)

Final answer: tan(25 pi / 6) = 1 / sqrt(3).

Question: Q3(iii). Evaluate without using a calculator: cosec(2130 deg).

Answer: Step 1: Remove full turns, 2130 deg = 5 x 360 deg + 330 deg, so cosec(2130 deg) = cosec(330 deg) = 1 / sin(330 deg)

Step 2: sin(330 deg) = sin(360 deg - 30 deg) = -sin(30 deg) = -1/2

Step 3: Take the reciprocal, 1 / (-1/2) = -2

Final answer: cosec(2130 deg) = -2.

Question: Q3(iv). Evaluate without using a calculator: sec(55 deg) / cosec(35 deg) + sin(48 deg) / cos(42 deg).

Answer: Step 1: Use complements, sec(55 deg) = sec(90 deg - 35 deg) = cosec(35 deg), so the first fraction is cosec(35 deg) / cosec(35 deg) = 1

Step 2: Also cos(42 deg) = cos(90 deg - 48 deg) = sin(48 deg), so the second fraction is sin(48 deg) / sin(48 deg) = 1

Step 3: Add the two results, 1 + 1

Final answer: The value is 2.

Question: Q3(v). Evaluate without using a calculator: 2 x [sin(56 deg) / cos(34 deg)] - cot(37 deg) / tan(53 deg) - sqrt(2) sin(45 deg).

Answer: Step 1: cos(34 deg) = cos(90 deg - 56 deg) = sin(56 deg), so the first bracket is 1 and 2 x 1 = 2

Step 2: tan(53 deg) = tan(90 deg - 37 deg) = cot(37 deg), so the second fraction is 1

Step 3: sqrt(2) sin(45 deg) = sqrt(2) x (1 / sqrt(2)) = 1

Step 4: Combine, 2 - 1 - 1 = 0

Final answer: The value is 0.

Question: Q3(vi). Evaluate without using a calculator: cos(1 deg) + cos(2 deg) + cos(3 deg) + ... + cos(180 deg).

Answer: Step 1: Pair terms that add to 180 deg, using cos(180 deg - theta) = -cos(theta), so cos(1 deg) + cos(179 deg) = 0, cos(2 deg) + cos(178 deg) = 0, and so on up to cos(89 deg) + cos(91 deg) = 0

Step 2: The middle term cos(90 deg) = 0 also contributes nothing

Step 3: The only unpaired term left is the last one, cos(180 deg) = -1

Step 4: So the whole sum reduces to -1

Final answer: The sum is -1.

Question: Q3(vii). Evaluate without using a calculator: tan(1 deg) x tan(2 deg) x tan(3 deg) x ... x tan(89 deg).

Answer: Step 1: Pair factors that add to 90 deg, using tan(90 deg - theta) = cot(theta) = 1 / tan(theta)

Step 2: Each pair, such as tan(1 deg) x tan(89 deg) = tan(1 deg) x cot(1

deg) = 1, and similarly for 2 deg with 88 deg, up to 44 deg with 46 deg

Step 3: The middle unpaired factor is tan(45 deg) = 1

Step 4: The product is therefore 1 x 1 x ... x 1

Final answer: The product is 1.

Question: Q4(i). Show that cos(45 deg - theta) = (1 / sqrt(2))(cos(theta) + sin(theta)).

Answer: Step 1: Expand the LHS with cos(alpha - beta) = cos(alpha)cos(beta) + sin(alpha)sin(beta)

Step 2: cos(45 deg - theta) = cos(45 deg)cos(theta) + sin(45 deg)sin(theta)

Step 3: Use cos(45 deg) = sin(45 deg) = 1 / sqrt(2), giving = (1 / sqrt(2))cos(theta) + (1 / sqrt(2))sin(theta)

Step 4: Factor out 1 / sqrt(2)

Final answer: cos(45 deg - theta) = (1 / sqrt(2))(cos(theta) + sin(theta)), as required.

Question: Q4(ii). Show that tan(45 deg + theta)tan(45 deg - theta) = 1.

Answer: Step 1: Note (45 deg - theta) is the complement of (45 deg + theta), because they add to 90 deg, so tan(45 deg - theta) = cot(45 deg + theta)

Step 2: Hence the product is tan(45 deg + theta) x cot(45 deg + theta)

Step 3: Since cot is the reciprocal of tan, tan(x) x cot(x) = 1 for x = 45 deg + theta

Step 4: Alternatively, expanding both factors with the tan sum and difference formulas gives [(1 + tan(theta)) / (1 - tan(theta))] x [(1 -

tan(theta)) / (1 + tan(theta))] = 1, confirming the result

Final answer: The product is 1, as required.

Question: Q5(i). Express 15 sin(theta) + 8 cos(theta) in the form r sin(theta + phi), where theta and phi are in the first quadrant.

Answer: Step 1: Compare with a sin(theta) + b cos(theta) = r sin(theta + phi), where a = r cos(phi) and b = r sin(phi), so here r cos(phi) = 15 and r sin(phi) = 8

Step 2: Square and add, r^2 = 15^2 + 8^2 = 225 + 64 = 289, so r = 17, taking the positive root

Step 3: Then cos(phi) = 15/17 and sin(phi) = 8/17, which indeed gives a first-quadrant phi with tan(phi) = 8/15

Step 4: Substitute r and phi back into the form

Final answer: 15 sin(theta) + 8 cos(theta) = 17 sin(theta + phi), where cos(phi) = 15/17 and sin(phi) = 8/17.

Question: Q5(ii). Express (sqrt(3) / 2) sin(theta) + (1/2) cos(theta) in the form r sin(theta + phi), where theta and phi are in the first quadrant.

Answer: Step 1: Here r cos(phi) = sqrt(3) / 2 and r sin(phi) = 1/2

Step 2: Square and add, r^2 = 3/4 + 1/4 = 1, so r = 1

Step 3: Then cos(phi) = sqrt(3) / 2 and sin(phi) = 1/2, the standard values for phi = 30 deg

Step 4: Substitute back into the form

Final answer: (sqrt(3) / 2) sin(theta) + (1/2) cos(theta) = sin(theta + 30 deg).

Question: Q5(iii). Express sin(theta) + cos(theta) in the form r sin(theta +

phi), where theta and phi are in the first quadrant.

Answer: Step 1: Here r cos(phi) = 1 and r sin(phi) = 1

Step 2: Square and add, r^2 = 1 + 1 = 2, so r = sqrt(2)

Step 3: Then cos(phi) = 1 / sqrt(2) and sin(phi) = 1 / sqrt(2), giving phi = 45 deg

Step 4: Substitute back into the form, and check by expanding, sqrt(2) sin(theta + 45 deg) = sqrt(2)[sin(theta)(1 / sqrt(2)) + cos(theta)(1 / sqrt(2))] = sin(theta) + cos(theta)

Final answer: sin(theta) + cos(theta) = sqrt(2) sin(theta + 45 deg).

Question: Q6. Prove that tan(alpha) / cot(90 deg - alpha) + cos(90 deg - alpha) / sin(alpha) = 2.

Answer: Step 1: Use the co-ratio cot(90 deg - alpha) = tan(alpha), so the first fraction is tan(alpha) / tan(alpha) = 1

Step 2: Use cos(90 deg - alpha) = sin(alpha), so the second fraction is sin(alpha) / sin(alpha) = 1

Step 3: Add the two fractions, 1 + 1

Final answer: LHS = 2 = RHS, so the identity is proved.

Question: Q7(i). If A + B + C = 180 deg, prove that tan(A) + tan(B) + tan(C) = tan(A)tan(B)tan(C).

Answer: Step 1: From A + B = 180 deg - C, take tan of both sides, tan(A + B) = tan(180 deg - C) = -tan(C)

Step 2: Expand the LHS, [tan(A) + tan(B)] / [1 - tan(A)tan(B)] = -tan(C)

Step 3: Cross-multiply, tan(A) + tan(B) = -tan(C) + tan(A)tan(B)tan(C)

Step 4: Bring -tan(C) to the left, giving tan(A) + tan(B) + tan(C) =

tan(A)tan(B)tan(C)

Final answer: The identity is proved.

Question: Q7(ii). If A + B + C = 180 deg, prove that cos((B + C) / 2) = sin(A / 2).

Answer: Step 1: From B + C = 180 deg - A, divide by 2, giving (B + C) / 2 = 90 deg - A / 2

Step 2: Substitute into the LHS, cos((B + C) / 2) = cos(90 deg - A / 2)

Step 3: Use the co-ratio cos(90 deg - x) = sin(x) with x = A / 2

Final answer: cos((B + C) / 2) = sin(A / 2), as required.

Question: Q7(iii). If A + B + C = 180 deg, prove that tan(A / 2)tan(B / 2) + tan(B / 2)tan(C / 2) + tan(C / 2)tan(A / 2) = 1.

Answer: Step 1: From (A + B) / 2 = 90 deg - C / 2, take tan of both sides, tan((A + B) / 2) = cot(C / 2) = 1 / tan(C / 2)

Step 2: Expand the LHS, [tan(A / 2) + tan(B / 2)] / [1 - tan(A / 2)tan(B / 2)] = 1 / tan(C / 2)

Step 3: Cross-multiply, tan(A / 2)tan(C / 2) + tan(B / 2)tan(C / 2) = 1 - tan(A / 2)tan(B / 2)

Step 4: Bring tan(A / 2)tan(B / 2) to the left, giving the sum of the three pairwise products equal to 1

Final answer: tan(A / 2)tan(B / 2) + tan(B / 2)tan(C / 2) + tan(C / 2)tan(A / 2) = 1, as required.

Exercise 10.3

Question: Q1. Prove the identity sin(2theta) / sin(theta) - cos(2theta) / cos(theta) = sec(theta).

Answer: Step 1: Use the double-angle formula sin(2theta) = 2 sin(theta)cos(theta), so the first term is 2 sin(theta)cos(theta) / sin(theta) = 2 cos(theta)

Step 2: Use cos(2theta) = 2 cos^2(theta) - 1, so the second term is [2 cos^2(theta) - 1] / cos(theta) = 2 cos(theta) - 1 / cos(theta) = 2 cos(theta) - sec(theta)

Step 3: Subtract, LHS = 2 cos(theta) - [2 cos(theta) - sec(theta)]

Step 4: The 2 cos(theta) terms cancel, leaving sec(theta)

Final answer: LHS = sec(theta) = RHS, so the identity is proved.

Question: Q2. Prove the identity sin(3alpha) / sin(alpha) - cos(3alpha) / cos(alpha) = 2.

Answer: Step 1: Use the triple-angle formulas sin(3alpha) = 3 sin(alpha) - 4 sin^3(alpha) and cos(3alpha) = 4 cos^3(alpha) - 3 cos(alpha)

Step 2: Divide the first by sin(alpha), giving 3 - 4 sin^2(alpha)

Step 3: Divide the second by cos(alpha), giving 4 cos^2(alpha) - 3

Step 4: Subtract, LHS = [3 - 4 sin^2(alpha)] - [4 cos^2(alpha) - 3] = 6 - 4[sin^2(alpha) + cos^2(alpha)]

Step 5: Use sin^2 + cos^2 = 1, giving LHS = 6 - 4 = 2

Final answer: LHS = 2 = RHS, so the identity is proved.

Question: Q3. Prove the identity sin(4theta) / sin(2theta) - cos(4theta) / cos(2theta) = sec(2theta).

Answer: Step 1: This is the same pattern as Q1 with theta replaced by

2theta

Step 2: Use sin(4theta) = 2 sin(2theta)cos(2theta), so the first term is 2 cos(2theta)

Step 3: Use cos(4theta) = 2 cos^2(2theta) - 1, so the second term is 2 cos(2theta) - sec(2theta)

Step 4: Subtract, LHS = 2 cos(2theta) - 2 cos(2theta) + sec(2theta)

Final answer: LHS = sec(2theta) = RHS, so the identity is proved.

Question: Q4. Prove the identity cos(2alpha) = [1 - tan^2(alpha)] / [1 + tan^2(alpha)].

Answer: Step 1: Start from the RHS and write tan(alpha) = sin(alpha) / cos(alpha), so RHS = [1 - sin^2(alpha) / cos^2(alpha)] / [1 + sin^2(alpha) / cos^2(alpha)]

Step 2: Multiply numerator and denominator by cos^2(alpha), giving RHS = [cos^2(alpha) - sin^2(alpha)] / [cos^2(alpha) + sin^2(alpha)]

Step 3: The numerator is the double-angle formula cos(2alpha) = cos^2(alpha) - sin^2(alpha)

Step 4: The denominator is sin^2 + cos^2 = 1

Step 5: So RHS = cos(2alpha) / 1 = cos(2alpha)

Final answer: RHS = LHS, so the identity is proved.

Question: Q5. Prove the identity [cos(theta / 2) - sin(theta / 2)]^2 = 1 - sin(theta).

Answer: Step 1: Expand the square on the LHS, = cos^2(theta / 2) - 2 sin(theta / 2)cos(theta / 2) + sin^2(theta / 2)

Step 2: Group the squared terms, cos^2(theta / 2) + sin^2(theta / 2) = 1

by sin^2 + cos^2 = 1

Step 3: The middle term 2 sin(theta / 2)cos(theta / 2) is the double-angle formula sin(theta), with theta / 2 doubled

Step 4: So LHS = 1 - sin(theta)

Final answer: LHS = RHS, so the identity is proved.

Question: Q6. Prove the identity sin(2alpha) = 2 tan(alpha) / [1 + tan^2(alpha)].

Answer: Step 1: Start from the RHS and write tan(alpha) = sin(alpha) / cos(alpha), so RHS = [2 sin(alpha) / cos(alpha)] / [1 + sin^2(alpha) / cos^2(alpha)]

Step 2: Multiply numerator and denominator by cos^2(alpha), giving RHS = 2 sin(alpha)cos(alpha) / [cos^2(alpha) + sin^2(alpha)]

Step 3: The denominator is 1, and the numerator is the double-angle formula sin(2alpha) = 2 sin(alpha)cos(alpha)

Final answer: RHS = sin(2alpha) = LHS, so the identity is proved.

Question: Q7. Prove the identity cos(2theta) = [cot^2(theta) - 1] / [cot^2(theta) + 1].

Answer: Step 1: Start from the RHS and write cot(theta) = cos(theta) / sin(theta), so RHS = [cos^2(theta) / sin^2(theta) - 1] / [cos^2(theta) / sin^2(theta) + 1]

Step 2: Multiply numerator and denominator by sin^2(theta), giving RHS = [cos^2(theta) - sin^2(theta)] / [cos^2(theta) + sin^2(theta)]

Step 3: The numerator is cos(2theta) and the denominator is 1

Final answer: RHS = cos(2theta) = LHS, so the identity is proved.

Question: Q8. Prove the identity sin(4theta) = 4 sin(theta)cos(theta)cos(2theta).

Answer: Step 1: Write 4theta as double of 2theta and use sin(2x) = 2 sin(x)cos(x) with x = 2theta, giving sin(4theta) = 2 sin(2theta)cos(2theta)

Step 2: Now use the double-angle formula again on sin(2theta) = 2 sin(theta)cos(theta)

Step 3: Substitute, sin(4theta) = 2 x [2 sin(theta)cos(theta)] x cos(2theta)

Step 4: Collect the constants, 2 x 2 = 4

Final answer: sin(4theta) = 4 sin(theta)cos(theta)cos(2theta) = RHS, so the identity is proved.

Question: Q9. Prove the identity cos(4theta) = 8 cos^4(theta) - 8 cos^2(theta) + 1.

Answer: Step 1: Use the double-angle formula cos(4theta) = 2 cos^2(2theta) - 1

Step 2: Use cos(2theta) = 2 cos^2(theta) - 1 inside it, giving cos(4theta) = 2[2 cos^2(theta) - 1]^2 - 1

Step 3: Expand the square, [2 cos^2(theta) - 1]^2 = 4 cos^4(theta) - 4 cos^2(theta) + 1

Step 4: Multiply by 2 and subtract 1, giving 8 cos^4(theta) - 8 cos^2(theta) + 2 - 1

Final answer: cos(4theta) = 8 cos^4(theta) - 8 cos^2(theta) + 1 = RHS, so the identity is proved.

Question: Q10. Prove the identity tan(alpha / 2) = cosec(alpha) - cot(alpha).

Answer: Step 1: Start from the RHS, cosec(alpha) - cot(alpha) = 1 /

sin(alpha) - cos(alpha) / sin(alpha) = [1 - cos(alpha)] / sin(alpha)

Step 2: Use the half-angle forms 1 - cos(alpha) = 2 sin^2(alpha / 2) and sin(alpha) = 2 sin(alpha / 2)cos(alpha / 2)

Step 3: Substitute, RHS = [2 sin^2(alpha / 2)] / [2 sin(alpha / 2)cos(alpha / 2)]

Step 4: Cancel 2 sin(alpha / 2), giving sin(alpha / 2) / cos(alpha / 2) = tan(alpha / 2)

Final answer: RHS = tan(alpha / 2) = LHS, so the identity is proved.

Question: Q11. Prove the identity cosec^2(theta / 2) = 2 sec(theta) / [sec(theta) - 1].

Answer: Step 1: Work on the RHS and write sec(theta) = 1 / cos(theta), so RHS = [2 / cos(theta)] / [1 / cos(theta) - 1]

Step 2: Simplify the denominator, 1 / cos(theta) - 1 = [1 - cos(theta)] / cos(theta), so RHS = 2 / [1 - cos(theta)]

Step 3: On the LHS, cosec^2(theta / 2) = 1 / sin^2(theta / 2), and the half-angle formula gives sin^2(theta / 2) = [1 - cos(theta)] / 2

Step 4: So LHS = 1 / ([1 - cos(theta)] / 2) = 2 / [1 - cos(theta)], the same as the RHS

Final answer: LHS = RHS, so the identity is proved.

Question: Q12. Prove the identity [sin(theta) + sin(2theta)] / [1 + cos(theta) + cos(2theta)] = tan(theta).

Answer: Step 1: In the numerator use sin(2theta) = 2 sin(theta)cos(theta) and factorise, sin(theta) + sin(2theta) = sin(theta)[1 + 2 cos(theta)]

Step 2: In the denominator use cos(2theta) = 2 cos^2(theta) - 1, so 1 +

cos(theta) + cos(2theta) = 1 + cos(theta) + 2 cos^2(theta) - 1 = cos(theta) + 2 cos^2(theta)

Step 3: Factorise the denominator, = cos(theta)[1 + 2 cos(theta)]

Step 4: Divide and cancel the common factor [1 + 2 cos(theta)], giving sin(theta) / cos(theta)

Final answer: LHS = tan(theta) = RHS, so the identity is proved.

Question: Q13. Prove the identity tan(4theta) = 4 tan(theta)[1 - tan^2(theta)] / [tan^4(theta) - 6 tan^2(theta) + 1].

Answer: Step 1: Let t = tan(theta) and first use tan(2x) = 2 tan(x) / [1 - tan^2(x)] to get tan(2theta) = 2t / (1 - t^2)

Step 2: Apply the same formula again, tan(4theta) = 2 tan(2theta) / [1 - tan^2(2theta)] = [4t / (1 - t^2)] / [1 - 4t^2 / (1 - t^2)^2]

Step 3: Multiply numerator and denominator by (1 - t^2)^2, giving numerator 4t(1 - t^2)

Step 4: The denominator becomes (1 - t^2)^2 - 4t^2 = 1 - 2t^2 + t^4 - 4t^2 = t^4 - 6t^2 + 1

Step 5: Put t = tan(theta) back

Final answer: tan(4theta) = 4 tan(theta)[1 - tan^2(theta)] / [tan^4(theta) - 6 tan^2(theta) + 1] = RHS, so the identity is proved.

Question: Q14. Prove the identity sec(theta) = [tan(theta / 2) + cot(theta / 2)] / [cot(theta / 2) - tan(theta / 2)].

Answer: Step 1: Write the numerator in sin and cos of theta / 2, tan(theta / 2) + cot(theta / 2) = sin(theta / 2) / cos(theta / 2) + cos(theta / 2) / sin(theta / 2) = [sin^2(theta / 2) + cos^2(theta / 2)] / [sin(theta / 2)cos(theta / 2)] = 1 / [sin(theta / 2)cos(theta / 2)]

Step 2: Since sin(theta) = 2 sin(theta / 2)cos(theta / 2), this numerator equals 2 / sin(theta)

Step 3: Similarly the denominator is [cos^2(theta / 2) - sin^2(theta / 2)] / [sin(theta / 2)cos(theta / 2)] = cos(theta) / [sin(theta / 2)cos(theta / 2)] = 2 cos(theta) / sin(theta), using the double-angle formula cos(theta) = cos^2(theta / 2) - sin^2(theta / 2)

Step 4: Divide numerator by denominator, [2 / sin(theta)] / [2 cos(theta) / sin(theta)] = 1 / cos(theta)

Final answer: RHS = sec(theta) = LHS, so the identity is proved.

Question: Q15. Prove the identity cot(2theta) = [cot(theta) - tan(theta)] / 2.

Answer: Step 1: Start from the RHS, cot(theta) - tan(theta) = cos(theta) / sin(theta) - sin(theta) / cos(theta) = [cos^2(theta) - sin^2(theta)] / [sin(theta)cos(theta)]

Step 2: The numerator is the double-angle formula cos(2theta)

Step 3: Divide by 2, RHS = cos(2theta) / [2 sin(theta)cos(theta)], and 2 sin(theta)cos(theta) = sin(2theta)

Step 4: So RHS = cos(2theta) / sin(2theta) = cot(2theta)

Final answer: RHS = LHS, so the identity is proved.

Question: Q16. Prove the identity tan^2(theta / 2) = [tan(theta) - sin(theta)] / [tan(theta) + sin(theta)].

Answer: Step 1: Write tan(theta) = sin(theta) / cos(theta) in the numerator, tan(theta) - sin(theta) = sin(theta)[1 / cos(theta) - 1] = sin(theta)[1 - cos(theta)] / cos(theta)

Step 2: Similarly the denominator is tan(theta) + sin(theta) = sin(theta)[1

+ cos(theta)] / cos(theta)

Step 3: Divide, the factors sin(theta) / cos(theta) cancel, giving RHS = [1 - cos(theta)] / [1 + cos(theta)]

Step 4: By the half-angle formulas 1 - cos(theta) = 2 sin^2(theta / 2) and 1 + cos(theta) = 2 cos^2(theta / 2), so this ratio is tan^2(theta / 2)

Final answer: RHS = tan^2(theta / 2) = LHS, so the identity is proved.

Question: Q17(i). Evaluate tan(22.5 deg).

Answer: Step 1: 22.5 deg is half of 45 deg, so use tan(theta / 2) = sin(theta) / [1 + cos(theta)] with theta = 45 deg

Step 2: tan(22.5 deg) = sin(45 deg) / [1 + cos(45 deg)] = (1 / sqrt(2)) / (1 + 1 / sqrt(2))

Step 3: Multiply top and bottom by sqrt(2), giving 1 / (sqrt(2) + 1)

Step 4: Rationalise by multiplying by (sqrt(2) - 1) / (sqrt(2) - 1), giving (sqrt(2) - 1) / (2 - 1)

Final answer: tan(22.5 deg) = sqrt(2) - 1.

Question: Q17(ii). Evaluate cos(15 deg).

Answer: Step 1: 15 deg is half of 30 deg, so use the half-angle formula cos(theta / 2) = sqrt([1 + cos(theta)] / 2) with theta = 30 deg, taking the positive root because 15 deg is acute

Step 2: cos(15 deg) = sqrt([1 + sqrt(3) / 2] / 2) = sqrt(2 + sqrt(3)) / 2

Step 3: This nested-root value is the same as the sum value found from cos(45 deg - 30 deg), namely (sqrt(3) + 1) / (2 sqrt(2)), and squaring either form gives (2 + sqrt(3)) / 4

Final answer: cos(15 deg) = (sqrt(3) + 1) / (2 sqrt(2)) = sqrt(2 + sqrt(3)) /

2.

Question: Q18(i). If cos(theta) = -4/5 and pi / 2 < theta < pi, find sin(theta / 2).

Answer: Step 1: theta is in the second quadrant, so theta / 2 is in the first quadrant and all its ratios are positive

Step 2: Use sin(theta / 2) = sqrt([1 - cos(theta)] / 2) = sqrt([1 - (-4/5)] / 2) = sqrt((9/5) / 2)

Step 3: Simplify, = sqrt(9/10) = 3 / sqrt(10)

Final answer: sin(theta / 2) = 3 / sqrt(10).

Question: Q18(ii). If cos(theta) = -4/5 and pi / 2 < theta < pi, find cos(theta / 2).

Answer: Step 1: As in part (i), theta / 2 is in the first quadrant, so take the positive root

Step 2: Use cos(theta / 2) = sqrt([1 + cos(theta)] / 2) = sqrt([1 + (-4/5)] / 2) = sqrt((1/5) / 2)

Step 3: Simplify, = sqrt(1/10)

Final answer: cos(theta / 2) = 1 / sqrt(10).

Question: Q18(iii). If cos(theta) = -4/5 and pi / 2 < theta < pi, find tan(theta / 2).

Answer: Step 1: Use tan(theta / 2) = sin(theta / 2) / cos(theta / 2), with the values from parts (i) and (ii)

Step 2: = (3 / sqrt(10)) / (1 / sqrt(10)), and the sqrt(10) factors cancel

Step 3: Check with the other form, tan(theta / 2) = sin(theta) / [1 + cos(theta)] = (3/5) / (1/5) = 3, using sin(theta) = 3/5 in the second

quadrant

Final answer: tan(theta / 2) = 3.

Question: Q18(iv). If cos(theta) = -4/5 and pi / 2 < theta < pi, find sin(2theta).

Answer: Step 1: First get sin(theta) from sin^2 + cos^2 = 1, sin(theta) = sqrt(1 - 16/25) = 3/5, positive because theta is in the second quadrant

Step 2: Use sin(2theta) = 2 sin(theta)cos(theta)

Step 3: Substitute, = 2 x (3/5) x (-4/5) = -24/25

Final answer: sin(2theta) = -24/25.

Question: Q18(v). If cos(theta) = -4/5 and pi / 2 < theta < pi, find cos(2theta).

Answer: Step 1: Use cos(2theta) = 2 cos^2(theta) - 1

Step 2: Substitute cos(theta) = -4/5, giving 2 x (16/25) - 1

Step 3: = 32/25 - 25/25

Final answer: cos(2theta) = 7/25.

Question: Q18(vi). If cos(theta) = -4/5 and pi / 2 < theta < pi, find tan(2theta).

Answer: Step 1: Use tan(2theta) = sin(2theta) / cos(2theta)

Step 2: Substitute the results of parts (iv) and (v), = (-24/25) / (7/25)

Step 3: The denominators cancel, giving -24/7

Final answer: tan(2theta) = -24/7.

Question: Q19(i). If sin(theta) = 12/13, where 0 < theta < pi / 2, find sin(theta / 2).

Answer: Step 1: First get cos(theta) = sqrt(1 - (12/13)^2) = 5/13, positive because theta is acute, and theta / 2 is also acute so the positive root is taken

Step 2: Use sin(theta / 2) = sqrt([1 - cos(theta)] / 2) = sqrt([1 - 5/13] / 2) = sqrt((8/13) / 2)

Step 3: Simplify, = sqrt(4/13)

Final answer: sin(theta / 2) = 2 / sqrt(13).

Question: Q19(ii). If sin(theta) = 12/13, where 0 < theta < pi / 2, find cos(theta / 2).

Answer: Step 1: Use cos(theta / 2) = sqrt([1 + cos(theta)] / 2) with cos(theta) = 5/13, taking the positive root

Step 2: = sqrt([1 + 5/13] / 2) = sqrt((18/13) / 2) = sqrt(9/13)

Step 3: Simplify, = 3 / sqrt(13)

Final answer: cos(theta / 2) = 3 / sqrt(13).

Question: Q19(iii). If sin(theta) = 12/13, where 0 < theta < pi / 2, find tan(theta / 2).

Answer: Step 1: Use tan(theta / 2) = sin(theta / 2) / cos(theta / 2) with parts (i) and (ii)

Step 2: = (2 / sqrt(13)) / (3 / sqrt(13)), and the sqrt(13) factors cancel

Final answer: tan(theta / 2) = 2/3.

Question: Q19(iv). If sin(theta) = 12/13, where 0 < theta < pi / 2, find sin(2theta).

Answer: Step 1: Use sin(2theta) = 2 sin(theta)cos(theta) with sin(theta) = 12/13 and cos(theta) = 5/13

Step 2: Substitute, = 2 x (12/13) x (5/13)

Step 3: = 120/169

Final answer: sin(2theta) = 120/169.

Question: Q19(v). If sin(theta) = 12/13, where 0 < theta < pi / 2, find cos(2theta).

Answer: Step 1: Use cos(2theta) = cos^2(theta) - sin^2(theta)

Step 2: Substitute, = (5/13)^2 - (12/13)^2 = 25/169 - 144/169

Step 3: = -119/169, negative because 2theta is in the second quadrant

Final answer: cos(2theta) = -119/169.

Question: Q19(vi). If sin(theta) = 12/13, where 0 < theta < pi / 2, find tan(2theta).

Answer: Step 1: Use tan(2theta) = sin(2theta) / cos(2theta)

Step 2: Substitute parts (iv) and (v), = (120/169) / (-119/169)

Step 3: The denominators cancel

Final answer: tan(2theta) = -120/119.

Question: Q20(i). If cos(theta) = 3/5, where 0 < theta < pi / 2, find sin(3theta).

Answer: Step 1: First get sin(theta) = sqrt(1 - (3/5)^2) = 4/5, positive because theta is acute

Step 2: Use the triple-angle formula sin(3theta) = 3 sin(theta) - 4 sin^3(theta)

Step 3: Substitute, = 3 x (4/5) - 4 x (64/125) = 12/5 - 256/125

Step 4: Convert to a common denominator, = 300/125 - 256/125

Final answer: sin(3theta) = 44/125.

Question: Q20(ii). If cos(theta) = 3/5, where 0 < theta < pi / 2, find cos(3theta).

Answer: Step 1: Use the triple-angle formula cos(3theta) = 4 cos^3(theta) - 3 cos(theta)

Step 2: Substitute cos(theta) = 3/5, giving 4 x (27/125) - 3 x (3/5) = 108/125 - 9/5

Step 3: Convert to a common denominator, = 108/125 - 225/125

Final answer: cos(3theta) = -117/125.

Question: Q20(iii). If cos(theta) = 3/5, where 0 < theta < pi / 2, find tan(3theta).

Answer: Step 1: Use tan(3theta) = sin(3theta) / cos(3theta)

Step 2: Substitute parts (i) and (ii), = (44/125) / (-117/125)

Step 3: The denominators cancel

Final answer: tan(3theta) = -44/117.

Exercise 10.4

Question: Q1(i). Express the product 2 sin(6theta)cos(3theta) as a sum or difference.

Answer: Step 1: Use 2 sin(A)cos(B) = sin(A + B) + sin(A - B) with A = 6theta and B = 3theta

Step 2: A + B = 9theta and A - B = 3theta

Step 3: Substitute into the formula

Final answer: 2 sin(6theta)cos(3theta) = sin(9theta) + sin(3theta).

Question: Q1(ii). Express the product cos(3theta)sin(6theta) as a sum or difference.

Answer: Step 1: Use cos(A)sin(B) = (1/2)[sin(A + B) - sin(A - B)] with A = 3theta and B = 6theta

Step 2: A + B = 9theta and A - B = -3theta

Step 3: Substitute, = (1/2)[sin(9theta) - sin(-3theta)], and sin(-3theta) = -sin(3theta)

Step 4: So the minus of a minus gives a plus

Final answer: cos(3theta)sin(6theta) = (1/2)[sin(9theta) + sin(3theta)].

Question: Q1(iii). Express the product 2 sin(alpha - beta)cos(alpha + beta) as a sum or difference.

Answer: Step 1: Use 2 sin(A)cos(B) = sin(A + B) + sin(A - B) with A = alpha - beta and B = alpha + beta

Step 2: A + B = (alpha - beta) + (alpha + beta) = 2alpha

Step 3: A - B = (alpha - beta) - (alpha + beta) = -2beta, and sin(-2beta) = -sin(2beta)

Step 4: Substitute

Final answer: 2 sin(alpha - beta)cos(alpha + beta) = sin(2alpha) - sin(2beta).

Question: Q1(iv). Express the product -2 sin((alpha + beta) / 2)sin((alpha - beta) / 2) as a sum or difference.

Answer: Step 1: Use -2 sin(P)sin(Q) = cos(P + Q) - cos(P - Q), from sin(P)sin(Q) = -(1/2)[cos(P + Q) - cos(P - Q)]

Step 2: Here P + Q = [(alpha + beta) + (alpha - beta)] / 2 = alpha

Step 3: And P - Q = [(alpha + beta) - (alpha - beta)] / 2 = beta

Step 4: Substitute

Final answer: -2 sin((alpha + beta) / 2)sin((alpha - beta) / 2) = cos(alpha) - cos(beta).

Question: Q1(v). Express the product sin(theta + 45 deg)sin(theta - 45 deg) as a sum or difference.

Answer: Step 1: Use sin(A)sin(B) = -(1/2)[cos(A + B) - cos(A - B)] with A = theta + 45 deg and B = theta - 45 deg

Step 2: A + B = 2theta and A - B = 90 deg

Step 3: Substitute, = -(1/2)[cos(2theta) - cos(90 deg)], and cos(90 deg) = 0

Step 4: Simplify

Final answer: sin(theta + 45 deg)sin(theta - 45 deg) = -(1/2)cos(2theta).

Question: Q1(vi). Express the product 2 cos(2theta + 60 deg)cos(2theta - 60 deg) as a sum or difference.

Answer: Step 1: Use 2 cos(A)cos(B) = cos(A + B) + cos(A - B) with A = 2theta + 60 deg and B = 2theta - 60 deg

Step 2: A + B = 4theta and A - B = 120 deg

Step 3: Substitute, = cos(4theta) + cos(120 deg)

Step 4: Use cos(120 deg) = -1/2

Final answer: 2 cos(2theta + 60 deg)cos(2theta - 60 deg) = cos(4theta) - 1/2.

Question: Q2(i). Express the sum or difference sin(2alpha) - sin(2beta) as a product.

Answer: Step 1: Use sin(u) - sin(v) = 2 cos((u + v) / 2)sin((u - v) / 2) with u = 2alpha and v = 2beta

Step 2: (u + v) / 2 = alpha + beta and (u - v) / 2 = alpha - beta

Step 3: Substitute

Final answer: sin(2alpha) - sin(2beta) = 2 cos(alpha + beta)sin(alpha - beta).

Question: Q2(ii). Express the sum or difference cos(5theta / 2) + cos(3theta / 2) as a product.

Answer: Step 1: Use cos(u) + cos(v) = 2 cos((u + v) / 2)cos((u - v) / 2)

Step 2: (5theta / 2 + 3theta / 2) / 2 = 2theta

Step 3: (5theta / 2 - 3theta / 2) / 2 = theta / 2

Step 4: Substitute

Final answer: cos(5theta / 2) + cos(3theta / 2) = 2 cos(2theta)cos(theta / 2).

Question: Q2(iii). Express the sum or difference cos(25 deg) + cos(65 deg) as a product.

Answer: Step 1: Use cos(u) + cos(v) = 2 cos((u + v) / 2)cos((u - v) / 2)

Step 2: (25 deg + 65 deg) / 2 = 45 deg and (65 deg - 25 deg) / 2 = 20 deg, using the positive half-difference

Step 3: Substitute, = 2 cos(45 deg)cos(20 deg)

Step 4: Use cos(45 deg) = 1 / sqrt(2), so 2 x (1 / sqrt(2)) = sqrt(2)

Final answer: cos(25 deg) + cos(65 deg) = sqrt(2) cos(20 deg).

Question: Q2(iv). Express the sum or difference cos(theta + 30 deg) - cos(theta - 30 deg) as a product.

Answer: Step 1: Use cos(u) - cos(v) = -2 sin((u + v) / 2)sin((u - v) / 2)

Step 2: (u + v) / 2 = theta and (u - v) / 2 = 30 deg

Step 3: Substitute, = -2 sin(theta)sin(30 deg)

Step 4: Use sin(30 deg) = 1/2, so -2 x (1/2) = -1

Final answer: cos(theta + 30 deg) - cos(theta - 30 deg) = -sin(theta).

Question: Q2(v). Express the sum or difference sin(pi / 2) - sin(pi / 4) as a product.

Answer: Step 1: Use sin(u) - sin(v) = 2 cos((u + v) / 2)sin((u - v) / 2) with u = pi / 2 and v = pi / 4

Step 2: (u + v) / 2 = (3 pi / 4) / 2 = 3 pi / 8

Step 3: (u - v) / 2 = (pi / 4) / 2 = pi / 8

Step 4: Substitute

Final answer: sin(pi / 2) - sin(pi / 4) = 2 cos(3 pi / 8)sin(pi / 8).

Question: Q2(vi). Express the sum or difference sin(2(theta + 40 deg)) + sin(2(theta - 40 deg)) as a product.

Answer: Step 1: First rewrite the terms as sin(2theta + 80 deg) + sin(2theta - 80 deg)

Step 2: Use sin(u) + sin(v) = 2 sin((u + v) / 2)cos((u - v) / 2)

Step 3: (u + v) / 2 = 2theta and (u - v) / 2 = 80 deg

Step 4: Substitute

Final answer: sin(2(theta + 40 deg)) + sin(2(theta - 40 deg)) = 2

sin(2theta)cos(80 deg).

Question: Q3(i). Prove the identity [sin(alpha) - sin(beta)] / [cos(alpha) + cos(beta)] = tan((alpha - beta) / 2).

Answer: Step 1: Convert the numerator with sin(alpha) - sin(beta) = 2 cos((alpha + beta) / 2)sin((alpha - beta) / 2)

Step 2: Convert the denominator with cos(alpha) + cos(beta) = 2 cos((alpha + beta) / 2)cos((alpha - beta) / 2)

Step 3: Divide and cancel the common factor 2 cos((alpha + beta) / 2)

Step 4: What remains is sin((alpha - beta) / 2) / cos((alpha - beta) / 2)

Final answer: LHS = tan((alpha - beta) / 2) = RHS, so the identity is proved.

Question: Q3(ii). Prove the identity [sin(alpha) + sin(beta)] / [sin(alpha) - sin(beta)] = tan((alpha + beta) / 2)cot((alpha - beta) / 2).

Answer: Step 1: Convert the numerator, sin(alpha) + sin(beta) = 2 sin((alpha + beta) / 2)cos((alpha - beta) / 2)

Step 2: Convert the denominator, sin(alpha) - sin(beta) = 2 cos((alpha + beta) / 2)sin((alpha - beta) / 2)

Step 3: Divide and cancel the factor 2, giving [sin((alpha + beta) / 2) / cos((alpha + beta) / 2)] x [cos((alpha - beta) / 2) / sin((alpha - beta) / 2)]

Step 4: Recognise the first bracket as tan((alpha + beta) / 2) and the second as cot((alpha - beta) / 2)

Final answer: LHS = tan((alpha + beta) / 2)cot((alpha - beta) / 2) = RHS, so the identity is proved.

Question: Q3(iii). Prove the identity [sin(6theta) + sin(4theta)] /

[cos(6theta) + cos(4theta)] = tan(5theta).

Answer: Step 1: Convert the numerator, sin(6theta) + sin(4theta) = 2 sin(5theta)cos(theta), since the half-sum is 5theta and the half-difference is theta

Step 2: Convert the denominator, cos(6theta) + cos(4theta) = 2 cos(5theta)cos(theta)

Step 3: Divide and cancel 2 cos(theta)

Step 4: What remains is sin(5theta) / cos(5theta)

Final answer: LHS = tan(5theta) = RHS, so the identity is proved.

Question: Q3(iv). Prove the identity [sin(2theta) + sin(4theta) + sin(6theta) + sin(8theta)] / [cos(2theta) + cos(4theta) + cos(6theta) + cos(8theta)] = tan(5theta).

Answer: Step 1: Group the numerator in pairs, sin(2theta) + sin(8theta) = 2 sin(5theta)cos(3theta) and sin(4theta) + sin(6theta) = 2 sin(5theta)cos(theta), so the numerator is 2 sin(5theta)[cos(3theta) + cos(theta)]

Step 2: Group the denominator in the same way, cos(2theta) + cos(8theta) = 2 cos(5theta)cos(3theta) and cos(4theta) + cos(6theta) = 2 cos(5theta)cos(theta), so the denominator is 2 cos(5theta)[cos(3theta) + cos(theta)]

Step 3: Divide and cancel 2[cos(3theta) + cos(theta)]

Step 4: What remains is sin(5theta) / cos(5theta)

Final answer: LHS = tan(5theta) = RHS, so the identity is proved.

Question: Q3(v). Prove the identity [sin(alpha) + sin(2alpha)] / [cos(alpha) - cos(2alpha)] = cot(alpha / 2).

Answer: Step 1: Convert the numerator, sin(alpha) + sin(2alpha) = 2 sin(3alpha / 2)cos(alpha / 2), since the half-sum is 3alpha / 2 and the half-difference is -alpha / 2, with cos even

Step 2: Convert the denominator, cos(alpha) - cos(2alpha) = -2 sin(3alpha / 2)sin(-alpha / 2) = 2 sin(3alpha / 2)sin(alpha / 2), since sin is odd

Step 3: Divide and cancel 2 sin(3alpha / 2)

Step 4: What remains is cos(alpha / 2) / sin(alpha / 2)

Final answer: LHS = cot(alpha / 2) = RHS, so the identity is proved.

Question: Q3(vi). Prove the identity [sin(theta + phi) - 2 sin(theta) + sin(theta - phi)] / [cos(theta + phi) - 2 cos(theta) + cos(theta - phi)] = tan(theta).

Answer: Step 1: Group the first and last terms of the numerator, sin(theta + phi) + sin(theta - phi) = 2 sin(theta)cos(phi), so the numerator is 2 sin(theta)cos(phi) - 2 sin(theta) = 2 sin(theta)[cos(phi) - 1]

Step 2: Group the denominator in the same way, cos(theta + phi) + cos(theta - phi) = 2 cos(theta)cos(phi), so the denominator is 2 cos(theta)[cos(phi) - 1]

Step 3: Divide and cancel 2[cos(phi) - 1], which is valid when cos(phi) != 1

Step 4: What remains is sin(theta) / cos(theta)

Final answer: LHS = tan(theta) = RHS, so the identity is proved.

Question: Q4(i). Evaluate sin(20 deg)sin(40 deg)sin(60 deg)sin(80 deg).

Answer: Step 1: Take sin(60 deg) = sqrt(3) / 2 out, leaving (sqrt(3) / 2) x sin(20 deg)sin(40 deg)sin(80 deg), and first work out the remaining product P = sin(20 deg)sin(40 deg)sin(80 deg)

Step 2: Convert one pair with sin(X)sin(Y) = (1/2)[cos(X - Y) - cos(X + Y)], giving sin(40 deg)sin(80 deg) = (1/2)[cos(40 deg) - cos(120 deg)] = (1/2)[cos(40 deg) + 1/2], since cos(120 deg) = -1/2

Step 3: So P = (1/2)sin(20 deg)cos(40 deg) + (1/4)sin(20 deg), and convert the remaining pair with sin(X)cos(Y) = (1/2)[sin(X + Y) + sin(X - Y)], giving sin(20 deg)cos(40 deg) = (1/2)[sin(60 deg) - sin(20 deg)] = (1/2)[sqrt(3) / 2 - sin(20 deg)]

Step 4: Substitute, P = (1/4)[sqrt(3) / 2 - sin(20 deg)] + (1/4)sin(20 deg) = sqrt(3) / 8, because the sin(20 deg) terms cancel

Step 5: Multiply by the factor taken out in Step 1, (sqrt(3) / 2) x (sqrt(3) / 8) = 3/16

Final answer: The value is 3/16.

Question: Q4(ii). Evaluate cos(20 deg)cos(40 deg)cos(60 deg)cos(80 deg).

Answer: Step 1: Take cos(60 deg) = 1/2 out, leaving (1/2) x P where P = cos(20 deg)cos(40 deg)cos(80 deg)

Step 2: Convert one pair with cos(X)cos(Y) = (1/2)[cos(X + Y) + cos(X - Y)], giving cos(40 deg)cos(80 deg) = (1/2)[cos(120 deg) + cos(40 deg)] = (1/2)[cos(40 deg) - 1/2], since cos(120 deg) = -1/2

Step 3: So P = (1/2)cos(20 deg)cos(40 deg) - (1/4)cos(20 deg), and convert the remaining pair, cos(20 deg)cos(40 deg) = (1/2)[cos(60 deg) + cos(20 deg)] = (1/2)[1/2 + cos(20 deg)]

Step 4: Substitute, P = (1/4)[1/2 + cos(20 deg)] - (1/4)cos(20 deg) = 1/8, because the cos(20 deg) terms cancel

Step 5: Multiply by the factor 1/2 from Step 1, (1/2) x (1/8) = 1/16

Final answer: The value is 1/16.

Question: Q5(i). Show that [cos(8 deg) - sin(8 deg)] / [cos(8 deg) + sin(8 deg)] = tan(37 deg).

Answer: Step 1: Divide numerator and denominator by cos(8 deg), giving [1 - tan(8 deg)] / [1 + tan(8 deg)]

Step 2: Write 1 as tan(45 deg), so this is [tan(45 deg) - tan(8 deg)] / [1 + tan(45 deg)tan(8 deg)]

Step 3: Recognise the tan difference formula tan(alpha - beta) = [tan(alpha) - tan(beta)] / [1 + tan(alpha)tan(beta)] with alpha = 45 deg and beta = 8 deg

Step 4: So the expression is tan(45 deg - 8 deg)

Final answer: The expression equals tan(37 deg), as required.

Question: Q5(ii). Show that [cos(11 deg) + sin(11 deg)] / [cos(11 deg) - sin(11 deg)] = tan(56 deg).

Answer: Step 1: Divide numerator and denominator by cos(11 deg), giving [1 + tan(11 deg)] / [1 - tan(11 deg)]

Step 2: Write 1 as tan(45 deg), so this is [tan(45 deg) + tan(11 deg)] / [1 - tan(45 deg)tan(11 deg)]

Step 3: Recognise the tan sum formula with alpha = 45 deg and beta = 11 deg

Step 4: So the expression is tan(45 deg + 11 deg)

Final answer: The expression equals tan(56 deg), as required.

Question: Q5(iii). Show that sin(50 deg) - sin(70 deg) + sin(10 deg) = 0.

Answer: Step 1: Group the first and last terms and convert to a product, sin(50 deg) + sin(10 deg) = 2 sin(30 deg)cos(20 deg), since the half-sum is 30 deg and the half-difference is 20 deg

Step 2: Use sin(30 deg) = 1/2, so this pair equals 2 x (1/2) x cos(20 deg) = cos(20 deg)

Step 3: The expression becomes cos(20 deg) - sin(70 deg)

Step 4: Use sin(70 deg) = sin(90 deg - 20 deg) = cos(20 deg), so cos(20 deg) - cos(20 deg) = 0

Final answer: The expression equals 0, as required.

Review Exercise 10

Question: Q1(i). Select the correct option: cos(alpha - beta) is equal to: (a) cos(alpha)cos(beta) + sin(alpha)sin(beta) (b) cos(alpha)cos(beta) - sin(alpha)sin(beta) (c) sin(alpha)cos(beta) + cos(alpha)sin(beta) (d) sin(alpha)cos(beta) - cos(alpha)sin(beta).

Answer: Step 1: Recall the fundamental law cos(alpha - beta) = cos(alpha)cos(beta) + sin(alpha)sin(beta), proved with the distance formula on the unit circle

Step 2: Option (b) is cos(alpha + beta), and options (c) and (d) are the sin sum and difference formulas

Final answer: (a) cos(alpha)cos(beta) + sin(alpha)sin(beta).

Question: Q1(ii). Select the correct option: Angles associated with a basic angle of measure theta to a right angle or its multiple are called: (a) Coterminal angles (b) Angles in standard positions (c) Allied angles (d) Obtuse angles.

Answer: Step 1: Angles of the form n(90 deg) +- theta are formed from the basic angle theta and multiples of a right angle

Step 2: By definition in this unit these are the allied angles of theta

Final answer: (c) Allied angles.

Question: Q1(iii). Select the correct option: sin(pi / 2 + theta) is equal to: (a) cos(theta) (b) sin(theta) (c) -cos(theta) (d) -sin(theta).

Answer: Step 1: Expand with the sin sum formula, sin(pi / 2)cos(theta) + cos(pi / 2)sin(theta)

Step 2: Use sin(pi / 2) = 1 and cos(pi / 2) = 0, giving cos(theta)

Final answer: (a) cos(theta).

Question: Q1(iv). Select the correct option: cos(pi - theta) is equal to: (a) sin(theta) (b) cos(theta) (c) -sin(theta) (d) -cos(theta).

Answer: Step 1: Expand with the cos difference formula, cos(pi)cos(theta) + sin(pi)sin(theta)

Step 2: Use cos(pi) = -1 and sin(pi) = 0, giving -cos(theta)

Final answer: (d) -cos(theta).

Question: Q1(v). Select the correct option: tan(pi + theta) is equal to: (a) tan(theta) (b) -cot(theta) (c) -tan(theta) (d) cot(theta).

Answer: Step 1: Use the tan sum formula, [tan(pi) + tan(theta)] / [1 - tan(pi)tan(theta)]

Step 2: Use tan(pi) = 0, giving tan(theta) / 1 = tan(theta)

Final answer: (a) tan(theta).

Question: Q1(vi). Select the correct option: cos(2 pi + theta) is equal to: (a) sin(theta) (b) cos(theta) (c) -sin(theta) (d) -cos(theta).

Answer: Step 1: Adding 2 pi is one full turn, so the angle has the same terminal side as theta

Step 2: Hence its cosine is the same as cos(theta), which the sum formula also gives with cos(2 pi) = 1 and sin(2 pi) = 0

Final answer: (b) cos(theta).

Question: Q1(vii). Select the correct option: sin(540 deg) is equal to: (a) 1 (b) 0 (c) 1 / sqrt(2) (d) sqrt(3) / 2.

Answer: Step 1: Remove a full turn, 540 deg - 360 deg = 180 deg

Step 2: sin(540 deg) = sin(180 deg) = 0

Final answer: (b) 0.

Question: Q1(viii). Select the correct option: sin(-300 deg) is equal to: (a) 1 (b) sqrt(3) / 2 (c) 0 (d) -1.

Answer: Step 1: Sine is odd, so sin(-300 deg) = -sin(300 deg)

Step 2: sin(300 deg) = sin(360 deg - 60 deg) = -sin(60 deg) = -sqrt(3) / 2

Step 3: Negate, -(-sqrt(3) / 2) = sqrt(3) / 2

Final answer: (b) sqrt(3) / 2.

Question: Q1(ix). Select the correct option: If alpha, beta and gamma are the angles of a triangle ABC, then sin(alpha + beta) is equal to: (a) sin(gamma) (b) -sin(gamma) (c) cos(gamma) (d) -cos(gamma).

Answer: Step 1: In a triangle alpha + beta = 180 deg - gamma

Step 2: So sin(alpha + beta) = sin(180 deg - gamma) = sin(gamma)

Final answer: (a) sin(gamma).

Question: Q1(x). Select the correct option: cos(2alpha) is equal to: (a) cos^2(alpha) - sin^2(alpha) (b) 2 cos^2(alpha) - 1 (c) 1 - 2 sin^2(alpha) (d) all of these.

Answer: Step 1: The basic double-angle form is cos(2alpha) =

cos^2(alpha) - sin^2(alpha), which is option (a)

Step 2: Replacing sin^2(alpha) by 1 - cos^2(alpha) gives 2 cos^2(alpha) - 1, option (b), and replacing cos^2(alpha) by 1 - sin^2(alpha) gives 1 - 2 sin^2(alpha), option (c)

Step 3: All three forms are correct

Final answer: (d) all of these.

Question: Q1(xi). Select the correct option: sin(alpha / 2) is equal to: (a) +- sqrt([1 + sin(alpha)] / 2) (b) +- sqrt([1 - cos(alpha)] / 2) (c) +- sqrt([1 + cos(alpha)] / 2) (d) +- sqrt([1 - sin(alpha)] / 2).

Answer: Step 1: From cos(alpha) = 1 - 2 sin^2(alpha / 2), rearrange to get sin^2(alpha / 2) = [1 - cos(alpha)] / 2

Step 2: Take square roots, keeping the +- sign because the sign depends on the quadrant of alpha / 2

Final answer: (b) +- sqrt([1 - cos(alpha)] / 2).

Question: Q1(xii). Select the correct option: cos(3alpha) is equal to: (a) 3 cos(alpha) - 4 cos(alpha) (b) 3 cos^3(alpha) + 4 cos(alpha) (c) 4 cos^3(alpha) - 3 cos(alpha) (d) 4 cos^3(alpha) + 4 cos(alpha).

Answer: Step 1: Expand cos(3alpha) = cos(2alpha + alpha) and then use the double-angle formulas

Step 2: The standard triple-angle result is cos(3alpha) = 4 cos^3(alpha) - 3 cos(alpha)

Final answer: (c) 4 cos^3(alpha) - 3 cos(alpha).

Question: Q1(xiii). Select the correct option: cos(alpha) - cos(beta) =: (a) 2 sin((alpha + beta) / 2)cos((alpha - beta) / 2) (b) 2 cos((alpha + beta) / 2)sin((alpha - beta) / 2) (c) 2 cos((alpha + beta) / 2)cos((alpha - beta) / 2)

(d) -2 sin((alpha + beta) / 2)sin((alpha - beta) / 2).

Answer: Step 1: Use the sum-to-product formula cos(u) - cos(v) = -2 sin((u + v) / 2)sin((u - v) / 2)

Step 2: With u = alpha and v = beta this matches option (d) exactly, the other options being the formulas for sin differences and cos sums

Final answer: (d) -2 sin((alpha + beta) / 2)sin((alpha - beta) / 2).

Unit 11: Application of Trigonometry

Exercise 11.1

Question: 1(i). Solve the right angled triangle ABC in which C = 90 deg, a = 4 m and b = 3 m. Find c, A and B.

Answer: Step 1: since C = 90 deg, c is the hypotenuse, so use Pythagoras theorem c^2 = a^2 + b^2

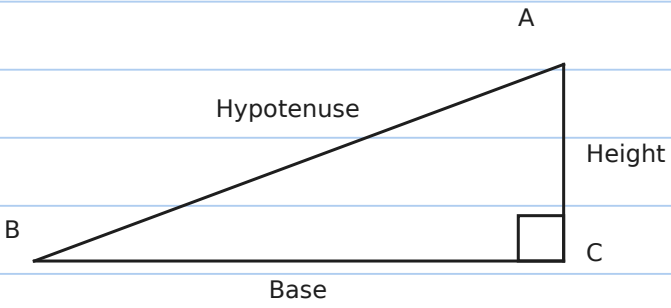
Step 2: substitute c^2 = 4^2 + 3^2 = 16 + 9 = 25

Step 3: take square root, c = sqrt(25) = 5 m

Step 4: for angle A, sin(A) = opposite/hypotenuse = a/c = 4/5 = 0.8, so A = sin^-1(0.8) = 53.13 deg

Step 5: B = 180 deg - 90 deg - 53.13 deg = 36.87 deg

Final answer: c = 5 m, A = 53.13 deg, B = 36.87 deg



Question: 1(ii). Solve the right angled triangle ABC in which C = 90 deg, a = 12 m and b = 5 m. Find c, A and B.

Answer: Step 1: since C = 90 deg, c is the hypotenuse, so use Pythagoras theorem c^2 = a^2 + b^2

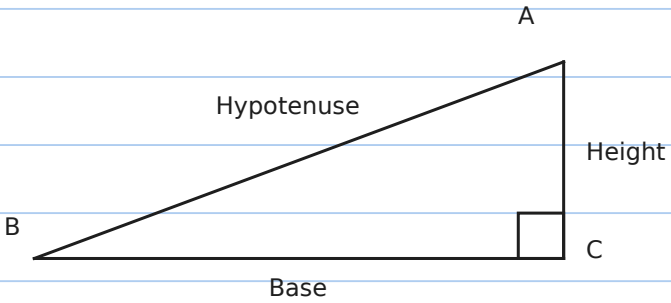
Step 2: substitute c^2 = 12^2 + 5^2 = 144 + 25 = 169

Step 3: take square root, c = sqrt(169) = 13 m

Step 4: sin(A) = a/c = 12/13 = 0.9231, so A = sin^-1(0.9231) = 67.38 deg

Step 5: B = 180 deg - 90 deg - 67.38 deg = 22.62 deg

Final answer: c = 13 m, A = 67.38 deg, B = 22.62 deg



Question: 1(iii). Solve the right angled triangle ABC in which C = 90 deg, a = 8 m and b = 6 m. Find c, A and B.

Answer: Step 1: since C = 90 deg, c is the hypotenuse, so use Pythagoras theorem c^2 = a^2 + b^2

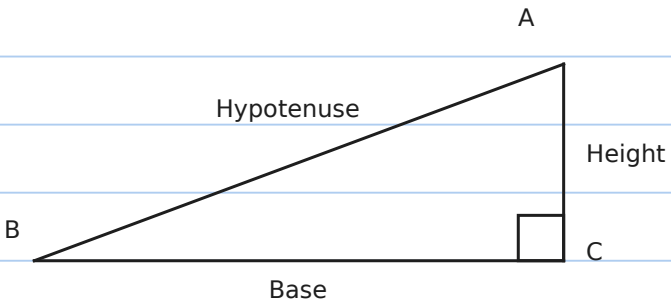
Step 2: substitute c^2 = 8^2 + 6^2 = 64 + 36 = 100

Step 3: take square root, c = sqrt(100) = 10 m

Step 4: sin(A) = a/c = 8/10 = 0.8, so A = sin^-1(0.8) = 53.13 deg

Step 5: B = 180 deg - 90 deg - 53.13 deg = 36.87 deg

Final answer: c = 10 m, A = 53.13 deg, B = 36.87 deg



Question: 1(iv). Solve the right angled triangle ABC in which C = 90 deg, c = 8 m and B = 42 deg 30 min. Find A, a and b.

Answer: Step 1: the angles add to 180 deg, so A = 180 deg - 90 deg - 42.5 deg = 47.5 deg, that is 47 deg 30 min

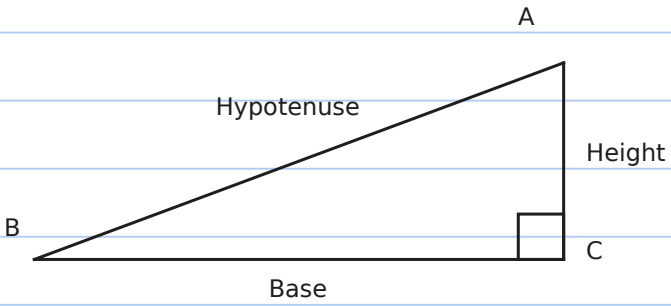
Step 2: c is the hypotenuse, and sin(A) = a/c, so a = c sin(A)

Step 3: substitute a = 8 x sin(47.5 deg) = 8 x 0.7373 = 5.90 m

Step 4: similarly b = c sin(B) = 8 x sin(42.5 deg) = 8 x 0.6756 = 5.40 m

Step 5: check by Pythagoras, sqrt(5.90^2 + 5.40^2) = sqrt(34.8 + 29.2) = sqrt(64) = 8 m, which agrees with c

Final answer: A = 47 deg 30 min, a = 5.90 m, b = 5.40 m



Question: 1(v). Solve the right angled triangle ABC in which C = 90 deg, a = 140 m and A = 38 deg 20 min. Find B, b and c.

Answer: Step 1: B = 180 deg - 90 deg - 38 deg 20 min = 51 deg 40 min, that is 51.67 deg

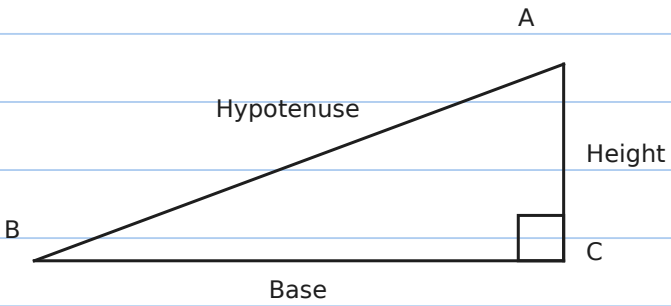
Step 2: sin(A) = a/c, so c = a/sin(A)

Step 3: substitute c = 140/sin(38.33 deg) = 140/0.6202 = 225.72 m

Step 4: b = c sin(B) = 225.72 x sin(51.67 deg) = 225.72 x 0.7846 = 177.06 m

Step 5: check, a/c = 140/225.72 = 0.6202 = sin(38.33 deg), which agrees

Final answer: B = 51 deg 40 min, b = 177.06 m, c = 225.72 m



Question: 1(vi). Solve the right angled triangle ABC in which C = 90 deg, b = 30.8 m and A = 41 deg 50 min. Find B, a and c.

Answer: Step 1: B = 180 deg - 90 deg - 41 deg 50 min = 48 deg 10 min, that is 48.17 deg

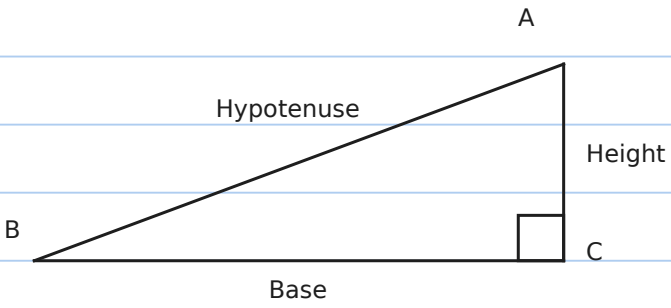
Step 2: sin(B) = b/c, so c = b/sin(B)

Step 3: substitute c = 30.8/sin(48.17 deg) = 30.8/0.7453 = 41.34 m

Step 4: a = c sin(A) = 41.34 x sin(41.83 deg) = 41.34 x 0.6670 = 27.57 m

Step 5: check by Pythagoras, sqrt(27.57^2 + 30.8^2) = sqrt(760.1 + 948.6) = sqrt(1708.7) = 41.34 m, which agrees with c

Final answer: B = 48 deg 10 min, a = 27.57 m, c = 41.34 m



Question: 2. A vertical stick 16 m long casts a 12 cm long shadow (as printed in the book). Find the angle of elevation of the sun.

Answer: Step 1: the stick, its shadow and the sun's ray form a right angled triangle, and the angle of elevation, say theta, is the angle between the shadow and the sun's ray at the tip of the shadow

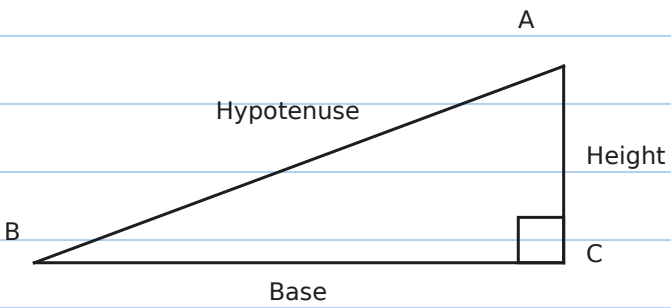
Step 2: convert the shadow to metres, 12 cm = 0.12 m

Step 3: tan(theta) = opposite/adjacent = height of stick/length of shadow = 16/0.12 = 133.33

Step 4: theta = tan^-1(133.33) = 89.57 deg

Step 5: note that this almost vertical result shows the printed "12 cm" is almost certainly a misprint for a 12 m shadow; with a 12 m shadow, tan(theta) = 16/12 = 1.3333 and theta = tan^-1(1.3333) = 53.13 deg, the standard textbook answer

Final answer: 89.57 deg for the values as printed (53.13 deg if the shadow is 12 m, as intended)



Question: 3. Find the distance of a man from the foot of a tower 169 m high, if the angle of depression of the man from its top is 48 deg 30 min.

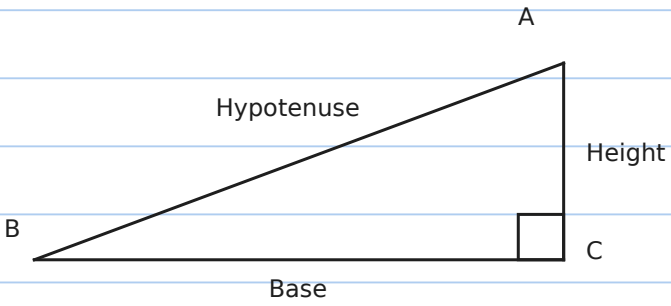
Answer: Step 1: the angle of depression from the top equals the angle of elevation from the man, 48 deg 30 min = 48.5 deg, by alternate angles

Step 2: let x be the required distance; the tower, the ground and the line of sight form a right angled triangle with height 169 m opposite the 48.5 deg angle

Step 3: tan(48.5 deg) = 169/x, so x = 169/tan(48.5 deg)

Step 4: substitute tan(48.5 deg) = 1.1303, so x = 169/1.1303 = 149.52 m

Final answer: the man is about 149.52 m from the foot of the tower



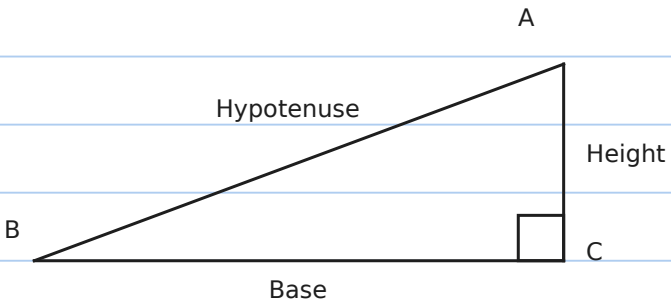
Question: 4. A string of a flying kite is 200 meters long and its angle of elevation is 58 deg 30 min. Find the height of the kite when the string is fully stretched.

Answer: Step 1: the stretched string is the hypotenuse (200 m) of a right angled triangle, and the height h of the kite is the side opposite the angle of elevation 58 deg 30 min = 58.5 deg

Step 2: use sin(angle) = opposite/hypotenuse, so sin(58.5 deg) = h/200

Step 3: h = 200 x sin(58.5 deg) = 200 x 0.8526 = 170.53 m

Final answer: the height of the kite is about 170.53 m



Question: 5. A man 1.8 m tall observes that the angle of elevation of the top of a tree at a distance of 12 m from him is 31 deg 45 min. What is the height of the tree?

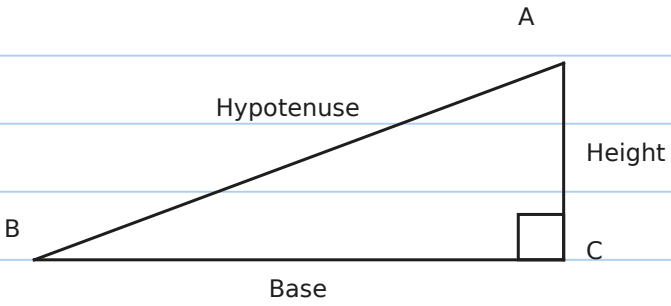
Answer: Step 1: draw a horizontal line from the man's eyes (1.8 m above the ground) to the tree; the part of the tree above this line, say h, forms a right angled triangle with the 12 m distance

Step 2: tan(31 deg 45 min) = h/12, with 31 deg 45 min = 31.75 deg

Step 3: h = 12 x tan(31.75 deg) = 12 x 0.6188 = 7.43 m

Step 4: add the man's height, because his eyes are 1.8 m above the ground: total height = 7.43 + 1.8 = 9.23 m

Final answer: the height of the tree is about 9.23 m



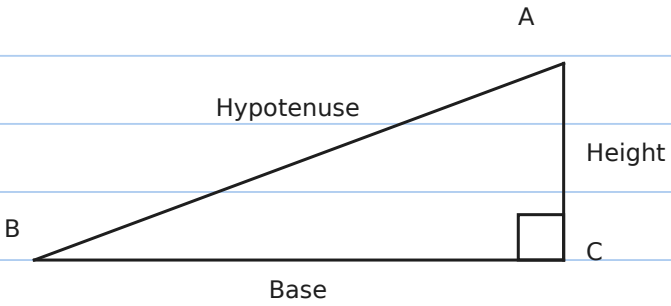
Question: 6. From the top of a cliff 80 m high, the angle of depression of a boat is 12 deg 30 min. How far is the boat from the cliff?

Answer: Step 1: the angle of depression equals the angle of elevation of the top of the cliff from the boat, 12 deg 30 min = 12.5 deg

Step 2: let x be the distance of the boat from the foot of the cliff; then tan(12.5 deg) = 80/x

Step 3: x = 80/tan(12.5 deg) = 80/0.2217 = 360.86 m

Final answer: the boat is about 360.86 m from the cliff



Question: 7. Two masts are 20 m and 12 m high. If the line joining their tops makes an angle of 35 deg with the horizontal, find the distance between them.

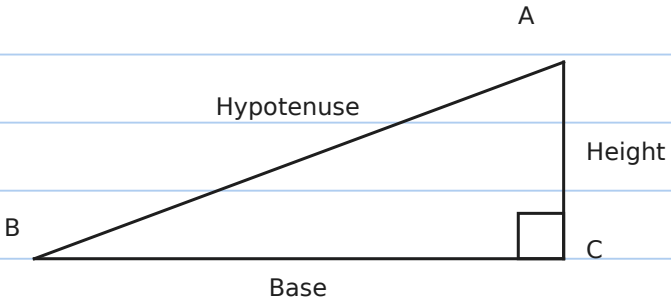
Answer: Step 1: join the tops of the masts and drop a horizontal line from the top of the shorter mast to the taller mast; this forms a right angled triangle

Step 2: the vertical side of this triangle is the difference of the heights, 20 - 12 = 8 m

Step 3: let d be the distance between the masts (the horizontal side); then tan(35 deg) = 8/d

Step 4: d = 8/tan(35 deg) = 8/0.7002 = 11.43 m

Final answer: the distance between the masts is about 11.43 m



Question: 8. A window washer is working in a hotel building. An observer at a distance of 20 m from the building finds the angle of elevation of the worker to be 30 deg. The worker climbs up 12 m and the observer moves 4 m further away from the building. Find the new angle of elevation of the worker.

Answer: Step 1: first find the worker's initial height h above the observer's eye level: tan(30 deg) = h/20, so h = 20 x tan(30 deg) = 20 x 0.5774 = 11.55 m

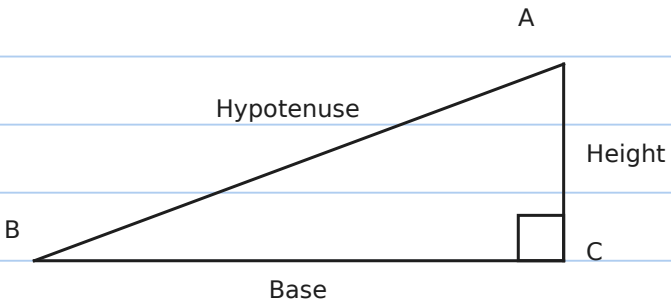
Step 2: after climbing 12 m, the worker's height above eye level is 11.55 + 12 = 23.55 m

Step 3: the observer moves 4 m further away, so the new distance from the building is 20 + 4 = 24 m

Step 4: if theta is the new angle of elevation, tan(theta) = 23.55/24 = 0.9811

Step 5: theta = tan^-1(0.9811) = 44.45 deg

Final answer: the new angle of elevation is about 44.45 deg



Exercise 11.2

Question: 1(i). Solve the triangle ABC by using the law of sines, given a = 70, b = 34 and A = 79 deg.

Answer: Step 1: by the law of sines, a/sin(A) = b/sin(B) = c/sin(C), so sin(B) = b sin(A)/a

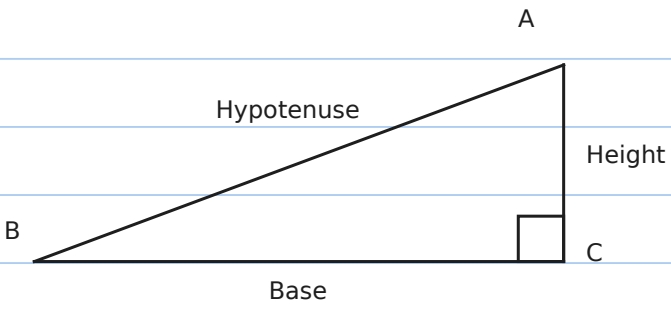
Step 2: substitute sin(B) = 34 x sin(79 deg)/70 = 34 x 0.9816/70 = 0.4768

Step 3: B = sin^-1(0.4768) = 28.48 deg (the supplement 151.52 deg is rejected because A + B would exceed 180 deg)

Step 4: C = 180 deg - 79 deg - 28.48 deg = 72.52 deg

Step 5: c = a sin(C)/sin(A) = 70 x sin(72.52 deg)/sin(79 deg) = 70 x 0.9539/0.9816 = 68.02

Final answer: B = 28.48 deg, C = 72.52 deg, c = 68.02



Question: 1(ii). Solve the triangle ABC by using the law of sines, given b =

136, C = 104.2 deg and A = 43.1 deg.

Answer: Step 1: B = 180 deg - A - C = 180 deg - 43.1 deg - 104.2 deg = 32.7 deg

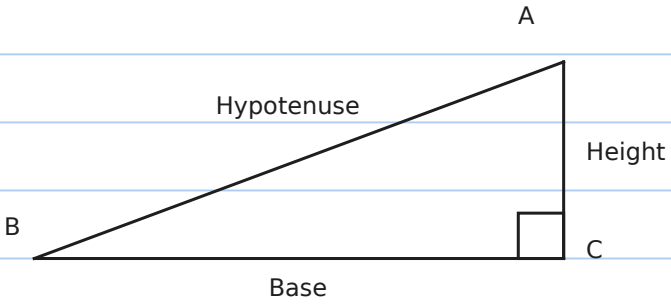
Step 2: by the law of sines, a = b sin(A)/sin(B)

Step 3: substitute a = 136 x sin(43.1 deg)/sin(32.7 deg) = 136 x 0.6833/0.5402 = 172.01

Step 4: similarly c = b sin(C)/sin(B) = 136 x sin(104.2 deg)/sin(32.7 deg) = 136 x 0.9690/0.5402 = 244.05

Step 5: check, a/sin(A) = 172.01/0.6833 = 251.7 and b/sin(B) = 136/0.5402 = 251.8, which agree

Final answer: B = 32.7 deg, a = 172.01, c = 244.05



Question: 1(iii). Solve the triangle ABC by using the law of sines, given A = 48 deg 45 min, B = 69 deg 15 min and c = 40.5.

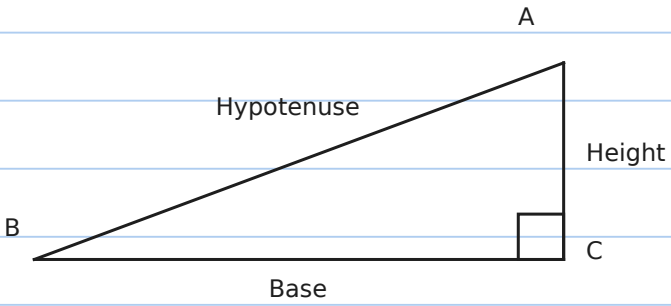
Answer: Step 1: C = 180 deg - 48.75 deg - 69.25 deg = 62 deg

Step 2: by the law of sines, a = c sin(A)/sin(C)

Step 3: substitute a = 40.5 x sin(48.75 deg)/sin(62 deg) = 40.5 x 0.7518/0.8829 = 34.49

Step 4: b = c sin(B)/sin(C) = 40.5 x sin(69.25 deg)/sin(62 deg) = 40.5 x 0.9350/0.8829 = 42.89

Final answer: C = 62 deg, a = 34.49, b = 42.89



Question: 2(i). Solve the triangle ABC by using the laws of cosines, given A = 78.2 deg, b = 62.4 and c = 150.

Answer: Step 1: A is the included angle between sides b and c, so use the law of cosines a^2 = b^2 + c^2 - 2bc cos(A)

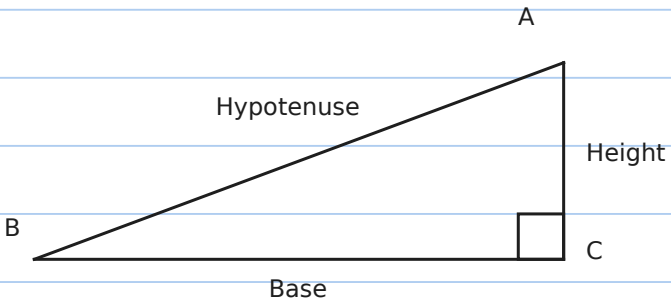
Step 2: substitute a^2 = 62.4^2 + 150^2 - 2(62.4)(150) cos(78.2 deg) = 3893.76 + 22500 - 18720 x 0.2045 = 22565.8

Step 3: a = sqrt(22565.8) = 150.22

Step 4: find B from cos(B) = (a^2 + c^2 - b^2)/(2ac) = (22565.8 + 22500 - 3893.76)/(2 x 150.22 x 150) = 41172/45066 = 0.9136, so B = cos^-1(0.9136) = 23.99 deg

Step 5: C = 180 deg - 78.2 deg - 23.99 deg = 77.81 deg

Final answer: a = 150.22, B = 23.99 deg, C = 77.81 deg



Question: 2(ii). Solve the triangle ABC by using the laws of cosines, given a = 48, b = 33.3 and C = 41 deg 30 min.

Answer: Step 1: C = 41.5 deg is the included angle between a and b, so c^2 = a^2 + b^2 - 2ab cos(C)

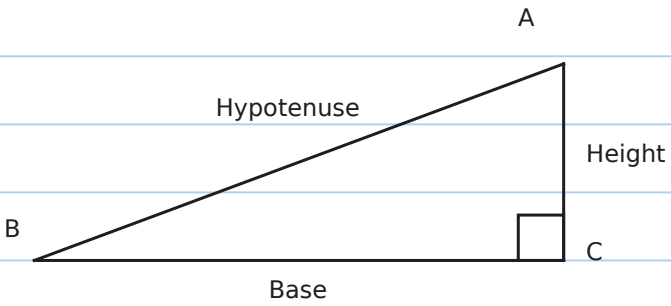
Step 2: substitute c^2 = 48^2 + 33.3^2 - 2(48)(33.3) cos(41.5 deg) = 2304 + 1108.89 - 3196.8 x 0.7490 = 1018.6

Step 3: c = sqrt(1018.6) = 31.92

Step 4: cos(A) = (b^2 + c^2 - a^2)/(2bc) = (1108.89 + 1018.6 - 2304)/(2 x 33.3 x 31.92) = (-176.5)/2125.9 = -0.0830, so A = cos^-1(-0.0830) = 94.76 deg

Step 5: B = 180 deg - 41.5 deg - 94.76 deg = 43.74 deg

Final answer: c = 31.92, A = 94.76 deg, B = 43.74 deg



Question: 2(iii). Solve the triangle ABC by using the laws of cosines, given a = 73, c = 40 and B = 76 deg 45 min.

Answer: Step 1: B = 76.75 deg is the included angle between a and c, so b^2 = a^2 + c^2 - 2ac cos(B)

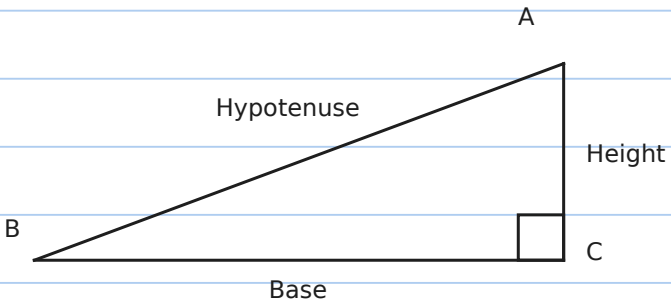
Step 2: substitute b^2 = 73^2 + 40^2 - 2(73)(40) cos(76.75 deg) = 5329 + 1600 - 5840 x 0.2296 = 5588.1

Step 3: b = sqrt(5588.1) = 74.77

Step 4: cos(A) = (b^2 + c^2 - a^2)/(2bc) = (5588.1 + 1600 - 5329)/(2 x 74.77 x 40) = 1859.1/5981.6 = 0.3108, so A = cos^-1(0.3108) = 71.87 deg

Step 5: C = 180 deg - 76.75 deg - 71.87 deg = 31.38 deg

Final answer: b = 74.77, A = 71.87 deg, C = 31.38 deg



Question: 3(i). Solve the triangle ABC by using the laws of tangents, given a = 432, c = 325 and B = 42 deg.

Answer: Step 1: A + C = 180 deg - 42 deg = 138 deg, so (A + C)/2 = 69 deg

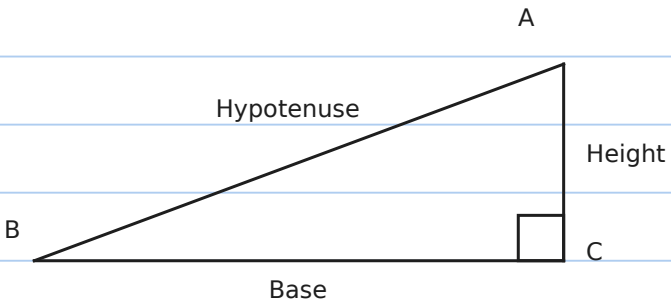
Step 2: by the law of tangents, tan((A - C)/2)/tan((A + C)/2) = (a - c)/(a + c) = (432 - 325)/(432 + 325) = 107/757 = 0.1413

Step 3: tan((A - C)/2) = 0.1413 x tan(69 deg) = 0.1413 x 2.6051 = 0.3683, so (A - C)/2 = tan^-1(0.3683) = 20.21 deg and A - C = 40.42 deg

Step 4: solving A + C = 138 deg with A - C = 40.42 deg gives A = (138 + 40.42)/2 = 89.21 deg and C = (138 - 40.42)/2 = 48.79 deg

Step 5: by the law of sines, b = a sin(B)/sin(A) = 432 x sin(42 deg)/sin(89.21 deg) = 432 x 0.6691/0.9999 = 289.09

Final answer: A = 89.21 deg, C = 48.79 deg, b = 289.09



Question: 3(ii). Solve the triangle ABC by using the laws of tangents, given b = 39, c = 35 and A = 75 deg.

Answer: Step 1: B + C = 180 deg - 75 deg = 105 deg, so (B + C)/2 = 52.5 deg

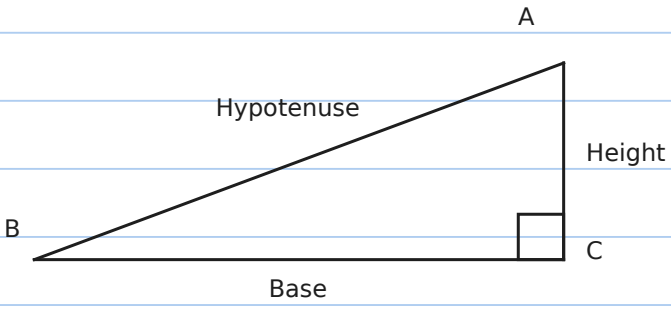
Step 2: by the law of tangents, tan((B - C)/2)/tan((B + C)/2) = (b - c)/(b + c) = (39 - 35)/(39 + 35) = 4/74 = 0.0541

Step 3: tan((B - C)/2) = 0.0541 x tan(52.5 deg) = 0.0541 x 1.3032 = 0.0705, so (B - C)/2 = tan^-1(0.0705) = 4.03 deg and B - C = 8.06 deg

Step 4: solving B + C = 105 deg with B - C = 8.06 deg gives B = 56.53 deg and C = 48.47 deg

Step 5: by the law of sines, a = b sin(A)/sin(B) = 39 x sin(75 deg)/sin(56.53 deg) = 39 x 0.9659/0.8343 = 45.16

Final answer: B = 56.53 deg, C = 48.47 deg, a = 45.16



Question: 3(iii). Solve the triangle ABC by using the laws of tangents, given a = 39.14, b = 34.21 and C = 78 deg 10 min.

Answer: Step 1: A + B = 180 deg - 78 deg 10 min = 101 deg 50 min = 101.83 deg, so (A + B)/2 = 50.92 deg

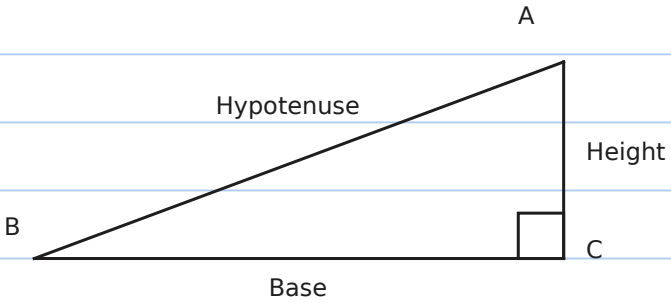
Step 2: by the law of tangents, tan((A - B)/2)/tan((A + B)/2) = (a - b)/(a + b) = (39.14 - 34.21)/(39.14 + 34.21) = 4.93/73.35 = 0.0672

Step 3: tan((A - B)/2) = 0.0672 x tan(50.92 deg) = 0.0672 x 1.2313 = 0.0828, so (A - B)/2 = tan^-1(0.0828) = 4.73 deg and A - B = 9.46 deg

Step 4: solving A + B = 101.83 deg with A - B = 9.46 deg gives A = 55.65 deg and B = 46.19 deg

Step 5: by the law of sines, c = a sin(C)/sin(A) = 39.14 x sin(78.17 deg)/sin(55.65 deg) = 39.14 x 0.9785/0.8257 = 46.40

Final answer: A = 55.65 deg, B = 46.19 deg, c = 46.40



Question: 4(i). Solve the triangle ABC by using suitable laws, given a = 9, b = 7 and c = 5.

Answer: Step 1: all three sides are known, so use the law of cosines for each angle, starting with the largest angle A opposite the longest side a = 9

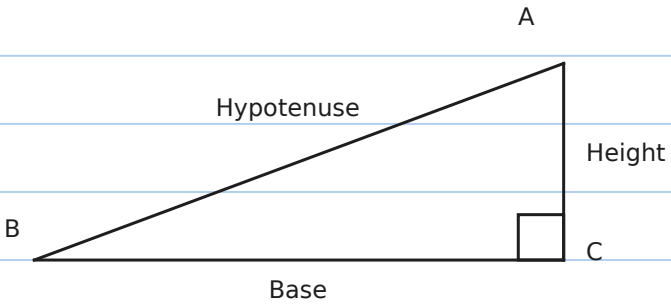
Step 2: cos(A) = (b^2 + c^2 - a^2)/(2bc) = (49 + 25 - 81)/(2 x 7 x 5) = -7/70 = -0.1, so A = cos^-1(-0.1) = 95.74 deg

Step 3: cos(B) = (a^2 + c^2 - b^2)/(2ac) = (81 + 25 - 49)/(2 x 9 x 5) = 57/90 = 0.6333, so B = cos^-1(0.6333) = 50.70 deg

Step 4: C = 180 deg - 95.74 deg - 50.70 deg = 33.56 deg

Step 5: check, cos(C) = (81 + 49 - 25)/(2 x 9 x 7) = 105/126 = 0.8333 gives C = 33.56 deg, which agrees

Final answer: A = 95.74 deg, B = 50.70 deg, C = 33.56 deg



Question: 4(ii). Solve the triangle ABC by using suitable laws, given b = 35, a = 37 and A = 23 deg 25 min.

Answer: Step 1: a side and its opposite angle are known, so use the law of sines: sin(B) = b sin(A)/a

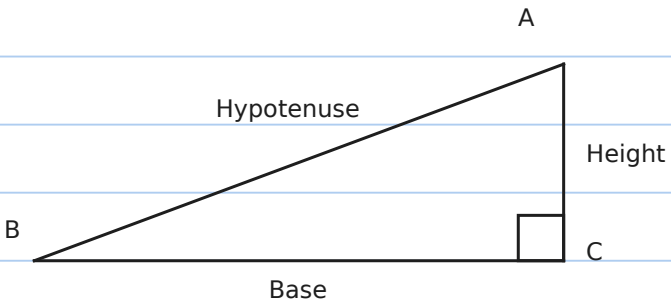
Step 2: substitute sin(B) = 35 x sin(23.42 deg)/37 = 35 x 0.3974/37 = 0.3759

Step 3: B = sin^-1(0.3759) = 22.08 deg (the supplement 157.92 deg is rejected because 23.42 deg + 157.92 deg exceeds 180 deg)

Step 4: C = 180 deg - 23.42 deg - 22.08 deg = 134.50 deg

Step 5: c = a sin(C)/sin(A) = 37 x sin(134.50 deg)/sin(23.42 deg) = 37 x 0.7133/0.3974 = 66.40

Final answer: B = 22.08 deg, C = 134.50 deg, c = 66.40



Question: 5(i). Solve the triangle ABC by using the half angle formulae of sines, given a = 55, c = 65 and b = 85.

Answer: Step 1: semi-perimeter s = (a + b + c)/2 = (55 + 85 + 65)/2 = 102.5, so s - a = 47.5, s - b = 17.5 and s - c = 37.5

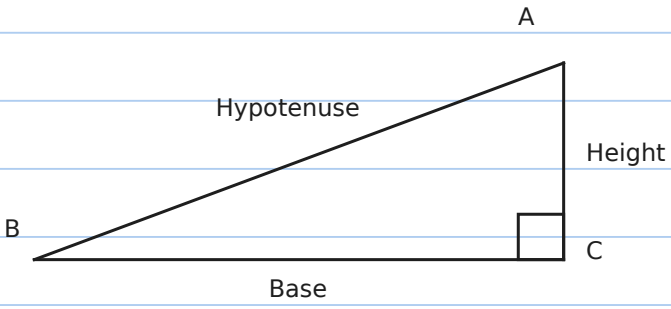
Step 2: half angle formula of sine for A, sin(A/2) = sqrt((s - b)(s - c)/(bc)) = sqrt(17.5 x 37.5/(85 x 65)) = sqrt(0.1188) = 0.3446, so A/2 = 20.16 deg and A = 40.32 deg

Step 3: sin(B/2) = sqrt((s - a)(s - c)/(ac)) = sqrt(47.5 x 37.5/(55 x 65)) = sqrt(0.4983) = 0.7059, so B/2 = 44.90 deg and B = 89.80 deg

Step 4: sin(C/2) = sqrt((s - a)(s - b)/(ab)) = sqrt(47.5 x 17.5/(55 x 85)) = sqrt(0.1778) = 0.4217, so C/2 = 24.94 deg and C = 49.88 deg

Step 5: check, 40.32 + 89.80 + 49.88 = 180 deg

Final answer: A = 40.32 deg, B = 89.80 deg, C = 49.88 deg



Question: 5(ii). Solve the triangle ABC by using the half angle formulae of sines, given b = 15, a = 20 and c = 14.

Answer: Step 1: s = (20 + 15 + 14)/2 = 24.5, so s - a = 4.5, s - b = 9.5 and s - c = 10.5

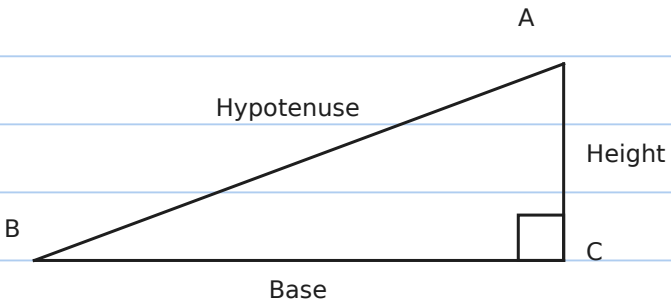
Step 2: sin(A/2) = sqrt((s - b)(s - c)/(bc)) = sqrt(9.5 x 10.5/(15 x 14)) = sqrt(0.4750) = 0.6892, so A/2 = 43.57 deg and A = 87.13 deg

Step 3: sin(B/2) = sqrt((s - a)(s - c)/(ac)) = sqrt(4.5 x 10.5/(20 x 14)) = sqrt(0.1688) = 0.4108, so B/2 = 24.26 deg and B = 48.51 deg

Step 4: sin(C/2) = sqrt((s - a)(s - b)/(ab)) = sqrt(4.5 x 9.5/(20 x 15)) = sqrt(0.1425) = 0.3775, so C/2 = 22.18 deg and C = 44.36 deg

Step 5: check, 87.13 + 48.51 + 44.36 = 180 deg

Final answer: A = 87.13 deg, B = 48.51 deg, C = 44.36 deg



Question: 5(iii). Solve the triangle ABC by using the half angle formulae of sines, given c = 1.3, b = 2.3 and a = 2.7.

Answer: Step 1: s = (2.7 + 2.3 + 1.3)/2 = 3.15, so s - a = 0.45, s - b = 0.85 and s - c = 1.85

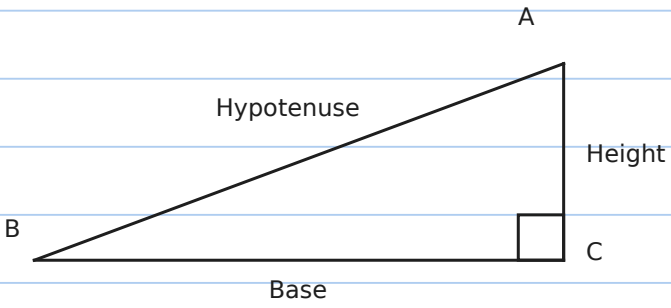
Step 2: sin(A/2) = sqrt((s - b)(s - c)/(bc)) = sqrt(0.85 x 1.85/(2.3 x 1.3)) = sqrt(0.5259) = 0.7252, so A/2 = 46.49 deg and A = 92.97 deg

Step 3: sin(B/2) = sqrt((s - a)(s - c)/(ac)) = sqrt(0.45 x 1.85/(2.7 x 1.3)) = sqrt(0.2372) = 0.4870, so B/2 = 29.15 deg and B = 58.29 deg

Step 4: sin(C/2) = sqrt((s - a)(s - b)/(ab)) = sqrt(0.45 x 0.85/(2.7 x 2.3)) = sqrt(0.0616) = 0.2482, so C/2 = 14.37 deg and C = 28.74 deg

Step 5: check, 92.97 + 58.29 + 28.74 = 180 deg

Final answer: A = 92.97 deg, B = 58.29 deg, C = 28.74 deg



Question: 6(i). Solve the triangle ABC by using the half angle formulae of cosines, given a = 95, b = 76 and c = 85.

Answer: Step 1: s = (95 + 76 + 85)/2 = 128, so s - a = 33, s - b = 52 and s - c = 43

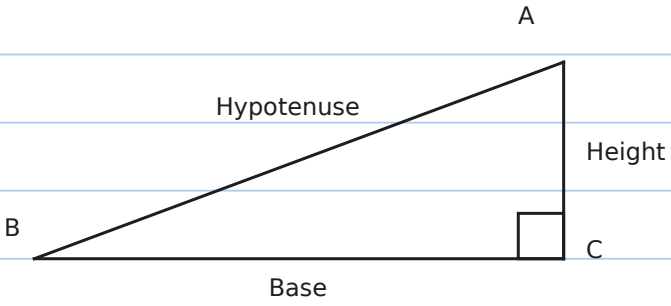
Step 2: half angle formula of cosine for A, cos(A/2) = sqrt(s(s - a)/(bc)) = sqrt(128 x 33/(76 x 85)) = sqrt(0.6540) = 0.8087, so A/2 = 36.04 deg and A = 72.08 deg

Step 3: cos(B/2) = sqrt(s(s - b)/(ac)) = sqrt(128 x 52/(95 x 85)) = sqrt(0.8243) = 0.9079, so B/2 = 24.79 deg and B = 49.57 deg

Step 4: cos(C/2) = sqrt(s(s - c)/(ab)) = sqrt(128 x 43/(95 x 76)) = sqrt(0.7625) = 0.8732, so C/2 = 29.18 deg and C = 58.36 deg

Step 5: check, 72.08 + 49.57 + 58.36 = 180 deg (rounding)

Final answer: A = 72.08 deg, B = 49.57 deg, C = 58.36 deg



Question: 6(ii). Solve the triangle ABC by using the half angle formulae of cosines, given a = 10, b = 7 and c = 5.

Answer: Step 1: s = (10 + 7 + 5)/2 = 11, so s - a = 1, s - b = 4 and s - c = 6

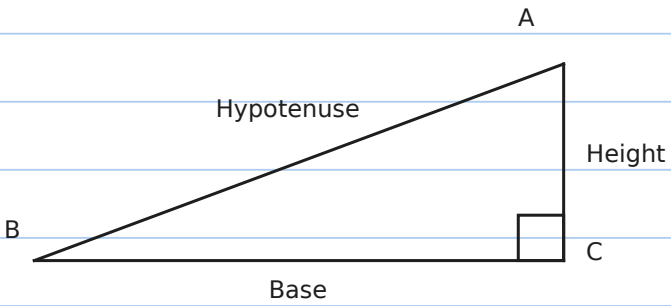
Step 2: cos(A/2) = sqrt(s(s - a)/(bc)) = sqrt(11 x 1/(7 x 5)) = sqrt(0.3143) = 0.5606, so A/2 = 55.90 deg and A = 111.80 deg

Step 3: cos(B/2) = sqrt(s(s - b)/(ac)) = sqrt(11 x 4/(10 x 5)) = sqrt(0.88) = 0.9381, so B/2 = 20.27 deg and B = 40.54 deg

Step 4: cos(C/2) = sqrt(s(s - c)/(ab)) = sqrt(11 x 6/(10 x 7)) = sqrt(0.9429) = 0.9710, so C/2 = 13.83 deg and C = 27.66 deg

Step 5: check, 111.80 + 40.54 + 27.66 = 180 deg

Final answer: A = 111.80 deg, B = 40.54 deg, C = 27.66 deg



Question: 6(iii). Solve the triangle ABC by using the half angle formulae of cosines, given c = 10, a = 15 and b = 7.

Answer: Step 1: s = (15 + 7 + 10)/2 = 16, so s - a = 1, s - b = 9 and s - c

= 6

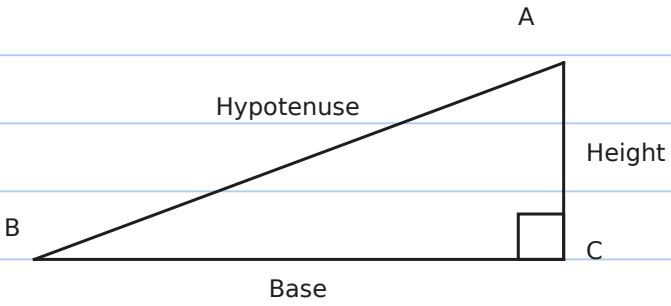
Step 2: cos(A/2) = sqrt(s(s - a)/(bc)) = sqrt(16 x 1/(7 x 10)) = sqrt(0.2286) = 0.4781, so A/2 = 61.44 deg and A = 122.88 deg

Step 3: cos(B/2) = sqrt(s(s - b)/(ac)) = sqrt(16 x 9/(15 x 10)) = sqrt(0.96) = 0.9798, so B/2 = 11.54 deg and B = 23.07 deg

Step 4: cos(C/2) = sqrt(s(s - c)/(ab)) = sqrt(16 x 6/(15 x 7)) = sqrt(0.9143) = 0.9562, so C/2 = 17.03 deg and C = 34.05 deg

Step 5: check, 122.88 + 23.07 + 34.05 = 180 deg

Final answer: A = 122.88 deg, B = 23.07 deg, C = 34.05 deg



Question: 7(i). Solve the triangle ABC by using the half angle formulae of tangents, given c = 23, a = 13 and b = 16.

Answer: Step 1: s = (13 + 16 + 23)/2 = 26, so s - a = 13, s - b = 10 and s - c = 3

Step 2: first find r from r^2 = (s - a)(s - b)(s - c)/s = (13 x 10 x 3)/26 = 15, so r = 3.873

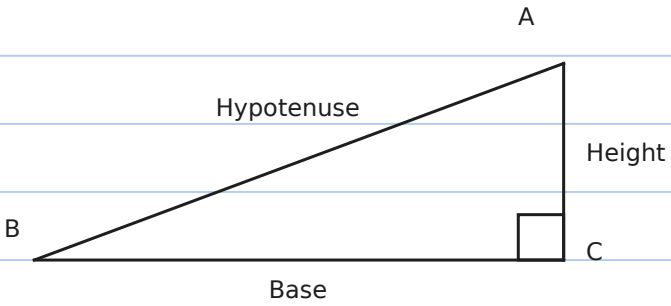
Step 3: half angle formula of tangent, tan(A/2) = r/(s - a) = 3.873/13 = 0.2979, so A/2 = 16.59 deg and A = 33.18 deg

Step 4: tan(B/2) = r/(s - b) = 3.873/10 = 0.3873, so B/2 = 21.17 deg and B = 42.34 deg

Step 5: tan(C/2) = r/(s - c) = 3.873/3 = 1.2910, so C/2 = 52.24 deg and C = 104.48 deg

Step 6: check, 33.18 + 42.34 + 104.48 = 180 deg

Final answer: A = 33.18 deg, B = 42.34 deg, C = 104.48 deg



Question: 7(ii). Solve the triangle ABC by using the half angle formulae of tangents, given a = 25, b = 20 and c = 18.

Answer: Step 1: s = (25 + 20 + 18)/2 = 31.5, so s - a = 6.5, s - b = 11.5 and s - c = 13.5

Step 2: r^2 = (s - a)(s - b)(s - c)/s = (6.5 x 11.5 x 13.5)/31.5 = 32.04, so r = 5.66

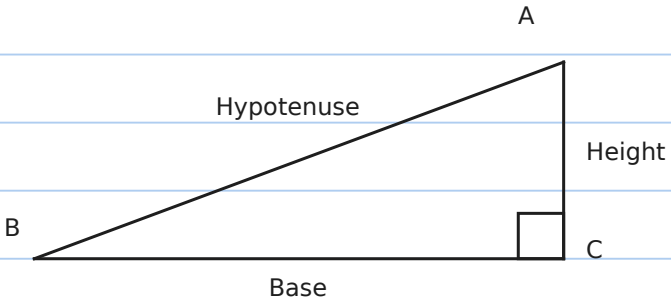
Step 3: tan(A/2) = r/(s - a) = 5.66/6.5 = 0.8708, so A/2 = 41.05 deg and A = 82.10 deg

Step 4: tan(B/2) = r/(s - b) = 5.66/11.5 = 0.4922, so B/2 = 26.21 deg and B = 52.41 deg

Step 5: tan(C/2) = r/(s - c) = 5.66/13.5 = 0.4193, so C/2 = 22.75 deg and C = 45.49 deg

Step 6: check, 82.10 + 52.41 + 45.49 = 180 deg

Final answer: A = 82.10 deg, B = 52.41 deg, C = 45.49 deg



Question: 7(iii). Solve the triangle ABC by using the half angle formulae of tangents, given a = 16, b = 11 and c = 13.

Answer: Step 1: s = (16 + 11 + 13)/2 = 20, so s - a = 4, s - b = 9 and s - c = 7

Step 2: r^2 = (s - a)(s - b)(s - c)/s = (4 x 9 x 7)/20 = 12.6, so r = 3.550

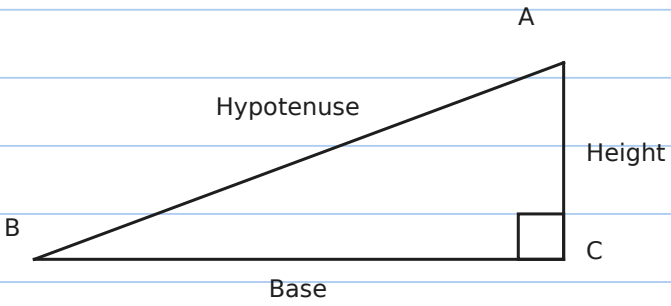
Step 3: tan(A/2) = r/(s - a) = 3.550/4 = 0.8874, so A/2 = 41.59 deg and A = 83.17 deg

Step 4: tan(B/2) = r/(s - b) = 3.550/9 = 0.3944, so B/2 = 21.53 deg and B = 43.05 deg

Step 5: tan(C/2) = r/(s - c) = 3.550/7 = 0.5071, so C/2 = 26.89 deg and C = 53.78 deg

Step 6: check, 83.17 + 43.05 + 53.78 = 180 deg

Final answer: A = 83.17 deg, B = 43.05 deg, C = 53.78 deg



Question: 8. Find the largest angle of triangle ABC when a = 6 cm, b = 8 cm and c = 9.4 cm.

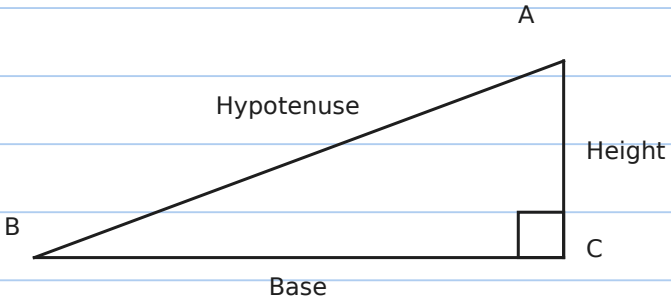
Answer: Step 1: the largest angle is opposite the longest side; the longest side is c = 9.4 cm, so find angle C

Step 2: by the law of cosines, cos(C) = (a^2 + b^2 - c^2)/(2ab)

Step 3: substitute cos(C) = (36 + 64 - 88.36)/(2 x 6 x 8) = 11.64/96 = 0.1213

Step 4: C = cos^-1(0.1213) = 83.04 deg

Final answer: the largest angle is C = 83.04 deg



Question: 9. Find the smallest angle in triangle ABC when a = 25 cm, b = 18 cm and c = 21 cm.

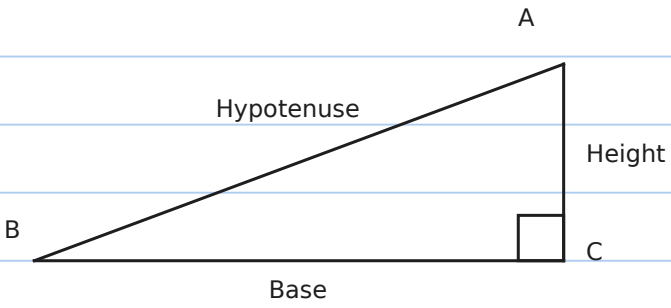
Answer: Step 1: the smallest angle is opposite the shortest side; the shortest side is b = 18 cm, so find angle B

Step 2: by the law of cosines, cos(B) = (a^2 + c^2 - b^2)/(2ac)

Step 3: substitute cos(B) = (625 + 441 - 324)/(2 x 25 x 21) = 742/1050 = 0.7067

Step 4: B = cos^-1(0.7067) = 45.04 deg

Final answer: the smallest angle is B = 45.04 deg



Question: 10. Find the length of the third side of a triangular building that faces 13.6 meters along one street and 13 meters along another street, the angle of intersection between the streets measuring 72 deg.

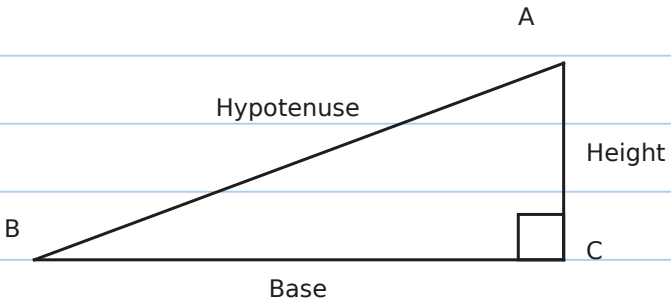
Answer: Step 1: the two street sides (13.6 m and 13 m) include the angle 72 deg between them, and the third side, say d, is opposite this angle

Step 2: by the law of cosines, d^2 = 13.6^2 + 13^2 - 2(13.6)(13) cos(72 deg)

Step 3: substitute d^2 = 184.96 + 169 - 353.6 x 0.3090 = 353.96 - 109.27 = 244.69

Step 4: d = sqrt(244.69) = 15.64 m

Final answer: the third side is about 15.64 m



Question: 11. If one side of a triangle is y units long, another side is 3 times as long, and the angle between the two sides measures 35 deg, find the measures of the other two angles and the third side.

Answer: Step 1: let the sides be a = y and b = 3y with included angle C = 35 deg; the third side c is opposite C

Step 2: by the law of cosines, c^2 = y^2 + (3y)^2 - 2(y)(3y) cos(35 deg) = y^2 + 9y^2 - 6y^2 x 0.8192 = 10y^2 - 4.915y^2 = 5.085y^2

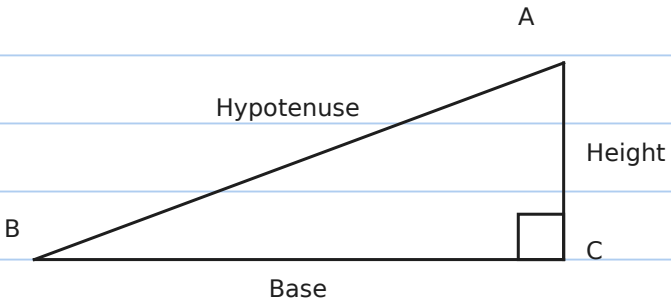
Step 3: c = y x sqrt(5.085) = 2.255y units

Step 4: A + B = 180 deg - 35 deg = 145 deg; by the law of tangents, tan((B - A)/2) = ((b - a)/(b + a)) x tan((B + A)/2) = (2y/4y) x tan(72.5 deg) = 0.5 x 3.1716 = 1.5858, so (B - A)/2 = 57.76 deg and B - A = 115.53 deg

Step 5: solving B + A = 145 deg with B - A = 115.53 deg gives B = 130.26 deg and A = 14.74 deg

Step 6: check by the law of sines, sin(A)/a = sin(14.74 deg)/y = 0.254/y and sin(B)/b = sin(130.26 deg)/(3y) = 0.763/(3y) = 0.254/y, which agree

Final answer: the third side is 2.255y units, and the other two angles are 14.74 deg (opposite the side y) and 130.26 deg (opposite the side 3y)



Question: 12. If the length of the larger side of a parallelogram is 55 cm and one diagonal of the parallelogram makes angles of measure 30 deg and 50 deg with a pair of adjacent sides, find the length of the diagonal.

Answer: Step 1: the diagonal and the two adjacent sides form a triangle; in this triangle the diagonal makes angles of 30 deg and 50 deg with the two sides, so the third angle of the triangle is 180 deg - 30 deg - 50 deg = 100 deg

Step 2: the larger side (55 cm) is opposite the larger of the two given

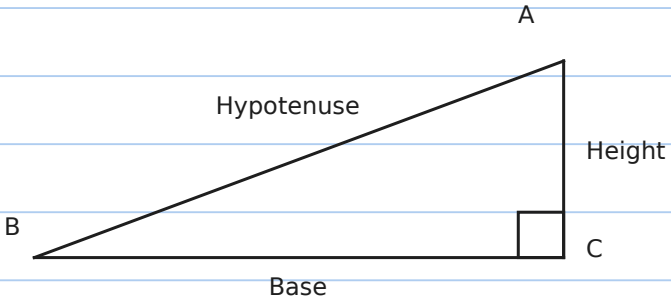
angles, 50 deg, because in a triangle the larger side lies opposite the larger angle

Step 3: the diagonal d is opposite the 100 deg angle

Step 4: by the law of sines, d/sin(100 deg) = 55/sin(50 deg)

Step 5: d = 55 x sin(100 deg)/sin(50 deg) = 55 x 0.9848/0.7660 = 70.71 cm

Final answer: the length of the diagonal is about 70.71 cm



Question: 13. The sides of a parallelogram are 25 cm and 35 cm long and one of its angles is 36 deg. Find the lengths of its diagonals.

Answer: Step 1: each diagonal splits the parallelogram into a triangle with sides 25 cm and 35 cm; one diagonal is opposite the 36 deg angle and the other is opposite the supplementary angle 180 deg - 36 deg = 144 deg

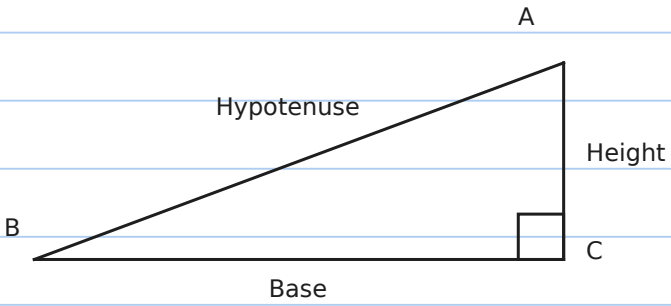
Step 2: for the shorter diagonal, by the law of cosines, d1^2 = 25^2 + 35^2 - 2(25)(35) cos(36 deg) = 625 + 1225 - 1750 x 0.8090 = 1850 - 1415.8 = 434.2

Step 3: d1 = sqrt(434.2) = 20.84 cm

Step 4: for the longer diagonal, d2^2 = 25^2 + 35^2 - 2(25)(35) cos(144 deg) = 1850 - 1750 x (-0.8090) = 1850 + 1415.8 = 3265.8

Step 5: d2 = sqrt(3265.8) = 57.15 cm

Final answer: the diagonals are about 20.84 cm and 57.15 cm



Question: 14. Two hikers start from the same point; one walks 9 km heading east, the other one 10 km heading 55 deg north east. How far apart are they at the end of their walks?

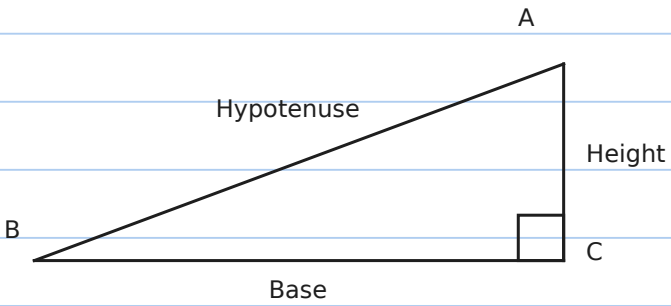
Answer: Step 1: the two paths and the line joining the hikers form a triangle with sides 9 km and 10 km; heading east and heading 55 deg north of east means the angle between the two paths is 55 deg

Step 2: let d be the distance between the hikers, opposite the 55 deg angle; by the law of cosines, d^2 = 9^2 + 10^2 - 2(9)(10) cos(55 deg)

Step 3: substitute d^2 = 81 + 100 - 180 x 0.5736 = 181 - 103.24 = 77.76

Step 4: d = sqrt(77.76) = 8.82 km

Final answer: the hikers are about 8.82 km apart



Question: 15. Two planes start from Karachi International Airport at the same time and fly in directions that make an angle of 127 deg with each other. Their speeds are 525 km/h. How far apart are they at the end of 2

hours of flying time?

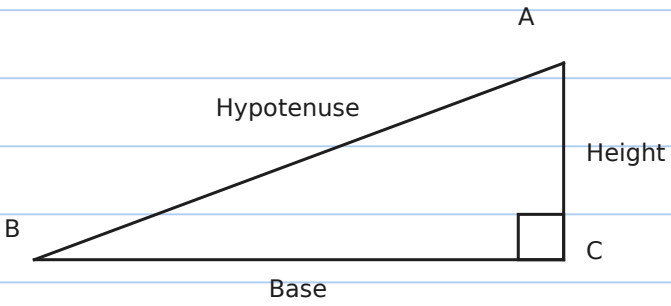
Answer: Step 1: in 2 hours each plane flies 525 x 2 = 1050 km, so the two flight paths are sides of 1050 km including an angle of 127 deg

Step 2: let d be the distance between the planes; by the law of cosines, d^2 = 1050^2 + 1050^2 - 2(1050)(1050) cos(127 deg)

Step 3: substitute cos(127 deg) = -0.6018, so d^2 = 1102500 + 1102500 - 2205000 x (-0.6018) = 2205000 + 1326969 = 3531969

Step 4: d = sqrt(3531969) = 1879.36 km

Final answer: the planes are about 1879.36 km apart after 2 hours



Exercise 11.3

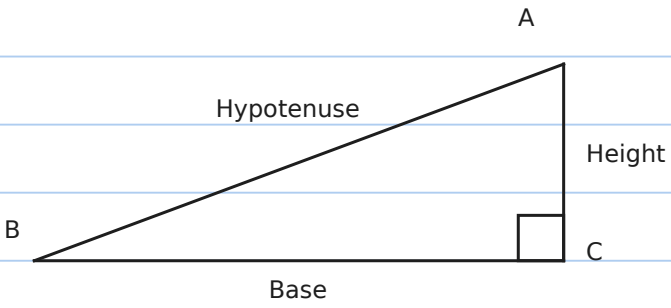
Question: 1(i). Find the area of triangle ABC when a = 8.6, b = 11.4 and C = 90 deg.

Answer: Step 1: sides a and b include angle C, so use Area = (1/2)ab sin(C)

Step 2: substitute Area = (1/2)(8.6)(11.4) sin(90 deg)

Step 3: sin(90 deg) = 1, so Area = (1/2)(98.04)(1) = 49.02

Final answer: Area = 49.02 square units



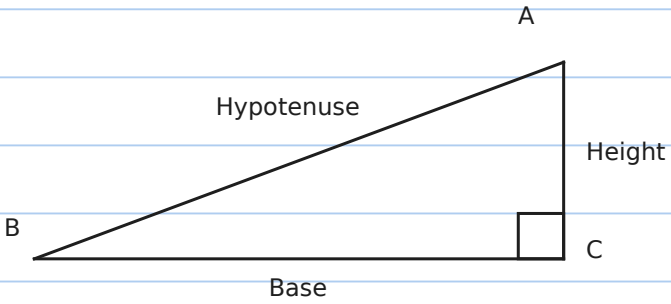
Question: 1(ii). Find the area of triangle ABC when b = 63, c = 17 and A = 120 deg.

Answer: Step 1: sides b and c include angle A, so use Area = (1/2)bc sin(A)

Step 2: substitute Area = (1/2)(63)(17) sin(120 deg)

Step 3: sin(120 deg) = 0.8660, so Area = (1/2)(1071)(0.8660) = 463.76

Final answer: Area = 463.76 square units



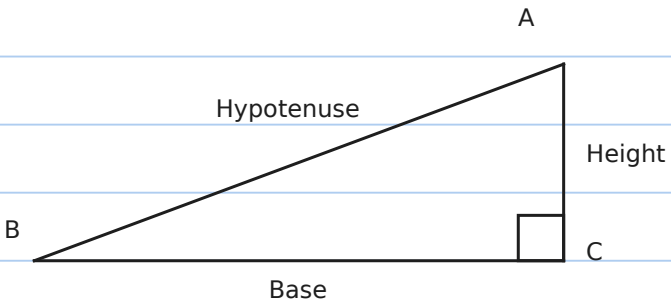
Question: 1(iii). Find the area of triangle ABC when c = 5.34, a = 9.30 and B = 53 deg 34 min.

Answer: Step 1: sides c and a include angle B, so use Area = (1/2)ca sin(B)

Step 2: substitute Area = (1/2)(5.34)(9.30) sin(53 deg 34 min) = (1/2)(49.662) sin(53.57 deg)

Step 3: sin(53.57 deg) = 0.8047, so Area = (1/2)(49.662)(0.8047) = 19.98

Final answer: Area = 19.98 square units



Question: 1(iv). Find the area of triangle ABC when c = 36, A = 46 deg and B = 66 deg.

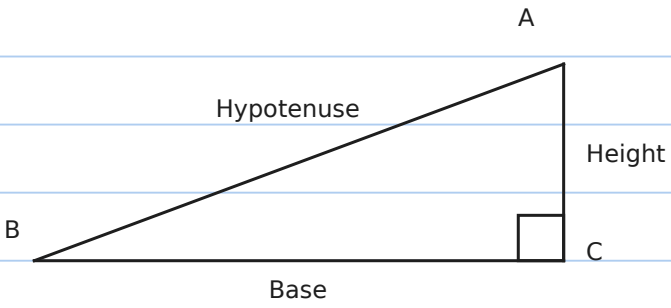
Answer: Step 1: first find C = 180 deg - 46 deg - 66 deg = 68 deg

Step 2: one side and two angles are known, so use Area = (1/2)c^2 sin(A) sin(B)/sin(C)

Step 3: substitute Area = (1/2)(36^2) x sin(46 deg) sin(66 deg)/sin(68 deg) = (1/2)(1296) x (0.7193 x 0.9135)/0.9272

Step 4: Area = 648 x 0.6571/0.9272 = 648 x 0.7087 = 459.28

Final answer: Area = 459.28 square units



Question: 1(v). Find the area of triangle ABC when a = 54, B = 37 deg and C = 54 deg.

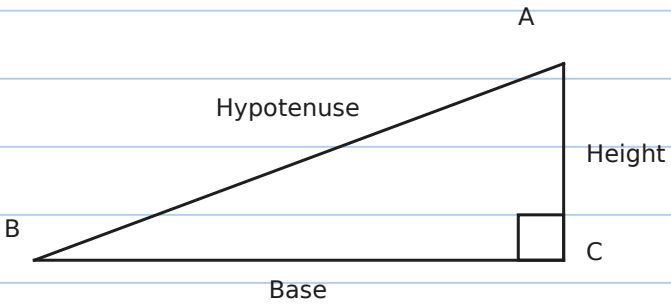
Answer: Step 1: first find A = 180 deg - 37 deg - 54 deg = 89 deg

Step 2: use Area = (1/2)a^2 sin(B) sin(C)/sin(A)

Step 3: substitute Area = (1/2)(54^2) x sin(37 deg) sin(54 deg)/sin(89 deg) = (1/2)(2916) x (0.6018 x 0.8090)/0.9998

Step 4: Area = 1458 x 0.4869/0.9998 = 709.98

Final answer: Area = 709.98 square units



Question: 1(vi). Find the area of triangle ABC when b = 4.8, A = 35 deg 9 min and C = 85 deg 31 min.

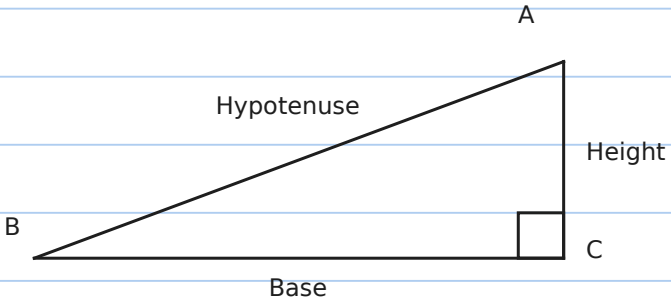
Answer: Step 1: first find B = 180 deg - 35 deg 9 min - 85 deg 31 min = 59 deg 20 min = 59.33 deg

Step 2: use Area = (1/2)b^2 sin(A) sin(C)/sin(B)

Step 3: substitute Area = (1/2)(4.8^2) x sin(35.15 deg) sin(85.52 deg)/sin(59.33 deg) = (1/2)(23.04) x (0.5760 x 0.9968)/0.8601

Step 4: Area = 11.52 x 0.5742/0.8601 = 7.69

Final answer: Area = 7.69 square units



Question: 1(vii). Find the area of triangle ABC when a = 37, b = 41 and c = 37.

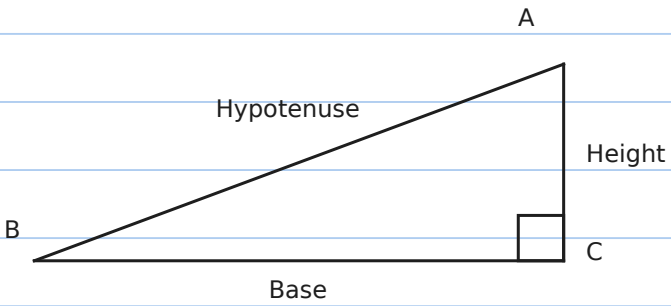
Answer: Step 1: all three sides are known, so use Heron's formula (Hero's formula) Area = sqrt(s(s - a)(s - b)(s - c))

Step 2: s = (37 + 41 + 37)/2 = 57.5, so s - a = 20.5, s - b = 16.5 and s - c = 20.5

Step 3: substitute Area = sqrt(57.5 x 20.5 x 16.5 x 20.5) = sqrt(398578)

Step 4: Area = 631.44

Final answer: Area = 631.44 square units



Question: 1(viii). Find the area of triangle ABC when a = 41.34, b = 35.65 and c = 56.81.

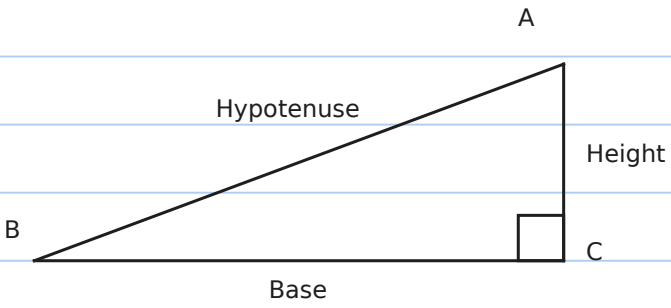
Answer: Step 1: all three sides are known, so use Heron's formula Area = sqrt(s(s - a)(s - b)(s - c))

Step 2: s = (41.34 + 35.65 + 56.81)/2 = 66.9, so s - a = 25.56, s - b = 31.25 and s - c = 10.09

Step 3: substitute Area = sqrt(66.9 x 25.56 x 31.25 x 10.09) = sqrt(539174)

Step 4: Area = 734.28

Final answer: Area = 734.28 square units



Question: 1(ix). Find the area of triangle ABC when a = 98, b = 120 and c = 200.

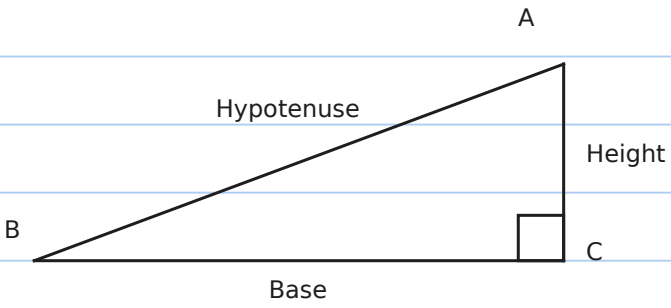
Answer: Step 1: all three sides are known, so use Heron's formula Area = sqrt(s(s - a)(s - b)(s - c))

Step 2: s = (98 + 120 + 200)/2 = 209, so s - a = 111, s - b = 89 and s - c = 9

Step 3: substitute Area = sqrt(209 x 111 x 89 x 9) = sqrt(18582399)

Step 4: Area = 4310.73

Final answer: Area = 4310.73 square units



Question: 2. The area of a triangle is 3.346 square units. If B = 20.9 deg and C = 117.2 deg, find a and angle A.

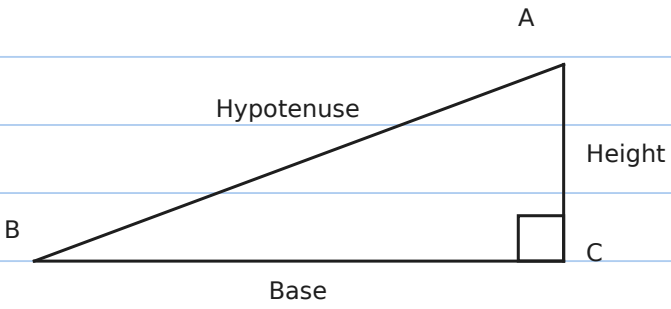
Answer: Step 1: A = 180 deg - B - C = 180 deg - 20.9 deg - 117.2 deg = 41.9 deg

Step 2: with side a and the two angles B and C, Area = (1/2)a^2 sin(B) sin(C)/sin(A), so a^2 = 2 x Area x sin(A)/(sin(B) sin(C))

Step 3: substitute a^2 = 2 x 3.346 x sin(41.9 deg)/(sin(20.9 deg) x sin(117.2 deg)) = 6.692 x 0.6676/(0.3568 x 0.8899) = 4.468/0.3175 = 14.07

Step 4: a = sqrt(14.07) = 3.75

Final answer: A = 41.9 deg and a = 3.75 units



Question: 3. The measures of two sides of a triangle are 4 and 5 units. Find the third side so that the area of the triangle is 6 square units.

Answer: Step 1: let theta be the included angle between the sides 4 and 5; then Area = (1/2)(4)(5) sin(theta) = 10 sin(theta)

Step 2: set 10 sin(theta) = 6, so sin(theta) = 0.6

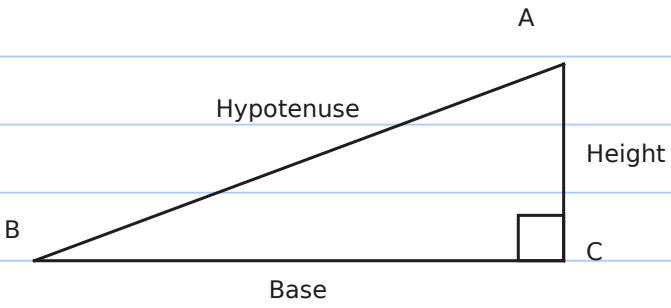
Step 3: theta = 36.87 deg or theta = 180 deg - 36.87 deg = 143.13 deg, and in both cases cos(theta) = 0.8 or cos(theta) = -0.8 respectively

Step 4: by the law of cosines, the third side c satisfies c^2 = 4^2 + 5^2 - 2(4)(5) cos(theta) = 41 - 40 cos(theta)

Step 5: for theta = 36.87 deg, c^2 = 41 - 40(0.8) = 41 - 32 = 9, so c = 3 units

Step 6: for theta = 143.13 deg, c^2 = 41 - 40(-0.8) = 41 + 32 = 73, so c = sqrt(73) = 8.54 units

Final answer: the third side is 3 units (acute included angle) or 8.54 units (obtuse included angle); both give area 6 square units



Question: 4. Find the area of the equilateral triangle whose each side is x units long.

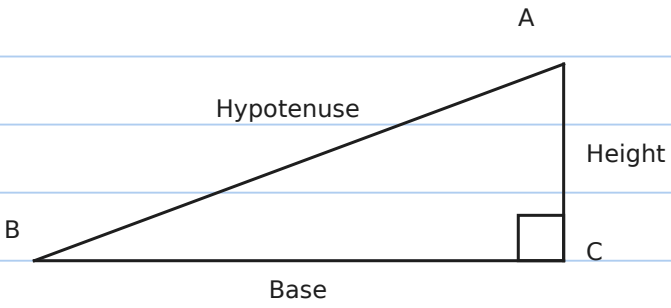
Answer: Step 1: in an equilateral triangle every angle is 60 deg, and any two sides include a 60 deg angle

Step 2: use Area = (1/2) x (side) x (side) x sin(included angle) = (1/2)(x)(x) sin(60 deg)

Step 3: sin(60 deg) = sqrt(3)/2, so Area = (1/2)x^2 x sqrt(3)/2

Step 4: simplify, Area = (sqrt(3)/4)x^2

Final answer: Area = (sqrt(3)/4)x^2 square units



Exercise 11.4

Question: 1(i). Find the values of R and r of triangle ABC when a = 5 cm, b = 10 cm and c = 12 cm.

Answer: Step 1: s = (5 + 10 + 12)/2 = 13.5 cm, so s - a = 8.5, s - b = 3.5

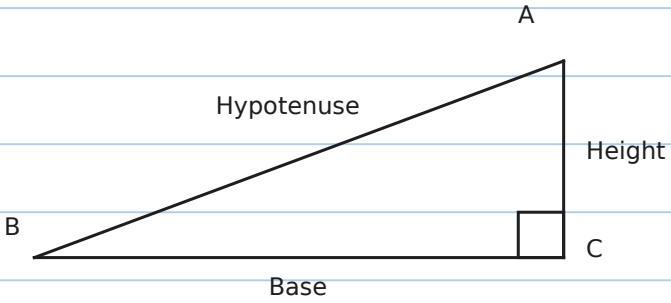
and s - c = 1.5

Step 2: by Heron's formula, Area = sqrt(s(s - a)(s - b)(s - c)) = sqrt(13.5 x 8.5 x 3.5 x 1.5) = sqrt(602.44) = 24.54 sq cm

Step 3: circumradius R = abc/(4 x Area) = (5 x 10 x 12)/(4 x 24.54) = 600/98.18 = 6.11 cm

Step 4: inradius r = Area/s = 24.54/13.5 = 1.82 cm

Final answer: R = 6.11 cm and r = 1.82 cm



Question: 1(ii). Find the values of R and r of triangle ABC when a = 10.5 cm, b = 11.5 cm and c = 20.5 cm.

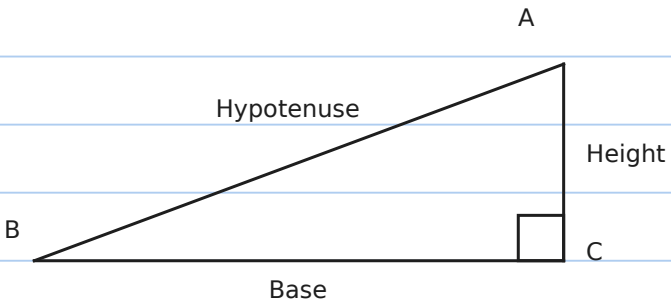
Answer: Step 1: s = (10.5 + 11.5 + 20.5)/2 = 21.25 cm, so s - a = 10.75, s - b = 9.75 and s - c = 0.75

Step 2: Area = sqrt(21.25 x 10.75 x 9.75 x 0.75) = sqrt(1670.6) = 40.87 sq cm

Step 3: R = abc/(4 x Area) = (10.5 x 11.5 x 20.5)/(4 x 40.87) = 2475.4/163.5 = 15.14 cm

Step 4: r = Area/s = 40.87/21.25 = 1.92 cm

Final answer: R = 15.14 cm and r = 1.92 cm



Question: 1(iii). Find the values of R and r of triangle ABC when a = 40 cm, b = 12 cm and c = 38 cm.

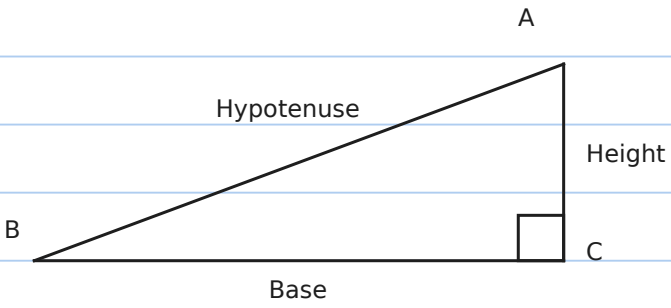
Answer: Step 1: s = (40 + 12 + 38)/2 = 45 cm, so s - a = 5, s - b = 33 and s - c = 7

Step 2: Area = sqrt(45 x 5 x 33 x 7) = sqrt(51975) = 227.98 sq cm

Step 3: R = abc/(4 x Area) = (40 x 12 x 38)/(4 x 227.98) = 18240/911.9 = 20.00 cm

Step 4: r = Area/s = 227.98/45 = 5.07 cm

Final answer: R = 20.00 cm and r = 5.07 cm



Question: 2(i). Find r1, r2 and r3 if the measures of the sides of triangle ABC are a = 23 cm, b = 24 cm and c = 28 cm.

Answer: Step 1: s = (23 + 24 + 28)/2 = 37.5 cm, so s - a = 14.5, s - b = 13.5 and s - c = 9.5

Step 2: Area = sqrt(37.5 x 14.5 x 13.5 x 9.5) = sqrt(69738) = 264.08 sq cm

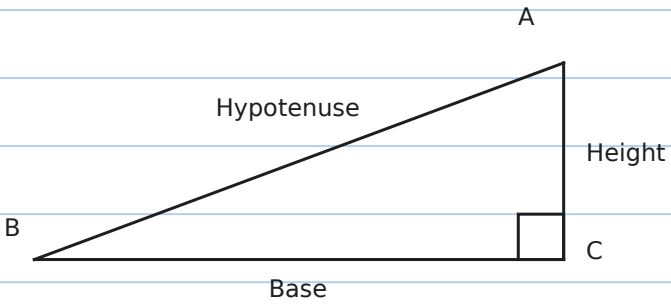
Step 3: the escribed radii are r1 = Area/(s - a), r2 = Area/(s - b) and r3 = Area/(s - c)

Step 4: r1 = 264.08/14.5 = 18.21 cm

Step 5: r2 = 264.08/13.5 = 19.56 cm

Step 6: r3 = 264.08/9.5 = 27.80 cm

Final answer: r1 = 18.21 cm, r2 = 19.56 cm, r3 = 27.80 cm



Question: 2(ii). Find r1, r2 and r3 if the measures of the sides of triangle ABC are a = 24.4 cm, b = 34.8 cm and c = 42.5 cm.

Answer: Step 1: s = (24.4 + 34.8 + 42.5)/2 = 50.85 cm, so s - a = 26.45, s - b = 16.05 and s - c = 8.35

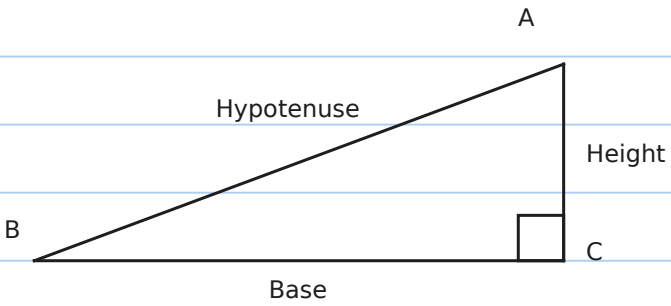
Step 2: Area = sqrt(50.85 x 26.45 x 16.05 x 8.35) = sqrt(180254) = 424.56 sq cm

Step 3: r1 = Area/(s - a) = 424.56/26.45 = 16.05 cm

Step 4: r2 = Area/(s - b) = 424.56/16.05 = 26.45 cm

Step 5: r3 = Area/(s - c) = 424.56/8.35 = 50.85 cm

Final answer: r1 = 16.05 cm, r2 = 26.45 cm, r3 = 50.85 cm



Question: 2(iii). Find r1, r2 and r3 if the measures of the sides of triangle ABC are a = 75 cm, b = 62 cm and c = 53 cm.

Answer: Step 1: s = (75 + 62 + 53)/2 = 95 cm, so s - a = 20, s - b = 33 and s - c = 42

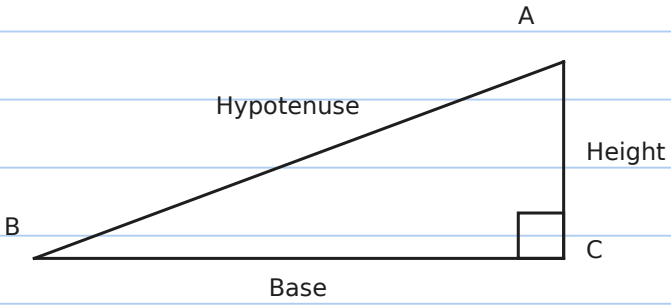
Step 2: Area = sqrt(95 x 20 x 33 x 42) = sqrt(2633400) = 1622.78 sq cm

Step 3: r1 = Area/(s - a) = 1622.78/20 = 81.14 cm

Step 4: r2 = Area/(s - b) = 1622.78/33 = 49.18 cm

Step 5: r3 = Area/(s - c) = 1622.78/42 = 38.64 cm

Final answer: r1 = 81.14 cm, r2 = 49.18 cm, r3 = 38.64 cm



Question: 3. If a = b = c, then prove that r1 : R : r = 3 : 2 : 1.

Answer: Step 1: let each side be x; the triangle is equilateral, so s = 3x/2 and s - a = 3x/2 - x = x/2

Step 2: Area = sqrt(s(s - a)(s - b)(s - c)) = sqrt((3x/2)(x/2)(x/2)(x/2)) = (sqrt(3)/4)x^2

Step 3: r = Area/s = ((sqrt(3)/4)x^2)/(3x/2) = (sqrt(3)/6)x

Step 4: R = abc/(4 x Area) = x^3/(4 x (sqrt(3)/4)x^2) = x/sqrt(3) = (sqrt(3)/3)x

Step 5: r1 = Area/(s - a) = ((sqrt(3)/4)x^2)/(x/2) = (sqrt(3)/2)x

Step 6: form the ratio r1 : R : r = (sqrt(3)/2)x : (sqrt(3)/3)x : (sqrt(3)/6)x; dividing each term by (sqrt(3)/6)x gives 3 : 2 : 1

Final answer: r1 : R : r = 3 : 2 : 1, proved

Question: 4(i). Show that r1 r2 r3 = r s^2.

Answer: Step 1: use the formulae r1 = Area/(s - a), r2 = Area/(s - b), r3 = Area/(s - c) and r = Area/s

Step 2: multiply, r1 r2 r3 = Area^3/((s - a)(s - b)(s - c))

Step 3: by Heron's formula Area^2 = s(s - a)(s - b)(s - c), so (s - a)(s - b)(s - c) = Area^2/s

Step 4: substitute, r1 r2 r3 = Area^3/(Area^2/s) = Area x s

Step 5: since r = Area/s, Area = r s, so r1 r2 r3 = (r s) x s = r s^2

Final answer: r1 r2 r3 = r s^2, proved

Question: 4(ii). Show that r = 4R sin(A/2) sin(B/2) sin(C/2).

Answer: Step 1: start from the half angle formulae in terms of R: a = 2R sin(A), and similarly for b and c; also Area = (1/2)bc sin(A)

Step 2: use r = Area/s and s = (a + b + c)/2; with a = 2R sin(A) etc., a standard product form is Area = 2R^2 sin(A) sin(B) sin(C), which follows from Area = abc/(4R) = (2R sin(A))(2R sin(B))(2R sin(C))/(4R)

Step 3: also s = (2R/2)(sin(A) + sin(B) + sin(C)) = R x 4 cos(A/2) cos(B/2) cos(C/2), using the identity sin(A) + sin(B) + sin(C) = 4 cos(A/2) cos(B/2) cos(C/2), valid because A + B + C = 180 deg

Step 4: write sin(A) = 2 sin(A/2) cos(A/2), and similarly for B and C, so Area = 2R^2 x 8 sin(A/2) cos(A/2) sin(B/2) cos(B/2) sin(C/2) cos(C/2) = 16R^2 sin(A/2) sin(B/2) sin(C/2) cos(A/2) cos(B/2) cos(C/2)

Step 5: divide, r = Area/s = [16R^2 sin(A/2) sin(B/2) sin(C/2) cos(A/2) cos(B/2) cos(C/2)]/[4R cos(A/2) cos(B/2) cos(C/2)] = 4R sin(A/2) sin(B/2) sin(C/2), the cosine factors cancelling

Final answer: r = 4R sin(A/2) sin(B/2) sin(C/2), proved

Question: 4(iii). Show that r1 = a cos(B/2) cos(C/2)/cos(A/2).

Answer: Step 1: the same method as part (ii), applied to the escribed circle opposite A, gives r1 = 4R sin(A/2) cos(B/2) cos(C/2)

Step 2: by the law of sines in the form a = 2R sin(A), and sin(A) = 2 sin(A/2) cos(A/2), so a = 4R sin(A/2) cos(A/2), hence 4R sin(A/2) = a/cos(A/2)

Step 3: substitute this in the expression of Step 1: r1 = [a/cos(A/2)] x cos(B/2) cos(C/2)

Step 4: rearrange, r1 = a cos(B/2) cos(C/2)/cos(A/2)

Final answer: r1 = a cos(B/2) cos(C/2)/cos(A/2), proved

Question: 5. Prove that 1/r^2 + 1/r1^2 + 1/r2^2 + 1/r3^2 = (a^2 + b^2 + c^2)/Area^2.

Answer: Step 1: from r = Area/s, r1 = Area/(s - a), r2 = Area/(s - b) and r3 = Area/(s - c), the reciprocals are 1/r = s/Area, 1/r1 = (s - a)/Area, 1/r2 = (s - b)/Area and 1/r3 = (s - c)/Area

Step 2: square and add: the left side equals [s^2 + (s - a)^2 + (s - b)^2 + (s - c)^2]/Area^2

Step 3: expand the numerator: s^2 + (s^2 - 2sa + a^2) + (s^2 - 2sb +

b^2) + (s^2 - 2sc + c^2) = 4s^2 - 2s(a + b + c) + (a^2 + b^2 + c^2)

Step 4: since 2s = a + b + c, we have 2s(a + b + c) = (a + b + c)^2 = 4s^2, so the first two terms cancel: 4s^2 - 4s^2 = 0

Step 5: the numerator is therefore exactly a^2 + b^2 + c^2

Final answer: 1/r^2 + 1/r1^2 + 1/r2^2 + 1/r3^2 = (a^2 + b^2 + c^2)/Area^2, proved

Question: 6. Show that 1/(2rR) = 1/(ab) + 1/(bc) + 1/(ca).

Answer: Step 1: combine the right side over the common denominator abc: 1/(ab) + 1/(bc) + 1/(ca) = (c + a + b)/(abc) = (a + b + c)/(abc) = 2s/(abc)

Step 2: for the left side, use r = Area/s and R = abc/(4 x Area)

Step 3: 2rR = 2 x (Area/s) x abc/(4 x Area) = abc/(2s), the Area factors cancelling

Step 4: take the reciprocal, 1/(2rR) = 2s/(abc)

Step 5: this is the same expression as Step 1, so the two sides are equal

Final answer: 1/(2rR) = 1/(ab) + 1/(bc) + 1/(ca), proved

Review Exercise 11

Question: 1(i). Select the correct answer: From the top of a cliff 80 m high the angle of depression of a boat is alpha. If the distance between the boat and the foot of the cliff is 80 sqrt(3) m, then angle alpha is: (a) pi/4 (b) pi/6 (c) pi/3 (d) 3pi/4

Answer: Step 1: the angle of depression equals the angle of elevation from

the boat, and tan(alpha) = height/distance = 80/(80 sqrt(3))

Step 2: simplify, tan(alpha) = 1/sqrt(3)

Step 3: alpha = tan^-1(1/sqrt(3)) = 30 deg = pi/6 radians

Final answer: (b) pi/6

Question: 1(ii). Select the correct answer: To solve an oblique triangle, we use: (a) law of sines (b) laws of cosines (c) laws of tangents (d) all of these

Answer: Step 1: an oblique triangle has no right angle, and depending on the given elements it is solved by the law of sines, the law of cosines or the law of tangents

Step 2: all three laws listed are used for oblique triangles

Final answer: (d) all of these

Question: 1(iii). Select the correct answer: A circle which touches all the sides of a triangle is called: (a) circum-circle (b) in-circle (c) ex-circle (d) tri-circle

Answer: Step 1: the circle drawn inside the triangle touching all three sides is the inscribed circle, that is the in-circle, with centre the incentre and radius r

Step 2: the circum-circle passes through the vertices instead, and an ex-circle touches one side externally, so they do not match the description

Final answer: (b) in-circle

Question: 1(iv). Select the correct answer: In triangle ABC, if B = 90 deg, then b^2 = c^2 + a^2 - 2ac cos(B) becomes: (a) law of sines (b) law of tangents (c) Pythagoras theorem (d) none of these

Answer: Step 1: substitute cos(90 deg) = 0 in b^2 = a^2 + c^2 - 2ac cos(B)

Step 2: the last term vanishes, giving b^2 = a^2 + c^2, which is exactly the statement of Pythagoras theorem for the hypotenuse b

Final answer: (c) Pythagoras theorem

Question: 1(v). Select the correct answer: In any triangle ABC, sqrt((s - a)(s - c)/(ac)) is equal to: (a) sin(A/2) (b) cos(A/2) (c) sin(B/2) (d) sin(C/2)

Answer: Step 1: recall the half angle formula of sine, sin(B/2) = sqrt((s - a)(s - c)/(ac)), because the factors (s - a) and (s - c) belong to the sides other than b, over the product of the sides a and c enclosing angle B

Step 2: the given expression matches this formula exactly

Final answer: (c) sin(B/2)

Question: 1(vi). Select the correct answer: In any triangle ABC, cos(C/2) is equal to: (a) sqrt(s(s - a)/(ab)) (b) sqrt(s(s - b)/(ac)) (c) sqrt(s(s - a)/(bc)) (d) sqrt(s(s - c)/(ab))

Answer: Step 1: recall the half angle formula of cosine, cos(C/2) = sqrt(s(s - c)/(ab)), with (s - c) in the numerator and the product of the two sides a and b enclosing angle C in the denominator

Step 2: only option (d) has this form

Final answer: (d) sqrt(s(s - c)/(ab))

Question: 1(vii). Select the correct answer: In any triangle ABC, with usual notations, s is equal to: (a) a + b + c (b) (a + b + c)/2 (c) (a + b + c)/3 (d) abc/2

Answer: Step 1: s denotes the semi-perimeter, half the perimeter of the triangle

Step 2: the perimeter is a + b + c, so s = (a + b + c)/2

Final answer: (b) (a + b + c)/2

Question: 1(viii). Select the correct answer: sqrt(s(s - b)/((s - a)(s - c))) is equal to: (a) sin(B/2) (b) cos(B/2) (c) tan(B/2) (d) cot(B/2)

Answer: Step 1: recall tan(B/2) = sqrt((s - a)(s - c)/(s(s - b)))

Step 2: the given expression is the reciprocal of tan(B/2), and the reciprocal of tan is cot

Step 3: therefore the expression equals cot(B/2)

Final answer: (d) cot(B/2)

Question: 1(ix). Select the correct answer: In any triangle ABC, sqrt((s - a)(s - b)/(s(s - c))) is equal to: (a) sin(C/2) (b) cos(C/2) (c) tan(C/2) (d) cot(C/2)

Answer: Step 1: recall the half angle formula of tangent, tan(C/2) = sqrt((s - a)(s - b)/(s(s - c))), because tan(C/2) = r/(s - c) and r^2 = (s - a)(s - b)(s - c)/s

Step 2: the given expression matches this formula exactly

Final answer: (c) tan(C/2)

Question: 1(x). Select the correct answer: We can solve an oblique triangle, if: (a) one side and two angles are known (b) three sides are known (c) two sides and their included angle are known (d) all of these

Answer: Step 1: each listed set of elements determines a triangle completely: one side with two angles (the third angle follows from the angle sum), three sides, and two sides with the included angle

Step 2: hence an oblique triangle can be solved in every listed case

Final answer: (d) all of these

Question: 1(xi). Select the correct answer: In triangle ABC, inradius = : (a) Area/(s - a) (b) Area/(s - b) (c) Area/(s - c) (d) Area/s

Answer: Step 1: the area of the triangle equals the sum of the areas of the three triangles formed with the incentre, giving Area = r s

Step 2: therefore r = Area/s, while Area/(s - a), Area/(s - b) and Area/(s - c) are the escribed radii r1, r2 and r3

Final answer: (d) Area/s

Question: 1(xii). Select the correct answer: In any triangle ABC, Area of triangle is: (a) bc sin(A) (b) (1/2)ca sin(A) (c) (1/2)ab sin(C) (d) (1/2)ab sin(B)

Answer: Step 1: the area is (1/2) x (product of two sides) x sin(their included angle)

Step 2: sides a and b include angle C, so Area = (1/2)ab sin(C)

Step 3: option (a) lacks the factor 1/2, option (b) pairs sides c and a with the wrong angle (they include B), and option (d) also uses the wrong angle for sides a and b

Final answer: (c) (1/2)ab sin(C)

Question: 1(xiii). Select the correct answer: Hero's formula is: (a) Area = s(s - a)(s - b)(s - c) (b) Area = sqrt((s - a)(s - b)(s - c)) (c) Area = sqrt(s(s - a)(s - b)(s - c)) (d) Area = (a + b + c)/2

Answer: Step 1: Hero's (Heron's) formula gives the area from the three sides as the square root of s(s - a)(s - b)(s - c), where s is the semi-perimeter

Step 2: option (a) misses the square root, option (b) misses the factor s, and option (d) is only the semi-perimeter doubled back, not an area

Final answer: (c) Area = sqrt(s(s - a)(s - b)(s - c))

Question: 1(xiv). Select the correct answer: The circle passing through the

three vertices of a triangle is called: (a) circum-circle (b) in-circle (c) ex-centre (d) escribed circle

Answer: Step 1: by definition, the circle through all three vertices A, B and C is the circum-circle, its centre is the circum-centre and its radius is R

Step 2: the in-circle and escribed circles touch the sides rather than pass through the vertices

Final answer: (a) circum-circle

Question: 1(xv). Select the correct answer: The point of intersection of the right bisectors (perpendicular bisectors) of the sides of a triangle is called: (a) circum-centre (b) in-centre (c) escribed centre (d) ortho-centre

Answer: Step 1: the perpendicular bisectors of the three sides meet at the centre of the circum-circle, which is equidistant from the three vertices

Step 2: that point is the circum-centre (the in-centre is instead the meeting point of the angle bisectors)

Final answer: (a) circum-centre

Question: 1(xvi). Select the correct answer: Radius of the circle which passes through all the vertices of a triangle is: (a) circum-radius (b) in-radius (c) e-radius (d) diameter

Answer: Step 1: the circle through the vertices is the circum-circle, and its radius is denoted by R and called the circum-radius, with R = abc/(4 x Area)

Final answer: (a) circum-radius

Question: 1(xvii). Select the correct answer: In any triangle ABC, with usual notations, abc = : (a) R (b) Rs (c) 4R x Area (d) Area/s

Answer: Step 1: the circumradius formula is R = abc/(4 x Area)

Step 2: multiply both sides by 4 x Area: abc = 4R x Area

Final answer: (c) 4R x Area

Question: 1(xviii). Select the correct answer: The point of intersection of the internal bisectors of angles of a triangle is: (a) in-centre (b) e-centre (c) circum-centre (d) ex-centre

Answer: Step 1: the three internal angle bisectors meet at one point inside the triangle, which is the centre of the in-circle and is equidistant from the three sides

Step 2: that point is the in-centre

Final answer: (a) in-centre

Question: 1(xix). Select the correct answer: In any equilateral triangle ABC, with usual notations, r : R : r1 = : (a) 1 : 2 : 3 (b) 3 : 2 : 1 (c) 1 : 3 : 2 (d) 1 : 1 : 1

Answer: Step 1: for an equilateral triangle of side x, r = (sqrt(3)/6)x, R = (sqrt(3)/3)x and r1 = (sqrt(3)/2)x (as found in Exercise 11.4 Question 3)

Step 2: divide r : R : r1 by (sqrt(3)/6)x: (sqrt(3)/6)x gives 1, (sqrt(3)/3)x gives 2 and (sqrt(3)/2)x gives 3

Step 3: so r : R : r1 = 1 : 2 : 3

Final answer: (a) 1 : 2 : 3

Question: 1(xx). Select the correct answer: Circum radius of triangle ABC is: (a) Area/s (b) Area/(s - b) (c) Area/(s - a) (d) abc/(4 x Area)

Answer: Step 1: the circumradius is R = abc/(4 x Area), derived from a = 2R sin(A) together with Area = (1/2)bc sin(A)

Step 2: Area/s is the inradius r, and Area/(s - a) and Area/(s - b) are escribed radii, so only option (d) is the circumradius

Final answer: (d) abc/(4 x Area)

Question: 2(i). Prove that r2 = b cos(A/2) cos(C/2)/cos(B/2).

Answer: Step 1: as in Exercise 11.4 Question 4(iii), the escribed radius opposite B is r2 = 4R sin(B/2) cos(A/2) cos(C/2)

Step 2: by the law of sines, b = 2R sin(B) = 2R x 2 sin(B/2) cos(B/2) = 4R sin(B/2) cos(B/2), so 4R sin(B/2) = b/cos(B/2)

Step 3: substitute in Step 1: r2 = [b/cos(B/2)] x cos(A/2) cos(C/2)

Step 4: rearrange, r2 = b cos(A/2) cos(C/2)/cos(B/2)

Final answer: r2 = b cos(A/2) cos(C/2)/cos(B/2), proved

Question: 2(ii). Prove that r1 r2 + r2 r3 + r3 r1 = s^2.

Answer: Step 1: use r1 = Area/(s - a), r2 = Area/(s - b) and r3 = Area/(s - c); then r1 r2 = Area^2/((s - a)(s - b)), r2 r3 = Area^2/((s - b)(s - c)) and r3 r1 = Area^2/((s - c)(s - a))

Step 2: add with the common denominator (s - a)(s - b)(s - c): the sum equals Area^2 x [(s - c) + (s - a) + (s - b)]/((s - a)(s - b)(s - c))

Step 3: simplify the bracket: (s - a) + (s - b) + (s - c) = 3s - (a + b + c) = 3s - 2s = s

Step 4: so the sum equals Area^2 x s/((s - a)(s - b)(s - c))

Step 5: by Heron's formula Area^2 = s(s - a)(s - b)(s - c), so the sum = s(s - a)(s - b)(s - c) x s/((s - a)(s - b)(s - c)) = s^2

Final answer: r1 r2 + r2 r3 + r3 r1 = s^2, proved

Question: 3(i). Find the area of triangle ABC when a = 5.2 cm, B = 30 deg and C = 40 deg.

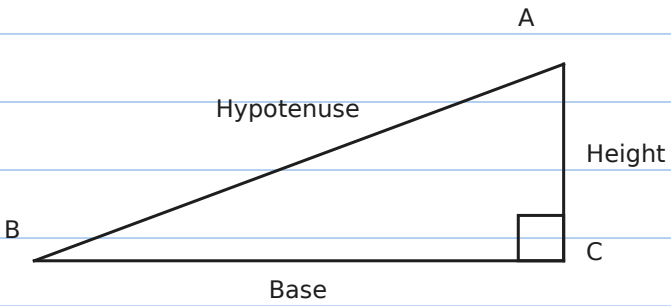
Answer: Step 1: first find A = 180 deg - 30 deg - 40 deg = 110 deg

Step 2: use Area = (1/2)a^2 sin(B) sin(C)/sin(A)

Step 3: substitute Area = (1/2)(5.2^2) x sin(30 deg) sin(40 deg)/sin(110 deg) = (1/2)(27.04) x (0.5 x 0.6428)/0.9397

Step 4: Area = 13.52 x 0.3214/0.9397 = 4.62

Final answer: Area = 4.62 sq cm



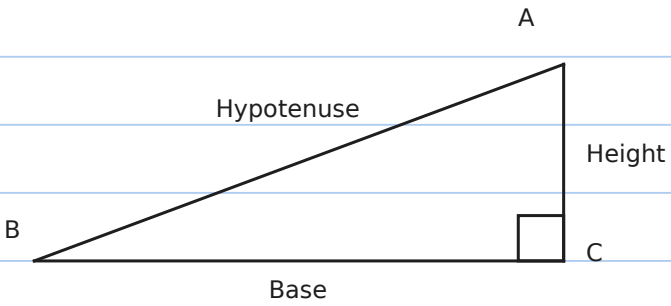
Question: 3(ii). Find the area of triangle ABC when a = 125 cm, b = 120 and C = 150 deg.

Answer: Step 1: sides a and b include angle C, so use Area = (1/2)ab sin(C)

Step 2: substitute Area = (1/2)(125)(120) sin(150 deg)

Step 3: sin(150 deg) = 0.5, so Area = (1/2)(15000)(0.5) = 3750

Final answer: Area = 3750 sq cm



Question: 4(i). Find the R and r of triangle ABC when a = 18.8 cm, b = 24.5 cm and c = 30.2 cm.

Answer: Step 1: s = (18.8 + 24.5 + 30.2)/2 = 36.75 cm, so s - a = 17.95, s

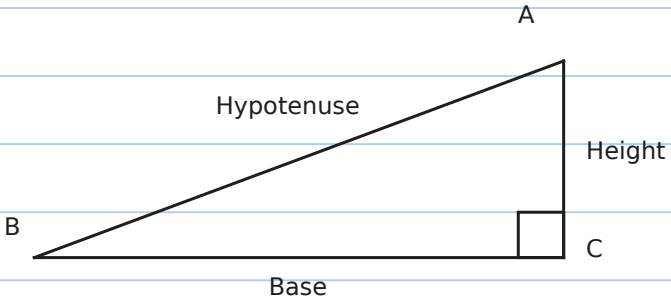
- b = 12.25 and s - c = 6.55

Step 2: Area = sqrt(36.75 x 17.95 x 12.25 x 6.55) = sqrt(52936) = 230.06 sq cm

Step 3: R = abc/(4 x Area) = (18.8 x 24.5 x 30.2)/(4 x 230.06) = 13910.1/920.2 = 15.12 cm

Step 4: r = Area/s = 230.06/36.75 = 6.26 cm

Final answer: R = 15.12 cm and r = 6.26 cm



Question: 4(ii). Find the R and r of triangle ABC when a = 13.8 cm, b = 13.8 cm and c = 10.4 cm.

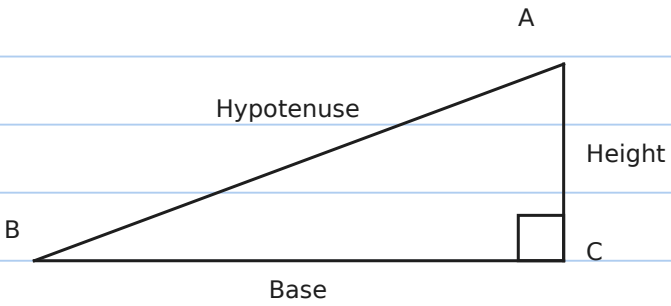
Answer: Step 1: s = (13.8 + 13.8 + 10.4)/2 = 19 cm, so s - a = 5.2, s - b = 5.2 and s - c = 8.6

Step 2: Area = sqrt(19 x 5.2 x 5.2 x 8.6) = sqrt(4418.4) = 66.47 sq cm

Step 3: R = abc/(4 x Area) = (13.8 x 13.8 x 10.4)/(4 x 66.47) = 1980.4/265.9 = 7.45 cm

Step 4: r = Area/s = 66.47/19 = 3.50 cm

Final answer: R = 7.45 cm and r = 3.50 cm



Question: 5. Find r1, r2 and r3 of the escribed circles if the measures of the sides of triangle ABC are a = 3 cm, b = 4 cm and c = 5 cm.

Answer: Step 1: s = (3 + 4 + 5)/2 = 6 cm, so s - a = 3, s - b = 2 and s - c = 1

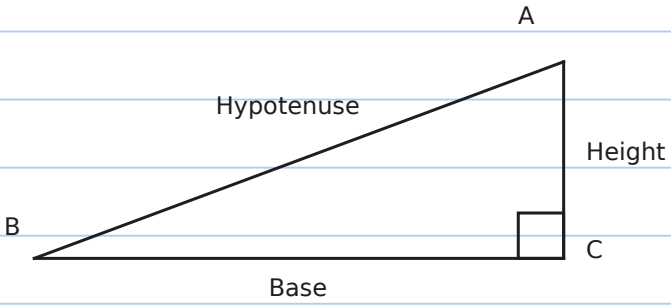
Step 2: Area = sqrt(6 x 3 x 2 x 1) = sqrt(36) = 6 sq cm

Step 3: r1 = Area/(s - a) = 6/3 = 2 cm

Step 4: r2 = Area/(s - b) = 6/2 = 3 cm

Step 5: r3 = Area/(s - c) = 6/1 = 6 cm

Final answer: r1 = 2 cm, r2 = 3 cm, r3 = 6 cm



Question: 6. A hiker walks due east at 4 km per hour and a second hiker, starting at the same point, walks 55 deg north-east at the rate of 5 km per hour. How far apart will they be after 3 hours?

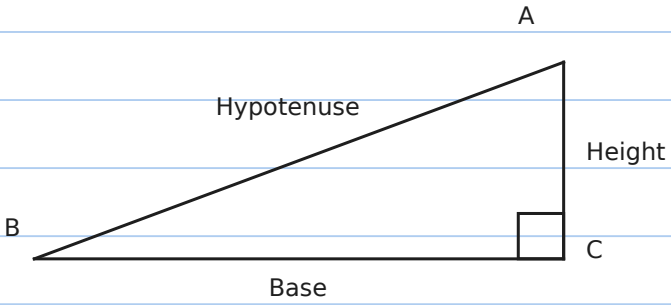
Answer: Step 1: after 3 hours the first hiker has walked 4 x 3 = 12 km due east and the second has walked 5 x 3 = 15 km in a direction 55 deg north of east, so the angle between their paths is 55 deg

Step 2: let d be the distance between them; by the law of cosines, d^2 = 12^2 + 15^2 - 2(12)(15) cos(55 deg)

Step 3: substitute d^2 = 144 + 225 - 360 x 0.5736 = 369 - 206.49 = 162.51

Step 4: d = sqrt(162.51) = 12.75 km

Final answer: the hikers will be about 12.75 km apart after 3 hours



Question: 7. Three points A, B and C form a triangle such that the ratio of the measures of their angles is 1 : 2 : 3. Find the ratio of the lengths of the sides.

Answer: Step 1: let the angles be x, 2x and 3x; then x + 2x + 3x = 180 deg, so 6x = 180 deg and x = 30 deg

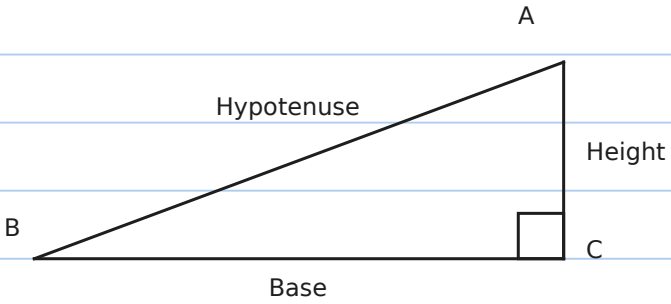
Step 2: the angles are therefore A = 30 deg, B = 60 deg and C = 90 deg

Step 3: by the law of sines, a : b : c = sin(A) : sin(B) : sin(C)

Step 4: substitute a : b : c = sin(30 deg) : sin(60 deg) : sin(90 deg) = 0.5 : sqrt(3)/2 : 1

Step 5: multiply every term by 2: a : b : c = 1 : sqrt(3) : 2

Final answer: the sides are in the ratio 1 : sqrt(3) : 2



Question: 8. A piece of plastic strip 1 meter long is bent to form an isosceles triangle with 95 deg as measure of its largest angle. Find the length of the sides.

Answer: Step 1: the 95 deg angle must be the vertex angle, because if a base angle were 95 deg the two base angles alone would total 190 deg, more than 180 deg; so the base angles are (180 deg - 95 deg)/2 = 42.5 deg each

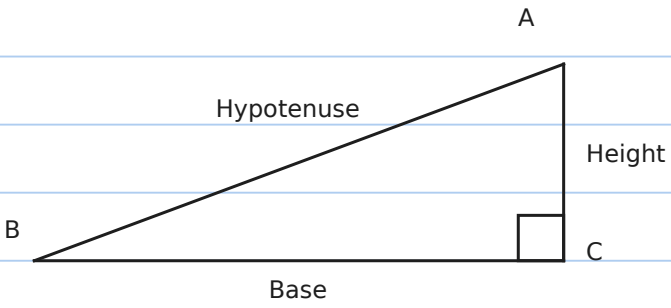
Step 2: let each equal side (leg) be L and the base be B; the perimeter gives 2L + B = 1

Step 3: by the law of sines, B/sin(95 deg) = L/sin(42.5 deg), so B = L x sin(95 deg)/sin(42.5 deg) = L x 0.9962/0.6756 = 1.4746L

Step 4: substitute in the perimeter: 2L + 1.4746L = 1, so 3.4746L = 1 and L = 0.2878 m

Step 5: B = 1.4746 x 0.2878 = 0.4244 m

Final answer: the equal sides are about 0.288 m (28.8 cm) each and the base is about 0.424 m (42.4 cm)



Question: 9. Two airplanes leave a field at the same time. One flies 30 deg east of north at 250 km/h, the other 45 deg east of south at 300 km/h. How far apart are they at the end of 2 hours?

Answer: Step 1: in 2 hours the first plane flies 250 x 2 = 500 km and the second flies 300 x 2 = 600 km

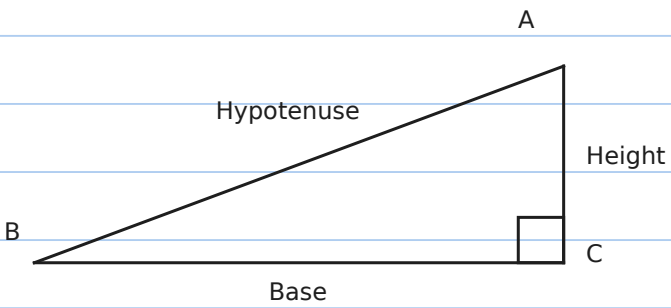
Step 2: find the angle between their directions: the first direction is 30 deg east of north (bearing 30 deg) and the second is 45 deg east of south (bearing 180 deg - 45 deg = 135 deg), so the angle between the paths is 135 deg - 30 deg = 105 deg

Step 3: by the law of cosines, d^2 = 500^2 + 600^2 - 2(500)(600) cos(105 deg), where d is the distance between the planes

Step 4: substitute cos(105 deg) = -0.2588, so d^2 = 250000 + 360000 - 600000 x (-0.2588) = 610000 + 155291 = 765291

Step 5: d = sqrt(765291) = 874.81 km

Final answer: the airplanes are about 874.81 km apart after 2 hours



Question: 10. Two men are on the opposite sides of a 100 m high tower. If the measures of the angles of elevation of the top of the tower are 18 deg and 22 deg respectively, find the distance between them.

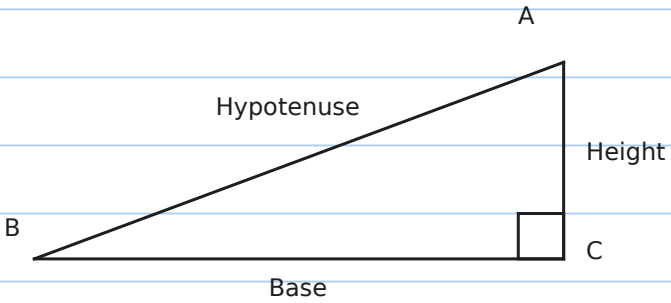
Answer: Step 1: because the men are on opposite sides of the tower, the distance between them is the sum of their distances from the foot of the tower

Step 2: for the first man, tan(18 deg) = 100/d1, so d1 = 100/tan(18 deg) = 100/0.3249 = 307.77 m

Step 3: for the second man, tan(22 deg) = 100/d2, so d2 = 100/tan(22 deg) = 100/0.4040 = 247.51 m

Step 4: add, distance = 307.77 + 247.51 = 555.28 m

Final answer: the distance between the two men is about 555.28 m



Question: 11. A man standing 60 m away from a tower notices that the angles of elevation of the top and the bottom of a flagstaff on the top of the tower are 64 deg and 62 deg respectively. Find the length of the flagstaff.

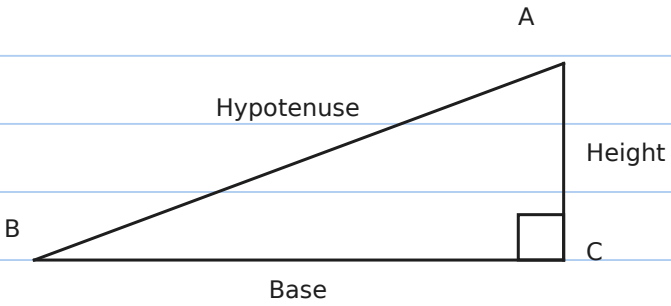
Answer: Step 1: let H be the height of the tower including the flagstaff, and h the height of the tower up to the bottom of the flagstaff

Step 2: from the angle of elevation 64 deg to the top, H = 60 tan(64 deg) = 60 x 2.0503 = 123.02 m

Step 3: from the angle of elevation 62 deg to the bottom of the flagstaff, h = 60 tan(62 deg) = 60 x 1.8807 = 112.84 m

Step 4: length of flagstaff = H - h = 123.02 - 112.84 = 10.17 m

Final answer: the length of the flagstaff is about 10.17 m



Question: 12. The angle of elevation of the top of a 60 m high tower from a point A, on the same level as the foot of the tower, is 25 deg. Find the angle of elevation of the top of the tower from a point B, 20 m nearer to the foot of the tower than A.

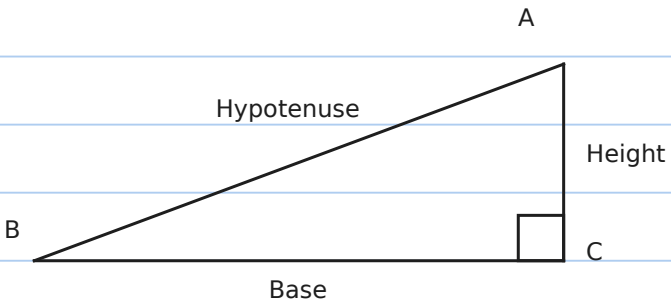
Answer: Step 1: find the distance of A from the foot: tan(25 deg) = 60/dA, so dA = 60/tan(25 deg) = 60/0.4663 = 128.67 m

Step 2: B is 20 m nearer the foot, so the distance of B from the foot is 128.67 - 20 = 108.67 m

Step 3: if theta is the angle of elevation from B, tan(theta) = 60/108.67 = 0.5521

Step 4: theta = tan^-1(0.5521) = 28.90 deg

Final answer: the angle of elevation from B is about 28.90 deg



Question: 13. Two buildings A and B are 100 m apart. The angle of elevation from the top of building A to the top of building B is 20 deg. The angle of elevation from the base of building B to the top of building A is 50 deg. Find the height of building B.

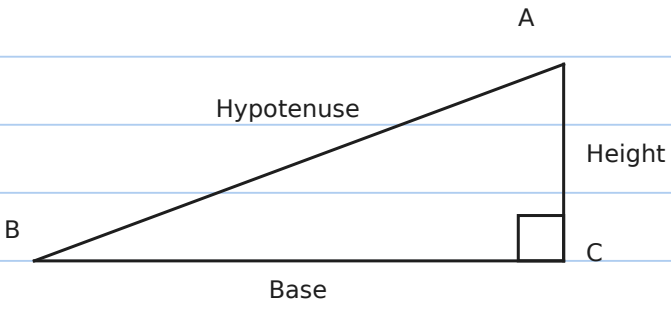
Answer: Step 1: let hA and hB be the heights of buildings A and B; from the base of B, tan(50 deg) = hA/100

Step 2: hA = 100 x tan(50 deg) = 100 x 1.1918 = 119.18 m

Step 3: the angle of elevation from the top of A to the top of B is 20 deg, so B is taller than A, and the excess height is hB - hA = 100 tan(20 deg) = 100 x 0.3640 = 36.40 m

Step 4: hB = hA + 36.40 = 119.18 + 36.40 = 155.57 m

Final answer: the height of building B is about 155.57 m



Unit 12: Graphs of Trigonometric and Inverse Trigonometric Functions and Solution of Trigonometric Equations

Exercise 12.1

Question: Exercise 12.1 Q1(i): Find the domain and range of the function 2sin(3x).

Answer: Step 1: The sine function is defined for every real angle, and multiplying the angle by 3 or the value by 2 creates no restriction, so the function is defined for all real x, that is, Domain = R.

Step 2: For every x, -1 <= sin(3x) <= 1.

Step 3: Multiply all parts by 2: -2 <= 2sin(3x) <= 2, so the function takes every value between -2 and 2.

Final answer: Domain = R, Range = {y | y in R and -2 <= y <= 2}.

Question: Exercise 12.1 Q1(ii): Find the domain and range of the function 5cos(4x).

Answer: Step 1: The cosine function is defined for every real angle, so the function is defined for all real x, that is, Domain = R.

Step 2: For every x, -1 <= cos(4x) <= 1.

Step 3: Multiply all parts by 5: -5 <= 5cos(4x) <= 5, so the range is the full interval from -5 to 5.

Final answer: Domain = R, Range = {y | y in R and -5 <= y <= 5}.

Question: Exercise 12.1 Q1(iii): Find the domain and range of the function 8tan(2x).

Answer: Step 1: tan(2x) = sin(2x)/cos(2x) is undefined where cos(2x) = 0.

Step 2: cos(2x) = 0 when 2x = n*pi/2 where n is an odd integer, so x = n*pi/4 where n is an odd integer; these values must be removed from R.

Step 3: The factor 8 stretches the values but tangent already takes every real value, so the range is unchanged.

Final answer: Domain = R - {n*pi/4 | n is an odd integer}, Range = R.

Question: Exercise 12.1 Q1(iv): Find the domain and range of the function cosec(6x).

Answer: Step 1: cosec(6x) = 1/sin(6x) is undefined where sin(6x) = 0.

Step 2: sin(6x) = 0 when 6x = n*pi, n in Z, so x = n*pi/6, n in Z; these values must be removed from R (this set is also written as {n*pi/12 | n is an even integer}).

Step 3: Since |sin(6x)| <= 1 and sin(6x) is never 0 in the domain, |cosec(6x)| >= 1, and cosec takes all such values.

Final answer: Domain = R - {n*pi/6 | n in Z}, Range = {y | y <= -1 or y >= 1}.

Question: Exercise 12.1 Q1(v): Find the domain and range of the function 5cot(2x).

Answer: Step 1: cot(2x) = cos(2x)/sin(2x) is undefined where sin(2x) = 0.

Step 2: sin(2x) = 0 when 2x = n*pi, n in Z, so x = n*pi/2, n in Z; these values must be removed from R (this set is also written as {n*pi/4 | n is an even integer}).

Step 3: Cotangent takes every real value, and the factor 5 does not change that, so the range is all real numbers.

Final answer: Domain = R - {n*pi/2 | n in Z}, Range = R.

Question: Exercise 12.1 Q1(vi): Find the domain and range of the function sin(x/3).

Answer: Step 1: The sine function is defined for every real angle, and x/3 is real for every real x, so Domain = R.

Step 2: As x runs over R, x/3 also runs over all real angles, so sin(x/3) takes the same values as sin(x).

Step 3: Hence -1 <= sin(x/3) <= 1 for all x.

Final answer: Domain = R, Range = {y | y in R and -1 <= y <= 1}.

Question: Exercise 12.1 Q1(vii): Find the domain and range of the function cosec(x/4).

Answer: Step 1: cosec(x/4) = 1/sin(x/4) is undefined where sin(x/4) = 0.

Step 2: sin(x/4) = 0 when x/4 = n*pi, n in Z, so x = 4n*pi, n in Z; these values must be removed from R.

Step 3: Since |sin(x/4)| <= 1, we get |cosec(x/4)| >= 1, and every such value is attained.

Final answer: Domain = R - {4n*pi | n in Z}, Range = {y | y <= -1 or y >= 1}.

Question: Exercise 12.1 Q1(viii): Find the domain and range of the function tan(x/7).

Answer: Step 1: tan(x/7) = sin(x/7)/cos(x/7) is undefined where cos(x/7) = 0.

Step 2: cos(x/7) = 0 when x/7 = n*pi/2 where n is an odd integer, so x = 7n*pi/2 where n is an odd integer; these values must be removed from R.

Step 3: Tangent takes every real value, so the range is all real numbers.

Final answer: Domain = R - {7n*pi/2 | n is an odd integer}, Range = R.

Question: Exercise 12.1 Q2(i): Determine whether the trigonometric function f(x) = sin(x)cos(x) is even, odd or neither.

Answer: Step 1: Replace x by -x: f(-x) = sin(-x)cos(-x).

Step 2: Use sin(-x) = -sin(x) and cos(-x) = cos(x): f(-x) = (-sin(x))(cos(x)) = -sin(x)cos(x).

Step 3: Hence f(-x) = -f(x) for all x in the domain, which is exactly the condition for an odd function.

Final answer: f(x) = sin(x)cos(x) is an odd function.

Question: Exercise 12.1 Q2(ii): Determine whether the trigonometric function g(x) = sin^2(x)/(1 + tan(x)) is even, odd or neither.

Answer: Step 1: Replace x by -x: g(-x) = sin^2(-x)/(1 + tan(-x)).

Step 2: Use sin^2(-x) = sin^2(x) (because sin(-x) = -sin(x) and the square removes the sign) and tan(-x) = -tan(x): g(-x) = sin^2(x)/(1 - tan(x)).

Step 3: Compare with g(x) = sin^2(x)/(1 + tan(x)) and -g(x) = -sin^2(x)/(1 + tan(x)): g(-x) is equal to neither g(x) nor -g(x) in general (for example at x = pi/6 the three values are all different).

Final answer: g(x) is neither even nor odd.

Question: Exercise 12.1 Q2(iii): Determine whether the trigonometric function h(x) = tan(x)/(x + sin(x)) is even, odd or neither.

Answer: Step 1: Replace x by -x: h(-x) = tan(-x)/(-x + sin(-x)).

Step 2: Use tan(-x) = -tan(x) and sin(-x) = -sin(x): h(-x) = (-tan(x))/(-x - sin(x)).

Step 3: Take -1 out of the denominator: (-tan(x))/(-(x + sin(x))) = tan(x)/(x + sin(x)) = h(x).

Step 4: Hence h(-x) = h(x) for all x in the domain, which is exactly the condition for an even function.

Final answer: h(x) = tan(x)/(x + sin(x)) is an even function.

Question: Exercise 12.1 Q2(iv): Determine whether the trigonometric function k(x) = x^3(sin(x) + cos(x)) is even, odd or neither.

Answer: Step 1: Replace x by -x: k(-x) = (-x)^3(sin(-x) + cos(-x)).

Step 2: Use (-x)^3 = -x^3, sin(-x) = -sin(x), cos(-x) = cos(x): k(-x) = -x^3(-sin(x) + cos(x)) = x^3(sin(x) - cos(x)).

Step 3: Compare with k(x) = x^3(sin(x) + cos(x)) and -k(x) = x^3(-sin(x) - cos(x)): k(-x) matches neither, because the cos(x) term has the wrong sign for both.

Final answer: k(x) is neither even nor odd.

Question: Exercise 12.1 Q3(i): Find the period of the function sin(3x).

Answer: Step 1: The function is of the form sin(kx) with k = 3.

Step 2: The period of sin(kx) is 2*pi/|k|.

Step 3: Period = 2*pi/3; check: sin(3(x + 2*pi/3)) = sin(3x + 2*pi) =

sin(3x).

Final answer: The period of sin(3x) is 2*pi/3.

Question: Exercise 12.1 Q3(ii): Find the period of the function cos(4x).

Answer: Step 1: The function is of the form cos(kx) with k = 4.

Step 2: The period of cos(kx) is 2*pi/|k|.

Step 3: Period = 2*pi/4 = pi/2; check: cos(4(x + pi/2)) = cos(4x + 2*pi) = cos(4x).

Final answer: The period of cos(4x) is pi/2.

Question: Exercise 12.1 Q3(iii): Find the period of the function tan(x/3).

Answer: Step 1: The function is of the form tan(kx) with k = 1/3.

Step 2: The period of tan(kx) is pi/|k| (tangent has basic period pi, not 2*pi).

Step 3: Period = pi/(1/3) = 3*pi; check: tan((x + 3*pi)/3) = tan(x/3 + pi) = tan(x/3).

Final answer: The period of tan(x/3) is 3*pi.

Question: Exercise 12.1 Q3(iv): Find the period of the function sec(x/5).

Answer: Step 1: The function is of the form sec(kx) with k = 1/5.

Step 2: The period of sec(kx) is 2*pi/|k|, the same as cosine.

Step 3: Period = 2*pi/(1/5) = 10*pi; check: sec((x + 10*pi)/5) = sec(x/5 + 2*pi) = sec(x/5).

Final answer: The period of sec(x/5) is 10*pi.

Question: Exercise 12.1 Q3(v): Find the period of the function cosec(8x).

Answer: Step 1: The function is of the form cosec(kx) with k = 8.

Step 2: The period of cosec(kx) is 2*pi/|k|, the same as sine.

Step 3: Period = 2*pi/8 = pi/4; check: cosec(8(x + pi/4)) = cosec(8x + 2*pi) = cosec(8x).

Final answer: The period of cosec(8x) is pi/4.

Question: Exercise 12.1 Q3(vi): Find the period of the function cot(x/6).

Answer: Step 1: The function is of the form cot(kx) with k = 1/6.

Step 2: The period of cot(kx) is pi/|k|, the same as tangent.

Step 3: Period = pi/(1/6) = 6*pi; check: cot((x + 6*pi)/6) = cot(x/6 + pi) = cot(x/6).

Final answer: The period of cot(x/6) is 6*pi.

Question: Exercise 12.1 Q3(vii): Find the period of the function sqrt(5)cos(3x/2).

Answer: Step 1: The constant factor sqrt(5) changes only the amplitude, not the period; the angle coefficient is k = 3/2.

Step 2: The period of cos(kx) is 2*pi/|k|.

Step 3: Period = 2*pi/(3/2) = 4*pi/3; check: sqrt(5)cos((3/2)(x + 4*pi/3)) = sqrt(5)cos(3x/2 + 2*pi) = sqrt(5)cos(3x/2).

Final answer: The period of sqrt(5)cos(3x/2) is 4*pi/3.

Question: Exercise 12.1 Q3(viii): Find the period of the function cot(sqrt(2)x).

Answer: Step 1: The function is of the form cot(kx) with k = sqrt(2).

Step 2: The period of cot(kx) is pi/|k|.

Step 3: Period = pi/sqrt(2); check: cot(sqrt(2)(x + pi/sqrt(2))) = cot(sqrt(2)x + pi) = cot(sqrt(2)x).

Final answer: The period of cot(sqrt(2)x) is pi/sqrt(2).

Question: Exercise 12.1 Q3(ix): Find the period of the function sin(x/3).

Answer: Step 1: The function is of the form sin(kx) with k = 1/3.

Step 2: The period of sin(kx) is 2*pi/|k|.

Step 3: Period = 2*pi/(1/3) = 6*pi; check: sin((x + 6*pi)/3) = sin(x/3 + 2*pi) = sin(x/3).

Final answer: The period of sin(x/3) is 6*pi.

Question: Exercise 12.1 Q3(x): Find the period of the function cos(x/4).

Answer: Step 1: The function is of the form cos(kx) with k = 1/4.

Step 2: The period of cos(kx) is 2*pi/|k|.

Step 3: Period = 2*pi/(1/4) = 8*pi; check: cos((x + 8*pi)/4) = cos(x/4 + 2*pi) = cos(x/4).

Final answer: The period of cos(x/4) is 8*pi.

Question: Exercise 12.1 Q3(xi): Find the period of the function (7/2)cot(2*pi*x/3).

Answer: Step 1: The constant factor 7/2 changes only the size of the values; the angle coefficient is k = 2*pi/3.

Step 2: The period of cot(kx) is pi/|k|.

Step 3: Period = pi/(2*pi/3) = 3/2; check: cot((2*pi/3)(x + 3/2)) = cot(2*pi*x/3 + pi) = cot(2*pi*x/3).

Final answer: The period of (7/2)cot(2*pi*x/3) is 3/2.

Question: Exercise 12.1 Q3(xii): Find the period of the function -(2/5)sec(3x/pi).

Answer: Step 1: The constant factor -2/5 changes only the amplitude and

sign; the angle coefficient is k = 3/pi.

Step 2: The period of sec(kx) is 2*pi/|k|.

Step 3: Period = 2*pi/(3/pi) = 2*pi^2/3; check: sec((3/pi)(x + 2*pi^2/3)) = sec(3x/pi + 2*pi) = sec(3x/pi).

Final answer: The period of -(2/5)sec(3x/pi) is 2*pi^2/3.

Question: Exercise 12.1 Q4(i): Find the maximum and minimum values of the function y = 4 + 3sin(theta).

Answer: Step 1: Write the function in the form a + b*sin(theta) with a = 4 and b = 3.

Step 2: Since -1 <= sin(theta) <= 1, the maximum value is M = a + |b| = 4 + 3 = 7, attained at sin(theta) = 1.

Step 3: The minimum value is m = a - |b| = 4 - 3 = 1, attained at sin(theta) = -1.

Final answer: Maximum value = 7, minimum value = 1.

Question: Exercise 12.1 Q4(ii): Find the maximum and minimum values of the function y = 2/3 - 4cos(theta).

Answer: Step 1: Write the function in the form a + b*cos(theta) with a = 2/3 and b = -4, so |b| = 4.

Step 2: The maximum value is M = a + |b| = 2/3 + 4 = 14/3, attained at cos(theta) = -1 (because b is negative).

Step 3: The minimum value is m = a - |b| = 2/3 - 4 = -10/3, attained at cos(theta) = 1.

Final answer: Maximum value = 14/3, minimum value = -10/3.

Question: Exercise 12.1 Q4(iii): Find the maximum and minimum values of

the function y = 6 - (1/3)sin(3*theta + 2).

Answer: Step 1: Write the function in the form a + b*sin(c*theta + d) with a = 6 and b = -1/3, so |b| = 1/3; the constants c = 3 and d = 2 shift and compress the graph but do not change its extreme values.

Step 2: The maximum value is M = a + |b| = 6 + 1/3 = 19/3.

Step 3: The minimum value is m = a - |b| = 6 - 1/3 = 17/3.

Final answer: Maximum value = 19/3, minimum value = 17/3.

Question: Exercise 12.1 Q4(iv): Find the maximum and minimum values of the function y = 8 + 5cos(theta - 25).

Answer: Step 1: Write the function in the form a + b*cos(theta + d) with a = 8 and b = 5; the phase shift 25 only moves the graph sideways and does not change its extreme values.

Step 2: The maximum value is M = a + |b| = 8 + 5 = 13.

Step 3: The minimum value is m = a - |b| = 8 - 5 = 3.

Final answer: Maximum value = 13, minimum value = 3.

Question: Exercise 12.1 Q4(v): Find the maximum and minimum values of the function y = 1/(25 - 12sin(3*theta - 2)).

Answer: Step 1: First find the extremes of the denominator, which is of the form a + b*sin(c*theta + d) with a = 25 and b = -12: maximum of denominator M = 25 + 12 = 37 and minimum m = 25 - 12 = 13.

Step 2: Both M and m are positive, so taking reciprocals reverses the order: the largest value of y comes from the smallest denominator.

Step 3: Maximum of y = 1/m = 1/13, and minimum of y = 1/M = 1/37.

Final answer: Maximum value = 1/13, minimum value = 1/37.

Question: Exercise 12.1 Q4(vi): Find the maximum and minimum values of the function y = 1/(1 + 6cos(5*theta - 4)).

Answer: Step 1: First find the extremes of the denominator, which is of the form a + b*cos(c*theta + d) with a = 1 and b = 6: maximum of denominator M = 1 + 6 = 7 and minimum m = 1 - 6 = -5.

Step 2: Here m < 0 < M, so for the reciprocal the maximum value is 1/M and the minimum value is 1/m (the reciprocal function jumps across the zero of the denominator, and the extreme values are these two).

Step 3: Maximum of y = 1/7, and minimum of y = 1/(-5) = -1/5.

Final answer: Maximum value = 1/7, minimum value = -1/5.

Exercise 12.2

Question: Exercise 12.2 Q1: Draw the graph of y = -sin(x) for 0 <= x <= 2*pi.

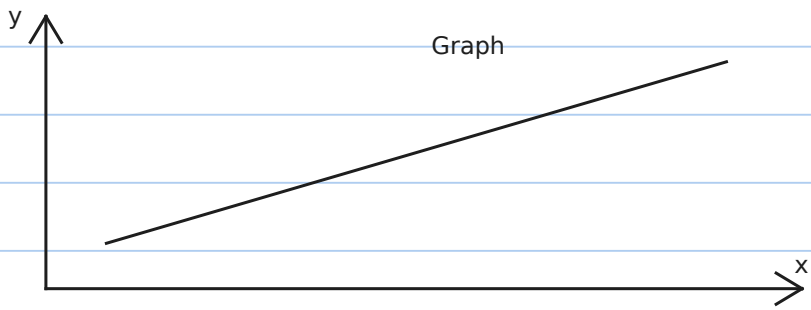
Answer: Step 1: The graph of y = -sin(x) is the graph of y = sin(x) reflected in the x-axis, so every value changes sign.

Step 2: Table of key values: at x = 0, y = 0; at x = pi/2, y = -1; at x = pi, y = 0; at x = 3*pi/2, y = 1; at x = 2*pi, y = 0 (finer values: at pi/6, -0.5; at pi/3, -0.87; at 5*pi/6, -0.5; at 7*pi/6, 0.5; at 4*pi/3, 0.87; at 5*pi/3, 0.87; at 11*pi/6, 0.5).

Step 3: Plot these points and join them with a smooth wave that first dips below the axis and then rises above it.

Step 4: Period = 2*pi, amplitude = 1, maximum value = 1 at x = 3*pi/2, minimum value = -1 at x = pi/2.

Final answer: The graph is the sine wave reflected in the x-axis over [0, 2*pi], with period 2*pi, amplitude 1, maximum 1 and minimum -1.



Question: Exercise 12.2 Q2: Draw the graph of y = cos(x) for 0 <= x <= 2*pi.

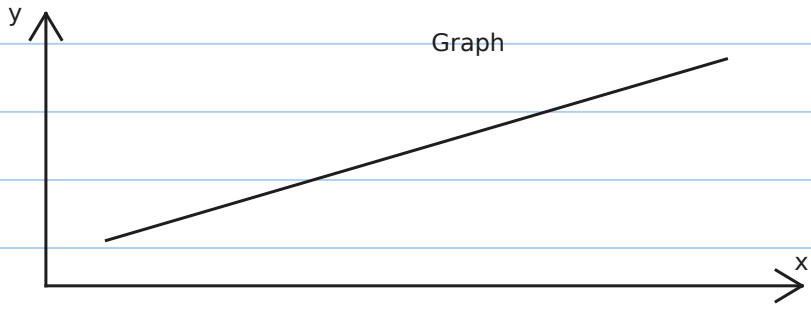
Answer: Step 1: Use the standard cosine values on [0, 2*pi].

Step 2: Table of key values: at x = 0, y = 1; at x = pi/2, y = 0; at x = pi, y = -1; at x = 3*pi/2, y = 0; at x = 2*pi, y = 1 (finer values: at pi/3, 0.5; at 2*pi/3, -0.5; at 4*pi/3, -0.5; at 5*pi/3, 0.5).

Step 3: Plot the points and join them with a smooth wave starting at its highest point, falling to -1 at x = pi and rising back to 1.

Step 4: Period = 2*pi, amplitude = 1, maximum value = 1 at x = 0 and x = 2*pi, minimum value = -1 at x = pi.

Final answer: The graph is one complete cosine wave over [0, 2*pi], with period 2*pi, amplitude 1, maximum 1 and minimum -1.



Question: Exercise 12.2 Q3: Draw the graph of y = sin(x/2) for 0 <= x <=

2*pi.

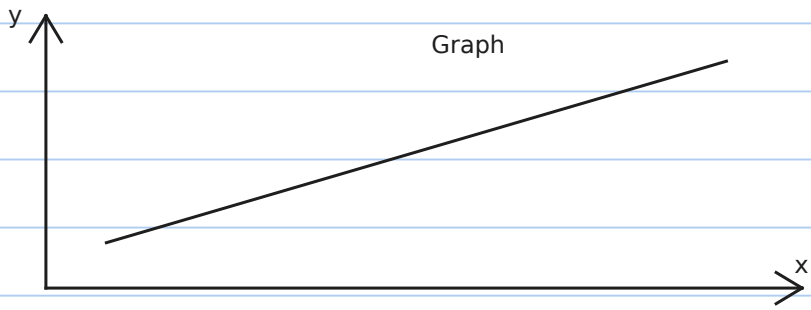
Answer: Step 1: The coefficient 1/2 stretches the sine graph horizontally, so only half of one cycle appears in [0, 2*pi].

Step 2: Table of key values: at x = 0, y = sin(0) = 0; at x = pi/2, y = sin(pi/4) = 0.71; at x = pi, y = sin(pi/2) = 1; at x = 3*pi/2, y = sin(3*pi/4) = 0.71; at x = 2*pi, y = sin(pi) = 0.

Step 3: Plot the points and join them with a smooth half-wave rising to its peak at x = pi and returning to 0 at x = 2*pi.

Step 4: Period = 2*pi/(1/2) = 4*pi, amplitude = 1, maximum value = 1 at x = pi; on this interval the minimum value is 0 (the full graph reaches -1 at x = 3*pi).

Final answer: The graph is the expanded (half-cycle) sine wave over [0, 2*pi], with period 4*pi and amplitude 1.



Question: Exercise 12.2 Q4: Draw the graph of y = tan(2x) for -pi <= x <= 2*pi.

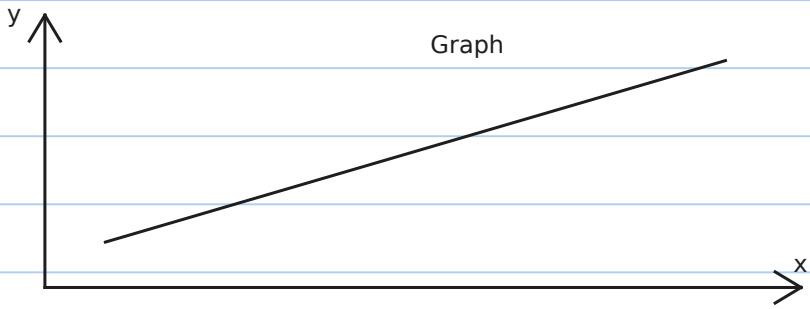
Answer: Step 1: The period of tan(2x) is pi/2, so the usual tangent branch repeats every pi/2 and there are six branches in [-pi, 2*pi].

Step 2: The graph is undefined (vertical asymptotes) where 2x = pi/2 + n*pi, that is at x = pi/4 + n*pi/2: ..., -3*pi/4, -pi/4, pi/4, 3*pi/4, 5*pi/4, 7*pi/4.

Step 3: Key values on each branch: y = 0 at x = -pi, -pi/2, 0, pi/2, pi, 3*pi/2, 2*pi; y = 1 at x = pi/8 + n*pi/2 (for example x = pi/8); y = -1 at x = -pi/8 + n*pi/2 (for example x = -pi/8); each branch rises from -infinity to +infinity.

Step 4: Tangent has no amplitude and no maximum or minimum value; its range is all real numbers.

Final answer: The graph consists of identical rising tangent branches of period pi/2 with asymptotes at x = pi/4 + n*pi/2, and it is unbounded (no maximum or minimum).



Question: Exercise 12.2 Q5: Draw the graph of y = 5sec(x/3) for -2*pi <= x <= 2*pi.

Answer: Step 1: The period of sec(x/3) is 2*pi/(1/3) = 6*pi, so the interval [-2*pi, 2*pi] shows only the middle part of one period; the factor 5 stretches all values vertically.

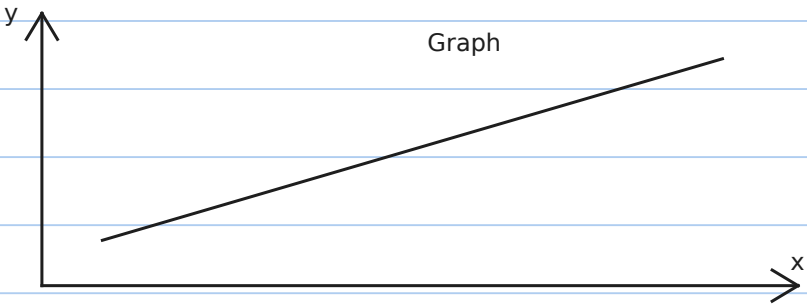
Step 2: The graph is undefined where cos(x/3) = 0, that is x/3 = +/- pi/2, giving vertical asymptotes at x = -3*pi/2 and x = 3*pi/2 inside the interval.

Step 3: Table of key values: at x = -2*pi, y = 5sec(-2*pi/3) = -10; at x = -pi, y = 5sec(-pi/3) = 10; at x = -pi/2, y = 5sec(-pi/6) = 10/sqrt(3), about 5.77; at x = 0, y = 5; at x = pi/2, y = 10/sqrt(3), about 5.77; at x = pi, y = 10; at x = 2*pi, y = 5sec(2*pi/3) = -10.

Step 4: Between the asymptotes the curve stays at y >= 5 with its lowest

point (0, 5); outside them it lies at y <= -5, passing through (-pi, 10) on the upper side of the left part shown and reaching -10 at both endpoints; the range is y <= -5 or y >= 5, and there is no overall maximum or minimum because the curve runs to +/- infinity near the asymptotes.

Final answer: The graph is the stretched secant curve of period 6*pi with asymptotes at x = +/- 3*pi/2, middle branch minimum 5 at x = 0, and values -10 at x = +/- 2*pi.



Question: Exercise 12.2 Q6: Draw the graph of y = -4cot(x) for -pi <= x <= 2*pi.

Answer: Step 1: The period of cot(x) is pi, so there are three branches in [-pi, 2*pi]; the factor -4 reflects the cotangent curve and stretches it vertically.

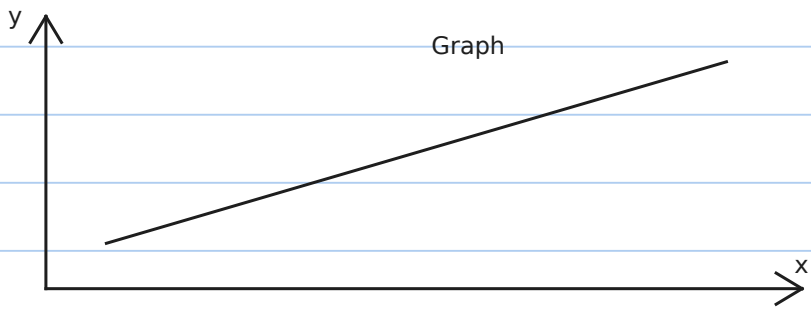
Step 2: The graph is undefined where sin(x) = 0, giving vertical asymptotes at x = -pi, 0, pi, 2*pi.

Step 3: Table of key values: at x = -3*pi/4, y = -4cot(-3*pi/4) = -4; at x = -pi/2, y = 0; at x = -pi/4, y = 4; at x = pi/4, y = -4; at x = pi/2, y = 0; at x = 3*pi/4, y = 4; at x = 5*pi/4, y = -4; at x = 3*pi/2, y = 0; at x = 7*pi/4, y = 4; each branch falls from +infinity to -infinity as x increases.

Step 4: Cotangent has no amplitude and no maximum or minimum value; its range is all real numbers.

Final answer: The graph consists of falling cotangent branches of period pi

with asymptotes at x = -pi, 0, pi, 2*pi, crossing zero at x = -pi/2, pi/2, 3*pi/2.



Question: Exercise 12.2 Q7(i): Guess and draw the graph of y = cos(4*theta) for 0 <= theta <= 2*pi, without using a table of values.

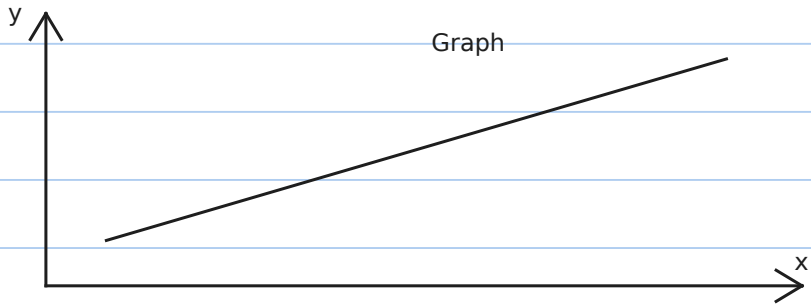
Answer: Step 1: Compare with y = cos(theta): the coefficient 4 compresses the graph horizontally, giving period = 2*pi/4 = pi/2.

Step 2: In [0, 2*pi] there are 2*pi/(pi/2) = 4 complete cycles, and the amplitude is 1.

Step 3: Key points of the first cycle: at theta = 0, y = 1; at theta = pi/8, y = 0; at theta = pi/4, y = -1; at theta = 3*pi/8, y = 0; at theta = pi/2, y = 1; the same cycle then repeats at pi/2, pi, 3*pi/2 and 2*pi.

Step 4: Maximum value = 1 and minimum value = -1.

Final answer: The graph is a cosine wave compressed to period pi/2 with 4 cycles in [0, 2*pi] and amplitude 1.



Question: Exercise 12.2 Q7(ii): Guess and draw the graph of y =

2sin(theta/2) for 0 <= theta <= 2*pi, without using a table of values.

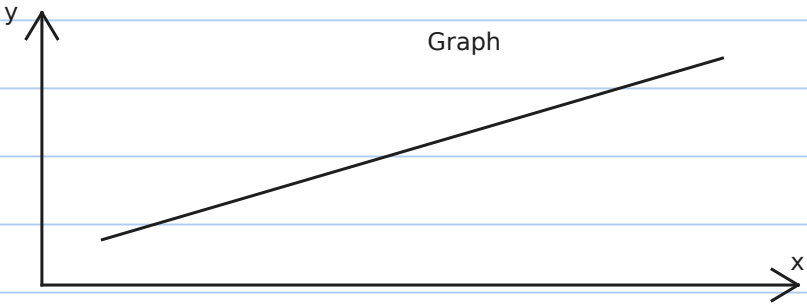
Answer: Step 1: Compare with y = sin(theta): the coefficient 1/2 expands the graph horizontally, giving period = 2*pi/(1/2) = 4*pi, and the factor 2 gives amplitude 2.

Step 2: In [0, 2*pi] only half a cycle appears (1/2 cycle).

Step 3: Key points: at theta = 0, y = 0; at theta = pi/2, y = 2sin(pi/4) = 1.41; at theta = pi, y = 2; at theta = 3*pi/2, y = 2sin(3*pi/4) = 1.41; at theta = 2*pi, y = 0.

Step 4: Maximum value on this interval = 2 at theta = pi, minimum value = 0 at the endpoints.

Final answer: The graph is an expanded half sine wave of period 4*pi and amplitude 2, peaking at (pi, 2).



Question: Exercise 12.2 Q7(iii): Guess and draw the graph of y = 3cos(theta/2) for 0 <= theta <= 2*pi, without using a table of values.

Answer: Step 1: Compare with y = cos(theta): the coefficient 1/2 expands the graph horizontally, giving period = 4*pi, and the factor 3 gives amplitude 3.

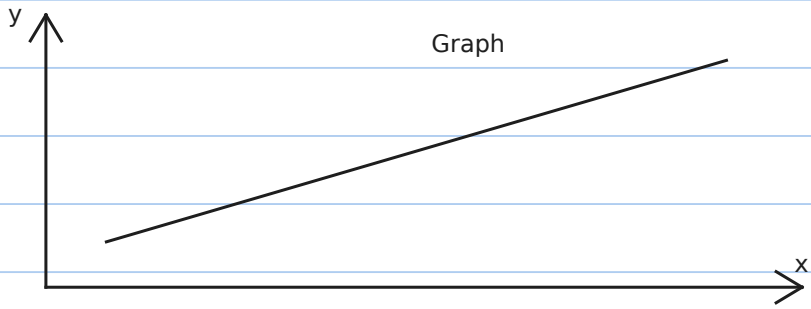
Step 2: In [0, 2*pi] only half a cycle appears (1/2 cycle).

Step 3: Key points: at theta = 0, y = 3; at theta = pi/2, y = 3cos(pi/4) = 2.12; at theta = pi, y = 0; at theta = 3*pi/2, y = 3cos(3*pi/4) = -2.12; at

theta = 2*pi, y = -3.

Step 4: Maximum value = 3 at theta = 0 and minimum value = -3 at theta = 2*pi on this interval.

Final answer: The graph is an expanded half cosine wave of period 4*pi and amplitude 3, falling from 3 to -3 over [0, 2*pi].



Question: Exercise 12.2 Q7(iv): Guess and draw the graph of y = 2sin(3*theta) for 0 <= theta <= 2*pi, without using a table of values.

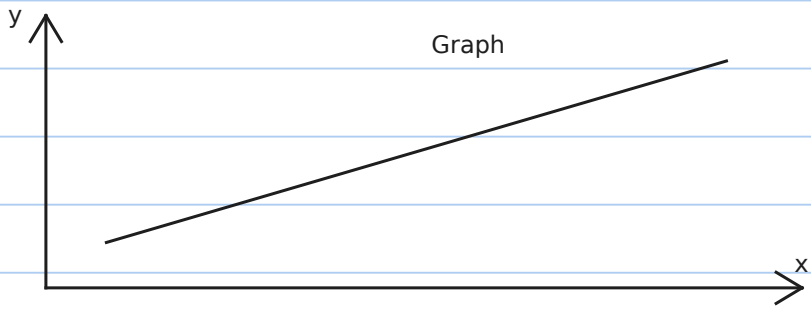
Answer: Step 1: Compare with y = sin(theta): the coefficient 3 compresses the graph horizontally, giving period = 2*pi/3, and the factor 2 gives amplitude 2.

Step 2: In [0, 2*pi] there are 2*pi/(2*pi/3) = 3 complete cycles.

Step 3: Key points of the first cycle: at theta = 0, y = 0; at theta = pi/6, y = 2; at theta = pi/3, y = 0; at theta = pi/2, y = -2; at theta = 2*pi/3, y = 0; the cycle repeats twice more up to 2*pi.

Step 4: Maximum value = 2 and minimum value = -2.

Final answer: The graph is a sine wave compressed to period 2*pi/3 with 3 cycles in [0, 2*pi] and amplitude 2.



Question: Exercise 12.2 Q7(v): Guess and draw the graph of y = cos(theta/3) for 0 <= theta <= 2*pi, without using a table of values.

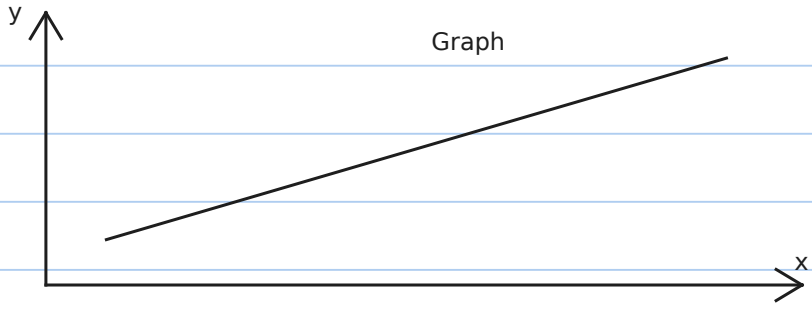
Answer: Step 1: Compare with y = cos(theta): the coefficient 1/3 expands the graph horizontally, giving period = 2*pi/(1/3) = 6*pi and amplitude 1.

Step 2: In [0, 2*pi] only 1/3 of a cycle appears.

Step 3: Key points: at theta = 0, y = 1; at theta = pi/2, y = cos(pi/6) = 0.87; at theta = pi, y = cos(pi/3) = 0.5; at theta = 3*pi/2, y = cos(pi/2) = 0; at theta = 2*pi, y = cos(2*pi/3) = -0.5.

Step 4: Maximum value = 1 at theta = 0; the curve is still falling at theta = 2*pi and would reach -1 only at theta = 3*pi.

Final answer: The graph is a greatly expanded cosine curve of period 6*pi, showing the first third of a cycle over [0, 2*pi].



Question: Exercise 12.2 Q7(vi): Guess and draw the graph of y = sin(2*theta) for 0 <= theta <= 2*pi, without using a table of values.

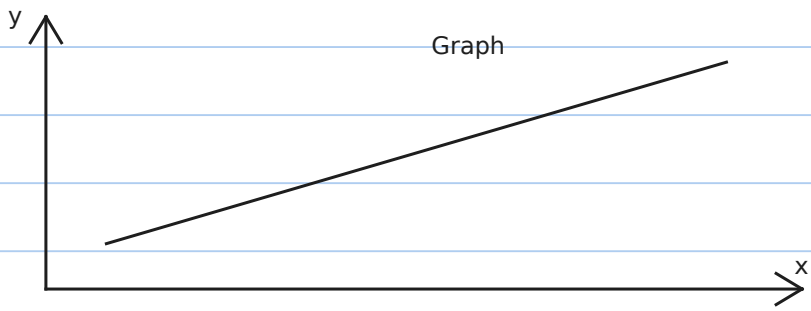
Answer: Step 1: Compare with y = sin(theta): the coefficient 2 compresses the graph horizontally, giving period = 2*pi/2 = pi and amplitude 1.

Step 2: In [0, 2*pi] there are 2 complete cycles.

Step 3: Key points of the first cycle: at theta = 0, y = 0; at theta = pi/4, y = 1; at theta = pi/2, y = 0; at theta = 3*pi/4, y = -1; at theta = pi, y = 0; the second cycle repeats the same shape from pi to 2*pi.

Step 4: Maximum value = 1 and minimum value = -1.

Final answer: The graph is a sine wave compressed to period pi with 2 cycles in [0, 2*pi] and amplitude 1.



Question: Exercise 12.2 Q8(i): Using the properties of the graphs of sine, cosine and tangent, show that cos(pi/2 + theta) = -sin(theta).

Answer: Step 1: By the translation property, the graph of y = cos(theta + pi/2) is the cosine graph shifted pi/2 units to the left.

Step 2: The shifted cosine curve coincides exactly with the graph of y = -sin(theta) (it starts at 0, falls to -1 at theta = pi/2 and returns to 0 at theta = pi), so the two functions are equal.

Step 3: Formula check: cos(pi/2 + theta) = cos(pi/2)cos(theta) - sin(pi/2)sin(theta) = (0)cos(theta) - (1)sin(theta) = -sin(theta).

Final answer: cos(pi/2 + theta) = -sin(theta), as required.

Question: Exercise 12.2 Q8(ii): Using the properties of the graphs of sine, cosine and tangent, show the identity printed as sin(3*pi/2 + theta) = cos(theta).

Answer: Step 1: By the translation property, the graph of y = sin(theta + 3*pi/2) is the sine graph shifted 3*pi/2 units to the left.

Step 2: The shifted sine curve coincides with the graph of y = -cos(theta), not with +cos(theta): at theta = 0 it gives sin(3*pi/2) = -1, while cos(0) = 1.

Step 3: Formula check: sin(3*pi/2 + theta) = sin(3*pi/2)cos(theta) + cos(3*pi/2)sin(theta) = (-1)cos(theta) + (0)sin(theta) = -cos(theta).

Final answer: The correct identity is sin(3*pi/2 + theta) = -cos(theta); the right-hand side printed in the book misses the minus sign.

Question: Exercise 12.2 Q8(iii): Using the properties of the graphs of sine, cosine and tangent, show that cos(pi - theta) = -cos(theta).

Answer: Step 1: By the translation property, cos(pi - theta) is obtained from the cosine graph by the shift associated with pi, which turns the cosine curve into its reflection in the x-axis.

Step 2: The reflected curve is exactly the graph of y = -cos(theta), so the two functions agree at every point (for example at theta = 0: cos(pi) = -1 = -cos(0)).

Step 3: Formula check: cos(pi - theta) = cos(pi)cos(theta) + sin(pi)sin(theta) = (-1)cos(theta) + (0)sin(theta) = -cos(theta).

Final answer: cos(pi - theta) = -cos(theta), as required.

Question: Exercise 12.2 Q8(iv): Using the properties of the graphs of sine, cosine and tangent, show that tan(pi + theta) = tan(theta).

Answer: Step 1: By the periodic property, tangent has period pi, so shifting its graph by pi brings every branch onto itself.

Step 2: Hence the graph of y = tan(theta + pi) is identical with the graph

of y = tan(theta).

Step 3: Formula check: tan(pi + theta) = (tan(pi) + tan(theta))/(1 - tan(pi)tan(theta)) = (0 + tan(theta))/(1 - 0) = tan(theta).

Final answer: tan(pi + theta) = tan(theta), as required.

Question: Exercise 12.2 Q8(v): Using the properties of the graphs of sine, cosine and tangent, show that sin(3*pi/2 - theta) = -cos(theta).

Answer: Step 1: By the translation property, the graph of y = sin(3*pi/2 - theta) is obtained from the sine graph by the 3*pi/2 shift together with the reversal theta to -theta.

Step 2: The resulting curve coincides with the graph of y = -cos(theta) (for example at theta = 0 it gives sin(3*pi/2) = -1 = -cos(0), and at theta = pi/2 it gives sin(pi) = 0 = -cos(pi/2)).

Step 3: Formula check: sin(3*pi/2 - theta) = sin(3*pi/2)cos(theta) - cos(3*pi/2)sin(theta) = (-1)cos(theta) - (0)sin(theta) = -cos(theta).

Final answer: sin(3*pi/2 - theta) = -cos(theta), as required.

Question: Exercise 12.2 Q8(vi): Using the properties of the graphs of sine, cosine and tangent, show that cos(2*pi + theta) = cos(theta).

Answer: Step 1: By the periodic property, cosine has period 2*pi, so shifting its graph by 2*pi brings the curve onto itself.

Step 2: Hence the graph of y = cos(theta + 2*pi) is identical with the graph of y = cos(theta).

Step 3: Formula check: cos(2*pi + theta) = cos(2*pi)cos(theta) - sin(2*pi)sin(theta) = (1)cos(theta) - (0)sin(theta) = cos(theta).

Final answer: cos(2*pi + theta) = cos(theta), as required.

Question: Exercise 12.2 Q9(i): For any integer k, deduce that cos(theta + 2k*pi) = cos(theta).

Answer: Step 1: For any integer k, cos(2k*pi) = 1 and sin(2k*pi) = 0, because 2k*pi is a whole number of complete revolutions.

Step 2: Expand with the addition formula: cos(theta + 2k*pi) = cos(theta)cos(2k*pi) - sin(theta)sin(2k*pi).

Step 3: Substitute the values from Step 1: cos(theta + 2k*pi) = cos(theta)(1) - sin(theta)(0) = cos(theta).

Final answer: cos(theta + 2k*pi) = cos(theta) for every integer k, which is the periodic property of cosine.

Question: Exercise 12.2 Q9(ii): For any integer k, deduce that cosec(theta + 2k*pi) = cosec(theta).

Answer: Step 1: First, sin(theta + 2k*pi) = sin(theta)cos(2k*pi) + cos(theta)sin(2k*pi) = sin(theta)(1) + cos(theta)(0) = sin(theta), using cos(2k*pi) = 1 and sin(2k*pi) = 0.

Step 2: Cosecant is the reciprocal of sine: cosec(theta + 2k*pi) = 1/sin(theta + 2k*pi).

Step 3: By Step 1 this equals 1/sin(theta) = cosec(theta), wherever sin(theta) is not 0.

Final answer: cosec(theta + 2k*pi) = cosec(theta) for every integer k.

Question: Exercise 12.2 Q9(iii): For any integer k, deduce that tan(theta + 2k*pi) = tan(theta).

Answer: Step 1: For any integer k, tan(2k*pi) = 0, since sin(2k*pi) = 0 and cos(2k*pi) = 1.

Step 2: Expand with the addition formula: tan(theta + 2k*pi) = (tan(theta)

+ tan(2k*pi))/(1 - tan(theta)tan(2k*pi)).

Step 3: Substitute tan(2k*pi) = 0: tan(theta + 2k*pi) = (tan(theta) + 0)/(1 - 0) = tan(theta).

Final answer: tan(theta + 2k*pi) = tan(theta) for every integer k (in fact tangent already repeats after pi).

Question: Exercise 12.2 Q9(iv): For any integer k, deduce that cot(theta + 2k*pi) = cot(theta).

Answer: Step 1: From the periodic properties proved above, sin(theta + 2k*pi) = sin(theta) and cos(theta + 2k*pi) = cos(theta) for every integer k.

Step 2: Cotangent is the ratio cot(u) = cos(u)/sin(u).

Step 3: Therefore cot(theta + 2k*pi) = cos(theta + 2k*pi)/sin(theta + 2k*pi) = cos(theta)/sin(theta) = cot(theta), wherever sin(theta) is not 0.

Final answer: cot(theta + 2k*pi) = cot(theta) for every integer k.

Exercise 12.3

Question: Exercise 12.3 Q1: Find the solutions of sin(x) - 1/2 = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: sin(x) = 1/2.

Step 2: The reference (principal) value is x = pi/6, because sin(pi/6) = 1/2.

Step 3: Sine is also positive in the second quadrant, giving x = pi - pi/6 = 5*pi/6, and sine has period 2*pi, so the general solution is x = pi/6 + 2n*pi or x = 5*pi/6 + 2n*pi, n in Z.

Step 4: Keep only the values inside -pi/2 <= x <= pi/2: 5*pi/6 is too large

and 5*pi/6 - 2*pi = -7*pi/6 is too small, so only x = pi/6 remains; check: sin(pi/6) - 1/2 = 0.

Final answer: In [-pi/2, pi/2] the solution is x = pi/6 (general solution: x = pi/6 + 2n*pi or x = 5*pi/6 + 2n*pi).

Question: Exercise 12.3 Q2: Find the solutions of cos(x) - 1/2 = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: cos(x) = 1/2.

Step 2: The reference value is x = pi/3, because cos(pi/3) = 1/2.

Step 3: Cosine is an even function and has period 2*pi, so the general solution is x = pi/3 + 2n*pi or x = 5*pi/3 + 2n*pi, n in Z (equivalently x = +/- pi/3 + 2n*pi).

Step 4: Inside -pi/2 <= x <= pi/2 we get x = pi/3 and, from 5*pi/3 - 2*pi, x = -pi/3; check: cos(-pi/3) = cos(pi/3) = 1/2.

Final answer: In [-pi/2, pi/2] the solutions are x = pi/3 and x = -pi/3 (general solution: x = pi/3 + 2n*pi or x = 5*pi/3 + 2n*pi).

Question: Exercise 12.3 Q3: Find the solutions of cos(x) + 1/sqrt(2) = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: cos(x) = -1/sqrt(2).

Step 2: The reference angle is pi/4, because cos(pi/4) = 1/sqrt(2); cosine is negative in the second and third quadrants.

Step 3: The solutions in [0, 2*pi] are x = pi - pi/4 = 3*pi/4 and x = pi + pi/4 = 5*pi/4, and cosine has period 2*pi, so the general solution is x = 3*pi/4 + 2n*pi or x = 5*pi/4 + 2n*pi, n in Z.

Step 4: On the interval -pi/2 <= x <= pi/2, cos(x) >= 0 always, so cos(x) = -1/sqrt(2) (a negative number) can never occur there; check the

endpoints: cos(pi/2) = 0, still greater than -1/sqrt(2).

Final answer: There is no solution inside [-pi/2, pi/2]; the general solution is x = 3*pi/4 + 2n*pi or x = 5*pi/4 + 2n*pi.

Question: Exercise 12.3 Q4: Find the solutions of sin(x) + 1/sqrt(2) = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: sin(x) = -1/sqrt(2).

Step 2: The reference angle is pi/4, because sin(pi/4) = 1/sqrt(2); sine is negative in the third and fourth quadrants.

Step 3: The solutions in [0, 2*pi] are x = pi + pi/4 = 5*pi/4 and x = 2*pi - pi/4 = 7*pi/4, and sine has period 2*pi, so the general solution is x = 5*pi/4 + 2n*pi or x = 7*pi/4 + 2n*pi, n in Z.

Step 4: Inside -pi/2 <= x <= pi/2 only 7*pi/4 - 2*pi = -pi/4 qualifies; check: sin(-pi/4) = -1/sqrt(2).

Final answer: In [-pi/2, pi/2] the solution is x = -pi/4 (general solution: x = 5*pi/4 + 2n*pi or x = 7*pi/4 + 2n*pi).

Question: Exercise 12.3 Q5: Find the solutions of tan(x) - 1/sqrt(3) = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: tan(x) = 1/sqrt(3).

Step 2: The reference value is x = pi/6, because tan(pi/6) = 1/sqrt(3).

Step 3: Tangent has period pi, so the general solution is x = pi/6 + n*pi, n in Z (also written as x = pi/6 + 2n*pi or x = 7*pi/6 + 2n*pi).

Step 4: Inside -pi/2 <= x <= pi/2 only x = pi/6 qualifies (pi/6 - pi = -5*pi/6 is outside); check: tan(pi/6) - 1/sqrt(3) = 0.

Final answer: In [-pi/2, pi/2] the solution is x = pi/6 (general solution: x =

pi/6 + n*pi).

Question: Exercise 12.3 Q6: Find the solutions of tan(x) - sqrt(3) = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: tan(x) = sqrt(3).

Step 2: The reference value is x = pi/3, because tan(pi/3) = sqrt(3).

Step 3: Tangent has period pi, so the general solution is x = pi/3 + n*pi, n in Z.

Step 4: Inside -pi/2 <= x <= pi/2 only x = pi/3 qualifies (pi/3 - pi = -2*pi/3 is outside); check: tan(pi/3) - sqrt(3) = 0.

Final answer: In [-pi/2, pi/2] the solution is x = pi/3 (general solution: x = pi/3 + n*pi).

Question: Exercise 12.3 Q7: Find the solutions of sin(x) + sqrt(3)/2 = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: sin(x) = -sqrt(3)/2.

Step 2: The reference angle is pi/3, because sin(pi/3) = sqrt(3)/2; sine is negative in the third and fourth quadrants.

Step 3: The solutions in [0, 2*pi] are x = pi + pi/3 = 4*pi/3 and x = 2*pi - pi/3 = 5*pi/3, and sine has period 2*pi, so the general solution is x = 4*pi/3 + 2n*pi or x = 5*pi/3 + 2n*pi, n in Z.

Step 4: Inside -pi/2 <= x <= pi/2 only 5*pi/3 - 2*pi = -pi/3 qualifies (4*pi/3 - 2*pi = -2*pi/3 is outside); check: sin(-pi/3) = -sqrt(3)/2.

Final answer: In [-pi/2, pi/2] the solution is x = -pi/3 (general solution: x = 4*pi/3 + 2n*pi or x = 5*pi/3 + 2n*pi).

Question: Exercise 12.3 Q8: Find the solutions of cos(x) - 1 = 0 by using

properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: cos(x) = 1.

Step 2: The reference value is x = 0, because cos(0) = 1 and cosine equals 1 only at whole revolutions.

Step 3: Cosine has period 2*pi, so the general solution is x = 2n*pi, n in Z.

Step 4: Inside -pi/2 <= x <= pi/2 only x = 0 qualifies; check: cos(0) - 1 = 0.

Final answer: In [-pi/2, pi/2] the solution is x = 0 (general solution: x = 2n*pi).

Question: Exercise 12.3 Q9: Find the solutions of tan(x) + 1/sqrt(3) = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: tan(x) = -1/sqrt(3).

Step 2: The reference angle is pi/6, because tan(pi/6) = 1/sqrt(3); tangent is negative in the second and fourth quadrants.

Step 3: Tangent has period pi, so the general solution is x = -pi/6 + n*pi, n in Z (the same set as x = 5*pi/6 + n*pi).

Step 4: Inside -pi/2 <= x <= pi/2 only x = -pi/6 qualifies; check: tan(-pi/6) = -1/sqrt(3).

Final answer: In [-pi/2, pi/2] the solution is x = -pi/6 (general solution: x = -pi/6 + n*pi).

Question: Exercise 12.3 Q10: Find the solutions of tan(x) - 1 = 0 by using properties, when -pi/2 <= x <= pi/2.

Answer: Step 1: Isolate the trigonometric function: tan(x) = 1.

Step 2: The reference value is x = pi/4, because tan(pi/4) = 1.

Step 3: Tangent has period pi, so the general solution is x = pi/4 + n*pi, n in Z.

Step 4: Inside -pi/2 <= x <= pi/2 only x = pi/4 qualifies (pi/4 - pi = -3*pi/4 is outside); check: tan(pi/4) - 1 = 0.

Final answer: In [-pi/2, pi/2] the solution is x = pi/4 (general solution: x = pi/4 + n*pi).

Question: Exercise 12.3 Q11: By using a graph, find the solutions of sin(x) - 2x/pi = 0 (for -pi/2 <= x <= pi/2).

Answer: Step 1: Split the equation into two graphs on the same axes: y = sin(x) and the straight line y = 2x/pi; the solutions are the x-coordinates of their intersection points.

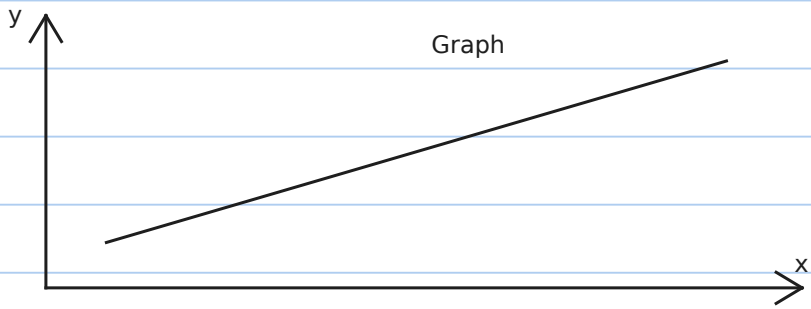
Step 2: Table for y = sin(x): at x = -pi/2, -pi/4, 0, pi/4, pi/2 the values are -1, -0.71, 0, 0.71, 1.

Step 3: Table for y = 2x/pi: at the same x values the line gives -1, -0.5, 0, 0.5, 1.

Step 4: The two graphs meet at the two endpoints and at the origin: at x = -pi/2 both give -1, at x = 0 both give 0, and at x = pi/2 both give 1; between these points the sine curve stays strictly on one side of the line, so there are no other crossings.

Step 5: Verify by substitution: sin(pi/2) - 2(pi/2)/pi = 1 - 1 = 0 and sin(-pi/2) - 2(-pi/2)/pi = -1 + 1 = 0.

Final answer: The solutions in [-pi/2, pi/2] are x = -pi/2, x = 0 and x = pi/2.



Question: Exercise 12.3 Q12: By using a graph, find the solutions of cos(x) + x/pi = 0 (for -pi/2 <= x <= pi/2).

Answer: Step 1: Rewrite the equation as cos(x) = -x/pi and draw y = cos(x) and the straight line y = -x/pi on the same axes; the solution is the x-coordinate of the intersection point.

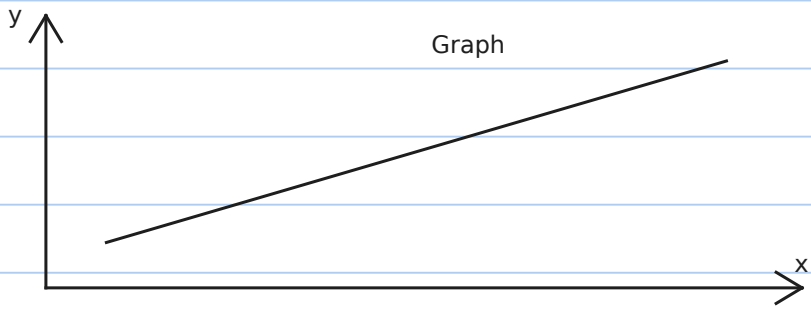
Step 2: Table for y = cos(x): at x = -pi/2, -pi/4, 0, pi/4, pi/2 the values are 0, 0.71, 1, 0.71, 0.

Step 3: Table for y = -x/pi: at the same x values the line gives 0.5, 0.25, 0, -0.25, -0.5.

Step 4: For x >= 0 the line is <= 0 while cos(x) >= 0, and they could only meet if both were 0, which does not happen; for x < 0 the falling line crosses the cosine curve exactly once, between -pi/2 and 0. Reading the crossing from the graph (and refining it) gives x = -1.18 approximately.

Step 5: Verify: cos(-1.18) = 0.38 and -(-1.18)/pi = 0.38, so cos(x) + x/pi = 0 up to the accuracy of the graph (the accurate value is x = -1.184).

Final answer: The solution in [-pi/2, pi/2] is x = -1.18 approximately. (Outside this interval, x = pi is an exact solution, since cos(pi) + pi/pi = -1 + 1 = 0.)



Question: Exercise 12.3 Q13: By using a graph, find the solutions of tan(x) - 4x/pi = 0 (for -pi/2 <= x <= pi/2).

Answer: Step 1: Split the equation into y = tan(x) and the straight line y = 4x/pi on the same axes; the solutions are the x-coordinates of the intersection points.

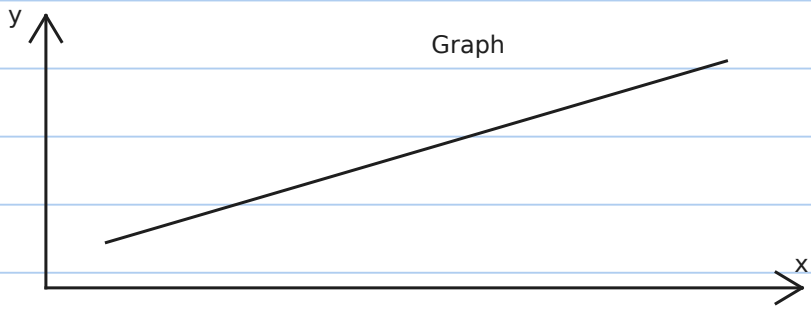
Step 2: Table for y = tan(x): at x = -pi/4, 0, pi/4 the values are -1, 0, 1, and tan(x) runs to -infinity at -pi/2 and +infinity at pi/2.

Step 3: Table for y = 4x/pi: at x = -pi/2, -pi/4, 0, pi/4, pi/2 the line gives -2, -1, 0, 1, 2.

Step 4: The graphs meet where the tables agree: at x = -pi/4 (both -1), at x = 0 (both 0) and at x = pi/4 (both 1); just right of 0 the line lies above the tangent curve and just left of pi/4 the tangent curve rises above the line, crossing only at pi/4 itself, and symmetrically on the negative side.

Step 5: Verify by substitution: tan(pi/4) - 4(pi/4)/pi = 1 - 1 = 0 and tan(-pi/4) - 4(-pi/4)/pi = -1 + 1 = 0.

Final answer: The solutions in [-pi/2, pi/2] are x = -pi/4, x = 0 and x = pi/4.



Question: Exercise 12.3 Q14: By using a graph, find the solutions of sin(x) - 3x/pi = 0 (for -pi/2 <= x <= pi/2).

Answer: Step 1: Split the equation into y = sin(x) and the straight line y = 3x/pi on the same axes; the solutions are the x-coordinates of the intersection points.

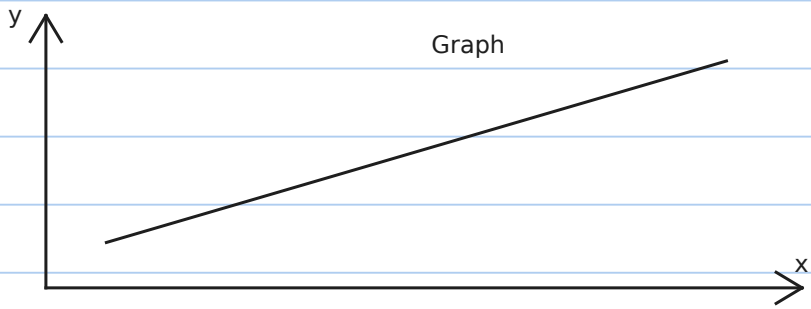
Step 2: Table for y = sin(x): at x = -pi/2, -pi/4, 0, pi/4, pi/2 the values are -1, -0.71, 0, 0.71, 1.

Step 3: Table for y = 3x/pi: at the same x values the line gives -1.5, -0.75, 0, 0.75, 1.5.

Step 4: At the origin the sine curve has slope 1, which is slightly steeper than the line's slope 3/pi (about 0.95), so just right of 0 the sine curve lies above the line; but at x = pi/4 the line (0.75) has already overtaken the sine curve (0.71), so the graphs cross once more at a positive x. Reading the graph, this crossing is exactly at x = pi/6, and by the symmetry of both graphs there is also a crossing at x = -pi/6, besides the obvious crossing at the origin.

Step 5: Verify by substitution: sin(pi/6) - 3(pi/6)/pi = 1/2 - 1/2 = 0 and sin(-pi/6) - 3(-pi/6)/pi = -1/2 + 1/2 = 0.

Final answer: The solutions in [-pi/2, pi/2] are x = -pi/6, x = 0 and x = pi/6.



Question: Exercise 12.3 Q15: By using a graph, find the solutions of cos(x) - 3x/(2*pi) = 0 (for -pi/2 <= x <= pi/2).

Answer: Step 1: Split the equation into y = cos(x) and the straight line y = 3x/(2*pi) on the same axes; the solution is the x-coordinate of the intersection point.

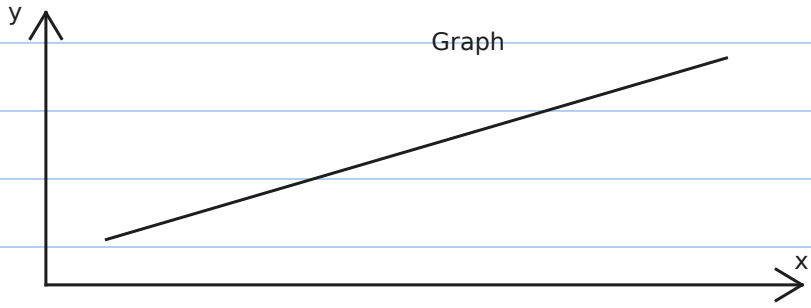
Step 2: Table for y = cos(x): at x = -pi/2, -pi/4, 0, pi/4, pi/2 the values are 0, 0.71, 1, 0.71, 0.

Step 3: Table for y = 3x/(2*pi): at the same x values the line gives -0.75, -0.375, 0, 0.375, 0.75.

Step 4: For x < 0 the line is negative while cos(x) is positive, so there is no crossing; for x > 0 the cosine curve falls while the line rises, so they cross exactly once, and the graph shows the crossing at x = pi/3 (where cos(x) = 0.5 and the line also gives 3(pi/3)/(2*pi) = 0.5).

Step 5: Verify by substitution: cos(pi/3) - 3(pi/3)/(2*pi) = 1/2 - 1/2 = 0.

Final answer: The solution in [-pi/2, pi/2] is x = pi/3.



Question: Exercise 12.3 Q16: By using a graph, find the solutions of tan(x) + x = 0 (for -pi/2 <= x <= pi/2).

Answer: Step 1: Rewrite the equation as tan(x) = -x and draw y = tan(x) and the straight line y = -x on the same axes; the solutions are the x-coordinates of the intersection points.

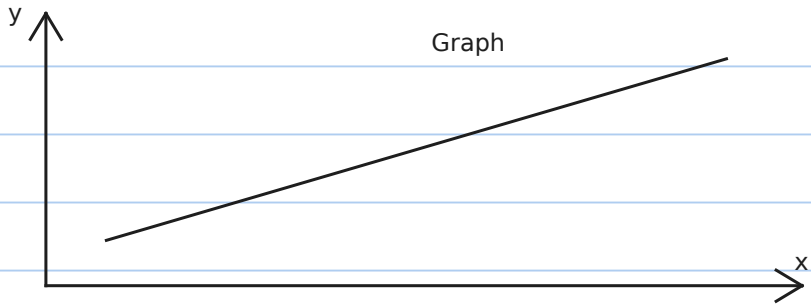
Step 2: Table for y = tan(x): at x = -pi/4, 0, pi/4 the values are -1, 0, 1.

Step 3: Table for y = -x: at x = -pi/4, 0, pi/4 the line gives 0.79, 0, -0.79.

Step 4: For x > 0, tan(x) > 0 > -x, so the curve stays above the line; for x < 0, tan(x) < 0 < -x, so the curve stays below the line; hence the only meeting point is where both pass through the origin.

Step 5: Verify by substitution: tan(0) + 0 = 0.

Final answer: The only solution in [-pi/2, pi/2] is x = 0.



Exercise 12.4

Question: Exercise 12.4 Q1(i): Without using a calculator, find the value of sin^-1(1).

Answer: Step 1: Let y = sin^-1(1); then sin(y) = 1, and y must lie in the principal range of sin^-1, -pi/2 <= y <= pi/2.

Step 2: The angle in this range whose sine is 1 is y = pi/2.

Step 3: Check: sin(pi/2) = 1 and pi/2 is in the principal range.

Final answer: sin^-1(1) = pi/2.

Question: Exercise 12.4 Q1(ii): Without using a calculator, find the value of sin^-1(-1).

Answer: Step 1: Let y = sin^-1(-1); then sin(y) = -1, and y must lie in the principal range -pi/2 <= y <= pi/2.

Step 2: The angle in this range whose sine is -1 is y = -pi/2.

Step 3: Check: sin(-pi/2) = -1 and -pi/2 is in the principal range.

Final answer: sin^-1(-1) = -pi/2.

Question: Exercise 12.4 Q1(iii): Without using a calculator, find the value of cos^-1(sqrt(3)/2).

Answer: Step 1: Let y = cos^-1(sqrt(3)/2); then cos(y) = sqrt(3)/2, and y must lie in the principal range of cos^-1, 0 <= y <= pi.

Step 2: The angle in this range whose cosine is sqrt(3)/2 is y = pi/6.

Step 3: Check: cos(pi/6) = sqrt(3)/2 and pi/6 is in [0, pi].

Final answer: cos^-1(sqrt(3)/2) = pi/6.

Question: Exercise 12.4 Q1(iv): Without using a calculator, find the value of cot^-1(-1).

Answer: Step 1: Let y = cot^-1(-1); then cot(y) = -1, and y must lie in the principal range of cot^-1, 0 < y < pi (as in the domain-range table of this unit).

Step 2: cot(y) = -1 means tan(y) = -1; the angle in (0, pi) with tangent -1 is y = 3*pi/4, since tan(3*pi/4) = -1.

Step 3: Check: cot(3*pi/4) = cos(3*pi/4)/sin(3*pi/4) =

(-1/sqrt(2))/(1/sqrt(2)) = -1.

Final answer: cot^-1(-1) = 3*pi/4. (Note: the worked example in the book's text uses the other branch -pi/2 < y < pi/2 and gets -pi/4; with the principal range 0 < y < pi used in the unit's own table, the value is 3*pi/4.)

Question: Exercise 12.4 Q1(v): Without using a calculator, find the value of cos^-1(1/2).

Answer: Step 1: Let y = cos^-1(1/2); then cos(y) = 1/2, and y must lie in the principal range 0 <= y <= pi.

Step 2: The angle in this range whose cosine is 1/2 is y = pi/3.

Step 3: Check: cos(pi/3) = 1/2 and pi/3 is in [0, pi].

Final answer: cos^-1(1/2) = pi/3.

Question: Exercise 12.4 Q1(vi): Without using a calculator, find the value of tan^-1(-1/sqrt(3)).

Answer: Step 1: Let y = tan^-1(-1/sqrt(3)); then tan(y) = -1/sqrt(3), and y must lie in the principal range of tan^-1, -pi/2 < y < pi/2.

Step 2: Since tan(pi/6) = 1/sqrt(3) and tangent is odd, the angle is y = -pi/6.

Step 3: Check: tan(-pi/6) = -1/sqrt(3) and -pi/6 is in the principal range.

Final answer: tan^-1(-1/sqrt(3)) = -pi/6.

Question: Exercise 12.4 Q1(vii): Without using a calculator, find the value of tan^-1(1/sqrt(3)).

Answer: Step 1: Let y = tan^-1(1/sqrt(3)); then tan(y) = 1/sqrt(3), and y must lie in the principal range -pi/2 < y < pi/2.

Step 2: The angle in this range whose tangent is 1/sqrt(3) is y = pi/6.

Step 3: Check: tan(pi/6) = 1/sqrt(3).

Final answer: tan^-1(1/sqrt(3)) = pi/6.

Question: Exercise 12.4 Q1(viii): Without using a calculator, find the value of cosec^-1(-2/sqrt(3)).

Answer: Step 1: Let y = cosec^-1(-2/sqrt(3)); then cosec(y) = -2/sqrt(3), and y must lie in the principal range of cosec^-1, -pi/2 <= y <= pi/2 with y not 0.

Step 2: Take reciprocals: sin(y) = -sqrt(3)/2.

Step 3: The angle in the principal range whose sine is -sqrt(3)/2 is y = -pi/3.

Step 4: Check: cosec(-pi/3) = 1/sin(-pi/3) = -2/sqrt(3).

Final answer: cosec^-1(-2/sqrt(3)) = -pi/3.

Question: Exercise 12.4 Q1(ix): Without using a calculator, find the value of sin^-1(-1/sqrt(2)).

Answer: Step 1: Let y = sin^-1(-1/sqrt(2)); then sin(y) = -1/sqrt(2), and y must lie in the principal range -pi/2 <= y <= pi/2.

Step 2: Since sin(pi/4) = 1/sqrt(2) and sine is odd, the angle is y = -pi/4.

Step 3: Check: sin(-pi/4) = -1/sqrt(2) and -pi/4 is in the principal range.

Final answer: sin^-1(-1/sqrt(2)) = -pi/4.

Question: Exercise 12.4 Q1(x): Without using a calculator, find the value of cot^-1(1/sqrt(3)).

Answer: Step 1: Let y = cot^-1(1/sqrt(3)); then cot(y) = 1/sqrt(3), and y must lie in the principal range 0 < y < pi.

Step 2: cot(y) = 1/sqrt(3) means tan(y) = sqrt(3).

Step 3: The angle in (0, pi) whose tangent is sqrt(3) is y = pi/3.

Step 4: Check: cot(pi/3) = 1/tan(pi/3) = 1/sqrt(3).

Final answer: cot^-1(1/sqrt(3)) = pi/3.

Question: Exercise 12.4 Q1(xi): Without using a calculator, find the value of cot^-1(-1).

Answer: Step 1: This is the same value asked in Q1(iv) (the book repeats it): let y = cot^-1(-1), so cot(y) = -1 with y in the principal range 0 < y < pi.

Step 2: cot(y) = -1 means tan(y) = -1, and the angle in (0, pi) with tangent -1 is y = 3*pi/4.

Step 3: Check: cot(3*pi/4) = -1 and 3*pi/4 lies in (0, pi).

Final answer: cot^-1(-1) = 3*pi/4.

Question: Exercise 12.4 Q1(xii): Without using a calculator, find the value of sec^-1(2).

Answer: Step 1: Let y = sec^-1(2); then sec(y) = 2, and y must lie in the principal range of sec^-1, 0 <= y <= pi with y not pi/2.

Step 2: Take reciprocals: cos(y) = 1/2.

Step 3: The angle in the principal range whose cosine is 1/2 is y = pi/3.

Step 4: Check: sec(pi/3) = 1/cos(pi/3) = 2.

Final answer: sec^-1(2) = pi/3.

Question: Exercise 12.4 Q1(xiii): Without using a calculator, find the value of sec^-1(1).

Answer: Step 1: Let y = sec^-1(1); then sec(y) = 1, and y must lie in the principal range 0 <= y <= pi with y not pi/2.

Step 2: Take reciprocals: cos(y) = 1.

Step 3: The angle in the principal range whose cosine is 1 is y = 0.

Step 4: Check: sec(0) = 1/cos(0) = 1.

Final answer: sec^-1(1) = 0.

Question: Exercise 12.4 Q1(xiv): Without using a calculator, find the value of cosec^-1(2/sqrt(3)).

Answer: Step 1: Let y = cosec^-1(2/sqrt(3)); then cosec(y) = 2/sqrt(3), and y must lie in the principal range -pi/2 <= y <= pi/2 with y not 0.

Step 2: Take reciprocals: sin(y) = sqrt(3)/2.

Step 3: The angle in the principal range whose sine is sqrt(3)/2 is y = pi/3.

Step 4: Check: cosec(pi/3) = 1/sin(pi/3) = 2/sqrt(3).

Final answer: cosec^-1(2/sqrt(3)) = pi/3.

Question: Exercise 12.4 Q1(xv): Without using a calculator, find the value of cosec^-1(1).

Answer: Step 1: Let y = cosec^-1(1); then cosec(y) = 1, and y must lie in the principal range -pi/2 <= y <= pi/2 with y not 0.

Step 2: Take reciprocals: sin(y) = 1.

Step 3: The angle in the principal range whose sine is 1 is y = pi/2.

Step 4: Check: cosec(pi/2) = 1/sin(pi/2) = 1.

Final answer: cosec^-1(1) = pi/2.

Question: Exercise 12.4 Q2(i): Find the value of cos(sin^-1(sqrt(3)/2)).

Answer: Step 1: First evaluate the inner inverse function: let y = sin^-1(sqrt(3)/2), so sin(y) = sqrt(3)/2 with -pi/2 <= y <= pi/2, giving y =

pi/3.

Step 2: Now the required value is cos(y) = cos(pi/3).

Step 3: cos(pi/3) = 1/2.

Final answer: cos(sin^-1(sqrt(3)/2)) = 1/2.

Question: Exercise 12.4 Q2(ii): Find the value of cosec(cos^-1(1/2)).

Answer: Step 1: First evaluate the inner inverse function: let y = cos^-1(1/2), so cos(y) = 1/2 with 0 <= y <= pi, giving y = pi/3.

Step 2: Now the required value is cosec(y) = cosec(pi/3) = 1/sin(pi/3).

Step 3: sin(pi/3) = sqrt(3)/2, so cosec(pi/3) = 2/sqrt(3).

Final answer: cosec(cos^-1(1/2)) = 2/sqrt(3).

Question: Exercise 12.4 Q2(iii): Find the value of tan(cos^-1(-1)).

Answer: Step 1: First evaluate the inner inverse function: let y = cos^-1(-1), so cos(y) = -1 with 0 <= y <= pi, giving y = pi.

Step 2: Now the required value is tan(y) = tan(pi).

Step 3: tan(pi) = sin(pi)/cos(pi) = 0/(-1) = 0.

Final answer: tan(cos^-1(-1)) = 0.

Question: Exercise 12.4 Q2(iv): Find the value of sec(tan^-1(1)).

Answer: Step 1: First evaluate the inner inverse function: let y = tan^-1(1), so tan(y) = 1 with -pi/2 < y < pi/2, giving y = pi/4.

Step 2: Now the required value is sec(y) = sec(pi/4) = 1/cos(pi/4).

Step 3: cos(pi/4) = 1/sqrt(2), so sec(pi/4) = sqrt(2).

Final answer: sec(tan^-1(1)) = sqrt(2).

Question: Exercise 12.4 Q2(v): Find the value of cot(tan^-1(-sqrt(3))).

Answer: Step 1: First evaluate the inner inverse function: let y = tan^-1(-sqrt(3)), so tan(y) = -sqrt(3) with -pi/2 < y < pi/2, giving y = -pi/3 (because tan(pi/3) = sqrt(3) and tangent is odd).

Step 2: Now the required value is cot(y) = cot(-pi/3) = 1/tan(-pi/3).

Step 3: tan(-pi/3) = -sqrt(3), so cot(-pi/3) = -1/sqrt(3).

Final answer: cot(tan^-1(-sqrt(3))) = -1/sqrt(3).

Question: Exercise 12.4 Q2(vi): Find the value of sin(sec^-1(sqrt(2))).

Answer: Step 1: First evaluate the inner inverse function: let y = sec^-1(sqrt(2)), so sec(y) = sqrt(2) with 0 <= y <= pi, y not pi/2, giving cos(y) = 1/sqrt(2) and hence y = pi/4.

Step 2: Now the required value is sin(y) = sin(pi/4).

Step 3: sin(pi/4) = 1/sqrt(2).

Final answer: sin(sec^-1(sqrt(2))) = 1/sqrt(2).

Question: Exercise 12.4 Q3(i): Show that tan^-1(12/5) = cot^-1(5/12).

Answer: Step 1: Let A = tan^-1(12/5), so tan(A) = 12/5 and A is acute (0 < A < pi/2).

Step 2: Take the reciprocal ratio: cot(A) = 1/tan(A) = 5/12.

Step 3: Apply cot^-1 to both sides: A = cot^-1(5/12), and this is valid because A lies in (0, pi/2), which is inside the principal range (0, pi) of cot^-1.

Step 4: Replace A by its definition.

Final answer: tan^-1(12/5) = cot^-1(5/12), as required.

Question: Exercise 12.4 Q3(ii): Show that tan^-1(sqrt(3)) = sin^-1(sqrt(3)/2).

Answer: Step 1: Evaluate the left side: let A = tan^-1(sqrt(3)); tan(A) = sqrt(3) with -pi/2 < A < pi/2 gives A = pi/3.

Step 2: Evaluate the right side: let B = sin^-1(sqrt(3)/2); sin(B) = sqrt(3)/2 with -pi/2 <= B <= pi/2 gives B = pi/3.

Step 3: Both sides equal the same angle pi/3, so they are equal.

Final answer: tan^-1(sqrt(3)) = sin^-1(sqrt(3)/2) = pi/3, as required.

Question: Exercise 12.4 Q3(iii): Show that cosec^-1(2/sqrt(3)) = cos^-1(1/2).

Answer: Step 1: Evaluate the left side: let A = cosec^-1(2/sqrt(3)); then sin(A) = sqrt(3)/2 with -pi/2 <= A <= pi/2, A not 0, giving A = pi/3.

Step 2: Evaluate the right side: let B = cos^-1(1/2); then cos(B) = 1/2 with 0 <= B <= pi, giving B = pi/3.

Step 3: Both sides equal pi/3, so they are equal.

Final answer: cosec^-1(2/sqrt(3)) = cos^-1(1/2) = pi/3, as required.

Question: Exercise 12.4 Q3(iv): Show that cos^-1(4/9) = cot^-1(4/sqrt(65)).

Answer: Step 1: Let A = cos^-1(4/9), so cos(A) = 4/9 with 0 <= A <= pi; A is acute because cos(A) > 0.

Step 2: Find sin(A) from sin^2(A) + cos^2(A) = 1: sin(A) = sqrt(1 - 16/81) = sqrt(65/81) = sqrt(65)/9, taking the positive root because A is acute.

Step 3: Form cot(A) = cos(A)/sin(A) = (4/9)/(sqrt(65)/9) = 4/sqrt(65).

Step 4: Apply cot^-1: A = cot^-1(4/sqrt(65)), valid because A lies in (0, pi), the principal range of cot^-1.

Final answer: cos^-1(4/9) = cot^-1(4/sqrt(65)), as required.

Question: Exercise 12.4 Q4(i): Show that sin^-1(77/85) + sin^-1(3/5) = cos^-1(87/425) (as printed in the book).

Answer: Step 1: Let A = sin^-1(77/85) and B = sin^-1(3/5). Then sin(A) = 77/85 and sin(B) = 3/5, with A and B acute.

Step 2: Find the cosines: cos(A) = sqrt(1 - (77/85)^2) = sqrt((7225 - 5929)/7225) = sqrt(1296)/85 = 36/85, and cos(B) = sqrt(1 - (3/5)^2) = 4/5.

Step 3: Use cos(A + B) = cos(A)cos(B) - sin(A)sin(B) = (36/85)(4/5) - (77/85)(3/5) = 144/425 - 231/425 = -87/425.

Step 4: Since A + B lies between 0 and pi, A + B = cos^-1(-87/425).

Final answer: sin^-1(77/85) + sin^-1(3/5) = cos^-1(-87/425). The identity as printed in the book drops the minus sign: the correct right-hand side is cos^-1(-87/425), because the sum (about 101.8 deg) is obtuse and its cosine is negative.

Question: Exercise 12.4 Q4(ii): Show that cos^-1(2/sqrt(5)) + tan^-1(1/3) = pi/4.

Answer: Step 1: Convert the first term to a tan^-1: let A = cos^-1(2/sqrt(5)), so cos(A) = 2/sqrt(5) and sin(A) = sqrt(1 - 4/5) = 1/sqrt(5), giving tan(A) = (1/sqrt(5))/(2/sqrt(5)) = 1/2, so A = tan^-1(1/2).

Step 2: The sum becomes tan^-1(1/2) + tan^-1(1/3).

Step 3: Apply tan^-1(P) + tan^-1(Q) = tan^-1((P + Q)/(1 - PQ)): (1/2 + 1/3)/(1 - (1/2)(1/3)) = (5/6)/(5/6) = 1.

Step 4: So the sum is tan^-1(1) = pi/4.

Final answer: cos^-1(2/sqrt(5)) + tan^-1(1/3) = pi/4, as required.

Question: Exercise 12.4 Q4(iii): Show that cos^-1(63/65) + 2tan^-1(1/5) =

cos^-1(4/5).

Answer: Step 1: Double the second term first: 2tan^-1(1/5) = tan^-1((2*(1/5))/(1 - (1/5)^2)) = tan^-1((2/5)/(24/25)) = tan^-1(10/24) = tan^-1(5/12).

Step 2: Convert tan^-1(5/12) to a cos^-1: if tan(C) = 5/12 then, from the 5-12-13 triangle, cos(C) = 12/13, so 2tan^-1(1/5) = cos^-1(12/13).

Step 3: Now apply cos^-1(P) + cos^-1(Q) = cos^-1(PQ - sqrt((1 - P^2)(1 - Q^2))) with P = 63/65 and Q = 12/13: PQ = 756/845, and sqrt((1 - P^2)(1 - Q^2)) = (16/65)(5/13) = 80/845.

Step 4: The sum is cos^-1(756/845 - 80/845) = cos^-1(676/845) = cos^-1(4/5), since 676/845 simplifies to 4/5 (dividing by 169).

Final answer: cos^-1(63/65) + 2tan^-1(1/5) = cos^-1(4/5), as required.

Question: Exercise 12.4 Q4(iv): Show that cos^-1(3/5) - cos^-1(8/17) = cos^-1(84/85).

Answer: Step 1: Use the formula cos^-1(P) - cos^-1(Q) = cos^-1(PQ + sqrt((1 - P^2)(1 - Q^2))) with P = 3/5 and Q = 8/17.

Step 2: Compute PQ = (3/5)(8/17) = 24/85.

Step 3: Compute sqrt((1 - P^2)(1 - Q^2)) = sqrt(1 - 9/25)*sqrt(1 - 64/289) = (4/5)(15/17) = 60/85.

Step 4: Add: 24/85 + 60/85 = 84/85.

Final answer: cos^-1(3/5) - cos^-1(8/17) = cos^-1(84/85), as required.

Question: Exercise 12.4 Q4(v): Show that 2tan^-1(2/3) = cos^-1(5/13).

Answer: Step 1: Let A = tan^-1(2/3), so tan(A) = 2/3.

Step 2: Double the angle: tan(2A) = 2tan(A)/(1 - tan^2(A)) = (4/3)/(1 - 4/9)

= (4/3)/(5/9) = 12/5, so 2A = tan^-1(12/5).

Step 3: Convert to cos^-1: if tan(2A) = 12/5 then, from the 5-12-13 triangle, cos(2A) = 5/13 (adjacent 5, hypotenuse 13), and 2A is acute, so 2A = cos^-1(5/13).

Final answer: 2tan^-1(2/3) = cos^-1(5/13), as required.

Question: Exercise 12.4 Q4(vi): Show that sec^-1(sqrt(5)/2) + cot^-1(3) = pi/4.

Answer: Step 1: Convert the first term: let A = sec^-1(sqrt(5)/2), so cos(A) = 2/sqrt(5), sin(A) = 1/sqrt(5) and tan(A) = 1/2, giving A = tan^-1(1/2).

Step 2: Convert the second term: cot^-1(3) = tan^-1(1/3), because cot and tan are reciprocals and both angles are acute.

Step 3: The sum is tan^-1(1/2) + tan^-1(1/3) = tan^-1((1/2 + 1/3)/(1 - (1/2)(1/3))) = tan^-1((5/6)/(5/6)) = tan^-1(1).

Step 4: tan^-1(1) = pi/4.

Final answer: sec^-1(sqrt(5)/2) + cot^-1(3) = pi/4, as required.

Question: Exercise 12.4 Q4(vii): Show that tan^-1(3/4) + tan^-1(3/5) - tan^-1(8/19) = pi/4.

Answer: Step 1: Combine the first two terms: tan^-1(3/4) + tan^-1(3/5) = tan^-1((3/4 + 3/5)/(1 - (3/4)(3/5))) = tan^-1((27/20)/(11/20)) = tan^-1(27/11).

Step 2: Now subtract the third term using tan^-1(P) - tan^-1(Q) = tan^-1((P - Q)/(1 + PQ)) with P = 27/11 and Q = 8/19: numerator = 27/11 - 8/19 = (513 - 88)/209 = 425/209.

Step 3: Denominator = 1 + (27/11)(8/19) = 1 + 216/209 = 425/209.

Step 4: The quotient is (425/209)/(425/209) = 1, so the whole expression is tan^-1(1) = pi/4.

Final answer: tan^-1(3/4) + tan^-1(3/5) - tan^-1(8/19) = pi/4, as required.

Question: Exercise 12.4 Q4(viii): Show that tan^-1(1/3) + tan^-1(1/2) = tan^-1(1/11) + tan^-1(5/6).

Answer: Step 1: Evaluate the left side: tan^-1(1/3) + tan^-1(1/2) = tan^-1((1/3 + 1/2)/(1 - (1/3)(1/2))) = tan^-1((5/6)/(5/6)) = tan^-1(1) = pi/4.

Step 2: Evaluate the right side: tan^-1(1/11) + tan^-1(5/6) = tan^-1((1/11 + 5/6)/(1 - (1/11)(5/6))) = tan^-1((61/66)/(61/66)) = tan^-1(1) = pi/4.

Step 3: Both sides equal pi/4, so they are equal to each other.

Final answer: tan^-1(1/3) + tan^-1(1/2) = tan^-1(1/11) + tan^-1(5/6), as required.

Question: Exercise 12.4 Q5(i): Prove that sin(cos^-1(y)) = sqrt(1 - y^2).

Answer: Step 1: Let A = cos^-1(y), so cos(A) = y and 0 <= A <= pi.

Step 2: Use sin^2(A) + cos^2(A) = 1: sin(A) = sqrt(1 - cos^2(A)) = sqrt(1 - y^2).

Step 3: The positive square root is taken because sin(A) >= 0 for A in [0, pi].

Step 4: Replace A by cos^-1(y).

Final answer: sin(cos^-1(y)) = sqrt(1 - y^2), as required.

Question: Exercise 12.4 Q5(ii): Prove that tan^-1(y) + tan^-1(-y) = 0.

Answer: Step 1: Start with the left side and use the formula tan^-1(P) + tan^-1(Q) = tan^-1((P + Q)/(1 - PQ)) with P = y and Q = -y.

Step 2: Numerator: y + (-y) = 0; denominator: 1 - (y)(-y) = 1 + y^2.

Step 3: So the sum is tan^-1(0/(1 + y^2)) = tan^-1(0) = 0.

Final answer: tan^-1(y) + tan^-1(-y) = 0, as required (this is the odd property of tan^-1).

Question: Exercise 12.4 Q5(iii): Prove that cos^-1(y) + cos^-1(-y) = pi.

Answer: Step 1: Let A = cos^-1(y), so cos(A) = y and 0 <= A <= pi.

Step 2: Consider pi - A: it also lies in [0, pi], and cos(pi - A) = -cos(A) = -y.

Step 3: Therefore pi - A is exactly the principal value cos^-1(-y), that is, cos^-1(-y) = pi - A.

Step 4: Add A to both sides: A + cos^-1(-y) = pi, and replacing A gives the result.

Final answer: cos^-1(y) + cos^-1(-y) = pi, as required.

Question: Exercise 12.4 Q5(iv): Prove that sin(2cos^-1(y)) = 2y*sqrt(1 - y^2).

Answer: Step 1: Let A = cos^-1(y), so cos(A) = y and 0 <= A <= pi; the required expression is sin(2A).

Step 2: Use the double-angle formula: sin(2A) = 2sin(A)cos(A).

Step 3: From Q5(i), sin(A) = sin(cos^-1(y)) = sqrt(1 - y^2), and cos(A) = y.

Step 4: Substitute: sin(2A) = 2*sqrt(1 - y^2)*y = 2y*sqrt(1 - y^2).

Final answer: sin(2cos^-1(y)) = 2y*sqrt(1 - y^2), as required.

Exercise 12.5

Question: Exercise 12.5 Q1: Find the general solution of the trigonometric equation cos(x) + 1/2 = 0.

Answer: Step 1: Isolate the trigonometric function: cos(x) = -1/2.

Step 2: The reference angle is pi/3, because cos(pi/3) = 1/2; cosine is negative in the second and third quadrants, giving the particular solutions x = pi - pi/3 = 2*pi/3 and x = pi + pi/3 = 4*pi/3 in [0, 2*pi].

Step 3: Cosine has period 2*pi, so add 2n*pi, n in Z, to each particular solution.

Step 4: Verify: cos(2*pi/3) + 1/2 = -1/2 + 1/2 = 0 and cos(4*pi/3) + 1/2 = -1/2 + 1/2 = 0.

Final answer: x = 2*pi/3 + 2n*pi or x = 4*pi/3 + 2n*pi, n in Z.

Question: Exercise 12.5 Q2: Find the general solution of the trigonometric equation tan(x) + 1/sqrt(3) = 0.

Answer: Step 1: Isolate the trigonometric function: tan(x) = -1/sqrt(3).

Step 2: The reference angle is pi/6, because tan(pi/6) = 1/sqrt(3); tangent is negative in the second and fourth quadrants, and a convenient particular solution is x = -pi/6 (equivalently 5*pi/6 in [0, 2*pi]).

Step 3: Tangent has period pi, so add n*pi, n in Z.

Step 4: Verify: tan(5*pi/6) + 1/sqrt(3) = -1/sqrt(3) + 1/sqrt(3) = 0.

Final answer: x = 5*pi/6 + n*pi, n in Z.

Question: Exercise 12.5 Q3: Find the general solution of the trigonometric equation cosec(3x/2) + 2 = 0.

Answer: Step 1: Isolate the trigonometric function: cosec(3x/2) = -2, that is, sin(3x/2) = -1/2.

Step 2: The reference angle is pi/6; sine is negative in the third and fourth quadrants, so 3x/2 = pi + pi/6 = 7*pi/6 or 3x/2 = 2*pi - pi/6 = 11*pi/6.

Step 3: The period of sin(3x/2) is 2*pi/(3/2) = 4*pi/3, so 3x/2 = 7*pi/6 + 2n*pi or 3x/2 = 11*pi/6 + 2n*pi, n in Z.

Step 4: Divide by 3/2 (multiply by 2/3): x = 7*pi/9 + 4n*pi/3 or x = 11*pi/9 + 4n*pi/3.

Step 5: Verify with x = 7*pi/9: 3x/2 = 7*pi/6, cosec(7*pi/6) = 1/sin(7*pi/6) = -2, so cosec(3x/2) + 2 = 0.

Final answer: x = 7*pi/9 + 4n*pi/3 or x = 11*pi/9 + 4n*pi/3, n in Z.

Question: Exercise 12.5 Q4: Find the general solution of the trigonometric equation sec^2(theta) - 4/3 = 0.

Answer: Step 1: Isolate the squared function: sec^2(theta) = 4/3, so cos^2(theta) = 3/4 and cos(theta) = +/- sqrt(3)/2.

Step 2: The reference angle is pi/6, because cos(pi/6) = sqrt(3)/2.

Step 3: Cosine takes the value sqrt(3)/2 at theta = pi/6 and 11*pi/6, and the value -sqrt(3)/2 at theta = 5*pi/6 and 7*pi/6, in [0, 2*pi].

Step 4: Cosine has period 2*pi, so add 2n*pi, n in Z, to each value.

Step 5: Verify with theta = 5*pi/6: sec^2(5*pi/6) = 1/cos^2(5*pi/6) = 1/(3/4) = 4/3, so sec^2(theta) - 4/3 = 0.

Final answer: theta = pi/6 + 2n*pi, 5*pi/6 + 2n*pi, 7*pi/6 + 2n*pi or 11*pi/6 + 2n*pi, n in Z.

Question: Exercise 12.5 Q5: Find the general solution of the trigonometric equation 4sin^2(x) = 1.

Answer: Step 1: Isolate the squared function: sin^2(x) = 1/4, so sin(x) =

+/- 1/2.

Step 2: The reference angle is pi/6, because sin(pi/6) = 1/2.

Step 3: Sine takes the value 1/2 at x = pi/6 and 5*pi/6, and the value -1/2 at x = 7*pi/6 and 11*pi/6, in [0, 2*pi].

Step 4: Sine has period 2*pi, so add 2n*pi, n in Z, to each value.

Step 5: Verify with x = 7*pi/6: 4sin^2(7*pi/6) = 4(1/4) = 1.

Final answer: x = pi/6 + 2n*pi, 5*pi/6 + 2n*pi, 7*pi/6 + 2n*pi or 11*pi/6 + 2n*pi, n in Z.

Question: Exercise 12.5 Q6: Find the general solution of the trigonometric equation sec^2(x) = 4.

Answer: Step 1: Isolate the squared function: cos^2(x) = 1/4, so cos(x) = +/- 1/2.

Step 2: The reference angle is pi/3, because cos(pi/3) = 1/2.

Step 3: Cosine takes the value 1/2 at x = pi/3 and 5*pi/3, and the value -1/2 at x = 2*pi/3 and 4*pi/3, in [0, 2*pi].

Step 4: Cosine has period 2*pi, so add 2n*pi, n in Z, to each value.

Step 5: Verify with x = 2*pi/3: sec^2(2*pi/3) = 1/cos^2(2*pi/3) = 1/(1/4) = 4.

Final answer: x = pi/3 + 2n*pi, 2*pi/3 + 2n*pi, 4*pi/3 + 2n*pi or 5*pi/3 + 2n*pi, n in Z.

Question: Exercise 12.5 Q7: Find the general solution of the trigonometric equation 2cos(x) + sin^2(x) = 1.

Answer: Step 1: Replace sin^2(x) by 1 - cos^2(x): 2cos(x) + 1 - cos^2(x) = 1.

Step 2: Simplify: 2cos(x) - cos^2(x) = 0, that is, cos(x)(2 - cos(x)) = 0.

Step 3: So cos(x) = 0 or cos(x) = 2; the second is impossible because cos(x) <= 1, so only cos(x) = 0 remains.

Step 4: cos(x) = 0 at x = pi/2 and 3*pi/2 in [0, 2*pi], and cosine has period 2*pi, which combines into x = pi/2 + n*pi, n in Z (also written as x = 2n*pi +/- pi/2).

Step 5: Verify with x = pi/2: 2cos(pi/2) + sin^2(pi/2) = 0 + 1 = 1.

Final answer: x = pi/2 + n*pi, n in Z.

Question: Exercise 12.5 Q8: Find the general solution of the trigonometric equation 4cos^2(x) - 8sin(x) + 1 = 0.

Answer: Step 1: Replace cos^2(x) by 1 - sin^2(x): 4(1 - sin^2(x)) - 8sin(x) + 1 = 0, that is, -4sin^2(x) - 8sin(x) + 5 = 0.

Step 2: Multiply by -1: 4sin^2(x) + 8sin(x) - 5 = 0, a quadratic in sin(x).

Step 3: Solve by the quadratic formula: sin(x) = (-8 +/- sqrt(64 + 80))/8 = (-8 +/- 12)/8, giving sin(x) = 1/2 or sin(x) = -5/2.

Step 4: Reject sin(x) = -5/2 because sine cannot be less than -1; keep sin(x) = 1/2.

Step 5: The reference angle is pi/6; sine is positive in the first and second quadrants, giving x = pi/6 and x = 5*pi/6 in [0, 2*pi], and sine has period 2*pi.

Step 6: Verify with x = pi/6: 4cos^2(pi/6) - 8sin(pi/6) + 1 = 4(3/4) - 8(1/2) + 1 = 3 - 4 + 1 = 0.

Final answer: x = pi/6 + 2n*pi or x = 5*pi/6 + 2n*pi, n in Z.

Question: Exercise 12.5 Q9: Find the general solution of the trigonometric

equation 3tan^2(x) + 2*sqrt(3)tan(x) + 1 = 0.

Answer: Step 1: Recognise the left side as a perfect square in tan(x): 3tan^2(x) + 2*sqrt(3)tan(x) + 1 = (sqrt(3)tan(x) + 1)^2.

Step 2: So (sqrt(3)tan(x) + 1)^2 = 0, giving sqrt(3)tan(x) + 1 = 0.

Step 3: Isolate: tan(x) = -1/sqrt(3).

Step 4: The reference angle is pi/6; a particular solution is x = 5*pi/6 in [0, 2*pi], and tangent has period pi.

Step 5: Verify with x = 5*pi/6: tan(5*pi/6) = -1/sqrt(3), so 3(1/3) + 2*sqrt(3)(-1/sqrt(3)) + 1 = 1 - 2 + 1 = 0.

Final answer: x = 5*pi/6 + n*pi, n in Z.

Question: Exercise 12.5 Q10: Find the general solution of the trigonometric equation (cosec(x) + 2)(2cos(x) - 1) = 0.

Answer: Step 1: A product is zero when one factor is zero: cosec(x) + 2 = 0 or 2cos(x) - 1 = 0.

Step 2: First factor: cosec(x) = -2, that is, sin(x) = -1/2, giving x = 7*pi/6 or x = 11*pi/6 in [0, 2*pi] (reference angle pi/6, sine negative in the third and fourth quadrants).

Step 3: Second factor: cos(x) = 1/2, giving x = pi/3 or x = 5*pi/3 in [0, 2*pi].

Step 4: Both functions repeat with the relevant periods, so add 2n*pi, n in Z, to each particular solution (note sin(x) is not 0 at any of these values, so cosec(x) is defined).

Step 5: Verify: at x = pi/3, cosec(x) + 2 is not zero but 2cos(pi/3) - 1 = 0; at x = 7*pi/6, cosec(7*pi/6) + 2 = -2 + 2 = 0; in both cases the product is 0.

Final answer: x = pi/3 + 2n*pi, 5*pi/3 + 2n*pi, 7*pi/6 + 2n*pi or 11*pi/6 + 2n*pi, n in Z.

Question: Exercise 12.5 Q11: Find the general solution of the trigonometric equation tan(theta) = 2sin(theta).

Answer: Step 1: Write tan(theta) = sin(theta)/cos(theta) (with cos(theta) not 0): sin(theta)/cos(theta) = 2sin(theta).

Step 2: Bring all to one side and factor: sin(theta) - 2sin(theta)cos(theta) = 0 after multiplying by cos(theta), that is, sin(theta)(1 - 2cos(theta)) = 0.

Step 3: First case sin(theta) = 0 gives theta = n*pi, n in Z.

Step 4: Second case cos(theta) = 1/2 gives theta = pi/3 + 2n*pi or theta = 5*pi/3 + 2n*pi, n in Z (reference angle pi/3).

Step 5: Verify: at theta = pi, tan(pi) = 0 and 2sin(pi) = 0; at theta = pi/3, tan(pi/3) = sqrt(3) and 2sin(pi/3) = sqrt(3).

Final answer: theta = n*pi, or theta = pi/3 + 2n*pi, or theta = 5*pi/3 + 2n*pi, n in Z.

Question: Exercise 12.5 Q12: Find the general solution of the trigonometric equation 4sin^2(x/2) - 3 = 0.

Answer: Step 1: Isolate the squared function: sin^2(x/2) = 3/4, so sin(x/2) = +/- sqrt(3)/2.

Step 2: The reference angle is pi/3, because sin(pi/3) = sqrt(3)/2.

Step 3: In one period of sin(x/2), which is 4*pi, x/2 takes the values pi/3, 2*pi/3, 4*pi/3, 5*pi/3 up to multiples of 2*pi; the distinct solution families for x/2 are x/2 = pi/3 + 2n*pi and x/2 = 2*pi/3 + 2n*pi together with their supplements, which reduce to x/2 = +/- pi/3 + n*pi.

Step 4: Multiply by 2: x = 2*pi/3 + 2n*pi or x = 4*pi/3 + 2n*pi, n in Z

(equivalently cos(x) = -1/2, since cos(x) = 1 - 2sin^2(x/2) = 1 - 3/2 = -1/2).

Step 5: Verify with x = 2*pi/3: 4sin^2(pi/3) - 3 = 4(3/4) - 3 = 0. (Note x = pi is not a solution: 4sin^2(pi/2) - 3 = 1, not 0.)

Final answer: x = 2*pi/3 + 2n*pi or x = 4*pi/3 + 2n*pi, n in Z.

Question: Exercise 12.5 Q13: Find the general solution of the trigonometric equation cos(2x) = cos(x).

Answer: Step 1: Use cos(2x) = 2cos^2(x) - 1: 2cos^2(x) - 1 = cos(x).

Step 2: Rearrange into a quadratic: 2cos^2(x) - cos(x) - 1 = 0.

Step 3: Factorise: (2cos(x) + 1)(cos(x) - 1) = 0, so cos(x) = 1 or cos(x) = -1/2.

Step 4: cos(x) = 1 gives x = 2n*pi; cos(x) = -1/2 (reference angle pi/3, second and third quadrants) gives x = 2*pi/3 + 2n*pi or x = 4*pi/3 + 2n*pi, n in Z.

Step 5: Verify with x = 2*pi/3: cos(4*pi/3) = -1/2 and cos(2*pi/3) = -1/2, so the two sides are equal.

Final answer: x = 2n*pi, or x = 2*pi/3 + 2n*pi, or x = 4*pi/3 + 2n*pi, n in Z.

Question: Exercise 12.5 Q14: Find the general solution of the trigonometric equation sin(4*theta) - sin(2*theta) = cos(3*theta).

Answer: Step 1: Use the sum-to-product formula on the left: sin(4*theta) - sin(2*theta) = 2cos(3*theta)sin(theta).

Step 2: The equation becomes 2cos(3*theta)sin(theta) = cos(3*theta), that is, cos(3*theta)(2sin(theta) - 1) = 0.

Step 3: First case cos(3*theta) = 0: 3*theta = pi/2 + n*pi, so theta = pi/6 + n*pi/3, n in Z.

Step 4: Second case sin(theta) = 1/2: theta = pi/6 + 2n*pi or theta = 5*pi/6 + 2n*pi; both of these are already contained in the first family (take n = 0 and n = 2 in pi/6 + n*pi/3), so no new solutions appear.

Step 5: Verify with theta = pi/2 (from the first family): sin(2*pi) - sin(pi) = 0 and cos(3*pi/2) = 0, so both sides are 0; also with theta = 7*pi/6: sin(14*pi/3) - sin(7*pi/3) = sin(2*pi/3) - sin(pi/3) = 0 and cos(7*pi/2) = cos(3*pi/2) = 0.

Final answer: theta = pi/6 + n*pi/3, n in Z.

Question: Exercise 12.5 Q15: Find the general solution of the trigonometric equation cos(theta) + cos(3*theta) + cos(5*theta) = 0.

Answer: Step 1: Group the first and last terms and use the sum-to-product formula: cos(theta) + cos(5*theta) = 2cos(3*theta)cos(2*theta).

Step 2: The equation becomes 2cos(3*theta)cos(2*theta) + cos(3*theta) = 0, that is, cos(3*theta)(2cos(2*theta) + 1) = 0.

Step 3: First case cos(3*theta) = 0: 3*theta = pi/2 + n*pi, so theta = pi/6 + n*pi/3, n in Z.

Step 4: Second case cos(2*theta) = -1/2: 2*theta = 2n*pi +/- 2*pi/3, so theta = n*pi +/- pi/3, n in Z (that is, theta = pi/3 + n*pi or theta = 2*pi/3 + n*pi).

Step 5: Verify with theta = pi/3 (second family): cos(pi/3) + cos(pi) + cos(5*pi/3) = 1/2 - 1 + 1/2 = 0; and with theta = pi/6 (first family): cos(pi/6) + cos(pi/2) + cos(5*pi/6) = sqrt(3)/2 + 0 - sqrt(3)/2 = 0.

Final answer: theta = pi/6 + n*pi/3, or theta = pi/3 + n*pi, or theta =

2*pi/3 + n*pi, n in Z.

Question: Exercise 12.5 Q16: Find the general solution of the trigonometric equation sin(x) + sin(3x) + sin(5x) + sin(7x) = 0.

Answer: Step 1: Pair the terms: sin(x) + sin(7x) = 2sin(4x)cos(3x) and sin(3x) + sin(5x) = 2sin(4x)cos(x), using the sum-to-product formula.

Step 2: The equation becomes 2sin(4x)(cos(3x) + cos(x)) = 0; use cos(3x) + cos(x) = 2cos(2x)cos(x) to get 4sin(4x)cos(2x)cos(x) = 0.

Step 3: First case sin(4x) = 0: 4x = n*pi, so x = n*pi/4, n in Z.

Step 4: Second case cos(2x) = 0 gives x = pi/4 + n*pi/2, and third case cos(x) = 0 gives x = pi/2 + n*pi; both families are already contained in x = n*pi/4 (odd and even multiples of pi/4 respectively), so they add nothing new.

Step 5: Verify with x = pi/4: sin(pi/4) + sin(3*pi/4) + sin(5*pi/4) + sin(7*pi/4) = 0.71 + 0.71 - 0.71 - 0.71 = 0; and with x = pi/2: 1 - 1 + 1 - 1 = 0.

Final answer: x = n*pi/4, n in Z.

Question: Exercise 12.5 Q17: Find the general solution of the trigonometric equation cosec(theta) = sqrt(3) + cot(theta).

Answer: Step 1: Note the domain first: cosec(theta) and cot(theta) require sin(theta) not 0, so theta = n*pi is excluded from the start.

Step 2: Write both functions over sin(theta): 1/sin(theta) - cos(theta)/sin(theta) = sqrt(3), that is, (1 - cos(theta))/sin(theta) = sqrt(3).

Step 3: Use the half-angle identity (1 - cos(theta))/sin(theta) = tan(theta/2): the equation becomes tan(theta/2) = sqrt(3).

Step 4: The reference angle is pi/3, and tangent has period pi, so theta/2 =

pi/3 + n*pi, giving theta = 2*pi/3 + 2n*pi, n in Z.

Step 5: Verify with theta = 2*pi/3: cosec(2*pi/3) = 2/sqrt(3), about 1.155, and sqrt(3) + cot(2*pi/3) = sqrt(3) - 1/sqrt(3) = 2/sqrt(3), the same value; no root was introduced or lost because we never squared.

Final answer: theta = 2*pi/3 + 2n*pi, n in Z.

Question: Exercise 12.5 Q18: Find the general solution of the trigonometric equation 2cos(x) + sin^2(x) - 1 = 0.

Answer: Step 1: Replace sin^2(x) by 1 - cos^2(x): 2cos(x) + (1 - cos^2(x)) - 1 = 0.

Step 2: Simplify: 2cos(x) - cos^2(x) = 0, that is, cos(x)(2 - cos(x)) = 0.

Step 3: So cos(x) = 0 or cos(x) = 2; the second is impossible because cos(x) <= 1, so only cos(x) = 0 remains.

Step 4: cos(x) = 0 gives x = pi/2 + n*pi, n in Z.

Step 5: Verify with x = pi/2: 2cos(pi/2) + sin^2(pi/2) - 1 = 0 + 1 - 1 = 0; also with x = 3*pi/2: 0 + 1 - 1 = 0.

Final answer: x = pi/2 + n*pi, n in Z.

Review Exercise 12

Question: Review Exercise 12 Q1(i): Select the correct answer: If y = Sin(x) (the principal sine function), then its domain is: (a) -pi/2 <= x <= pi/2 (b) 0 <= x <= pi (c) [0, x], x not pi/2 (d) None of these.

Answer: Step 1: The principal sine function is the restriction of sine to the interval on which it is one-to-one and onto [-1, 1], so that sin^-1 exists.

Step 2: On [-pi/2, pi/2] sine increases from -1 to 1 and takes each value exactly once, which is exactly the restriction used in this unit.

Final answer: (a) -pi/2 <= x <= pi/2.

Question: Review Exercise 12 Q1(ii): Select the correct answer: If y = Cos(x) (the principal cosine function), then its range is: (a) -1 <= y <= 1 (b) R (c) y <= -1 or y >= 1 (d) y <= -1 or y >= 1.

Answer: Step 1: Restricting the domain of cosine (to [0, pi] for the principal function) does not change the set of values it takes.

Step 2: Cosine always satisfies -1 <= cos(x) <= 1, and it attains every value in between.

Final answer: (a) -1 <= y <= 1.

Question: Review Exercise 12 Q1(iii): Select the correct answer: If y = tan^-1(x), then its domain is: (a) -1 <= x <= 1 (b) R (c) x >= -1 or x <= 1 (d) x <= -1 or x >= 1.

Answer: Step 1: The domain of an inverse function is the range of the original function.

Step 2: The range of tangent is all real numbers, since tan(x) grows from -infinity to +infinity on (-pi/2, pi/2).

Final answer: (b) R.

Question: Review Exercise 12 Q1(iv): Select the correct answer: If y = sec^-1(x), then its range is: (a) 0 <= y <= pi, y not pi/2 (b) -pi/2 <= y <= pi/2, y not 0 (c) 0 < y < pi (d) -pi/2 < y < pi/2.

Answer: Step 1: The range of an inverse function is the (restricted) domain of the original function.

Step 2: Secant is made one-to-one on [0, pi] with the point pi/2 removed

(secant is undefined there), and this is the principal range adopted in this unit.

Final answer: (a) 0 <= y <= pi, y not pi/2.

Question: Review Exercise 12 Q1(v): Select the correct answer: An inverse function exists if and only if the function is: (a) bijective (b) a (1-1) function (c) an onto function (d) an into function.

Answer: Step 1: An inverse must pair every output back to exactly one input, so the function must be both one-to-one (no output repeated) and onto (every output reached).

Step 2: A function that is both one-to-one and onto is called bijective.

Final answer: (a) bijective.

Question: Review Exercise 12 Q1(vi): Select the correct answer: cot^-1(x) = ... : (a) pi/2 - tan^-1(x) (b) pi/2 - cot^-1(x) (c) pi/2 + tan^-1(x) (d) pi/2 + cot^-1(x).

Answer: Step 1: Let A = tan^-1(x), so tan(A) = x with -pi/2 < A < pi/2.

Step 2: Then cot(pi/2 - A) = tan(A) = x, and pi/2 - A lies in (0, pi), the principal range of cot^-1, so pi/2 - A = cot^-1(x).

Step 3: Rearranged: cot^-1(x) = pi/2 - tan^-1(x).

Final answer: (a) pi/2 - tan^-1(x).

Question: Review Exercise 12 Q1(vii): Select the correct answer: sec[sin^-1(-1/2)] = ... : (a) 2/sqrt(3) (b) sqrt(3)/2 (c) 1/2 (d) -2/sqrt(3).

Answer: Step 1: Evaluate the inner value: sin^-1(-1/2) = -pi/6, because sin(-pi/6) = -1/2 and -pi/6 is in [-pi/2, pi/2].

Step 2: Now compute sec(-pi/6) = 1/cos(-pi/6) = 1/cos(pi/6) = 1/(sqrt(3)/2)

= 2/sqrt(3).

Final answer: (a) 2/sqrt(3).

Question: Review Exercise 12 Q1(viii): Select the correct answer: cos(sin^-1(1/sqrt(2))) = ... : (a) 2/sqrt(3) (b) sqrt(3)/2 (c) 1/2 (d) 1/sqrt(2).

Answer: Step 1: Evaluate the inner value: sin^-1(1/sqrt(2)) = pi/4, because sin(pi/4) = 1/sqrt(2).

Step 2: Now compute cos(pi/4) = 1/sqrt(2).

Final answer: (d) 1/sqrt(2).

Question: Review Exercise 12 Q1(ix): Select the correct answer: cos^-1(A) + cos^-1(B) = ... : (a) cos^-1(AB - sqrt((1 - A^2)(1 - B^2))) (b) cos^-1(AB + sqrt((1 - A^2)(1 - B^2))) (c) cos^-1(AB + sqrt((1 + A^2)(1 + B^2))) (d) cos^-1(AB - sqrt((1 + A^2)(1 + B^2))).

Answer: Step 1: Put A = cos(u) and B = cos(v); then sin(u) = sqrt(1 - A^2) and sin(v) = sqrt(1 - B^2).

Step 2: cos(u + v) = cos(u)cos(v) - sin(u)sin(v) = AB - sqrt((1 - A^2)(1 - B^2)), so u + v = cos^-1(AB - sqrt((1 - A^2)(1 - B^2))).

Final answer: (a) cos^-1(AB - sqrt((1 - A^2)(1 - B^2))).

Question: Review Exercise 12 Q1(x): Select the correct answer: tan^-1(-x) = ... : (a) -tan^-1(x) (b) pi - tan^-1(x) (c) cot^-1(x) (d) tan^-1(x).

Answer: Step 1: Tangent is an odd function, tan(-t) = -tan(t), and its inverse inherits the same property.

Step 2: If tan(A) = x then tan(-A) = -x, so tan^-1(-x) = -A = -tan^-1(x).

Final answer: (a) -tan^-1(x).

Question: Review Exercise 12 Q1(xi): Select the correct answer: tan^-1(-1) = ... : (a) pi/4 (b) -pi/4 (c) pi/2 (d) -pi/2.

Answer: Step 1: We need the angle in (-pi/2, pi/2) whose tangent is -1.

Step 2: tan(pi/4) = 1 and tangent is odd, so tan(-pi/4) = -1, giving tan^-1(-1) = -pi/4.

Final answer: (b) -pi/4.

Question: Review Exercise 12 Q1(xii): Select the correct answer: If tan(2x) = -1, then the solution in the interval [-pi/2, pi/2] is: (a) -pi/8 (b) -pi/4 (c) 3*pi/8 (d) 3*pi/4.

Answer: Step 1: Isolate the angle: tan(2x) = -1 gives 2x = -pi/4 + n*pi (reference angle pi/4, tangent period pi).

Step 2: Divide by 2: x = -pi/8 + n*pi/2.

Step 3: In [-pi/2, pi/2] this gives two values, x = -pi/8 (n = 0) and x = 3*pi/8 (n = 1); check: tan(2*(3*pi/8)) = tan(3*pi/4) = -1 and tan(2*(-pi/8)) = tan(-pi/4) = -1.

Final answer: (a) -pi/8 (the value listed in the book's answer key; x = 3*pi/8, option (c), also satisfies the equation in this interval).

Question: Review Exercise 12 Q1(xiii): Select the correct answer: The general solution of 4sin(x) - 8 = 0 is: (a) {pi + 2n*pi}, n in Z (b) {pi + n*pi}, n in Z (c) {-pi + n*pi}, n in Z (d) Not possible.

Answer: Step 1: Isolate the trigonometric function: 4sin(x) = 8, so sin(x) = 2.

Step 2: Sine never exceeds 1, so sin(x) = 2 has no solution at all, particular or general.

Final answer: (d) Not possible.

Question: Review Exercise 12 Q1(xiv): Select the correct answer: The solution of the equation 2sin(x) + sqrt(3) = 0 in the 4th quadrant is: (a) pi/3 (b) -pi/3 (c) -pi/6 (d) 11*pi/6.

Answer: Step 1: Isolate the trigonometric function: sin(x) = -sqrt(3)/2.

Step 2: The reference angle is pi/3; sine is negative in the third and fourth quadrants, and the fourth-quadrant angle (written as a negative angle) is x = -pi/3.

Step 3: Check: 2sin(-pi/3) + sqrt(3) = 2(-sqrt(3)/2) + sqrt(3) = 0, and -pi/3 is in the fourth quadrant.

Final answer: (b) -pi/3 (the same direction is written 11*pi/6, option (d), when angles are measured from 0 to 2*pi).

Question: Review Exercise 12 Q1(xv): Select the correct answer: All trigonometric functions are ..... functions: (a) periodic (b) even (c) injective (d) bijective.

Answer: Step 1: Every trigonometric function repeats its values after a fixed interval (2*pi for sin, cos, sec, cosec and pi for tan, cot).

Step 2: They are not all even (sine and tangent are odd), and on their full domains they are neither injective nor bijective.

Final answer: (a) periodic.

Question: Review Exercise 12 Q1(xvi): Select the correct answer: The general solution of every trigonometric equation consists of: (a) one solution only (b) two solutions (c) infinitely many solutions (d) no real solution.

Answer: Step 1: Trigonometric functions are periodic, so whenever an equation has one solution, adding whole periods produces more solutions.

Step 2: Hence a general solution is a family containing infinitely many values, indexed by n in Z.

Final answer: (c) infinitely many solutions.

Question: Review Exercise 12 Q1(xvii): Select the correct answer: The period of cos(theta) is: (a) pi/2 (b) pi (c) 2*pi (d) 4*pi.

Answer: Step 1: The cosine curve completes one full cycle when the angle increases by 2*pi: cos(theta + 2*pi) = cos(theta), and no smaller positive shift works.

Final answer: (c) 2*pi.

Question: Review Exercise 12 Q1(xviii): Select the correct answer: One solution of sec(x) = -2 is: (a) 2*pi/3 (b) pi/3 (c) 4*pi/5 (d) -pi/3.

Answer: Step 1: Take reciprocals: sec(x) = -2 means cos(x) = -1/2.

Step 2: Test the options: cos(2*pi/3) = -1/2, so sec(2*pi/3) = -2; the other options give cos = 1/2 (options b, d) or a different value (option c).

Final answer: (a) 2*pi/3.

Question: Review Exercise 12 Q1(xix): Select the correct answer: If cos(theta) = -1/2 and sin(theta) = -sqrt(3)/2, then theta is: (a) pi/3 (b) 2*pi/3 (c) 4*pi/3 (d) 5*pi/3.

Answer: Step 1: The reference angle for both values is pi/3.

Step 2: Cosine negative puts theta in the second or third quadrant, and sine negative puts it in the third or fourth quadrant; both together force the third quadrant.

Step 3: The third-quadrant angle is theta = pi + pi/3 = 4*pi/3; check: cos(4*pi/3) = -1/2 and sin(4*pi/3) = -sqrt(3)/2.

Final answer: (c) 4*pi/3.

Question: Review Exercise 12 Q1(xx): Select the correct answer: The period of sin(x/2) is: (a) 4*pi (b) pi (c) 2*pi (d) 3*pi.

Answer: Step 1: The function is sin(kx) with k = 1/2.

Step 2: Period = 2*pi/|k| = 2*pi/(1/2) = 4*pi.

Final answer: (a) 4*pi.

Question: Review Exercise 12 Q1(xxi): Select the correct answer: The value of cos(5*pi) is: (a) 0 (b) 1 (c) -1 (d) None of these.

Answer: Step 1: Reduce by the period: 5*pi = 2*(2*pi) + pi, so cos(5*pi) = cos(pi).

Step 2: cos(pi) = -1.

Final answer: (c) -1.

Question: Review Exercise 12 Q1(xxii): Select the correct answer: The period of 3sec(x/3) is: (a) 5*pi (b) pi (c) 2*pi (d) 6*pi.

Answer: Step 1: The factor 3 does not affect the period; the angle coefficient is k = 1/3.

Step 2: Period of sec(kx) = 2*pi/|k| = 2*pi/(1/3) = 6*pi.

Final answer: (d) 6*pi.

Question: Review Exercise 12 Q1(xxiii): Select the correct answer: The period of 7tan(7x) is: (a) 7*pi/3 (b) 7*pi (c) pi/7 (d) pi.

Answer: Step 1: The factor 7 outside does not affect the period; the angle coefficient is k = 7.

Step 2: Period of tan(kx) = pi/|k| = pi/7.

Final answer: (c) pi/7.

Question: Review Exercise 12 Q2(i): Find the maximum and minimum

values of the function y = 5 - 7cos(theta).

Answer: Step 1: Write the function in the form a + b*cos(theta) with a = 5 and b = -7, so |b| = 7.

Step 2: Maximum value M = a + |b| = 5 + 7 = 12, attained at cos(theta) = -1.

Step 3: Minimum value m = a - |b| = 5 - 7 = -2, attained at cos(theta) = 1.

Final answer: Maximum value = 12, minimum value = -2.

Question: Review Exercise 12 Q2(ii): Find the maximum and minimum values of the function y = 4 + 3sin(2*theta - 5).

Answer: Step 1: Write the function in the form a + b*sin(c*theta + d) with a = 4 and b = 3; the constants c = 2 and d = -5 do not affect the extreme values.

Step 2: Maximum value M = a + |b| = 4 + 3 = 7.

Step 3: Minimum value m = a - |b| = 4 - 3 = 1.

Final answer: Maximum value = 7, minimum value = 1.

Question: Review Exercise 12 Q3: By using a graph, find the solution of the equation 2cos(theta) - theta = 0.

Answer: Step 1: Rewrite the equation as cos(theta) = theta/2 and draw y = cos(theta) and the straight line y = theta/2 on the same axes; the solution is the theta-coordinate of the intersection point.

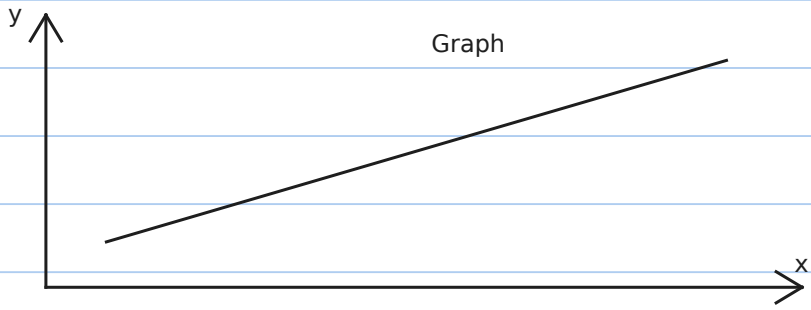
Step 2: Table for y = cos(theta): at theta = 0, pi/4, pi/3, pi/2 the values are 1, 0.71, 0.5, 0.

Step 3: Table for y = theta/2: at the same theta values the line gives 0, 0.39, 0.52, 0.79; for theta < 0 the line is negative while cos(theta) is positive, so there is no crossing on the negative side.

Step 4: On the positive side the falling cosine curve crosses the rising line once, just below theta = pi/3; reading the graph gives theta = pi/3 approximately (the accurate value is theta = 1.03, while pi/3 = 1.047).

Step 5: Verify approximately: 2cos(1.03) - 1.03 = 2(0.515) - 1.03 = 0.

Final answer: theta = pi/3 approximately (more accurately theta = 1.03 radians).



Question: Review Exercise 12 Q4(i): Without using a calculator, show that cosec^-1(13/12) = tan^-1(12/15) (as printed in the book).

Answer: Step 1: Let A = cosec^-1(13/12), so cosec(A) = 13/12 and sin(A) = 12/13, with A acute.

Step 2: Find cos(A) = sqrt(1 - (12/13)^2) = sqrt(25/169) = 5/13.

Step 3: Form tan(A) = sin(A)/cos(A) = (12/13)/(5/13) = 12/5, so A = tan^-1(12/5).

Final answer: cosec^-1(13/12) = tan^-1(12/5). The denominator printed in the book, 15, is a misprint for 5: tan^-1(12/15) is about 38.7 deg, whereas cosec^-1(13/12) is about 67.4 deg, equal to tan^-1(12/5).

Question: Review Exercise 12 Q4(ii): Without using a calculator, show that cos^-1(sqrt(3)/2) = tan^-1(1/sqrt(3)).

Answer: Step 1: Evaluate the left side: cos^-1(sqrt(3)/2) = pi/6, because cos(pi/6) = sqrt(3)/2 and pi/6 is in [0, pi].

Step 2: Evaluate the right side: tan^-1(1/sqrt(3)) = pi/6, because tan(pi/6) = 1/sqrt(3) and pi/6 is in (-pi/2, pi/2).

Step 3: Both sides equal pi/6, so they are equal.

Final answer: cos^-1(sqrt(3)/2) = tan^-1(1/sqrt(3)) = pi/6, as required.

Question: Review Exercise 12 Q5: Show that cos^-1(7/25) - cos^-1(5/13) = cos^-1(323/325).

Answer: Step 1: Use the formula cos^-1(P) - cos^-1(Q) = cos^-1(PQ + sqrt((1 - P^2)(1 - Q^2))) with P = 7/25 and Q = 5/13.

Step 2: Compute PQ = (7/25)(5/13) = 35/325.

Step 3: Compute sqrt((1 - P^2)(1 - Q^2)) = sqrt(1 - 49/625)*sqrt(1 - 25/169) = (24/25)(12/13) = 288/325.

Step 4: Add: 35/325 + 288/325 = 323/325.

Final answer: cos^-1(7/25) - cos^-1(5/13) = cos^-1(323/325), as required.

Question: Review Exercise 12 Q6: Find the solution sets of the equation sec(3x) = sec(x).

Answer: Step 1: Take reciprocals (where both sides are defined): cos(3x) = cos(x), with the domain restriction cos(x) not 0 and cos(3x) not 0.

Step 2: cos(3x) = cos(x) gives two families: 3x = 2n*pi + x, so x = n*pi; or 3x = 2n*pi - x, so 4x = 2n*pi and x = n*pi/2, n in Z.

Step 3: Combine: the family x = n*pi/2 contains x = n*pi, so all candidates are x = n*pi/2.

Step 4: Discard the values where the original equation is undefined: at x = pi/2 + n*pi (the odd multiples of pi/2 inside the family), cos(x) = 0, so sec(x) does not exist and these values must be rejected.

Step 5: Verify the survivors: at x = 0, sec(0) = 1 = sec(0); at x = pi, sec(3*pi) = -1 = sec(pi).

Final answer: The solution set is {n*pi | n in Z}.